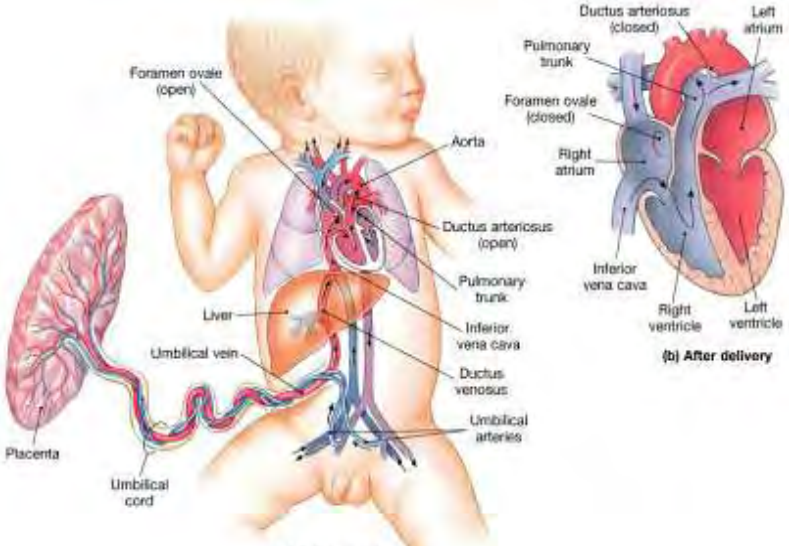
CVS by Dr.Asma Hadida

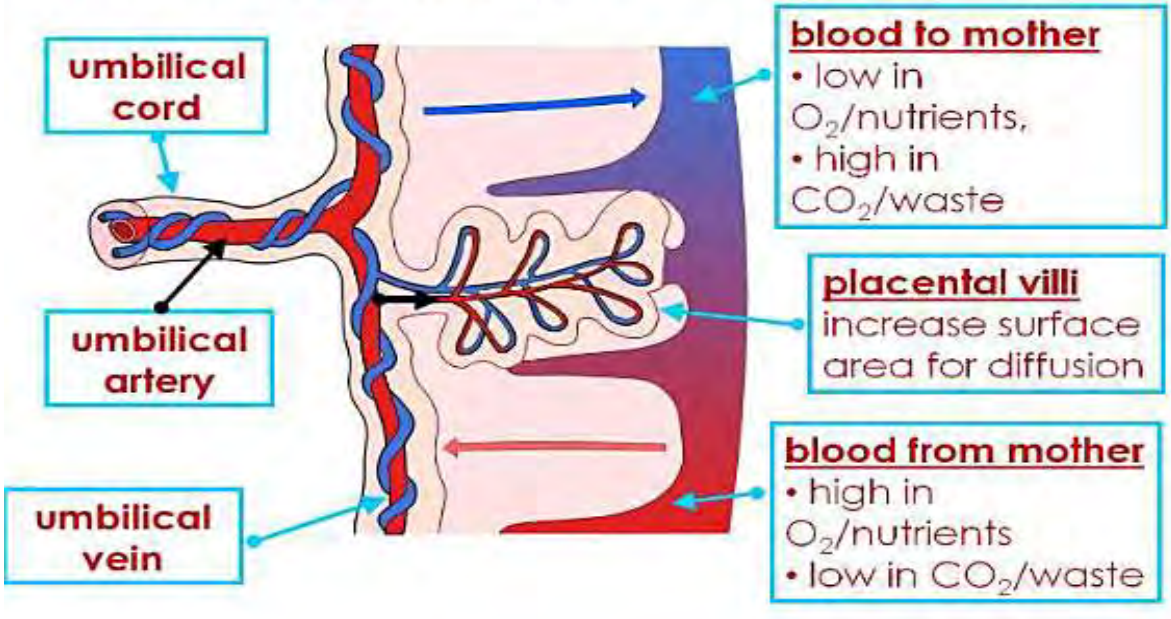
2015_2016

Fetal circulation



DIFFUSION AND THE PLACENTA

How does the placenta work?



Fetal and neonatal circulation

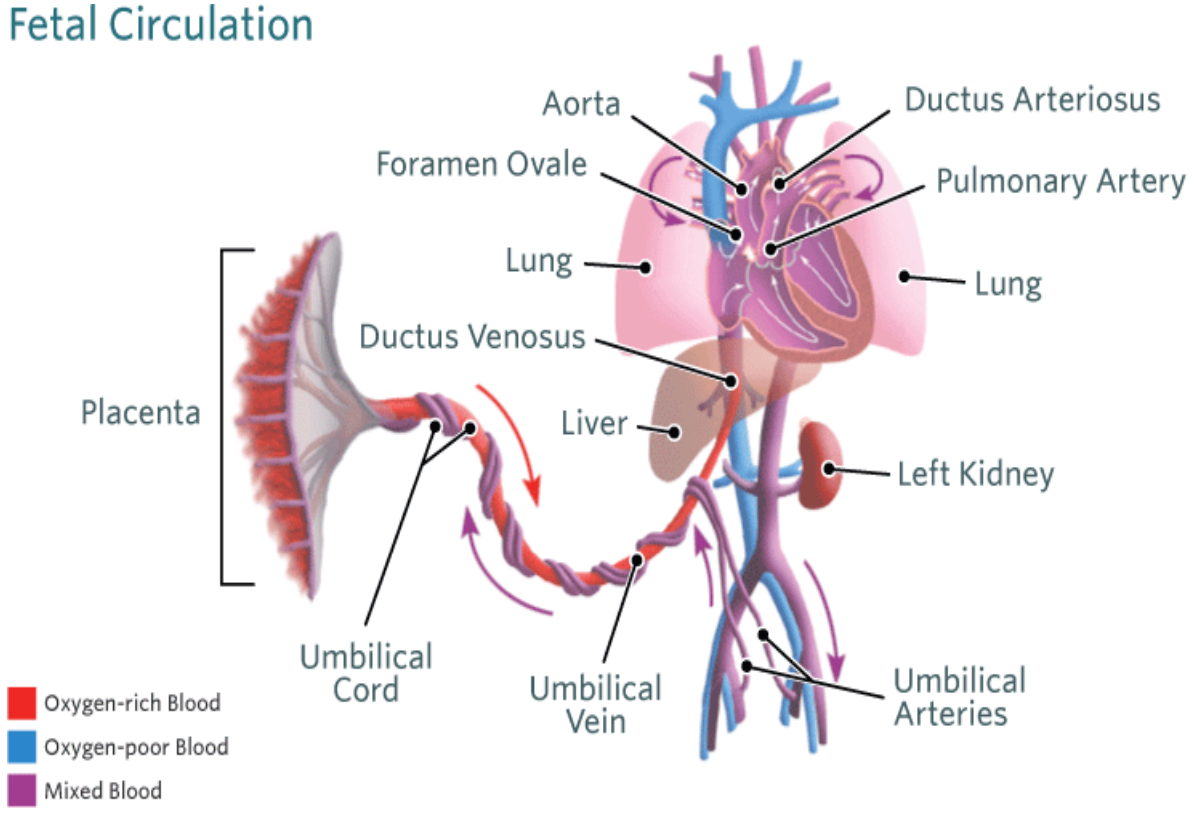
Patency of three structures ***(the foramen ovale , ductus artriosus , and ductus venosus)*** is a critical features of the cardiovascular anatomy and physiology of the fetus .

a. Normal physiology :

1. Fetal circulation :

- Fetal blood is oxygenated in the placenta and then enters the umbilical vein:
  - One portion of the oxygenated blood perfuse **the liver** and proceeds to the inferior vena cava **via** the hepatic vein
  - Another portion enters the **ductus venosus** , which empties directly into inferior vena cava
- Together with venous return from the lower part of the body , this blood flows into the **right atrium :**

- In right atrium large portion of blood **80%** to → left atrium **via foramen oval** then to the systemic circulation.
- Small amount of blood in right atrium **20%** will pass and enter right ventricle with about< **10%** reach the **lung** and the rest will enter the systemic circulation **via ductus arteriousus** to the descending aorta ***(because of high pulmonary vascular resistance and low systemic vascular resistance)***



2. **Transition to neonatal circulation :**
   - At birth , the infant s first breaths cause an increase in arterial oxygen tension (PO2) this **lowers** pulmonary vascular resistance , resulting in **increase** pulmonary blood flow . the increased pulmonary venous return to the left atrium causes the pressure to rise and results in **functional closure of foramen ovale**
   - **Systemic vascular resistance is** increased by elimination of the low resistance circuit of the placenta at birth
   - **Closure of ductus arteriousus** occurs shortly after birth , first functionally as a result of a rise in (PO2) and decline in circulating prostaglandin (E) and then anatomically as a result of fibrosis
3. **Normal changes in pulmonary vascular resistance :**
   - Pulmonary vascular resistance drops dramatically within 24 h and then continuous to fall slowly , reaching normal adult levels by approximately 6 weeks , it is inversely related to the diameter of the small pulmonary arterioles

Abnormalities of pulmonary circulation

- **Persistent pulmonary hypertension of the newborn :** pulmonary vascular resistance can remain high after birth if constriction of the arteriolar Lumina by pathological process occurs (hypoxemia – acidosis – or some unidentified factors) . the resultant pulmonary hypertension leads to right to left shunt at the ductus or foramen ovale

Per natal /maternal risk factors associated with cardiac disease:

1. **DM :**
   • Lt. ventricular obstruction e.g. Aortic stenosis
   • Transposition of great arties
   • VSD.
   •**cardiomyopathy**.
2. **Rubella** : • PDA. • PS.
3. **SLE :** • Complete HB. • Pericarditis. • Endomyocardial fibrosis.
4. **Alcohol use :** • PS. • VSD.
5. **Lethium :** • Ebstain anomalies.
6. **Aspirin :** • persistent pulmonary hypertension.
7. **Coxaci B virus** :• Myocarditis.

⁂ Syndromes associated with CHD :

1. **Down syndrome** : • VSD or AV canal **"AVSD"**.
2. **Edward syndrome** : • VSD or double outlet Rt. V.
3. **Patue syndrome** : • VSD.
4. **Turner syndrome** : • **Coarectation of aorta .**
5. **Noonan syndrome** : • Cardiomyopathy . • PS • ASD.
6. **Tubersclerosis** : • Cardiac rhabdomyoma.
7. **Alagille syndrome** : • peripheral pul. A stenosis.
8. **Marfan syndrome** :• Aortic incompetence.
   • Mitral valve prolapsed.
   • Aortic aneurysm and aortic root dilation & rupture
9. **Digeorge syndrome** : • Aortic arch anomaly. • TOF. Inter irrupted aortic arch , truncus artriosus • . **(CATCH 22) ,**
10. **William syndrome :** supravalvular aortic stenosis and diffuse vascular stenosis

evaluation of cardiovascular system

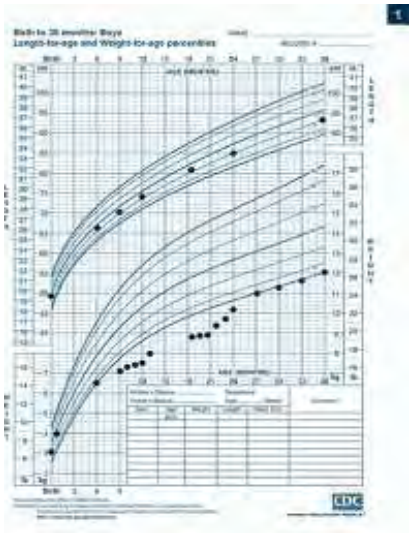
History :

1. **Cyanosis** :
   a. **Peripheral** : bluish discoloration around the amount and eyelid , hands and feet but not MM
   b. **Central** : in mucous membrane (tip of tongue) indicate right to left shunt
2. **Shortness of breath** : may be manifested as difficult feeding and sweating during feeding
3. **Family history** : some congenital heart disease may be familial
4. **Chest pain** : is common in pediatric age group especially in adolescent but rarely due to cardiac cause

Physical examination :

1. **General observation :**

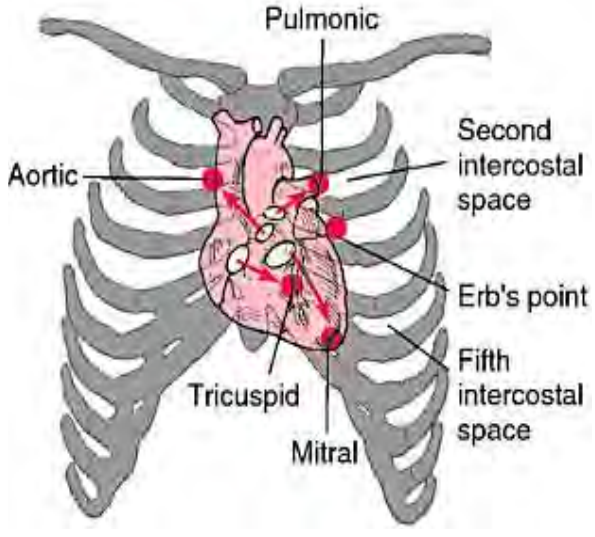
- **Weight :** decrease wt gain may indicate cardiac disease
- **Cyanosis and clubbing** of the finger and toe are diagnostic to right to left shunt
- Dysmorphic features
- Squatting position **(TOF)**



Physical examination :

2. pulse : brachial or radial arteries in both arms simultaneously in order to determine timing and volume . also radio femoral to exclude radio femoral delay **(in coarction of aorta)**

3. **blood pressure** : measured over the brachial and popliteal arteries. **Imp**



Physical examination :

4. pericardial palpation : Normal lies on higher plane than adult.

- **Apex beat** :
  - **Infant** : 3th - 4th ICS just outside mid clavicular line.
  - **2-5 year** : 4th ICS at mid clavicular line.
  - **Over 5 years** : 5th ICS mid clavicular line.

, a thrill or (palpable murmur) which defines an area of maximal turbulence . a diffuse impulse in the parasternal region (heave) may indicate right ventricular enlargement

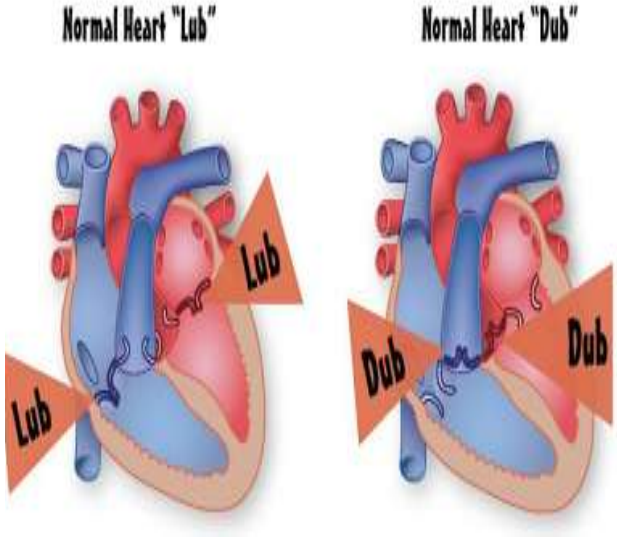
5. **cardiac auscultation :**

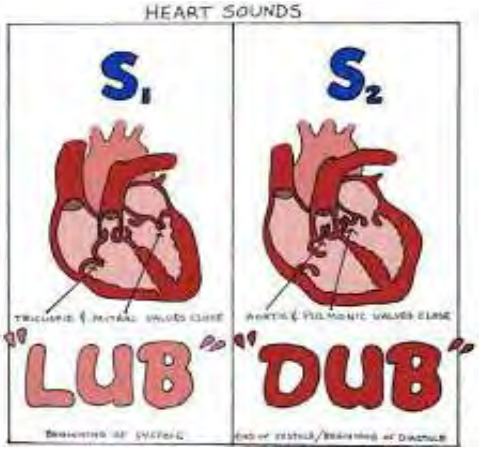
- **heart sounds :**

1. **1st HS S1** may be single or split
2. **2nd HS S2** is split during inspiration , abnormally wide splitting occurs with right ventricular overload and prolonged right ventricular emptying (ASD)
3. **3rd HS** in 1/3 of all children may be physiological or pathological

- ***Rt axis deviation in the EEG is normal finding in infant and young children***.

4. **4th HS is always pathological**





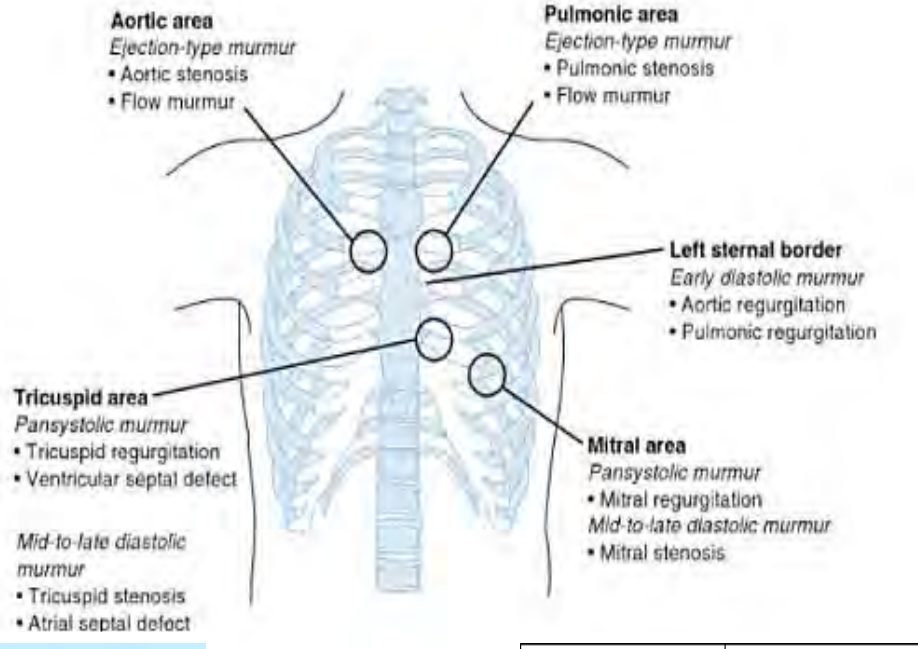
Toha_Tohty

Comparing the 3rd and 4th heart sounds

LearnTheHeart.com

S3 - "ventricular gallop"	S4 - "atrial gallop"
Occurs in early diastole	Occurs in late diastole
Occurs during passive LV filling	Occurs during active LV filling
May be normal at times	Almost always abnormal
Requires a very compliant LV	Requires a non-compliant LV
Can be a sign of systolic CHF	Can be a sign of diastolic CHF

See more
In CVS
Examination



Grading systolic murmurs

Intensity	Description
Grade I/VI	Barely audible
Grade II/VI	Audible, but soft
Grade III/VI	Easily audible
Grade IV/VI	Easily audible and associated with a thrill
Grade V/VI	Easily audible, associated with a thrill, and still heard with the stethoscope only lightly on the chest
Grade VI/VI	Easily audible, associated with a thrill, and still heard with the stethoscope off of the chest

Systolic murmurs	Diastolic murmurs
Midsystolic Aortic stenosis Pulmonic stenosis Atrial septal defect HOCM Holosystolic Mitral regurgitation Tricuspid regurgitation VSD Late systolic murmur Mitral valve prolapse	Early diastolic Aortic regurgitation Pulmonic regurgitation Austin-Flint Mid/late diastolic Mitral stenosis Tricuspid stenosis Other rare murmurs Patent ductus arteriosus

Innocent vs. Pathologic Murmurs

Innocent	Pathologic
☐ Systolic	☐ Diastolic
☐ Ejection	☐ Holosystolic
☐ Soft or vibratory	☐ Harsh
☐ Grade 1-2/6	☐ Grade ≥ 3/6
☐ Normal S1, S2	☐ Abnormal split S2
☐ No extra sounds	☐ Extra sounds **"click"**
☐ Louder supine	☐ Louder with standing

Diagnostic tests:

1. **Chest X-ray**:-
   i. Heart size.
   ii. Lung field.
2. ECG.
3. **Echo**: **"The best"**

➢ **Lung field**:

- Normally pulmonary vascular marking should be present up **to zone II.**
- **Zone III → Plethoric /** hyperemic field.
- **Zone I → Oligemic** lung field.

∗ **Causes of oligemic lung**:

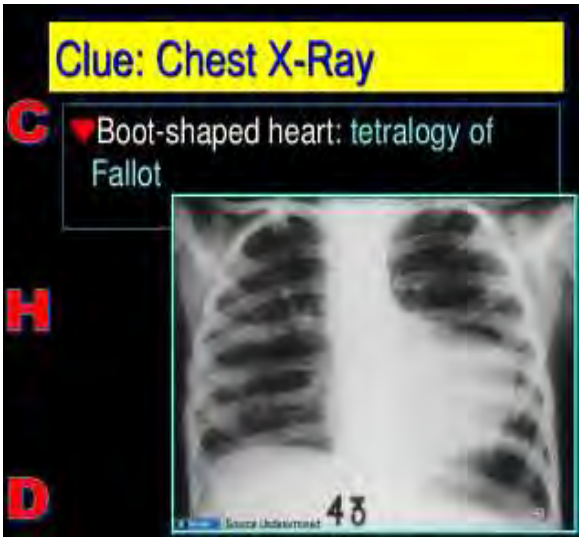
1. **A cyanotic CHD** : P.S , pul. Hypertention.
2. **Cyanotic CHD** : TOF , Tricusped atresia , Ebstein anomalies.

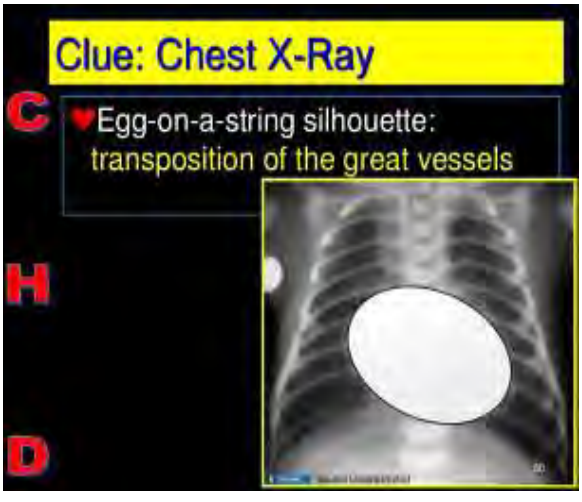
∗ **Causes of plethoric lung**:

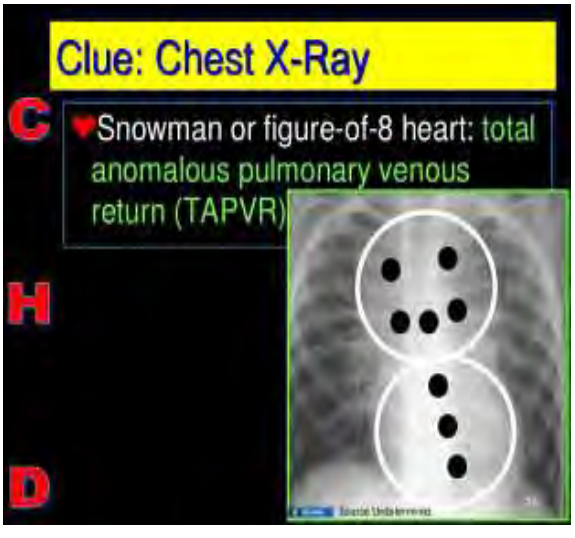
1. **A cyanotic CHD** : ASD , VSD , and PDA.
2. **Cyanotic CHD** : TGA , TAPVR , truncus arteiosus , hypoplastic left heart

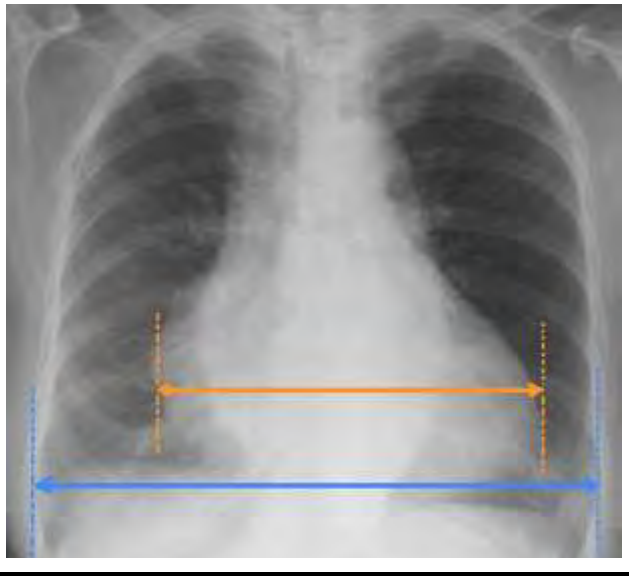
➢ **Specific appearance or finding on chest x-ray:**

- **TGA** → Egg shaped heart.
- **TOF** → Boat shaped heart.
- **TAPVR** → Figure of 8 or snow man shaped.
- **COA** → Rib notching.









Congenital heart disease
CHD

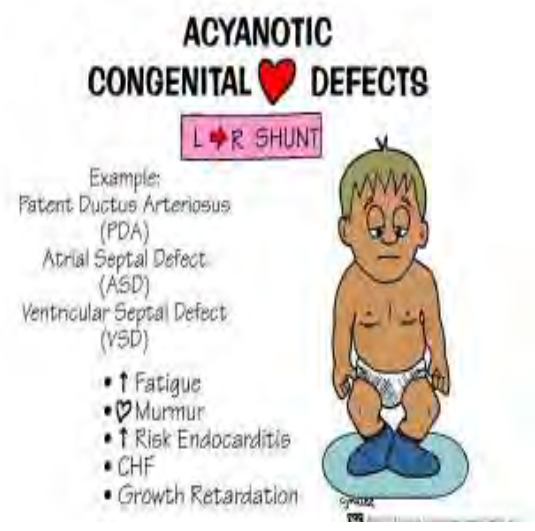
- **0.5-0.8** per 100 live birth.
- Most lesion occur between **15-80 day of gestation**.
- Diagnosed by **1 month of age**.
- Murmur **may not be heard in early life because increased pul. Vascular resistant**.
- **10-20%** associated with other congenital anomalies.

Etiology:

- **Most are unknown**:
  - Teratogen e.g. Rubella and alcohol.
  - Genetic factor 10%.
  - Environmental factor 5%.
  - Multifactorial 85%.
- **10-15%** of CHD more than one defect on same patient.
- Commonest **is VSD** then PDA then COA then PS and ASD.
- Commonest Cyanotic CHD is **TOF** and Commonest Cyanotic CHD in neonate is **TGA.**
- Most common structural abnormalities in children **CHD**.

General complication of CHD :

1. Congestive heart failure.
2. Pulmonary hypertension.
3. Infective endocarditis.
4. Growth retardation.
5. **Chronic hypoxic leading to:**
   - Polycythemia even in iron deficiency.
   - Hyperviscousity of blood.
   - Thrombocytopenia.
   - C.V.thrombosis.
   - Cerebral abscess "rare".
   - Hyperuricemia.



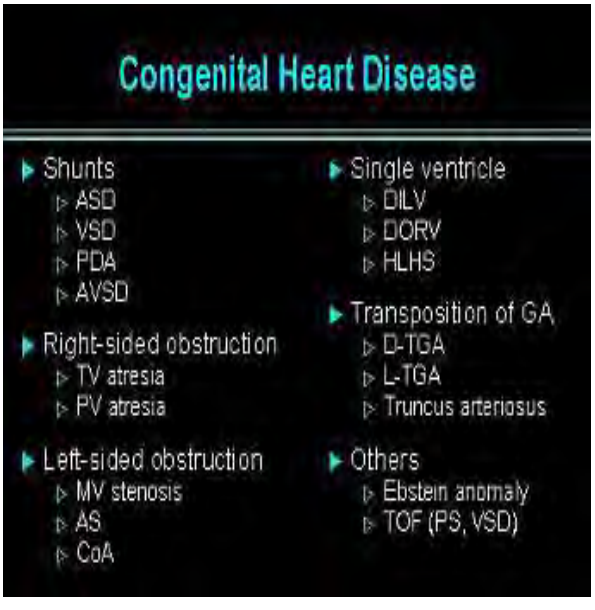
Classification of CHD :-

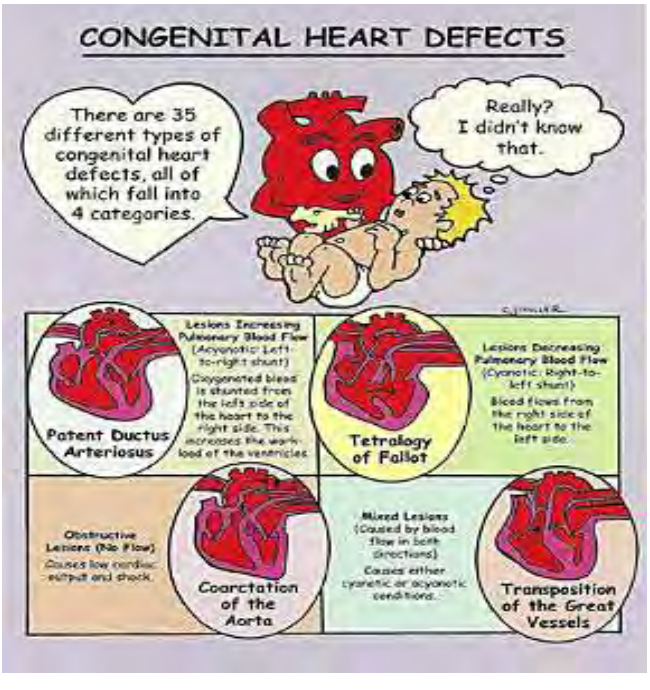
1. Stenotic :

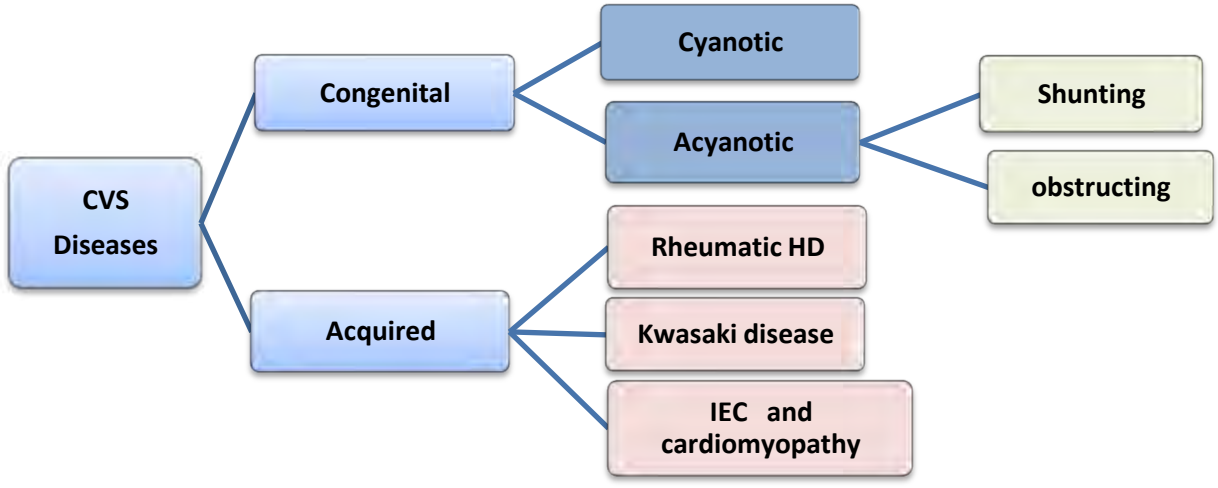
- Aortic stenosis
- Pulmonary stenosis
- Coarectation of aorta

2. Shunting :

Rt to Lt :	**Lt to Rt** :	**Mixing** :
TOF	PDA	Truncs arteriosis
TGA	VSD	Total anomalis pulmonary venous return
T.atresia.	ASD	







CYANOTIC CHD

CCHD with ↓ PBF

ē RVH

1. TOF.
2. Pul. atresia.
3. Double outlet Rt ventricle ē P.S

ē LVH

1. Pul.atresia ēout VSD.
2. Tricusped atresia.
3. Ebstein anomaly

Criteria

1. Cyanotic spells "common"
2. HF "rare"
3. P_2 of S_2 decreased.
4. **Chest x -ray → Oligemic lung.**

CCHD with ↑ PBF

ē RVH

1. TGA.
2. TAPVR.
3. Double outlet Rt ventricle ēout P.S

ē Bil.VH

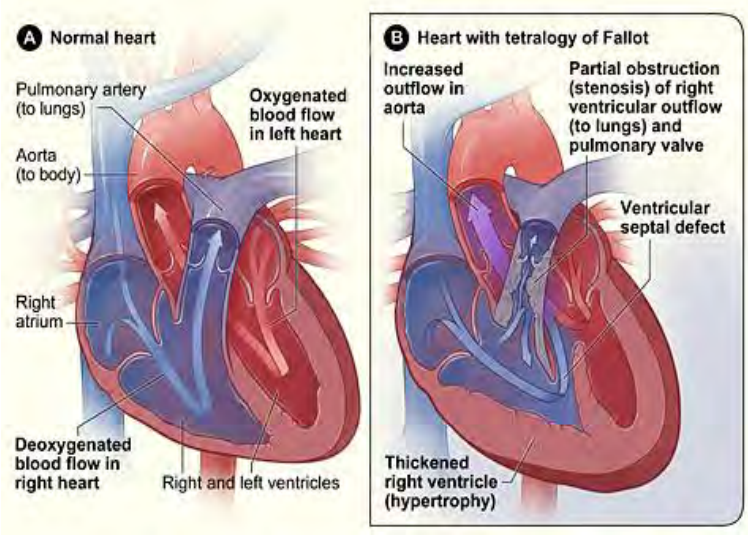
1. Truncus arteriosus.
2. Single ventricle.

Criteria

1. Poor feeding → dyspnea on suckling.
2. Recurrent chest infection.
3. Recurrent HF.
4. P_2 increased.
5. **Chest x-ray → Plethoric lung.**

Fallot tetrology

- Most common Cyanotic CHD >
- **Composed of :**
  1. **Pul. stenosis** :
     - → Infundibular 50%.
     - → Vulvular 10%.
     - → Both 30%.
  2. **RVH.**
  3. **VSD(large below aortic valve)**
  4. **Overriding of the aorta** :
     - Dextro-position.
     - Rt aortic arch in 25% receive mixed blood.

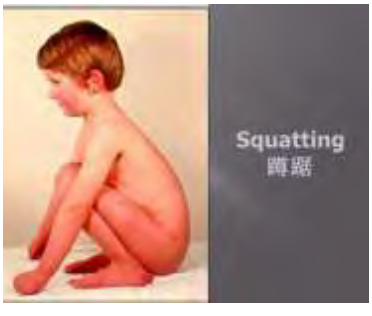


Clinical pictures: (6m-12 m)

1. **Cyanosis :** due to hypoxemia resulting from Rt to Lt shunt.
   *** Depend on the degree of P.S**:
   i. **Sever P.S**: cyanosis is deep and appear first week of life.
   ii. **Moderate P.S:** Balanced shunt → delayed cyanosis.
   iii. **Mild P.S**: Picture of Lt to Rt shunt.
   - *Most patient cyanosis no present at birth, and develop after few months of life since with time there is increasing hypertrophy of the infundibulum and the Rt ventricular outflow obstruction increases.*
   - **Body compensate by:**
     - Polycythemia
     - Aortic pul. collateral arteries later on.

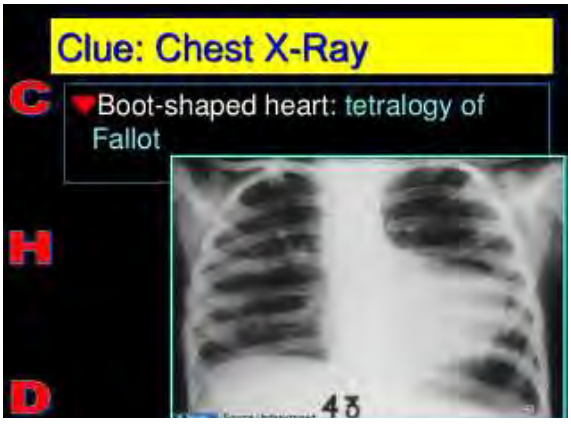
2. **Exercise intolerance and dyspnea on excertion:** due to increase venous return.
3. **Squtting :** After physical effort → dyspnea may occur → the child assume squatting position → ↑ sys. Vas. Resistant → ↑aortic pressure → ↑ pul. blood flow → atrial to increase blood oxygenated in lung.
   ↓ venous return from legs which has very low O_2 sat. after exercise.
4. **Clubbing :** related to the degree and duration of cyanosis.

5. **Paraxysmal hypercyanotic "hypoxic spells":**
   - Due to **infundibular spasm** ± ↓ systemic vascular resistant , which when prolonged results in hypoxia and metabolic acidosis, the resultant **hypoxia, acidosis and increased PCO_2** stimulate the respiratory center and hyperpnoea occur.
   - **Decrease** of already **reduced** pul. B.F.
   - Occur during first 2 years of life. In morning after crying, feeding or defecation.
   - Attack of increasing cyanosis.
     (Hyperpnea, **rapid deep breathing** → Resp.acidosis) (**Irritability** due to hypoxemia).
   - **↓ or absent of systolic murmur** due to sever infundibular spasm and decrease flow to the lung.
     So when murmur is present, it is good sign → there is more blood going to the lung.
   - Sever spell may lead to convulsion, CVA even death (due to polycythemia).
6. **FTT.**
7. **Cardiac examination:**
   - Normal size heart.
   - Single S_2 on pul. area
     (A_2 only heard)
   - Systolic ejection murmur heart max. in the third Lt I.C.S i.e. murmur is not produced by VSD.



Investigation:

1. **Chest x-ray**:
   - Oligemic lung.
   - Boat shaped heart.
   - Rt sided aortic arch in 25%.
2. **Echo:** Diagnostic.
3. **ECG** : RVH.

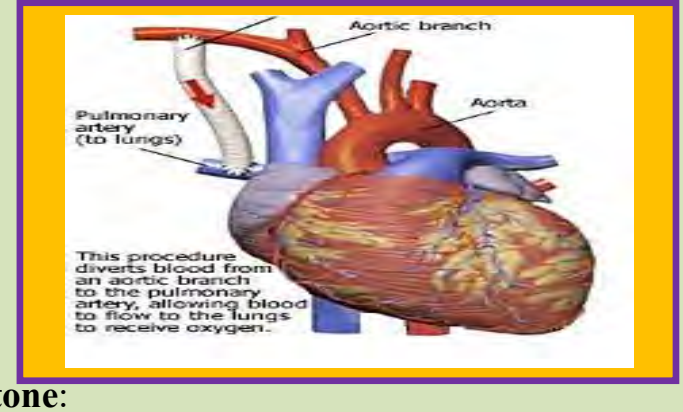


Complication:

- **CNS complication:**
  - ➥ **CVA** : due to polycythemia.
  - ➥ **Brain abscess**:
    Due to septic emboli , lack of pulmonary filtration
- Infective endocarditis
- HF (rare).
  - May occur in iron deficiency anemia with marked decrease O_2 carrying capacity.
- **Pul. TB** : due to oligemic lung.

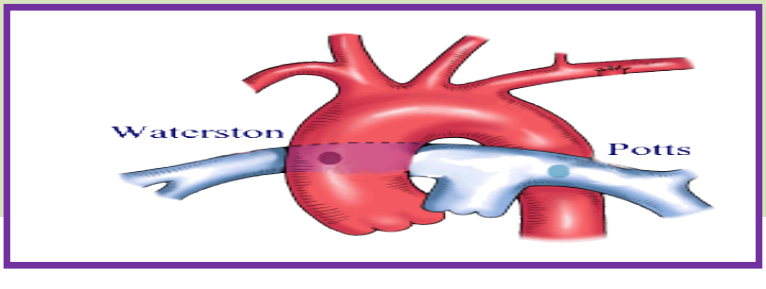
Types of Pallitative shunt:

1) **Modified blalock -taussing**:
   - Anastomosis b/w subclavian artery and ipsilateral pul. artery.



2) **Water stone**:
   - Anastomosis b/w ascending aorta and Rt pul. artery.

3) Potts:
   - Anastomosis b/w descending aorta and left pul. artery.



Types of TOF:

(1) Classic type as before.
(2) Fallot with pul. atresia "cyanosis at birth".
(3) Fallot with absent pul. valve → mild pul. stenosis and pul. regurge.

Treatment:

1. <u>Medical</u> :-

➲ **<u>Treatment of hypoxic spells:</u>**
- Hold baby in the knee chest position and O_2 inha.
- Morphine to suppress respiratory center.
- NaHCO3 "slow" to correct acidosis.
- Propranolol to ↓ infundibular spasm.

➲ **<u>After the attack:</u>**
- Oral propranolol prophylaxis.
- Avoid digitalis as it may induce infundibular spasm.

➲ **Avoid cerebral thrombosis by**:
- Treatment of iron deficiency anemia.
- Avoid dehydration.
- Partial exchange.

➲ In Fallot with **sever cyanosis at birth, keep PDA open by PGE_1 infusion**.

➲ Prophylactic against I.E.C.

2. <u>Surgical</u> :-

a. **Pallitative shunt:**
- **Anastomosis b/w aorta and pul. to increase pul. blood flow**.
- In cyanotic infant < 3 month with poorly controlled hypoxic spells.

b. **Total correction:**
- 4 month – 2 years according to severity.

D.D of TOF :

1. Double outlet Rt ventricle with P.S:

- Aorta and pul. arise from Rt ventricle.
- Mixing of blood via VSD - PDA.
- Baby born cyanotic and cyanosis increase with closure of DA.

2. Pul. atresia:

a) **With VSD** :
- Rt to Lt shunt via VSD , mixing.
- Pul. blood flow depend on PDA or collateral b/w aorta and pul.

b) **Without VSD** :
- Mixing via PFO which is essential for life.

C/P: - Cyanosis at birth → increase intensity with PDA closure.
- **S_2 → single "A_2".**
- Murmur of PDA or collateral.

Dx: Echo.

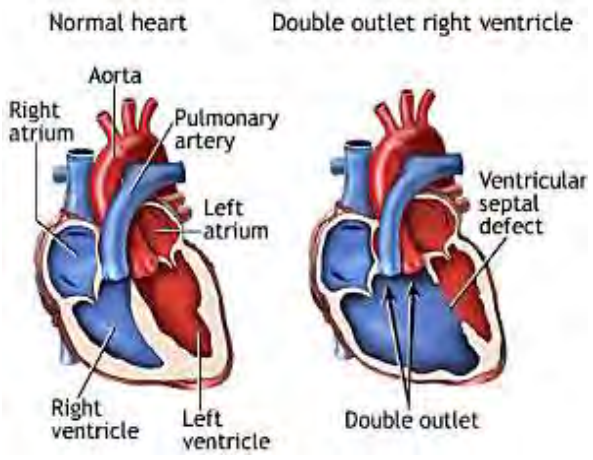
3. Tricusped atresia:

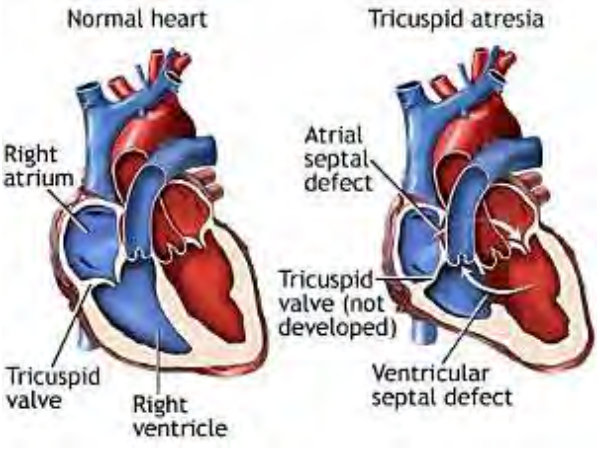
- Rt to Lt via POF "mixing and cyanosis".
- Pul. blood flow depend on VSD or PDA.
- Rt ventricle is hypoplastic.

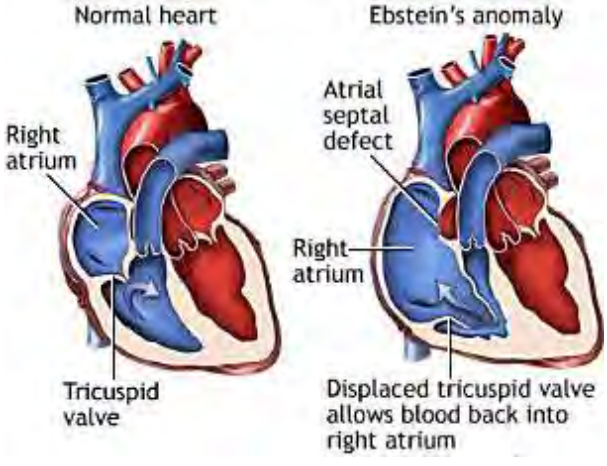
Treatment of 1 and 2 :
- ✓ PGE1 infusion.
- ✓ Palliative shunt.
- ✓ Total correction.

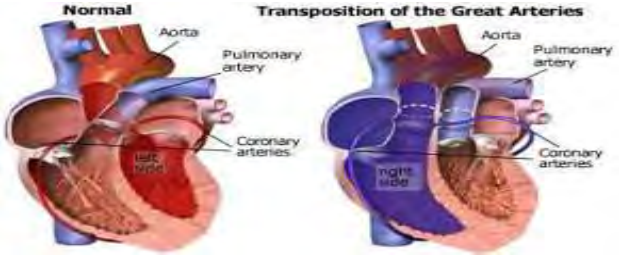
4. Ebstein anomally:

- Rt to Lt via PFO → mixing.
- **Composed of :**
  1) Huge Rt atrium.
  2) Downward displacement of T valve.
  3) Tricuspid regurge is common.
  4) Small Rt ventricle.
- **C/P**:
  - May asymptomatic.
  - Splinting of $S_1$and S_2.
  - **May be :** • Mild cyanosis .
    • Atrial arrhythmias.
    • PSM, may be HF.









CCHD with increased pulmonary blood flow

1. Transposition of great Blood vessels

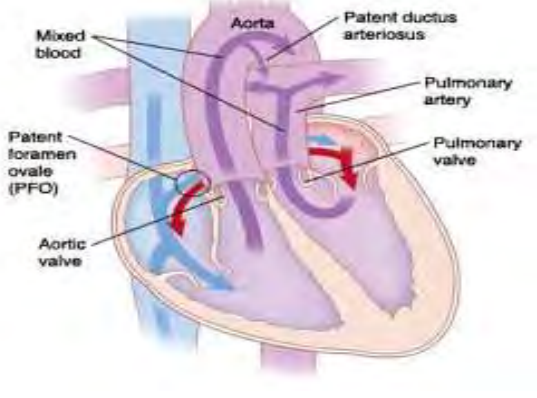
The aorta arise from Rt ventricle and pulmonary artery from left ventricle , deoxygenated blood to systemic circ and oxygenated blood to the lung , unless there is mixing of blood b/w them this condition incompatible with life .

Types:

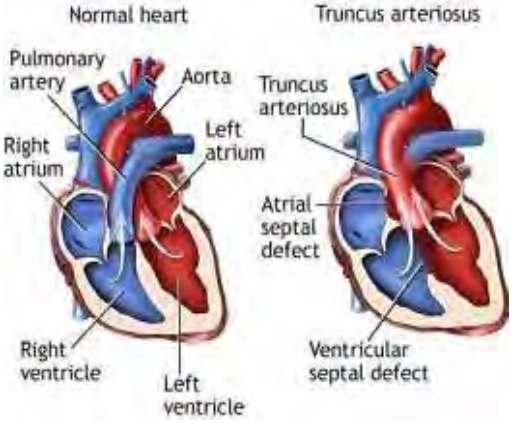
1. Isolated intact VSD :
- Sever cyanosis at birth.
- Acidosis and hypoglycemia **"emergency".**
- No murmur.
- Single S2.
- **Chest x-ray:**
  - Normal lung field.
  - **Heart** : RVH **"egg on side"**.
- **Treatment:**
  1) Keep PDA open by PGE_1 and avoid O_2.
  2) **Palliative operation** : Rashkind operation balloon atrial septostomy → create large ASD → free intracardiac mixing.
  3) **Total correction :** Arterial switch or atrial switch.

2. TGA with VSD :
- Mild cyanosis.
- ↑ B.F to lung.
- VSD murmur.
- **Chest x-ray**:
  - Plethoric lung.
  - **Heart** : RVH "egg on side".
- Echo diagnostic.
- **Treatment:**
  - Atrial switch operation.

3. TGA with VSD and PS:
- Minimal stenosis.
- Mimic TOF.
- Murmur of PS.
- **Chest x-ray:**
  - Oligemic lung.
  - **Heart :** RVH "egg on side".
- ECG: → BVH.
- **Treatment**:
  - Medical treatment as TOF.
  - Total correction.



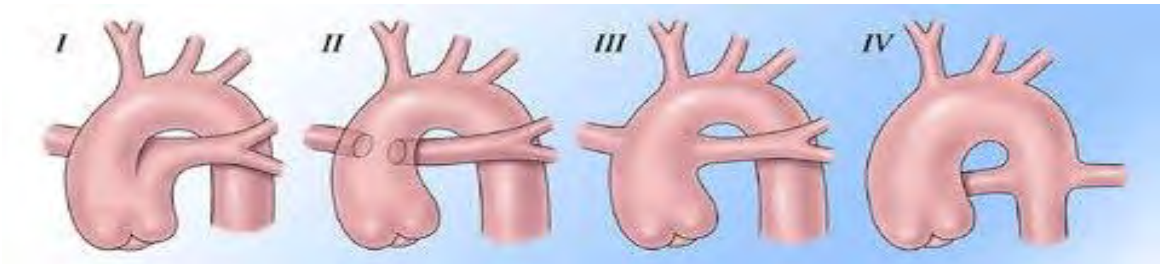
2. Truncus arteriosus



- One artery trunk leave the heart → give rise to both aorta and pul. artery , one semilunar vulve "**Trunchal valve**"
- Large VSD below the trunk.
- **C/P :**
  - Cyanosis usually minimal.
  - Features of ↑ pul. blood flow.
  - S_2 → single.
  - VSD → murmur.
- **Chest x-ray:** Rt side aortic arch in 50% of cases.
- **Treatment**: Surgical correction.

Types :

Type I	Type II	Type III	Type IV
Single pul. artery from Lt side.	Two pul. artery from the post. wall.	Two pul. artery from lat. wall.	Peudotruncus arteries arising from descending aorta.

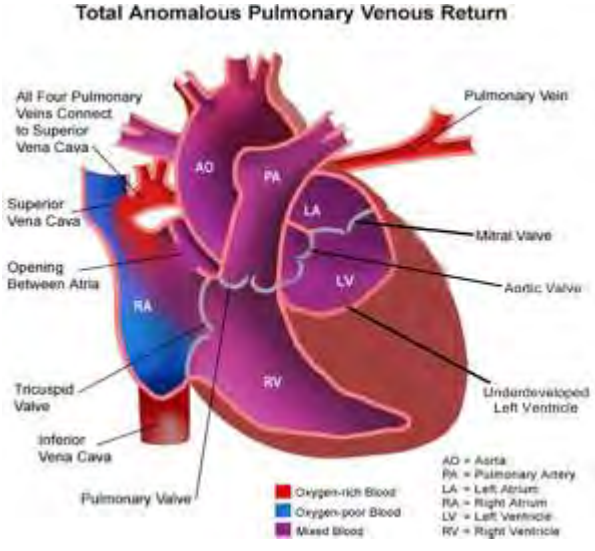


3. Single ventricle

- Absent interventricular septum → both aorta and pul. artery arise from common ventricle → free mixing of blood.
- Degree of cyanosis depend on wither pul. valve is stenotic or not which determine pul. blood flow.

4. Total anomalies pul. venous return

- Pul. vein drain into Rt atrium.
- ASD allow blood to pass to Lt atrium → mixing.
- **Chest x-ray:**
  - Snow man or figure 8 shaped heart.



Pulmonary stenosis:

- The commonest Rt side lesion.
- Most common **valvular** which may be tricuspid or bicuspid **associated with Noonan's syndrome**. (♂ = ♀).
- **Peripheral PS not valvular :**
  - ↳ Cong. Rubella syndrome.
  - ↳ Williams syndrome.
- **Examination:**
  - Ejection systolic murmur.
  - Pul component of S_2 is weak.
- **Investigation:**
  - **Chest x-ray**: Oligemic lung – RVH.
- **Treatment**:
  - Balloon valvuplasty in moderate to sever cases.
  - Open valvotomy in sever cases.

Complication:

1. CHF.
2. Myocardial injury and fibrosis.
3. Syncope.
4. Endocarditis.
5. Calcific aortic stenosis.
6. Systemic embolization.
7. Arrhythmias.
8. Sudden death.

Congenital aortic stenosis:

- **May be:**
  1. **Valvular**: Bicuspid aortic valve.
  2. **Supra valvular:** with **William's** syndrome.
  3. **Sub valvular:** membrane or septal hypertrophy.
- Hemodynamic → ↓COP.
- **C/P:**
  - Asymptomatic in mild stenosis.
  - HF in infancy if sever.
  - Low COP → dizziness, fainting, angina pain.
- **Complication**: IEC , sudden death.
- **O/E:**
  - Pulse → small volume.
  - Systolic BP → low.
- **Examination**:
  - LVH.
  - Trill at lower Lt sternal border, neck, carotid.
  - Ejection systolic murmur.
  - Delay A_2.
- **Diagnosis**: Echo.
- **Treatment:**
  - Balloon valvuplasty.
  - AV replacement.

ACyanotic CHD

★ **Types:**

1. **Ostium primum:** Defect in lower part of septum **"less common but serious",** b/c may extend to involve AV valve with abnormal mitral valve and tricuspid valve lead to regurge.
2. **Ostium secundum**: Defect in upper part of septum "more common" in region of fossa ovale.

★ Hemodynamic: Shunt from Lt to Rt → increased amount of blood passing to lung and RVH.

★ **C/P:** ♂ > ♀ presented at 3 year of life.

- Small defect: Asymptomatic.
- Large defect: - Recurrent chest infection.
  - Dyspnea on exertion, FTT.
  - ↓COP.

(>3 decade → arrhythmias)

★ **O/E:** Load S_1 ***wide fixed spiltting of S_2:***

↳ Normally the 2nd HS on pul. area is splited on inspiration due to prolongation of Rt ventricle emptying caused by ↑ VR. The fixed spiltting of 2nd HS is due to the Rt ventricle being filled to its maximum in inspiration and expiration. By Lt to Rt shunt due to constant filling of Rt. ventricle in all phases of respiration .

★ **Investigation:**

1. **Echo:** Diagnostic.
2. **Chest x-ray**: Cardiomegaly - plethoric lung.
3. **ECG:**
   - Rt axis deviation → Ostium secondum.
   - Lt axis deviation → Ostium primum.
   - Rt VH and Rt BBB.

★ **Treatment:**

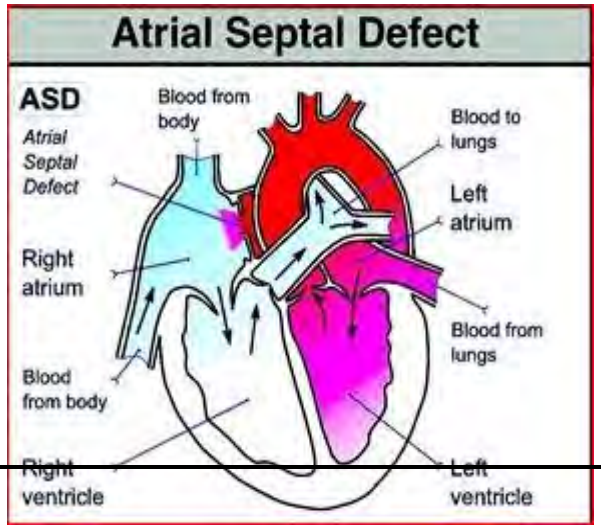
- Most infant close spontenously.
- Surgical ligation at 3-5 years.

★ **Complication:**

1) CHF.
2) FTT.
3) Esinmenger syndrome.

N.B:

- No need for prophylaxis against IEC in case of O. secondum → B/C the pressure across the defect is low and dosen't damage the endocardium.
- No Ab prophylaxis needed

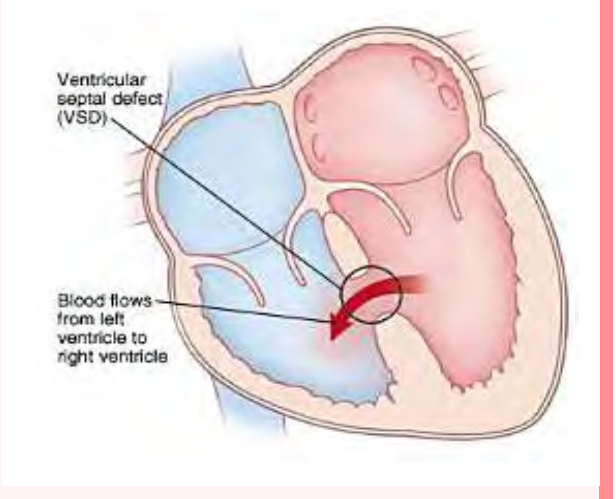


Most common CHD
25-30% of all cases of CHD

★ **Types:**
1. Membranus 70%.
2. Muscular part.
3. Interventricular septam.

★ **Hemodynamic:** Lt to Rt shunt b/c Lt side higher in pressure leads to volume over load. B/C the blood passes almost directly into pul. vein, the effect on pul. vasculature is generally greater than effect on Rt ventricle.

★ **C/P:**
- **Small VSD:** Asymptomatic.
- **Large VSD:** - Dyspnea with feeding, difficulty.
  - Repeated chest infection.
  - FTT. - HF.



★ **O/E:**

➲ **General examination**:
- Growth → FTT.
- Tachypnea.
- Pulse and BP → small volume.

➲ **Heart examination**:
- Pericardial bulge.
- Lt ventricular apex "forcefull not sustained"
- Systolic trill in 3rd - 4th spaces, parasternally.
- HS → accentuated split 2nd S_2 on pul. area.
- PSM → Lt 3rd + 4th spaces parasternally propagated all over the pericardium.
- Mild diastolic murmur → inflow murmur on mitral valve or ejection SM on pul. area. "functional"

➲ **Chest examination**:
- I.C.R and S.C.R.

★ **Investigation:**
1. **Chest x-ray**: Cardiomegaly - plethoric lung.
2. **ECG**: Lt VH or BVH.
3. **Echo**: Diagnostic

★ **Treatment:**
- Muscular small VSD are more likely to close in first 1-2 years than membranous.
- Surgical closure should be performed **3-5 years**.

★ **Complication:**
1) CHF.
2) FTT.
3) Recurrent chest infection.
4) Infective endocarditis
5) Esinmenger syndrome

⁂ Early closure in first year indicated in:

1. FTT not corrected medically.
2. Infant 6 m - 12 m with large defect and PHT.

Medical treatment :

- Oral diuretic.
- Digoxin.
- Ab.

Mangement depend on pressure of pul. :

1. Pul. pressure is ½ syst. BP → medical.
2. Pul. BP ⅔ syst. BP → surgery if medical failed.
3. Pul. BP = syst. BP → surgery.

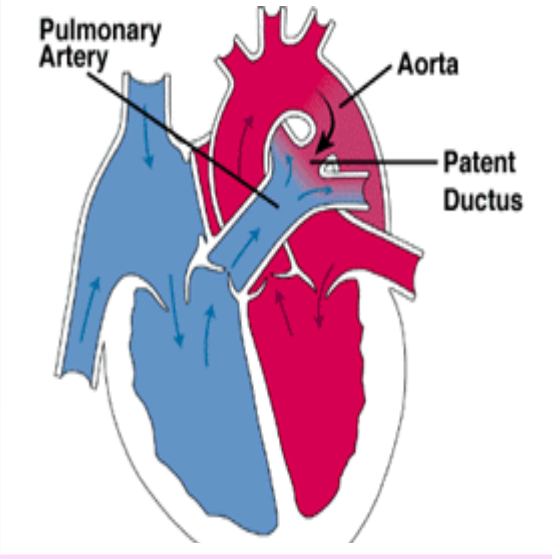
Patent Ductus Arteriosus (PDA)

- DA is close physiologically at first 24 hr while anatomically in 72 hr. (F:M 2:1).
- With congenital Rubella and preterm.

★ **Hemodynamic**: Blood pass from aorta to pul. resulting to increase blood flow to lung and Lt sided HF.
Lt atrium and Lt ventricle dilate in response to increased blood flow.

★ **C/P:**

- **The duct is small:** Infant asymptomatic.
- **The duct is large:**
  - FTT.
  - Dyspnea on exertion..
  - Recurrent chest infection.
  - HF (rare).



★ **O/E:**

➲ **General examination**:
- FTT.
- Tachypnea.
- Weak cry.
- Pulse & BP: Water humer pulse = collapsing.
- Bounding pulse: Higher pulse.

* Pulse pressure = SBP - DBP > 50.

➲ **Chest examination**:
- I.C.R and S.C.R.

➲ **Heart examination**:
- Lt ventricular hyperkinetic apex "forceful not sustained"
- Trilll max. on Lt 2nd I.C.S propagated to left clavicle, systolic may be continous.
- Continuous murmur is heard with max. on Lt 2nd I.C.S and propagate to clavicle and down to Lt sterna border to the apex.
- Murmur → continuous → aortic P > pul. P in diastole and systole.

★ **Diagnosis:**
1. Chest x-ray: Cardiomegaly - plethoric lung.
2. ECG:
   - Lt VH, Lt axis deviation.
   - Tall R wave in V6.

★ **Treatment:**
- Preterm infant: give indomethacin.
- Older patient: surgical ligation.

N.B:
- If PDA doesn't close within one week it is unlikely to close.
- PDA in term babies will not close spontaneously B/C defect in constrictor mechanism of duct.
- Premature → close spontaneously.

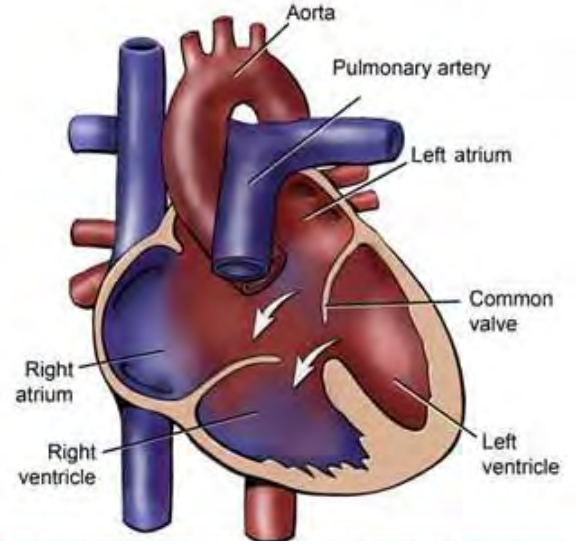
Endocarial cushing defect

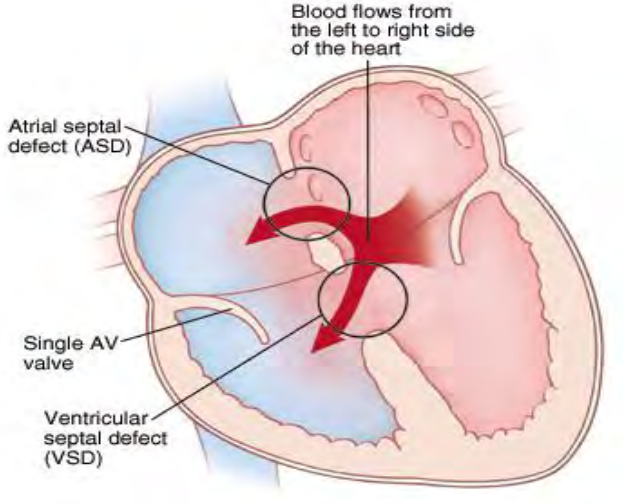
- AV canal or AVSD.
- Common with Down syndrome.
- AVSD with AVV abnormal.
- Lt to Rt shunt at both atrial and ventricular level with some Rt to Lt shunting with desaturation → mild intermittent cyanosis.
- It is the commonest cause of CHF early in neonatal period.

★ **Examination:**
- HF.
- FTT.
- Eisenmenger physiology occur earlier.
- Pericardial pulge.
- Load S_1 , widely fixed split S_2 "as ASD".
- Pul. ejection systolic murmur.
- Low pitched diastolic m. at Lt sterna border and apex.
- PSM → Lt lower sternal

★ **Diagnosis:**
1. **Chest x-ray**: Cardiomegaly - plethoric lung.
2. **ECG:** Rt VH mainly.
3. **Echo**: Diagnostic.

★ **Treatment:**
- Rt of CHF and prophylaxis against I.E.C.
- Surgical closure at 3-5 months may be earlier.





• Localized narrowing of the descending aorta usually occur just below the origin of the left subclavian artery. Sometimes above origin of the left subclavian artery.

★ **Hemodynamic:** Obstruction of the aorta usually results in the development of collateral circulation b/w the branches of subclavian artery - descending aortal and FA.

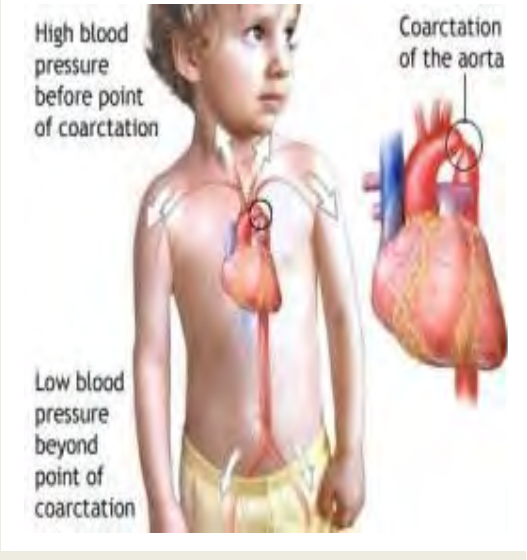
• **Pressure**:-

➥ In the proximal part of the aorta is **high** → Left VH and high BP in upper part of the body.

➥ Pressure in the distal part of aorta is **low** → low BP in lower part of the body.

★ **C/P:**

- Common with **Turner syndrome.**
- ♂ : ♀ → 2:1.
- Usually asymptomatic.
- **Early**: After closure of PDA 1st week of life.
  - Poor feeding.
  - Tachypnea.
  - Lethergic and progress to CHF end by shock.
  - It is worse if there is VSD.
- **Late:** After N. period come with HPN.
  - Headache, chest pain, fatigue and I.C.Hge.
  - Or murmur during evaluation.



★ **O/E:**

➲ **Early presentation**:

- Tachypnea, - Tachycardia, Distressed → CHF → Shock.
- Diff. BP → Upper and lower limb.
- Reduce or absent of L.L pulses "femoral pulse".
- **Radio-femoral delay,** pink UL and cyanotic LL.

➲ **Late presentation**:

- HPN (Lt arm < Rt arm)
- Murmur (best heard b/w scapula ↑ ē expiration).
- Delay radio-femoral pulse.
- Trill present in suprasternal notch and abdominal bruit in abdominal coarction.

★ **Investigation:**

1. **Chest x-ray:** Lt V. dilation.In children rib notching may be seen "enlarged I.C.arteries have eroded the under sided of the heart" → Reversed 3 signs.
2. **Echo**: Diagnostic.
3. **ECG:** - **Early neonate :** Rt axis deviation , Rt VH.
   - **Late** : Lt axis deviation , Lt VH.

N.B:
Normally BP is higher in lower limb by 10-20

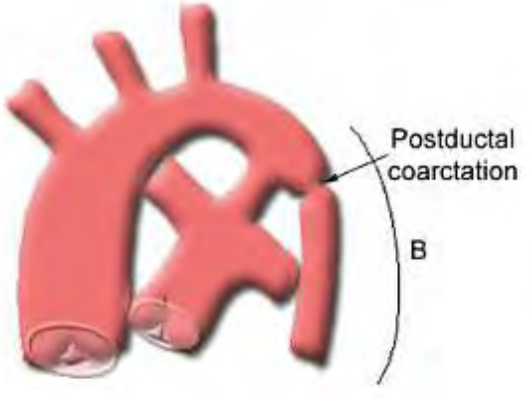
★ **Treatment:**

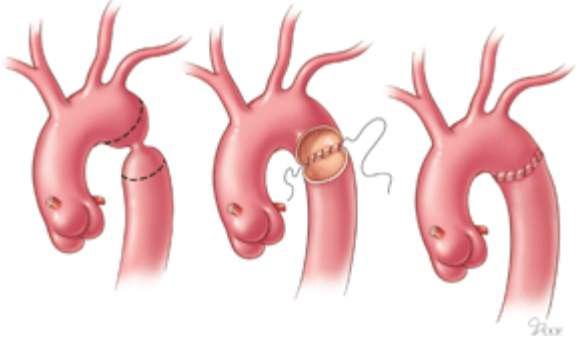
1) Early:
   - Treatment of CHF (Lasix, Captopril, Digoxin).
   - Prostaglandin E_1.
   - Correction of acidosis.
   - Surgical correction.

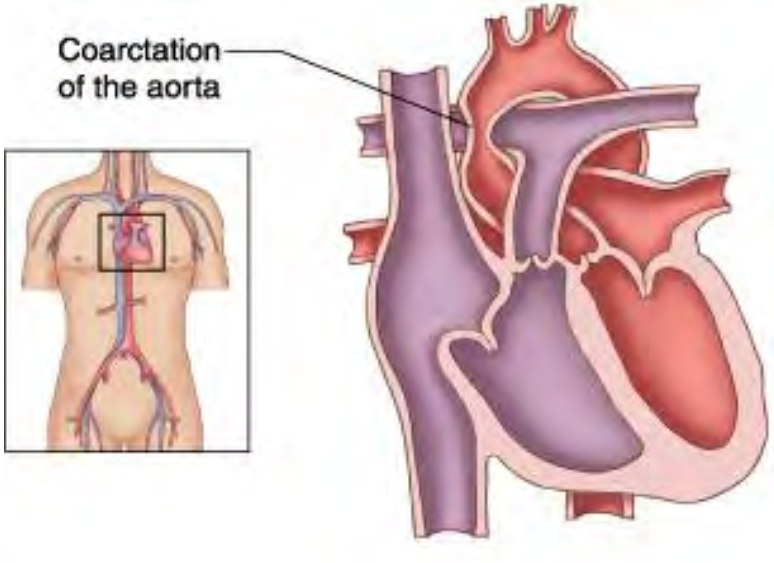
2) Late:
   - Treatment of HPN.
   - Surgical correction.

★ **Complication:**

1. Recurrent contraction.
2. Aortic aneurysm.
3. Persistent HPN.
4. Cerebral aneurysm "berry aneurysm of circle of will's".
5. Paralysis due to spinal cord ischemia.
6. Cardiomyopathy.



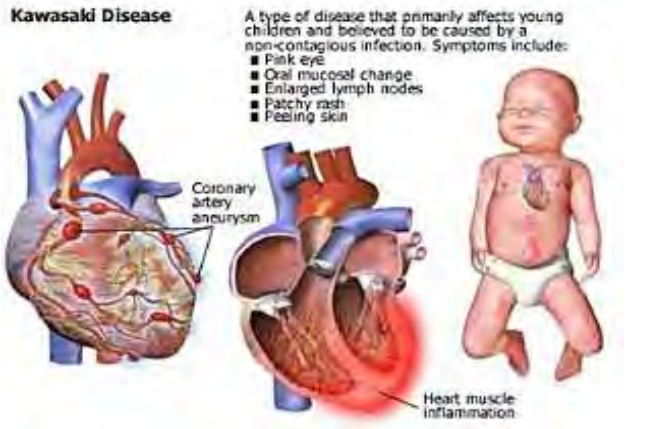




Kawasaki Disease

Muco-cutaneous lymph node syndrome (MCLNS)

- It is a **sever acute vasculitis** affecting all blood vessels mostly medium sized arteries especially coronary.
- It is the most common cause of acquired heart disease in developed counties and without treatment 20% will develop coronary artery abnormality.
- It affects children < 5 years.
- It is common in Asian children.



Diagnosis:

✓ **Fever for ≥ 5 days and 4 of the following criteria:**

1. Bilateral non purulent conjunctivitis.
2. Non purulent / supportive cervical lymphadenitis usually asymmetrical > 1.5cm.
3. Straw berry tongue / cracked lips.
4. Pleomorphic skin rash.
5. Erythema and swelling of hand and feet with desquamate of finger tips 1-3 weeks after onset.

Other finding:

- Diarrhea.
- Hepatitis.
- Hydrops of gall bladder.
- Extreme irritability.
- Aseptic meningitis.
- Sterile pyuria.
- Otitis media.
- Urethritis.
- Arthritis.$^{+++}$

Cardiac finding:

- Myocarditis "50%" → Tachycardia, poor ventricular function.
- Pericarditis → CHF.
- **Coronary artery aneurysm "2nd - 3rd week".**

Lab. Results:

1. WBC ↑ or normal and mainly neutrophil.
2. ↑ ESR , ↑ CRP.
3. Normocytic anemia.
4. **Plt :** at 1st week normal.
   2nd - 3rd week ↑ up to million/mm3.
5. **R/E of urine :** sterile pyuria.

Risk factors:

1. Male and age < 1 year.
2. Fever > 14 days.
3. Hb < 10.
4. WBC > 30,000 (mainly neutrophil).
5. Hypoalbumenia.

Treatment:

- IV immunoglobulin.
- High dose of aspirin "moniter by ESR".
- **Steroids** : only in case of persistent fever.

➲ All patient diagnosed with Kawasaki must do an Echo for aneurysm.

Rheumatic Fever

Autoimmune inflammatory disease involving mainly the heart, joint, and less frequently the central nervous system, skin and subcutaneous tissue.

Incidence:

- All age groups except early infancy with peak incidence between 5-15 years, low socioeconomic status, genetic.
- Rheumatic chorea is more common in females.

Etiology:

- The disease is related to URTI with group A, B hemolytic streptococcus "not all cases have a history of proceeding infection".
- Alatent period 2-3 weeks is always present between the onset of infection and the onset of disease, the disease has tendency to recur.

Pathogenesis:

- The most accepted theory is that is an autoimmune disease. Antibodies formed against streptococcal antigen can react with cardiac muscles "cross reactivity".

Diagnosis:

Major criteria

- Polyarthritis.
- Carditis.
- Chorea.
- Erythema marginatum.
- S/C nodules.

Minor criteria

- Fever: usually present at onset (38.5-40°C)
- Arthralgia

"can't be used in presence of arthritis"

- Previous RF.
- ↑ acute phase reactants: ↑ESR, ↑CRP, ↑WBC.

Eveidence of recent Strepto. Infection

- Recent scarlet fever.
- +ve throat culture.
- Anti strep. Ab.
- ASOT
- ASK.

Diagnosis of RF is based on modified Jones criteria:

- The presence of 2 major criteria or 1 major and 2 minor in presence of evidence of recent streptococcal infection.
- Exception of the above include:
  1. Chorea if other causes have been excluded.
  2. Insidious or late onset carditis with no other explanation.
  3. Rheumatic recurrence in patient with documented rheumatic heart disease. "minor criteria + evidence of strep. Infection"

Diagnosis based on 2 major criteria is stronger than that based on 1 major and 2 minor criteria.

★ **Major criteria.**

A) Polyarthritis:

↳ The most common **presentation 75%.**

- Affecting more than one joint.
- **Large joint affection** : knee, ankles, wrist, elbow, less commonly the hip.
- The affected joint is hot, red, tender, swollen with limitation of movement.
- It migrates from one joint to other *"one joint is improving as the other becomes worse".*
- Dramatic response to **salicylates within 48hr**.
- It leave the joints completely free "**no permanent deformity**", untreated arthritis may persist for 3-4 weeks then disappear spontaneously.

B) Carditis:

↳ The 2nd most common presentation after arthritis "50% in initial attack"..

- It can be detected on initial examination or appear 1-2 weeks later, may be late onset 6weeks-6 months.
- It is pancarditis.

➲ Pericarditis:

- Pericardial pain "stitching chest pain".
- Pericardial rub by auscultation.
- Pericardial effusion "uncommon

Dull aching pain, dullness outside the apex, distant heart sound, low voltage ECG.

➲ Myocarditis:

- Tachycardia disproportionate with fever.
- Muffled heart sound.
- Heart failure may be present at the onset or may occur during the course.
- Rarely bradycardia due to heart block.

➲ **Endocarditis:**

- Left side valve: mitral valve affection is more than aortic valve and both may be affected together.
  - **Mitral valve regurgitation:** "Pancystolic murmur at the apex" is present in the acute stage of the disease, later in the chronic stage scarring of the mitral valve may lead to stenosis → "mid diastolic murmur at apex".
  - **Aortic valve regurgitation**: is detected by early diastolic murmur along the left sterna border "isolated aortic valve affection is rare".

- In general symptoms are evident if percarditis or heart failure are present otherwise manifestation may be vague e.g. loss of appetite, pallor and easy fatigue; average duration of carditis varies **from 6 weeks - 6 months**.

C) Chorea "Sydenham chorea":

- It occur much later than other RF manifestation, the latent period following streptococcal pharyngitis may be as longer as several months.
- Associated with other rheumatic manifestation in 10% of cases usually with carditis it may be the only manifestation of RF. It is caused by destruction of basal ganglia by AntNAb.
- **Manifestation include:**
  1. **Choretic movements:** Involuntary, static, irregular dysrhythmic sudde jerk pseudopurposful movements of any part of the body "Proximal more than distal" or may facial grimace. ↑ with emotional stress and ↓ by sleep lasts for months.
  2. **Hypotonia:** Is evident by pendular knee jerk, inability to maintain extended arm or hand grip and speech changes "due to tongue hypotonia and fasciculation".
  3. Emotional lability.
  4. Hyper extended joint, ↓ reflexes.

D) Erythema marginatum: <5%
(Erythema annular)

- Red non pruritic macules that fade centerally.
- Appear over the trunk and inner surface of legs and arms,. "may be seen in sepsis, during reaction and GN".
- It disappear rapidly or it may come and go for several months.

E) Subcutaneous nodules:

- Small rounded hard and freely movable, painless, over the bony prominences e.g. elbow "extensor surface".
- Their presence indicates sever carditis.
  "they appear several weeks after the attack".

- Throat swab is usually negative.
- Investigation are normal in isolated chorea due to long latent period

Carditis is the most serious manifestation of RF as it may be fatal during the acute stage and later on it may result in chronic valve lesion.

Investigation:

Laboratory:
- Evidence of streptococcal infection.
- Acute phase reation.

Imaging:
1. ECG.
2. X-ray : Cardiomegally.
3. Echo : For valvular lesion.

Treatment of acute attack:

a. **Supportive treatment:**
Rest: patient with carditis (till HF is controlled and ESR is normalized) should have absolute bed rest for at least 4 weeks. Daily examination is important to detect carditis that usually presents within 2 weeks of onset.

b. **Specific treatment:**

1. **Arthritis:** "or carditis without cardiomegaly:
   - **Salicylate**
   - **Indication**: The above + during steroid withdrawal.
   - Gradual withdrawal monitored by decline in ESR and CRP.
   - **Steroids**.

   Indication: moderate to sever carditis, HF.
2. **Carditis with cardiomegaly or failure:**
   - Prednisone
   - Salicylate.
3. **Chorea:** Phenobarbitone
   - Haloperidol "in patient > 12 years".

Treatment:

Prevention : "very imp."
- Prevention of strep. Infection e.g. proper ventilation.
- Early Dx of streptococcal pharyngitis then adequate treatment by single injection of Benzathine penicillin 1.200.00 IM "1.2 million" or oral course of Penicillin for at least 10 days.

★ Prevention of RF activity in patient with history of RF:
- **Benzathine penicillin 1.200.00 IM every 3-4 weeks.**
  - For at least 5 years after last episode for cases without carditis.
  - For at least 10 years after last episode for cases with carditis without residuals.
  - For life or till age of 40 year for cases with carditis with residuals.

Alternative: Daily oral penicillin V or erythromycin

Complications:

1. CHF :

a. Acute due to myocarditis.

b. Chronic with episodes of acute failure precipitated by:

i. Rheumatic activity.

ii. Infective endocarditis.

iii. Chest infection.

2. Chronic valve lesion :

MR, AR, AS, AR or combinated valve lesion which can complicated by:

- Pulmonary hypertension.
- Arrhythmia.
- IEC.
- Thromboembolism.

3. Rheumatic activity "recurrence" :

Suggested by:

- Fever with arthritis or arthralgia.
- Changes in character of already existing murmur.
- Appearance of new murmur.
- Carditis with HF.
- Pericardial with effusion.

4. Pulmonary hypertension:

Due to long standing MV lesion.

- Symptoms:
  Dyspnea, fatigue, may syncope.
- Signs:
  Pul. Pulsation, soft ejection systolic murmur over pul. area, load S_2.

The major complication of RF is the development of **rheumatic valvular heart** disease, non of other manifestation result in chronic disease.

c. Treatment of complication:

- Mild cases: Complete bed rest, O_2, fluid restriction and steroids.
- Sever cases:
  - Preload reducing agent: Diuretic e.g. Furosemide 2mg/kg/day.
  - Inotropes: Digoxin
  - After load reducing agent: Vasodilator e.g. Captopril.

Infective Endocarditis

Etiology:

1. **Bacteremia:**
   - Staph. aureus "The most imp. " 30%
   - Strepto viridians 50% "follow dental surgery"
   - Others:
     - G –ve organism e.g H. influenza "common in neonate and immunocomprimized".
     - Fungal (sever & bad prognosis) "In immunodeficiency and post open heart surgery".
     - Pseudomonus.
     - Enterococci : (rare but dangerous B/C has resistant to antibiotic). Follow GIT and GU surgery or instrumentation.
2. **Abnormal heart:**
   - Rheumatic heart.
   - Congenital heart disease (VSD, PDA) "Significant pressure gradient" .

Predisposing factors:

- Tooth extraction.
- Surgical procedures.
- Cardiac catheterization.
- Prosthatic valve.

Clinical pictures :

➲ **General manifestation :**
- Fever 85-99%.
- Poor appetite leading to weight loss.
- Pallor.
- Purpura.
- Pale clubbing.
- Palpable spleen 15-20%.

➲ **Rare manifestation :**

Due to vasculitis "due to immune complexes":-
- Osler nodules: firm, tender intradermal nodules in pades of fingers and toes.
- Janeway lesion: Painless erythema in the palms and soles.
- Splinter Hge: Linear hemorrhagic streaks beneath nails.
- Petechiae 20-40%.

➲ **Cardiac manifestation :**
- Appearance of new murmur.
- Changes in the character of previous murmur.
- Seagull murmur "musical" due to rupture of valve leaflets.
- Heart failure.

➲ **Embolic manifestation :**
1. Neurological:
   - Embolic strok "seizures".
   - Cerebral abscess.
   - Mycotic aneurysm → I.C.Hge.
2. Retinal Hge: " Roth spot " oval with pale center.
3. Pulmonary embolism.
4. Renal infarction. (Hematuria & RF)
5. Peripheral embolism.

Pathogenesis:

Implantation of the organism in diseased endocarditis → local inflammation and formation of friable vegetation composed of platelet fibrin, inflammatory cells, and organism.

Complication :

1. **Cardiac :**
   - Toxic myocarditis.
   - Heart failure.
   - Arrhythmia and HB.
   - Myocardial abscess.
   - Valvular lesion or abscess.
2. **Immune complex lesion :**
   - Acute GN.
   - Arthritis.
3. **Embolic manifestation.**
4. **Remote infection.**
   e.g. meningitis, osteomyelitis.

Investigation :

- **Blood culture** :
  3-5 blood sample before start of antibiotic.
- **Echo :**
  - Detect vegetation and underlying cardiac lesion.
  - May predict embolization.
- **CBC :**
  - Anemia "normocytic normochromic 70-90%
  - ↑WBC. 20-30%.
- **ESR** : high.
- Microscopic hematuria is found in 30% of patient.
- Rheumatoid factor 40-50% > 6 weeks duration.
- **Urine analysis:** proteinuria in 50-60%.
- **MRI** : Aortic root aneurysm.

Diagnosis :

Modified Duke Criteria :

It is based on pathological or clinical finding 2 major criteria or 1 major and 3 minor or 5 minor.

Major criteria:

1. +ve blood C/S persistence or evidence of infection.
2. +ve Echo finding "vegetation, para valvular abscess.
3. New valvular regurgitation.

Minor criteria:

1. Predisposition "H/O IV drug use or CHD".
2. Fever >38°C.
3. Vascular phenomena:
   Arterial embolism – Septic pul. infarction.
   - I.C.Hge.
   - Conjunctival Hge.
   - Janeway lesion.
4. Immunological phenomina:
   GN, oster nodes, +ve result of rheumatic factor.
5. +ve blod culture without any infection manifestation.

Treatment :

1. Prophylaxis:

- Oral and dental care.
- Patient with risk factors who will be exposed to surgical procedures must receive prophylactic antibiotic as follow:

Procedure	1 hr before surgery	6 hr after surgery
Oral and dental	Oral Amoxicillin 50 mg/kg	Oral Amoxicillin 25 mg/kg
GIT and GU	Parenatal Ampicillin 50 mg/kg and Gentamycin 2 mg/kg	

Erythromycin, Clindamycin, Vancomycin are alternative for cases already receiving long acting Penicillin prophylaxis or Penicillin sensitive.

2. Curative:

➲ **Medical** :

1) Hospitalization.
2) Bed rest.
3) Treatment of heart failure.
4) IV antibiotic combination is starts immediately while waiting for results of blood cultures.
   - Penicillin G, Gentamycin, Cloxacillin.

OR
- Vancomycin, Amphotricin B, 5-F lorocytosine for fungal endocarditis.

➲ **Surgical** :

Goal: - Remove vegetation.
- May be valve replacement.

Indication:

1. Sever heart valve lesion with uncontrolled HF.
2. Fungal endocarditis.
3. Failed medical treatment.

Heart Failure

Inability of the heart to pump sufficient blood needed for body requirements despite adequate venous return.

Causes :

Fetal :
- Severe anemia e.g. hemolysis.
- Supraventricular tachycardia.
- Ventricular tachycardia.
- Complete heart block.

Premature neonate:
- Fluid overload.
- PDA, VSD.
- Cor pulmonale in bronchopulmonary dysplasia.

Full term neonate:
- Asphyxial cardiomyopathy.
- A-V malformation.
- Coarctation of the aorta.
- Hypoplastic left side of the heart.
- Single ventricle, truncus arteriosus.
- Viral mvocarditis.

Infant and toddler:
- Large left to right shunts as big VSD.
- Viral myocarditis.
- Supraventricular tachycardia.
- Acute hypertension "hemolytic uremic syndrome".
- Kawasaki disease.

Children and adolescents:
- Rheumatic heart disease.
- Cor pulmonale.
- Acute hypertension "GN".
- Endocarditis.Viral myocarditis.
- Chemotherapy.
- Hemosidrosis.Thyrotoxicosis.
- Cardiomyopathy

C/P :

1. In newe

2. born and infants:
- Dyspnea on feeding and inability to cry.
- Irritability, pallor or cyanosis.
- Edema is a late sign, it appear first in the face then genitalia and lumbosacral region before feet B/C of supine posture.
- The congested neck veins are usually lacking due to short neck and excess fat of the neck.
- Enlarged tender liver.
- Fine basal crepitation and wheezes.
- The heart is frequent enlarged with tachycardia and may be gallop rhythm.

2. In older children:
- Dyspnea, orthopnea or PND.
- Pallor, tachycardia, peripheral cyanosis, cold extremities, profuse sweating.
- Congested neck veins, enlarged tender liver and edema of both lower limbs.
- Wheeze, crepitations due to pulmonary congestion and edema.
- In severe cases acute pulmonary edema with cyanosis and coarse crepitation will be evident, PE may occur.
- Tachycardia, gallop rhythm.

Treatment:

1) Bed rest: In semi-upright position.
2) Humidified oxygen: 30-40% in pulmonary edema.
3) Diet: ↓ salt and fluid intake, ↑ calori and NGT feeding may be helpful.
4) Diuretic : ↓cardiac preload.
   - Furesemide " lasix " IV in pulmonary congestion .
   - Chronic furesemide therapy is then prescribed.
   - KCL supplementation or spironolacton orally may be administrated in conjugation with chronic diuretic therapy to preserve K

   ⏩ S/E of lasix :
   - Hypokalemia
   - Alkalosis
   - May increase digitalis toxicity.
5) Digxin : IV or orally if not critically ill "↑myocardial contractility = inotropic "
6) The underlying cause of cardiac failure must be managed .
7) Treatment of associated infection is essential e.g. chest infection.
8) Vasodilator therapy : act by ↓cardiac after load →↑strok volume used in :
   - ➲ Hypertensive HF.
   - ➲ Large left to right shunt.
   - ➲ DCM.
   - ➲ Sever MR & AR.

   ⏩ **Types :**
   - ACE inhibitor e.g captopril , enalpril.
   - Hydralazine.
   - Nitroglyceride IV infusion in acute pulmonary edema.

Digitalization :

1. Get baseline ECG & electrolyte "especially K "
2. Loading dose is given within 24hr
   - 1/2 of total dose TDD immediate
   - 1/4 TDD after 8hr →1/4 after another 8hr .
3. Maintenance dose =1/4 TDD is given in two divided doses after 12hr .

Absolute C/I to digitalis :

1) Hypertrophic cardiomyopathy.
2) HB.
3) TOF.

Digitalis toxicity :

- Diagnosis is based on rhythm and conduction disturbances.
- N/V are not frequent in the pediatric pt . it occur may be in older children.
- Any form of arrhythmia occurring following the institution of digitalis therapy must be considered to be drug related until proved otherwise.

⏵ Factors which ↑digitalis toxicity :
- Renal impairment.
- ↑ myocardial sensitivity e.g hypokalemia & rheumatic carditis, hypomagnesemia, hypercalcemia, myocarditis.
- Prematurity.

Treatment :

- Continuous ECG monitoring
- Stop digitalis
- Correct hypokalemia
- Correct arrhythmia (atropine , lidocaine for VA)
- ***Digiband*** is the antidote .

Supraventricular tachycardia (SVT)

- Caused by re-entry circuit through the A-V node .
- SVT may occur in the normal or diseased myocardium and attacks are characterized by their abrupt onset and cessation.
- HR typically is 180-320 /min and does not vary with respiration .

On ECG :

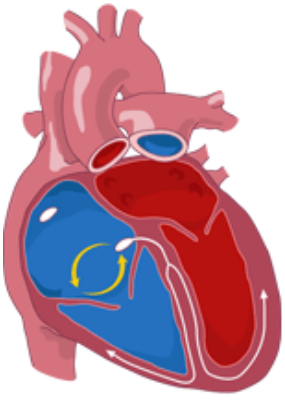
- P wave may be present 50% of the time although their morphology and axis may be abnormal.
- QRS complexes may be normal or abnormal in cases in which aberrant conduction is present , in these cases differentiation from VT is difficult but essential.
- The complete absence of P wave and of wide QRS complexes that are dissimilar to the QRS complexes found during sinus rhythm are diagnostic of VT.

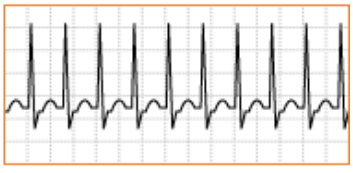
C/P :

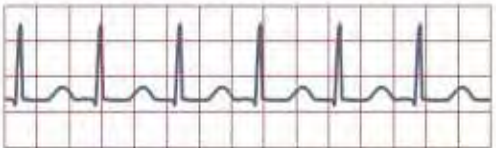
1. **In newborn and infant :**
   - Sudden pallor, tachycardia, refusal to feed, vomiting .
   - Looks ill, restless with grayish cyanosis , breathlessness , and sub costal recession .
   - Cardiomegally and hepatomegally are marked .
   - Crepitating over the lung may be due to HF or to an associated resp. infection which sometimes is the triggering mechanism .
   - A systolic murmur may be heard which disappear after the attack is over .
   - Dieresis occur after termination of the attack and return to normal sinus rhythm
2. **In older children :**
   - The child aware of the sudden onset , become quit does not wish to move .
   - He may appear pale and anxious , pulse is very rapid regular & weak .

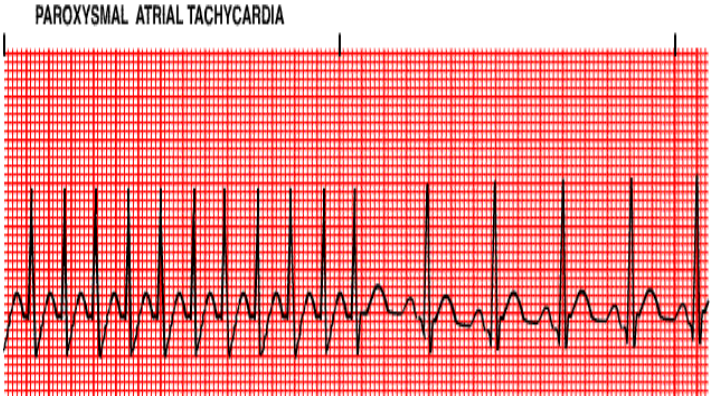
Treatment :

- Depend on the symptom of the child.
- In urgent situation the pt should be electrical cardioverted with 1-2 watt-sec/kg of electrical current .
- Asymptomatic patient may be treated with various vagal maneuver , such as placement of iced saline on the face , and neck , breath holding or a valsalva maneuver or pharmacology therapy may be used .
- Digoxin is the treatment , may use quinidine , propranolol and verapamil.








PAROXYSMAL ATRIAL TACHYCARDIA

Developmental assessment by Asma hadida

2015 _2016

Fields of development

There are four fields of developmental skills to consider whenever a young child is seen. **These are:**

- gross motor .
- vision and fine motor.
- hearing, speech and language.
- Social, emotional and behavioral.

Gross motor skills are the most obvious initial area of developmental progress. **As fine motor skills** require good vision, these are grouped together; similarly, **normal speech and language** development depends on reasonable **hearing** and so these are also considered together. **Social, emotional and behavioural** skills are a spectrum of psychological development.

The acquisition of developmental abilities for each skill field follows a remarkably constant pattern between children, but may vary in rate.

A deficiency in any one skill area can have an impact on other areas. a hearing impairment may affect a child's language, social and communication skills and behaviour. As a child grows, additional skills become important, such as attention and concentration and how an individual child manages to integrate his skills.

Developmental milestones

- ***Limit ages*** are the age by which they should have been achieved. **Limit ages are usually 2 standard deviations from the mean.** They are more useful as a guide to whether a child's development is normal than the median ages. Failure to meet them gives guidance for action regarding more detailed assessment, investigation or intervention.

When considering developmental milestones:

- ***median age*** is the age when half of a standard population of children achieve that level; it serves as a guide to when stages of development are likely to be reached but does not tell us if the child's skills are outside the normal range.

Variation in the pattern of motor development

There is variation in the pattern of motor development between children. most achieve mobility by crawling (83%), some bottom-shuffle and others crawl with their abdomen on the floor, so-called **commando crawling** (creeping).
A very few just stand up and walk. The locomotor pattern (crawling, creeping, shuffling, just standing up) determines the age of sitting, standing or walking.

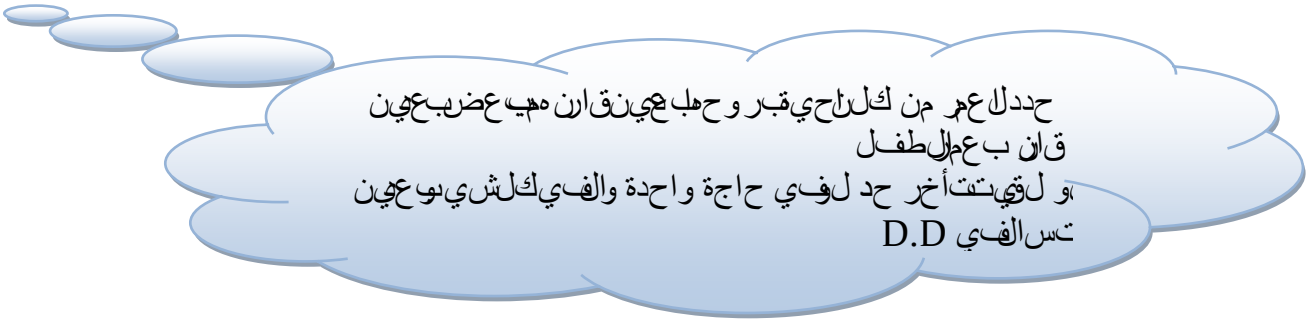
Adjusting for prematurity

If a child has been born preterm, this should be allowed for when assessing developmental age by calculating it from the expected date of delivery. Thus the anticipated developmental skills of a 9-month baby (**chronological age**) born 3 months early at 28 weeks' gestation are more like those of a 6-month baby (**corrected age**). Correction is not required after about 2 years of age when the number of weeks early the child was born no longer represents a significant proportion of the child's life.

Is development normal?

- Concentrate on each field of development (gross motor; vision and fine motor; hearing and speech/language; social, emotional and behavioural) **separately**.
- Consider the pattern of development reached by thinking longitudinally about each developmental field. Ask about the sequence of development already achieved.
- Determine the stage the child has reached for each skill field.
- Relate the progress of each developmental field to the others.
- Then relate the child's developmental achievements to his age (**chronological or corrected).**

This will enable you to decide if the child's developmental progress is **normal or delayed**. If there is developmental delay, does it affect all four developmental fields (**global delay**), or one or more developmental field only (specific developmental delay)? As children grow older and acquire further skills, it becomes easier to make a more accurate assessment of their abilities and developmental status.



Cognitive development

Some primitive reflexes present at birth. (These should disappear by 4-6 months)

Reflex - mode of eliciting it	**Description**
Moro - **sudden head extension.**	Symmetrical extension, then flexion of all limbs.
Grasp - **an object is placed in the palm at the base of the fingers.**	Flexion of the fingers of the hand.
Rooting - **stimulus near the mouth**	Turning of the head towards the stimulus
Placing - **infant held vertically and the dorsum of the feet brought into contact with a surface.**	Lifts first one foot, placing it on the surface, followed by the other.
Positive supporting reflex - **infant held vertically, feet on a surface.**	Legs take body weight, may push up against gravity.
A tonic neck reflex **(ATNR) -**	Infant adopts a 'fencing' posture,

lying supine, the head is turned by the examiner to one side.	with the arm outstretched on the side to which the head is turned.

- **gross motor development**: an explosion of skills during the first year of life.
- **vision and fine motor development**: more evident acquisition of skills from 1 year onwards.
- **hearing, speech and language**: a big expansion of skills from 18 months
- **social, emotional and behavioural development**: expansion in skills is most obvious from 2.5 years.

Developmental questioning needs to cover the whole area of developmental progression but this more focused way of taking a developmental history allows a quicker and more appropriate assessment. It directs the assessment to current abilities instead of concentrating on parents trying to remember the age when their child acquired developmental milestones sometime in the past.

Equipment for developmental testing

-Simple basic equipment is all that is needed for most developmental assessment.
- Equipment is aimed at bringing out the child's skills using play. Cubes, a ball, picture book, doll and miniature toys such as a tea-set, crayons and paper.

Developmental assessment steps

Before the examination of the four areas, history must be taken from the mother: when the child sits? Smile? Walk?
Also ask the mother about any antenatal or postnatal problems which may interfere with developmental e.g. infection, birth asphyxia.

General examination
Comment on any dysmorphic features "any child who doesn't looks normal when compared to other Children".
Measure the OFC "gives an idea on the brain developmental, child's behavior.

In the exam:
✓Ask mother the name of the child and call the child with it.
✓Start by inspection:
E.g. the child is sitting without support.
Keep child and mother comfortable: do not stand pt. till you complete thing of sitting "do not go systemic"
Keep small paper beside you and write short note.
Give a range do not give specific age, even if you know age of the child.
Initial age of development assessment is 6weeks and after 6 years it will be dependent of school performance and intellange.
After finishing gross motor, fine motor, speech and social you may do primitive reflexes to help you know the age.
Hearing test.

Age	Gross motor	Fine motor and vision	Social	Language and hearing
Newborn	**Prone** : flex knees under abdomen and rotate head to side **"term ",preterm** : will lie flat with flexed knees but not under the abdomen .	Follow the object from midline through range less than 45c from **birth to one month** , follow from midline to 90c **from one month to 1.5** month and turn to light also .		Cries . Respond to sound .
6weeks	**Prone** : raises abdomen up head briefly , knees not under abdomen "hip extended ". **Ventral suspension:** the head will be at same level with the body briefly		**Social smiling**	
3months	**Prone** : bears his weight on his forearms "flex UL" so lead to raise the head above level of ground **"head support "**. **Ventral suspension :** head control ., and pulling to sit ,	Follow object from midline through range of 180c , hold rattle if placed in his hands **, grasp reflex ,** 4_5m can reach small object ,	Smiles spontaneously , react with excitement , and stares to his one hand and laugh .	Quietness to voice and stares, turn to sound at level of ear , bubbling ,
6month	**Prone :** bears his weight on his hands with extended arms and raise of head and chest **. sits supported** **5m** : rolls supine to	Handing object and transfer from hand to hand to mouth**(mouthing)**	Recognize mother , displeasure if toy removed	Turn head to voice Say "ma , ba" , Turn to sound at 0.5 m

	prone .			
7months	Sit **without** support .	**Shakes** a rattle , bangs toys on table surface ,	Persistent to obtain toys out of reach	Da _Da and Ma_Ma sound
8months	Rolls from front to supine , sit **without** support with straight back .	Tries picking up with fingers .	Hold bottle	
9months	Crawling on the abdomen , stand with support , Roles back to front , pull to stand .	Can use thumb and fingers to pick cubs . **Immature pencil grasp.** Point with index fingers and looking for falling object . Very observant .	Responds to his name wary at strangers and become sensitive to separation from his mother , Feeding himself biscuit and also understanding NO and Bye Bye.	Da _Da , Ma_Ma, Turn to voice Do hearing destructive test .
10months	Creeps on hands and knees **"walk around furniture " stand supported**		Initiate waving bye bye	
11months	Stands unsupported .	Can pick small object.		
21months	Walk supported stand alone **Sit from lying position .**	**Stops** mouthing of objects , bangs 2 cups together . **Mature pincer grasp**	Wave bye bye. Eating by spoon Drink from cups and find hidden object . Clap hands .	Say 2_3words meaning also can respond to order
15months	Able to get up alone from sitting position . Walk alone steady ,	Linear scribbling and build tower of 2 cubs		
18months	Creeps down stairs and climbs up stairs step by step and push and pull toys	Circular scribbling and tower of 3 cubs . Point to the	Feeds independently Point to body part "nose	Says up to 20 words .

	and doors .	picture in the books . turn books on pages .	_mouth ". Symbolic play	
2years	Walks up and down stairs with 2 feet per step "immature " also kicks a balls .	Copies **ventral** line and tower of 6 .	Feeds with fork and spoon .	Says 2_3words in sentences , I ME
3yaers	Walks up stairs on foot per step and down with 2 feet also can drive tricycle .	Copies **of circular** and build tower 9 and a bridge	Able to use toilet wash hands and brush teeth	Able to tell full name , known sex male or female .
4years	Walk up and down stairs by one foot per step as adult .	Copies **of cross line** and build step of tower .	Able to undress and combing hair .	Mention address also count to 10 .
5years	Runs on toys and catch ball	**4_5 yeas** : draw a **square .** **5years** : copies **triangle** .	Able to use knife and fork Able cloths and do large button	Ask about of words and who and when .

Growth

Definition :

Growth : increase in the size and number of cells it includes aspects as wt. , length , and HC .

Development : maturation of functions and gaining of various skills increase in complexity , it includes aspects as : motor development , mental development and sexual development .

Both growth and development go parallel to each other .

Growth stages :

Intrauterine stage :

Embryonic stage : the first 8 weeks .

Fetal stage : characterized by rapid growth and development .

Early : 9-24weeks .

Late : 25-40 weeks .

Extrauterine stage post natal :

Neonate : first 4 weeks of life " 28 days "

Infant : birth to 1y include :

Neonate : " birth to 28days "

post neonate : " 28days to 1year "

infancy period of most rapid physical growth and mental development .

preschool : 1-4years

school child "5-12years "

adolescence"12-20 years ":the period of passage from childhood to adulthood .

✱ *Factors affecting growth and development :*

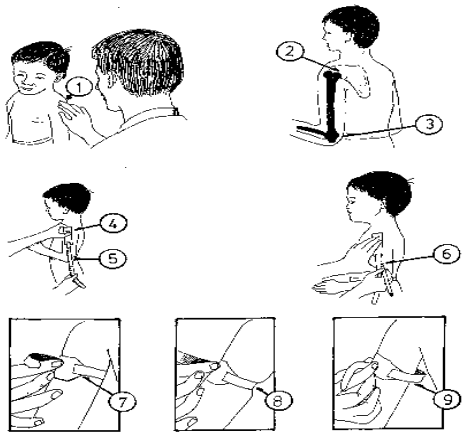
1. Genetic factor , race .
2. **Endocrinal factors** : the growth hormone, thyroid hormone and sex hormones are essential for normal growth and development.
3. **Environmental factors :** socioeconomic state , health of mother .
3. **Nutritional factors** : e.g PEM , cong . anomalies, physical factors, chronic disease. Imp.
4. Intelligent **" most important factor "**
5. **Sex** : girls grows faster than boys from 7m to 4y then equally .girls start puberty at younger age than boys .

Assessment of growth :

1. Anthropometric measurement .
2. Teething .
3. Osseous maturation .

Wt. : average 3-3.5 kg . there is an initial period of loss of wt. in the first 3-4days .
10 % due to fluid redistribution .

- **During the 1st 4months** : 3/4 kg /m (4m double birth wt.).
- **2nd 4months** : 0.5 kg /month .
- **3rh 4 months** : 0.25kg /month (at 12m triple birth wt.)
- **During early childhood 2-6 years** : 2kg /years . (wt. = age in years x 2+8)
- **During late childhood "6-12years "**: 2.5 kg /years .



Length and height :

Under 3years the length is measured in supine position .
Over 3years the height is measured in standing position .

- **At birth** : 50cm .
- **During the 1st year of life :**
- 3cm every month in the 1st 3months .
- 2cm every month in between 3-6months .
- 1.5 cm every month in between 6-12months.
- **1st year** : 75cm .
- **1y_2y** : 1cm /month .
- **After 2y = age in years x 5+80 .**
- **Or 4years** : 100cm " double birth length ".
- **4_8y** : 7cm /year .
- **9_12y** : 5cm /year .
- **12year** : 150 cm " triple birth length " .

*** HC :**

At birth 35cm .
6m → 43cm .
1y → 47cm .
2y → 49cm .
4y → 50cm .
6y → 51cm .
12y → 53cm .

Upper /lower segment ratio :

Birth : 1.7 : 1 .
3years : 1.3 :1 .
5years : 1.25 :1 .
Puberty : 1:1 .

****Mid arm circumference :***

- Normally is between 13.5 - 14.5cm in children 1-5y old .
- Those between 12.5 - 13.5cm have border line nutrition .
- Those below 12.5cm are overtly malnutrion.

Chest circumference :

It is usually measured in mid respiration at the level of the xiphoid . it is usually related to the head .

At birth ; the head is larger by 2cm

Between 1-2years : both are equal .

After 2years: the chest is larger than the head .

☺ **Teething :**

Primary or deciduous teeth :

- Central incisors 5-7m
- Lateral incisors7-9m
- 1st molar 10-16m
- Canines 16-20m
- 2nd molar 20-30m .

Secondary or permanent teeth :

- 1st molar 6y .
- Central incisor 7y .
- Lateral incisors 8y .
- 1st premolar 9y .
- Canines 11y .
- 2nd molar 12y .
- 3rd molar 18y .

Osseous maturation :

- ***At birth*** : ossify center around the knee joint are well developed .
- ***In early childhood "2-6y "*** : one ossify carpal center per year .
- ***In late childhood and adolescence*** : fusion of epiphysis .

Normally bone age corresponding to the age of the child .

Growth chart :

Growth chart :
Graphical method by which physical growth of the child can be assessed .
There is wide range variation among normal children of the same age .
Each child has his Owen inherited pattern of growth .

Types: the most imp . and commonly used are percentile GC .

Criteria :

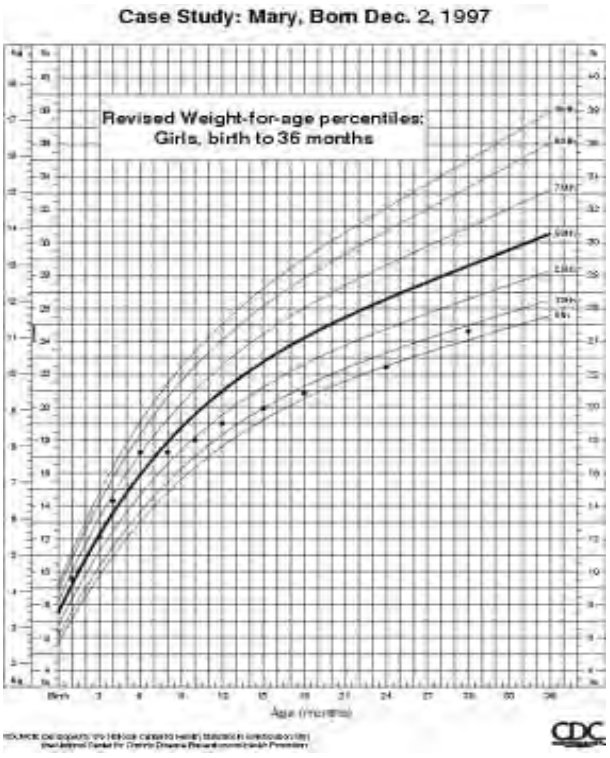
Made for different aspects of physical growth : ht , wt , and HC .

Vary from : boys to girls , one country to another .

Each chart composed of 7 curves :

50th percentile _average .
25th " _ below average .
10th " _ low normal .
5th " _ lowest normal .
75th " _ above average .
90th " _ high normal .
95th " _ highest normal .

Normal child on percentile :
Should lie between the 5th and 95th percentile curve . one single reading values below 5th or above 95th are abnormal .
Should follow the same percentile level throughout the growth period .
One serial measurement deviation of the child from his own percentile curve is abnormal .



⁂**Hearing destruction test** in which 7-9m
هم دقيا تصخشيفوكي هم رجح يفلفطلا
هوصبشم مهمل ةعقلكرحجاح ياهيفيهلي
دع هاريال ثيحبلظلا هفخ نم دحأ يأيف
:لمعي لفطلاذأ ىوتسم

- High pitched sound " spoon and cup "
- Low pitched sound سمهيوا رفصي

ولروهش6 هرمع رتم5.0 يوتسميلعمسول
يلع عمسولروهش9 هرمع رتميوتسم يلعمس
.ةنسرمع هسأرقوفيوتسم

Normal hearing eudiometry could be done at age of 4y .

N.B :

- ✓ ***Parachot reflex "protective reflex "*** : starting at 6-8m and will be mature at 9m and continuous for life.
- ✓ Pupil of newborn infant react to light since birth .
- ✓ Binocular vision develop in 1st 2years of life so squint is usually normal below
- ✓ 6m of age .
- ✓ The 1st 4m of life important period for retinal stimulation .
- ✓ Tears usually start at4m of age ***" child cry without tears before that "*** .
- ✓ Full visual acuity will develop at 4y in which the child will able to do ***E.snellen test***.

➲Causes of delayed walking :

- Physiological delay "familial "
- PEM.
- Rickets.
- Cerebral palsy.
- Down syndrome , cretinism.
- Chronic systemic diseases .
- Muscle disease as cong. Myopathy.

➲ Causes of delayed speech :

- Physiological delay
- Deafness.
- Emotional and social deprivation.
- MR.
- Bilingualism “tow language are spoken at home”
- Paralysis of muscle of articulation and tongue.
- Autism.

➲ Causes of delayed smiling :

- Lack of stimulation .
- Blindness.
- MR.
- Autism.

How to calculate IQ ?

IQ =mental age /chorological age

⇨ 70-75% border line

⇨ 50-70% mild learning difficulty

⇨ 35-50% moderate "

⇨ 20-35% sever "

⇨ <20% profound "total care "

Normal growth and puberty

➥From birth to adolescence , ***growth proceeds in biologically determined cycles that fall into 4 distinct periods*** :

1. A rapid increase in length during the first 2 years .
2. A slow period from 2 years to preadolescence .
3. A rapid period from puberty to 15 or 16 years of age , during which the growth rate increase two folds to reach a peak ***(the adolescent growth spurt) .***
4. A sharp deceleration in growth velocity until growth ceases at puberty , final height is achieved once there is bony fusion of the epiphysis .

➥ Growth normally ceases around 15 years in girls and 16 years in boys .

➥ One of the most powerful determinants for growth in the normal child , is the genetic influence of the parents height . ***(mid parental height percentile)*** .

➥ ***Puberty*** : start in girls at 11.4 years and 12 years in boys .

Boys	Girls
Testicular growth	Breast development
Pubic hair	Pubic hair
Growth of penis and scrotum	Maturation of external genitalia
Muscle bulk increase	Female body habitus
Beard , axillary and body hair	Axillary hair
First seminal discharge	Menstruation

Boys genital development :

Stage 1 : testes , penis and scrotum as in early childhood.

Stage 2 : testes grow , skin of scrotum reddens and becomes coarser .

Stage 3 : penile enlargement , further growth of testes and scrotum .

Stage 4 : penis broadens , glans enlarges , and scrotal skin darkens .

Stage 5 : adult genitalia .

∗Boys testicular volume :

The pre- pubertal testes are 1-2ml in volume , increasing to 20-25 ml in the adult . a prader orchid meter can be used to assess volume .

Girls breast development:
Stage 1 : raised nipples only .
Stage 2 : breast bud development .
Stage 3 : further enlargement of breast and areola.
Stage 4 : projection of areola and nipple .
Stage 5 : nipple projects , areola flat , adult breast contour .

Pubic hair (both sexes) :
Stage 1: nil .
Stage 2: sparse growth of long hair .
Stage 3: hair darker , coarse and curlier .
Stage 4 : distribution limited to immediate surrounds of genitalia .
Stage 5: adult type and distribution .

➢**Abnormal growth and puberty :**

Short stature :

many children will be seen to grow , in length normally below the 5th percentile (3% of population) . estimation of mid – parental height , will often clarify the situation. on other hand the following children need referral to a pediatrician :

1. children whose height is less than 90% of the 50th percentile for age (stunting) .
2. children crossing height centile in downward direction especially over 2 years .

➢ ***Assessment of such short children involves a number of essentials*** *:*

1. ***history :*** exclude social pathology and neglect , chronic illness e.g heart disease or malnutrition , and drugs e.g steroids .
2. ***examination :***
   dysmorphic features e.g down syndrome or turner .
   disproportion e.g short limbs e.gachondroplasia .
   fundi-papilledema /optic atrophy e.gcraniopharyngioma .
   evidence of systemic diseases e.g CHD.
   Assessment of nutritional status e.g malnutrition
3. ***basic investigation*** : bone age .
4. ***further investigations*** : GH , pit. Function test .

classification of short stature:

a. **Proportionate short stature** :

- Normal variant .
- Chronic malnutrition .
- CRF, cardiac and respiratory diseases .
- Endocrinal disorders.e.g. hypopitutrism .
- Chromosomal anomalies .

b. **disproportionate short stature**:

- rickets .
- achondroplasia .
- hypothyroidism .

➢ **Normal variant of short stature**:

a. Constitutional growth delay .
b. Genetic short stature (familial).

a. <u>Constitutional short stature :</u>

Birth length is normal and infants grow normally for 4-12 m then linear growth and weight gain slow down , associated with bone age delay of <3years , the child grow at a normal velocity parallel to and to below the 3rd percentile line, but is small for parental height . the condition is often familial and is associated with delayed puberty .bone age remains

roughly equivalent to height age . growth will continue after 15-16 years when his / her peers have stopped growing , and the prognosis for a normal final height is good .

b. <u>Genetic short stature (familial) :</u>

This is a common cause of short stature and the parents will be found to have a low mid-parental height percentile . low birth length and normal age of puberty .

Causes of pathological short stature :

Fat child	Thin child (normal appearance)	Dysmophic
Hypothyroidism	asthma	Turner syndrome
Growth hormone	Social deprivation	Down syndrome
Pseudohypoparathyroidism	Celiac disease	Prader-willi syndrome
Cushing syndrome	Cystic fibrosis	Skeletal dysplasia
Panhypopituitarism	Renal disease	

<u>Tall stature (gigantism) :</u>

The assessment is carried out as follows :

1. Plot measurement .
2. Calculate mid parental height percentile .
3. Exclude obesity .
4. Exclude dysmorphic syndrome e.gmarfans syndrome , kleinfelters syndrome and cerebral gigantism .

   Familial tall stature and obesity make up the majority of tall children .

Poor weight gain /failure to thrive (FTT):

Early puberty :

Commoner in girls , often constitutional in nature , defined as development of secondary sexual characteristics in girls before 7 years , and boys before 9 years . growth should be checked to see wither

accelerated , and examination carried out to differentiate between true precocious puberty and virilization(false precocious puberty) .

Characteristic	True precocious puberty	Virilization
Secondary sexual characters	Present	Present (hirsutism ++)
Breast development	Present	Absent
Penis and testis	Both enlarge	Penis only
Clitoris	Normal development	Excessive enlargement
Spermatogenesis /ovulation	Present	Absent
Family history	Frequent	Absent

Delayed puberty :

- This is normal constitutional and associated with growth delay familial .
- Defined as no sexual development by the age of 15years of boys and 14years in girls .
- Careful examination for puberty staging is essential in order to avoid mistakes .in boys the testicular and in girls breast bud development are the most useful physical signs .
- It is always wise to check chromosome karyotype to exclude turner syndrome(in female) and kleinfilter syndrome in (male) .

Endocrine by Asma Hadida

2016_2015

Introduction

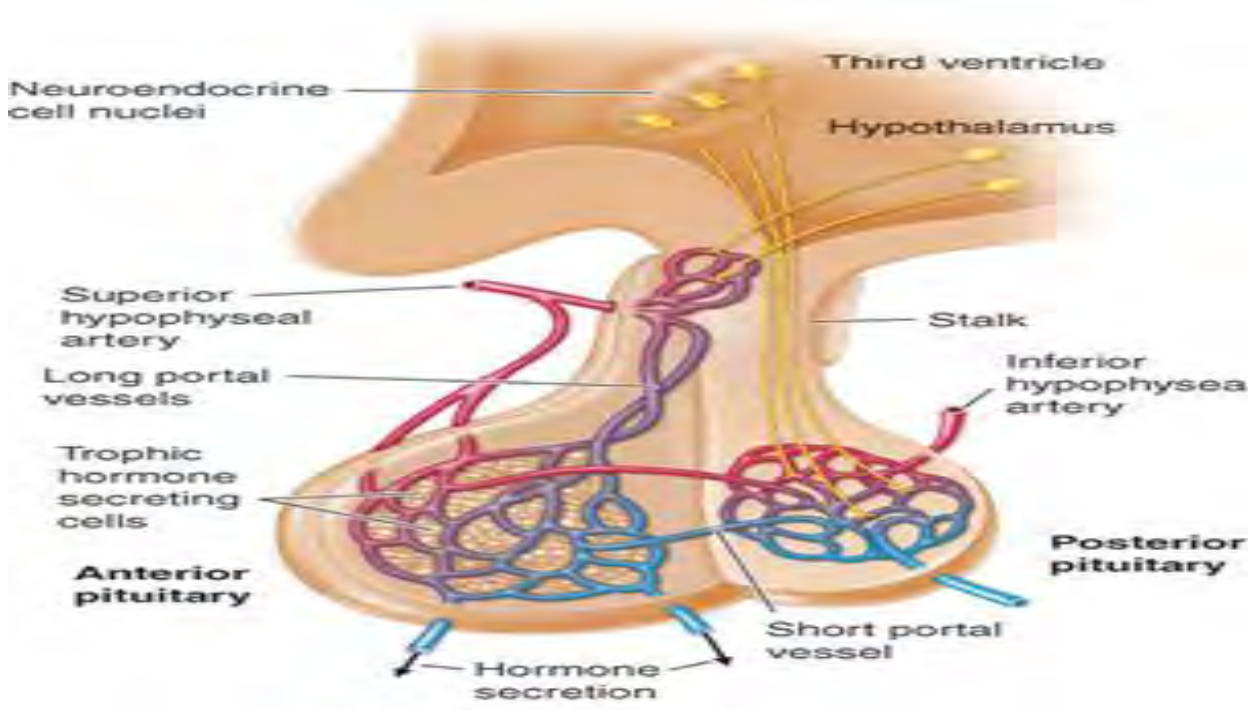
Nervous and endocrine system inter reation : **(in the hypothalamus)**

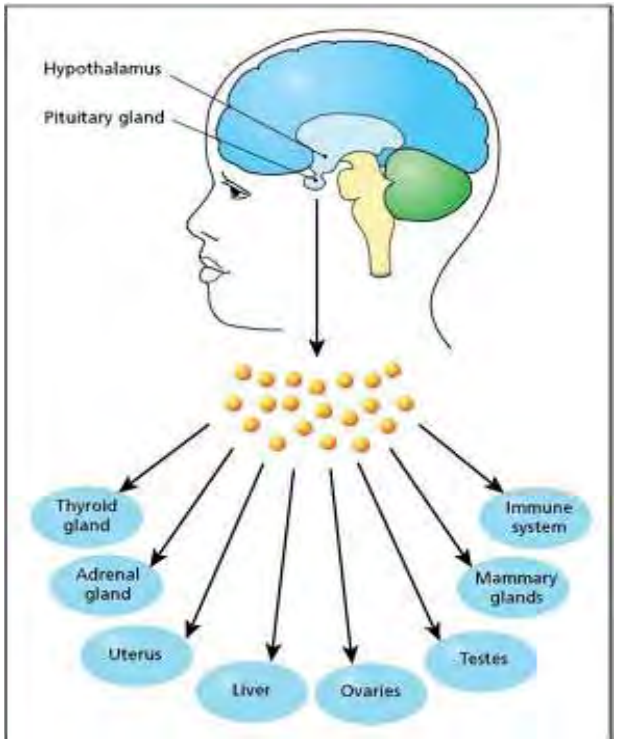
1. Hypothalamus hypophyseal portal circulation :
Between tha hypothalamus & anterior pituitary ventromedial , arcauate , preoptic & paraventricular nuclei of the hypothalamus →releasing or inhibiting hormones →portal circulation →affect the secretion of anterior pituitary gland e.g TSH , ACTH , FSH , & LH

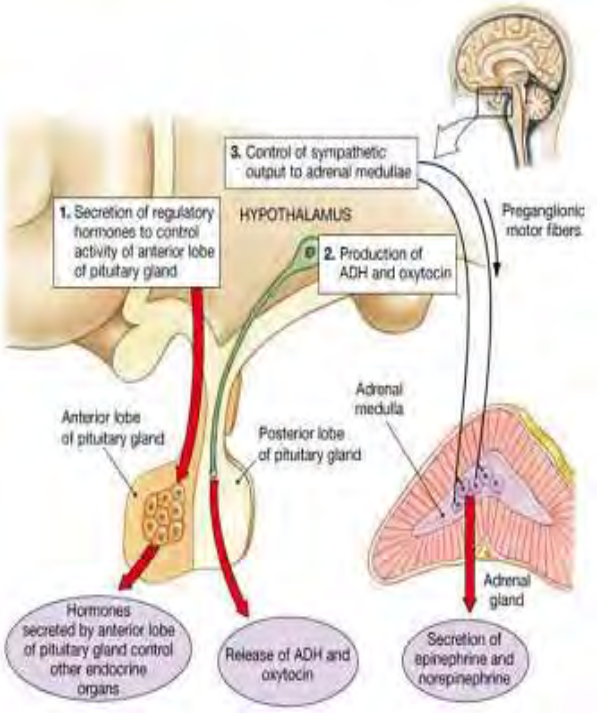
2. hypothalamus hypophyseal tract :
Between hypothalamus & poaterior pituitary , supra optic & para ventricular nuclei of the hypothalamus →form ADH & oxytocin hormone →released and secreted from the nerve ending in the posterior pituitary

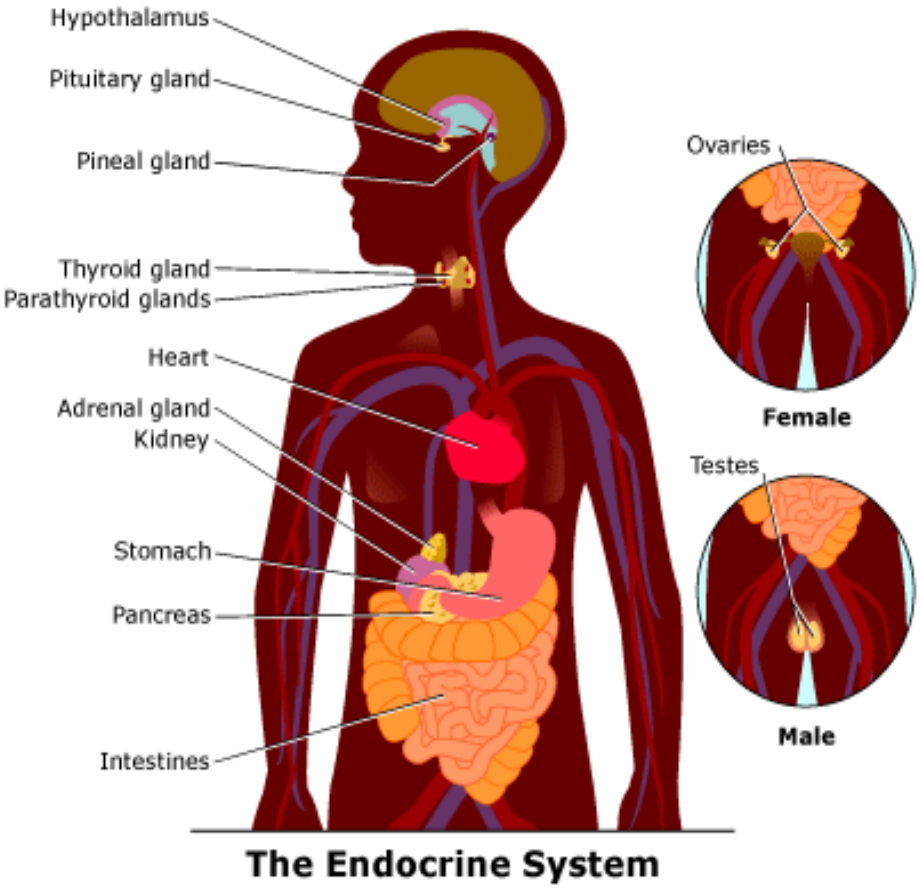
3. **Direct connection between hypothalamus & suprarenal medulla :**
→affects the secretion of epinephrine & norepinephrine

The endocrine system can affect the nervous system :
Parathyroid gland →control blood ca level →affect the excitability of nerves , Thyroid hormones can also affect the excitability of the nervous system









The Endocrine System

Endocrine glands :

glands that secrete hormones **directly** in blood →affect target tissues & organs all over the body

- Pituitary glands is the master of endocrine glands
- **3 endocrine glands under the control of pituitary gland** : thyroid , supra renal cortex & gonads
- **3 endoocrine glands not under control of pituitary gland** : parathyroid gland , supra renal medulla & pancrease

The endocrine hormones

- Some hormones affects many different types of cells e.g GH , & thyroxine such hormones are very important in regulating almost all body functions (metabolism , growth , & development , reproduction ….. etc
- Other hormones affect only specific target tissues e.g ACTH on adrenal cortex & ovarian hormones on female sex organs & secondary sex chatacters

Chemistry :

1. **P**rotein & **p**olypeptides : **p**ituitary , **p**ancreatic , **p**arathyroid & hypothalamic hormones
2. **S**teroids hormones : **S**uprarenal cortex hormones , **S**ex hormones & vit D derivatives
3. **T**yrosine derivatives : **T**hyroid hormones & catecholamine .

Pitutary gland
)Hypophysis cerebri)

✶ Hormones af anterior pituitary :(adenohypophysis)

1. GH
2. TSH
3. ACTH
4. Gonadotropins **(FSH , LH)**
5. Prolactin

✶ Hormones of posterior pituitary : (neurohypophysis)

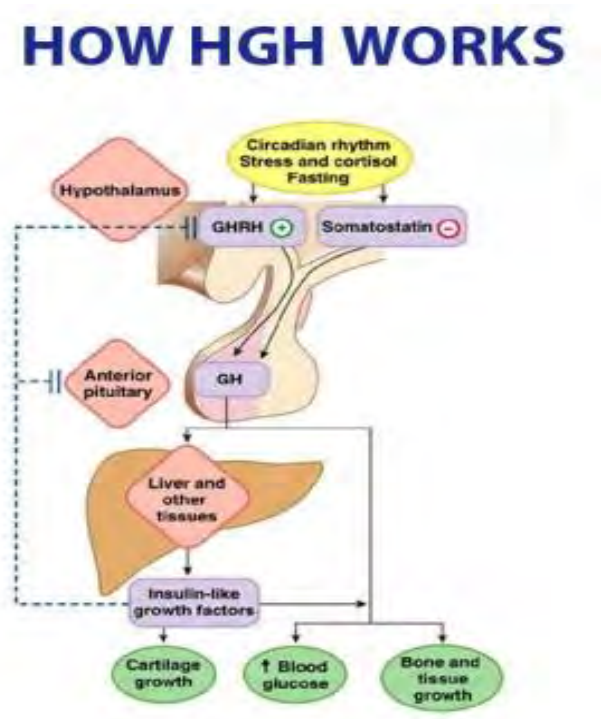
1. Oxytocin .
2. ADH

GH deficiency

GH deficiency occur either isolated or with other hormones of anterior pituitary (**panhypopititrism**).

Physiology of GH:

GH secreted by ant. Pit. Under control of hypothalamus **(GHRH)** , which **is pulsatile** (eight peak per day with low basal level in B/W) 50% of daily GH secretion occur in early hrs of **the night** .



Factors that increase secretion of GH:

1. Stress and exercise
2. Sleep
3. Hypoglycemia
4. Starvation
5. α adrenergic stimulation

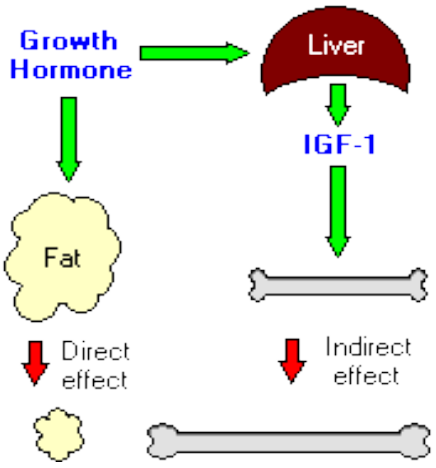
Factors that decrease secretion of GH:

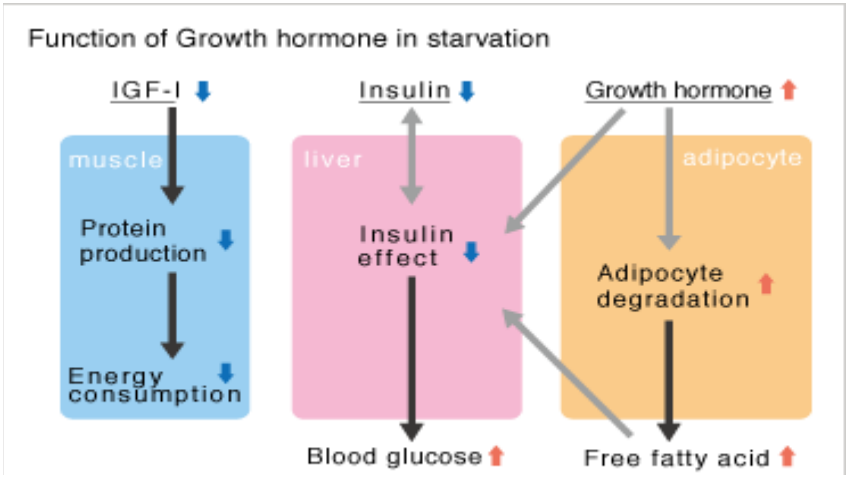
1. hyperglycemia
2. GH therapy
3. Somatostatin
4. β adrenergic stimulation

Action of GH :

1. Anabolic (especially on long bones)
2. Increase protein synthesis.
3. Anti insulin effect.

Action of GH is mediated by insulin like growth factor I (**IGEI**) produced by the liver .





∗ ***Causes of hypopituitrism*** :

1. **Isolated GH def. :**
   - ⇨ **Genetic :**
     - ◗ Mutation of GH or GHRH genes
     - ◗ End organ resistance :
       - abnormal GH receptors **(laron syndrome)**
       - inactive GH .
       - abnormal IGFI.
   - ⇨ **Acquired :**
     - ◗ post cranial irradiation .
     - ◗ idiopathic **(most common)**

2. Multiple pit. Hormones def. **(panhypopituirism)**:
   1. **Genetic** :mutation of genes
   2. **Congenital :**pit agenesis or hypoplasia , usually ass. With mid facial anomalies as cleft palate
   3. **Acquired :**
      - ◈ Tumors as craniopharnygioma
      - ◈ Truma
      - ◈ Infiltration as histocytosis

∗***Clinical picture of GH deficiency :***

1. **At birth :**
   - ⇨ Normal wt and ht in full term baby (**GH is not essential for IUG)**
   - ⇨ Microphallus is of diagnostic clue
   - ⇨ Sever hypoglycemia .
   - ⇨ Associated with mid facial anomalies e.g cleft palate or cleft lip , single central incisor , septo optic dysplasia .

Later on :

- ⇨**Short stature** : proportionate short stature appear at the end of 2nd year , growth velocity <5cm/year.
- ⇨**Childish face** : immature appearance and small face , nose , mandible , wide ant fontanels and fine hair , delayed teething and delayed bone age .
- ⇨**Hypogonadism** : genitalia are under developed for age .
- ⇨Hypoglycemia
- ⇨ Normal intelligence .
- ⇨Delayed puberty .
- ⇨ Features associated with other hormones def. e,g hypothyroidism

Investigation:

1. **GH level :**

   By GH provocation test agents :

   - Clonidine
   - Insulin

- Exercise - Arginine
- l.dopa .

To confirm diagnosis tow tests should be done b/c single blood sample GH measurement is not informative .Also measure IGFI not sensitive B/C give **false +ve in liver diseases** .

2. **Measurement of IGF binding protein** : more accurate (not related to liver disease) and good indicator of GH state .
3. Level of other ant . pit hormones
4. GHRH stimulation test (to differentiate pit . from hypothalamic causes).
5. Laron syndrome **(GH insensitivity**): IGFI low and GH is high ." treatment by IGFI recombinants ".
6. **For the cause** : skull x-ray , CT &MRI for pit . tumor , aplasia or hypoplasia .
7. **For effect** : delayed bone age .

Treatment

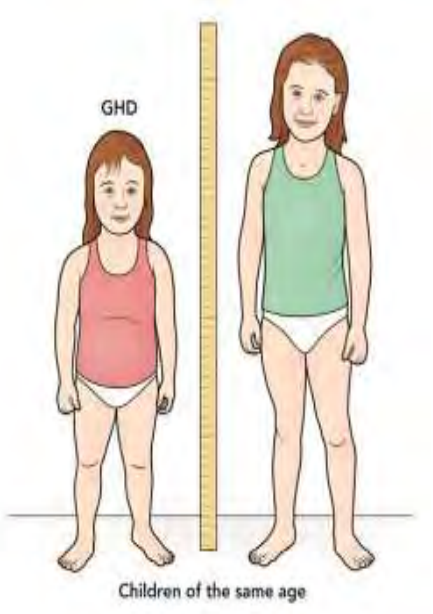
1. **Recombinant GH :** S/C 6 days / wk , IM alternative 3 days / wk
   Dose =12u / wk
   At bed time to mimic normal peak .used till complete fusion of ossification center or not increase in growth " rate of growth < 1inch per year or when bone age > 14 in female and > 16 in male ".
2. Replacement of other hormone deficiency .

Side effects of GR therapy :

1. Pseudotumour cerebri .
2. Silpped femoral epiphysis .
3. Risk for leukemia .

Indication of GH :

1. GH def .
2. Turner syndrome .
3. CRF.
4. Prader willy syndrome .



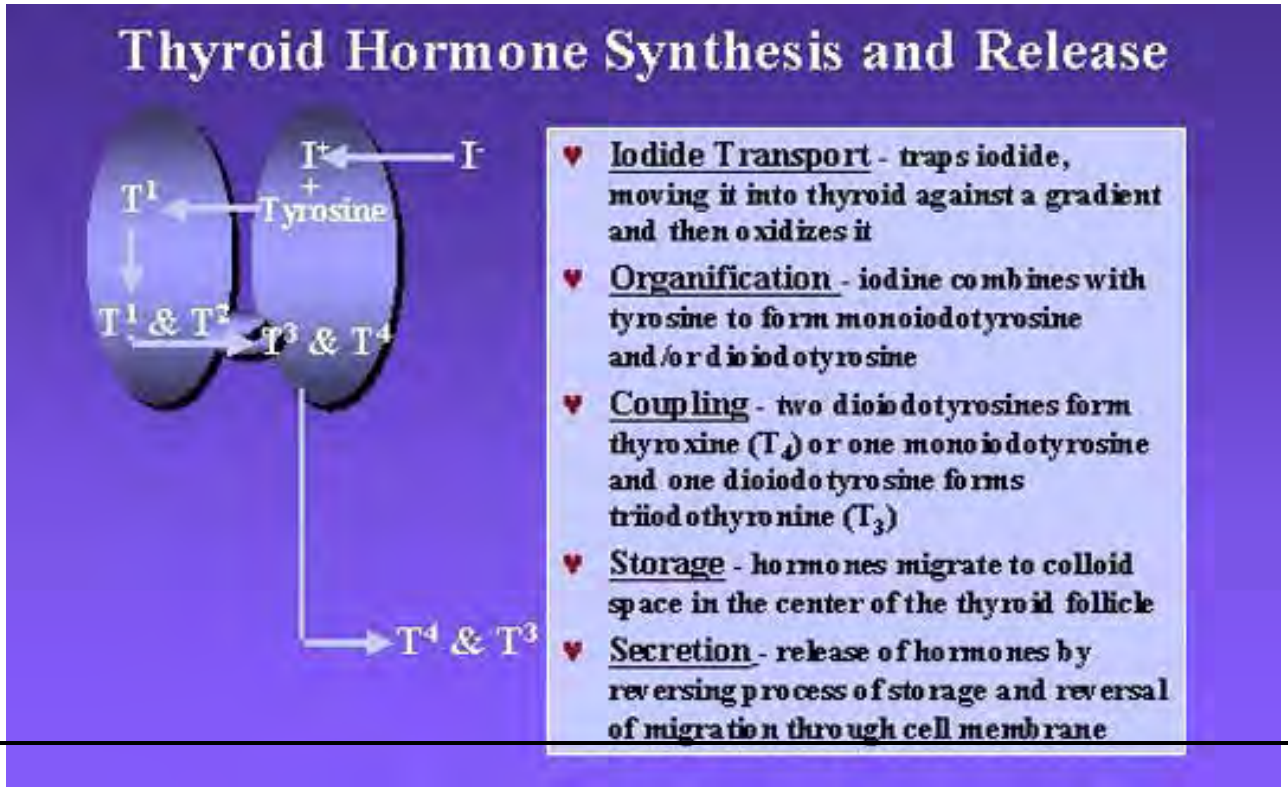
Thyroid gland

∗ **Thyroid gland secrete** :

1. **Thyroid hormone :** thyroxine T4 and T3 tri idothyronine .
2. **Calcitonine** : which deposit ca salts in bone .

∗ **Function of the thyroid hormone** :

1. Normal maturation of the growing brain in 1st year of life .
2. Normal skeletal growth " bone growth " and sexual maturation .
3. Oxidative metabolism & heat production in all cells **" increase BMR ":**
   - ↑glycogenolysis .
   - ↑gluconeogenesis.
   - ↑lipolysis
4. **B. Adrenergic effect** : ↑ all cardiac function , ↑sensitivity of B aderenergic receptors to cathcholamine (+ve inotropic &+ve chronotropic)
5. **GIT :** ↑motility .
6. **Blood** : hemopoietic factor (↑RBC).
7. Carotene to vitamin A by thyroxin hormone .



Only 20% of circulating T3 is produced by thyroid while 80% is produced by peripheral conversion of T4 by deiodinase .

Regulation :

1. **Thyrotropin releasing hormone "TRH"** : secreted from hypothalamus to modulate TSH function " +ve feedback".
2. **TSH:** secreted from pituitary gland to regulate thyroid function .
3. **Thyroid hormone** : exert –ve feed back inhibition at the level of pituitary gland and hypothalamus .

TSH does not cross placenta , T3&T4 can cross in limited amount , ant thyroid drugs can cross placenta

Types of hypothyrodisim :

➲ **According to the time of onset** :

a. **Congenital hypothyroidism** :present early in neonatal period and early infancy .
b. **Acquired hypothyroidism :** present in childhood after period of normal growth .

➲ **According to the site of lesion** :

a. **1ry hypothyroidism** : thyroid gland lesion .
b. **2ry hypothyroidism** : in the pituitary
c. **3ry hypothyroidism:** defect in the hypothalamus .

Congenital hypothyroidism =cretinism

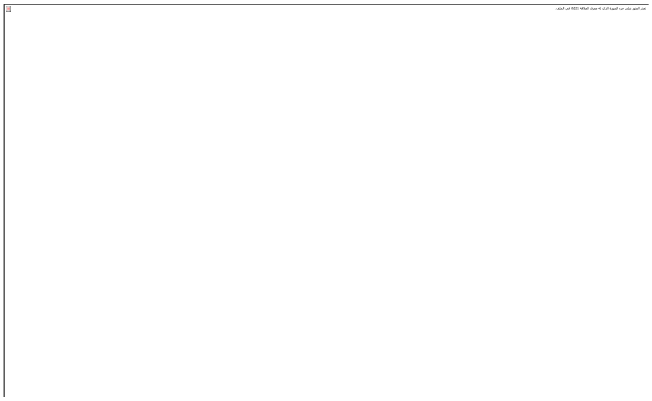
Serious B/C it affect the developing brain , early detection by neonatal thyroid screening is important to prevent MR .**more common in female (2:1)**

Causes

1. **Primary in the thyroid gland** :
   a. **Thyroid dysgenesis** : the commonest cause 90% , aplasia , hypoplasia or ectopic gland , no goiter .
   b. **Defective thyroid hormone synthesis "dyshormonogenesis":**
   -the 2nd common .
   -all are AR disorder .
   - associated with goiter
   c. **Transient hypothyroidism**: transplasental passage of maternal : TSH receptor blocking AB .
   d. **d.maternal iodine deficiency** : endemic goiter .
   e. End organ unresponsiveness" pseudo hypothyroidism " : with goiter (↑T4, ↑T3, ↑TSH)
2. **2ry hypothyroidism** : in the ant . pituitary due to TSH deficiency
3. **3ry hypothyroidism** : due to TSHRH deficiency and defect in the hypothalamus .

C/P

1. **In neonatal period** : **non – specific** manifestation so screening is essential . may be there is :
   GA >42wk , birth wt > 4kg
   a. Prolonged physiological jaundice .
   b. Lethargy , cry little sleep much .
   c. Poor feeding , lack of interest , chocking spells during feeding .
   d. Wide post and ant fontanel
   e. Noisy breathing due to large tongue
   f. Distended abdomen , constipation with umbilical hernia .
   g. Heavy at birth
   h. Subnormal temperature (cold mottled skin)
2. **If untreated cases the picture will be full blown by 3_6m of age** :
   • Motor retardation (delayed sitting , standing , and walking)
   • MR delayed mother recognition and speech



- **measurement** : short stature with inappropriate upper and lower segment ratio , body retain infantile proportion 1.7/1 (thyroxin is necessary for long bones growth)
- **head** : normal size with widely opened AF &PF
- **hair** : coarse , brittle with low ant hair line .
- **fore head** : short and wrinkled .
- **eye :** swollen eyelids due to deposited myxoedematous tissue .
- depressed nasal bridge .
- **mouth :** kept open with large thick broad tongue protruded from it , lip is pallor with delayed dentations.
- **neck** : short and thick .
- **Trunk:** distended abdomen with umbilical hernia.
- **skin** : pallor " anemia " and yellow discoloration " hypercarotenemia" , dry scaly skin " vit A def."
- **limb**: short limb , broad hand , generalized hypotonic
- **genitalia** : delayed sexual maturation rarely precocious puberty .
- **cardiac :** bradycardia , pericardial effusion , cardiomyopathy

Investigation

1. **Diagnosis** : T4 low (normal 5-12 microgram / 100ml)
   - High TSH " normal 0.5 - 4 micro unite " the most sensitive test for 1ry hypothyroidism .
   -In 1ry hypothyroidism there is compensatory ↑in peripheral conversion of T4→ T3" so measuring of T3 may be misleading .
   -TSH low in 2ry or 3ry hypothyroidism .
   -In pseudo hypothyroidism " T3 , T4 , TSH ALL ↑"
2. **For the effect** :

a. **X-ray :**

-Delayed bone age

-**Skull x-ray** : large AF , PF , and delayed teething

-**Chest x-ray** : cardiomegaly

- Cardiac : ECG " bradycardia and low voltage QRS"

b. **Echo :** cardiac enlargement and effusion

Treatment

➲ Long life therapy with oral thyroid hormone " sodium L. thyroxin "
➲ **Monitoring** : by clinical assessment :
- HT every 3m .
- Mental development and IQ.
- Bone age every year .
- Pubertal development should be within normal .

➲ Laboratory assessment :
T4 → High or normal
T3 → Normal
TSH → Below 5microunite .

Prognosis

Diagnosis and treatment before 3m → good prognosis
- If at 3m -6m → variable response
- If > 6m → permanent MR

Pendered syndrome goiter associated with deafness due to decrease of peroxidase enzyme that essential for uilization of iodine

Acquired hypothyroidism

Juvenile hypothyroidism with manifestation appearing after period of normal thyroid function .

Causes

1. **Thyroiditis :** autoimmune thyroditis " hashimoto disease " , the commonest cause , either isolated or part of autoimmune poly glandular disorder .
2. **Infiltration** : in hemochromatosis , tumor
3. **Injury to thyroid :** trauma , surgery , irradiation.
4. **Iodine containing drugs .**

C/P

1. Short stature .
2. School under achievement
3. Goiter
4. Delayed puberty may be precocious
5. Constipation
6. Cold intolerance
7. Cold pale thick skin

Investigation

As before +Search for auto AB for hashimoto thyroditis e.g thyroid antiperoxidase AB, Check for associated autoimmune disease e.g DM Addison

Hyperthyroidism

- Usually result from graves' disease " autoimmune thyroditis " 2ry to production of thyroid stimulating AB .
- The C/P similar to that of adult but eye manifestation **is less common** Common in teenage girl

C/P :
-anxiety and wt loss with increase appetite
-rapid growth in ht " advanced bone age "
- sweating and diarrhea
-short stature when the early fusion of epiphyseal palate occur
-tachycardia and wide pulse pressure
-warm vasodilator peripheries
-tremor
-learning difficulties
-behavioral problem , psychosis

Treatment :

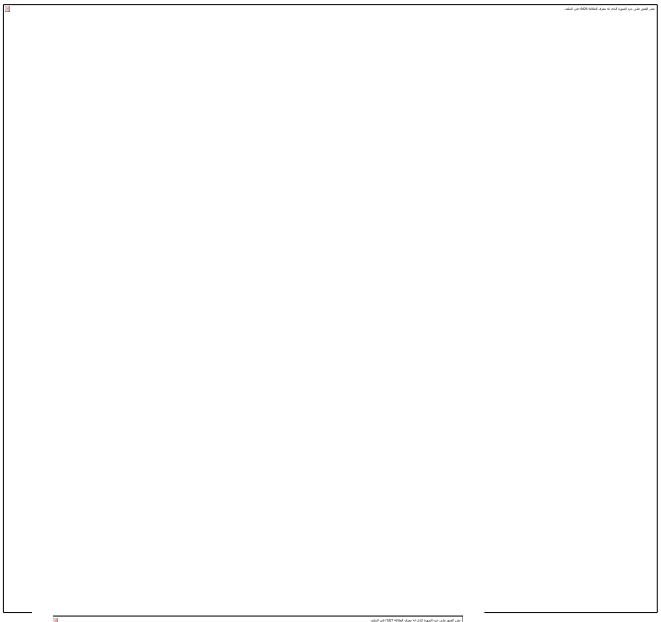
1. **Medical :** for 2years :
   a. carbimazole or propylthiouracil which interfere with thyroid hormone synthesis .
   b. **B Blocker symptomatic relief :** anxiety , tremor , and tachycardia .
   Eye signs may not resolve .40-75% relapse after stop treatment
2. **Surgery :** subtotal thyroidectomy
3. Radioiodine I 131

Preterm baby may have very low level of T4 for the 1st few weeks of life while the TSH normal so thyroxin is not required .

Adrenal gland

Adrenal gland consisit of (cortex and medull).

- **The cortex 3 zones :**
  - Outer (zona glomerulosa) ; secret mineralocorticoids (aldosterone)
  - Middle (zona fasiculate): glucocorticoida (cortisone)
  - Inner (zona reticularis) : sex steroids (androgen)
- The medulla ➲ epinephrine and nor epinephrine (cathecholamines)



- **Adrenal gland diseases :**
  - ↑secretion of androgen (adrenogenital syndrome)
  - ↑secretion of cortisol (cushing syndrome)
  - ↑secretion of aldosterone (hyperaldosteronism)

Adrenal Insufficiency

a. 1ry adrenal failure : reduced both cortisol and aldosterone &↑↑↑ACTH

✔ Causes of 1ry (most commom):

i.acquired :
1. autoimmune (addisons disease)
2. adrenal hge
3. adrenal infection e.g. TB
4. iatrogeneic

ii. congenital :
1. Congenital adrenal hyperplasia
2. Congenital adrenal hypoplasia
3. Familial glucorticoid deficiency

iii. Idiopathic

b. 2ry adrenal failure : dye to reduced ACTH lead to ↓ in cortisol level only

✔ Causes of 2ry adrenal failure :

i. Defects in hypothalamic pituiry :
1. Congenital panhypopituatrism
2. Intracranial mass
3. Intracranial inflammation
4. Intracranial radiotherapy
5. Post surgery or trauma injury

ii. suppression of hypothalamic – pituitary –adrenal axis :
1. Glucocorticoids therapy
2. Cushing disease (after pituitary tumour removal)

C/P 1. **Acute (adrenal crisis) :**
- Circulatory shock with sever hypotension and collapse
- Hypoglycemia , hyponatremia , hyperkalemia , metabolic acidosis
- Muscle spasm , nausea, vomiting , diarrhea

2. Chronic forms :
- Generalized fatigue , weakness
- Anorexia , nausea , vomiting , dizziness , wt loss
- Postural hypotention , pigmention of palm creases &mouth (b/c ↑ACTH)
- Delayed puberty 2ry to low androgen

N.B: addison disease commonly associated with autiimmune diseases e.g vitiligo , alopecia, perinious anemia

Diagnosis :

1. Basal serum cortisol &ACTH (ear;ly mornining)
ACTH ⇨high (1ry)
⇨low (2ry)

2. Adrenal stimulation test :
- Insulin tolerance test (gold standared): serum cortisol response to hypoglycemia >550nmol /l is normal
- **ACTH simulation test (synacthen test):**
1. Short test to diagnose adrenal failure :
- Serum cortisone is measured as base line
- Give IV synthetic ACTH
- ~~Measure cortisol after 30, 60min : serum cortisol response >550nmol/l at 60nim~~
2. Long synacthen test : ACTH for 3days

Treatment :

1. 1ry adrenal failure ; glucocorticoids & mineralocorticods replacement therapy
2. In 2ry : glucocortcoids replacement only

N.B: ↑dose of gluvocorticod by 2-3times in case of infection , sress , trauma , surgery .

Cushing disease

Causes :

1. 1ry (ACTH) independent :
   - Adrenal tumour
   - Adrenal hyperplasia
   - Mccune Albright syndrome
2. Iatrogenic : (most common in pediarric)
3. 2ry (ACTH)depended :

C/P :

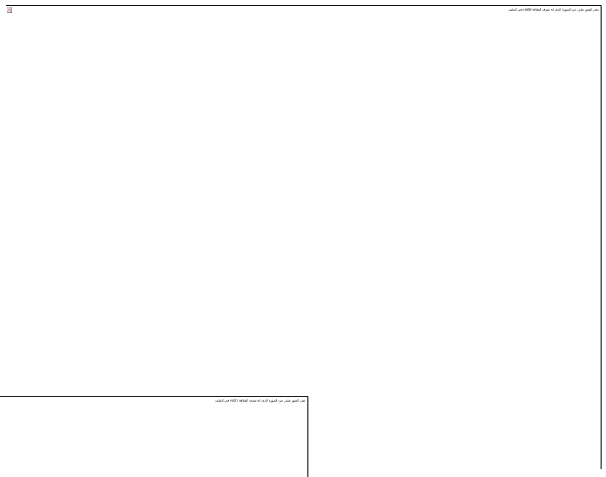
-moon face truncal obesity with central distribution of fat
-impaired growth , striae purpuric , delayed puperty &amenorrhea
-hyperglycemia
-hypertention common
-osteoprosis with pathological fracture
-Wasting &weakness of proximal tigh

Diagnosis :

1. 24hr urine cortisone high
2. 24hr serum cortisone level : high with loss of diurnal variation
3. Dexamethasone suppression test
4. CT MRI
5. U/E &ABG : hypokalemic metabolic alkalosis

Treatment :

Treatment of underlying cause



Pheochromocythom

Rare tumour usually benign of medulla of adrenal gland

Role of 10 :

- 10 % Malignant
- 10% Extraadrenal
- 10% Incidence
- 10% Bilateral

Associated with neurofibromatosis or

C/P :

- Headache
- Profuse Sweating
- Palpitation
- Abd Pain
- N&V
- Fluctuating non sustained hypertention with --tachycardia

Diagnosis : 24 hr urine condition for:
-venyl mandalic acid
-homovalenic acid
U/S, CT scan , MRI
Treatment : surgery with β & α blocker to suppress the cathcholamine pre-during –post surgery

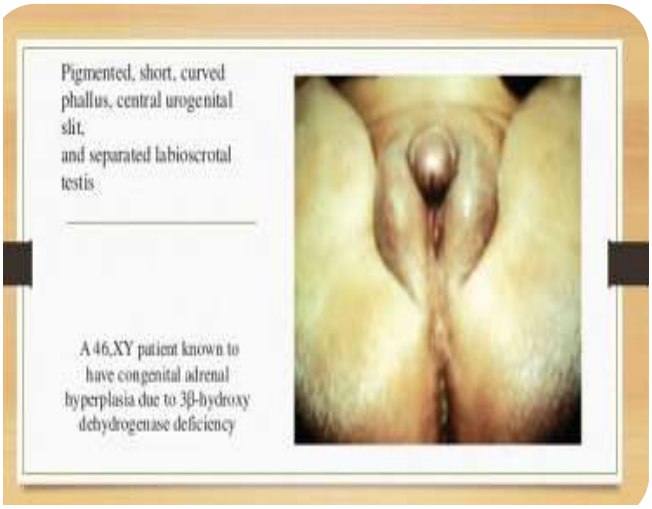
Ambiguous genitalia

- Unclear sex by inspection of genitalia

- **Types** :
  1. Female pseudohermaphrodism "XX chromosome " but virilized or musculinized :
     - It is the commonest type e.g cong . adrenal hyperplasia , adrenal tumours , maternal ingestion of androgen , adiopathic
  2. Male psedohermaphrodism "XY chromosome " but feminized or incomplete male :
     - E.g testicular problema TORCH infection
     - Hypothalamic /pituitary problem
     - 5alfa reductase def.
     - Testicular feminization " androgen insensitivity " due to

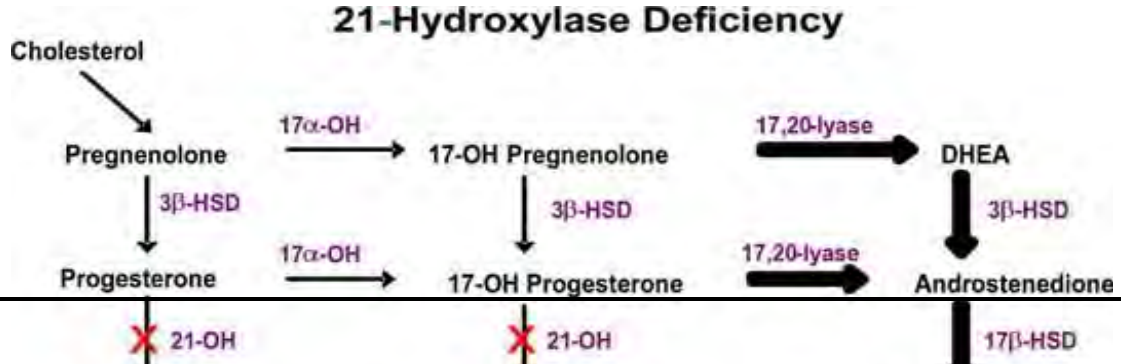
Management :
- Depend on time of presentation , anatomy of genitalia , family opinion
- If presentation at birth : delay announcement of birth &sex of baby , delay naming the baby or choose a name that goes with both sexes

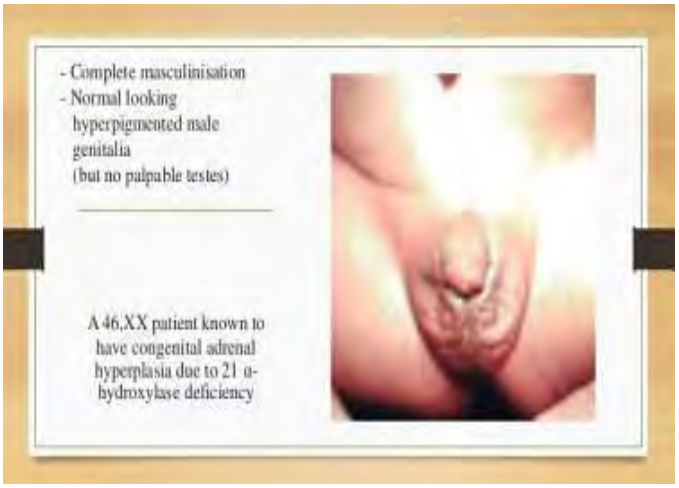
Investigation :
➲ U/S of internal organs and gonads
➲ Measurment of adrenal &sex hormone " 17αhydrocortisone "
➲ Chromosomal study " karyotyping "



Congenital adrenal hyperplasia =adrenogenital syndrome

- Its an autosomal resessive"chromosome 6" disorder of adrenal gland (hydrocortisone & miniralocortisone biosynthesis)
- The commonest enzyme abnormality is :
  - > 90% 21 hydroxylase deficiency
  - 11B hydroxylase (2nd common)
  - Other rare forms of CAH :
    * 3B hydroxysteriod dehydrogenase
    * 20-22 desmolase
    * 17&18 hydroxylase
    * 17&20 desmolase





- **Congenital adrenal hyperplasia (CAH)**
- **is a group of inherited autosomal-recessive disorders in which a genetic defect results in the deficiency of an enzyme essential for synthesis of cortisol and, at times, aldosterone. There are several forms of CAH, the most common of which is 21-hydroxylase (21-OH) deficiency, occurring in over 90% of all cases**

Clinical picture :

The presentation usually in neonatal period :

⇨ in female : (ambiguous genitalia) virilization of external genitalia (hypertrophy of clitoris and labial fusion) b/c of ↑androgen diagnosed early , internal organ normal

⇨in male : normal , the penis may be enlarged and the scrotum is hyper pigmented so may be missed.

- Salt losing crisis presentation is about 70-80% 0f 21 hydroxylase deficiency which may occur 3-4 day up to 3weeks .
  - ➲ Present with (hypotension , feature of dehydration , vomiting , shock , collapse)
  - ➲ 20 % non salt losing : precocious puberty (tall stature , muscular built , pubic hair , acne)at 4-6 y of age
  - ➲ There may FH of neonatal death if salt losing crisis was not diagnosed & treated.

Laboratory diagnosis :

- Precursor steroid " 17 –OH "progesterone ↑.
- Hypoglycemia , hyponatermia , hyperkalemia , metabolic acidosis
- Low cortisol , ↑androgen
- ↑ renin and ↓ aldosterone
- Definitive test : measure 17 OH progesterone before and after an IV bolus of ACTH

Management :

- In salt losing crisis : give normal saline I.V , dextrose , hydrocortisone I.V
- Long term management : long life glucocorticoids to suppress ACTH level and allow normal growth & maturation , long life mineralocorticoids " fludrocortisones " if there is salt loss .
- Affected female will need corrective surgery to the external genitalia .
- Follow up is important in CAH :
  - Growth " ht & wt " : children with CAH will be short as adult due to premature closure of epiphysis .
  - The dose of hydrocortisone should be correct .
  - If replacement ↓ ⇨↑ACTH , ↑ androgen ⇨rapid initial growth , ↓ final height.
  - If replacement is ↑⇨slow & delayed growth.
  - Monitoring of skeletal maturity : X – ray to show bone age.

N.B :

Prenatal diagnosis is possible by amniotic fluid analysis " 17 ALFA hydroxylase progesterone level " or chorionic villi sampling " DNA analysis " .dexamethasone can be given to the mother at 7th week of gestation to suppress fetal ACTH .

- Other types of CAH :
  - 11 B Hydroxylase deficiency : female ambiguous genitailia , hypertension due to ↑DOC which is weak mineralocorticoid like action , it is absent in the 1st few years of life , ↑androgen , hypokalemia , ↑DHEA and testosterone, reduce plasma rennin
  - 3B OH steroid deficiency : Salt losing " male and female ambiguous genitalia " ↑DHEA " ↑17 OH pregenolone
  - 17alfa OH " 17, 20 lase " deficiency : Male ambiguous genitalia , hypertension , ↑DOC , 18 –OH DOC , hypokalemia .

N.B :

- ✓ Exogenous administration of glucocorticoids suppresses secretion of ACTH this causes adrenal atrophy , cortisol secretion is also suppressed , abrupt withdrawal can cause plasma glucorticoids levels to be severely reduced recovery of secretion of cortisol may require months .
- ✓ Hyperpigmentation and darking of skin is the result of ↑ ACTH due to melanocyte stimulating properties .

Diabetes Mellitus

• Metabolic disorder resulting from insulin deficiency or it's action due to islet B cell damage, with subsequent defect in metabolism of CHO, protein, lipids.

• **Types :**

Type I : Insulin dependent DM in children.

Type II : Non insulin dependent DM in adult.

Etiology:

1. **Genetic susceptibility:**
   Genes related to HLA region on chromosome 6.
2. **Enviromental predisposing factors:**
   The most important are :
   a. Viruses: The following virus may be triggering factors for B cell

Pathophysiology:

Action of insulin:

- ↓ Blood glucose by :
  - ↑ Glucose uptake by cells.
  - Gluconeogenesis.
  - ↓ Glycogenolysis.
- ↑ Lipogenesis.
- Anabolic effect.
- ↑ protein synthesis.

↓ Insulin
↑ Anti insulin hormones

CHO
- ↑ Glycogenolysis
- ↑ Gluconeogenesis
- ↓ Glucose utilization

↓
- Hyperglycemia.
- Glucosuria.
- Polyuria "osmotic".
- Water and

Lipid
- ↑ Lipolysis
- ↑ FFT
- ↑ Ketone bodies "acetone, Acetonacetic acid, B hydroxyl buteric acid"

Protein
- ↓ Protein synthesis
- ↑ protein catabolism

↓
- Weight loss
- Growth failure

C/P:

1. Polyuria, polydipsia and recent weight loss
   (the main presentation)
   The duration of symptoms usually less than one month.
2. Secondry nocturnal enuresis due to polyuria.
3. DKA represent the initial presentation in 20-30%:
   Early manifestation:
   - Vomiting and abdominal pain "excretion of KB in the stomach"

   Late manifestation:
   - Dehydration and may be shock due to vomiting and polyuira.
   - Metabolic acidosis "Ketoacidosis" → deep rapid breathing followed by cerebral and myocardial depression.
   - Coma.

N.B:
- Anti insulin hormones "cortisol, glucagon, epinephrine"
- D.M: FBS > 126.
  2hours PP > 200.
- Impaired GT:

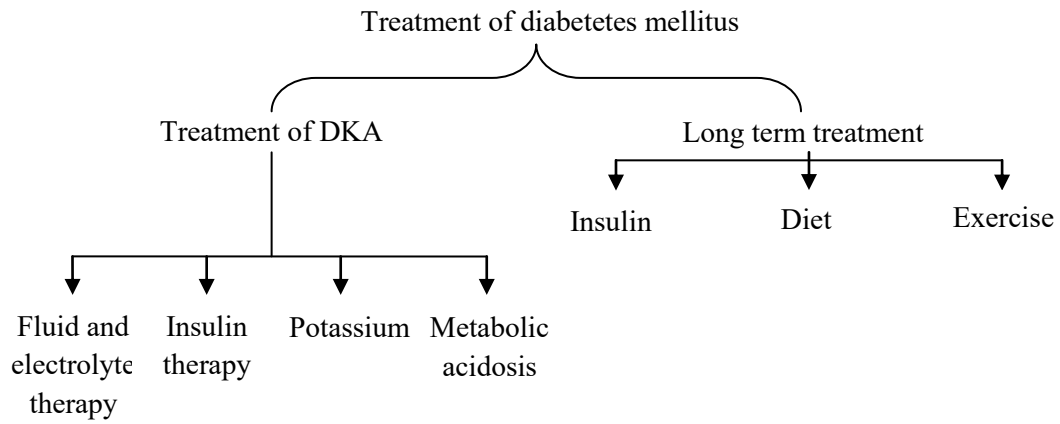
Investigation:
(Laboratory diagnostic criteria)

- Ordinary cause:
  - FBS: above 126 mg/dl.
  - 2 hrs post prandial: above 200 mg/dl.
  - RBG sample: more than 200 mg/dl in the presence of diabetes symptom.
- DKA:
  - Hyperglycemia "blood sugar >300mg/dl".
  - Metabolic acidosis "low PH and bicarbonate"
  - Glucosuria and ketonuria.
  - Cardiac monitor for

Complication:

1. Acute complication:
   a. Coma: DKA, hyperosmolar coma.
   b. Complication of treatment:
      - Hypoglycemia and hypoglycemia coma.
      - Insulin resistence.
      - Hypersensitivity to insulin.
2. Chronic complication:
   - Retinopathy, nephropathy, neuropathy, "sensory and autonomic".
   - Macroangiopathy:
     - Ischmic disease of heart brain and leg (not common in pediatric).

Treatment:



1. Fluid and electrolyte therapy:

- According to the degree of dehydration which is usually sever or shocked.
- Type Of fluid: Normal saline than glucose 5%.
- The correction should be grandual to avoid cerebral edema.
- Monitor fluid input and output, electrolyte and create acid base balance.
- Insert central venous line and urinary cathter if shocked "2 IV line".
- NGT could be inserted if there is vomiting .
- Shock therapy = 20 ml/kg over 20 min.
- Corrected slowly over 48-72 hour.
  1st day = D ½ + M ⅔ - Treatment of shock.
  2nd day = ⅓ M + ½ D.
- When blood glucose reaches 250mg/dl add glucose 5% to normal saline to avoid hypoglycemia ($^{1}/_{5}$ dextrose saline).

2. Insulin therapy:

- Insulin infusion is started, adjust dose according to blood glucose, blood glucose should be reduced gradually.
- Insulin infusion 1st without bolus dose.
- ↓ insulin infusion rate when:
  - Patient become conscious.
  - Blood glucose drop below 300 mg/dl to avoid hypoglycemia.
- Shift to S/C insulin (0.2-0.4) U/kg every

  6-8 hours when:
  - PH > 7.30 .
  - HCO_3 > 16 mEq/L.
  - Sodium is stable b/w 135-145 mEq/L.
  - No vomiting.

3. Potassium:

- The initial plasma K^+ is high, but it will fall after treatment with insuli and rehydration.
- K^+ replacement should be started when urine is passed.
- Continous cardiac monitoring and regular K^+ concentration are done until K^+ is stable.

4. Metabolic acidosis:

- It will be self corrected with fluid and insulin therapy "body can use glucose → ↓ Ketones.
- Bicarbonate is given only when there is sever acidosis.
  - PH < 7 or if patient is shocked and not responding to therapy.
  - Symptomatic hyperkalemia.

When patient managed for DKA:
Flow sheet :
- Electrolyte.
- Blood sugar.
- PH "ABG".
- Fluid balance every 1-4 hours.
- Vital signs.

Stop insulin infusion, start S/C insulin give oral fluid and normal diet.
If the cause is infection → give antibiotic.

Long term managemet of insulin dependent DM:

- Newly diagnosis cases better hospitilized to adjust insulin dose and educate the child and parent's the principle of D.M management.
- Child could be managed at home if lives near hospital, parent can use glucometer and give insulin, when child is stable.
- Offer education about insulin, montoring of glycemia control, diet, excersize, illness and hypoglycemia.
- **Aim**: To keep normal glyemia control, screening for associated diseases "Thyroid and celiac disease".

Insulin therapy:
1. Soluble "short acting" insulin :
   - Lispro : Rapid acting or Humulin-R.
   - Regular "Actrapid".
2. Intramediated acting :
   - Monotard, Insulated "NPH", Humulin N.
3. Long acting :
   - Ultralente.
   - Glargine "steady delivery over 24 hr".
4. Mixtard :
   - Short acting + intermediate acting.
   - Usually divided to 70/30 "8 unite mixtard contain 2 soluble, 6 NPH".

Type of insulin	Onset	Peak action	Duration
Very short acting (Lispro - Aspart)	10-20 minute	30-90 minute	3 hours
Short acting (Regular)	30 minute-hour	2- 4 hours	6-10 hours
Intermediate acting	1-2 hours	4-12 hours	16-24 hours

(NPH)			
Long acting (Ultralente - Glargine)	4-6 hours	8-20 hours	24-36 hours

Insulin regimens :

1. **Two injection regimen**: (Twice dialy)
   - Morning "before breakfast" : ⅔ (⅓ regular , ⅔ NPH).
   - Evening "before evening meal" : ⅓ (⅓ regular , ⅔ NPH).
2. **Multiple dialy Injection** "Intensive isulin therapy":
   - Premeal regular insulin.
   - NPH given in the morning and before bed time.
3. **Three injection regimen**:
   - NPH ad rapid acting analogue bolus at breakfast.
   - Rapid acting analog bolus at supper.
   - NPH at bed time.

Dose adjustment:

- There is no fixed for insulin.
- The dose is adjusted according to repeated blood glucose estimation.
- Glucose level in the blood should be kept between:
  - 80-100 mg% Fasting
  - 120-140 mg% postprandial.

Monitoring of glycaemic control:

★**Short term**:

- Frequent self monitoring of blood glucose, adjust insulin dose according to blood glucose measurement.
- Blood glucose should be measured more than 4 times/day during any illness, because the patient may vomit and take insulin without ~~eating → hypoglycemia or patient~~ may nt take inslin → DKA.

Ongoing care and follow up:

- **Measure HbA1c** : 2-4 times/year, current level should be available for use in follow up clinic.
- **Check injection sites** : Periumbilical, arms, thighs, advise to change site of injection "it becomes resistant to insulin by time".
- **Monitoing of growth :** Ht, Wt, BMI.
- Check for retinopathy.
- Check for diabetic nephropathy

Diat control:

✓ It should be matched to the insulin regimen :

- On the twice dialy regimen: Food intake is divided into 3 main meals with snacks between the meals to prevent hypoglycemia.
- On the flexible regimen: Peteint can eat more flexibly, snaks not needed "only if planning to do exercize" .

✓ A healthy diet is recommended:

- High in complex CHO and fiber "they give sustained release of glucose".
- Refined CHO cause rapid increase in blood glucose followed by a fast drop.
- The diet shoul be low in fat content → CHO sould provide :
  - 55% of calories.
  - 35% of fat.
  - 15% of protein.

N.B:

Dialy CHO reuirement =
100gm + (age in years × 10)

Somogyi phenomena :

Early morning hyperglycemia

- It is due to hypoglycemia during the night "as a result of ↑ dose of insulin" leads to ↑ secrection of GH "antagonizes" leading to hyperglycemia in the morning.
- Advice : ↓ insulin dose "evening dose".

Honeymoon

- In newly diagnosed cases of DM insulin requirement are small initially due to continued secretion of small amounts of endogenous insulin.
- This period is transient "may extend to 10 years".
- Pateint is adviced not to stop insulin "range of insulin dose during this period is

Dawn phenomena :

Early morning hyperglycemia

- It is due to giving the evening dose of intermediate insulin early in the evening and the normal ↑ in GH secretion in the morning → ↓ effect of insulin leading to hyperglycemia.
- Advice : ↑ evening dose of insulin , or give at bed time.

Diabetes Insipidus

- Pass of tastless urine because low Na+ content "inability to produce concentrated urine" due to either:

1. **Neurogenic DI** : Deacrease ADH production "vasopressin".

2. **Nephrogenic DI :** Lack of response of renal tubules of ADH (AR, AD, X-linked).
   Age : AR < 1 year of age.
   AD > 1 year of age.

Clinically

- During infancy poor feeding.
- Constipation.
- FTT.
- Irritability "due to polydipsia, wet diapers".
- Nocturia 2ry.
- Fever without focus "no sweating".
- Sever dehydration.
- Polyuria "urine output 4-10 liter/day".
- May be features of the cause e.g. ↑ICP in craniopharyngioma.

Causes :

1. Central causes:
- Idiopathic.
- Trauma.
- 2ry to I.C.Hge.
- Cerebral hypoxia.

2. Nephrogenic:
↓ of tubular sensitivity
- CRF.
- Nephritis.
- Reflux nephropathy.
- Congenital renal

Investigation :

- Urine :
  - Specific gravity : 1002-1005 "Diluted".
  - Osmolarity low: 50-200 m.osmol/L.
- Plasma osmolarity :
  - High > 295 m.osmol/L.
    - High serum Na^+ > 170 mmol/L and low K^+.
    - High osmolarity > 300 m.osmol/L.
- Water deprivation test : (Not recommended)
  - Vasopression "synthetic ADH" stimulation test : to differentiate between nephrogenic and neurogenic.

Treatment :

Neurogeni ⇒ Desmopression intranasal twice dialy

Nephrogeni ⇒
- Adequate hydration.
- **Correct hypokalemia by :**
  - Oral K^+.
  - K+ sparing diuretic.
- **Indomethacine :** stimulate receptor in nephrogenic.

N.B

Diabetes Insipidus may be 2ry to :

- Sickle cell disease.
- Phynitoin , Amphotricin.
- Sarcoidosis.
- Hypokalemia, Hypercalcemia.

D.D of Polyuria and Polydipsia :-

Polyuria: Passagee of excessive urine output > 2 liter/m^2.

Causes:

1. **Endocrinal**: DM , DI.
2. **Renal**:
   - Hypokalemia < 2.5 meq/L.
   - Renal tubular acidosis.
   - Bather syndrome.
   - Hypercalcemia > 13 mg/dl.
   - CRF.
3. **Psychogenic:** Compulsory water ingestion.

Feeding of infant by Dr.Asma Hadida

2015_2016

Feeding of infant

⁂ Weaning

Danger of starting complementary feeding before 6m :

1. Reduced production of breast milk
2. Exposure to the infectious agent present in food and fluid .
3. Increase the risk of food allergy e.g cow milk allergy .

✓ ***Principle of weaning*** :

1. Gradual to prevent GI disturbance, sudden weaning may also causes a psychological trauma to the young .
2. Should not start in summer b/c risk of GE .
3. Not start if the infant is under wt to avoid GI disturbances.

✓ ***Introduction of semi-solid food with breast feeding Exclusive breast feeding is recommended for 6m of age*** .

- **Aim of weaning and mixed feeding** :
  1. The child cannot tolerate more than 1 liter of fluid per day . so . with his increasing needs for energy more solid foods should be added.
  2. With growth , the baby require more foods items , minerals and vitamins , which cannot be supplied by milk alone .
  3. To train the GIT to digest starch and other solid foods .
  4. To educate the child independence by using spoon and cup .
  5. To train the child on spoon feeding, chewing and swallowing of solids .

Technique of weaning :

1. It should at age of 4-6m by replacing one milk feed by any foreign food .
2. Continued gradually by replacing other milk feed by foreign food until all feeds are replaced at age 1-2 y .
3. New food is given gradually and in small amount at first, if accepted by the baby the amount can be increased , never force the infant to take a new food .
4. The presentation of food is important, so better use colorful attractive spoon and plates to stimulate baby's interest and appetite .
5. New food items should be introduced one at a time

When to introduce different new food items

1. **4-6m** : cereal meals as rice pudding or mehalabia .
2. **5-7m** : vegetable soup "potatoes , carrots , squash", soft cheese.
3. **6-8m** : chicken ,liver , fish , rabbit , egg ,yolk
4. **9-12m** : custard , cheese , beans , breads , biscuits , fruits.
5. **1y :** whole egg , whole milk , red meat, cooked rice .
6. **1.5 -2y :** ordinary family diet.

Food to be avoided included :

1. That cause chocking as nuts , fruits with seeds ,
2. Highly spiced food .
3. Food additives

Number of feeds per day :

1. 1st 6m of life ⇨ 6 meals per day
2. Over 6 m of life ⇨ 4 meals per day +B. feeding .
3. 2-3y old ⇨ 3 feeds /day + 2 small snack " light meals ".

Formula feeding

Indications :

1. All C/I of breast feeding .
2. Insufficient B, feeding .
3. Twins .
4. Metabolic diseases .e.g. galactosemia, PKU

Types of formula feeding :

1. **Substitutive :** formula fed only .
2. Both breast and formula can be given together :
   a. **Complementary :** mainly breast fed .
   b. **Supplementary :** mainly bottle .

Humanized milk : modification of cow's milk :

1. **Protein :** casein is decreased and replaced by soluble whey protein to mimic composition of human milk .
2. **Fat :** milk fat is removed and replaced by a mixture of milk and vegetable fat to increase its unsaturated F.A .
3. **CHO :**like human milk , only major CHO is lactose .
4. Amount of mineral decrease .
5. Iron and vitamins are added .

Modified special formulas :

1. Lactose free formula e.g isomil given in lactose intolerance and galactosmia
2. Phyenyl alanine free formula for PKU
3. Special formula for neonate and low birth wt .
4. Low ca milk for hypercalcemia

:

How to prepare the formula **:**

1. The bottle and teat used for feeding should be clean and sterilized by boiling 5_10min .
2. Amount of water is powered in the bottle .
3. When the temperature of water become suitable add scoop of artificial milk :
   a. **American** :1scoop for each 60ml of water .
   b. **European milk :** 1 scoop for each 30ml of water .
4. Close the bottle and shake well .

Breast feeding

Advantages

1. ***For the mother :***
   a. Easy need no preparation .
   b. Cheapest food .
   c. Available at any time .
   d. Help involution of uterus .
   e. Decrease breast ca incidence .

2. ***For the baby :***
   a. Self-regulated .
   b. Always fresh .
   c. Proper temperature .
   d. Easy digested and absorbed .
   e. Protection against infection .
   f. Anti-allergic properties .
   g. High lactose provide galactolipid and essential poly unsaturated FA which essential for brain growth .
   h. High content of cholesterol helps myelination of NS and protect against atherosclerosis in later life .
   i. Most of protein are easily digested whey protein and the lipase present can digest 90%of the milk triglyceride .
   j. Rick's incidence decrease .
   k. Iron def. anemia less likely due to better absorption and no iron loss .

☑ **Anti infective properties** :

1. **Humeral :** AB secretory A , IgG ,E,D,
2. **Lysozyme :** bacteriolytic against enter bacteria and G +ve .
3. **Lactoperoxidase :** bacteria killing enzyme .
4. **Lctoferrin** : iron binding protein " bacteriostatic " by depriving the bacteria from the iron necessary for their growth and effective against candida .
5. **Anti-staph** : in lipid fraction .
6. Interferon .
7. **Bifidus factor** : mucopolysacchrides facilitate the growth of lactobacillus bifidus which form 99.8% of bacterial flora of the intestine .
8. **Cellular factors :**
   a. MQ :kill bacteria and fungi , they secrete lysozyme and lactoferrin and complement .
   b. Lympthocyte : B cell and Tcell

Technique of breast feeding :

1. **Good positioning** "close to mother, turned toward the mother , the whole body is well supported, neck straight or bent slightly back .
2. **Good attachment :** infant s chin is touching the breast and mouth is widely opened .lower lip is turned outward.

Criteria of effective suckling :

a. Infant sucks with slow deep sucks and sometimes pauses.
b. Cheeks are round "not tense or pulled in ".
c. Sounds of swallowing are heard.

Continue to feed until baby release nipple spontaneously, baby is then held vertically and tapped gently on back to drive off any swallowing air.

Baby is then placed in bed either lying on its rt. Side or prone.

- Give no food or drink: also no artificial teats or pacifiers.
- Baby feed on demand.

What is colostrum?

Secretion produced by the breast in 1st 3-5days.
It is lemon yellow viscous fluid more alkaline than milk and higher specific gravity .
It is followed by **traditional milk** then mature milk which is secreted on the 10th -20th day after delivery .

- It contain more protein , minerals , and fat soluble vit . and less fat and CHOthan mature milk .
- Rich in Igesp .IGA .
- More cholesterol than mature milk .
- More rich in macro minerals "Na ,K, CL, " and micro minerals "zinc _copper " .
- Rich in MQ , lymphocyte , antibacterial factors "lactoferrin and lysozyme " .
- More in vit .A than mature milk .
- Laxative action help in passage of meconium .

Minerals , vitamins and water supplementation :

- Exclusively breast fed infants do not require iron supplementation for the 1st 6m .30 mg vit c , 400IU vit .D can be given .

Previously boiled water can be offered at any time .

Fat content of last amount of milk in each feed "**hind milk** " is 3 times that in 1st drops "**fore milk** ", this helps in achieving satiety and thus preventing overfeeding and obesity in breast fed infant .

N.B

- ***Criteria of adequacy of breast feeding :***

1. Baby feeds at least **8times /day**, day and night .
2. Infant is calm and satisfied after feeds.
3. Normal motion "stools" with no constipation stool of breast fed 'semi formed , yellow in color , soft , acidic in reaction and frequent "each feed may be followed by passage of stool "
4. Normal amount of urine , 6times /day
5. Normal wt. Gain **"20-30gram /day** ".

- ***Under feeding in breast fed infant criteria***
  - Always crying - during and after feed with suckling of fist between feeds .
  - Stay on breast for long time more than 30 min .
  - Sleeplessness or very short sleep .
  - Constipation .
  - Air swallowing leads to colic , vomiting and may be hunger diarrhea " frequent , frothy , green , stools , small amounts "
  - Decrease amount of urine .
  - Decrease wt gain.

- ***Overfeeding criteria :***
  - Regurge is frequent.
  - Vomiting and colic .
  - Large bulky stools containing undigested curds.
  - Abd. Distention .
  - Polyuria .
  - Excessive sweating .
  - Wt. "plump and over wt ".

C/I of breast feeding :

a. **Absolute C/I** :

- Inborn errors of metabolism "galagtosemia , PKU ".
- Inborn errors of digestion e.g lactose intolerance .
- Maternal hepatitis .
- Maternal HIV ."controversial".
- Intake of dangerous toxic drugs e.g anticoagulant.

b. **Temporary C/I :**

- Sever cleft palate, micrognatia.
- Oral HS to infant.
- Maternal infection :
  - HS on breast, chicken pox: zoster IG. Given to non-infected neonate.
  - Active T.B: INH is given io infant.
  - Breast abscess.
  - Septicemia, typhoid fever,.
- Underdeveloped breast and severely inverted nipple.
- Pregnancy not C/I. But b/c mother cannot afford combined nutritional demands.

genetic by Dr .Asma Hadida

2015_2016

Introduction

Basic of genetic :

- Chromosome structure :
  - Each chromosome is composed of 2 chromatids .
  - The 2 chromatides are connected at centromere
  - Each chromosome has 2 short arms and 2 long arms
  - Each chromatides is composed of DNA in protein framework.

- **Chromosome number :**
  - In somatic cells : 46 chromosome (diploid number)
    44 autosome & 2 sex chromosome (XX in female & XY in male)
  - In germ cells : 23chromosome (haploid number)
    22 autosome &one sex chromosome (X in ovum &Xor Y in sperm)

➲ **Structure of the gene** :

1. Gene : part of DNA that code for synthesis of single polypeptide chain.
2. Every trait (character or feature) is determined by 2 gene , one from each parent :
   - If both genes are are similar (homozygous e.g AAor aa)
   - If both genes are different (heterozygous e.g Aa)
3. Dominant gene : expresses itself whether in homozygous or heterozygous
4. Recessive gene : express itself only when homozygous

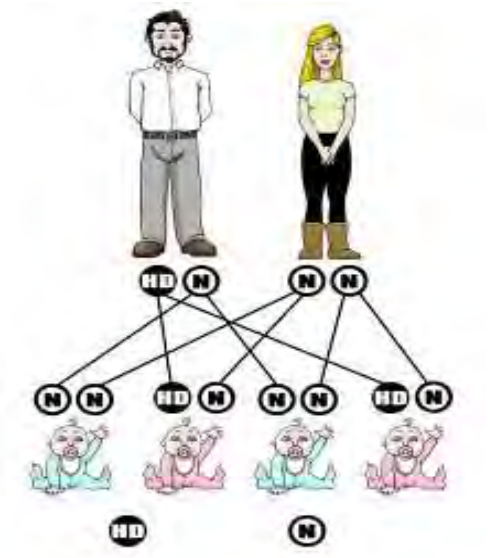
Mode of inheritance

A. Mendelian inheritance :

1. **Autosomal dominant (AD)** :

Criteria :

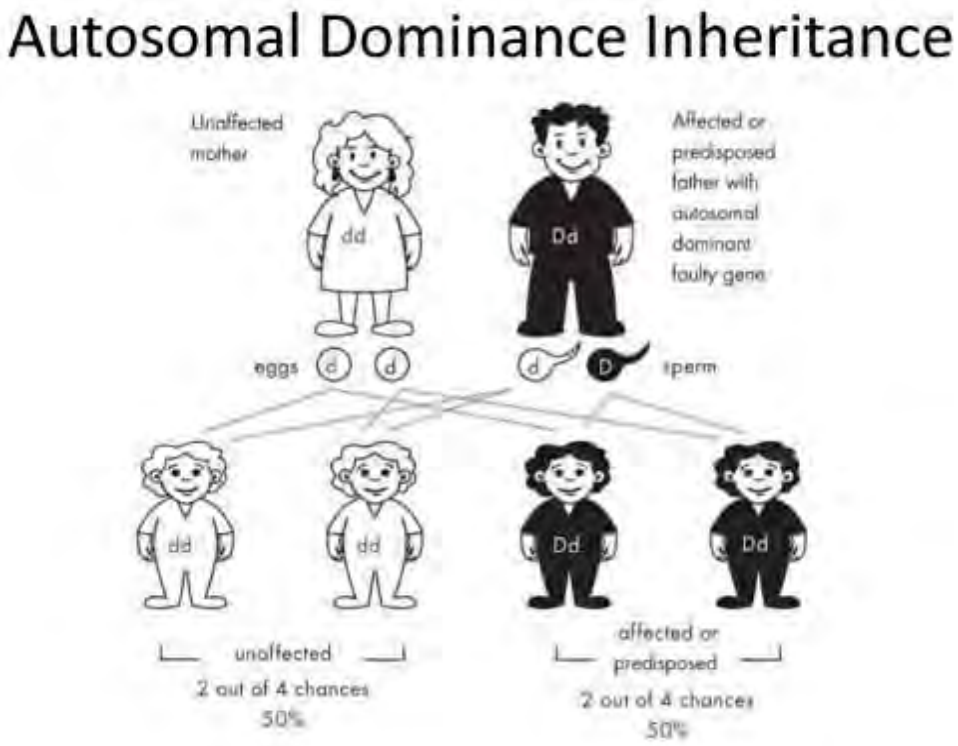
- The triat expressed in homozygous or heterozygous
- Affected person has an affected parent
- Disease occur in all generation
- New mutation is common

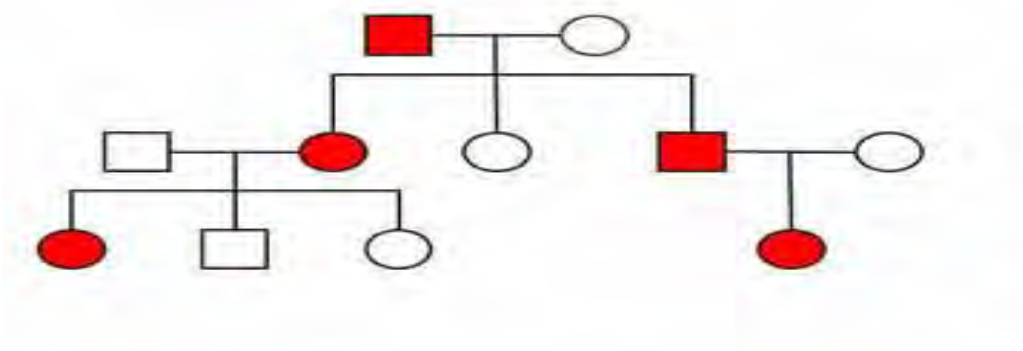


BB normal , Aa diseased : 50% normal and
50% diseased

Examples :

1. Neurofibromatosis.
2. Tubersclaerosis .
3. Herditory sherocytosis .
4. Acodroplasia .
5. Crigler najjar syndrome II, Gilbert .
6. Von well brand disease .
7. Myotonia dystrophica .
8. Noonan syndrome .





2. ***Autosomal recessive (AR) :***

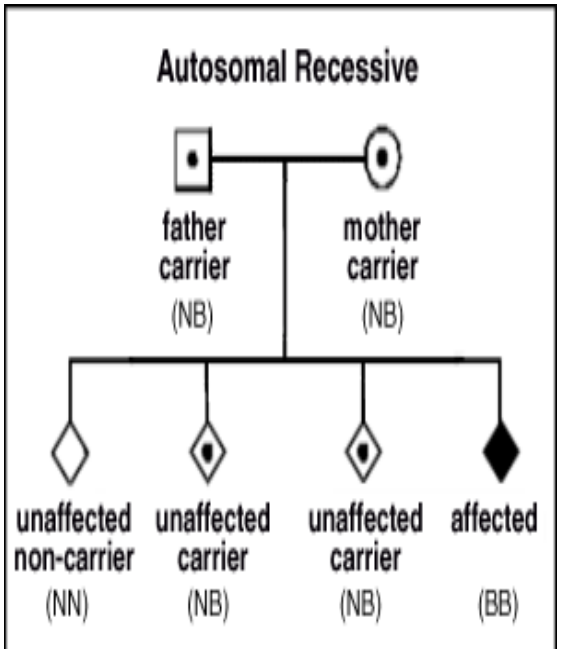
autosomal = not related to sex, recessive = carrier

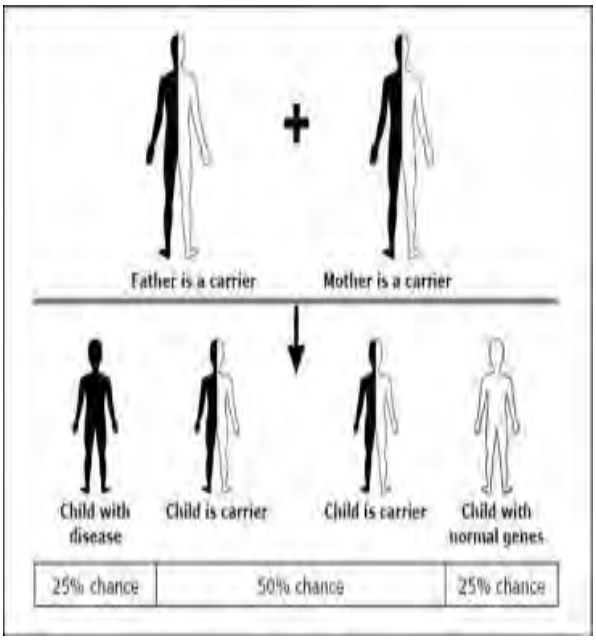
Criteria :

- The trait manifests only in homozygous state
- Both parents are carriers (consanguius marriage increase the incidence)
- Offspring : 25% free, 25% affected , 50% are carriers

Examples :

1. Cystic fibrosis
2. Cong. Adrenal hyperplasia
3. Galactosemia
4. PKU
5. Werding Hoffman disease
6. Crigler najjar syndrome I
7. Fridreich ataxia , Wilson disease
8. Glycogen storage disease , mucopolysacchridosis

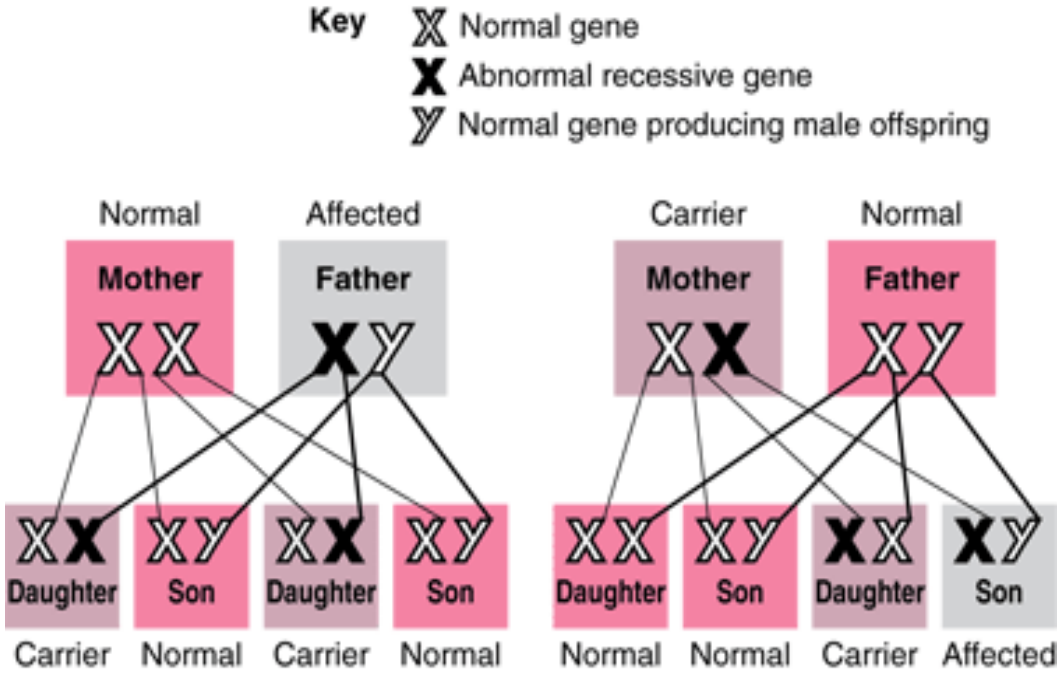


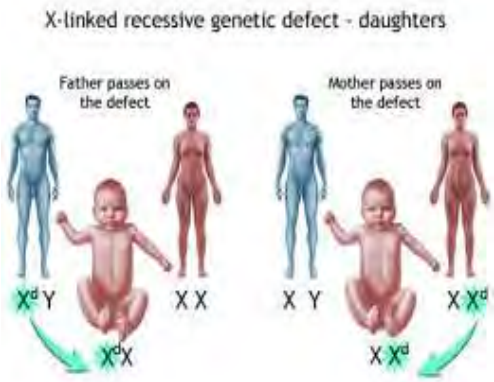


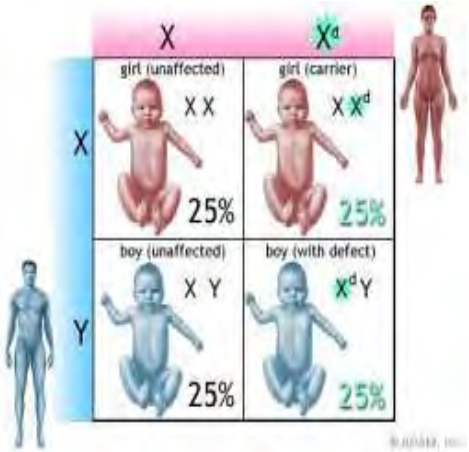
3. **Sex – linked recessive** :

Criteria :

1. Female carriers have : 1/2 of her sons affected &1/2of her daurther carrier.
2. Affected father have all his dauther carriers but there is no father to son transmission







Example :

1. G6PD deficiency.
2. Haemophilia A&B.
3. Hunter syndrome.
4. Ducchane muscular dystrophy.
5. Fragile X syndrome , rett syndrome.
6. Ocular albinism &colour blindness.

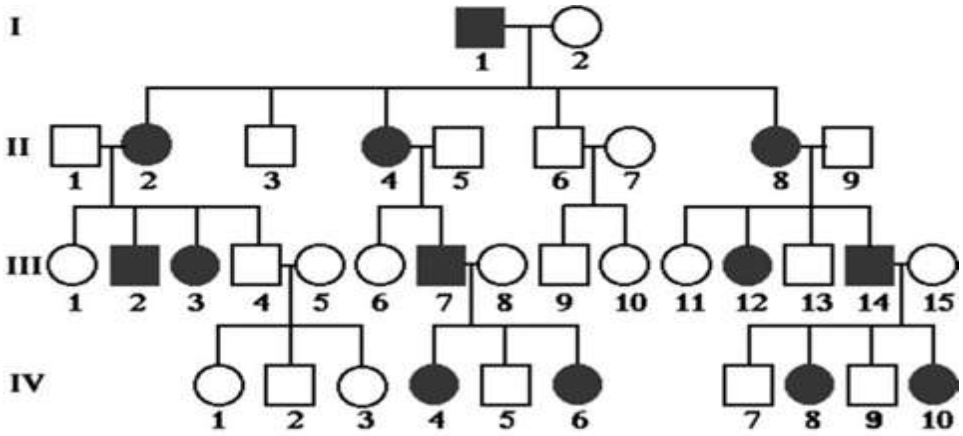
4. sex linked dominant (XD):

Criteria :

1. Affected father transmit the disease to all daughters but never to his sons .
2. Affected mother transmit the disease to 1/2 of her offspring whether male or female.

   examples :

   1. Familial hypophosphatamia.
   2. Alport syndrome.
   3. Prone belly syndrome.



B. Non mendelian modes of inheritance :

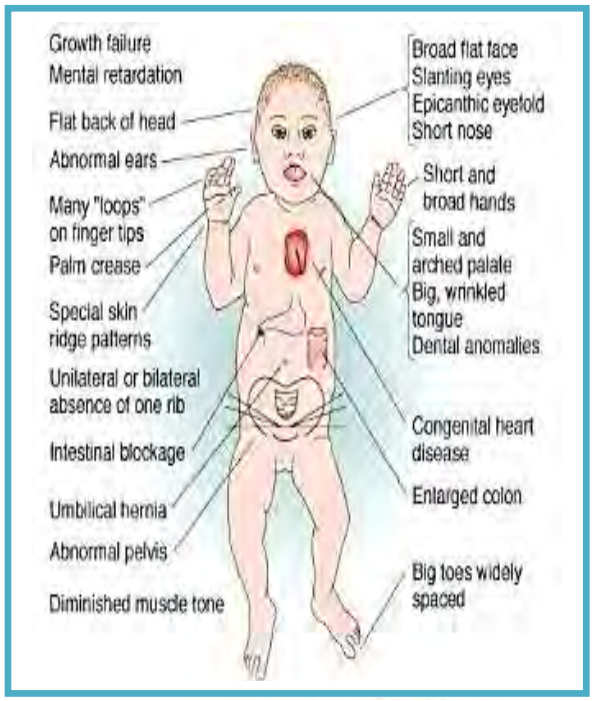
a. Mutifactorial : e.g cleft lip and palate , HPS

b. Non traditional mode of inheritance :

Mithchondrial DNA mutation : e.g : cardiomypathy , deafness .

Down syndrome

Down syndrome is the commonest chromosomal abnormalities , the extrachromosomes usually **maternal** , the incidence increase with maternal age 2% at age of 40y, ***causing*** learning disability , mild to moderate developmental delay .
The incidence (without antenatal screening) in live-born infants is about 1 in 650.



- ***Mild microcephaly "brachycephaly"***: small short anteroposteriolly .
- Wide ant. and post. Fontanelle.
- Fine silky hair
- Low set ear, small nose and depressed nasal bridge .
- Small mouth, protruding fissuring "**scrotal tongue in child more than 6 years "**.
- Hypertelorism, epicantal fold, upward slant of eyes, brush field spot "**speckled iris".**
- Brachydactly "***short fingers***", **simian creases**, broad hand, clindactyly "***curved* little finger**".
- Big space b/w first and second toes "**sandle gap**"

C/P:

- ✓ Delayed mental milestone **"MR"**
- ✓ Delayed motor milestone "**hypotonic**": hyper flexible joint "**acrobat sign** "

- ✓ ***Heart:*** CHD in 50% of cases, endocardial Cushing defect "*AV-canal, ASD and VSD*".
- ✓ ***Abdomen:*** distended with umbilical hernia, constipation , visceroptosis "
- ✓ ***Genitalia:*** small sized **"hypogonadism"**, undescended testis is frequent .

Complication :

- Immunodeficiency: recurrent infection "chest, serous OM "
- Neurological: MR "**accidental trauma**", atlanto_axial instability with risk of spinal cord injury, autism spectrum disorders, early Alzheimer, strabismus, cataract, nystagmus .
- Cardiac: CHD **"recurrent HF and chest infection** "
- Respiratory: recurrent chest infection, obstructive sleep apnea.
- Renal anomalies.
- Hematological: acute leukemia "AML" witches 20 times more common.
- Autoimmune endocinopathies: hypothyroidism, DM, Addison disease.
- GIT: anomalies (duodenal Artesia, Hirshsprung disease, imperforated anus), celiac disease .

Characteristic clinical manifestations of Down syndrome

Typical craniofacial appearance

- Round face and flat nasal bridge
- Upslanted palpebral fissures
- Epicanthic folds (a fold of skin running across the inner edge of the palpebral fissure)
- Brushfield spots in iris (pigmented spots)
- Small mouth and protruding tongue
- Small ears
- Flat occiput and third fontanelle

Other anomalies

- Short neck
- Single palmar creases, incurved fifth finger and wide 'sandal' gap between toes
- Hypotonia
- Congenital heart defects (40%)
- Duodenal atresia
- Hirschsprung disease

Later medical problems

- Delayed motor milestones
- Moderate to severe learning difficulties
- Small stature
- Increased susceptibility to infections
- Hearing impairment from secretory otitis media
- Visual impairment from cataracts, squints, myopia
- Increased risk of leukaemia and solid tumours
- Risk of atlanto-axial instability
- Increased risk of hypothyroidism and coeliac disease
- Epilepsy
- Alzheimer's disease.

Investigation :

a.. perinatal diagnosis :

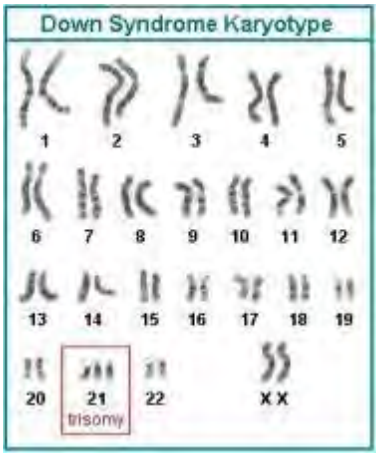
1. *integrated screening* :can detect up to 95% of down syndrome pregnancies using :
   - Maternal age .
   - Increase fetal nuchal translucency thickness.
   - Decrease pregnancy ass. plasma protein A .
   - ***Triple test "sample from maternal "shows*** :
     - Decrease alfafetoprotein .
     - Decrease unconjugated estriol .
     - Increase free B-HCG .
2. karyotyping for maternal amniotic fluid cells.

b. post natal diagnosis :

1. clinical .
2. ***karyotyping*** :
   A. **For the baby** :to confirm down syndrome and detect the type and risk of reccurence .
   B. **For the parents** :if the parents if the baby translocation type .

Types of down syndrome

1. ***Full trisomy 21 "regular mogol":*** 95% non disjunction "extra 21", 47xx or xy . no role for parental karyotyping, recurrence low 1%
2. ***Balanced translocation 4%:*** 14/21, 46xx or xy , higher risk of recurrence 1/10 .
   Half of translocated Mongols are hereditary , mostly maternal and other half is new .
3. ***Mosaic's 1%:*** mixed normal and abnormal cells karyotype "**less sever intellectual ability** " may not show all features of mongolism .



Health supervision of down syndrome:

✓ **Multidisplinary** care approach is mainstay for management .

✓ Screen for and manage complications .

1) **CHD**: screen at birth and young adult for acquired valve diseases.
2) **strabismus cataract and nystagmus**: screen at birth or by 6m, check vision annually .
3) **hearing impairment or loss**: screen at birth or by 3m with auditory brain stem response or otoacoustic emission testing , check hearing 6m -1y .
4) **celiac disease** :at 2y or with symptoms "**ATTG AB**".
5) **hypothyrodism** :at birth , repeat at 6m -12m and annually.
6) **obstructive sleep apnea** : start at 1y and at each visit , monitor for snoring , restless sleep .
7) **atlantoaxialsublaxation** :10-30% : radiograph at 3-5y or when planning to participate in contact sports ,diving , swimming , radiograph indicated wherever neurological symptoms are present even if transient "neck pain , torticals , gaitdisturbance , weakness"

How to calculate IQ ?

IQ = Mental age / Chorological age

- ⇨ 70-75% Border line
- ⇨ 50-70% Mild learning difficulty
- ⇨ 35-50% Moderate "
- ⇨ 20-35% Sever "
- ⇨ < 20% Profound "total care "

Edward syndrome (Trisomy 18)

- Incidence : it affects 1:6000 live birth , 2nd most common after down syndrome
- Karyoryping : (47XX, +18) or (47XY, +18)
- ***Clinical picture*** :
  1. LBWT or IUGR
  2. Head &face ⇨ microcephaly , prominent occipit , low set ear , hypertolirism , micrognatia & cleft lip and palate
  3. Exterimities ⇨closed fist with overlapping fingers , rocker bottom heal , clindactyle , hypoplastic nail
  4. Chest : CHD ⇨VSD, PDA, ASD.
  5. Abdomen : diaphragmatic hernia , hernia , renal anomalies ,
  6. General : sever developmental delay , preterm birth
  7. Major cause of death : apnea and cardiac anomaly .

N.B:

Edward syndrome has very low survival rate only 50% of patient live up to 2 months & only 5% of patient live up to 1 year .

Patau syndrome (Trisomy 13)

Incidence : 1: 10000

- Karyotyping (47XX+13)or (47XY+13)
- ***Clinical picture*** :
  1. **Head & face** : microcephaly , scalp defect , brain malformation , cleft lip and palate , capillary heamangioma , deafness.
  2. **Chest :** CHD(80%) VSD, PDA, ASD, thin posterior ribs (missing ribs)
  3. Exterimities : polydactyle , overlapping fingers & toeas , clindactyle , hypoplastic hyperconvex nails
  4. **General** : MR, sever developmental delay , growth retardation , renal anomaly

Edwards syndome and Patau syndrome

Clinical features of Edwards syndrome (trisomy 18)

- Low birthweight
- Prominent occiput
- Small mouth and chin
- Short sternum
- Flexed, overlapping fingers
- 'Rocker-bottom' feet
- Cardiac and renal malformations.

Overlapping of the fingers in Edwards syndrome.

Clinical features of Patau syndrome (trisomy 13)

- Structural defect of brain
- Scalp defects
- Small eyes (microphthalmia) and other eye defects
- Cleft lip and palate
- Polydactyly
- Cardiac and renal malformations.

Turner syndrome (45X0)

Incidence : 1:5000

- **Etiology :**
  1. Classical form 45XO.
  2. Deletion of short arm of one X chromosome.
  3. Turner mosaic 45XO , 46 XX.
- ***Clinical pictures:***

➲ At birth :

1. Transient lymphoedema in the dorsum of hand.
2. Low birth weight.
3. Loose skin, extra skin fold.

➲ Later on :

- **Head and face**:
  1. Ugly short stature female with normal mentally.
  2. Low posterior hair line and low set ear.
  3. Webbing neck.
  4. Recurrent otitis media leading to hearing defect.
- **Chest**:
  1. CHD :
     - Coarctation of aorta, non stenotic bicuspid valve and aortic.
  2. Widely spaced nipple and broad chest.
- **Abdomin**:
  1. IBD (crohn's and ulcerative colitis).
  2. Celiac disease.
  3. GIT bleeding.
- **Genito urinary**:
  1. Renal anomalies >30% as horse shoe kidney.
  2. Ovarian dysgenesis (streak gonads) with failure to develop 2ry sexual character and infertility.
  3. Café aliat spot and cystic ygroma.
- **Extrimities** :
  1. Cubitus valgus (wide carrying angle) , short 4th metatarsal &metacarpal bone & spooned shaped nails.
- More liable to hypothyroidism (↑risk by 5 times)
- Delayed puberty
- Pregnancy in turner female is possible with IVF with donated ovum .
- **Mosaic turner** : may have only short stature with amenorrhea without dysmorphism.

- **Diagnosis :** Karyotyping.
- **Treatment** : GH , estrogen replacement at 14-15years.

Fragile X syndrome

Incidence : 1:5500

- **Etiology:** X-linked recessive.
- ***Clinical pictures:***
  - Mental retardation.
  - Developmental delay.
  - Elongated face with large everted ear.
  - Large testis.
  - Hyper extension of joints.
  - Mitral valve prolapsed.
  - Most children hyperactive and exhibit infantile Autism.

N.B:

➲ Fragile syndrome is the most common cause of mental retardation.

A child with fragile X syndrome. At this age, the main physical feature is often the prominent ears.

Clinical findings in males in fragile X syndrome

- Moderate–severe learning difficulty (IQ 20–80, mean 50)
- Macrocephaly
- Macro-orchidism – postpubertal
- Characteristic facies – long face, large everted ears, prominent mandible and broad forehead, most evident in affected adults
- Other features – mitral valve prolapse, joint laxity, scoliosis, autism, hyperactivity.

Syndromes recognised by 'Gestalt' (clinical recognition)

Noonan syndrome
affects males and females. There are some similarities to the phenotype in Turner syndrome, but it is caused by mutation in an autosomal dominant gene and the karyotype is normal.

Clinical features of Noonan syndrome

- Characteristic facies
- Occasional mild learning difficulties
- Short webbed neck with trident hair line
- Pectus excavatum
- Short stature
- Congenital heart disease (especially pulmonary stenosis, atrial septal defect).

Williams syndrome
is usually sporadic.

Clinical features of Williams syndrome

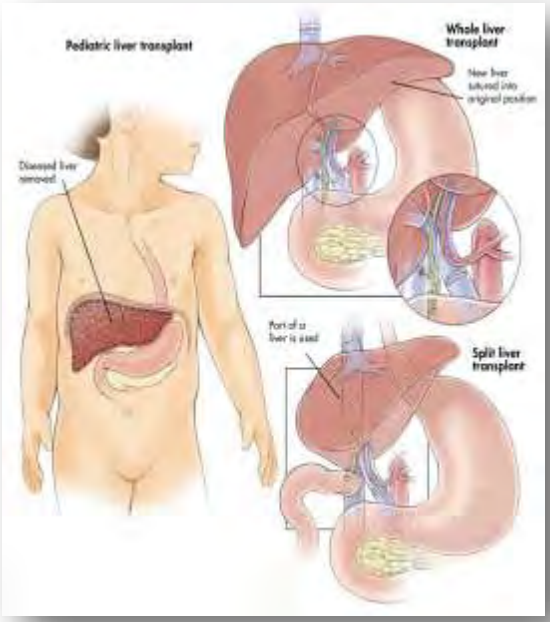
- Short stature
- Characteristic facies
- Transient neonatal hypercalcaemia (occasionally)
- Congenital heart disease (supravalvular aortic stenosis)
- Mild to moderate learning difficulties.

Prader–Willi syndrome.

Clinical features of Prader–Willi syndrome

- Characteristic facies
- Hypotonia
- Neonatal feeding difficulties
- Failure to thrive in infancy
- Obesity in later childhood
- Hypogonadism
- Developmental delay
- Learning difficulties.

GIT by Dr.Asma Hadida



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Symptoms and Signs of Digestive Tract Disorders

Vomiting

Forceful expulsion of food and secretions from esophagus or stomach through the mouth, it may be accompanied by nausea and forceful contraction of Abd. Muscles.

Causes

C. In the new born.
1. Physiological vomiting.
2. G.E.R "3rd – 10th days.
3. Cong.esophageal atresia.
4. Cong. Pyloric stenosis.
5. Intestinal obstruction.
6. Cerebral birth injuries.
7. I.C. Infection e.g., meningitis.
8. Imperforate anus.

B. In infancy.
1. Feeding difficulties.
2. Infections as O.M, pyelonephritis, pneumonia.
3. G.E.R.D.
4. Acute abd. Emergencies, e.g., intussusceptions.
5. **Cerebral causes:**
   a. Meningitis.
   b. Subdural Hge.
   c. Hydrocephalus

A. In older children.
1. **Acute infections:**
   - Food poisoning.
   - UTI.
   - Appendicitis.
2. **Drugs:**
   - Broad spectrum antibiotic.
   - Aspirin.
3. **CNS diseases:**
   - Meningitis, Encephalitis.
4. **Metabolic:**
   - A.R.F, D.K.A.
   - Adrenal failure.
   - Reye's syndrome.

Systemic disease that can cause persistent vomiting.
- Adrenal insufficiency.
- Metabolic disorder.
- CNS disorder with increase I.C.P.
- Hepatobiliary disorder.
- Renal disease.

Acidosis, vomiting, FTT, suggest endocrine cause or metabolic e.g,.
- D.K.A.
- Adrenal insufficiency.
- Adrenogenital syndrome.
- Glycogen storage disease.
- Galactosemia.

Diarrhea

The increase in the stool frequency and fluidity from the normal habit, passage of three or more watery or loose motion per day or single motion containing blood. excessive daily stool liquid volume (>10 mL stool/kg body weight/day)

Normal pattern of motion per day.

- **From birth to 4th months** → breast feed : 1-7 motions per day.
  → formula feed : 2-3 motions per day.
- **From 4th to end of 1st year** → 1 - 3 motions per day.
- **Above 1 year** → 1-2 motions per day "adult like".

note: Breast-fed infants may not pass stools for several days and be entirely healthy.

Classification of Diarrhea

A. According to Etiology:
- Infective diarrhea "**G.E**".
- Non-infective diarrhea.

B. According to the duration:
- **Acute diarrhea:** less than 2 weeks which starts acutely.
- **Persistent diarrhea:** start as acute diarrhea but persist more than 14 days.
- **Chronic diarrhea:** It is gradual onset lasting > 1 month or recurrent due to non-infectious cause.

C. Change in intestinal motility:
- **Increase motility** → decrease transient time e.g., thyrotoxicosis.
- **Decrease motility** → increased bacterial overgrowth e.g., stagnant loop syndrome.

D.According to Mechanism:

1. **Osmotic diarrhea:**
   - Due to presence of non-absorbable solutes in GIT → asmotic load → shift of water to intestinal lumen.
- **Examples:**
  - Lactase def. **"either primary or secondary to GE "lactose intolerance**.
  - Cong. Glucose - galactose mal absorption.
  - Ingestion of non-absorbable solutes e.g., lactulose, sorbitol.
- >volume: below 200 ml\day
- Effect of fasting: stopped
- Food type: related
- Stool ph: below 5
- Reducing substance: +\-
- Fecal Na and Cl: low

D. <u>According to Mechanism</u>:

2. <u>Secretory Diarrhea</u>:

<u>Due to either:</u>

i. Damaged absorptive villi cells with intact secretory crypt cells that migrate to cover the raw villi → excessive secretions and diminished absorption.

<u>Causes:</u>

- **Viral diarrhea** e.g., Rota Virus.
- **Bacterial** e.g., shigella, Entero invasive E.Coli.
- **Parasitic** e.g., Giardia Lamblia "induce mucosal adhesion.

ii. Entero toxins release → stimulate adenyle cyclase in crypt cells → excessive intestinal secretion.

<u>Causes:</u>

- Vibrio cholera.
- Entero toxigenic E.Coli.

3. <u>Decrease Intestinal Surface area</u>:

- **Example** : Short bowel syndrome, both osmatic & motility disorders.

- **secretory**:
- volume: above 200 ml\day
- effect of fasting: no effect
- food type: unrelated
- stool ph: above 6
- reducing substance: negative
- fecal Na and CL :high

4. <u>According to age</u>:

- **Toddler diarrhea: "Peas-carrot syndrome"** it is the commonest cause of persistent loose stool in pre-school age. It is chronic non specific diarrhea.

<u>C/P:</u>

- Stool consistency varies from loose to well formed stool occasionally and presence of undigested food with normal growth and physical examination. (3-6 times per/day, only at day time).
- It is suspected due to immature intestinal motility and disappear by 5 yrs of age.

<u>Diagnosis</u> by exclusion.

Acute Diarrhea

A. Acute non-effective diarrhea:

- **Dietetic error:**
  - Over feeding.
  - Under feeding.
  - Bad feeding: Change in milk type or concentration, new unsuitable food.
- **Drug Induced**:
  - Oral antibiotic.
  - Laxative.
- **Parental diarrhea:**
  - Diarrhea due to cause outside G.I.T.(otitis media , UTI)

B. Acute infective diarrhea "Gastro-enteritis":
Involve enteritis and reflex gastritis presented with vomiting diarrhea.

Severity

Mild = 4-6 motions/day.
Moderate = 6-10 motions/day.
Severe = > /10 motion/day.

Gastro-Enteritis

Risk Factor:

1. Failure of breast feeding.
2. Infant in age 6-24 months.
3. Malnutrition.
4. Impaired immunity.
5. Incomplete vaccination.
6. Unsanitary storing of food, food handling & disposal of feces.

Causes

Viral : 60%.

- Rota Virus.
- Adenovirus.
- Enterovirus e.g., Echo & Coxachie.

Criteria:

- Age usually < 2 years.
- Common in winter.
- May associated with URTI.
- Pyrexia if present usually < 38.5°C.

Diarrhea is mild to moderate, transient 5-7 days, watery & odorless.

Bacterial :
Criteria:

- Common in summer.
- High fever.
- Cramping abdomen pain.
- Usually severe diarrhea with may be :
  - **Bloody:**
    - With salmonella, shiegella, entero invasive strain and Entero hemorrhagic strain of E-Coli.
  - **Watery:**
    - With shiegella and entero toxogenic and entero pathogenic strain of E.Coli and vibro cholera.
    - **Watery Offensive** for 2-4 days then turn bloody e.g., campylobacter jejuni.

Protozoal:
G.Lambelia: Watery, Offensive, no fever or vomiting.
E.Histolytica: Bloody ± tenesmus, no fever or vomiting.
Cryptosporodium Parvum: Infect mainly the immune deficients, diarrhea is watery with vomiting & fever.

DD OF BLOODY DIARRHEA.

Infectious
- Shigella
- Salmonella
- E.Coli:
  Entero Hemorrhgic
  Entero Invasive
- Yersinia
- Campylobacter pylori
- E.Histolytica.
- Cryptosporidium

Non-infectious
- Intussception.
- Cow milk allergy.
- I.B.D.

Complication of G.E

1. **Dehydration:** The main cause of death in G.E.
2. Shock.
3. Acute Renal Failure.
4. Metabolic acidosis.
5. **Electrolyte disturbances:**
   - Hypokalemia.
   - Hypocalcemia.
   - Hypo & hyper natremia.
6. Convulsion.
7. **CNS infection:** Meningities & encephalitis due to shigella or neurotropic virus.
8. **Bleeding:** DIC, Hypoprothrombinemia.
9. Persistent diarrhea.
10. Malnutrition.

11. Others: Hepatitis.

Shock:

Hypovolemic & Septic

Clinically:

1. Hypotension & rapid thread pulse.
2. Decrease in vital organ perfusion.
   - Brain → Lethergic
   - Kidney → Oliguria.
3. Decrease Peripheral perfusion.
   - Paller.
   - Skin mottling.
   - Cold extremities.

Acute Renal Failure:

Due to :

1. Hypovolemia → decrease renal blood flow "pre-renal failure".
2. Untreated pre-renal failure → tubular necrosis lead to intrinsic renal failure.

Clinically:

- Oliguria, Acidotic breathing and Edema.

Metabolic Acidosis:

Due to :

- Loss of bicarbonate in stool.
- A.R.F.

Clinically:

- Acidotic breath.
- Disturbed consciousness.
- ABG lead to"↓pH, ↓ $PaCO_2$, ↓ HCO_3"

Hypokalemia: < 3 meq/l.

Due to :

- Loss of K^+ in stool in severe G.E.
- Aggravated by vigorous correction of acidosis.

Clinically:

- "Disturbed consc.".
- Cardiac arrhythmias.
- Abd. Distension "paralytic ileus".
- Hypotonia "atony".

Hypocalcemia:

Due to :

- Rapid correction of metabolic acidosis → Ca. Shift to bones and decrease ionized Ca. "post acidotic tetany".

Clinically:

- Tetany or convulsion.

Convulsions:

Cause of convulsion in case of GE:

i. Hypoglycemia.
ii. Febrile convulsions.
iii. Hyper & hypo-natremia.
iv. Toxic e.g., with shigella. "cause ↓ Na."
v. CNS infection e.g., meningitis.
vi. Post-acidotic hypocalcemia.
vii. Sinus thrombosis due to severe dehydration.
viii. Brain edema due to over hydration especially in hypernatremic dehydration.

Workup of GE

1. **For the cause :**
   - Stool analysis and culture.
2. **For the complications:**
   - Urea and electrolytes
   - Coagulation screen, pT, pTT, FDP, pLt.
   - ABG for MA.
   - R.F.T.
   - L.P.
   - Abd. X-ray.

Management of GE

Preventive:

1. Promote breast feeding.
2. Proper weaning practice.
3. Measles vaccines.
4. Hygienic measures e.g., protect water source from contamination.

Curative:

1. Supportive.
2. Specific therapy.
3. Additional therapy.

Supportive

A. G.E without dehydration:

Consisting of plenty of fluid, plenty of food and follow-up.

- **Fluid therapy:** Main aim of treatment is to avoid dehydration, the best fluid is ORS "Oral Rehydration Solution".
- **Food:** For infants >6 months or weaned, rice water, soup, yogurt, drinks.
- **Avoid:** Hyper osmolar fluid as it increases diarrhea.
- Continue breast feeding, potatoes, cereal, rice, bananas "supply K".
- Feeds given 6 times a day.
- Give extra meal each day for 2 weeks.
- **Follow-up:** Inform mother to seek medical consultation, if :
  - No improvement for 3 days.
  - Presence of warning signs:
    - High fever.
    - Refusal of oral fluid or feeding.
    - Frequent vomiting.
    - Marked thirst.
    - Bloody motion.

- Frequent watery motion.

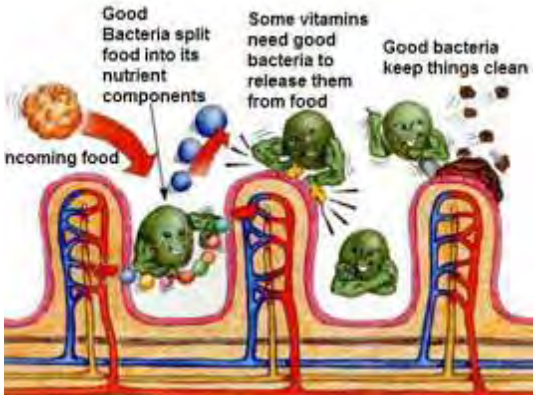
B. G.E. with dehydration: (see later)

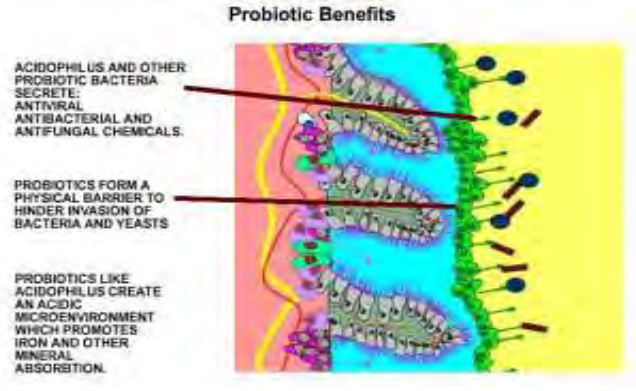
Specific therapy

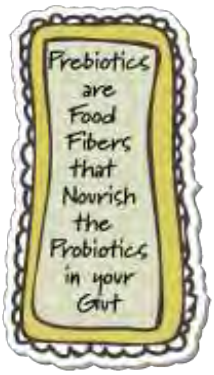
1. **Antibiotic: not indicated routinely in treatment**
   Indicated in :-
   - High grade fever.
   - Bloody diarrhea.
   - Immunocompromised patient.
   - Associated bacterial infection e.g., O.M, pneumonia.
2. **Anti-parasitic:** Metronidazole
3. **Treatment of complications:**
   - Acute renal failure (ARF): Respond to volume expansion.
   - Metabolic acidosis: If mild improve with adequate hydration with ORS, if severe, give $N_a HCO_3$.
   - **Hypocalcemia:** Ca. Gluconate.
   - **Hyponateremia & Hypokalemia**: Can respond to ORS.
   - **Hypoglycemia:** Give 20% glucose.
   - **Convulsion:** Treat the cause.

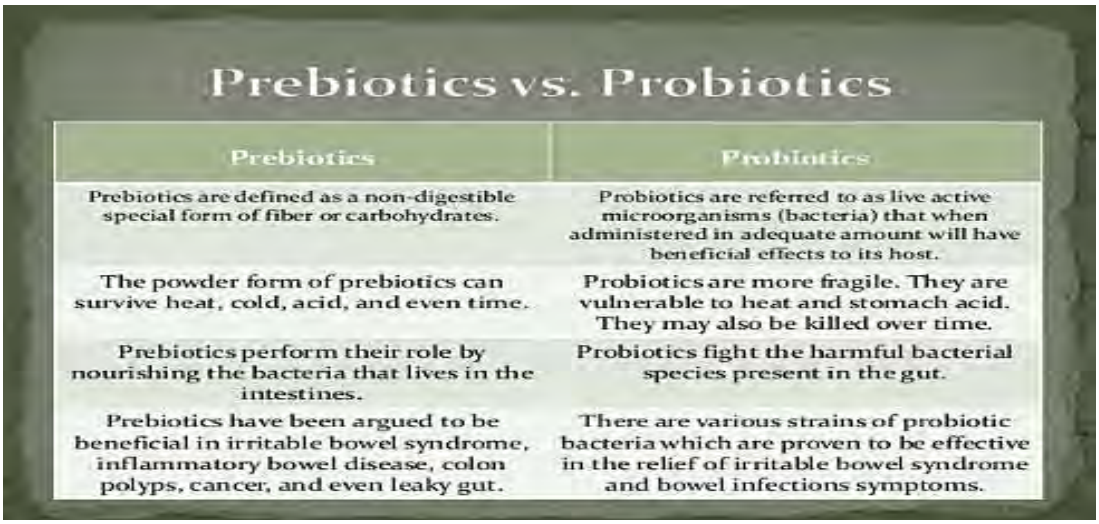
Additional therapy

1. **Pro-biotic:** Normal bacterial flora e.g., Lactobacillus & bifidobacillus.
   Action: Turn acidity of the gut and prevent adhesion of micro organisms, enhance immunity by combination with bacteria on binding sites.
2. **Pre-biotic:** "Oligosaccarides", enhancement of normal bacterial and increase immunity.
3. **Zinc sulphate:** Decrease period of diarrhea and prevents next attack, helpful for immunity, increase integrity of M.M of the Gut.









Persistent Diarrhea

Started as acute diarrhea "water or dysentery" but persist more than 14 days.

Etiology

1. **Persistent infection:** Gardia lamblia is commonest cause of persistent watery diarrhea.
2. **Post-Enterities Malabsorption:** Due to mucosal damage which leads to damaged villi with secondary digestive enzymes deficiency.
3. **Recent Introduction of animal milk or formula.**
4. **Impaired immunity.**

Diagnosis

1. **For post enteritis malabsorption:**
   - Symptoms resolve with suspected milk elimination.
   - Reducing substance in stool.
   - Stool pH < 5.
   - Small intestinal biopsy shows villous atrophy.
2. **For persistent infections:**
   - Stool analysis and cultures.

C/P

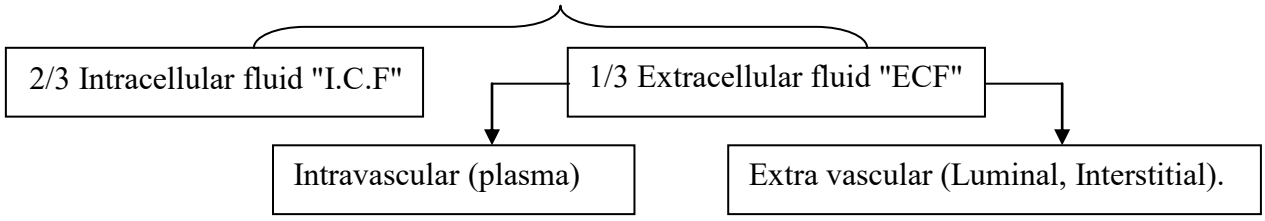
- Diarrhea & vomiting.
- Abdomen distension.\
- Perianal sortness due to watery acidic motions. (Excoriation mark).

Management

1. Adequate hydration.
2. Treat the cause.
3. Nutritional therapy.
   - Replace milk with fermented milk product such as yoghurt, increased breast feeding.
   - Give semi-solid food as rice.
   - Feeds given in frequent small meals at least 6 times daily.
   - If lactose intolerance is suspected, use lactose free milk for 1-2 weeks "till villi regeneration occur"

Dehydration

Body water is distributed as:



Definition: Dehydration means loss of water and electrolytes from ECF, the ICF may be secondarily affected.

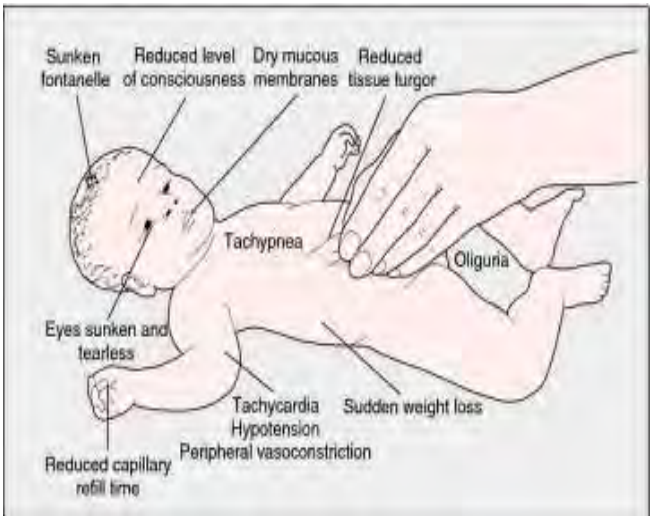
Etiology

1. **Diarrheal disease.**
2. **Decrease intake** e.g., starvation, coma.
3. **Vomiting** e.g., cong. pyloric stenosis, intestinal obstruction.
4. **Hyperventilation.**
5. **Excessive sweating.**
6. **Polyuria** e.g., DM, DI, CRF.

Assessment of Degree of Dehydration

	Mild	Moderate	Severe
Infant	5%	10%	15%
Adolescent	3%	6%	9%
Infants & young children	Thirsty, alert, restless	Thirsty, restless or lethargic, irritable	Drowsy, limp, cold, sweaty, cyanotic extremities; may be comatose.
Older children	Thirsty, alert.	Thirsty, alert (usually)	Usually conscious (but at reduced level), cold, sweaty, cyanotic extremities; wrinkled skin on fingers and toes, muscle cramps.
Tachycardia	Absent	Present	Present
Palpable	Present	Present (weak)	Decreased

pulses			
Blood pressure	Normal	Orthostatic hypotension.	Hypotension
Cutaneous perfusion	Normal	Normal	Reduced & mottled
Skin turgor	Normal	Slight reduction	Reduced
Fontanel	Normal	Slightly depressed	Sunken
Mucous membrane	Moist	Dry	Very dry
Tears	Present	Present or absent	Absent
Respiration	Normal	Deep, may be rapid	Deep & rapid.
Urine output	Normal	Oliguria	Anuria & severe oliguria



Key sign:

- General appearance
- Thirst
- Poor skin pinch
- One character (mild).
- If no tear + tachycardia "moderate"
- Poor pulse + cold extremities + severe ill (severe)

Hypotonic	Isotonic	Hypertonic
Serum Na <130 m eq/l.	130 – 150 meq/l	More than 150
Osmolality <275 m osml/l	275 – 295 m osml/l	More than 295
Excessive intake of water or hypotonic fluid during diarrhea	Balanced loss of N_a and Water	Excessive intake of hypertonic fluids during diarrhea
Net N_a^+ loss greater than water loss	No change of cellular hydration	Net water loss greater than Na loss
Over hydrated cells		Dehydrated cells
Marked collapse of ECF	Collapsed ECF	No collapse in ECF

	Hypotonic	Isotonic	Hypertonic
Manifestation of I.C.F affection			
Tongue	Moist	Dry, thirsty	Very dry "woody" thirst
Brain	Lethergic	Irritable	Convulsion, irritable, hyper reflexia
Manifestation of ECF affection			
Shock	Rapid	Slowly	Absent
Skin turgor	Marked loss	Moderate loss	Normal or doughy
Fontanelle	Marked depressed	Moderately	Mild sunken

Replacement Therapy

1. **Mild and moderate dehydration:** Treated with ORS.
2. **Severe dehydration:** Treated with IV rehydration.
   - IV fluid requirement:
     - Treat shock.
     - Deficit, maintenance, ongoing losses.

1. Mild to moderate dehydration:

- Type of fluid ORS.
- Amount 75 ml/kg "but if child wants more give more over 4 hours.
- How to give ORS?
  - One teaspoon full, 1-2 minutes for child below 2 years.
  - Freq. sips from cup for older child.
- **Mild dehydration:** 50 ml/kg over 4 hours then 10 ml/kg after each passed diarrhea or vomiting, 1st dose to correct dehydration, 2nd dose to avoid dehydration.
- **Moderate dehydration:** 100 ml/kg over 6 hours, then 10 ml/kg after each passed diarrhea or vomiting "you can manage at home but better at hospital".

C/I of ORS:

- Coma.
- Persistent vomiting.
- Abd. Distension.
- Paralytic ileus.
- Rapid loss of stool.
- Severe metabolic acidosis.
- CHO mal absorption.

If baby vomits, after giving ORS, wait 10 minutes and give at slower rate spoon every 2-3 minutes, if still vomit, may need IV fluid.

Liquid: e.g., juice, tea, are not suitable for use in ORS therapy because they may cause osmotic worsening of diarrhea and they contribute to hyponatremia, so water only is used.

Advantage of ORS:

- Cheap, easy to administer, used for all types of dehydration, can be given to any age, also any type of diarrhea.
- During diarrhea, do not stop breast feeding or any other type of feeding, because in GE 60% of food will be absorbed.
- Stop ORS when child doesn't want it any more.
- ORS should not be given after 24 hours of preparation.

Old formula of ORS	New formula of ORS (WHO)
Na = 90 mmol/l. Cl = 80 mmol/l. Bicarbonate = 30 mmol/l or citrate. K = 20 mmol/l. Glucose = 111 mmol/l *Dissolve in one liter of water, when added together, osmolarity 341 higher than body osmolarity, which may cause more diarrhea.*	Na = 70-75 mmol/l Cl = 65 mmol/l Bicarbonate or citrate = 20 mmol/l K = 10 mmol/l Glucose = 75 mmol/l. *Total osmolarity 245 less than body osmolarity, so less diarrhea.*

Note:

Glucose is need for N_a^+ absorption "also Cl and AA, and is used as energy with activation of N_a – K atpase pump lead to absorption of water and N_a."

If new formula not available, take old formula and dissolve in double amount of water.

Admission needed if:

- No home care.
- Continuous vomiting, ORS refusal or ORS not enough.
- Other illness (meningitis, pneumonia).
- Severe dehydration.
- Diagnosis not clear.

Calculation of IV fluid

1. Maintenance: Daily requirement

Less than 1 year	More than 1 year
1st day = 60-75 ml/kg/24 hours.	1st 10 kg = 100 ml/kg/24 hours.
2nd day = 90 ml/kg/24 hours.	2nd 10 kg = 50 ml/kg/24 hours.
3rd day = 90 ml/kg/24 hours.	Subsequent kg = 20 ml/kg/24 hours.
4th day = 120 ml/kg/24 hours.	
5th day = 150 ml/kg/24 hours.	
5th day to 1 year = 150 ml/kg/24 hours.	

2. Deficit: %age of body loss

	Mild	Moderate	Severe
Less than 1 year	5%	10%	15%
More than 1 year	3%	6%	9%
D = % x weight x 10			

3. Ongoing loss = number of motions x

Day 1 = M + D over 24 hours
Day 2 = M + Ongoing loss
} In hypo and isotonic dehydration

Hypertonic dehydration:

Overly rapid treatment of hypernatremic dehydration may cause significant morbidity and mortality. ***Idiogenic osmoles*** are generated within the brain during the development of hypernatremia. These idogenic osmoles increase the ***osmolality*** within the cells of the brain, providing protection against ***brain cell shrinkage*** secondary to movement of water out of cells into the hypertonic extracellular fluid. These idiogenic osmoles dissipate slowly during correction of hypernatremia. With rapid lowering of the extracellular ***osmolality*** during correction of hypernatremia, there may be a new gradient created that causes water movement from the extracellular space into the cells of the brain, producing ***cerebral edema.*** Possible manifestations of the resultant ***cerebral edema*** include seizures, brain herniation, and death.

Correction of hypertonic:

- The serum sodium concentration should not decrease more than 12 mEq/l every 24 hours. The deficits in severe hypernatremic dehydration may need to be corrected over 2 to 4 days.
- 1st day = 2/3 M + 1/2 D.
- 2nd day= 2/3 M + 1/2 D.
- **Type of IV Fluid:**
  - D $\frac{1}{5}$ saline.

Management of shocked patient:

- **ABC :**
  - Call for help.
  - Cleaning air way.
  - Breathing by O_2.
  - Insert 2 large cannula for IV fluid.
  - Give 20 ml/kg IV fluid (normal saline over 20 minutes), it can be repeated 3 times, if not improved call ICU doctor.
- 1st day = M+D – treatment of shock.
- 2nd day = M + Ongoing loss.

Types of IV Fluids:

- Depend on type of dehydration:
  - **Hypotonic dehydration:** 1/2 D/s if N_a = 120-130, but if N_a less 120, normal saline.
  - **Hypertonic:** 1/2 D/s or 1/5 D/s.
  - **Isotonic:** 1/5 D/s.
- **Normal saline:** 0.9% N_aCl means every 100 cc of water contain 0.9 gr N_aCl. Na ions = 150 mmol/l, Cl ions = 150 mmol/l.
- **1/2 Dextrose saline:** Each 100 cc water contain 0.45 gr N_aCl and 5 gr dextrose, Na = 77 mmol/l, Cl = 77

Complication of dehydration:

1. Shock.
2. ARF.
3. MA **due to:**
   a. Bicarbonate loss in motions.
   b. Renal impairment.
   c. Ketosis from starvation.
   d. Lactic acidosis from hypovolemic shock.
4. **Metabolic alkalosis:** If vomiting more severe than diarrhea.
5. Electrolytes disturbances.
6. Convulsion.
7. DIC.
8. Complication of fluid therapy.

Complication of fluid therapy:

- Hypervolemia lead to edema and HF.
- Electrolytes disorders:
  - **Hypo K:** Aggravated by rapid correction of acidosis lead to intracellular shift of K.
  - **Hyper K:** Aggravated by acidosis and excessive K infusion in presence of renal impairment.

For Hypo K:

Give KCl 15% in saline bottle.

- Never give KCl by I.V route directly.
- Do not give K unless serum K level is known.
- Do not give K unless child passes urine.

Treatment of metabolic acidosis:

Give N_a bicarbonate N_aHCO_3 = base deficit x 0.3 x body weight (From ABG).

- Do not mix NaHCO3 with Ca-gluconate, because it leads to reaction.
- N_aHCO_3 = 1/2 of it given slowly IV and other 1/2 in drip.
- If rapid infusion lead to transformation of bicarb to carbonic acid and give water + CO_2. CO_2 leads to sudden vasodilation and rupture of blood vessels (inter ventricular Hg).

Monitoring Therapy

- **Vital signs.**
  - Pulse
  - Blood pressure
- **Intake and output**
  - Fluid balance
  - Urine output and specific gravity
- **Physical examination**
  - Weight
  - Clinical signs of depletion or overload
- **Electrolytes.**

Infants are more susceptible to dehydration than adults, Why?

- Relative excess of total body water.
- Higher daily requirement of water in infant (150 ml/kg/24 hr).
- Higher freq. of diarrheal diseases in infants and children.
- Smaller metabolic reserve of water and electrolytes.
- Limited power of the kidney to concentrate urine.
- Greater surface area:weight ratio leading to greater insensible water loss.

Chronic

Diarrhea of gradual onset lasting more than 1 month or recurrent, due to mostly non infectious cause.

Etiology: "D.D of chronic diarrhea".

1. **Malabsorption Syndrome:**
   a. **Impaired digestion**:
      - Hepatic: biliary atresia, chronic hepatitis.
      - Pencreatitis : Cystic fibrosis.

   b. **Intestinal stasis:** P.E.M "acini atrophy", I.B.D and irritable bowel syndrome.

   c. Impaired absorption:
      - **Chronic infection**: Gardiasis, T.B.
      - **Food intolerance:** Lactonse intolerance, cow milk protein, celiac disease ,acrodermatitis enteropathica.
2. **Endocrinal:** e.g., thyrotoxicosis.
3. Immuno deficiency.
4. **Neoplasia:** e.g., Neuroblastoma due to vaso-active intestinal peptide "VIP".

C/P

1. **Pattern of motion may be :**
   - Watery.
   - Steatorrhea → with fat malabsorption → pale, bulky, offensive, greasy stool.
   - Bloody.
2. **Manifestation of malabsorption:**
   - General ill health with pallor & weakness.
   - Abd. Distention and flatulence.
   - Muscle wasting.
   - Loss of s/c fat & loss of weight.
   - Nutritional edema.
   - Mouth ulcer and glossitis.
   - Hypoglycemia, vitamins and mineral deficiencies.
3. **Manifestation of the cause:** e.g., hepatomegaly and jaundice in chronic liver disease.

Diagnosis

1. **Phase One:**
   i. History and examination.
   ii. Stool examination:
      a. To prove malabsorption:
         - **For CHO malabsorption:**
            - Stool pH "may be acidic"
            - Reducing substance in stool.
            - Breath hydrogen test.
         - **For Fat malabsorption:**
            - Increase stool fat globules.
            - Increase stool fat contents.
         - **For protein malabsorption:**
            - Increase Fecal α_1 anti- trypsin.
      b. Detect ova, parasites, leucocytes.
   iii. Stool culture.
   iv. **Blood studies:** CBC, ESR, Electrolytes, urea, creatinine.
2. **Phase Two:** Sweat chloride test, stool electrolyte.
3. **Three:** Endosopic study, small & large bowel biopsy, barium studies.
4. **Phase Four:** Hormonal studies e.g., VIP and gastrin, celiac workup, immunologic assay.

Giardiasis

Caused by G.Lamblia which is flagellated protozoa.

Mode of Infections:

- Feco-oral route.
- Ingestion of contaminated water & food with G.Lambia Cyst "infective stage".

I.P: 2-28 days ≅ 7 days.

C/P

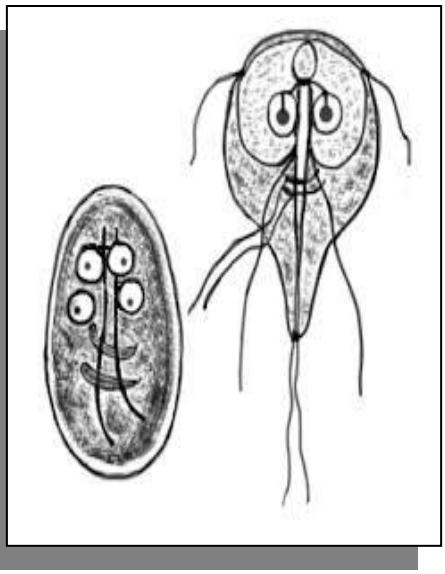
- Majority of cases are asymptomatic "carriers".
- **May present by :**
  - Anorexia, Abd. Distension, Nausea, Abd. Pain & diarrhea " watery or statoreahea", due to villous atropy.

Diagnosis

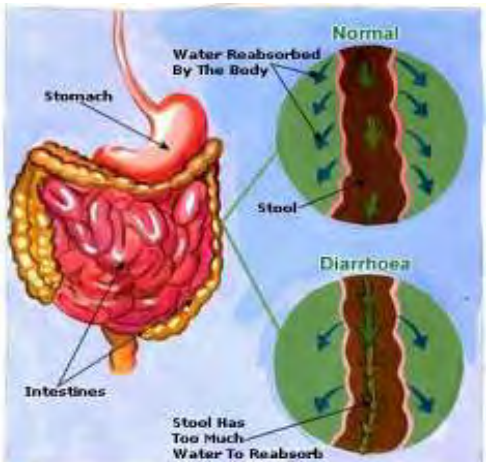
i. Single stool examination for cyst or throphozoite, get gardiasis 50-70%.
   Normal stool analysis on reading does not exclude gardiasis.
ii. 3 Stool examination ideally 48 hours apart getting gardiasis in 95%.
iii. 1 Stool examination and stool ELIZA, for gardiasis antigen get gardiasis in 95%.
iv. Duodenal biopsy **"Golden Standard"** get gardiasis 100%.

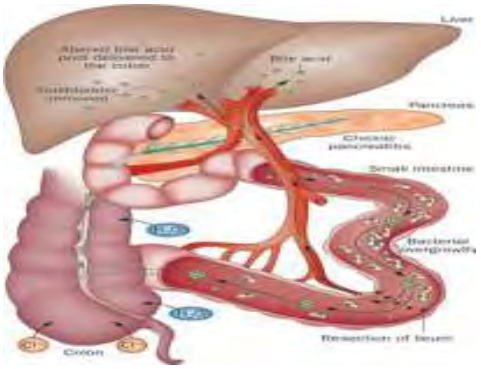
Treatment

Mitronidazole.



Note:
Gardiasis can mimic celiac disease in presentation and biopsy. Gardiasis can't cause brain abscess, but E.histolytica can cause brain abscess.

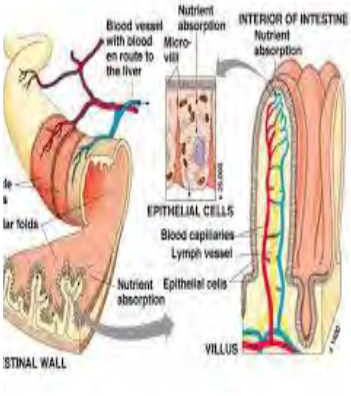




Celiac Disease = Gluten sensitive enteropathy

An enteropathy disorder of small intestine in which there is a damaging immunological response in the proximal small intestine, mainly induced by gluten. It is not food allergy, it is a T-cell mediated response "***type IV Hypersensitivity – cell medicated"***.

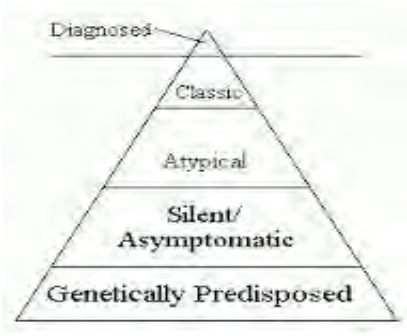
Pathogenesis



Ingestion of cereal products that contain glutine protein will lead to chronic exposure of small insteine to **gliadine fractions** **"structural unite of glutine".** This will result in sensitization of lymphocyte **"mainly T-Cell"** in the lamina propria of small intestine causing villus atrophy & distension of intestinal surface = Imn. Mediated damage.

There is positive family history "1st degree relative", also there is a **genetic factor** strong association with HLA-DQ_2, HLA-DQ_8, HLA-B_1. Also **environmental factor** contribute to celiac disease.

Clinical Picture.



Can occur at any age "children and adults", characterized clinically by malabsorption and pathologically by abnormal small intestine mucosa. The damage of small intestine is different from child to another, which leads to difference in presentation, celiac disease is only diagnosed when gluten is introduced into diet "not in 1st month of life".

Clinical presentation depends on :

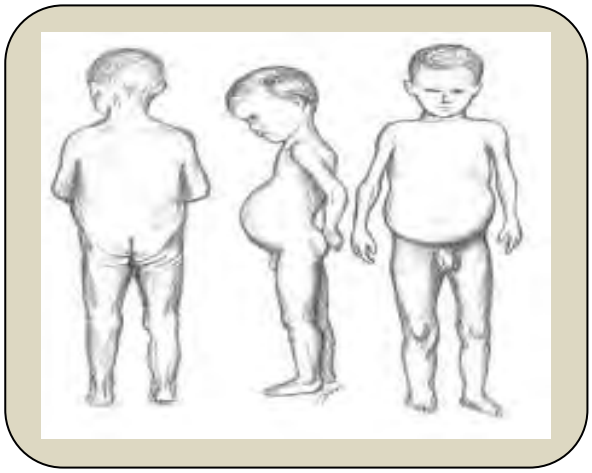
- Age
- Duration of disease
- Degree of small intestine involvement

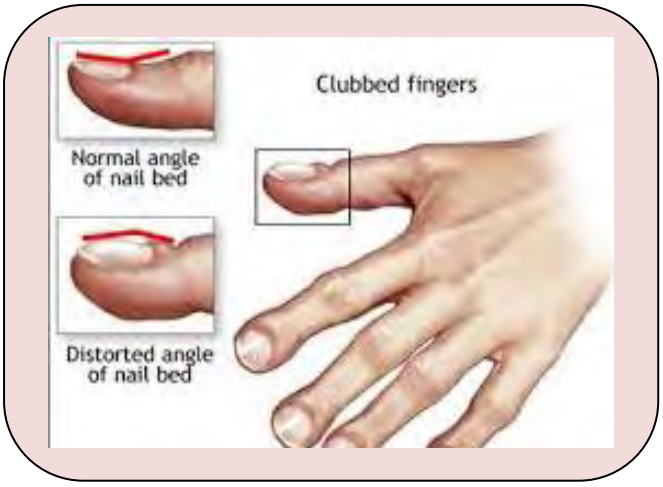
GIT: Usually start after 6 months, S&S are:

- Recurrent chronic diarrhea.
- Steatorrhea "bulky, greasy, offensive, foul, smelling stool.".
- Abd.distention.
- Weight loss and vomiting.
- Abd. Pain.
- Aneorexia and lethargic.
- Proximal muscle wasting.
- **Wasting of buttocks.**
- Recurrent aphthous ulcers in mouth.
- **FTT** most common presentation.
- **Irritability.**
- **Constipation.**
- Figure clubbing.
- Edema due to hypoproteinemia. may cause rectal prolapsed intussusceptions.

Extra-Intestinal: Usually at older age and adults, which are :

- Dental hypoplasia.
- Osteopenia & Osteaporosis due to Vit.D deficiency.
- Delayed puberty.
- Iron deficiency or folate deficiency "Megaloblastic" anemia, "pallor".
- Also poor splenic function "hyposplenism".





Gluten is present in:

- Wheat
- Oats
- Barley
- Rice
- Rye

Screening of children at risk of Celiac:

- "Down Syndrome
- D.M
- Thyroidits
- IgA Deficiency

Conditions associated with Coeliac Disease:

- IDDM "screening every 3 years".
- Down syndrome.
- Turner syndrome.
- Wiliam syndrome.
- Selective IgA deficiency (2-8 %).
- Autoimmune condition e.g., thyroditis, pernicious anemia.
- Small bowel malignancy "Lymphoma".
- Dermatitis herpetiformes.

Investigations

- **Serological Markers**: If the titer is high this indicates the need for a jejunal biopsy, used for screening of asymptomatic or high risk groups.
- Supports the diagnosis and monitors patients compliance with treatment.
- **Three types of AB**
  - Anti-gliadin "IgA, IgG".
  - Anti-endomysial "IgA, IgG".
  - Anti-tissue transglutaminase TTG: The best, but total IgA should be measured also because some patients are IgA deficient "or measure IgG" .
- AB testing **doesn't** confirm the diagnosis, it is only for screening or indicating Jejunal biopsy.
- **To confirm diagnosis:** Jejunal biopsy or duodenal biopsy.
- **Pathology of biopsy:** Flat villi, crypt hyperplasia intra-epithelial lymphocyte and plasma cell infiltration "*Total or subtotal villous atrophy".*

Management:

- **Gluten free diet** and supplementation of nutritional deficiency.
- Monitor response: By serological mark, they will decrease within a year of gluten free diet.
- If disease presents 2 years before age of diagnosis or if diagnosis is doubtful: Gluten challenge is done later in childhood, if no increase in serology and no damage to jejuna mucosa, gluten could be re-introduced gradually.
- But if serology abnormal: Withdraw gluten.

Response to treatment clinically:
Resolution of symptoms, increase in weight, child become happy, Gluten free diet should be permanent "***lifelong".***

Rickets rarely develop in Celiac disease because there is no growing of bone "***Rickets occurs in growing bones",*** it only occurs during treatment with gluten free diet without Ca^{+2} and Vit D supplement, supplementation of trace elements is important e.g., Ca^{+2}, Vit D, Folic acid, iron.

Complication of Celiac:

i. Small bowel lymphoma **"main cause of death".**
ii. Autoimmune thyroditis.
iii. Pernicious anemia.
iv. Iron deficiency enemia.
v. Osteoprusis.
vi. Ricket's after treatment.

D.D of villous atrophy:

i. Celiac disease.
ii. Transient glutin intolerance.
iii. Cow milk intolerance.
iv. Soya milk intolerance.
v. GE with post, GE Syndrome.
vi. Autoimmunce enteropathy.
vii. Acquired hypogammaglobulenemia
viii. Protein entergy malnutrition "PEM"

Dietary Protein Intolerances

They are usually **transient,** the commonest is Cow's milk but could also occur with Soya.

Cow's milk protein intolerance:

The protein in Cow's milk may cause an immune mediated inflammatory injury of small bowel "**villous atrophy**" and large bowel "**acute colitis**" usually start in the first month of li

Symptoms and sign:

- **G.I.T:** Chronic diarrhea may be bloody.
  Abdominal pain or distension.
  Vomiting.
  FTT.
- **Skin**: Infantile eczema.
  Sudden erythematus rash.
- Respiratory recurrent **wheezy chest.**
- Crying & irritability.
- Edema due to protein loss, sometimes it may lead to acute anaphylactic reaction with **urticaria**, strider, bronchospasm.

NOTE:

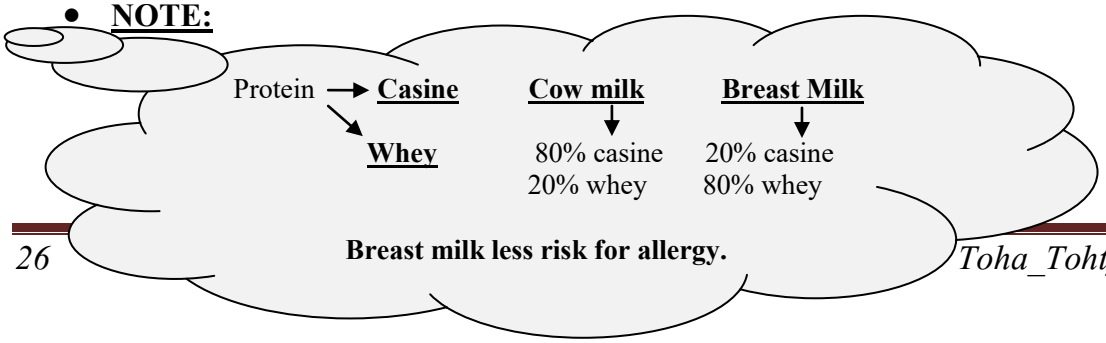
- 50% of cow milk allergy will develop an allergy to the other food protein **"egg, soya, peanut".**
- 50% - 80% of cow milk protein allergy will develop an allergy against one or more inhalant allergens e.g., house dust mite.
  - Cow milk allergy has a higher risk to develop other allergic diseases e.g., asthma, eczema.
  - Small amount of cow's milk protein can pass through the breast milk may cause allergic sign & symptoms in this case mother must exclude all dietary product of cow milk protein from her diet.

Investigations

1. **Serum for B-lactoglobulin:** If high level indicate allergy from cow milk protein, but not diagnostic.
2. **Elimination challenge test:** Should be done in hospital to avoid anaphylactic shock, stop all diet contain cow milk protein for 48 – 72 hours if symptoms resolved, return intake cow milk protein, if re-appear indicate allergy, it is diagnostic.
3. **Skin break test:** To detect allergy to other substances.
4. Blood film for Eosinophilia, IgE concentration in plasma.
5. Stool for Occult Blood.

Treatment

- **Hydrolysed formula (Hypo allergic milk):** It is milk formula in which the protein is already digested, so it loses its sensitivity.
- 80-90% resolve within 2-3 years about 10% remain allergic long life.
- **NOTE:**



Food Intolerances

CHO
- Oligosacc. "Glucose, Fructose, Galactose".
- Disacc. "Lactose, Sucarose, Maltose".
- Polysacc. "Starch".

Type of CHO in milk is Lactose:

- Lactose $\frac{Lactose}{Enzyme}$ glucose + galactose.

Note:

- Any food with lactose avoided in mother with galactosemia to prevent accumulation of galactose.
- **Another type of CHO :**
  - Reducing substance "lactose, maltose, isomaltose".
  - Non-reducing.

Lactose Intolerance:

Inability to digest significant amount of lactose, it is caused by shortage of the **enzyme lactase.**

Causes:

May be :

- Primary (Cong.)
- Secondary:
  - Most common secondary post G.E.

Lactose, a disaccharide molecule found in milk and dairy products, cannot be directly absorbed through the wall of the small intestine into the bloodstream so, in the absence of lactase, passes intact into the colon. Bacteria in the colon can metabolise lactose, and the resulting fermentation produces copious amounts of gas (a mixture of hydrogen, carbon dioxide and methane) that causes the various abdominal symptoms. The unabsorbed sugars and fermentation products also raise the osmotic pressure of the colon, causing an increased flow of water into the bowels (diarrhea).

C/P:

- Chronic diarrhea > 2 weeks (expulsive diarrhea).
- Nausea.
- Abd. Cramps.
- Abd. Distension.
- Excoriation mark in buttock due to increase in acidity.

Diagnosis:

- Stool pH acidic < 5.5.
- Stool reducing substance > 5 mg/gram stool.
- Hydrogen breath test: An oral CHO challenge is done, if there is CHO malabsorption, lead to bacterial fermentation of malabsorped CHO lead to increase in excretion of hydrogen in breath. (It is not done because lead to severe area diarrhea).
- Direct measurement of the enzyme level (rare).

Treatment

- Lactose free milk for 2-3 weeks but if the cause is congenital, continuous for life.

NOTE:
Preterm babies are more likely to be lactose intolerance because lactase level do not increase until the 3rd trimaster of pregnancy.

Acrodermatitis enteropathy

(AR) due to cong. Defect in zinc transport in small intestine. It could be also acquired due to inadequate zinc intake.

C/P

- Symmetrical erymthematous rash mainly on mucocutaneous junctions e.g., around mouth and anus.
- Chronic severe diarrhea.
- Hair loss "hair reddish tint".
- FTT.
- Super infection of candida is common as paronechea "fungal infection of nail".
- Cause of Napkin Rash.
- Poor wound healing.
- Occular changes lead to biliphritis, photophobia.

- Serum zinc level inside WBC.
- Plasma methalothioneum level "zinc binding protein" is low.

Treatment

- Zinc sulfate

How to differentiate monilial Rash & amonial dermatitis?

- If skin rask including skin fold with stellate lesion, it is monelial rash "candida".
- **Treatment:** Nystatine cream.
- Amonial Rash not involve fold.
- **Treatment:** By advising mother dryness and frequent change of napkins, amonial cream.

Note: D.D. of napkin rash.

- Monellial dermatitis.
- Amonial dermatitis.
- Contact dermatitis.
- Psoriatic dermatitis.
- Seporic dermatitis.
- Zinc deficiency.
- Lactose intolerance.

Inflammatory bowel diseases

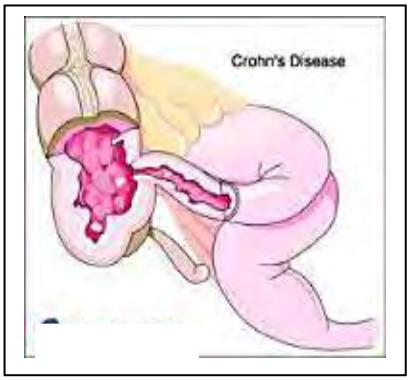
Crohn's & ulcerative colitis.

It is characterized by exacerbation and remission.

- **Etiology:** Unknown but is found to be in Jews, whites and tend to run in families indicating a **genetic predisposition.**
- **Incidence:** Common around adolescence.

Crohn's disease

It may affect GIT anywhere from mouth to anus in skipping patterns, involving the whole thickness of the wall "transmural".

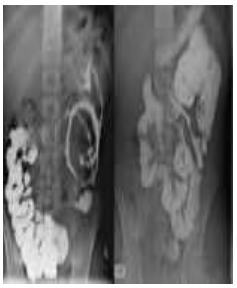


C/P

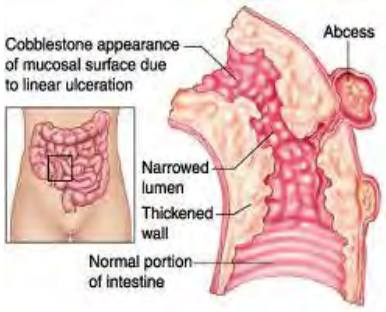
- Abd.pain.
- Weight loss.
- Anorexia.
- Diarrhea (may be bloody).
- Perianal involvement (common):
  - Abscess
  - Anal fissures
- Stricture and adhesions.
- Extra-intestinal manifestation.

Commonly affect distal illium & proximal colon.

Diagnosis



- It may take years due to its non-specific presentation.
  1. **Plain abd. X-ray** (partial intestinal obstruction).
  2. **Barium study** (string sing "area of thick wall & narrow leumen").
  3. **Endoscopy and biopsy** (golden standard) skip lesion and cooble stone appearance.
  4. **CBC:** increase WBC, increase PLT.
  5. **ANCa** (Anti-neutrophic cytoplasmic Ab "10-20%").



Treatment

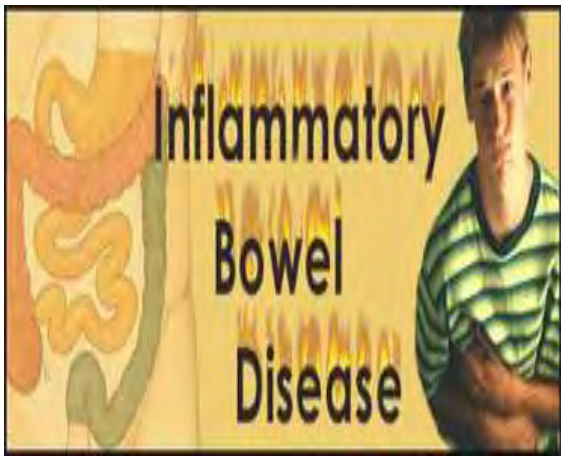
1. **Medical:**
   Steriod, Aminosalicylates, Azathioprin, Cyclosporin, TNF and Monoclonal AB "infilxamb".
   Antibiotic are used as infectious cause of diarrhea is difficult to be excluded.

High caloric intake to promote as normal growth as possible.

2. **Surgery:** For complication.

Complications:

- Intestinal obstruction.
- Fistula.
- Adhesions.
- Abscess.
- Poor growth & anemia.



Ulcerative Colitis

Disease affecting the large bowel & characterized by continuous mucosal ulceration affect mucosa and submucosa.

C/P

Bloody diarrhea with mucous, abd. Pain ± thenesmes.

- **Mild UC:** 90% of patients have <6 bowel motion/day, no fever, no anemia.
- **Moderate UC:** >6 bowel motion/day, anemia, fever, hypo albuminemia.
- **Severe UC:** Severe anemia, high fever, leukocytosis, tachycardia.

Commonly affect recto sigmoidal mucosa.

Diagnosis

Endoscopy & biopsy: Continuous lesion with superficial ulceration only mucosa and submucosa, mucosa is friable and may bleed easily.
Abd. X-ray: Toxic megacolon "C/I to do endoscopy".

Complication

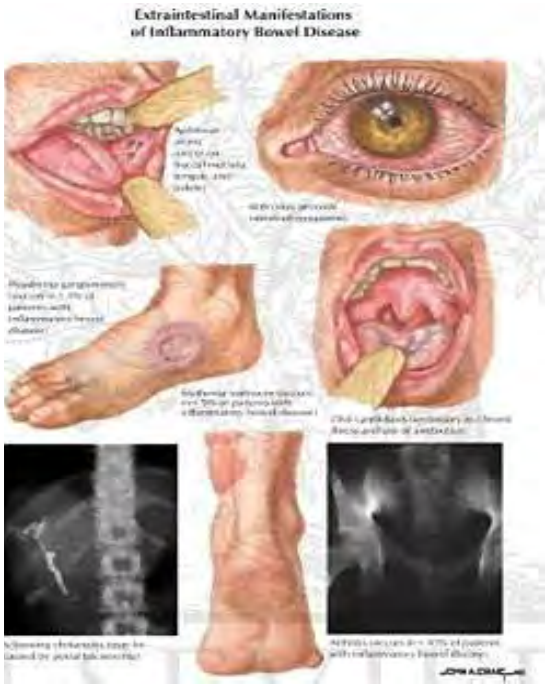
1. Increase risk of developing colonic cancer later.
2. Toxic megacolon with perforation.

Treatment

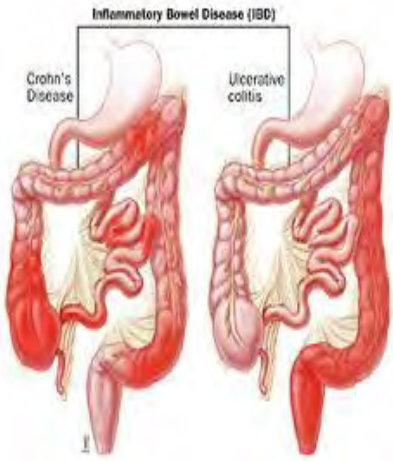
- Aminosalicalytes, sulfasalazine, steroids orally/enema.
- In severe cases may be treated

Extra intestinal manifestation of IBD

1. **Eye:** Uveitis, scleritis, epi-scleritis.
2. **Skin:** Erythema nodosum, pyoderma gangrenosum.
3. **Liver:** Autoimmune hepatitis, fatty liver, cirrhosis, ascending sclerosing cholangitis, gall stone.
4. **Kidney:** Amyloidosis, oxalate stone.
5. **Joint & bone:** Arthralgia, sacroiliatis, metabolic bone disease "osteoporosis".
6. **Thrombosis:** Sup. Mescentric thrombosis, portal vein thrombosis.



	Crohn's	U.C.
Rectal bleeding	Sometimes	Common
Diarrhea	Variable	Common
Perianal disease	Common	Unusual
Mouth ulceration	Common	Rare
Strictures	Common	Unusual
Fistula	Common	Unusual
Fissures	Common	none
Toxic megacolon	None	Yes
Skip lesion	Common	None
Transmural	Common	Unusual
Crypt abscess	Less common	Common
Granulomas	Common	Unusual



Characterized by Chronic or recurrent abd.pain, altered bowel habit, bloating in absence of structural and biochemical abnormalities.
Age: 10 – 15 years. Male = female

C/P

- Abd.pain "dull achy, colicky or sharp pain occur anywhere in abdomen, commonly hypogastric or periumbilical may aggrevated by food and stress and relieved by defecation.
- Constipation or diarrhea.
- Abdomen distention.
- Heart burn, N/V.
- Back pain.

Investigatio

d incomplete empty of bladder.

All investigation of chronic diarrhea.

Treatment

- No complete cure needs educational and social support.
- Lopramide which decrease intestinal motility.
- Cholestyramine which decrease diarrhea.

Causes of Abd.pain:

a) *Acute abd.pain:*
- **Surgical :** Acute appendicitis, Intussusception, Inflamed Meckel's diverticulum, pancreatitis.
- **Medical :** GE, mesenteric adenitis, UTI, kidney stone, Henchischonlien purpura, DKA, sickle cell disease.

b) *Chronic Abd.pain:*
- Dysfunctional (non-organic) pain >90%.
- Organic pain 10%:
  - **Common causes:** Parasitic infestations, chronic constipation, lactose intolerance, drugs.
  - **Un-common causes:** Sickle cell anemia, renal stone, ulcerative colitis, peptic ulcer.

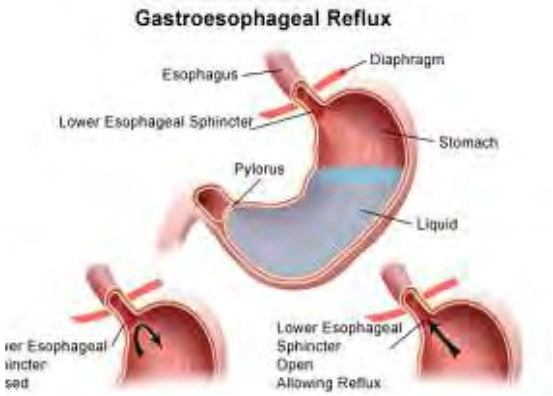
Dysfunctional psychogenic or non-specific recurrent abd. Pain.	Organic recurrent Abd.pain.
Etiology: Unknown may be due to : • Stress, family problem, school phobia. • **C/P:** Age usually > 6 years. • **Site :** periumbililcal can be located properly by the child. • **Severity:** Not severe, not interfering with activity. • **Nature:** Vague, subsides spontaneously within <20 min or by stooling. • No other associated complain. • Patient's parent usually come to complain after several weeks. • **Treatment:** Re-assurance.	**Etiology**: As above. • **C/P:** pain away from umbililcus, the site may suggest the cause, may be severe, e.g., renal stone. • **Associated complain:** e.g., hematuria and dysuria, bleeding per rectum. • **Treatment:** Treat the cause.

Gastro Esophageal Reflux

Definition: Abnormal retrograde of gastric content into esophagus due to persistent relaxation of lower esophageal sphincter and delayed gastric emptying.

Many infants have a minor degree of reflux. 1:300 infants have significant reflux and clinical manifestations. Infants GERD become evident in the first few months of life, peaks at **4 months** and resolves in most by **12 months** and nearly all by **24 months.**

GER: It is very common in infancy, the lower esophageal valvular mechanism depends on the intra-abd.segment of the esophagus and the length of the muscular component of the sphincter both may be relatively deficient in early life allowing for reflux. It is regarded physiological when it does not interfere with weight gain or health. It resolved spontaneously due to maturation of LES, upright posture and more solids in diet.



Reflux is common in :

- Children with cerebral palsy.
- Neonate with bronchopulmonary dysplasia.
- Following surgery for oesophageal atresia or diaphragmatic hernia.
- Down syndrome.

C/P

1.GIT Manifestation

A. **During infancy:** Regurgitation ***(especially postprandially),*** signs of esophagitis (irritability, arching, choking, gagging, feeding aversion), and resulting failure to thrive.

B. **During childhood:** Regurgitation during the preschool years, complaints of abdominal and chest pain ***(esophagitis)*** supervene during later childhood and adolescence, and occasionally, children present with neck contortions ***(turning of head and arching),*** designated ***Sandifer*** **syndrome.**

2. Respiratory manifestations

A. **Infantile reflux:** As obstructive apnea or as stridor or lower airway disease in which reflux complicates primary airway disease such as laryngomalacia or bronchopulmonary dysplasia, wheezing due to aspiration or vagal stimulation.

B. **Childhood reflux:** Frequently related to asthma or to otolaryngologic disease such as laryngitis, otitis media, or sinusitis. Asthma co-occurs with GERD in about 50% of asthmatic children.

Complications

1. **Esophageal complications:** Esophagitis and stricture, Barrett esophagus, and adenocarcinoma, esophagitis rarely lead to hematemesis and anemia.
2. **Nutritional:** Failure to thrive because of caloric deficits.
3. **Extraaesophageal complications (respiratory "atypical" presentation):**
   a. Aspiration pneumonia (5% of cases).
   b. Apnea and stridor.
   c. Reflux laryngitis.
   d. Asthma.
   e. Sinusitis.

Diagnosis

1. **Barium radiography: (Gastrographin study).** Bottle feed with dye frequent x-ray after 5 minutes, another x-ray repeat until detect dye completely in stomach then trendlenberg position "stomach up and esophagus down" then repeat x-ray, if dye return to esophagus, patient with GER, not diagnostic 100%. Many infants will have a normal esphogogram because GERD is intermittent.

2. **The best diagnostic test is 24 hours oesophageal pH monitoring:** Normally there is reflux of stomach content but for less than 4% of 24 hours period, if the pH is <4 and >4% of 24 hours, diagnostic.
3. Endoscopy and esophageal biopsy.
4. Esophageal manometery.

Management

1. **Positioning measures:**
   a. **Infants:** Keeping the infant prone during the day and nonprone position (supine) during sleep (to reduce risk of SDIS).
   b. **Children:** Left side positon and head elevation during sleep. Head elevation should utilize elevation of the head of the bed, rather than excess pillows, to avoid abdominal flexion and compression that might worsen reflux.
2. **Dietary measures:**
   a. **Infants:** Normalization of feeding techniques, volumes, and frequency if abnormal. Thickening the feedings with cereals results in fewer regurgitation episodes and greater caloric density, also may replace with A.R, formula.
   b. **Children:** Avoid acidic foods (tomatoes, chocolate, mint) and beverages (juice, carbonated and caffeinated drinks). Weight reduction for obese.
3. **Pharmacotherapy:**
   a. **Antacids:**e.g., MgHo & Alumenum HO (Gavascone).
   b. **Histamine-2: Receptor antagonists (H2RAs),** e.g., cimetidine, ranitidine used as antisecretory agents that act by selective inhibition of histamine receptors on gastric parietal cells.
   c. **Proton pump inhibitors (PPIs):** e.g., Omeprazole.
   d. **Prokinetic agents:** e.g., Domperidone = motilum (Local dopamine antagonist).
4. **Surgical:** Nissen fundoplication:

Telescopic of one portion of the intestine "proximal" into a more distal portion may result in impairment of the blood supply leading to necrosis of the involved segment of the bowel "gangrene".

- It is rare in neonate. It is the most common cause of intestinal obstruction between 3 months and 6 years of age. 60% of patients are younger than 1 year, and 80% of the cases before 24 months. Male:female ratio is 4:1.

Etiology

NOTE:

It is commonly ileu-cecal, could be ileo-ileal or colico-colic "rare".

Unknown but may be due to:

- Viral "adenovirus" or rotavirus.
- Introduction of new food proteins (weaning process).
- **Intestinal disorders causing lead points for intussusceptions:**
  - Meckel diverticulum.
  - Intestinal polyp.
  - Neurofibroma.
  - Intestinal duplication.
  - Hemangioma & lymphoma.
  - Henoch-schonlein purpura.
  - Cystic fibrosis.
  - Hemolytic uremic syndrome.

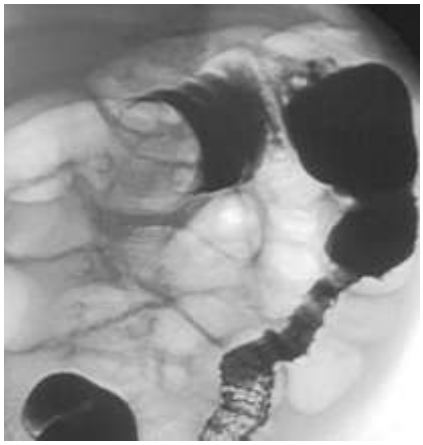
C/P

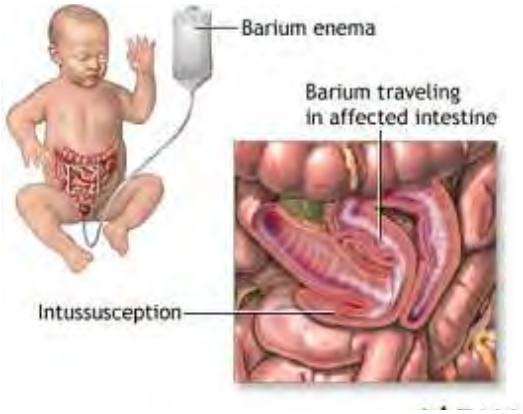
The onset is sudden in a previously well child.

- **Severe paroxysmal colicky pain:** it recurs at frequent intervals and is accompanied by straining efforts with legs and knees flexed and loud cries. In infants, pain may be manifested only by moaning sounds.
- **Vomiting:** It becomes bile stained in later phase.
- **Fecal excretions:** First few hours of symptoms, stool appears normal, later on fecal excretions are small or more often do not occur and little or no flatus is passed. Blood generally is passed in the first 12 hours but at times not for 1-2 days and infrequently not at all **(60% of infants pass a stool containing red blood and mucus, the current jelly stool).**
- **Shock-like state:** The infant becomes progressively weaker and lethargic, the pulse becomes weak and thready, respirations become shallow.
- **Abdominal examination.** Sausage-shaped mass (30% of patients do not have a palpable mass), in the right upper abdomen.
- **Rectal examination.** The presence of bloody mucus on finger as it is withdrawn after rectal examination.

Diagnosis

- Ultrasonography:
- Plain abdominal radiographs:
- Barium enema (coiled spring sign).
- Air enema.





Treatmen

- I.V volume expansion by fluids.
- **Reduction:** Pneumatic reduction and hydrostatic reduction.
  - **Contraindications:** prolonged intussusceptions with signs of shock, peritoneal irritation, intestinal perforation.
  - **Complications:** Perforation, peritonitis and death.
  - **The success rate:** 70-90% if reduction is done within the 1st 48 hours and 50% if symptoms are present longer than 48 hours.
- **Surgical Reduction:** If:
  - Evidence of peridoneal irritation.
  - Failure of reduction.

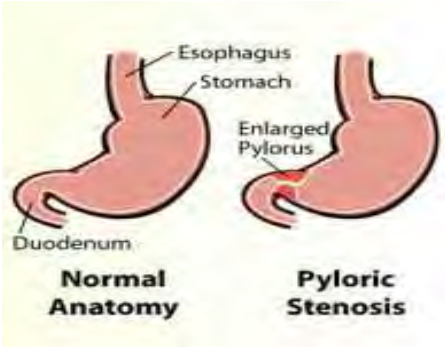
Hypertrophyic pyloric stenosis

Most of the cases are male and first born, increased incidence is seen in infants with type B and O blood group. Age: between 2 weeks – 2 months.

Etiology

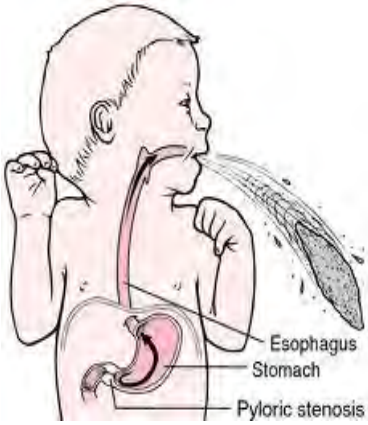
Multifactorial inheritance:

- Genetic factors.
- Environmental factors.



Pathology and Pathophysiology

- Diffuse hypertrophy and hyperplasia of the smooth muscles in the wall of the antrum of the stomach that ends abruptly where the duodenum begins.
- The antral region is thickened and elongated with a narrow lumen.
- The stomach musculature becomes uniformly hypertrophied in response to outflow obstruction.
- Vomiting may lead to loss of fluids, hydrogen ions, and chloride leading to dehydration and hypchloremic hypokalemic metabolic alkalosis.
- Thus, it consists of hypertrophy of the circular muscle layer of the pylorus.

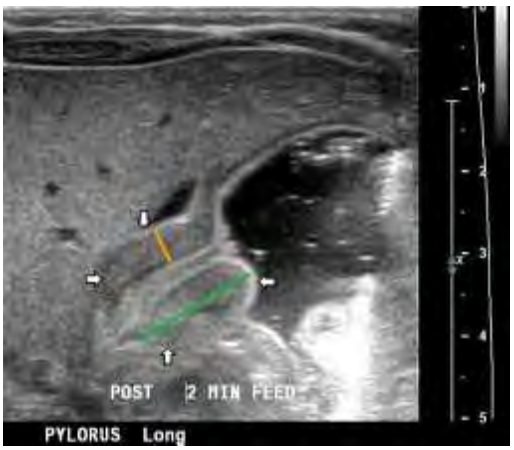


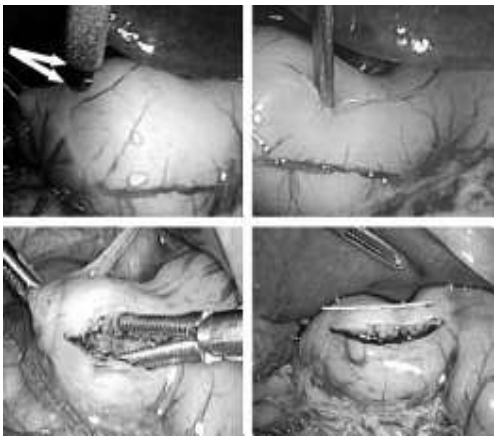
C/P

- **Symptoms:** Nonbilious projectile vomiting: During or shortly after each feed or it may be intermittent. The infant is hungry after vomiting, the stools are small and infrequent.
- **Physical examination:**
  - Signs of dehydration.
  - Weight loss and in untreated cases, chronic malnutrition.
  - Visible gastric peristalsis (proceeding from the left upper quadrant toward the pylorus in the right upper quadrant of the abdomen, soon after feeding).
  - A mass is palpated in the epigastrium above and to the right of the umbilicus beneath the liver edge. It is firm, movable, olive shape, and about 2 cm in length.
  - Prolonged jaundice: 5% may be due to decreased glucoronyle transferase enzyme activity and increased enterohepatic circulation.

Diagnosis

1. **Abdominal ultrasonography:** Pyloric muscle wall thickness >3-4 mm and a pyloric channel length more than 14 mm.
2. **Barium Study:** It shows an attenuated elongated pyloric canal, a bulge of the pyloric muscle into the antrum **(shoulder sign),** parallel streaks of barium seen in the narrowed channel, producing a **"double tract sign"**, and delayed or no gastric emptying.
3. **U/E and ABG:** Hypochloremic hypokalemic and hyponatremic metabolic alkalosis.





Treatmen

- Correct dehydration, electrolyte and acid base-balance.
- Fredet – Ramstedt operation "**pyloromyotomy"**

Constipation

Passage of hard, infrequent stool (it is common).

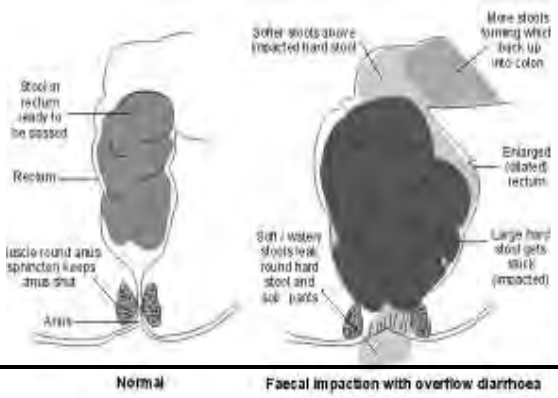
Causes

1. **Functional constipation:** 90-95% of the constipation beyond neonatal period, is improper toilet training or pain on defecation, due to voluntary avoiding defecation.
   Complication of stool retention, rectum becomes full and over distended leads to involuntary soiling (overflow diarrhea due to leakage around fecal mass).
   May also lead to UTI be extrinsic pressure on the urethra.

Management:

- Patient education: A regular bowel training program include sitting on the toilet for 5-10 min after meals to establish a regular bowel habit.
- Relief of impaction: An enema is usually required.
- Softening of the stool: Include the use of lactulose and polyethylene glycol preparation.
- Increases intake of juices, fruits and increase dietary fiber contents.

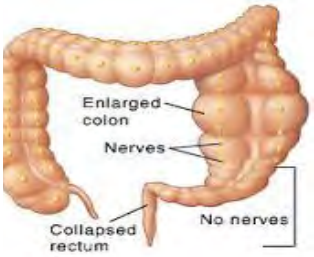
2. **Organic causes:**
   - **i. Anatomical:**
     a. Intestinal stricture.
     b. Imperforated anus.
     c. Abnormal muscle tone.
   - ii. **Bowel nerve abnormality:** hirschspring disease.
   - **iii. Spinal cord defect:**
     a. Trauma.
     b. Spina bifida.
   - iv. **Drugs and poison:** Lead poisoning and anti-cholenergic drug.
   - **v. Metabolic:**
     a. Hyper Ca^{+}.
     b. Hypo K^{+}.
     c. Hypothyrodism.



Normal Faecal impaction with overflow diarrhoea

1. Hirschprung's Disease = Cong.Megacolon

Absence of ganglion cells from myenteric and submucosal plexuses of part of the large intestine result in a narrow contrac colon.



- Rectosegmoid 75% and involve entire colon in 10%.
- **Incidence:** Male – female (4: 1).
- **Positive family history:** May be present.
- Defect in chromosome 10.
- The most common cause of lower intestinal obstruction.
- Highly associated with **down syndrome**

C/P

1. **Neonatal 80%:** Delay passage of meconium beyond 48 hours. May be acute obstruction.
2. **In older children:** Chronic constipation and abdominal distension, large fecal mass felt in left lower abdomen with empty rectum.
3. **Complicated cases:**
   a. Enterocolitis: Due to accumulation of feces, fluid and gases, which will cause obstruction of blood vessel lead to iskemia of intestinal wall and increase bacterial over growth that increase bacterial toxin damaging intestinal mucosa. Its infection with clostridia difficile staph, and anaerobes presented with bloody diarrhea and toxemia.
   b. Intermittent attach of I.O.
   c. Failure to thrive.
4. **On P/R:** Empty rectum and removal of finger passage of air and stool.

Diagnosis

1. **Anorectal manometry:** increase the pressure.
2. **Rectal suction biopsy**: Reveal absent ganglia.
3. **Barium enema:** Narrow a ganglionic segment "funnel shaped", distended proximal colon, distal segment narrow.

Treatment

Transient colostomy to 6-12 months of age, then permanent resection and anastomosis.

	Hirschspring	Functional constipation
History	At birth	>2 years usually school age
Encopresis	No	Yes
Enterocolitis	Yes	No
Poor growth	Yes	No
Faulty training	No	Yes
Abd. Distention	Yes	No
Anal tone	Normal	Normal
Stool in rectum.	Empty " No "	Yes (Full)
Biopsy	Diagnostic	Not done.

Note:

Encopresis: Daytime or night time involuntary soiling by formed stool beyond the age of expected toilet training. Treatment by toilet training.

Abdominal Masses

Acute abdominal masses.	Chronic abdominal masses.
Intussusception.	Hepatosplenomegaly.
Fecal masses.	Supra-renal masses.(Neuroblastoma)
Distended bladder.	Renal masses (Wilms tumor).
	Pancreatic masses (Pancreatic pseudocyst.
	Intestinal masses (Mesenteric cyst, Crohn's disease.
	Retroperitoneal masses (Teratoma, fibrosarcoma).
	Female genital tract (Ovarian cyst or tumor).

Gastrointestinal Bleeding

- **Hematemesis:** Passage of blood stained vomitis with indicate bleeding proximal to ligament of Tirietz.
- **Melena:** Passage of black coloured tary stool foul smelling and difficult to flush down in toilet. Digested blood from upper GI bleeding.
- **Hematochezia:** Passage of fresh bright red blood in stool usually it is from large bowel or from upper GIT if transient time is short.

Causes of bleeding per rectum:

- **Ileum :** Intussesception, Meckel's diverticulum.
- **Colon :** Dysentry "Shigella and salmonella".
- **Rectum :** Polyp, prolapse.
- **Anus:** Fissure, sexual abuse.

Meckle's diverticulum

It is the most common abnormality of GIT and most common cause of painless bleeding per rectum in pediatric. It is remnant of embryonic yolk sac "omphalomesentric or vitelline duct". **It is known as disease of (2):**

- 2 mucosa (gastric and pancreatic).
- 2% of infant affected.
- 2 year of age commonly presented.
- 2 inch long.
- 2 feet proximal from ileo-secal junction.

C/P

- Painless bleeding per rectum.
- Diverticulitis, I.O.
- Intussusception, perforation, peritonitis.

Diagnosis

Meckle's scan "Technitium 99

Treatment

Surgical removal.

NOTE:

Infant colic (midnight colic): Common symptom which occurs during the 1st few months of life, characterized by :

- Paraxysmal inconsolable crying or screaming and drawing up the knee.
- The condition is resolved by 4 months of age.

NOTE:

Stool analysis: Normally brown soft will formed no blood mucous or pus cells. pH = 6, contain <2mg/dl reducing substance.

Hematology by Asma Hadida

2015_2016

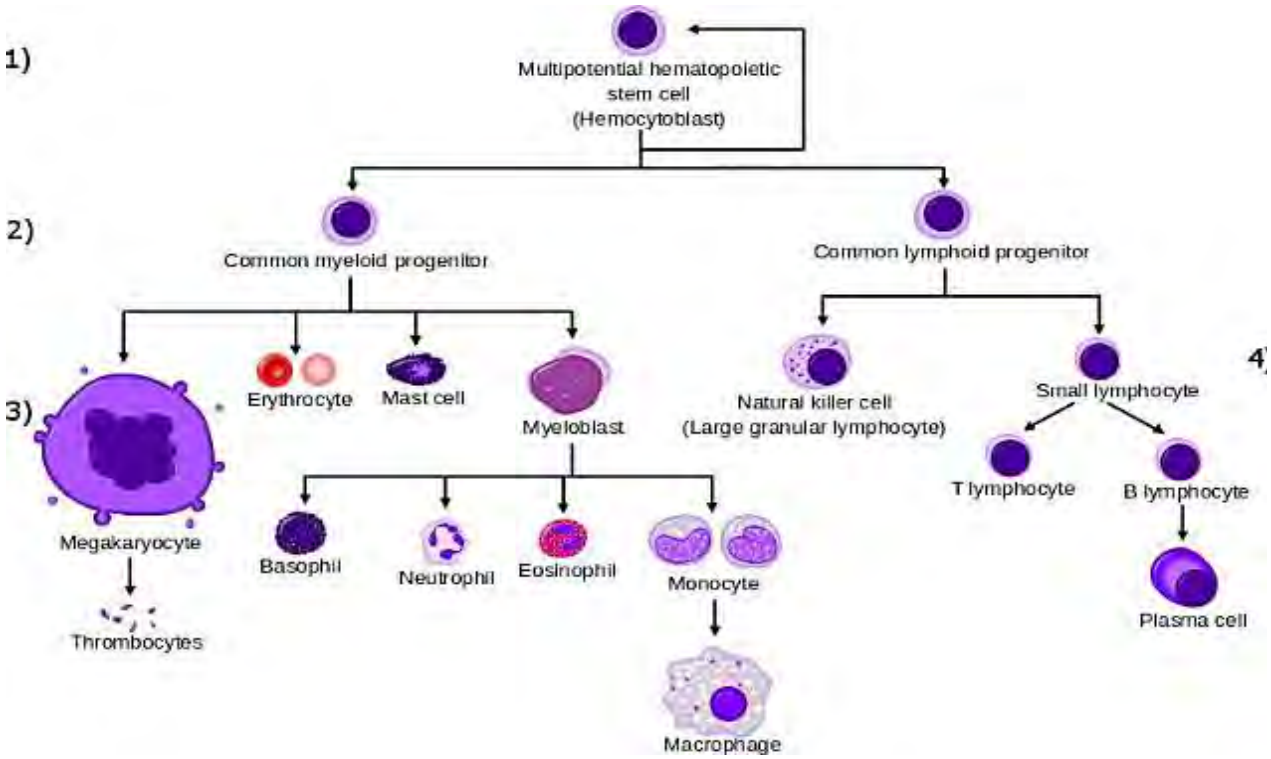
Introduction

▪ The main hemopioetic organs are **bone marrow, liver, spleen , lymph node, and the thymus.**

- The blood cells are formed from the **mensenchymal tissue** before the 2nd month intrauterine life from the 2nd month the **liver** become the main site of blood formation at about the 6th month of fetal life hemopiosis shifts **gradually to B.M**.
- During infancy the **whole bone marrow** forms the blood cell.
- During childhood the marrow activity **resides in the red marrow "ends of long bones , vertebra skull sternum and ribs** the inactive bone marrow yellow marrow can restore it activity in certain conditions as **chronic hemolytic anemia**.

Table 1. Fetal Hematopoiesis

	SITES	TIME	PRODUC
Measoblastic hematopoisis	Yolk Sac	10-14th day 3-4wk:primitive blasts 10-12wk:ceased	Erythroid
Hepatic hematopoiesis	Liver	6-8wk:appear 12-16wk:active 6mo:diminish/ stop at birth	Erythroid
Myeloid hematopoiesis	Bone marrow	4mo:start 6mo:increase/steady after birth: the only	Erythroid Neutrophils Macrophages



Types of Hb :

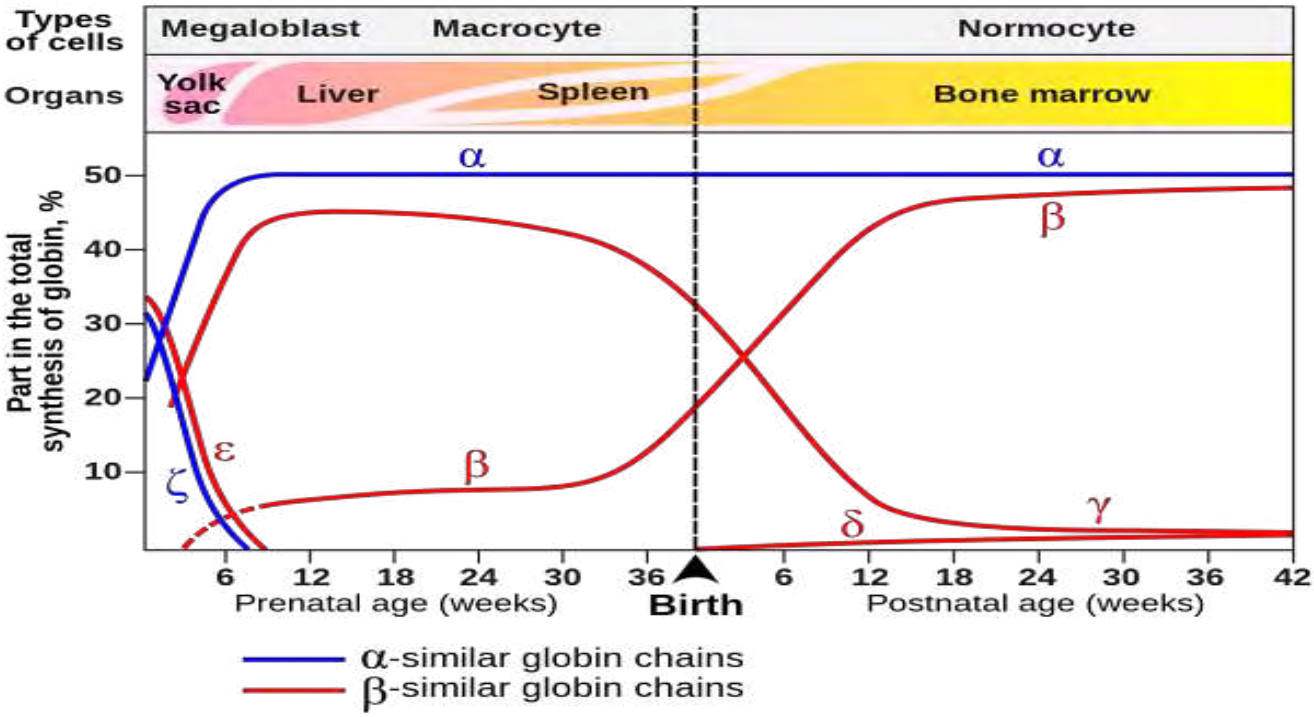
		% at birth	% at 12m
Hb F (fetal) ➡	2 Alpha - 2 Gamma	70%	< 2%
Hb A1 adult ➡	2 Alpha - 2 Beta	30%	> 95%
Hb A2 adult ➡	2 Alpha - 2 delta	less than 1%	2-3%

Hb F: is the Predominant Hb in the fetal life it is characterized by being alkali resistant.

Hb A : is the major adult Hb At 3rd to 6th month normal switch from gamma to beta chain production occurs.

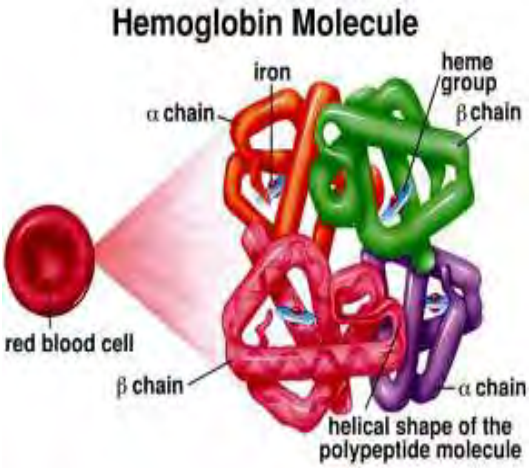
Blood indices :

1. **Hb content :**
   - 1st 2wks =14 to 20 gm /dl due to intrauterine hypoxia.
   - 1m = 10-18gm/dl.
   - **2m = 9-14gm/dl**.
   - 6m = 9.5-13.5gm/dl.
   - 1yr = 10.5-13.5 gm/dl.
   - 2-6yr = 11-14gm/dl .
   - 6-12y = 11.5-15gm/dl.
   - >12y: Female = 12-16g/dl.
     Male = 13-18g/dl.
2. **RBC count :**
   - In new born: 6 million/mm^3.
   - After ward: 4-6 million/mm^3.
3. **Hematocrite value** : % of RBC in 100 ml blood + 40-50.
4. **MCV = Mean corpuscular volume** ≅72-79 femtoliter.
   - If < 70 microcytic.
   - If > 85 macrocytic.
5. **MCH: Mean corpuscular hemoglobin** ≅ 30 pg.
6. **MCHC: Mean corpuscular Hb concentration** ≅ 33%.
   - Increase MCHC in H.spherocytosis
7. **Platelet count** : 150.00-400.000/mm^3.
8. **Reticulocytic cout** = < 1%
   - **Reticulocytosis occur in:-**
     - Hemolytic anemia.
     - Hge.
     - Response to hematinics e.g. iron folic vit.B12.
     - Recovery of B.M from suppression.



Normal major types of haemoglobin

Type	Composition	% of total haemoglobin
HbA_1	$\alpha_2\beta_2$	90%
HbA_2	$\alpha_2\ \delta_2$	<5%
HbF	$\alpha_2\gamma_2$	<2%
HbA_{1c}	$\alpha_2\beta_2$-glucose	<5%



DD of microcytic hypochrmic anemia

	Serum iron	Transferrtin saturation	Serum ferittin	
Iron def. anemia	Low	Low	Low	Response to iron
Thalassemia	Normal	Normal	Normal	Hb electrophoresis ↑HbF and A2
Anemia of chronic illness	Low	Normal	High	Treat the 1ry cases
Lead poisoning	High or normal	High	High	↑ lead in urine
Sidroblastic anemia	High	High	High	Sidroblast in B.M

- Reduction of Hb concentration and or hematocrite below the normal range for age and sex interfering with O_2 carrying capacity of the blood.

Classification : "causes"

1. **Decreased red cell production :**
   a) **Dyshemopoietic anemia due to deficiency of factor important for erythropoiesis:**
      - **Iron deficiency** : Iron deficiency anemia."The most common cause"
      - **B12 and folic acid deficiency**: megablastic anemia
      - Protein deficiency.
   b) **Anemia of bone marrow origin :**
      - **Bone marrow depression** : aplastic anemia.
        - Idiopathic depression:-
          - RBC only pure red cell anemia.
          - All blood elements "pancytopenia".
        - 2ry to infection toxin and irradiation.
      - **Bone marrow replacement by abnormal cell:**
        - Malignant cells e.g. leukemia cells.
        - Metabolic disease e.g. Gaucher cells.
        - Anemia of chronic illness, CRF " ↓ production deposit normal RBC precursor ".

2. **Increased red cell loss:**
   a) **Hemolytic anemia "may be acute or chronic"**
      - Membrane defect e.g. hereditary **spherocytosis** and **hereditary eliptocytosis**.
      - Hb defect e.g. **thalassemia** and **sickle cell anemia.**
      - Enzyme defect **G6PDHD**, **pyruvate kinase deficiency.**
      - **Immunology defect**: **+ve coombs test.**
        - passively acquired Ab: Rh and ABO incompatibility or incompatible blood transfusion
        - Actively formed Ab:
          - autoimmune hemolytic anemia "Idiopathic "
          - SLE
          - Malignancy
   b) **Hemorrhagic anemia:**
      - **Acute blood loss**: trauma accidents and surgical causes .
      - **Chronic blood loss:** parasite "bilharziasis" and ankylostoma gasrointestinal bleeding.
   c) **hypersplenism:**
      - Usually presents with pancytopenia.
   d) **Microangiopathic HA**: DIC, HUS.
   e) Malaria.
   f) Wilson disease.

Clinical picture of anemia:

General clinical picture of anemia:

➲ **Symptoms**

- General:
  - Headache.
  - Easy fatigue.
  - Lack of concentration.
- Cardiac:
  - Extertional dyspnea.
  - Palpitation.

➲ **Signs :**

- General:
  - Pallor nail beds and lips the most important clinical finding.
- Cardiac:
  - Hyper dynamic circulation e.g. collapsing pulse.
  - Hemic murmur.
  - HF in severe cases.

Clinical picture specific for each type of anemia

✶ **Chronic hemolytic anemia:**
Mild jaundice hepatosplenomegaly or splenomegaly, mongoliod feature and H/O repeated blood transfusion.

✶ **Acute hemolytic anemia:**
Sudden pallor sudden jaundice and dark urine "hemoglobinuria".

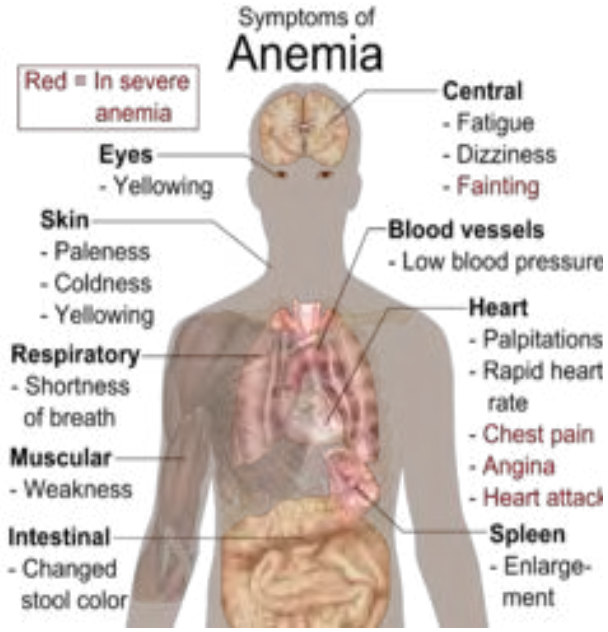
✶ **Iron def anemia :**
Irritability , anorexia and pica.

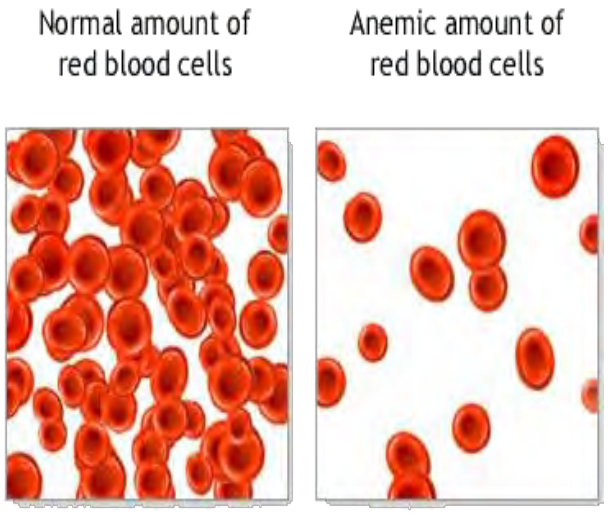
So; History of anemia :

- **Duration and speed of onset** of pallor associated symptoms (lethargic, irritability, fever recent trauma along history bruising joint pain jaundice dark urine).
- **Birth &neonatal history:**
  - Low birth weight, prematurity, SGA, neonatal blood loss, exchange transfusion, twin pregnancy, prolonged neonatal illness.
  - Onset of neonatal jaundice
- Dietary history.
- H/O drug &toxin exposure.
- F/H of blood diseases.
- PMH for chronic disease e.g. celiac disease, IBD.

Other causes of anemia in infancy and childhood:

- GIT bleeding
  - GERD.
  - Meckel's diverticulum.
  - Intussusceptions.
  - Cow's milk protein intolerance.
  - Bleeding disorder.
- Mal absorption syndrome, celiac disease
- IBD, RF, lead poisoning





Hemolytic anemia

Acute hemolytic

Causes:

Intracorpuscular

Enzyme def G6PD def

extracorpscular

- Non immunological "Coomb's –ve"
  - MAHA
    - HUS
    - DIC
    - RVT
    - Intracardiac prosthesis
  - Septicemia
    - Staph.
    - Strep
    - clostridia
  - Wilson disease
  - Malaria
- Immunological "Coomb's +ve"
  - Isoimmun
    - Hemolytic disease of newborn
    - Incompatible blood
  - Autoimmune HA

General C/F :

1. acute pallor.
2. Dark urine "hemoglobinuria".
3. Pyrexia.
4. Jaundice.
5. loin pain.

General investigation:

(1) Anemia :
Low Hb and Ht value.

(2) ↓ RBC survival :
- ↑ plasma hemoglobin.
- ↓ hepatoglobin and hemopxin.
- ↑ unconj. Bilirubin.
- Hemoglobinuria

(3) ↑ Erythropiosis :
- ↑ normoblast
- ↑ reticulocytic count > 5-15

Specific investigation:

(1) Commb's test :
- Diagnose immunliogic hemolytic anemia.

(2) Blood film :
- Henz body in G6PD deficiency.
- MAHA.
- Malaria.

(3) blood culture for septicemia.

Chronic hemolytic anemia

Causes:

Intracorpuscula

Membrance defect:
Heriditory sperocytosis.
- Heriditory eliptocytosis.

Hemoglubinopathy
- α-thalasemia.
- β-thalasemia.
- sickle cell anemia.

enzymatic defect:
Pyruvate kinase deficiency.

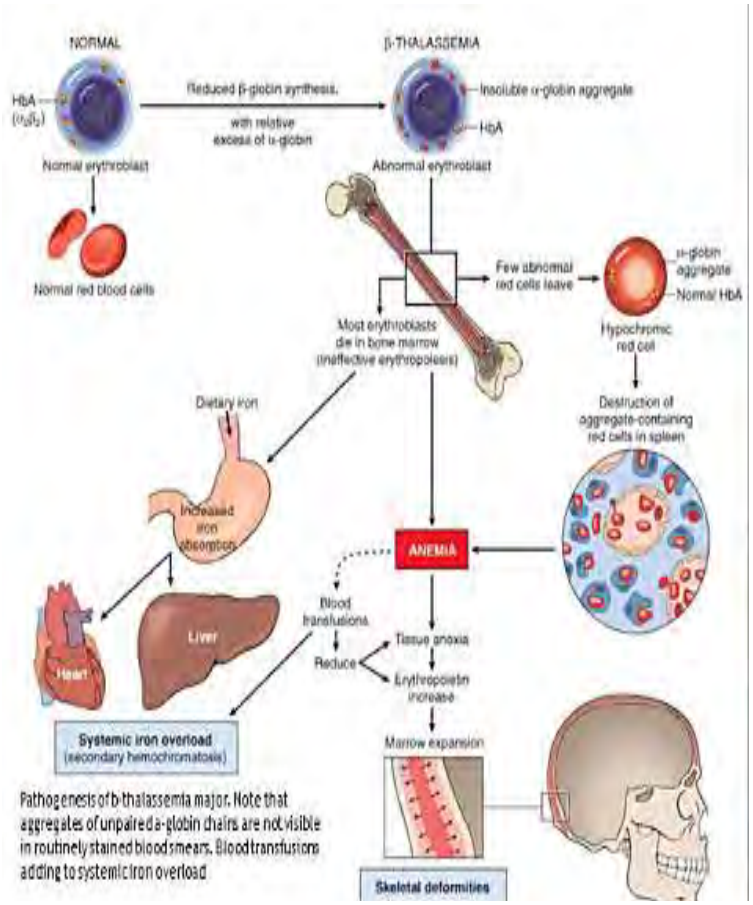
extracorpscular

Immunologic
⬇
Coomb's +ve Autoimmune

Non immunologic
⬇
Hypersplenism

General C/F

1. Features of anemia.
2. Jaundice.
3. Splenomegaly due to extra medullary hematopiosis.
4. **Skeletal changes due to BM expansion :**
   - Head:
     - Macrocephaly.
     - Depressed nasal bridge.
     - Prominent maxilla.
     - Prognathism.
   - General osteoporosis.



ACUTE HEMOLYTIC ANEMIA

(1) Glucose 6 phosphate dehydrogenase deficiency

- Hereditary transmitted: X-linked recessive more common in male.

Pathogenesis

G6PD deficiency involved in HMP shunt which responsible for generation of NADPH and reduced glutathione that protects RBC from oxidation with deficiency of G6PD exposure to oxidant agent result in oxidation of Hb to methehemoglobine that precipitate inside RBC as Heinz bodies this abnormal RBC rapidly removed from the circulation leading to acute anemia.

C/P :

Picture of acute hemolysis 1-2 days after ingestion of an oxidizing agent:-

- **Antipyretic** : Aspirin.
- **Antibiotic** : Chloromphinicol nalidixic acid.
- Anti malarial drugs chloroquine and sulfonamides
- **Food :** fava bean "favism".
- Infection.

- Sudden pallor "acute hemolysis".
- **Dark urine** :pink to red " hemoglobinuria".
- Tachypnea and tachycardia "due to acute hypoxia".

Investigation:

➲ To approve that anemia is acute hemolytic anemia:

- Acute anemia with fragmented RBC, Reticulocytosis above 5%.
- ncreased unconjugated biliurbin, hemoglobinuria.

➲ **To approve that anemia due to G6PD deficiency:**

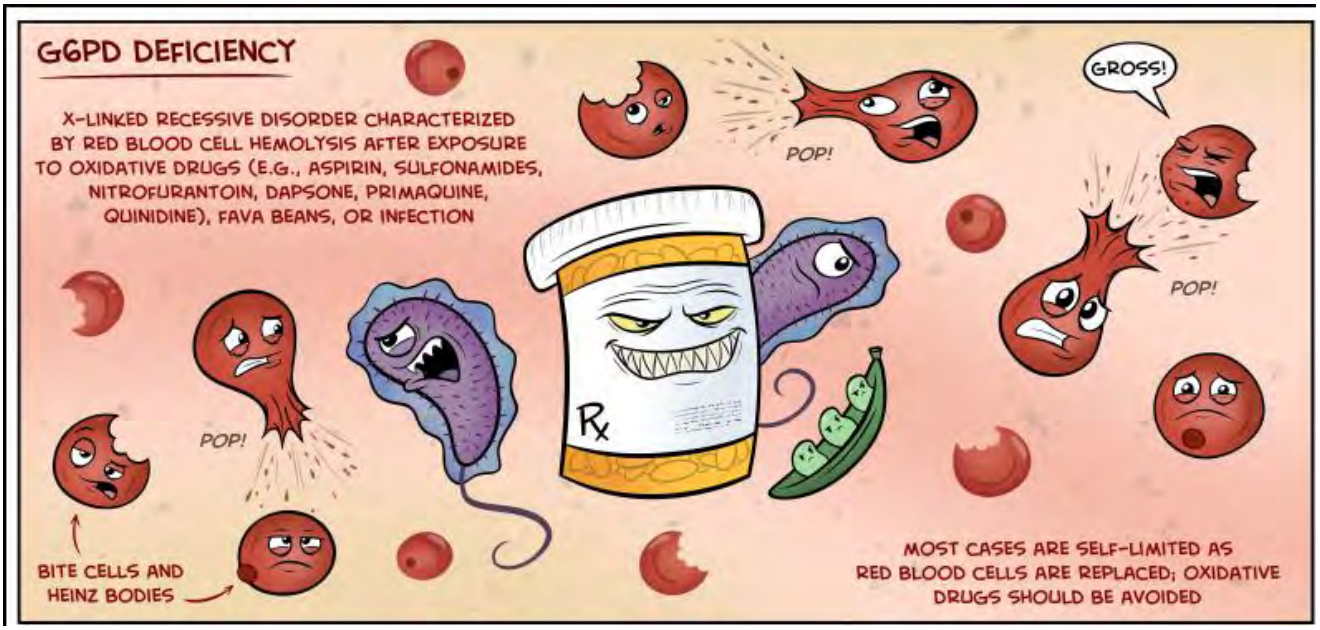
- Estimation of the activity of the enzyme after 3 wks of the attack B/C immediately after the attack there is high level of immature RBCs with relatively higher enzymatic activity that may give false normal result.

Treatment:

- ✔ Avoid all oxidization agent, parent should keep a list containing these agents.
- ✔ **During the attack**: packed red cell transfusion 10ml/kg.

(2) Other causes of acute hemolytic

1. **Hemolytic uremic syndrome**: ARF, thrombocytopenia, MAHA.
2. **Crises of chronic hemolytic anemia:** Splenomeglly or hepatosplenomegally.
3. **Autoimmiune hemolytic anemia:** Splenomeglly, marked Reticulocytosis, +ve coomb's test.
4. Rh and ABO in compatibility or incompatible blood transfusion.
5. **Infection or sepsis** :fever with high ESR and +ve CRP.
6. Metabolic hemolytic anemia e.g. Wilson disease.



Thalassemia

Group of genetically determined chronic hemolytic anemia characterized by defective synthesis of one of the globins' chains forming the major adult hemoglobin.

Classification:

1. **β-Thalassemia :** inability to form β chain of Hb .
   It occur in three fromes:
   1) Thalassemia major "cooley anemia": sever illness.
   2) Thalassemia intermedia: late onset and moderate.
   3) Thalassemia minor: mild form and usually asymptomatic.
2. **α-Thalassemia :** deficient synthesis of α chain of Hb "rare".
   - 4 gene involved → incompatible with life.
   - 3 gene involved → α thalassemia major.
   - 2 gene involved → α thalassemia minor.

ßThalassemia major

The commonest cause of chronic hemolytic anemia.

Etiology: ⇒ Hereditary transmitted as AR (2gene involves) on chromosome 11

Pathophysiology:

- Red cell destruction → Hb loss → anemia, jaundice and hemosidosis.
- Trail for compensation:
  - Bone marrow → reticulocytosis, hyperplasia, and expansion of marrow space.
  - Reticuloendothelial system → hepatosplenomegaly and lynphadenopathy.
- It present in the last part of the 1st year when gamma chain synthesis stops and beta production does not occur.
- It is undetectable in the early months of life due to relatively large amount of Hb F.

C/P:

➲ **History:**

- **Onset:** Last part of the 1st year
  " > 6months ".
- Progressive pallor mild jaundice and abd. distension "Hepatosplenomegaly".
- H/O repeated blood transfusion.
- FTT in early childhood is a common presentation.
- **H/O of crises:** Vaso-occlusive occur only in sickle cell anemia, while other crises can occur in any type of chronic hemolytic anemia.

Examination

1. General examination:

- **Measurements:** Stunted growth due to chronic anemia and endocrinal disturbances.
  Large head due to expansion of medullary spaces.
- **Vital signs**:
  Manifestation of hyper dynamic circulation e.g. collapsing pulse.
- **Head:**
  - Pallor, mild jaundice "due to slow rate of destruction and increased conjugation.
- **Skin color:** Greenish brown "pallor + jaundice + hemosidrosis".
- **Mongoloid features:** Due to expansion of marrow spaces in the face :-
  - Depressed nasal bridge.
  - Prominent maxilla.
  - Protruding upper central incisors.
- **Neck:** Lymphadenopathy.

2. Systemic examination:

- **Abdominal:** Huge splenomegally and hepatomegally as they are sites for :
  - Distraction of RBCs.
  - Formation new RBC.
  - Deposition of iron over load..
- **Heart**: Hemic murmur cardiomyopathy.

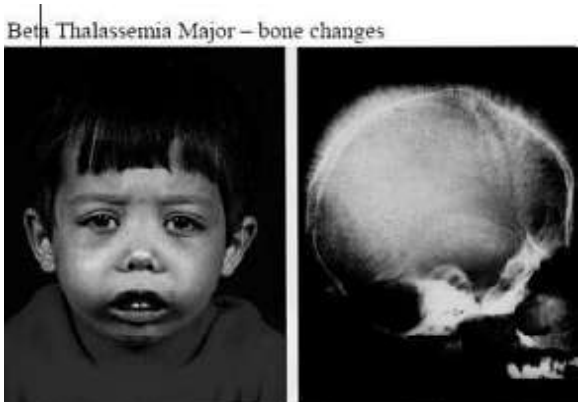
Complication: Are mainly due to anemia and iron overload "from repeated blood transfusion"

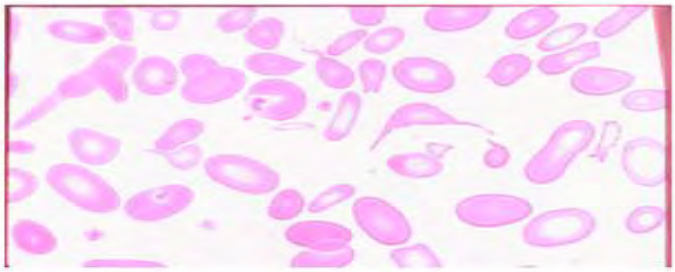
Systemic:

Cardiac : Anemia, heart failure, and arrhythmia due to hemosidrosis and anemia cardiomyopathy.

Hepatic : Liver cirrhosis and failure from iron overload and blood borne infection.

Endocrinal : DM and delayed puberty.





Investigation

1. To diagnose anemia as chronic hemolytic anemia :

➲ **Laboratory:**

- **blood picture:** chronic anemia "persistently low Hb and low RBC count"

reticulocytosis : reticulocytosis above 2% (normal 0.5-2%)

➲ **Blood chemistry:**

- Increase serum iron.
- Decrease iron binding capacity.
- High serum ferritin.
- Increase uncojugation biliurubin

➲ **Imaging:**

- Skull x-ray (late and not imp. finding for diagnosis)

2. To diagnose that the cause of chronic hemolytic anemia is thalassemia.
Microcytic hypochromic anemia with anisocytosis poikilocytosisand target cell

N.B:

➲ **Anisocytosis** : Variation in size
➲ **Poikilocytosis** :Variation in shape
➲ **Target cell** :Central area of density Hb with surrounding pallor
➲ Hb electrophoresis increase HbF increase HbA2
➲ High HbF (10-90%) and reduced major adult Hb

Complication of blood transfusion :
Reaction
Disease transmission

D/D :

- Other cause of chronic hemolytic anemia.
- Sickle cell anemia : vaso-occlusion crisis Hb electrophoresis Hbs.
- Spherocytosis : sherocyte in the blood film and +ve osmotic fragility test.
- Other causes of microcytic hypochromic anemia.

Treatment :

* β.thalassemia is the most common cause of transfusion dependent anemia.

1) **Correction of anemia** : Transfusion of packed RBC : 15-20 ml/kg to maintain Hb between 10.5-11g/dl
   - Every 4-5 wks ((hyper transfusion))

★ Advantage of hyper transfusion therapy :

1. Allows normal activity.
2. Prevent progress marrow expansion, decreased the cosmetic complication.
3. Minimizes cardiac dilatation liver and splenic enlargement.

Folic acid daily to prevent megaloblastic changes in the B.M

N.B: Accurate measurement of blood transfusion= actual Hb - pt Hp × wt × 5
Or packed RBC 10-15ml/kg and while blood 20ml/kg.

Removal and prevention of iron overload by iron chelating agents:
Desferioxamine "desferal" S/C by pump over 10 hrs 5-6might /wk.

2. splenectomy:

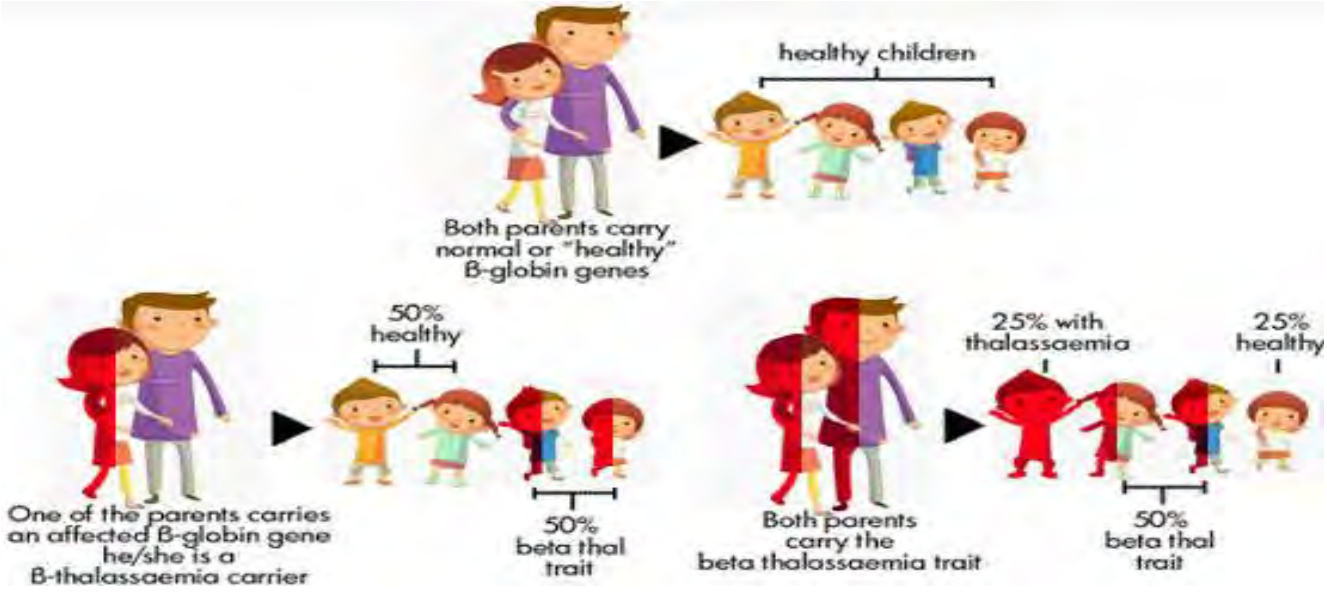
☑ **Indication:**
   i. Huge spleenism
   ii. Huge spleen cause pressure symptoms

☑ **Timing** : Should Not removed before 4y for to avoid sepsis.

☑ **Care before splenectomy :**
   - Vaccination against pneumococci, meningococci, and haemophilus influenza B.
   - Long acting pencillin monthly long life.

B.M:

- Transplantation using marrow cells or peripheral stem cells.
- Induction of fetal Hb production by drugs e.g :l carnitin hydroxyurea
  ↓ alpha chain
  ↓ hemolysis



ßThalassemia minor

- Picture of microcytic hypochromic anemia with no obvious signs of hemolysis.
- Hb electrophoresis increased HbA2 "minor adult".
- **The condition is suspected when a pt with microcytic hypochromic anemia fails to respond to iron therapy**

α Thalassemia

Impaired α chain production due to deletion of one or more of 4 alpha chain globin gene on chromosome 16.(two genes on each chromosome)

A. One gene :silent carrier

- Asymptomatic without any hematological abnormalities.
- **Electrophoresis** in neonatal period show < 3% bart'sHb **(4 Gamachian)**
- 4 globin chain which disappear few months later.

B. Two gene deletion

- Alpha thalasemia triat.
- Microcytic commonly mistaken as iron deficiency anemia
- Normal iron indices
- **Electrophoresis show :**
  - 3-8 bart's hemoglobin in neonatal period which disappear in childhood.
- Normal level of HF and HA2 for age.
- **Dx**: by exclusion
- To confirm DNA analysis

C. Three gene deletion

Hb H disease ((4ß chian))

Mild to moderate microcytic anemia at birth Evidence of chronic hemolysis

Electrophersis show

15-30 bart's Hb in neonatal period

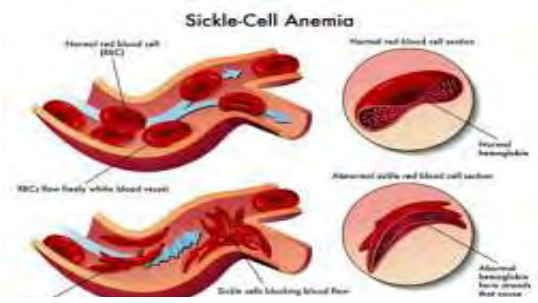
D. Four gene deletion

"fetal hydrops = sever intraction"

- Anemia and anemic HF resulting IUFD stillbirth or early neonatal death
- **Electrophoresis:**
  Dominant bart's Hb with complete absence of normal fetal and adult Hb
  i.e. : require B.M transplantion

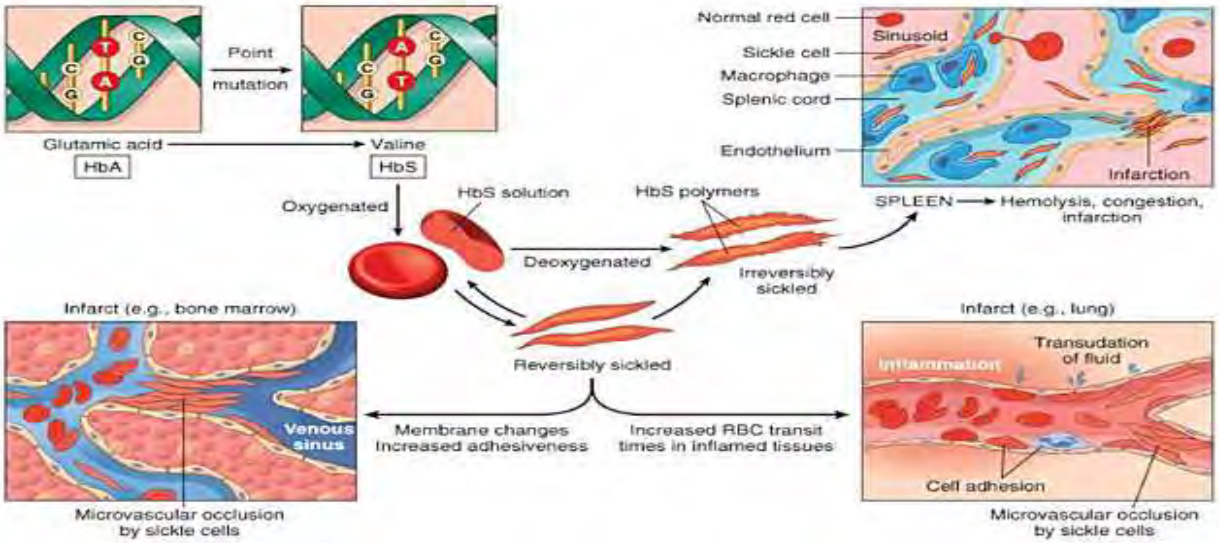
Sickle cell anemia

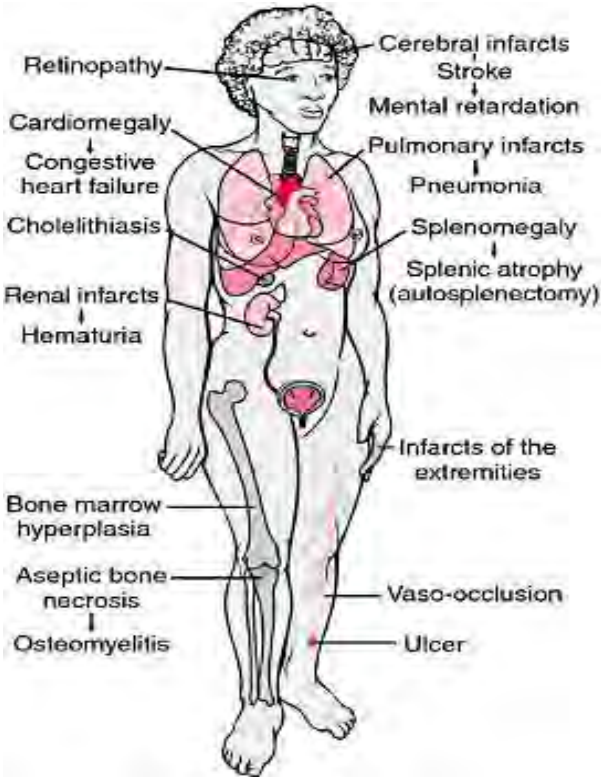
- **Incidence:** more common in black Africans.
- **Etiology**: transmitted as AR.
  - Homozygous form "two abnormal genes": sickle cell anemia
  - Heterozygous form "one abnormal genes ": sickle cell triat.



Pathogenesis

- Single amino acid substitution in the beta chain "a,a number 6 is valine instead of glutamic acid" → Hb S"sickle Hb" is formed instead of HbA"major adult Hb".
- Hbs in the deoxygenated state forms insoluble spindle shaped crystals that distort the RBCs shape "sickle shape instead of biconcave"
- Sickle cell are prematurely destroyed "being rigid" and increase blood viscosity, resulting in vascular thrombosis and infection.





C/P :

Onset:

- It presents in the last part of the 1st year when gamma chain synthesis stops and abnormal HbS is produced in excessive amounts.
- It is undetectable in the early month of life due to relatively large amount of HbF.
- **General C/P of chronic hemolytic anemia:-**
  - Chronic anemia "pallor", jaundice and hepatomegaly.
  - Spleen initially enlarges but repeated episode of infarcation cause its fibrosis by age of 5-6 "autosplenectomy"

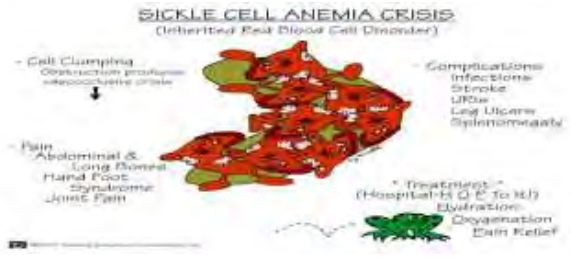
★ **Crisis:** one or more of different crises may occur:

1. **Vaso-occlusive crisis:**
   - **Incidence:** the most frequent "peculiar for sickle anemia".
   - **Predisposing factor**: infection, dehydration, and hypoxia.
   - Manifestation are due to vascular thrombosis and infarction in different sites:-
     - Bones "metacarpals and metatarsals ": painful swelling of the hands and feet "dactylitis"→ hand foot syndrome.
   - **Abdomen**: sever abd pain.
   - **Spleen**: repeated episodes of infarctions cause its fibrosis → autosplenectomy.
   - **Treatment** :
     - Hydration.
     - Analgesics.
     - Oxygen.
     - Exchange transfusion.

2. **Sequestration crisis:**
   For unknown cause large amount of blood become acutely pooled In the liver and spleen shock and marked enlargement of the liver and spleen.

3. **Aplastic crisis:**
   Transient episode of B.M failure that lead to severe anemia, there is failure RBCs production and reticulocytopenia.

4. **Hyper hemolytic crisis:**
   Patient with sickle cell anemia who have addition G6PD deficiency, it lead to acute severe anemia, hemoglobinuria "dark urine" and marked reticulocytosis.

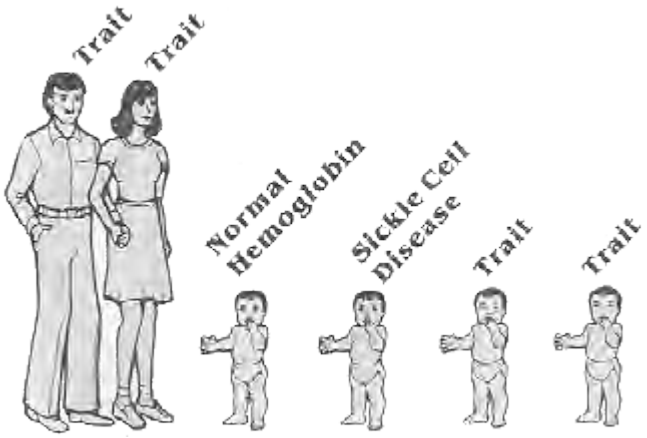


Treatment :

- General lines of treatment as thalassemia is "packed RBC transfusion iron chelating agents" but **no place for splenectomy as there is autosplenectomy**".
- Bone marrow transplantation in severe cases.
- Hydroxy urea in sever case.

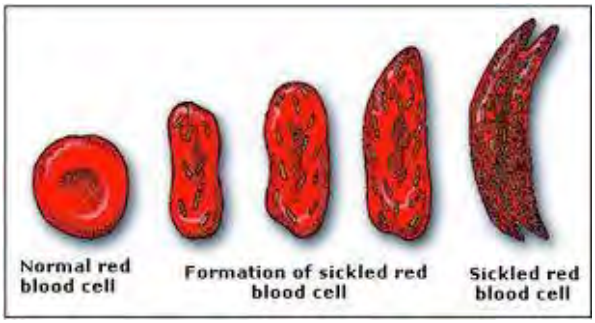
★ *Sickle cell trait (hertozygous form):*

- Sickling does not occur under physiological conditions but in sever hypoxia, the patient is usually asymptomatic.
- HbS constitute 20-40% of total Hb mass and HbA 60-80%.



Investigation :

- ➲ To prove that anemia is chronic hemolytic.
- ➲ To diagnose that the cause is due to sickle cell anemia.
- ➲ **Blood film** sickle shaped cell.
- ➲ **Hb electrophoresis**: increased Hbs "90-100%" , abscent HbA "major adult".
- ➲ Sickling test.



Hereditary spherocytosis

Hereditary transmitted as an AD

Pathogenesis :

- Cell membrane defect.
- Red cell membrane is more permeable to Na^+ → spherocytic red cell, increased utilization of energy to drive Na-outside the cell "Na pump" → premature senility and destrucation. (deficiency of spectrin or ankyrin).

C/P :

Onset: may be as early as the 1st few days of life with anemia and neonatal jaundice.

- Picture of chronic hemolysis "pallor jaundice, Hapatosplenomegly and mongoloid feature".
- + FH , Gallstone occur in 50 % of un splenctomized by 4-5years

Investigation :

➲ To prove that anemia is chronic hemolytic anemia"see before".

➲ **To diagnose spherocytosis** :

- Blood film : spherocytes small rounded without central pale
- Osmotic fragility test reveals increased fragility.

Treatment :

The same as thalassemia, **splenoectomy** improves the condition markedly "prevent hemolysis, crises and gall bladder stone".

Osomatic fragility test

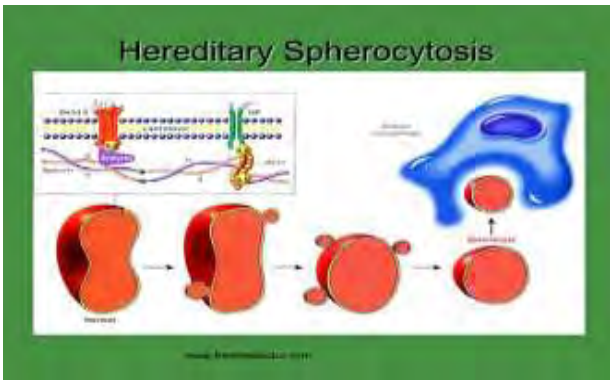
- Normally exposure of RBCs to hypotonic solution causes it to swell and rupture
- Spherocyte "already swollen cell" lyse more readily than normal biconcave cell in hypotonic solutions.

Deficiency of cell wall skeleton protein "spectrin or ankyrin"

Increase water content inside RBCs → cell become spherical with rigid cell wall

Over work of Na/K pump to keep Na out the cell increase "ATP and glucose consumption".

RBC are trapped and destroyed in spleen



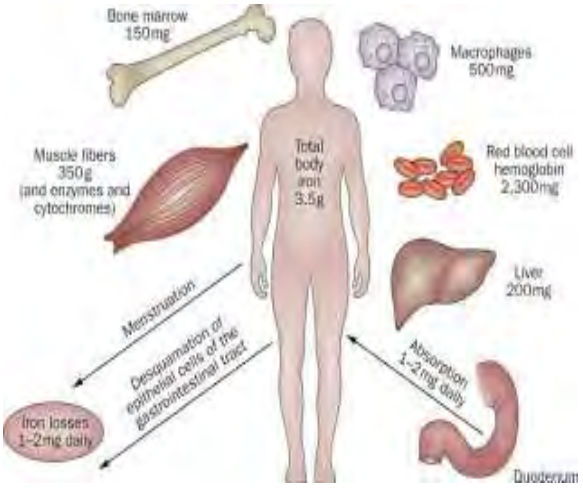
The most common cause of anemia in infancy more common at 9-24 month.

Physiology consideration : (Iron metabolism)

- **Daily requirement** : 10mg/day (10% absorbed : 1mg/dl).
- **Body content** : 0.5gram in neonate, 5gram in adult.

▸ Iron is absorbed from breast milk more efficiently than cow's milk. (But low iron content in breast milk).

- **Digestion :** Begins in stomach (combined action of HCl and vit. C) and continue in the small intestine.
  Enters the duodenal cell in the ferrous form, oxidized to ferric and become attached to apoferritin forming **ferritin → storage form**
- **Absorption :** Iron is transferred in the plasma in the ferric state, combined with transferring "iron binding capacity is the amount of free transferring ready to combine with iron 250-300 mg%"
- **Storage:** Iron is then delivered to the liver



C/P :

☑ C/P of anemia :

➲ Symptoms:

- ▸ General: Headache, easy fatigue, lack of concentration.
- ▸ Cardiac: exertional dyspnea and palpitation in severe cases.

➲ Signs:

- ▸ General: Pallor "nail beds and lips" the most important clinical finding, koilonychias "spooning".
- ▸ Cardiac: hyperdynamic circulation, hemic murmur and HF "sever cases".

☑ Specific manifestation of iron def. anemia :

- Irritability and anorexia in advanced cases.
- Pica pagophagia: Eating strange thing e.g dirt.
- Spleen enlarged in 10-15% of cases.

Etiology :

- **Inadequate supply.**
- **Inadequate store at birth:-**
  - Sever iron def anemia in the mother during pregnancy.
  - Fetal loss of blood at or before delivery.
  - Premature and low birth weight infants or twins.
- **Inadequate intake (delayed weaning):-**
  - Prolonged breast feeding after age of 6m without supplementation with iron containing food :
    The most important cause Protein calorie malnutrition and consuming food poor in iron.
- **Inadequate absorption**:-
  - Chronic diarrhea.
  - **Malabsorption:** celiac disease.
- Increased iron loss:- (Meckl's diverticulem)
  - **Hemorrhage**: acute or chronic bleeding -allergy to cow's milk → Iron, lactoferrin, heat labile protein may induce blood loss.
  - Parasite :anchylostoma and bilharziasis.
- **Increased iron requirement:-**
  - Rapid growth in prematurity and adolescents especially girls "menstruation".
  - CHD due to polycythemia.

Investigation :

1. **To diagnose anemia as iron def. anemia** :
   - **Blood picture**:
     - Microcytic hypo chromic anemia.
     - Hb level is low and RBC count is low.
     - Reticulocytic count is normal or slightly increased, it increased in initiation of therapy.
     - MCV is low "microcytic".
     - Reduction in Hb is more marked than RBC count.
   - **Blood chemistry** :
     - ↓ Serum iron (normal :90-150 microgram%).
     - ↑ Serum iron binding capacity: 250-350 microgram%.
     - ↓ Serum ferritin "reflects body iron stores".
   - **Bone marrow picture** :
     - Erythroid hyperplasia "not performed".
2. **To diagnose the cause :**
   - **Stool** : occult blood or parasites.
   - Investigations for GIT bleeding or Malabsorption.

Notes

- As the high Hb conc. of the newborn falls during 3rd month of life considerable amount of iron stores which is sufficient for blood formation in **the 1st 6-9 month ;** So it is unusual for iron def anemia to occur before 6 month but common **after 9-24month of age**.
- Absorption of iron in duodenum in form of "ferrous" then bind to trans ferittin in storage in liver.

Response of iron therapy :

1. **After 12-24**: Intracellular store "decrease irritability, increase appetite".
2. **36-48 hr:** Initial BM response "erythroid hyperplasia".
3. **48-72 hr:** Reticulocyte response peak "5th -7th days".
4. **4-30 day:** Increased Hb conc. "0.25-0.5gram/dl/day".
5. **1-3 month**: Stores repletion

Treatment :

✓ **Prevention :**
- Adequate supply of iron to the mother during pregnancy
- Proper weaning : Iron containing food (green vegetables or meat product) should be give the infant from 6th month.
- Early Dx and treatment of the causes e.g. parasitic infestation bleeding

✓ **Specific treatment:**

➲ **Iron therapy:**

1. Oral :
- Indications : It is the usual route.
- Preparation: Ferrous sulfate - ferrous gluconat.
- Dose: 6 mg/kg/day 3doses b/w meals.
- Course: 4-6 weeks after normalization of all blood values to replete the stores

2. IM :
- **Indications**: Failure of oval route "oral intolerance or mal absorption".
- **Preparation**: Iron dextran.
- **Dose**: 1ml (50 mg) in infant
  2 ml (100 mg) in young children.
- **Course**: 3-5 days.

✓ **Supportive treatment:**

➲ Blood transfusion: (Packed red cell :10ml/kg slowly) in impending heart failure or when there is serious infection.

- Absorption of iron ↑ by: Vit. C and citrate.
- Absorption of iron ↓ by: Antacid, tea, and phytate.

Causes of failure of iron therapy

1. Non _compliance
2. Continuing blood loss
3. Impaired absorption
4. Continuing use of cow milk
5. Coexisting of disease renal infection
6. In correct diagnosis.

Anemia with megloblastic in BM and macrocyte in peripheral blood

Causes :

1. Vit B_{12} "cobalamin".
2. Folic acid.

Pathogenesis :

Folic acid and B_{12} are essential for DNA synthesis in stem cell of RBC , platelets and WBCs.

So

Nucleus can't divide ⬇ ↓ RBC production

while RNA is normal ⬇ ↑↑ Cytoplasm ⬇

- Macrocyte In blood ⬇ Trapped in spleen ⬇ Extramedullary lysis
- Can't leave BM ⬇ Intramedullary lysis

Metabolism :

Vitamin B12

- Source: Animal origin only e.g milk meat .
- Requirement: 5-20 mg/day.
- Absorption: gastric parietal cell release intrinsic factor which binds to B_{12} → B_{12} if complex then absorbed for terminal illium.
- Causes of Vit B_{12} deficiency :

1) ***Dcrease intake***: very rare except breast feeding of vegetarian mothers.
2) ***Decrease absorption :***
   1. Generalized malabsorption
   2. Pernicious anemia:
      - Congenital: no or abnormal IF.
      - Juvenile: Ab against IF and partial cells.
   3. B12/IF consumption by diphylobothrium latum or bacterial over growth.
   4. Illeul disease or resection.
3) ***other:*** defect in transport e.g colalamin II def

Folic acid

- Source: animal &plant &green leaves,fruits
- Requirement: 20-50mcg /day.
- Absorption: From proximal jejunum "need Vit C".
- Causes of deficiency :

1) ***Dcrease intake:*** un common may occur in infant fed on goat milk.
2) ***Decrease absorption:***. Same ; Vit C deficiency.
3) ***Increase metabolism:***
   - Cytotoxic drugs → metatrexate.
   - Anticonvulsant → phynytoin , Valproate.
4) ***other:*** Increase requirement
   e.g. - premature "↓ stores".
   - Hemolytic anemia.

C/P :

a) anemia : anorexia, pallor …etc.
b) GIT manifestation esp. folat def. :
   - Red glazed tongue and glositis.
   - Abd. pain &chronic diarrhea.
c) Neurological manifestation : subacute combined degeneration "SCD".
d) Hyperpigmentation &wt loss

***May occur only in Vit B_{12} deficiency** :

1. Post .Column degeneration → sensory ataxia.
2. Pyramidal tract degeneration → delayed motor mile stone.(Diminished deep reflexes, +ve Babinski sign, +ve column).
3. peripheral nerve degeneration → parasthesia.

Investigation :

1. Is it megaloblastic anemia ?

- CBC :
  - Anemia: ↓ Hb%&Hc value.
  - Macrocytic: ↑ MCV >100ft.
  - Normochromic → MCHC normal.
  - Hyper segmented netrophils "contain 4-5 lobes" early finding.
  - WBCs and platelet are moderately reduced in advanced cases .
- BM : megaloblast.

2. What is the cause ?

a) Folic acid deficiency:
- Folic acid level in RBC → low.
- Low serum folic acid < 3 mg/ml (Normal 5-20mg/ml).

b) B12 deficiency:
- Low serum Vit B_{12} < 100mg/dl.
- Schilling test:small amount of radioactive vit B_{12} is given orally.
- 1mg non radioactive Vit B_{12} is given I.M to saturate Vit B_{12} binding proteins.
- Normally 10-30% of Vit B_{12} is excreted in urine.
- In Vit B_{12} malabsorption → < 2% is excreted.
- If the test is corrected by oral IF → consider IF deficiency.
- If the test is not corrected by oral IF consider illeal disease.

Treatment

1. Folic acid deficiency:
- Folic tablets 0.5 -1mg/day for 3-4 wks.
- If Dx of folic acid deficiency is doubtful use smaller dose 0.1 mg/day for a week and look reticulocytosis "therapeutic test".

(Large dose of folic acid worsen the neurological manifestation of B_{12} deficiency)

2. Vitamin B_{12} deficiency:

a) Without SCD:
- B_{12} IM monthly for life.

b) With SCD :
- 1 mg IM for 2 weeks than 1mg monthly for life.

With successful treatment reticulocytosis is seen in 2-4 days

3. Other :
- gastric function test
- IF Ab
- Small bowel biopsy

- Defect iron utilization.
- X-linked may be congenital or acquired 2ry to :
  Drugs: chloramphenicol
  - ↳ Grey baby syndrome.
  - ↳ Bone marrow suppression.

↳ So impaired synthesis:
- ➲ Iron retention in mitchondrea.
- ➲ Ringed sidroblasts in RBCs precursors in B.M.
- ➲ High serum Iron &ferritin.
- ➲ May respond to vitamin 6.

Investigation
- **CBC:** Hb low, MCV and MCHC low
  "microcytic hypochromic anemia"
- Serum iron high.
- Serum ferrtin high.
- Total iron binding capacity normal.
- **Bone marrow aspiration**:
  Ring sidroblastoma in BM.

Sidroblastic anemia can lead to acute leukemia in small % of patient.

7

- Rare and **AR** disorder mainly affect north European patient has mild anemia and fever symptoms.
- **In sever case:**
  - Sever anemia.
  - Bone marrow hyperplasia
- **Diagnosis** : By specific enzymatic assay "pyravate kinase level".
- **Treatment :**
  - Supportive treatment.
  - Splenectomy in splenectomy in sever case.

Abnormal AB in the blood produced by the body cause hemolysis

Etiology :

- Idiopathic
- 2ry to :
  1. Infection : viral infection e.g infectious mononucleosis, hepatitis B ,CMV, Tb, mycoplasma
  2. Collagen disease as SLE or RA.
  3. Malignancy as leukemia or lymphoma.
  4. Drugs as alpha methyl dopa.
  5. Immune deficiency: e.g. IgA , deficiency, wiskott -aldrich syndrome.

C/P :

- Manifestation of etiological disease.
- Manifestation of the hemolytic anemia which may be acute or chronic with acute exacerbation "pallor, jaundice dark urine, splenomegly ± hepatomegally".

Investigation :

1. Normochromic normocytic anemia with reticulocytosis.
2. Spherocyte are usually seen in blood film.
3. ↓ Haptoglobin.
4. Coomb's +ve.
5. BM is hyperplasia

Treatment :

- Packed RBCs transfusion.
- Treatment of underlying cause.
- In 1ry cause "idiopathic":
  Corticosteroids, cytotoxic drugs as cyclophosphamide, or splenectomy.

D/D of treatment with Ig:

1. Gullian barr syndrome.
2. Mysthenia gravis.
3. ITP.
4. Worm type of HA.
5. Kawasaki disease.

✶ Types of auto immune hemolytic anemia:

	Cold type	Worm type
Temp	< 4°C	> 37°C
Immunglobin	IgM	IgG
Incidence	20%	80%
Cause	• Idiopathic • Lymphoma • Ulcerative colitis • Infection "mycoplasma"	• Idiopathic • Lymphoma • Chromic lymphoblastic leukemia • Drug: ᾱ methyl dopa
Destruction	Liver	Spleen
Treatment	Without treatment self limiting	• Steroid • IV immunoglobulin • Splenoectomy

Hypersplenism

- Functional disorder and not an anatomic change in the spleen, it is usually associated with some degree of splenomeglly, hypersplinsm is associated with netropenia, thrombocytopenia and anemia either singly or combination, B.M is normal or hypercellular

Etiology:
1. Pirmary
2. 2ry to:
   - Bilharziasis.
   - Thalasemia.
   - Hodgkin's disease.
   - Chronic leukemia.

Mechanism :
Sequestration and increase destruction of blood cell in the spleen.

Treatment:
- Treatment of underlying cause.
- Splenectomy.

- Aplastic of the precursors of 3 blood element in BM leading to pancytopenia in the peripheral blood "anemia, purpura, infection".
- Splenomeglly and lymphadenopathy are **absent** (Important differentiating point from leukemia).

1. Congenital or constitutional aplastic anemia: (Fanconi anemia)

- Hereditary ,transmitted as AR defect in DNA synthesis.

C/P :

1) **General manifestation:**
   - **Skeletal anomalies:** present in 50 % of cases, microcephally, short stature, absent thumb and radius, congenital hip dislocation.
   - Microophthalmia and generalized hyper pigmentation also caf. aulit's spot.
   - CHD.
   - Ptosis.
2) Hematological manifestation
   - Present usually above the age of 3 years with an average 6-8 y.
   - Thrombocytopenia start 1st ,followed by progressive severe anemia, and then leucopenia, the disease progresses slowly

Investigation :

- **CBC:** pancytopenia, macrocytosis, ↑ MCV.
- **BM aspiration and biopsy** : aplastic BM "hypocellular with depressed precursors of the 3 blood elements".
- **Chromosomal study** : increased chromosomal breakage.

Treatment :

1. **Supportive treatment**:
   a. <u>Control infection</u>:
   - By good oral hygiene.
   - IV Ab according to C/S.
   - In case of granulocytopenia "netrophil less than 500 cell/mm^3" → give granulocyte transfusion or granulocyte colony stimulating factor.

   b. <u>Contral bleeding</u>:
   - Avoid IM injection.
   - Platelet transfusion if plt count <20,000
   - Contral anemia:
     - Erythropoitin "S/C".
     - Packed RBC if Hb <7 gm/dl.
2. **Specific treatment:**
   - Oral predinsolone with or without androgen → both give effective 50%.
   - M.O.A
     - Androgen: stimulate erythropiosis.
     - Steroid:
       - Release erythrocyte form storage site & maintain capillary integrity.
       - Release storage element from BM.
   - BM transplantation.
     - **Prognosis:** survival may years death form complication.

N.B: Any patient with fanconi anemia should done U/S abdomen B/C may complained of congenital renal anomalies e.g. agenesis or ectopic kidney or horse show kidney.
Fanconi anemia have high risk of leukemia mainly acute myeloblastic leukemia .

2. Acquired a plastic anemia

Etiology :

- Idiopathic.
- Infection e.g. I.M.N, hepatitis C&B.
- Drug e.g chloramphinicol
- Radiation.
- Chemical e.g. benzin containing products.

C/P :

The history is relatively short with purpura anemia and repeated infections.

Treatment :

- **Bone marrow transplantation** is treatment of choice.
- **Immunosuppresion :**
  Drugs as cyclosporine can be used in same cases "immune suppression of hematoposis is suggest to be responsible case with post infectious and idiopathic aplastic pancytopenia".

Diagnosis :

By B.M aspiration.

★ **Pure cell aplastiv : diamond - black fan syndrome**

- Present in the 1st few months of life with anemia.
- Some child response to cortisone but may become transfustion dependent.

★ **Shwach man diamond syndrome**

- Pancreatic exocrine failure with skeletal abnormalities.
- Bone marrow dysfunction "pancytopenia or isolated neutropenia" change to leukemia up to 25 of cases.

Hemorrhagic disorders

1. **Platelet disorders :**
   a. Decrease number (thrombocytopenia):

i. Decrease number plt production:

B.M Depression

Congenital

- Fanconi anemia
- TAR syndrome
- Wiskott Aldrich syndrome.

Acquired

- Drugs" Chloramphincol".
- Idiopathic.

Bone marrow infiltration

- Leukemia
- Neuroblastoma

ii. Increase plt destruction:

Immune

- ITP
- Collagen vascular as SLE, marfan's.
- Infections: meningococcal infection.

Non immune

- ↑ Plt consumption e.g DIC, MAHA "HUS"
- Hypersplenism
- Big hemangioma

"Plt destructed in small capillaries" e.g. kasabach merit syndrome

b. Abnormal Plt function:

i. **Hereditary**:
- Glanzmann thrombasthenia.
- Bernard soulier disease.
- Vonwill brand disease.

ii. Acquired:
- Drugs as aspirin.
- CRF.
- Malignancy as leukemia.

2. Vascular disorders :

ii. Hereditary:
- Hemorrhagic telangiectasia.

i. Acquired:
- CRF.
- Collagen disease as SLE
- Septicemia as meningococcemia
- Nutritional e.g. scurvy, PEM
- Allergic : HLP.
- Whooping cough.
- Infection endocarditis

Coagulation disorders

Congenital
- Hemophilia A,B,C.
- Afibrinogenemia.
- Von willibrand disease.

Acquired
- DIC.
- Liver disease.
- Hemorrhagic disease of newborn
- Deficiency of vit. K e.g. in malabsorpation syndromes.

- The commonest plt disorder is ITP
- The commonest coagulation are DIC and hemophilia

★ Tests for evaluation of the haemostatic mechanism:

A. <u>Tests for vascular and plt disorder</u> :

1. Bleeding time:
   - To assess vascular Integrity and plt Adequacy as regareds the number and function.
   - Normal value 4-8 min
2. CBC : plt number (normal =150 to 450x10^3/mm^2).
3. BM examination: to assess the state of megakryoctes.
4. plt morphology and function: By measment of clot retraction special test for plt adhesion and aggregation.

B. <u>Test for coagulation factors</u> :

1. Clotting time.
2. Thrombin time:
   - Normal thrombin time 15-20sec.
   - Prolongation occur in Deficiency of factor II, V, VII, X alone or combination.
3. Partial thromboplastin time "PTT".
   - Normally = 25-40 sec.
   - Prolongation occurs in def of factor VIII,IX and XI.
4. Specific assays for individual clotting factors:
   - To detect the deficiency factor and degree of deficiency.

	Bleeding defect	Coagulation defect
Purpura	+	-
Ecchymosis	+	++
Hemathrosis	-	+
Precipiting factor	Spontaneous	Minor trauma
Bleeding time	Prolonged	Normal

Coagulation time	Normal	Prolonged

Allergic or anaphylactic purpura
Henoch schonelein purpura

Etiology :

- It is unknown , however it is considered as hypersensitivity vasculitis involving small blood vessel , joints, skin ,gut, kidney.
- Most cases there is history of preceding upper respiratory tract infection.
- Hypersensitivity vasculitis may be due to virus antigen antibody reaction.
- Age 2-8 years.
- Male > female 3:2.

C/P :

(1) Skin rash : "100% of cases".
- Recurrent attack of urticarial and maculopapular lesion which change to petechial and purpuric rash.
- Distribution: in **extensor surface** mainly knee, elbow and buttock.
- Sometimes erythema multiform.
- Sometimes angioneurotic edema.

(2) Abdominal manifestation: 65%
- They result from petechial Hge and/or edema in bowell wall.
- Sever coliky abd. pain vomiting is common and occasionally associated with hematesis, melana or occult blood in stool may be present.
- The most important complication is **intussecepation.**

(3) Joint manifestation: "65% of cases" :
"artralgia or arthritis"
- It may poly or monoarthritis, ankle and knee are the commonest sites.
- Joint finding may resemble rheumatic arthritis (transitory, affect big joint which become swollen and tender no deformity)

(4) Renal manifestation: 20% of cases
- The clinical picture resembles that of acute post streptococcal GN (hypertension, edema, oliguria, and hematuria).
- Serum complement is normal.
- Some cases may progress to renal failure "the main cause of death".

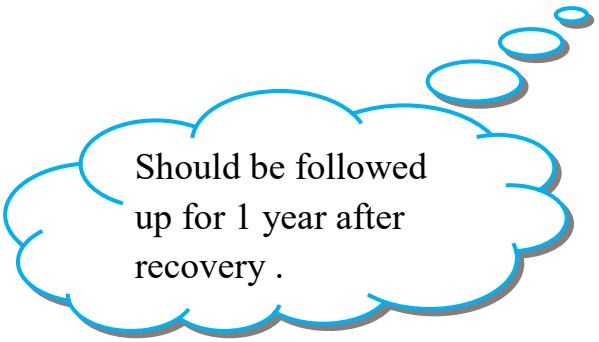
(5) CNS manifestation: "not common"
- Facial nerve palsy convulsion hemi paresis and coma, most of these cases have been attributed by hypertensive encephalopathy.

Libratory finding :

- Plt normal (number &function)
- Normal bleeding time .
- Normal clotting time .
- Normal serum complement.
- Serum IgA ↑ in 50% of cases .

➲ Other test to evaluate the cause:

- Occulted blood in stool.
- U/S abd. for intussusceptions.
- Urine analysis and RFT.



Treatment

- The treatment mainly symptomatic to control joint abd. manifestation.
  - Analgesic for colicky abd. pain.
  - Prednisone "2mg/kg/day" is indicated in severe cases and if CNS manifestation occurs.
  - RF and intussusceptions when present should managed accordingly

Prognosis:
Complete recovery is the rule unless sever renal disease occurs.

Idiopathic Thrombocytopenia Purpura (ITP)

- The commonest thrombocytopenia in childhood, it is characterized by :
  1- Thrombocytopenia (platelet count less than 100,000/mm^3).
  2- Short life span of platelet.
  3- Normal or increase number of megakaryocytic in BM.
  4- Absence of primary disease.
  5- Presence of anti platelet antibody in plasma.

The condition may be acute , chronic or recurrent:

➲ **In the acute form**:
- The platelet count returns to normal within 6 months after diagnosis and relapse does not occur.

➲ **In the chronic form**:
- The platelet count dose not returns to normal after 6 months, in the recurrent form the platelet count decrease after having returned to normal levels.

➲ **In adult the chronic form**: is most usual in children the acute form is most usual.

Etiology and pathogenesis :

1. Acute ITP:

Viral infection "as measles, IMN" → Ab formation against the virus → formation of circulating immune complexes, if there is defective phagocytosis of these complexes → deposition of the complexes on the Plt → destruction of the Plt by spleen.

2. Chronic ITP:

- **Autoimmune**: Presence of anti Plt antibody i.e. Ab against one of the Plt antigen.
- **The spleen has two roles in the pathogenesis of ITP:**
  a. It is the major site for anti plt Ab production
  b. It is the major site for destruction " or phagocytosis of the plt".

C/P :

- **Age** : 2-8 y
- **Sex** : Male = Female
- The disease present with acute onset of spontaneous or induced bleeding :
  1) **Skin**: Ecchymosis or petechiae usually found in the surface on the anti surface of lower extremities and over bony prominences as ribs, scapula, shoulder, leg, and pubic area but may be present anywhere.
  2) **MM**: petechiae may be found in the subconjuctival buccal mucosa and soft palate, bleeding from the nose, gums, MM of GIT or urinary tract are not uncommon. Menorrhagia may occur and may be sever
  3) **Internal organs**:
     - **CNS :** serious bleeding that may be associated with bleeding from mucous membranes
     - Retinal Heg
     - ***Deep muscle hematoma or hemarthrosis are rare, it is more commonly with coagulation disorder***s.

  With exception of Hge manifestation, the physical examination is not significant. Pallor is usually absent unless there has been significant bleeding. The tip of spleen is palpable in not more than 10% of patient.
  The presence of splenomegally suggest the probability of leukemia, collagen disease…etc.

Libratory finding :

1. CBC :
- Plt count always below 100.000/mm^3 (in mild to moderate bleeding, the account usually below 500.000/mm^3), in patient with severe bleeding, it is usually below 20.000/mm^3.
- Normocytic normochromic anemia if there has been sever bleeding.
- No blast cell.

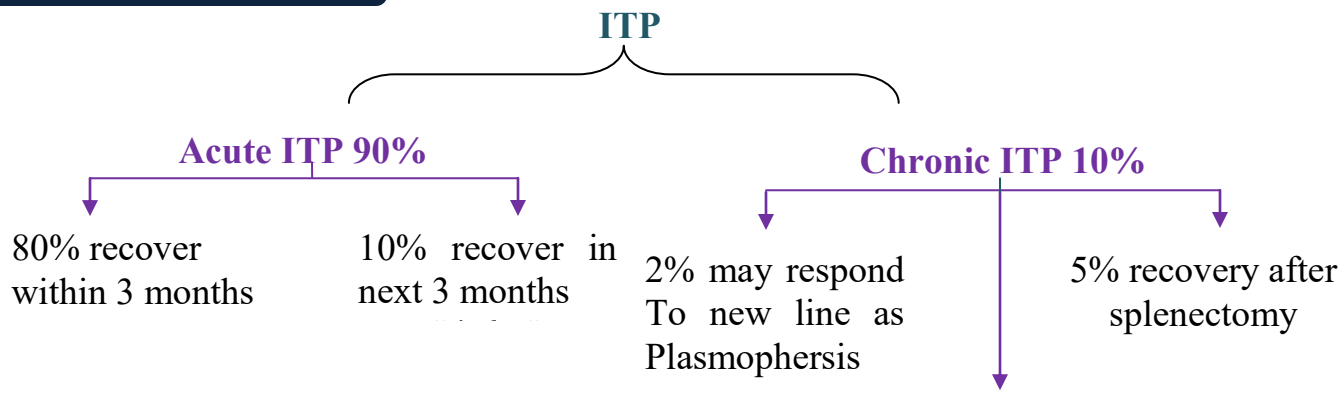
2. **Bleeding time:** Prolonged.
3. **PT, PTT and CT** : Normal.
4. **B.M examination :**
   - Increased number of megakaryocytes, some of which may show little productive activity.

Treatment :

1) No treatment is required if child has mild bruising with no evidence of MM bleeding, avoid use of salicylate and violent sport.
2) In case of severe bleeding, bleeding from MM and if patient count below 50.000/mm^3 Prednisone , or IV immunoglobulins are given to shorten the period of sever thrombocytopenia.
   - **Prednisone:**
     • Inhibit phagocytosis of Plt in the spleen.
     • Inhibits Ab production and improve capillary resistance against leaking Plt.
   - **I.V Ige :**
     • Block the receptor of phagocytes.
     • ↓ Plt Ab and help in clearance of persistent viral infection.
3) **In cases of dangerous bleeding as IC Hge:**
   - Massive doses of Plt + high dose IV methyl prednisolone + IV Ig and emergency splenectomy should be carried out.
4) **In chronic ITP** : Cytoxic drugs as cyclophospham or splenectomy.
5) **Anti D Ig:** can be used in moderate cases.

› **The diagnosis of ITP is** usually straight forward and may be require BM examination, and if there are atypical finding as lymphadenopathy , hepatosplenomegaly, anemia, netropenia or neutrophilia, B.M examination is necessary to exclude serious condition as leukemia.

Course and prognosis:



3% recovery by
ic drug

N.B

★ Splenctomy is indicated in:
- Emergency condition of I.C.Hge
- Chronic ITP not responding to medical treatment.
- ITP does not cause splenomegaly.

D/D from other causes of purpura

Thrombocytopenic
↓ Platelet

Non thrombocytopenic

↓ Production

↑ Destruction

Vascular
Defect:
- HSLP
- Infection (menigococemia)
- Scurvy

Thrombosthenia
- Drug
- Systemic disease e.g. RF.

↓ Production

BM depression
a. Congintal as:
Fanconi anemia.
b. Acquired as
Aplastic anemia
Megalcocytoplasia

BM replacement as
- Leukemia
- Neuroblastoma

Defecinecy
- Vit B12
- Folic
- Thromboplastic.

↑ Destruction

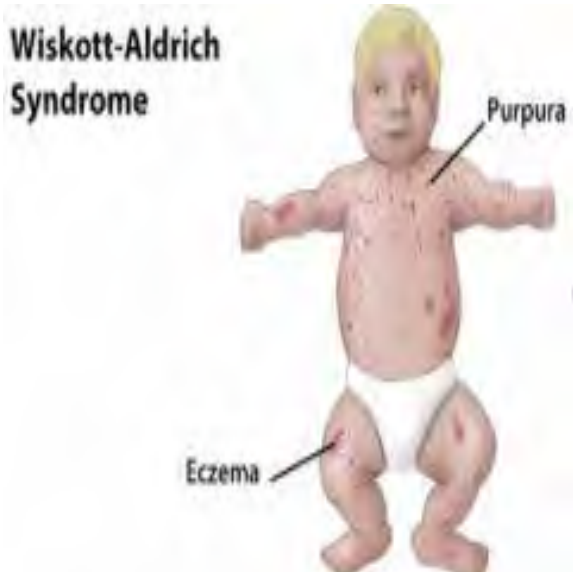
Immune as
- ITP

Non immune as
- DIC
- ITP HUS
- Infection
- Drugs
- Hypersplenism

N.B

In fallow up should ask about :

- Petechia increased or decreased
- Well or un well.
- Level of consious "oriented or not".
- Color of urine.
- Joint pain or swelling.
- Erythemia.
- Retinal affection .
- Any bleeding from any where.



Wiskott Aldrich syndrome :

- X-linked R characterized by triad of :
  - Thrombocytopenia.
  - Eczema.
  - Recurrent URTI.
- It is disease with the abnormal gene located on the X-chromosome that affect the lymphocyte & megakarycyte cell line.

Dx :

- Ig A , Ig E are high.
- Ig G normal or low.
- Ig M low.

C/P :

- Petechia / purpura.
- Eczema.
- Recurrent URTI commonly OM.
- Patient may have hepatosplenomeglly.
- Cervical lymphadenopathy.

Treatment :

- Splenectomy may correct the thrombocytopenia but expose the patient to life threatening infection with encapsulated microorganism so prophylactic Abs is mandatory.
- R.O.C is B.M transplantation

N.B:

- Life span of Plt = 7-10days
- Prolonged use of steroid > 3month BM depression.

Giant hemangioma "kasabach meritti syndrome"

- Large cavernous hemangioma commonly present on skin and rarely in viscera.
- Cause sequestration of plt (thrombocytopenia)
- **Rx :**
  - Surgical removal.
  - Intra lesion steroid injection.
  - Lesser therapy.

- Hemophilia A factor VIII deficiency "**Anti hemophilic globin".**
- Hemophilia B factor IX deficiency **"Christmas factor".**
- Hemophilia C factor XI deficiency "**plasma thromboplastin antecedent".**

Hemophilia A

- The commonest hereditary coagulation disorder, it is due to deficiency or functional abnormality of factor 8, transmitted as **X-linked recessive**.

C/P :

Affected male predominantly, the patient mother is asymptomatic carrier while his brother's and maternal grandfather and maternal uncles are often affected (**+FH is present in 70% of cases**)

The clinical manifestation depend on the degree of def of factor 8:-

- If factor 8 less than 1% of normal → **Sever.**
- If factor 8 less than 1-5% of normal → **Moderate.**
- If factor 8 less than 6-30% of normal → **Mild.**

Laboratory finding :

- Prolonged clotting time.
- Prolonged PTT.
- Normal PT .
- Decrease level of factor VIII by functional assay.
- Prolonged thromboplastin generation time which become corrected by plasma.

- Patient with sever hemophilia suffer from recurrent hemorrhage occurring spontaneously or following minor trauma.
- It occasionally presents in neonatal period with intracranial hemorrhage.
- They are often diagnosed when prolonged bleeding occur after circumcision.
- During early infancy excessive bruising and large IM hematoma result from minor trauma.
- **In childhood**:
  - Epistaxis.
  - GIT bleeding.
  - Hematuria .
  - Intraocular bleeding may occur.
  - **Hemoarthrosis** "bleeding inside the joints" may cause limping and pseudoparalysis of an extremities → IC Hge and bleeding into the neck are life threatening emergencies.
- Moderate hemophilia usually have Hge after trauma.
- Mild hemophilia sustain moderate or severe trauma in order to experience Hge

Complication :

1. Joint deformity 2ry to hemoathrosis
2. Spontaneous or post traumatic IC hge.
3. Hepatitis or AIDS.
4. Inhibitor to factor VIII develop about in 10% of all hemophiliacs leading to poor clinical response to therapy.

Treatment :

1) General measurement :

- Avoid sever injury, encourage non violent exercise e.g. swimming.
- Carful dental hygiene and regular dental examination.
- Avoid I.M injection.
- Avoid aspirin.
- Vaccine against HBV.
- Surgery is avoided but if necessary it should be carried out under cover of replacement therapy and in a hospital which has the laboratory facilities for monitoring the response to this therapy.
- Local measures may be used to arrest the bleeding from small wounds by pressure, immobilization and cold compresses.

2) **Replacement therapy:**

a. **Fresh frozen plasma**: used in emergency since it can be ready available, every 1ml contain 1unit factor VIII.
b. **Cryoprecipitate**: each bag contains 125 unit factor VIII.
c. **Factor VIII concentrations**.
d. **Desmopressin** : vasopressin analog stimulates a transient increase in factor VIII levels. it is useful in mild case.
e. **Anti fibryinolytic therapy** as epsilon aminocaproic acid : used for the prevention or treatment of oral Hge B/C saliva in rich in fibrinolytic enzymes.

Hemophilia B

- Factor IX deficiency.
- X-linked recessive trait.
- **C/P :** Similar to hemophilia A.
- **Treatment** : Fresh frozen plasma or factor IX concentrate .

* It is 6 times less common than hemophilia A.

Hemophilia C

- Factor XI deficiency.
- Auto somal recessive. (M=F).
- **C/P :** Similar to hemophilia A.
- **Treatment** : Fresh frozen plasma.

Von Will brand disease

■ Autosomal dominant, there is deficiency of coagulant component of factor VIII + Von will brand factor has the **following functions**:-

1. Carrier of factor VIII.
2. Platelet adhesion to endothelium.
3. Platelet aggregation.

C/P :

- VWB is released from endothelial cell and plt.
- Good history demonstrate strong FH of similar condition is essential for diagnosis.
- Bleeding is spontaneous from M.M such as gum, nose, menrrhagia and sever bleeding from wound site.
- Hemoarthrosis is rare but may occur in sever disease .

Investigation :

- **CBC**: anemia.
- Prolonged bleeding time.
- Prolonged PPT due to relatively ↓ factor VIII.
- **Factor assay**: low VWF and low F VIII.

Treatment :

- Stop bleeding.
- Give VW factor 40-60 u/kg.
- Give fresh frozen plasma.
- Start ferrous sulphat and vitamin C.
- Sometimes use of anti fibrynolytic drugs.

Disseminated Intravascular Coagulation (DIC)

- **Condition that result from Widespread activation of coagulation in the circulation result in :**
  1) Consumption of some coagulation factor and platelet.
  2) Fibrin deposition and 2ry fibrinolysis.

Pathogenesis :

- The wide spread activation of coagulation is initiated by one of the following mechanism:

A. Endothelial damage of the small B.V.
B. Release of a substance and 2ry fibrolysis.

Etiology :

- Usually 2ry to other disease e.g. :
  - Sever infection e.g. meningococcemia
  - Extensive tissue damage in burns and major surgery.
  - Sock.
  - GE.
  - Acute liver failure.
  - Sever hemolytic reaction.

Laboratory

➲ **CBC :**
- Normocytic normochromic anemia with increase reticulocyte count may present if sever Hge or hemolysis occur.
- Thrombocytopenia.
- **RBC morphology: *fragmented and helmet shape of RBCs.***

➲ Prolonged PT, PTT, TT and BT.
➲ Deficiency of factor I, II, and III.
➲ Blood film show fragmented RBC.
➲ Increased conc. of fibrin degradation product "evidence of ↑ed fibrinolysis".

C/P :

DIC can be diagnosed clinically by the following criteria:

1. Presence of underline disease.
2. **Acute hg manifestation due to consumption of clotting factor:**
   - Bleeding from injection site.
   - I.C.Hge.
   - Melena.
   - Hematemsis.
3. **Ischemia &necrosis**: "due to thrombosis"
   - **Skin** : gangrene
   - **CNS**: disturbed consciousness, convulsion and paralysis.
   - **Kidneys** : acute cortical necrosis leading to RF.
   - **Mesenteric**: illus and intestinal obstruction.
4. **Hemolytic anemia may occur "due to mechanical destruction of the RBCs when they pass through fibrin networks":** pallor + red urine

Treatment :

- Correction of underlying cause.
- Infusion of fresh frozen plasma or platelet.
- Heparin: small dose "100u/6hr" is used in cases with widespread thrombosis as incompatible blood transfusion.
- Anti fibrinolytic drugs as epsilon aminocrpoic acid.

D/D of spherocytosis"

1. H.spherocytosis.
2. Pyruvate kinas deficiency.
3. Autoimmune H.A.
4. ABO incompatibility.
5. MAHA.
6. Hypersplenism.
7. Burns.
8. Sickle cell anemia

Causes of lymphadnopathy :

A) Localized :

1. Acute nonspecific infection.
2. T.B "chronic specific".
3. Hodgkin's early disease.
4. Metastasis.

B) **Generalized** :

- **Viral infection:**
  - Measles , Rubella.
- **Bacterial infection:**
  - TB , Brucellosis, Typhoid.
- **Parasitic infestation:**
  - Toxoplasmosis.
- **CT disease** :
  - SLE.
  - Juvenile RA.
  - Leukemia.
  - Hodgkin's "late disease".
  - Non Hodgkin's lymphoma.
  - Sarcoidosis.

Target cells:

1. Thalasemia
2. Sever iron def A
3. Hyposplenism

Infectious disease By Dr.Asma Hadida

2015_2016

[اكتب اسم الشركة]

Measles (Rubeola)

Etiology: RNA virus "Paramyxovirus" : One antigenic type → One attack give life long immunity.

- Transmitted by droplet infection.
- Incubation period : 1-2 weeks.
- Infectivity period : 5 days before and 2 days after rash.

C/P

1. **Catarrhal stage :**
   - High fever, non purulent conjunctiviy, coryza, cough, sore thorat.
   - **Koplick's spots** "pathognomic" : It is grayish white dot's on baccal mucosa opposite the lower molar teeth, disappear 1-2 days after rash.
2. **Eruptive stage :**
   - Macolopopular rash usually appear on 4th the day of fever, starts hehind the ear near hair line, and spread alone the body in 24 hour.
   - In sever case; petechial or hemorrhagic rash "black measles" followed by desquamation.
3. **Convalescent stage :**
   - Rash fade in order of appearance with desquamation except palm and sole.

Rubella (German measles)

Etiology: RNA virus , one antigentic type give life long immunity.

- Transmitted by droplet infection and transplacental (Rubella syndrome).
- Incubation period : 2-3 weeks.
- Infectivity period : 2 days before and 7 days after rash.

C/P

1. **Catarrhal stage :**
   - Mild fever, nasopharyngitis, enlargement of post. cervical and post. auricular lymph node.
2. **Eruptive stage :**
   - Maculo popular rash less than in measles, starts in face and then involve trunk and limbs.
   - Usually disappear within 3 days (3rd day measls).
   - **Forch heimer spots :** Rose colored spots on the soft palat develop in 20% of patients and may appear before the rash.

Complication :

- Secondary bacterial pneumonia.
- Otitis media.
- Viral encephalitis.
- Subacute sclerosing pan encephalitis.
- Aseptic meningitis and GBS.
- Myocarditis.
- Mesentric lymphadenitis.
- Post measles diarrhea and malnutrition.

Treatment :

- ➲ Suppotive :
  - Adeqaute hydration.
  - Antipyretic.
  - High dose of vitamin A supplement.
  - Treat the complication.
- ➲ Prevention :
  - MMR vaccine.

Complication :

1. Congenital rubella syndrome.
2. Encepalitis.
3. Arthritis.
4. Thrombocytopenia (rare).

Treatment :

➲ Prevention : By MMR vaccine.

Mumps

Etiology: RNA paramyxovirus, affecting the salivary gland.

- Transmitted by droplet infection from human case.
- Incubation period : 2-3 weeks.

C/P

- About 25-30% are subclinical.
- Prodromal symptoms : Fever, maliaise, myalgia).
- Acute non supportive inflammation of salivary glands : Parotid gland usually prece the other, tender swelling push the ear forward and outward , increased by pain, teeth clinching , and swelling, increased maximum over 3 days and less 5 days.

Complication :

1. Orchitis.
2. Oophritiss.
3. Pancreatitis.
4. Thyroiditis.
5. Aseptic meningitis and meningoencephalitis. (F>M)
6. Others : Hearing loss, myocarditis.

Treatment :

- ➲ Isolation for 5 days after onset of clinical symptoms.
- ➲ Supportive treatment :
  - Analgesic and antipyretic.
  - Good diet.
- ➲ Treat the complication.

Roseola infantum

(Sixth disease , exanthem subitum)

Etiology: *Human herpus virus* type 6 (DNA virus) and HHV 7.

- Transmitted by droplet infection.
- Incubation period : 1-2 weeks.
- Peaks age : 6-12 month.

C/P

- **Fever** (often > 40°C) last 3-5 days (Febrile convulsion).
- 1-2 days after fever drops → rash appear (maculopapular and rose like), start in the trunk, then neck, face , and limbs , and rapidly disappear in 2 days.
- Most children with roseola are irritable and appear toxic.

Treatment : Supportive.

✓ Prognosis is excellent

Erythema infectiosum

(Fifth disease , slapped-cheek syndrome)

Etiology: *Human parovirus* B19 (DNA virus).

- Transmitted by droplet infection and transplacental.
- Incubation period : 1-2 weeks (4-14 day).

C/P

- Mild catarrhal stage: Sudden erythema of cheeks, maculopapular rash start in trunk then fade with central clearing giving adistinctive lacy, reticulated rash.

Complication

1. Trunsient arthritis (arthralgia).
2. Transient aplastic crisis in patient with chronic heamolytic anemia.
3. Fetal disease (fetal hydrops).

Treatment : Supportive .

Infectious mononucleosis (Glandolar fever)

Etiology: *Epestein barr virus* "EBV" (DNA oncogenic virus).

- Transmitted by droplet infection and rarely by blood. (virus infect the epithelium then establish in B lymphocyte).
- Incubation period : 1-2 months.

C/P :

- Fever and sore throat.
- Tonsillopharyngitis.
- Lymphadenopathy 90% (commonly cervical group).
- Splenomegally 50% "mild and late finding".
- Hepatomegally 30% and may be hepatitis.
- Maculopapular rash in 5% of patient and 80% of patient who received Ampicillin or Amoxicillin treatment.

Investigation :

- CBC: ↑ WBC , absolute lymphocytosis, and atypical lymphocytosis > 10%.
- Positive heterophole antibody test (Pavl Bvnel test).
- EBV Ig M antibody.

Treatment : ⇒ Supportive

Complication :

1. Upper airway obstruction.
2. Rupture spleen even with minor trauma.
3. Pneumonia
4. Aseptic meningitis, encephalitis, GBS..
5. Nasophyrngeal carcinoma and Burkitt's lymphoma.

Chicken pox

Etiology: *Varcilla zoster virus* (DNA human herpes).

- Transmitted by droplet infection from cases and contact with skin lesion from cases.
- Incubation period : 2-3 weeks.
- Agent : School age group.
- Infectivity period : 2 days before the rash and till the rash crusted.

C/P :

- **Prodroma** : **Fever**, malaise, anorexia.
- **Rash** :
  - Appear 1st on the scalp, face or trunk, then distributed centripetal with little involvement of the limbs.
  - Pattern: Erythematous macules → papule → vesicle → crust.
  - Characterized by itching.
  - All stages of rash present in the same time of vesicles.

Complication :

1. 2ry bacterial infection.
2. Viral pneumonia.
3. 2ry bacterial pneumonia.
4. Rey's syndrome.
5. Meningoencephalitis.
6. Congenital varricella.

Treatment :

➲ Supportive treatment :
- Anti pyretic.
- Anti viral.

➲ Prevention :
- Chicken pox vaccine.

Hand – Foot and mouth disease

- **Etiology:** *Coxachie A virus* (Entero virus) .
- **Transmitted by** oral-oral or feco-oral routes.
- C/P : - Small vesicles and painful ulcer in oral mucosa.
  - In palmer and sole : Red macule, vesicles and papule.
- **Treatment** : Supportive.

BACTERIAL INFECTIONS

Scarlet fever

- Etiology : *Group A - B hemolytic streptococci*.
- Transmitted by droplet infection.
- Incubation period : less than 5 days.

C/P :

- **Sudden onset of fever and sore throat.**
- Tongue :
  - 1st 2 days : white strawberry tongue.
  - On 5th day : red strawberry tongue.
- **Skin rash** :
  - Appear on 2nd day of the disease.
  - Maculopapular rash start around the neck , trunk then extremittes.
  - Rash is more intense in deep creases, don't blanch on pressure **(Pastia's lines).**
  - Dry rough texture to the skin resemble the feel of course **sand paper.**
  - In the face it spares the perioral area, flushed face with circum oral pallor.
  - Rash fades with **peeling** at fingers and toes after 3-7 days.

Investigation :

- CBC : ↑ WBC (neutrophilia).
- positive throat culture.
- ↑ ASO titer.

Complication :

1. Early : Otitis media, retrophyrngeal abscess, meningitis, pneumonia, and arthritis.
2. Late : Acute rheumatic fever, acute GN).

Treatment :

- Supportive treatment.
- Specific treatment : Oral penicillin.

Pertussis (Whooping cough)

- Etiology : *Bordetella pertusis and Bordetella parapertussis.*
- Transmitted by droplet infection.
- Incubation period : 1-2 weeks.
- Affected infant age group less than 1 year (peak < 4 months).

C/P :

- **Catarrhal stage** (1-2wks): Most infectious stage, mild symptoms with fever, cough, and coryza.
- **Paroxysmal stage** (2-6wks): Severe paroxysmal cough, followed by inspiratory whoop and vomiting (forcible inspiration against narrow glottis), in infants, apnea rather than whoop, which is potentially dangerous
- **Convalescent stage** (2-4wks): Gradual deline in severity of paroxysms but cough may last for months.

Diagnosis :

- ➲ Clinically : Sign of RD, apnea, cyanosis.
- ➲ CBC : ↑ WBC (lymphocytosis).
- ➲ ryngeal swab.

Treatment :

- Isolation.
- Care for feeding.
- O_2 supply.
- IV fluid.
- Antibiotic : **Macrolids**, erythromycin, clarithromycin is drug of choice.

Complication :

- Brochopneumonia.
- Otitis media.
- Atelactesis.
- Convulsion.
- I.C.Hge.
- Hernia , rectal prolapsed

Prevention : DPT vaccine.

RICKETTSIAL DISEASE

- Gram negative **coccobacilli** that resemble bacteria but have ***incomplete cell wall***. they contain both RNA and DNA but are obligatory intracellular parasites.
- Rickettsia infect arthropod vector and with the exception of ***louse borne epidemic typhus***, are transmitted to human only .
- *Rocky mountain spotted fever is the most common type*.

C/F:

Fever, **rash**, headache , and resp. S&S , patient with **Q fever** caused by "**coxiella burntti**" do not have rash , but may have pneumonia and hepatitis .

Diagnosis: (Mainly clinically)

- Biopsy "usually from skin"
- Serological tests **"AB IgG , IgM".**
- Weil -felix reaction not recommended .

Treatment :

Chloramphenicol or tetracycline .

★ Rocy mountain spoted fever:

- Transmitted by **ticks**.
- Reservoir "dogs and rodent , and sheep.
- **IP** =1-8 days
- **C/P:** nonspecific with headache , malaise , and fever . **maculopapular rash appears in 90% of cases** , it spread to involve the entire body including **palms and soles**. in early rash blanches on pressure and later becomes **petechial and pupuric eruption**. The patient may have myalgia , thrombocytopenia and hyponatremia are common , splenomegaly and hepatomegaly "33%", complication may also include meningitis encephalitis and hepatitis ,ARF, DIC.

Types	Causes	Vector	Hosts
Spotted fever rocky mountain	R. rickettsii	Tick	Dogs , rodent
Boutonneuse fever (Mediterranean spotted fever)	R. conorii	Tick	Dogs, rodent
Rickettsia pox	R. akarii	Mite	Mice
epidemic typhus / brill-Zinsser disease	R .prowazaki	Body louse	Human
Murine (endemic) typhus	R. typhi	Cat or rat flea , rat louse	Rats
Scrub typhus (scrub typhus)	R. tsutsugamushi	Mite	Rodent
Q fever	Coxiellaburnetii	Tick	Cattle , sheep, goats , cats

Regarding rickettsia (rocky mountain spotted fever) T and F :

a. Pneumocysticcarinii is the causative organism .
b. Ticks are the principle vector
c. Skin rash involves the palms and soles.
d. The weil_felix agglutination reaction is a helpful serological test.
e. Chloramphenicol is the treatment of choice.

Mediterranean spotted fever:

a. Is caused by spirochaetes .
b. Ticks are vectors and dogs and sheep are reservoir .
c. Tache noire is the site of tick bite.
d. Is endemic locally .
e. Doxycycline or chloramphenicol is effective therapy .

Parasitic infections

Amebiasis

Caused by **E.histolytica**, protozoa has two stages infective "**cyst**" and pathogenic stage "**trophozoite**".
Human are the natural host and reservoir .
It is transmitted by food or water and person to person by contact .
IP =7 days ."2days - 4months".

C/P:

- It is produce characteristic flask-shaped ulcer in the intestine .
- Most are asymptomatic .acute diarrhea 6-8 times per day . associated with tenesmus , bloody diarrhea with mucus with few leucocyte. abd. Pain and headache and fever .
- Ameboma "**a mass due to ameba's**" exrtaintestinal . intestinal perforation and hemorrhage .
- Hepatic amebiasis "**amebic liver abscess**"

Diagnosis :

- Stool sample , sigmiodscopically , tissue biopsy.
- Direct hemoagglutinine test .

Treatment :metronidazole .

Malaria

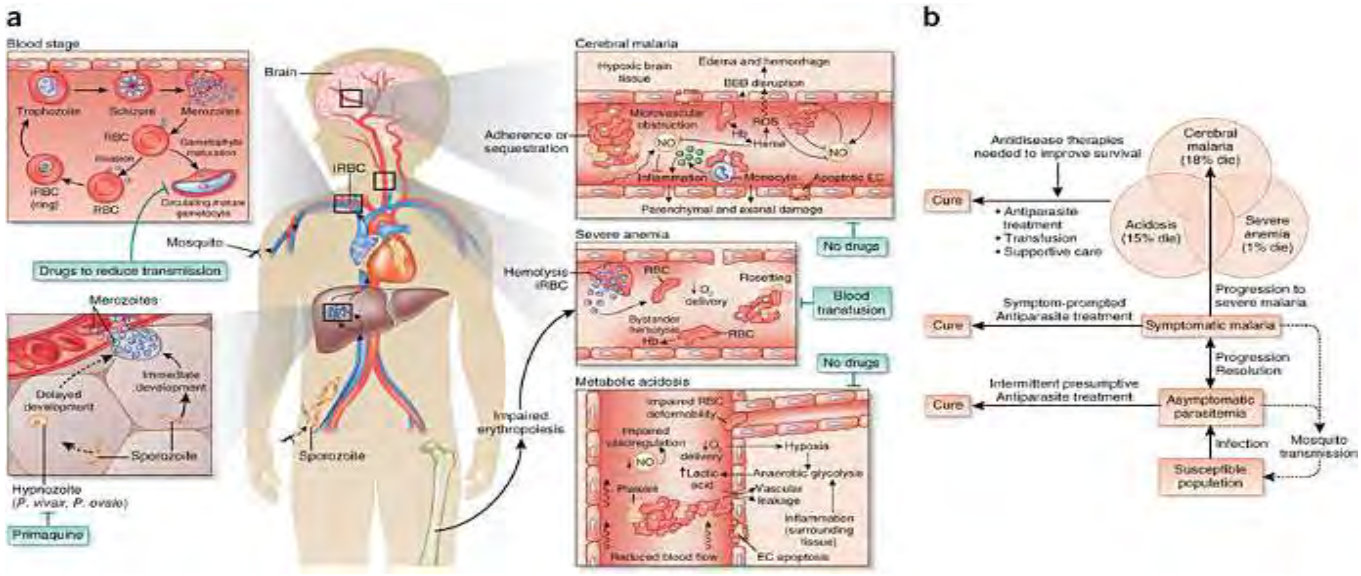
Acquired by bite of an infected female anopheles mosquito , and less commonly translucently and blood transfusion .
IP: 6-16days

Diagnosis :
Thick and thin blood film or smear .

Prevention :
Chemoprophylaxis "from 2 weeks before visit and 8 weeks after leaving area ", chloroquine or primquine used , mefloquine or doxycycline in resistant cases".

C/F:

- Nonspecific , then high fever with chills , abrupt return to normal temperature occur after 2-12hr , fever ass. With headache , abd. And back pain , nausea and often splenomegaly .
- **p. falciparum** cause encephalitis pneumonitis , enteritis and nephritis and disturbance of renal function .
- **nephrotic syndrome** is caused by P.malaria "prognosis poor"
- the most serious form of malaria is due to sever and sudden intravascular hemolysis "**black water**" ass. With heavy p. falciparum infection .



- **Childhood malaria** :
  a. May be transplasentally
  b. May present as pneumonia .
  c. Pt . with sickle cell disease have natural protection against malaria .
  d. Chloroquine or fansidar is effective therapy .

- **Which of the following conditions have a protection against malaria:**
  a. Fetal HB in utero .
  b. G6PDdef .
  c. Sickle cell trial .
  d. Sickle cell disease.
  e. HIV infection .

Toxoplasmosis

Caused by **toxoplasma gondi** which is intracellular protozoa , and infection is acquired per orally , transplacentally or rarely by blood.
IP=7days rare below 5 years of age.

C/P:

Cog . "see neonate"

Acquired :

- The condition usually asymptomatic infection. but symptom include fever , lymphadenopathy , hepatosplenomegally , disseminated infection including myocarditis, pneumonia , hepatitis , chororetintis , arthritis, encephalitis , are more common in immunosuppressed patient , hydrocephaly .
- Localized lymphadenopathy that is difficult to differentiated from lymphoma, is one of more common manifestation of toxoplasmosis .

Diagnosis :

Serology , skin test and histology .

Treatment :

- Pyrimethamine and sulfadiazine which act synergistically B/C these compound are folic acid inhibitors , folinic acid “calcium leukovorin” should be given .
- Corticosteroid reserved for pt. with CNS or ocular infection .

Toxoplasmosis:

a. Is caused by toxoplasma gondi.
b. Cats are most likely host .
c. May be transmitted transplacentally .
d. Causes congenital malformation .
e. Concomitant HIV infection should always be excluded.

Ascariasis

Caused by **ascarislumbericoid**

- Most common in children of preschool or early school age .
- After ingestion by human host larvae are released from the eggs via the venous circulation .
- Then they break through pul. Tissue in to the alveolar spaces , ascend the bronchial tree and trachea and are swallowed , upon their arrival the small intestine , the larvae develop into mature adult worms .
- Each female have life span 1-2years and is capable of producing 200000 egg /day .
- **M.O.T:** hand to mouth

C/P :

- Pulmonary ascararis "cough , blood stained sputum , eosinophilia and transit infiltrate on chest x_ray .when adult larva in small intestine may cause abd. Pain and distension also it may cause intestinal obstruction .
- Also may cause Biliary duct obstruction , may cause steatorrhea and decrease vitamin A absorption .
- Asymptomatic infection are common .

Diagnosis :

Stool examination characteristic egg is diagnostic .

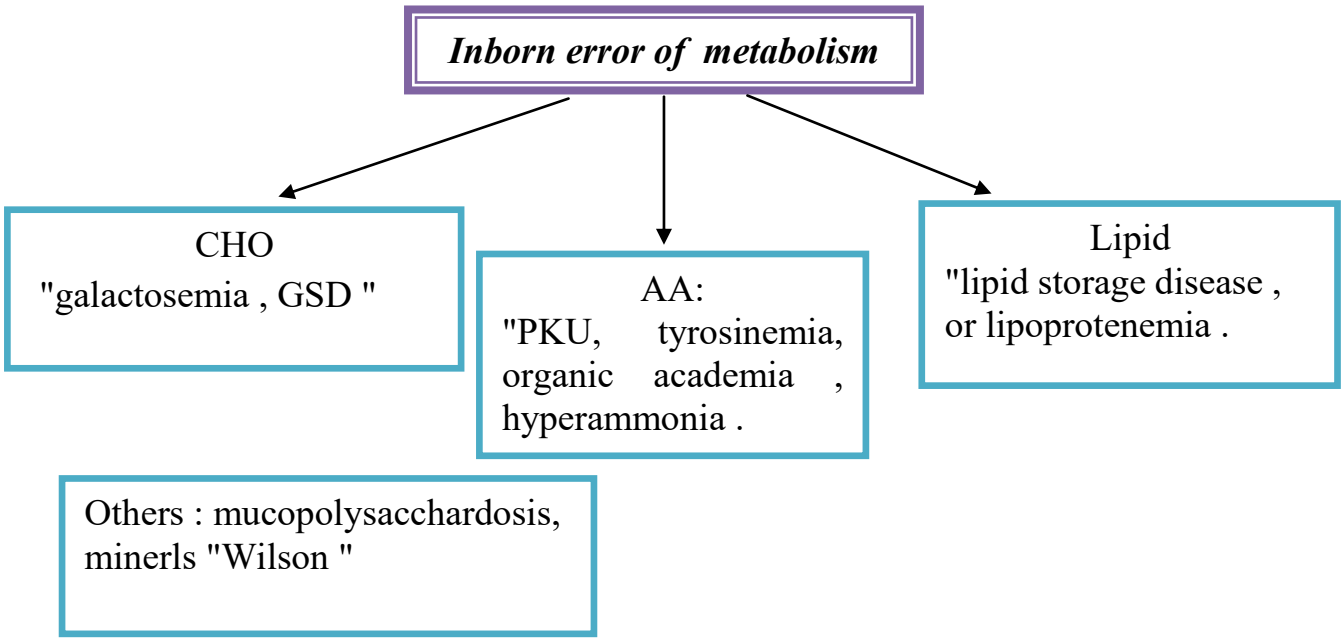
Treatment : mebendazole is drug of choice .

Inborn error of metabolism by Asma Hadida

2015_2016

Inborn error of metabolism

- Inherited metabolic disease "mostly AR " due to enzymatic or membrane transport defect .
- Detected by increased substrate or its metabolite in fluid or blood or urine or tissue.



Features : should be suspected in case of :

- FTT.
- Recurrent vomiting .
- Psychological motor or mental disorder .
- Coma , seizures , cataract or lens dislocation
- Acidosis.
- Unexplained liver , renal ,bone disease .
- Unexplained odor .
- Sever or infection esp . "candidiiasis"

Diagnosis :

Blood : guthric test , sugar and AA chromatography, enzyme assay , increase substrate.

Urine Clinitest : galactosemia , fructosemia .

Ferric CL test : PKU , tyrosinemia , histidininemia ,

Dinitrophenyl _ hydralazine : for PKU , thyrosenemia , histidenemia .

Cyanid nitroproside test : for cystinosis or homocystinuria .

Others : as reducing substance , ketones , porphyria .

Tissue: Biopsy material for enzymatic or biochemical studies .

Prenatal : amniotic fluid , chronic villi biopsy , also fetal blood for chemistry , enzymatic assay , and genetic studies

Treatment: avoidance or restriction of harmful or precipitating substance.
Replacement and supplement of the deficiency

Prevention :
Early detection **" screening " .**
Genetic counseling " education , carrier detection , prenatal diagnosis".

CHO disorder

Galctosemia

Hereditary galactose intolerance **"AR"** due to deficiency of uridyle transferase enzyme that stimulate the production of glucose from galactose or lactose **"milk sugar"** which accumulated with its metabolite "**galactolol"** in blood or tissue causing harmful effect in addition to the effect **of hypoglycemia** .

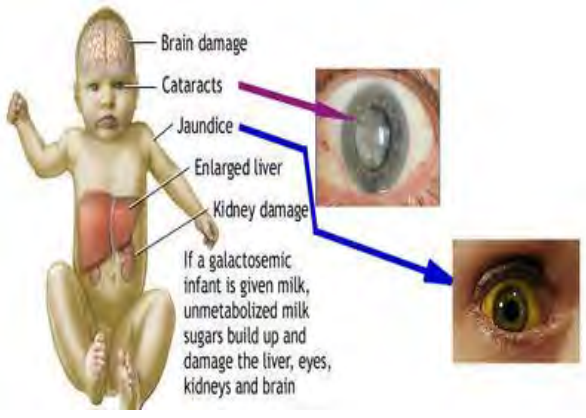
The baby : normal at birth develop feature in the first week of life after ingestion of lactose or galactose " milk " **including** :

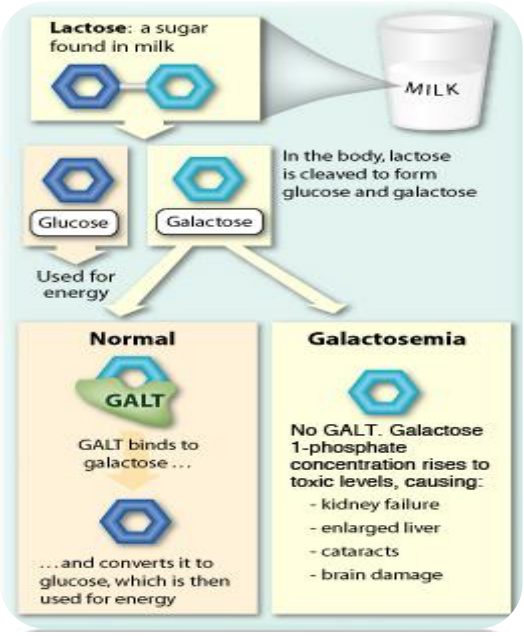
- FTT
- Hepatosplenomegaly .
- Persistent jaundice .
- Vomiting .
- Cataract "**toxic effect"** usually 2nd week .
- Hypotonia .
- Convulsion .

Diagnosis :

- Sugar chromatography in urine "clinistix " for reducing substance .
- **Blood** : high galactose and galactolol with hypoglycemia .
- **Enzyme assay** : decrease uridyl transferase .

Treatment :
lactose free diet or milk for life .





Glycogen storage disease :

- Caused by genetic defect in the enzymes responsible for destruction of glycogen leading to its deposition in tissue specially "**liver and spleen**".
- 8 known types **all AR** except **viii "x-linked recessive"** , affect muscle **"II, V, VII"** , liver "**I, III, VI, VIII**".
- The commonest type is **type I " von-gierkes** " due to deficiency of **G 6 phosphatase** .
- **Type II pompe** due to decrease acid maltase.
- **Type III " coris "** due to decrease de-branching enzyme .

C/P :

- Hypoglycemia or convulsion
- Hepatomagaly
- Ftt
- Macroglossia
- Hypotonia
- Pain or cramps after exercise

Diagnosis :

- High lactate , pyruvate .
- Hypoglycemia in hepatic type
- **High CPK in** muscle type .
- Poor or non response to glucagon stimulation test
- **Confirmed by :** enzyme assay in hepatic type
- Muscle or fibroblast biopsy

Treatment :

Giving frequent high glucose or CHO feed **" 50%-70%"** specially after exercise in muscle type or fructose feed.

Mucocpolysaccharidosis:

A group of storage disease due to genetic defect of sulfataze enzyme "¥ iduronidase" usually which is responsible for degradation of MPS "heparin , dermatan , or keratin" leading to accumulation in tissue and urine resulting in **corneal clouding** , **somatic feature** :

- Hepatosplenomegaly
- Skeletal deformity
- Coarse feature
- Gargoyle like feature
- Mental and motor detritions which progressive in nature .
- 8 known types **all ar** except : **type ii " x l r "**
- Mixed " **somatic and cns** " ⇨ i, ii
- **Somatic** only iv , v, vi
- **Cns** mainly iii
- The most imp . Is type i **(hurler syndrome**) : occur in infant in severe form and marked cns involvement .
- Type ii : **hunter (xlr)** moderate in severity with skin nodules but **no corneal clouding**

Diagnosis :

- Presence of MPS in urine
- Enzyme deficiency in fibroblast culture
- **Skeletal survy** : ovoid hooked hypoplastic vertebra , broad shaped ribs, macrocephaly and J shaped sella tursica with short neck and limb

Treatment:

- No specific treatment .
- Genetic counseling and bone marrow transplantation

Neurolipidosis :

Lipid storage disease associated with childhood dementia and blindness **after a period of normal development** due to deficiency of enzyme responsible for lipid destruction

∗ **The most imp. Types are :**

1. **Taysachs disease** : most important ganglioside due to def. of enzyme hexasaminidase A Lead to CNS manifestation **, cherry red spot** and **no visceral involvement** .
2. **Niemann pick disease** : due to def. of sphyngomyelinase leading to visceral , CNS , and eye involvement **" cherry red spot "**
3. **Gauchers disease** : due to def. of B-GLUCOCEREBROSIDASE enzyme with earlier neurological and late visceral but **no cherry red spot**.

Amino acid disorders

Phenylketoneuria : PKU

- **AR** deficiency of phenylalanine hydroxylase enzyme which is responsible for transformation of phenylalanine to tyrosine.
- Accumulation of PHE interfere with AA transport , lipid and protein synthesis " lead to ↓mylination of CNS " and decrease tyrosine leading to decrease pigmentation **in basal ganglia , skin and hair**.
- The baby who is normal at birth start to develop the feature after introduction of protein "PHE" Which include **developmental delay :**
  - ➲ MR
  - ➲ Microcephaly
  - ➲ Fair skin &hair
  - ➲ Blue eyes
  - ➲ FTT
  - ➲ Vomiting
  - ➲ Mousy smell

Diagnosis :-

- High PHE in serum
- Urine for ferric test
- Enzyme assay

Treatment :

Early detection " **Guthrie** " .
Phenylalanine free diet for life

Maple syrup urine disease

- Familial cerebral degenerative disease caused by failure of oxidative decarbooxylation of branched chain AA **"leucine, isolenin, and valine"**, due to genetic defect in decarboxylase enzyme lead to its secretion in urine causing maple syrup **"sweet malt or caramel like odor"** and in tissue "**specially brain**" causing damage .
- The main feature develop **in 1st week** of life with feeding and neurological damage & special odor .
- **Diagnosis :** by guteric test , DNPH , enzyme assay
- **Treatment**: restriction of branched chain AA avoid dehydration and give thiamine

Hereditary tyrosinemia : AR

- **Def .** of fumarylactoacetase enzyme responsible for tyrosine metabolism leading to its accumulation in blood and succinyl acetone in urine causing progressive hepatocellular damage , renal tubular acidosis and FTT
- **Diagnosis :**
  AA and its metabolites, enzyme assay

Cystine disorder

✱ ***Homocystinuria*** : AR characterized by ↓cystathionine synthasatase enzyme responsible for transformation of homocystine to cystathionine lead to accumulation of homocystine in urine "**homocystinuria**" and ↑methianine in blood .

Features :

- **CVS**
  - ↳ vascular thrombosis and CVA
- **Skeletal**
  - ↳ collagen synthesis disruption by homocysyine causing :
    - ➲ Osteoporosis .
    - ➲ Deformity
- **Tall stature**
- **Eye :** ectopic lens with lens dislocation down not before 3years in addition to skin and hair abnormalities .

Diagnosis :

- Cyanide nitroproside test .
- Guthrie test.

∗ ***Cystinuria*** **:** AR characterized by ↑ excretion of cystine ornithine , arginine and lysine in urine and cystin stone formation in kidney "**hexagonal radio opaque**" which is pathognomic for the condition .

Treatment :

- D- penicillamine .
- Alkalinization of urine.
- Renal transplant.

∗ ***Cystinosis*** : AR Condition associated with ↑ intracellular accumulation of cystine in many tissue .

- **Childhood type " early "** : usually renal commonest cause ⇨ fanconi syndrome
- **Later type " adult "** : cornea , thyroid , bone marrow but not renal .

Diagnosis :

- Normal level of AA Including cystine but ↑ urinary excretion of AA Protein , phosphate and glucose .
- Measurement of cystine in the cell " fibroblast or leukocyte ".

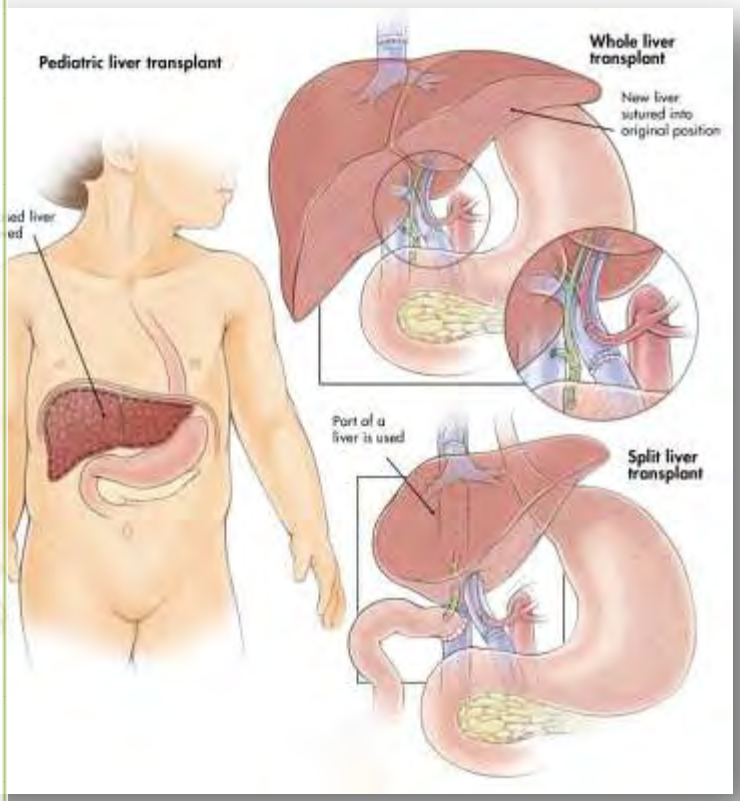
Treatment :

As treatment of renal tubular disorder.

Hyper ammonia :

- Amonia usually result from de-amination of AA Converted into urea **"through urea cycle"** which occur in mitochondria of liver , kidney , gut by action of certain enzyme *, high ammonia occur in* :
    1. liver disease
    2. immature liver
    3. reyes syndrome
    4. rett syndrome(mitochondrial)
    5. acidosis or urea cycle defect "**commonest due to def. of ornithine transcarboxylase** enzyme which **is x-linked** .
- **Main feature :**
    - FTT.
    - Lethargy
    - Vomiting
    - CNS manifestation " mental –seizure , ataxia , abnormal EEG specially after ingestion of protein
- **Diagnosis :**
    - High ammonia and AA
    - Abnormal LFT
    - Enzyme assay
- **Treatment :**
    - Restriction of protein to ↓ ammonia " sterilization of gut "
    - Na benzoate
    - Dialysis , treatment of complication

Liver diseases by Dr.Asma Hadida



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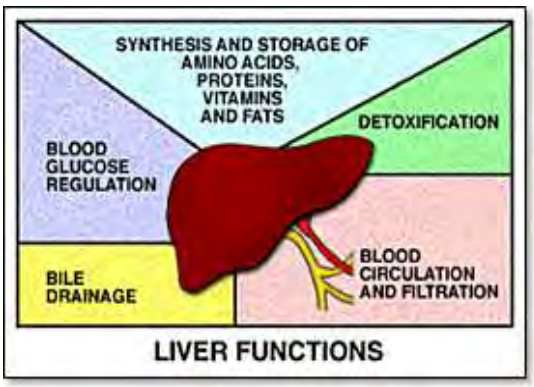
LIVER DISEASES

☑ **Function of liver**:

1. Metabolism and hemostasis.
2. Synthesis of protein (except gamma globulin and F13).
3. Bile acid metabolism.
4. Bilirubin metabolism.
5. Detoxication reaction e.g. convert ammonia to urea.
6. Synthesis and excretion of cholesterol.



Liver diseases come with:

- Hepatomegaly.
- Jaundice.
- Hepatitis.
- Liver failure.
- Failure of synthesis , function → bleeding and edema.
- failure of detoxication and wast elimination "**encephalopathy**".
- Obstruction of blood flow through the portal blood flow → **portal hypertension.**

Chronic liver diseases:-

- It is the persistent elevation of liver enzymes 3 years or more after prosumed acute infection.
- **May be manifested by:**
  - History of recurrent hepatitis.
  - Prolonged jaundice.
  - Lethergic and anorexia.
  - Brusing, pruritis, and poor wt gain.
- **O/E:**
  - Muscle wasting.
  - Low wt.
  - Clubbing, edema, "ascitis".
  - Spide naevi.
  - Splenomegaly.

Causes of hepatomegaly:

1. Hepatitis **(infection, autoimmune).**
2. Drug reaction.
3. Metastasis.
4. Hemolytic anemia.
5. Glycogen storge disease.
6. CHF
7. Hepatic vein thrombosis.

Liver span:

Age	span
At birth	5.5 cm
1 year	6 cm
2 year	6.5 cm
3 year	7 cm
4 year	7.5 cm
5 year	8 cm
12 year	9 cm

Hepatitis

Causes

Metabolic
1. Galactosemia.
2. Wilson disease.
3. Alfa 1 anti trypsin deficiency

Infections

Toxic
Drugs e.g.
- INH, Rifampacine.
- Paracetamol over dose.

Bacterial
- Septicemia
- Enteric fever
- Brucellosis

Protozoa
- Malaria
- Toxoplasma

Viral

Parasitic
- Bilharsiasis
- Hydatid cyst

Hepatotrophic
Viruses attack the liver mainly

Enteral transmitted
- Hepatitis A
- Hepatitis E

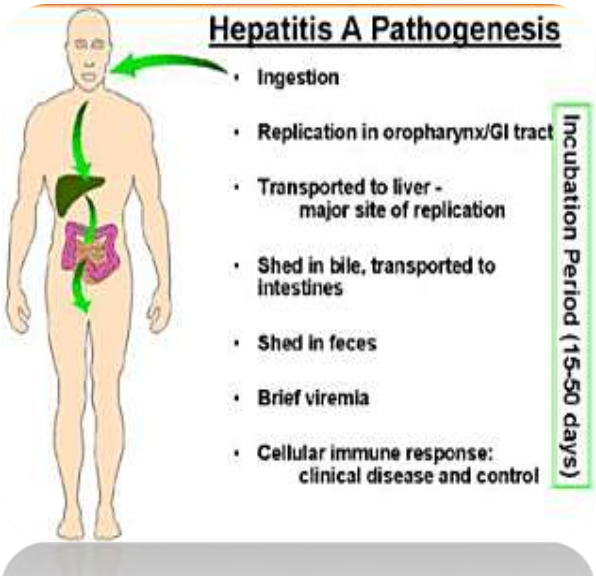
Parenteral transmitted
- Hepatitis B
- Hepatitis C
- Hepatitis D

Non hepatotrophic
Viruses cause hepatitis during its course, e.g.
- CMV.
- Herpes viruses.
- EBV.
- Rubella

Acute viral hepatitis

1. Hepatitis A and E :

- Excreted only in stool "**not in body fluid**".
- Non enveloped RNA viruses.
- Short I.P 30 day.
- Infectivity: During I.P and 1st week of jaundice.
- HAV is commonest cause of acute viral hepatitisespically in school age, HEV is rare in children.
- **Outcome:**
  - Complete recovery, no carrier state, no chronic hepatitis.
  - Fulminant hepatic failure may rarely occur.



2. Hepatitis B,C, D :

Criteria	HBV	HCV	HDV
Nature	Enveloped DNA virus.	Non enveloped RNA virus.	Non enveloped RNA virus, Needs HBV caot to be infective as HBV.
Route	• Parental. • Contaminated blood product. • Perinatal. • Sexual.	As HBV **mainly post blood transfusion.**	As HBV.
I.P	2 - 6 months	1 - 4 months	1 - 4 months

- **Outcome:**
  - Chronic carrier.
  - Chronic hepatitis 20% in HBV , 50% HCV may be **complicated by** :
    - Cirrhosis and liver failure.
    - Hepatocellular Cancer.
  - Fulminant hepatic failure 1-3% .

C/P of acute hepatitis:

1. Many cases asymptomatic.
2. Symptomatic cases pass in following phases:
   a) **Pre-icteric phase "2-4 wks" :** Fever, ,malaise, anorexia, nausea, vomiting, abd. pain.
   b) **Icteric phase "2-4 wks"** :
   - Improved previous symptoms with appearance of : Jaundice, tendet hepatomegaly, "Dark urine, pale stool".
   - Splenomegaly and lymphadenopathy are common with HBV.

   c) **Convalescent phase** :
   - Complete resolution "especially in HAV and HEV".
   - Other types "HBV, HCV, HDV" may pass to chronic hepatitis or carrier state.

Complication:

- As before plus:

1. Extra hepatic manifestation:

- Due to circulating immune complexes mainly in HBV, HCV:
- **Common presentation**:
  - Arthralgia.
  - Aplastic anemia.
- **Rare presentation**:
  - Myocarditis.
  - Glomerulonephritis.
  - Acute pancreatitis.
  - Vasculitis.

2. Fulminant hepatic failure:

- Common with HBV especially if there's co-infection or super infection with HDV.
- Uncommon with HCV.
- Very rare with HAV.

▸ **Clinically:**
- Deep progressive jaundice.
- Bleeding tendency "↓coagulation factors"
- Generalized edema with ascitis.
- Disturbed sleep rhythm, stupor and coma"rapidly developed coma".

▸ **Fate:**
- MR about 30%.

▸ **Treatment:**
- Liver transplantation.

Investigation:

1. To prove acute hepatitis:

❶ LFT :- **ALT and AST → very high.**
- **Serum bilirubin → moderate ↑↑ "conjug".**
- **Alkaline phosphatase → mild ↑.**
- **Albumin → usually normal.**

❷ Urine : - **Dark color due to ↑ cholebilirubin.**

❸ Stool : - **Pale color due to ↓ sterocobilinogen.**

In fulminant hepatitis:
- ALT and AST rise initially then decline.
- Rising bilirubin. (> 10 mg/dl)
- Prolonged PT. (> 20 second)
- Low albumin. (below 3 mg/dl)
- Hypoglycemia and hyperammonemia.
- Hypokalemia, hyponatremia and metabolic

2. For the cause: **Viral serology.**

1) HAV markers:-
- **Anti HAV "Ig M" → Recent HAV infection.**
- **Anti HAV "Ig G" → Previous HAV infection or HAV vaccination.**

2) HBV markers:-
- **HBs Ag → Acute infection.**
  → If persist > 6 months → Indicate chronic hepatitis.
- **HBe Ag → Acute infection "high infectivity"**
- **Ig M anti HBc Ag → Acute infection.**
- **Ig G anti HBc Ag → Infection acute or chronic.**
- Anti HBs Ag :
  → If present alone, it indicate previous vaccination.
  → If present with anti HBc Ag → Resolved infection.

3. For HCV, HDV, HEV infection : **PCR.**

N.B:
★ HBc Ag is present only in hepatocytes.
★ HBe Ag is not structural antigen but it is produced by self-cleavage of core anti-gen

Rey'es syndrome

Acute encephalopathy with fatty degeneration of the liver.

Clinical pictures:

1. Occur in previously healthy child.
2. **Association:** Aspirin treatment and viral infection"varicella".
3. Prodromal viral URTI in 90% of cases or chicken pox infection in 10%.
4. Sever profound vomiting 5-7 days following viral infection.
5. Hepatomegaly without jaundice.
6. Evidence of increased ICP:
   - Delirium and stupor
   - Generalized fits.
   - CSF normal but with ↑ pressure.
   - Focal neurological signs are absent in Ray's sundrome.

Staged into 5 stages according to clinical status:

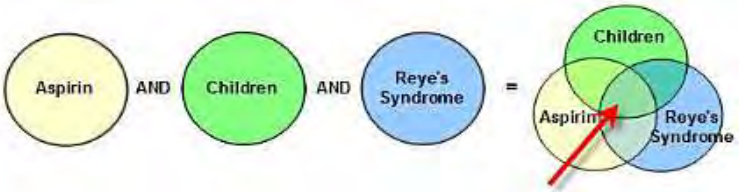
I	Lethergic – vomiting – sleepy.
II	Lethergic – confused.
III	Light coma – decorticate.
IV	Deep coma – decerebrate.
V	Flacid – apnic – reactive pupils.

Diagnosis:

1. **LFT** : Impaired + ↑↑ ammonia and hypoglycemia.
2. **CT brain** : Brain edema.
3. **Liver biopsy** : Diagnostic.
   - Light microscopy: Microvesicular fatty infiltration.
   - Electron microscopy: Mitochondrial damage.

Treatment:

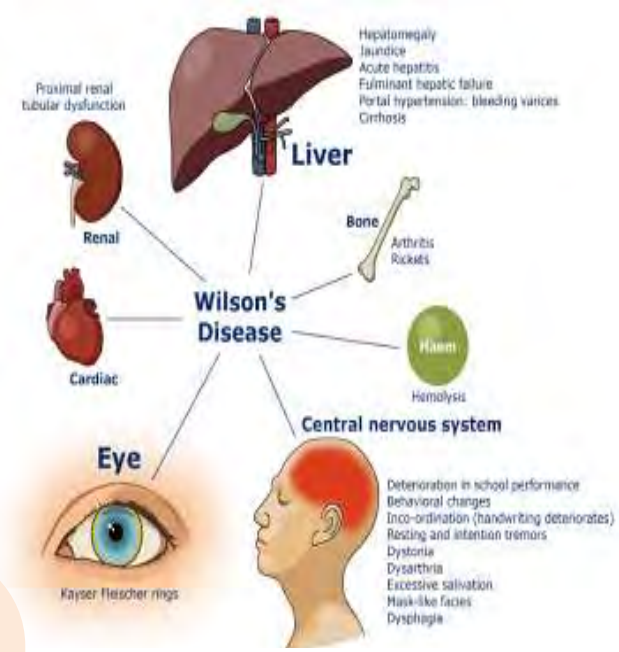
Supportive "liver support and care of coma".



Wilson diseases

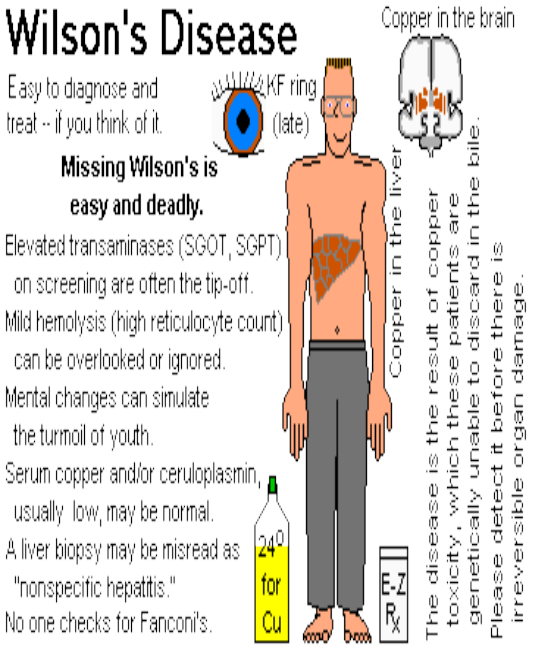
(Hepato lenticular degeneration)

- AR defect in ceruloplasmin "copper carrying protein" → copper accumulate in:
  - **Liver:** Hepatitis and cirrhosis.
  - **Basal ganglia**: Behavior and speech disorder.
  - **Cornea**: Kayser flisher ring.
  - **Renal tubules**: Tubular defects "Fanconi like".
    ↳ Vitamin D resistant Rickt's.
  - **Red blood cells**: Hemolytic anemia.



They rarely develop before 3 years of age:

- **At 7 years:** • Kayser fleisher ring "copper in outer cornea".
  • Sun flower cataracts.
- **< 12 years:** • Liver involvement with acute hepatitis.
  • Fulminant hepatitis.
  • Cirrhosis.
  • Portal hypertension.
- **20 years:** • Neurological manifestation:
  - Delayed school performance.
  - Behavioral changes.
  - Extra pyramidal manifestation.



Diagnostic triad:

- ↓ serum ceruloplamin.
- ↑ urinary copper after loading dose of D-penicillamine.
- Liver biopsy : excess copper deposition.

Treatment:

- D.penicillamine.
- Liver transplant.

Portal Hypertension

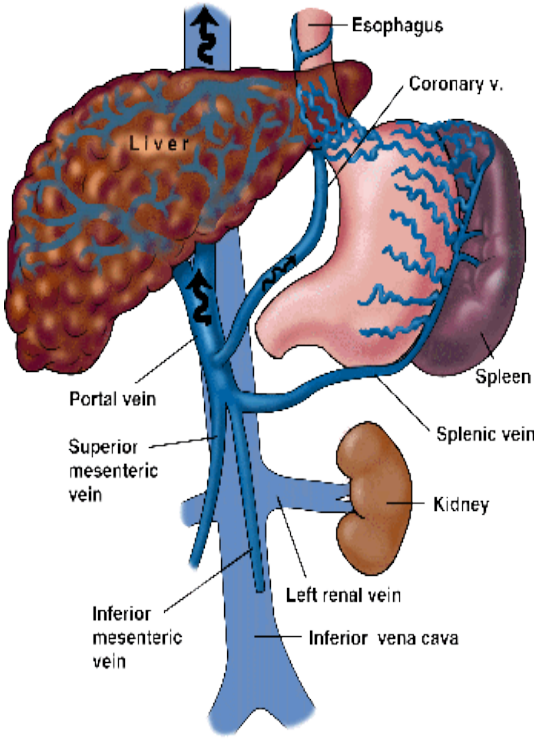
Increased pressure in the portal circulation > 12 mmHg.
(Normal = 7-11 mmHg)

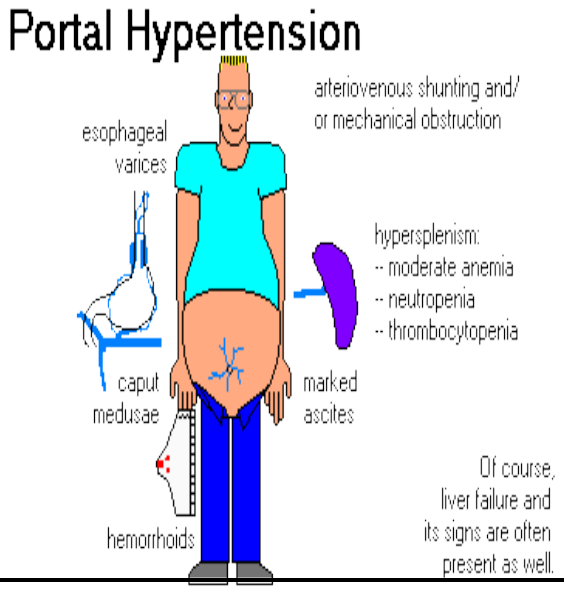
Etiology:

1. Pre-hepatic :
- Causes in portal vein or splenic vein thrombosis.
  - Umbilical vein catheterization.
  - Umbilical vein infusion.
  - Umbilical sepsis.

2. Hepatic :
- Pre-sinusoidal :
  - Causes in the portal tract.
  - Congenital hepatic fibrosis.
- Sinusoidal : Liver cirrhosis due to:
  - Chronic hepatitis.
  - Alpha 1 anti-trypsin deficiency.
  - Wilson disease.
  - Cystic fibrosis.
  - Biliary tract disease.
    - Extra-hepatic biliary atresia.
    - Sclerosing cholangitis.
- Post-sinusoidal : VOD.

3. Post-hepatic :
- Budd-Chiari syndrome.
- Constrictive pericarditis.





Clinical presentation:

➲ **Hematemesis:**
- From esophageal varices which is may be the 1st presenting symptom.

➲ **Abd. examination:**
- Dilated abd. wall veins "site of portosystemic anastomosis".
- Ascitis become evident with advanced cases.

➲ **Organomegally:**

- **Pre-hepatic** :
  - **Spleen:** Significant early splenomegaly "hypersplenism may occur resulting in pancytopenia"
  - **Liver**: Normal.
- **Hepatic** :
  - **Spleen**: Mild to moderate splenomegaly.
  - **Liver:** Enlarged or shrunken according to the disease stage.
- **Post-hepatic** :
  - **Liver:** Marked hepatomegaly.
  - **Spleen**: Normal may be enlarged in late stage.

Investigation:

1) Laboratory: Ascites fluid.
2) **Imaging:**
   - U/S : - Size of liver and spleen.
     - Ascites (amount)
   - Doppler.
3) Endoscopy.

Treatment:

Therapy is directed toward prevention and or management of variceal Hge.

1. Management of variceal Hge:-

Emergency for bleeding	Emergency endoscopy + Injection of sclerotherapy	Emergency shunt ⬇ Portosystemic shunt
- ABC. - Correction of coagulopathy (vit. K, fresh plasma, NGT, vasopressin infusion.		

2. Presentation of bleeding from varices :

- Avoid aspirin and NSAID
- B adrenergic blocker " propranolol " to lower pressure in the portal area.
- Prophylactic sclerotherapy.

Prevention of re bleeding : the above + surgical porto systemic shunt and liver transplantation .

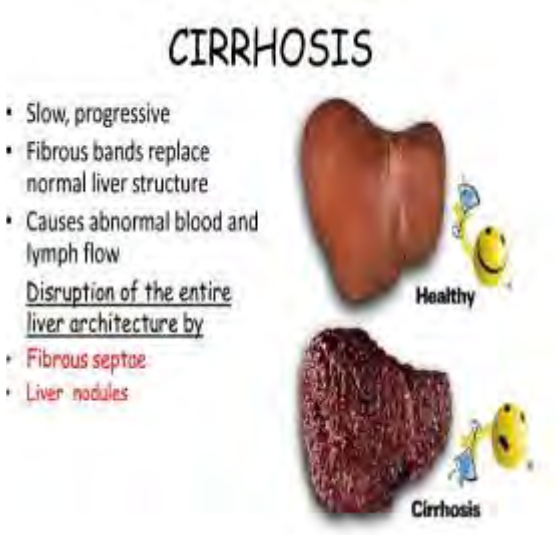
Liver cirrhosis

Chronic liver disease with triad of :

- Hepatocyte necrosis .
- Regeneration nodules .
- Lost hepatic architecture .

Causes :

- Post hepatic
- **Metabolic** e.g Wilson and hemochromatosis .
- **Biliary** : 1ry or 2ry to bile flow obstruction .
- Chronic hepatic congestion (cardiac cirrho



C/P:

1. **Compensated :** C/P of the cause .
2. **Decompensated :** feature of liver cell failure :
   a. Jaundice
   b. Bleeding tendency : skin bruises , GIT bleeding : hematemesis , melena .
   c. Ascitis and generalized edema .
   d. Hepatic encephalopathy : due to increase ammonia and neurotoxin . manifested by : disturbed sleep rhythm , flapping tremor , coma .
3. **Hepatorenal syndrome :** Functional renal failure in patient with end stage liver disease due to intense renal vasoconstriction with systemic vasodilatation lead to renal hypoperfusion and pre renal failure .
4. **Hepato pulmonary syndrome :** Intrapulmonary vascular dilatation → right to left shunting of blood → hypoxemia , dyspnea , cyanosis and clubbing .
5. **Others :**
   - ❖ Feotor hepaticus
   - ❖ Palmar erythema
   - ❖ Spider nevi
   - ❖ Muscle wasting

Diagnosis :

1. **To prove cirrhosis** : abd. U/S and MRI, liver biopsy " avoided in decompansated "
2. **For the cause** : e.g viral markes .
3. **For the complication :** LFT.

Treatment :

1. Supportive :
- ➲ CHO and vitamins rich diet .
- ➲ Low salt diet " for cases with edema "
- ➲ Limit protein " for cases prone to encephalopathy ".

2. **Anti fibrotic** : Colchicine.
3. Treatment of complication and liver transplantation.

Cholestasis

Prolonged elevation of direct " conjugated " bilirubin more than 20% of the total bilirubin , together with other substances that normally excreted in bile "cholesterol and bile acid".

Etiology :

1. Intrahepatic disease :

☑ **Liver cell injury** :

- infection e.g. TORCH , sepsis .
- metabolic e.g. CHO metabolism " galactosemia , α-1 anti trypsin deficiency.
- idiopathic neonatal hepatitis.

☑ **Intrahepatic bile duct lesion** :

- intrahepatic biliary hypoplasia
- progressive familial intrahepatic cholestasis
- benign recurrent intrahepatic cholestasis
- alagille syndrome " cong . heart disease , cholestasis , and abd. features ".

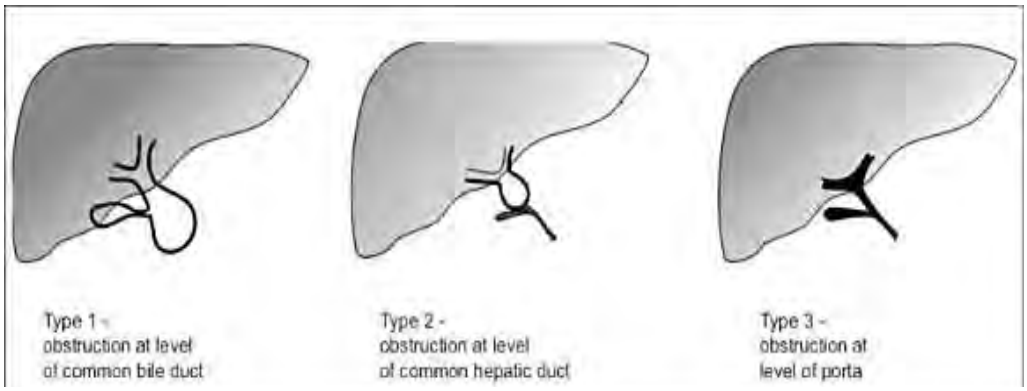
2. Extra hepatic disease :

a. Extrahepatic billiary atresia :

- Obstruction at or above porta hepatis : 85% of cases .
- Distal atresia with patent extra- hepatic duct up to porta hepatica " rare ".

b. Choledochal cyst .

- The commonest causes are idiopathic hepatitis and extra-hepatic biliary atresia .
- Early differentiation is important B/C in case of biliary atresia early surgical correction will prevent further hepatic damage .



Clinical diagnosis:

➲ **History**:

• **Present history**:

- Onset of jaundice: Cholestasis should be considered in every case with persistent jaundice for more than 2 weeks.
- **Clay colored stool.**
- Dark urine.

• **Prenatal history**:

- To exclude congenital infection.
- Family history of similar conditions.

➲ **Examination:**

1. Hepatomegaly is usually present.
2. **Manifestation due to retention and regurgitation of:**
   - Bile acid: Pruritus "itching marks".
   - Cholesterol: Xanthomatosis.
3. **Manifestation due to ↓ bile flow to the intestine**:
   - Malabsorption of fat → malnutrition → steatorrhea.
   - Malabsorption of fat soluble vitamine (AKED).
4. **Manifestation due to progressive liver cirrhosis in long standing cholestasis:**
   - Portal hyperyension.
   - Liver cell failure.
5. **Manifestation of specific causes** e.g.:
   - Cataract suggest congenital infection or galactosemia.
   - Low birth wt or microcephaly may be present with cong . infection .
   - Allagille syndrome " abnormal facies , cong . heart disease , vertebral anomalies ".

Laboratory approach of cholestasis :

1. Investigation to diagnose cholestasis :

- Increase total and direct bilirubin.
- Increase AST and ALT.
- Increase alkaline phosphates and gamma-glutamyl transpeptidase
- Decrease total serum protein and albumin.
- Prolonged PT.

2. Investigation to determine the cause :

a. search for treatable cause :
   - Galactosemia : reducing substance in urine.
   - Septicemia : CBC, CRP, ESR

b. Investigation for TORCH : total IGM.

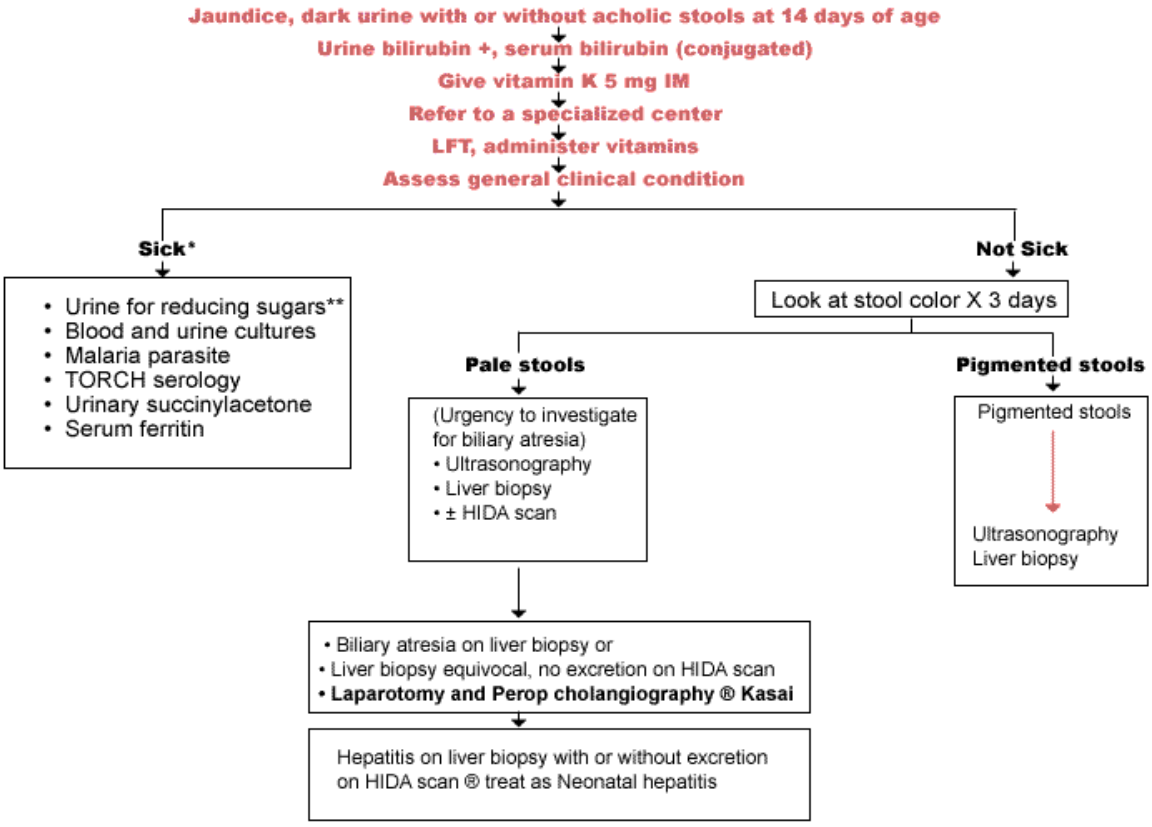
c. Search for metabolic condition .

d. Exclude for choledocal cyst : abd. U/S.

e. Differentiation B/W idiopathic hepatitis and extrahepatic billiary atresia :
   HIDA scan and liver biopsy

Treatment :

1. **Treatment of correctable condition :**
   Ab for septicemia , lactose free in galactosemia .
2. **Extrahepatic billiary tresia** : kasai " hepatoportoenterostomy " should be done before 60 days to obtain best result .
   Liver transplantation for end stage liver disease .
3. **Supportive treatment** :
   - **Nutritional support** : fat soluble vitamins , medium chain triglycerides , ca, zinc , and phosphorus .
   - **Purities**: phenobarbitone , cholestramine " bile acid binder".



neonate by Dr.Asma Hadida

2015_2016

FEATURES OF NORMAL NEWBORN

Measurement:

- Weight 2.5 – 3.5 kg at birth.
- Length 50 cm "48-53 cm".
- Head circumference "35 cm" (33-38 cm)
- Chest circumference 35 cm.
- "Measurement should be plotted on corresponding growth curves, normal 10^{th} – 90^{th} percentile"

Skull & head:

- Posterior fontanelle is mostly closed. "many not closed up to 2 months"
- Anterior fontanelle(Width + length) / 2 = < 5 cm.
- Sutures can be felt moulding normally.
- Caput succedaneum is obvious.
- " Especially after prolonged labor".

Skin:

- Covered by vernix caseosa with protective greasy layer.
- Colour pink, more pink if premature.
- Fine lanugo hair is present.
- Mongolian spots: bluish discoloration.
- "On sacral region and buttock ".

EYES:

- Pupillary reaction to light is present at birth.
- Movement are wandering & some times in dependently moving.
- Intermittent squint is common during the first weeks of life.

Abdomen & external genitalia:

- Abdomen is slightly distended.
- Liver is palpable 2 cm below costal margin.
- Umbilical stump dries and sloughs' in between 6-10 days of age.
- Testicles can be easily brought down into scrotum, if they are not already there.
- Labia majora are underdeveloped with prominent labia minora.
- Breast enlargement occurs in (male) and (female) and subside after several weeks, due to maternal hormones.

Chest & respiration:

- Respiratory rate is average 40 – 60 breath/cycle/min and irregular in rate and depth and is mainly abdominal varies according to feeding, crying, the rate should be counted over 1 min.
- Type of breathing (periodic breathing) breath sound is bronco vesicular.
- Obligatory nasal breathing, respiration is entirely diaphragmatic.

Cvs:

- Heart lies relatively transversely and apex in 3rd – 4th inter costal space just outside mid axillary line.
- HR is extremely variable according to state of baby and activity average 120-140 beat/min, may reach 160 beat/min by stethoscope.
- Normal B.P 80/55.

Physiological phenomena in the newborn:

- **Physiological jaundice.**
- **Physiological diarrhea.**
- **Physiological constipation.**
- **Physiological vomiting:** Most babies vomit mucus, often blood streaked in the 1st few hours after birth, if it persists, it is due to irritation of the stomach by amniotic material swallowed during delivery, gastric lavage with saline is done.
- **Physiological anemia.**
- **Physiological anuria:** The fetus usually urinates in utero; urine is present in the bladder at birth and sometimes infant many urinate shortly after birth, the majority do not void until more than 12 hours after birth.
- **Physiological breast enlargement:** The breast engorgement may occur in both (female) and (male) reaching maximum between 3rd – 5th day remain enlarged for several weeks. It is believed to be 2ndary to withdrawal of maternal hormones, secretion of milk substance may occur (it is called "witch milk") squeezing of breast is C/I may lead to mastitis.
- **Vaginal bleeding:** It is due to withdrawal of maternal hormones, it usually occurs on 3rd – 5th day, a milky white mucoid discharge sometimes blood streaked is observed at the vagina.
- **Physiological weight loss:** Most babies loses weight in the 1st three days "less than 10%", weight gain 25-35 gm/day, starts by the 4th day and progress thereafter, caused by labor and the transition to extrauterine

life. Also the extra fat stores are there because mother's mature milk does not come in for 3-5 days.

TRANSIENT CUTANEOUS MANIFISTATION OF THE NEWBORN:

1. **Mongolian spots:** Irregular area of blue pigmentation usually present in the sacral area and gluteal regions, but may be present also at any area of the body and extremities, they disappear after few years (1-2 years) and have no pathological significant, they result from ↑ melanin.
2. **Milia:** Retention of keratin and sebaceous material in the pilosebaceous follicles are seen as minute white papules on the cheeks, nose and chin, they disappear spontaneously in 2-3 weeks.
3. **Capillary hemangioma:** They are common over eyelids, forehead and back of the neck and the occiput.
4. **Erythema toxicum:** "Urticaria neonaturum" resembles papular urticana and commonly seen in the 1st week, it appears as irregular red blotches "Blotchy erythema" with firm yellow – white papules that are sterile on culture contain eosinophils, this condition is common, benign and self limited.
5. **Miliaria:** "Vesicular sweat rash": The collections of the tiny vesicles on the head and chest of babies who have been sweating between of illness or overheating, usually develop at 2-3 weeks.
6. **Seaborrhea of the scalp:** "Cardle cap": It is common towards the end of the neonatal period, the secretion of the sebaceous glands on the scalp accumulate as a grey/greasy crusted layer of irregular thickness, local hygiene with soap and water or 1% NaHco3 is sufficient to remove the crust in most cases.
7. **Minor abnormalities:**
   Natal teeth: Front lower incisors present at birth. Remove if loose to avoid the risk of aspiration.
   Ear Tags: Check that the ear and hearing is normal, if there is an ear anomaly, some centers ultrasound the kidneys as slight increased risk of renal abnormalities.
   Extra digits: Usually connected by a skin tag but may contain bone.

BIRTH ASPHYXIA AND RESUSCITATION

Birth asphyxia is defined as:

Failure of the newborn to establish spontaneous regular respiration immediately after birth leading to either death or survival with permanent neurological damage.

Normally : The first breath is stimulated by :

1. Drop of PaO_2 on cutting the umbilical cord.
2. Rise of $PaCO_2$.
3. Drop of baby temperature.
4. Tactile stimulation in the delivery room.

So… **birth asphyxia may define as** :

1. Profound academia " pH<7 on an umbilical sample.
2. APGAR score 0-3 longer than 5 minutes.
3. Neonatal neurological manifestation e.g., hypotonia.
4. Multisystem organ failure.

ASSESSMENT at BIRTH:

At birth the airway is cleared and at 1 minute, the baby's condition is assessed by APGAR scoring system.

APGAR SCORING SYSTEM

Parameter	Score (0)	Score (1)	Score (2)
Heart rate /min	Absent	< 100	>100
Respiratory effort	Apneic	Weak, irregular, gasping	Regular
Muscle tone	Flaccid	Some flexion	Good flexion of Arms & legs
Grimace reflex	None	Facial grimace	Cry, sneeze, cough
Appearance	Blue, Pale	Body pink, hands & feet blue.	Pink

N.B: If resuscitation is needed, the apgar score should be repeated at 5 and 10 minutes after birth.

Physiology of Perinatal asphyxia:

The initial response to hypoxia is an increase in frequency of respiration and a rise in heart rate and BP (increase sympathetic drive). The respiration then cease (primary apnea) as HR and BP begin to fall. This initial period of apnea lasts 30 – 60 seconds. Gasping respiration 3-6 minute followed by secondary apnea, then begin, while HR & BP gradually decline with mixed acidosis. Redistribution of blood flow from muscle, kidney and GIT to allow the perfusion to vital organ; heart, brain and adrenal.

C/P:

Depends on duration and severity of asphyxia.

a. **In the fetus**
   - Fetal monitor shows: slow, weak, irregular heart beat (type II deceleration).
   - Scalp pH less than 7.2.

   Action: Provide mother high O_2 concentration and prepare for immediate delivery.

b. **After delivery:**
   - Meconium staining of the newborn, amniotic fluid & vernixcaseosa.
   - Decrease consciousness with failure of spontaneous breathing.
   - Low apgar score with cyanosis and flaccidity.

c. **Late manifestation.**
   - Hypoxic, ischemic, encephalopathy.

SARNAT CLINICAL GRADING

Sign	Mild stage-1	Moderate stage-2	Severe stage-3
Consciousness	Hyper alert	Lethergic	Coma
Muscle tone	Normal	Hypotonia	Flaccid
Tendon reflex	Hyper active	Same	Absent
Moro reflex	Exaggerated	Weak	Absent
Pupils	Dilated	Miotic	Variable
Respiration	Regular	Periodic	Ataxic, apnec
Seizure	No	Frequent	Frequent
Outcome	Good	Variable	Death or sever defficit

Complications:

- **Cardiac :** HF, hypotension, tachycardia, myocardial necrosis.
- **Respiratory**: meconium aspiration, persistent pulmonary hypertension of newborn, RD.
- **Renal**: Acute tubular necrosis & hematuria.
- **GIT**: NEC & Intestinal perforation.

- **Metabolic**: hypoglycemia, hypocalcemia, hypomagnesemia, hyponatremia, lactic acidosis and syndrome of inappropriate secretion of ADH.
- **Liver**: Increase in enzymes.
- **Hematologic**: Thrombocytopenia and DIC.

Management:

A- Preventive: Prevention of R.F and neonatal resuscitation.

B- Curative: Re-warming.

Support respiration. → O_2 inhalation for hypoxia.
→ Mechanical ventilation for hypercapnea / apnea.

Support circulation. → I.V fluid.
→ Keep mean arterial pressure above 40 mm Hg.
→ Dopamine infusion.

Support nutrition. → May need TPN.

Symptomatic treatment for → Brain edema & convulsion.

Prognosis:

1. Normal MRI and EEG are associated with good outcome.
2. Death may occur in severe cases.
3. HIE may sever enough to cause permanent brain damage, e.g., CP, MR.

Resuscitation of Newborn

INDICATIONS

Antepartum evidence of fetal distress and labour and delivery conditions:

1. Serious heart rate abnormalities, e.g., sustained bradycardia.
2. Scalp pH of 7.20 or less.
3. Abnormal fetal presentation.
4. Significant vaginal bleeding.

Antepartum events:

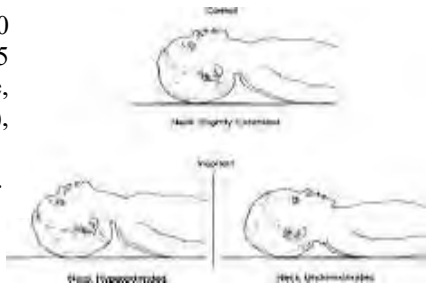
1. Meconium staining in amniotic fluid.
2. Prematurity < 37 wks, postmaturity > 42 weeks, low birth weight < 2.5 kg, or high birth weight > 4 kgs.
3. Multiple gestation.
4. Major congenital anomalies diagnosed perinatally.
5. Cord prolapsed.

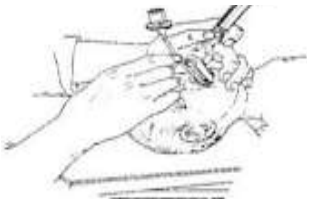
GOALS OF RESUSCITATION (ABC)

1. **Patent airway:** By proper positioning of the newborn, suctioning the upper airway and if necessary, perform endotracheal intubation.
2. **Breathing:** Initiating using tactile stimulation, positive pressure stimulation with a bag and mask, or through an E.T.T.
3. **Maintaining the circulation**: With chest compression and medication, if needed.

Equipments

1. Radiant warmer.
2. Oxygen source (100%) with adjustable flowmeter and adequate length of tubing.
3. **Bag and mask equipment:** (Ambu Bag) and the mask with appropriate size.
4. A bulb syringe for suctioning.
5. Stethoscope of appropriate size.
6. **Equipped emergency box:** Laryngoscope (No.0 & 1 blade), endotracheal tubes (2.5, 3, and 3.5 mm interneal diameter) drugs (epinephrine, sodium bicarbonate, naloxone, and saline 0.9%), umbilical catheterization and syringes).
7. Transport incubator with portable oxygen supply.





Resuscitating Technique:

1. Prepare for receiving the infant by proper hand washing & wearing protective gloves & gowns.
2. The infant should be dry, warm, and in good position to ensure patent airway.
3. **Airway suctioning:** Suction the mouth, oropharrynx and nares. *Deep or prolonged suctioning may cause arrhythmias via vagal stimulation. If thick meconium is present, suctioning of trachea (via endotracheal tube) before the onset of respiration is indicated.*
4. **Apply gentle tactile stimulation to initiate breathing:** Gentle slapping of the feet or rubbing of the back.
5. Assessment of the need for resuscitation **(Apgar scoring system):** Apgar score consists of total points assigned to 5 objective signs in the newborn. These are evaluated and given a **score of 0, 1 or 2 scores at 1 and 5 minutes, occasionally at 10 & 20 min are noted.** A score of 10 indicates an infant in best possible condition.
   - 1 minute apgar score, is an index of **intrapartum asphyxia** and shows the need for resuscitation.
   - **Monitoring during resuscitation:** Apgar scoring is not particularly useful during resuscitation. During these long tense moments, assessment of 3 features; **the respiratory activity** (spontaneous breathing or not), the **heart rate** (> 100 beats per min or less), and the **skin color** (the overall color is pink, acrocyanosis is normal).
   - **Apgar score beyond 1 minute (5, 10, 15 & 20)** are indicated every 5 min to assess the ade3quancy of the resuscitative efforts. Prolonged periods of severe depression **(with Apgar score around 3)** may **predict** abnormal long-term neurological outcome.

Sequence of Intervention

1. **No Asphyxia (initial Apgar score 8-10):** It is noticed in over 90% of newborns. The newborn needs only routine care.
2. **Mild Asphyxia (initial Apgar score 5-7):** The newborn require only stimulation and oxygen, stimulation by gentle slapping on feet or back rubbing, oxygen (100%) at rate of 5 ltr/min.
3. **Moderate Asphyxia (initial Apgar score 3-4):** The infant is apneic (no spontaneous breathing) despite tactile stimulation and facial oxygen and the heart rate is less than 100 beats/min, positive pressure ventilation (PPV) with oxygen 100% for 30 seconds using Ambu bag. The positioning of the newborn with head in the mid line and a slight neck extension. The mask should cover the chin & nose. Oxygen at rate 5-8 ltr/min. The initial inspiratory pressure: The initial breath should be delivered at a pressure of 30-40 cm H_2O to establish the functional residual capacity. The subsequent inflation will be effective at lower inspiratory pressures (15-20 cm/H_2O), successful ventilation is determined by observing good chest rise, listening for equal breath sounds.

Re-evaluate the newborn after 30 sec of positive pressure ventilation

- Still apneic & heart rate is >60/min: continue on positive pressure ventilation and re-check heart rate in 15-30 seconds, if heart rate becomes > 100 beats/min, color becomes pink, and with spontaneous breathing, observe the newborn and apply ongoing care.

- Still apneic, the heart rate is < 60/min, and color remains poor (no improvement on PPV) or the Apgar score falls to 0, 1 or 2: Pnemumothorax should be excluded (diminished air entry), endotracheal intubation should be considered.

Intubation: Suction is done first via the endotracheal intubation and then connected to ambu bag to provide PPV via the tube, the heart rate should increase to over 100 per/min and color should rapidly improve with adequate ventilation.

Circulation: Cardiac massage (chest compression): After intubation and 30 sec of ventilation with 100% Oxygen, the heart rate remains below 60/min, compress the sternum gently for 1 cm in depth at a rate of 120 times per min, 3 sternal compressions then ventilate once by ambu-bag **(the ratio of compressions to ventilation is 3:1, so the ventilator rate is 40/min).** Chest compression leads to increased intrathoracic pressure, resulting in forward flow through arteries **(aorta and pulmonary artery)**. During relaxation, the decreased intrathoracic pressure helps venous return via the vena cavae.

- If heart rate is increasing >60 min, chest compression should be discontinued and ventilation continued until respiration is spontaneous.
- If no improvement is noted, combined ventilation and chest compression for successive periods of 30 sec, interposed with 6 sec periods of heart rate assessment.

Medication: When heart rate remains <60/min after 30 seconds of combined ventilation and chest compression or during asystole.

- **Epinephrine** via endotracheal tube (after dilution with 2 ml saline) or rapidly via an I.V route, the hearth rate and blood pressure should be increased, dose may be repeated after 5 min.
- **Recheck** the effectiveness of ventilation & chest compression and **consider:**
    - Hypovolemia and severe metabolic acidosis.
        - **Treatment of severe metabolic acidosis :** Sodium bicarbonate given slowly I.V and can be repeated every 5-10 min when needed. Also can be mixed with equal volume of glucose 25% if hypoglycemia is suspected.
        - **Treatment of hypovolemia:** Volume expansion using normal saline, albumin 5%, plasma or whole blood.
        - **Dopamine or Dobutamine** are indicated as severe asphyxia may depress myocardial function.

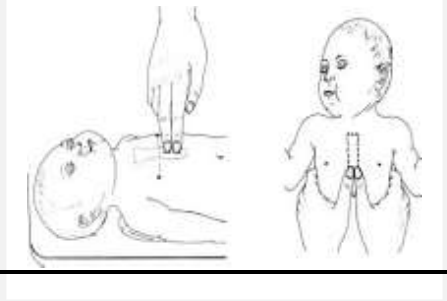
4. **Severe Asphyxia:** Infant who are born limb, cyanotic, apneic, and pulseless, or infants with Apgar score of 0-2 at any time. **These infants require immediate and vigorous resuscitation**. This process requires at least 3 trained people working together.
    - Perform initial stabilization (dry, warm, and positioning).
    - Suction of oropharynx and nasopharynx.
    - Start bag and mask ventilation (40-60 breaths per min).
    - Failure of positive pressure ventilation by bag and mask ventilation indicates endotracheal intubation and perform chest compressions.

Ongoing Care:

When an adequate airway has been established, adequate ventilation has been achieved and HR exceeds 100/min, the infant is best moved to the neonatal I.C.U where careful physical examination, close observation of vital signs, and chest X-ray should be done.

Indication of endotracheal intubation:

1. If bag and mask ventilation is ineffective.
2. If airway obstruction is suspected (e.g., goiter or micrognathia).
3. If meconium aspiration is present or suspected.
4. If external cardiac massage is necessary.
5. If prolonged ventilator support is anticipated (e.g., diaphragmatic hernia).



BIRTH INJURIES

Risk factor of birth injuries

Baby:
- Prematurity.
- Anomaly
- Mal-presentation.
- Macrosomia.

Mother:
- Oligohydraminos.
- Contracted pelvis.
- Prolonged labour.

Physician:
- Instrumental delivery.

Birth injuries are stated as below:

1. Cranial injuries:

A - Scalp lesion
- Caput succedaneum.
- Extra cranial hemorrhage.
  - Cephalohematoma
  - SubgalealHge.

B - Intracranial Hge.
- Subdural Hge.
- Subarachnoid Hge.
- Periventricular or inter ventricular or germinal matrix Hge.

C – Skull fracture
- Linear.
- Depressed.

2. Nerve injuries:

A – Facial nerve injury

B – Brachial plexes injury:
- Erb's palsy "Duchene".
- Klumpke palsy.

C – Pherinic Nerve Pulsy

D – Spinal coral injury

E – Reccurent laryngeal nervy palsy
- Unilateral
- Bilateral

3. Bone Injuries.

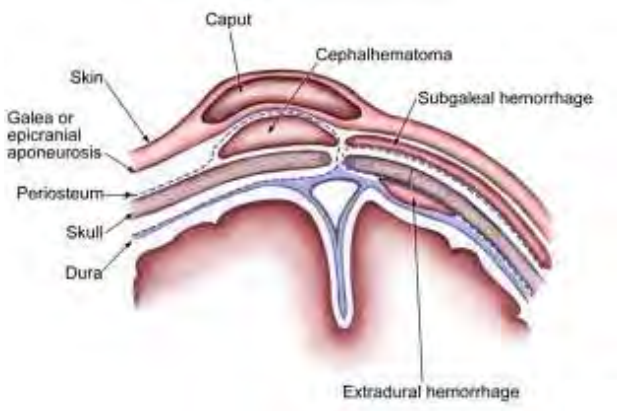
- **A – Clavicle Fracture**
- **B – Fracture Long bones**

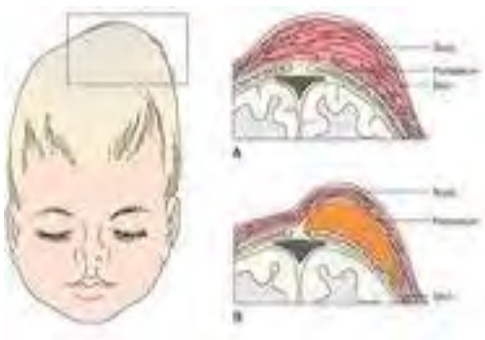
4. Soft tissue injuries

- **A – Sternomastoid muscle injury**
- **B – Visceral injury**
  - Liver and spleen.
  - Adrenal Hge.

1.AScalp lesion

Item	Caput succedaneum	Cephalhematoma
Definition	Diffuse edematous swelling of soft tissue may be with skin bruises and ecchymosis.	Subperiosteal heamorrhage
Site	Presenting part of fetus at birth, can cross suture lines.	Limited to one bone, does not cross suture lines.
Time of appearance.	Present at birth.	Not present at birth, gradually develops within hours.
Disappearance.	Within few days	Needs weeks (4-6)
Treatment and complication.	No treatment and no complication.	May be needed if there is depressed fracture, anemia or hyperbilirubinema.





- **Subgaleal Hge.** It is the blood that invaded the potential space between skull periosteum and galea aponeurotica, an area that extends posteriorly from the orbital ridges to the occiput and laterally to the ears, its growth may be insidious and not recognized for hours or days or it may present as hemorrhagic shock and even death and may associated with anemia, jaundice and may be more severe than cephalhematoma.

1.B Intracranial Hge.

- ✓ **Subdural Hge:** It does not always cause symptom immediately after birth, but with time the RBC undergoes hemolysis and water is drawn into the hemorrhage resulting in an expanding symptomatic lesion (anemia, vomiting, convulsion). Macrocephaly may occur 1-2 months.
  Treatment: Surgical evacuation.
- ✓ **Subarachnoid Hge:** Seizures are common presenting feature, may be spontaneous.
  Treatment: Symptomatic.
- ✓ **Periventricular or interventricular or germinal matrix Hge:** Mainly in preterm 90% occurring in the first 3 days after birth.
  C/P: Seizure, apnea, bradycardia, coma, metabolic acidosis, anemia and bulging fontanel.
  Complications: Peri-ventricular Hge & infarction, post hemorrhagic hydrocephaly.
  Treatment: Treatment is supportive.

2 A. Facial nerve injury.

It is usually a lower motor neuron facial palsy, characterized by asymmetric crying faces. The forehead and eyelid (open eyes) are affected.
C/P: Absent nasolabial fold. Asymmetric crying. Deviation of the mouth to healthy side. Inability to close the eyes.
Treatment: Care of the eyes, care of feeding, physiotherapy if persist more than 3 months, need neuroplasty.

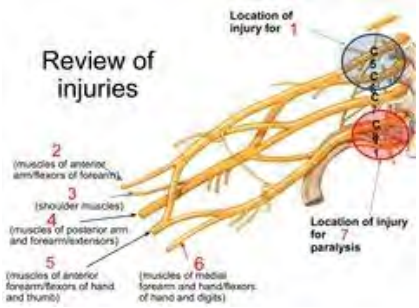
2 B. Brachial plexes injuries.

1. **Erb's palsy** "Duchene" "C5,6,7"
More than 90% of brachia palsies. The characteristic position of the arm like the arm of "waiter tip posture", consist of adduction, internal rotation of the arm with pronation of the forearm.
Reflexes: Absent moro and preserved grasp reflex on the affected side.

Association: Impaired the sensation over the external surface of the upper limb and pherenic nerve palsy in up to 75% of cases.

Management: Immobilization in position opposite to the lesion, physiotherapy after one week (after resolution of nerve edema) to prevent muscle contracture.

2. Klumpke palsy "C8,T1".



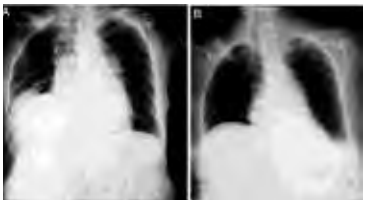
This rare form <1% of brachial palsies.

Hornex syndrome "ptosis, miosis and anhydrosis" occur in 30% if the sympathetic throracic root is also injured. Intrinsic muscle of the hand are affected and grasp reflex is absent "claw hand". Moro is preserved.

Treatment: Immobilization in hand writing position and physiotherapy.

Prognosis: If nerve roots are intact, full recovery will occur in more than 90% by 3 months, if no improvement, consult neurosurgery.

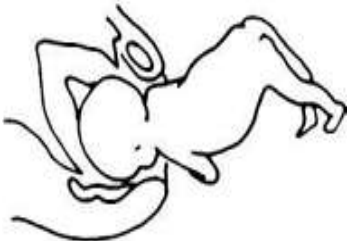
2C. Pherinic nerve palsy: "C3, 4,5".



C/P: Respiratory distress. Diminished breath sound on the affected side.

Investigation: Chest X-ray elevated copula of the diaphragm on the affected side with mediastinal shift to the other side.

2 D.Spinal Cord injury. Breech delivery is risk factor.



C/P: History of difficult delivery with low apgar score and apnea.

Neurological dysfunction:

Management: Immobilization on flat firm surface. CT & MRI may be needed.

2 E. Reccurent laryngeal nerve palsy.

C/P: Unilateral ⟶ hoarse cry & strider.

Bilateral ⟶ severe RD.

3. Bone injuries:

A.Clavicle fracture: Most common neonatal orthopedic injury. Moro reflex is absent on the affected side. Creptus and palpable bony irregularity "in green stick facture may be asymptomatic"

Treatment : Immobilization of arm and shoulder, figure 8 bandage.

B.Fracture long bones: Humerus and femur.

C/P: Loss of Moro flex, swelling and painful movement of the limb.

4. **Soft tissue injuries:**
   A. **Sternomastoid muscle injury. "Sternomastoid tumor",** "calcified muscular hematoma". Commonly present in the 2nd and 3rd week of life. Firm mass with secondary shortening of the muscle leading to torticles.
   Treatment: Physiotherapy.

Neonatal Jaundice

Jaundice is yellowish discoloration of skin and M.M due to increased serum bilirubin above normal level. "Normal cord bilirubin is less than 3 mg/dl. Jaundice is obvious clinically in neonate when serum bilirubin exceeds 5 mg/dl.

Bilirubin Metabolism:

Old RBC { Globin
Heme. "byhemeoxygenase { iron
Bilivirdin

Bilivirdin → Bilirubin "lipid soluble, water insoluble" Unconjugated.

Bilirubin "lipid soluble, water insoluble" Unconjugated. → Carried out on albumin

Carried out on albumin → Uptake by hepatocyte and bind to cytoplasmic ligandins "Z&Y" protein to deliver it to endoplasmic reticulum.

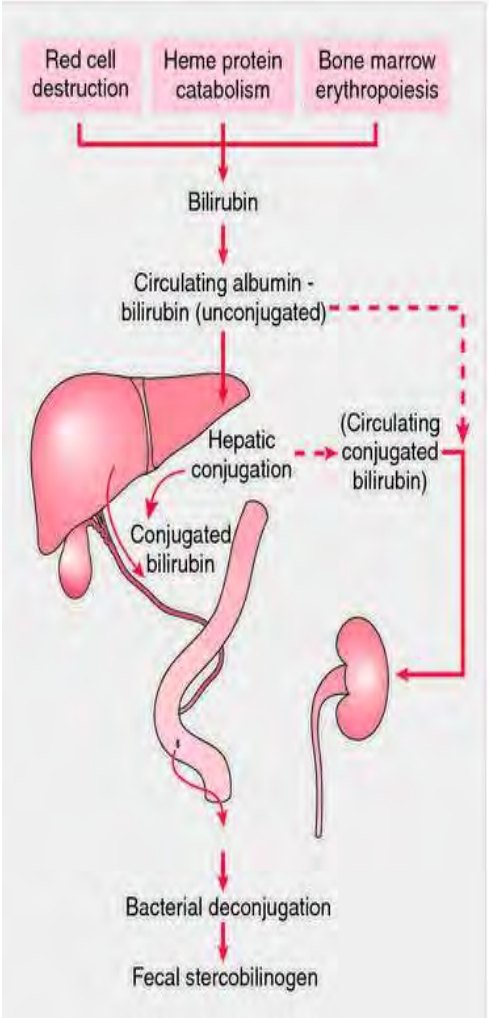
Uptake by hepatocyte and bind to cytoplasmic ligandins "Z&Y" protein to deliver it to endoplasmic reticulum. → **Conjugation** occur which is stimulated by **glucoronyltransferase enzyme.**

Conjugation occur which is stimulated by **glucoronyltransferase enzyme.** → Conjugated bilirubin "water soluble lipid insoluble"

Conjugated bilirubin "water soluble lipid insoluble" → **Active secretion** by liver cell in to bile canaliculi

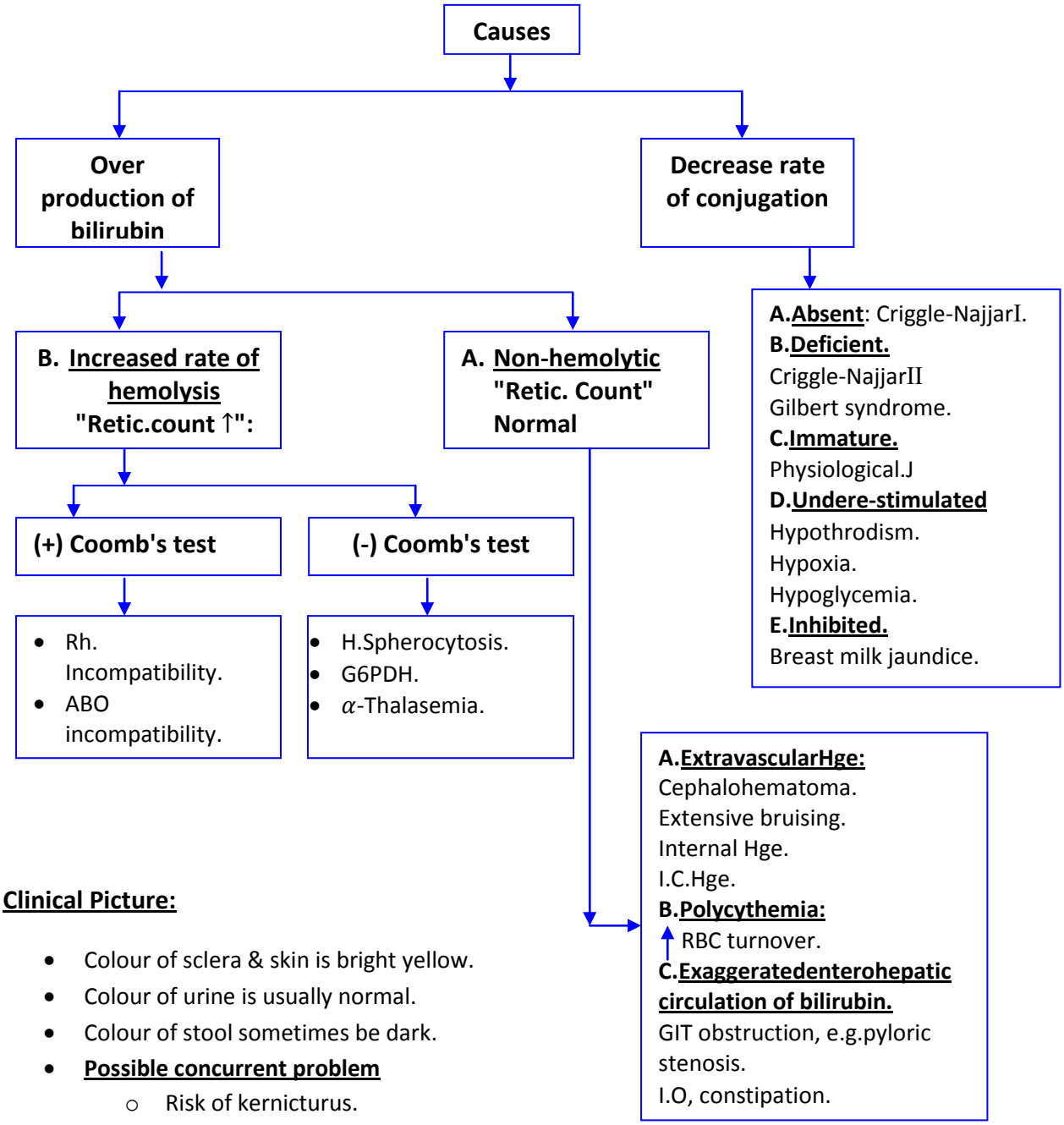
Active secretion by liver cell in to bile canaliculi → **Excretion** in to the intestine.

Excretion in to the intestine. →
- Some amount is deconjugated by B.glucoronidaseenzyme reabsorbed in to the liver "entero-hepatic circulation"
- Some amount is changed to steocobilinogen "stool", small amount of sterocobilinogen reach the systemic blood "urobilinogen" urine.



Unconjugated Hyperbilirubinemia.

- If total bilirubin above normal and conjugated fraction <15% of total bilirubin.



Clinical Picture:

- Colour of sclera & skin is bright yellow.
- Colour of urine is usually normal.
- Colour of stool sometimes be dark.
- **Possible concurrent problem**
  - Risk of kernicturus.
  - Risk of anemia.

N.B: Important.

- Jaundice is evident on face with serum bilirubin, 5 mg / dl.
- On mid abdomen with level around 10 mg/dl.
- On soles of feet with level around 20 mg/dl.

Timing of Clinical Jaundice

1st day	2nd-3rd day	After 1st week	Persistent during 1st month.
RH-incompatibility	Physiological J.	Breast milk J.	CNS I & II
ABO incompatibility	CNS I&II	Hemolytic anemia	Hypothyrodism
			GIT obstruction
			Breast milk J.

Characters	Physiological J	Pathological J
Type	Unconjugated	Conj. and unconjucated.
Rate of rise	1-3 mg/dl per day	>15 mg/dl per day Or > 0.5 mg/dl per hr.
Onset	Full term 2nd – 3rd day Preterm 3rd – 4th day	Any time even the 1st day.
Incidence	Full term 40-60% Preterm 60-80%	
Peak at	Full term 4th day Preterm 6th – 8th day	
Peaking level	Full term 12 mg/dl Preterm 15 mg/dl	More than 12mg/dl & more than 15 mg/dl
Disappear at	Full term end of 1st week Preterm end of 2nd week	Persistent > 1st week in Full term & 2nd week In preterm
Associated symptoms or Problems.	No Risk of kernicture. No anemia or organomegally	There is associated risk of kernicture, anemia and organomegaly
Diagnosis	By exclusion	According to the cause
Treatment	No need of treatment Phototherapy may be needed For every low birth weight.	According to the cause

- **Physiologic jaundice**

- The commonest cause of neonatal jaundice.
- **Etiology:**
  - Decrease glucoronyl transferase enzyme activity.
  - Short life span of neonatal RBC (90 days).
  - Reduced Z&Y protein "ligandins", during the 1st week.
  - Higher erythrocyte mass in neonate.

- **Breast Milk Jaundice.**
  - 2% of breast feed healthy term, recurrence 70% in subsequent pregnancy.
  - Start by the end of the 1st week. It continuous to rise with the maximum level 10 – 30 mg/dl.
  - Peaking at 10-15 days of age. Slowly decline by 3-12 weeks of age.
  - If nursing is interrupted bilirubin level falls quickly.

  Etiology:
  - Unknown but due to breast milk may contain pregnandiole **"progesterone metabolite"** and non-estrified F.A, which inhibit glucoronyl transferase enzyme.
  - B.glucoronidase in breast milk enhance entero hepatic circulation.

  Diagnosis:
  - By exclusion.

Gilbert disease.

Etiology: A.D. Decrease in hepatic glucoronyle-transferase level.
C/P: Mild hyper bilirubinemia, usually need no treatment.

Criggler Najjar Syndrome Type I.

Etiology: AR Absent glucoronyle – transferase level.
C/P: Severe disease.
Treatment: Supportive.

Criggler Najjar Syndrome Type II.

Etiology: AD Partial deficiency of glucoronyl transferase enzyme.
C/P: Less severe respond to phenobarbitone.

Investigation of indirect hyper-bilirubinemia.

1. Total bilirubin and direct fraction .
2. CBC, RBG,ABG.
3. **Coomb's Test**, → if positive, → check blood group of infant and mother.
4. Hb value, if high, → polycythemia.
   if normal or low, → check reticulocyte count.
5. **Reticulocyte count,** → if normal, → extra vascular Hge.

if high, → check RBC, shape by blood film. Osmotic fragility test.

6. **Others:**
   - Serum T4, T3, TSH.
   - Phenobarbitone trial.
   - Urine analysis.
   - Hb electrophoresis.
   - LFT.

M A N A G E M E N T :

- Admission to NICU.
- Cot or incubator care.
- I.V fluid.
- Correct hypoglycemia.
- Phototherapy.
- Exchange transfusion.

Phototherapy

- **Indication :**
  1. Rise of bilirubin below the critical level.
     Healthy full term from 15 – 20 mg/dl.
     In preterm and sick neonate → at lower levels.
  2. Waiting for exchange transfusion.
  3. **Prophylactic in:**
     - Very low birth weight.
     - Severely bruised neonate.
     - Immediate after birth if RH – incompatibility is suspected.

Avoided in : Direct hyper bilirubinemia lead to **Bronzed baby Syndrome.**

- Exposure to blue or white light with wave length 425 – 475 nanometer, convert insoluble unconjugated bilirubin to non-toxic soluble form by photoisomerization and photoxidation, which excreted in bile and urine, 50 cm above the baby "distance of light".

Procedure:

1. Baby is completely naked, except eyes and genitalia.
2. Change position.
3. Continuous exposure with short interval for feeding.
4. Monitor temperature every 6 hours.
5. Fluid requirement increase because phototherapy increase insensible water loss.

6. **Discharge** from phototherapy when bilirubin is low enough to avoid its toxic effect regarding age and condition.

Side Effect:

1. Loose stool due to excretion of bile salts and unconjugated bilirubin.
2. Skin rash and tanning of skin.
3. Hyperthermia or hypothermia.
4. Dehydration due to insensible water loss.
5. Damage to exposed eye **"retina damage"** or genitalia.

Exchange transfusion

- **Indication :**
  1. **RH incompatibility:**
     - Cord bilirubin more than 5mg/dl "normally < 3mg/dl.
     - Cord hemoglobin <10 gm/dl.
     - Rapid rise of bilirubin > 1mg/dl per hour despite phototherapy.
     - **Bilirubin level exceeding:**
       - 10 mg/dl at 1st day.
       - 15 mg/dl at 2nd day.
       - 20 mg/dl at any time.
     - History of kernicture in a sibling.
  2. **In other causes:** If serum bilirubin exceeds critical level.
     - Health full term more than 20 mg/dl.
     - Preterm and sick unit at lower level.

- **Idea:**
  - Remove excess unconjugated lipid soluble bilirubin.
  - Correct anemia.
  - Remove anti-body from the circulation.
  - Provide albumin.

- **Procedure:**
  - **Blood used is :** O-ve, compatible with both maternal and neonatal blood.
  - Fresh and warm.
  - **Amount:** double the neonate blood volume "2x85 ml/kg".
  - Small amount 10-20 ml are removed and replaced by equal amount of new blood.
  - I.V glucose and Ca gluconate are given at 100 ml blood intervals.

- **Complications:**

1. Complication of umbilical catheterization e.g., embolism, thrombosis, sepsis and portal hypertension in later life.
2. Heart failure **"volume overload".**
3. Hazards of blood transfusion.
4. Hypo Ca, hypoglycemia, hyper K.
5. Necrotizing enterocolitis.

Kernicture = Bilirubin Encephalopathy

Yellowish dis-coloration, staining of the cerebellar and cerebral nuclei "especially basal ganglia" due to deposition of unconjugated bilirubin resulting in neuronal necrosis.

Etiology

Level of serum bilirubin exceeding critical levels:

- > 10mg/dl in 1st day.
- > 15 mg/dl in 2nd day.
- > 25 mg/dl afterward.

Duration of exposure to high bilirubin level:

- The longer the duration the more risk.

However, Kernictures may occur at lower levels in presence of R.F which are:

Increase BBB permeability:

- Prematurity & low birth weight.
- Acidosis.
- Sepsis.
- Hypoxia.
- Anemia.

Displace bilirubin from albumin:

- Drugs "ampicilin, sulpha, aspirin".
- Hypothermia.
- Hypoalbuminemia.

C.P:

1. Acute bilirubin encephalopathy
2. Lucid Interval: There is apparent recovery or few symptoms.
3. Chronic bilirubin encephalopathy: Picture of cerebral palsy. Type: Chorio-asthetoid or spastic.

Phase 1:
In 1st -2nd day, poor suckling, lethargy, loss of mmoro reflex, hypotonia and seizure.

Phase 2:
In the middle of the 1st week, hypertonia /opisthotonus, fever, high pitched cry.

Phase 3:
After the 1st week, hypertonia and stiffness.

Management:

a. Prophylaxis.
b. **Treatment:** Not curable, need only supportive treatment for cerebral palsy.

Conjugated Hyperbilirubinemia.

Rise of total serum bilirubin with the conjugated fraction more than 15% of total of >2mg/dl.

Cholestasis: Retention of conjugated bilirubin as well as other constituents of bile e.g., bile salts or "prolonged elevation of serum level of conjugated bilirubin more than 14 days.

Causes

Defective secretion of conjugated bilirubin by hepatocyte.

a.Genetic:
- Rotor & Dubin Johanson syndrome.
- bile acid synthesis defects.
- Progressive familial intrahepatic circulation.

b.Acquired:
Neonatal hepatitis due to:
- **Infection:** TORCH, spsis, viral.
- **Metabolic:** Alpha1, antitrypsisn deficiency.
- Galactosemia. Tyrosenemia.

Defective excretion due to bile flow obstruction.

A.Intra hepatic:
- Congenital intrahepatic biliary atresea.
- Intrahepatic biliary hypoplasia, e.g., Allagile syndrome.

B.Extra hepatic:
- Congenital extrahepatic biliary atresea.
- Biliary stones or tumors.

C/P:

1. Colour of sclera is olive green.
2. Colour of urine is dark.
3. Colour of stool is pale.
4. **Possible concurrent association:**
   - Hepatosplenomegaly.
   - Malabsorption and FTT.
   - Liver cell dysfunction.
   - No risk for kernicture.
   - Underlyhing systemic diseases, e.g., sepsis.
5. **Timing:**
   - In 1st day of life "or in the 1st week of life" : TORCH
   - Late in the 1st week of life : Neonatal sepsis.
   - Persistent in or during 1st month: neonatal hepatitis, congenital biliary atresia.

Investigation:

- LFT.
- Liver biopsy.
- Metabolic screen, TORCH screen, septic screen.

Treatment:

1. **Curable causes:**
   i. **Sepsis :** Antibiotic.
   ii. **Galactosemia :** Lactose free diet.
   iii. **Extrahepatic biliary atresia:** Kasoni operation **"hepato-portoenterostomy".**
2. **Supportive:**
   i. Formula with medium chain triglyceride.
   ii. Fat soluble vitamin and water soluble vitamin.
   iii. Bile acid binders **"cholestyramine".**
   iv. Mineral **"Ca, Phosphate, zinc".**
   v. Liver transplantation in end stage liver failure.

Prematurity

Neonate whose birth occur before 37 complete weeks, regardless of the birth weight:

Etiology

Fetal:	Fetal distress, multiple pregnancy, Erythroblastosis, cong. Anomalies.
Maternal:	Pre-eclampsia, chronic medical illness, e.g., CHD, infections e.g., Group B streptococcus.
Placental:	Placental dysfunction, P.Previa, Abruptio placenta.
Uterine:	Incompetent cervix.
Others:	Premature ROM, polyhydraminos.

Physiological Handicaps	Complications

1. **Respiratory system:** Poorly developed, resp.muscle are weak, Gag & Cough reflex are weak, lung alveoli are poorly developed with decrease elastic tissue, decrease formation of surfactant, immature pul. Capillaries.	• Asphyxia neonatrum.
	• Apnea.
	• Aspiration pneumonia.
	• Lung collapse.
	• RDS.
	Complication of Treatment:
	  • Pneumothorax.
	  • Broncho-pul.dysplacia.

2. **Temperature regulation mechanism:** • Immature center. • Decrease sweating mechanism. • Decrease S/C fat. • Poor muscular activity. • Relatively large surface area.	• Cold injury **"sclerema neonatorm".** • Hypothermia.
3. **GIT tract:** • Weak suckling and swallowing reflexes, weak cardiac sphincter with active pyloric sphincter & small capacity of the stomach. • Poor developed lacteals **"lymphatic channels"** present in villi. • Hypotonia of muscle coat of color.	• GER & vomiting. • Decrease absorption of Fat and Fat soluble vitamins, leads to malnutrition and Rickt's. • Abdominal distension and constipation.

4. <u>Liver function:</u> • **<u>CHO:</u>** Decrease glycogen storage & immaturity of liver enzyme. • **<u>Protein :</u>** Immaturity of enzyme. • Immaturity of glucoronyl transferase enzyme.	• Hypoglycemia. • Hypoproteinemia and hypoprothrombinemia. • Phyciological jaundice. • Kernicture and Brain damage.
5. <u>Hemostatic:</u> • The capillaries are fragile, impaired coagulation.	• Liable to Hge. • I.C.H. (I.V.Hge.)
6. <u>Immune system:</u> • Immature with decrease phagocytosis and compliment, decrease IG, and CMI.	• Infection

<u>Another complication of prematurity:</u>

1. **<u>Brain damage:</u>** Due to : I.C.H, infection, hypoxia, kernicture, Hypoglycemia.
2. **<u>Kernicteures:</u>** Due to : uncong. Hyper bilirubinemia, low body Weight with diminished s/c fat, exposure to hypoglycemia, hy[oxia, acidosis and hypothermia.

3. <u>Necrotizing enterocolitis:</u>

<u>Causes:</u> Not known but there is risk factor.

i. Premature & very low birth weight & SGA.
ii. Hypotermia.
iii. Polycythemia.
iv. Exchange transfusion.
v. Cyanotic CHD.

<u>Pathogenesis:</u>

Decrease of blood supply of bowel → to decrease defence mechanism → good media for bacterial proliferation → necrosis → gangrene → perforation → Peritonitis. **<u>"Affect, large and small intestine".</u>**

<u>C/P:</u>

- Vomiting of bile stained fluid, distended Abd. With blood and mucus in the stool.
- Temperature instability.
- Apnea.
- Lethergic – Rectal bleeding 30%.

<u>Diagnosis:</u>

Abdominal x-ray shows intra mural gases, perforation of bowel "air under diaphragm" → Pneumotosis intestinal.

Treatment:

- Stop oral feeding.
- I.V. Fluid.
- I.V. Broad spectrum, antibiotic and metronidazole.
- Surgical resection of perforated and necrotic bowel may be needed.
- NGT → to avoid gases accumulation.

4. **Retinopathy of prematurity:**
   Due to oxygen at high concentration → high oxygen tension in arterial blood → lead to constriction of retinal blood vessels → proliferation of new fragile blood vessels → bleeding into vitreous humor with Fibrosis and membrane formation behind the lens **"Retro-lental Fibroplasia".** This ends by Retinal Detachment and blindness.

5. **Long term complication:**
   a. Malnutrition - Rickets - and anemia.
   b. Retinopathy of prematurity.
   c. CP and MR.
   d. Cong. Malformation.
   e. Chronic lung disease.
   f. PDA.

C/P of Prematurity:

- Low anthropometric.
- Hypoactive – weak cry.
- Head and abdomen appear large as compared with limbs.
- **Skin:** thin, red, shiny and translucent with excess languo hair and vernic caseosa.
- Slight edema.
- Hypotonia, **(frog like position).**
- Respiration irregular with attack of apnea, **"RR=35-45 cycle/min".**
- HR = 140 b/min with irregularity may be present with no significant.

Management of prematurity:

A. Immediate post-Natal care:
- Baby must be warmed.
- Drying and suction.
- Apgar & resuscitation.
- Complete examination.
- **Vit K.**
- If large > 2Kg with no critical illness → discharge.
- If < 2 Kg or critical illness → incubator.

B. Proper AN care.
- Avoid smoking, radiation and drugs.
- Assessment by ultra sound.
- Dexamethazone to mother.

C. Incubator care:

1. **Temperature**: **"36.5 – 37.2°C"** to reduce heat loss and Oxygen consumption.
2. **Humidity**: 40-60%, reduce heat loss and water loss from Bronchial tree.
3. **O_2 therapy**: Given at lowest conc. For shortest period.
4. **Prevention of infection:** Hand washing.
5. **Feeding:**

Oral feeding: By suckling if large premature, or by **Gavage tube** feeding with **NGT** in baby < 1.5Kg with mild R.D.
Start with small amount if no vomiting every 2-3 hours until reach 150-180 cal/kg per day.
If baby intolerance to oral feeding → start I.V.fluid.
I.V. Fluid: → In severe R.D, or intolerance to oral feeding.
1st day 60-80 ml/kg per day.
Type : Glucose 10%.
Then glucose 10%, saline calcium added.
If oral feeding cannot resumed initiate TPN.

6. **Treatment of Association:**
   Antibiotic for sepsis.
   Phototherapy.
7. **Discharge from incubator if :**
   i. Infant >1750 grams with good suckling.
   ii. Maintain his temperature outside the incubator.
   iii. No critical illness.
   iv. Normal respiration.

Instructions to the parents:

- Maintain body temperature.
- Keep infant away from infection and minimize handling and overcrowding.
- Schedule for feeding.
- Schedule for vaccination according to the date of birth.
- Encourage follow up visits.
- Ophthalmic examination.
-

Post maturity

Infant born after 42 weeks gestation regardless of his birth weight.

Causes :

- Unknown.
- High incidence with trisomies or anecephaly.

C/P:

Face: opened eyes and alert baby.

Skin: wrinkled, dry ± meconium staining.
Long nails.

Complications:

- Perinatal asphyxia ±meconijm aspiration.
- Hypoglycemia.
- Polycythemia.
- Hypocalcemia.

Small for gestational age

Types

- **Symmetric IUGR.**
  Weight, length and HC < 10th centile, mainly **due to :**
- Insults in early pregnancy.
- Cong. Infections.
- Chromosomal disorders.

- **Asymmetric IUGR.**
  Weight is much more affected than length and HC, mainly **due to :**
- Insults in late pregnancy.
- Maternal malnutrition.
- Placental insufficiency.
- Poor socioeconomic.
- Multiple gestation.

Assessment.

- Detect impaired Fetal growth by **prenatal ultra sound:**
- ↓ Biparietal diameter.
- ↓ Abdominal circumference.
- ↓ Femur length.

Neonatal :

- Low birth weight.
- Ballard score.

- **Complications IMP**
  - Hypoglycemia.
  - Polycythemia.
  - Coagulation disorders.
  - Pul. Hge.
  - Infection.
  - Hypothermia.
  - Poor feeding.
  - Jaundice.
  - Permanent physical and mental retardation.

Neonatal Respiratory Distress

Causes of N.R.D

Peripheral

Pulmonary:

1. **Lungs:**
   - R.D.S.
   - T.T.N.
   - Cong.Pneumonia.
   - Cong.Lobar emphysema.
   - Lung Collapse, cyst.
   - Hypoplasia.
2. **Airway:**
   - MAS.
   - Obstruction:bilateralchonal atresia, vascular ring, macro glossia.
3. **Pleura:**
   - Pneumothorax.
   - Pleural effusion.
   - Diaphragmatic hernia.

Extra pulmonary:

1. **HF "cardiac".**
2. **Hematologic:**
   - Severe anemia.
   - Polycythemia.
3. **Metabolic.**
   - Hypoglycemia.
   - Hypothermia.
   - Metabolic acidosis.

Central:

Central NSF, **due to:**

- Over sedation.
- Perinatal asphyxia.
- I.C. Hge.

Manifested by :

- Slow, irregular, gasping respiration.
- Apneic attack.
- Disturbed consciousness.
- Poor reflexes.

NB:

N.B. **Piere Robin syndrome:** It consists of micrognathia (Mandible hypoplasia) so tongue is easily displaced backward.

Respiratory Distress Syndrome
=
Hyaline Membrace Disease

Syndrome occurs in newborn due to surfactant deficiency.

Surfactant:

- **Composed mainly of :**
  - Dipalmitoylphospatidy/cholin **"lecithin".**
  - Phosphatidyl glycerol.

- Surfactant protein A,B,C,D

- **Produced by :** - Alveolar cells type II starting 21-23 weeks of gestation and Mature after 35th weeks **"Near term".**
- **Function:** - Reduced surface tension of fluid within the alveoli, so prevent their collapse at the end of expiration and reduce the lung stiffness and work of breathing " ↓ the negative intrathoracic pressure".
- **Pathophysiology** - ↓ Surfactant →↑ alveolar surface tension → diffuse alveolar collapse "atelectasis" during the expiration → high pressure is required to initiate lung inflation →↑ed work of breathing with **impaired gas exchange**

↓O_2 (Hypoxia) → Stimulate anerubic respiration → ↑Lactic Acid → Metabolic Acidosis

↑CO_2 (hypercapnea) → Respiratory acidosis → ↓Perfusion → Severe distress

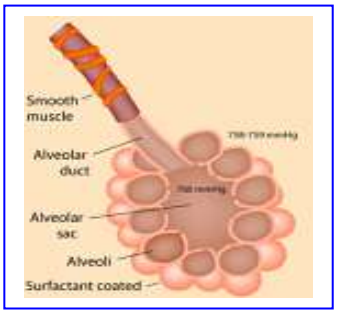
- **Hypoxemia**→ pulmonary vessel V.C → alveolar hypoperfusion →↓ metabolism of Alveolar cells type II → more surfactant deficiency → progressive at electasis.
- **Pathology :** - Diffuse atelectasis.
  - Esinophilic "hyaline" membrane lining alveoli & small bronchiole.

Causes

1. **Prematurity :** The small the G.A the higher incidence, about 60% of premature < 28 weeks develop RDS.
2. **Infant of diabetic mother:** Maternal hyperglycemia →Fetal hyperinsulinemia → reduced Fetal cortisone, Fetal cortisone is essential for surfactant production.
3. **C/S and precipitate labour:** Due to lack of stressful delivery → decrease fetal cortisone.
4. **Perinatal asphyxia:** Due to hypoxemia of alveolar cells type II.
5. **Others:** 2nd twin, male sex, RDS in sibling. Hypothermia, which cause decrease synthesis of surfactant.

C/P

- **Signs fof R.D:** After birth immediately, 4-6 hours post delivery progressively increase to reach peak at the 3rd day of life.
- **Auscultation:** In severe cases: Decrease air entry bilateral with bilateral fine basal crepitation.
- **Course:** In mild cases→ gradual improvement occurs after the 3rd day. In severe cases → ends in deaths or complications.



Complications

Of the disease:

- Hypoxia & acidosis leads to decrease myocardial contractility and cardiogenic shock.
- PDA due to hypoxia.
- Interventricular Heg.
- Sepsis.

Of the treatment:

- Pneumothorax.
- Pul. Hge.
- Chronic lung disease "broncho-pul" dysplasia in infant treated for long period by O_2.
- Retionopathy of prematurity.

Investigations

Prenatal Diagnosis: Done on samples from amniocentesis.

1. **Lecithin/sphingomyelin Ratio:**

- If > 2.5: mature lung , no risk of RDS.
- If 1.5 – 2 transitional lung, risk of RDS.
- If < 1.5, immature lung lead to severe RDS.

2. **Saturated phosphatidyle choline:**

- > 500 micro gram/dl means to immature lung.
- < 500 micro gram/dl means mature lung.

Risks of RDS is reduced by

1. **PROM.**
2. **Prenatal steroid.**
3. **Pre-eclampsia.**
4. **I.U.G.R.**

Post natal :

1. **X-ray: Mild RDS :** Diffuse Reticulo-nodular infiltrate "ground glass appearance", air bronchogram : outlie of air filled large airway against opaque lungs, small lung volumes.
   Severe RDS: White lung "opacification of both lungs".
2. **ABG:** In severe RDS, there is hypoxemia, hypercapnea&resp.acidosis.
3. **Sepsis workup:** To rule out early onset sepsis.
4. **Shake test:** Pul.aspirate before baby is one hour of age, add 0.5 ml pul.aspirate to 4 ml saline and 0.5 ml absolute alcohol and shake.

- **Absence of bubbles**: indicate absent surfactant.
- **Incomplete circles of bubbles:** intermediate risk of RDS.
- **Double rows of bubbles or more:** No risk of RDS " mature lungs".

Management

A.Supportive measures:

1. **Incubator care:** With frequent monitoring of vital signs and ABG.
2. **Respiratory Support:** Keep PaO_2 between 50-80 mm Hg.
3. **Support Circulation:** IV fluid, Correct hypotension, "plasma, albumin, dopamine".
4. **Support Nutrition:** If oral feeding can't be tolerated within 4-5 days advice TPN. **(Oral feeding in preterm with RDS may increase O_2 consumption & predispose to NEC).**
5. **Symptomatic Treatment:** Correct metabolic acidosis & anemia.

B .Specific Treatment:

1. **Antibiotic:** Because it's difficult to differentiate RDS from cong. pneumonia, give (Ampicillin & gentamycin).
2. **Curfactant:** Immediate after birth for very low birth weight "Prophylactic" and cases of established RDS "main treatment".

   Types :
   - **Synthetic:** Exosurf and recently surfaxin.
   - **Natural:** Lung extract of bovin "survanta", calf "infasurf".

   Doses:
   - 3-5 ml/kg per dose in ETT for 2-4 doses at 6-12 hr intervals.

Method of O_2 delivery:

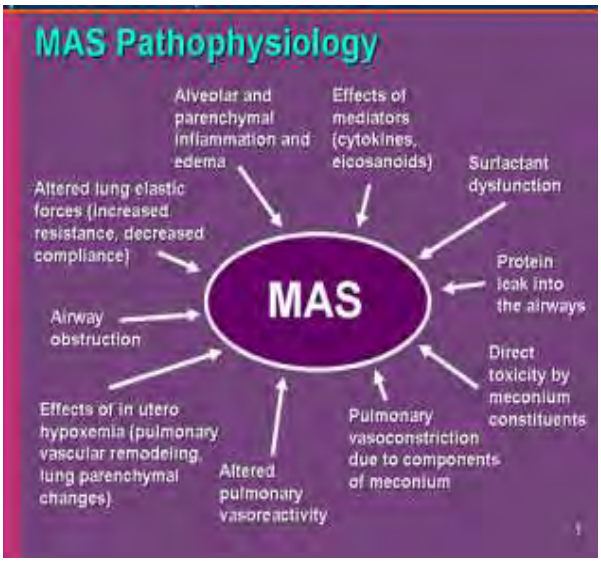
1. **Incubator O_2:** In infant > 1500 gm and 32 weeks and doesn't require more than 30% O_2.
2. **Head box :** In infant requiring more than 30% O_2 and less than 1500 gm or 32 weeks.
3. **Continuous positive airway pressure, "CPAP":** Only during expiratory phase.
   - PaO_2<50 mm Hg in 60% O_2 or more.
   - Prevent collapse in surfactant deficit alveoli.
4. **Mechanical Ventilation:** Pressure during inspiratory and expiratory phase.
   - In apnea.
   - Respiratory failure: PaO_2< 50 mm Hg, $PaCO_2$> 60

Transient Tachypnea of Newborn

- Transient R.D in full term.
- **Mechanism :** Delayed resorption of Fetal lung fluids by pul. Lymphatic.
- Alveoli in intrauterine Fetal life filled with fluid, this fluid normally decreased by :
    i. Squeezing of baby in birth canal during normal vaginal delivery.
    ii. Remaining fluid absorbed due to vasodilation.
- **Risk Factors:**
    i. Maternal Diabetes.
    ii. Perinatal asphyxia.
    iii. Excess maternal analgesia.
    iv. C. Section.
    v. C/P: Mild respiratory distress within few hours after birth. Spontaneous resolution occur within 2-3 days.
- **Chest X-ray:**
    i. Increased pul. Vascular marking.
    ii. Hyper inflation of the lung.
    iii. Fluid in costopherinic angel and in horizontal fissure.
- **Treatment:** No specific treatment, oxygen usually low concentration. Anti biotic and IP fluid may be needed.

Meconium Aspiration Syndrome

- Severe R.D in post mature and full term exposed to intrauterine asphyxia.
- **Mechanism:** I.U. asphyxia → paralysis of anal sphincter → meconium stained liquor → if meconium aspirated in lung at birth, **the area of :**
    i. Complete obstruction → patchy collapsy.
    ii. Incomplete obstruction → Air trapping.
    iii. Secondary infection and chemical pneumonitis.



C/P :

1. Severe R.D with grunting and cyanosis.
2. Meconium staining of the nails and umbilicus and skin.

Chest X-ray:

- Coarse infiltrate.
- Over inflation and may be pneumothorax.

Management:

A.Preventive: Antenatal care and Fetal monitoring, suction before the first breath and don't use over nose suction because it may stimulate parasympathetic system, which leads to apnic attack.

B. Curative:

1. Oxygen therapy.
2. Antibiotic.
3. May be mechanical ventilation.

N.B:

Meconium liquor is not a sign of meconium aspiration in breech presentation, because meconium will pass when low part of the body is passing through birth canal, so the head will be above it and not affected. Do not use ambu bag because it will push meconium more deep.

Pneumothorax

- **Causes :**
  1. RDS.
  2. M.A.S.
  3. Over Resuscitation.
  4. Staph. Pneumonia "Rare".
- **Diagnosis:** Asyptomatic in 1% of all neonate.
- **X-ray:** Jet black opacity and deviation of mediastinum to the opposite site.
- **Treatment:** Air aspiration then insert inter coastal tube with under water seal. In the second inter coastal space.

Neonatal Pneumonia:

As result of infection may be acquired by:

1. **Transplacental infection** : e.g., TORCH.
2. **Perinatally:** GIT or genito urinary maternal secretion.
   Micro-organism : Group BB, hemolytic streptococci, gram negative enteric aerobes and CMV.
   Factors increased risk : Premature, PROM, amnionitis.
3. **Post natal infection:**
   Hospital or community acquired.

Neonatal Apnea

Cessation of respiration more than 20 seconds associated with bradycardia or cyanosis.

Etiology

1. Prematurity.
2. Sepsis.
3. Hypoxia.
4. Hge (I.C. Hg).
5. Metabolic e.g., Hypoglycemia &Hypocalcemia.
6. Hypothermia.
7. G.E.R.
8. Meternal Drugs

Treatment

- Correct the cause.
- Physical stimulation during the attack.
- Resp. stimulation e.g., theophylin which is the treatment of choice in apnea of prematurity.

Hemorrhagic Disease of Newborn

Due to deficiency of vit.K dependent clotting factors. II, VII, IX, X, 10,7,2,9.

Incidence:

- About 2% of neonate not given vit.K at birth.
- Premature is more liable than full term.
- Breast feeders is more liable than Formula feeders "breast milk is deficient in Vit.K."

Causes and timing

1. **Classic form**: Due to:
   a. Depletion of transplacentalVit.K by the 2nd day.
   b. Delay endogenous Vit.K synthesis to the 7th day "due to lack of intestinal bacterial flora".
   c. Liver immaturity in preterm babies.

2. **Early onset hemorrhagic disease:** Present in first day of life due to maternal medication during the pregnancy inhabitingVit.K production, e.g., phynitoin, phenobarbitone.

3. **Late Onset Heg. Disease :**Present between 2nd week to the 6th month may occur due to Vit.K mal absorption e.g., cholestasis.

C/P :

1. Bleeding tendency:
   - Usually present between 2nd – 7th day of life, from GIT and internal hge. is rare.
   - Baby looks healthy except with severe hge.
2. Hemorrhagic anemia.

Investigation

1. Prolonged PT and PTT.
2. Deficiency of Vit K dependent factors.
3. Normal bleeding time and plt.count.

Management

1. **Prevention:**
   - Vit.K_1, 1 mg I.M at birth.
   - Vit K_1, 1 mg. I.V with history of maternal anti-conveulsant.
2. **Treatment:**
   - Vit. K_1 from 1-5 mg I.V.
   - Fresh plasma transfusion "For preterm and liver diseases" if PT greatly prolonged.
   - Fresh blood transfusion in severe bleeding.

Neonatal Polycythemia

Increase RBC count with venous HCT over 65%.

Etiology

1. Placental red cell transfusion e.g. delayed cord clamping.
2. Placental insufficiency "chronic I.U.hupoxia", which increase erythropiotine.
3. Others e.g. I.D.M, wilms tumor
    - Premature
    - CHD
    - FFT in twin

C/P

- Asymptomatic "only plethoric face"
- Symptomatic e.g. R.D., indirect hyperbilirubinemia, lethargic, hypoglycemia, NEC.

Treatment

Indication: symptomatic cases and those with HCT >75%.
Action: partial exchange transfusion with 5% albumin, saline or plasma.

Neonatal Bleeding

1. Bleeding in healthy baby:

a. Swallowed maternal blood
b. Hemorrhagic disease of newborn
c. Inherited coagulation defects e.g. hemophilias
d. Inherited thrombocytopenia e.g. TAR syndrome
e. Immune thrombocytopenia: due to transplacental Anti-plt.Ab
f. Trauma to the involved site e.g. thermometer "Rectal bleeding"

2. Bleeding in sick baby

a. Gastric stress ulcer
b. NEC
c. DIC
d. Surgical cause e.g. volvulus
e. Severe liver disease

Neonatal Anemia

2. Physiological anemia of newborn:

Hb concentration at birth in full term baby is 15 – 18 gm% due to relative intrauterine hypoxia stimulating erythropiosis, at birth this hypoxia is eliminated and erythropiosis is suppressed, at 9 – 12 weeks of age in full term and 6 weeks in preterm Hb level is lowered to level which erythropoiesis is again stimulated.

1. Pathological anemia:

✓ **Decrease production**

- B.M depression by sepsis
- Cong. Anemia
- Anemia of prematurity
- B.M replacement e.g. cong. Leukemia

✓ **Increase loss:**

1. **Hemolysis**
   - Hemolytic anemia, e.g., spherocytosis, G6 PD.
   - Hemolytic disease of newborn.
2. **Hge:** Internal: Feto-meternal or F.F transfusion, external, e.g., birth trauma or Hge, during or after delivery.

Alkali Denaturation Test

Fetal blood contains HbF which resists denaturation by alkali "Na Hydroxide", while maternal blood "HbA" is easily denaturated, HbA change from pink to yellow brown "alkaline hematin", while HbF stay pink.

Hemolytic Disease of Newborn

Erythroblastosis Fetalis

Result from transplacental passage of maternal antibodies active against fetal RBC.
R.H Iso-immunization: 15% of women are “Rh” negative. Despite the high rate of Rh-negative mothers, **Rh-iso-immunization is much less frequent due to :**

1. Immunization occurs late in pregnancy or during labor, so the 1st baby is un-affected.
2. The father, which is Rh positive may be D/d heterozygous, so the baby may be Rh negative.
3. ABO incompatibility between the baby and mother reduce incidence of Rh iso-immunization as anti-A and Anti-B Ig., already present in maternal plasma, so clears the blood from RBCs of the baby.

C/P

It depends on the severity of hemolysis.

A. Hydropsfetalis:

It is the most severe form with bad prognosis may be still born or die later.
Pallor due to severe anemia, generalized edema, anemic HF, Hepatosplenomegaly.

B. Ictenus gravis neonatonum.

Less severe form, anemia, hepatosplenomegaly, jaundice, die if not treated urgently.

C. Rh-Hemolytic anemia.

It is usually at its maximum at the end of the 3rd week, may need packed red cell transfusion.

Investigations

1. **Antenatal:**
   a. Screening all women for Rh – before pregnancy.
   b. During pregnancy for Rh negative mothers, measurement of maternal anti-D in maternal serum, if the titer is more than 1/64 or rising, amniocentesis is recommended.
   c. Amniocentesis and spectrophotometric measurement of optical density of amniotic fluid. The severity of the hemolysis can be assessed by the quantity of bilirubin transferred from the fetus to the amniotic fluid.
2. **After Delivery:** Cord blood sample for :
   a. Blood grouping,.
   b. Hb, anemia is present if Hb is < 14 gm/dl.
   c. Reticulocytosis.
   d. Positive Coomb's test.

Management

1. Preventive:

Give Anti-D for Rh negative women during pregnancy, after 28 weeks of gestation and a second dose within 72 hours after birth.

2. Curative:

A. Before birth:

If by amniocentesis, severe hemolysis is present, two choises:

- Induction of early delivery at 34 weeks.
- Fetal intrauterine intraperitoneal or intraumbilical blood transfusion of group O Rh negative blood, between 22 – 32 weeks of gestation indicated when fetal hematocrit of less than 25% or Hb less than 8 gm/dl or fetal hydrops.

B: After Birth:

The neonate should be admitted to "SCBU", phototherapy starts immediately and an exchange blood transfusion is performed, if bilirubin level reach 20 mg/dl.

ABO in compatibility

About 10% group O, women have IgG anti-A and Anti-B hemolysins which can cross the placenta (normally anti-A and anti-B are IGM and does not cross the placenta).

In comparison to Rh incompatibility:

1. The mother is usually group O and the baby A or B.
2. The first baby can be affected with milder presentation.
3. Direct Coomb's test is weak positive in contrast to Rh incompatibility is strong positive.

Neonatal Seizures

Paroxysmal alternation of neurologic functions including motor, behavioral and or autonomic changes.

Causes:

A. CNS : The commonest cause:

a. Hypoxic ischemic encephalopathy 40%.
b. I.C.Hg and CNs trauma 15%.
c. CNS infection including meningitis, encephalitis TORCH, tetanus 5%.
d. CNS malformation e.g., cerebral dysgenesis 5%.
e. Bilirubin encephalopathy "Kernictenis".
f. N.Cut syndromes e.g., NF, TS.

B. Metabolic :

a. Hypoglycenia.
b. Hypocalemia.
c. Hypomagnesemia.
d. Hyponatremia or hypernatremia.
e. Inborn error of metabolism e.g.,:

- Pyridoxin "Vit.B6" dependency.
- Phynylketonuria.
- Hyper ammonemia.
- Organic acidemia.

C. Other causes:

a. Drug withdrawal e.g., maternal narcotic or heroin addiction.
b. Theophylline at toxic dose.
c. Administration of local anesthesia during labor.
d. Neonatal epileptic syndromes e.g., benign familial neonatal seizures.

C/P

1. **Onset of convulsion:**
   a. **1st three days** of life e.g., HIE, drug withdrawal, I.V.Hge, Metabolic cause.
   b. **After 3 days** e.g., I.C.Hge and metabolic cause.
   c. **After 1st week** e.g., meningitis.

2. **Types of Seizures:**
   a. **Subtle seizures:** The commonest type occur alone or with other types. It may be:
      i. Eye movement, blinking, nystagmus or sustained eye opening.
      ii. Repititive oral movement, suckling, chewing or lips smaking.
      iii. Limb movement, bicycling, boxing.
      iv. Epileptic apnea with initial tachycardia and episodic oxygen desaturation.
   b. **Tonic seizures:** The commonest form in premature, sustained rigid posturating of the body, may be focal or generalized.
   c. **Clonic seizures**: Rapid alternating contraction and relaxation of the muscles, may be unifocal, of multifocal or rarely generalized.
   d. **Myoclonic seizures:** Non-rhythmic sudden fast shock like movement of limbs, may be unifocal or generalized.

Investigations

1. Check initially for blood glucose, serum calcium, magnesium, Sodium, CBC, Urine, CRP etc.
2. Septic screen.
3. Cranial U/s, CT, MRI.
4. EEG.
5. Metabolic screen.
6. TORCH screen.
7. Urine drug toxicology.

Jitteriness

Movement disorder "Not seizure", restricted to the new born, it is a tremor like movements of limbs "never affect the face".
Precipitated by sensory stimuli or by stretching of the joints, stopped by holding the limb or passive flexion. Not associated with HR or EE changes. May occur in normal infant, drug withdrawal, hypocalcemia and hypoglycemia.

Management

1. **Step One:**
   a. Aim; stabilized; vital function.
   b. Admit neonate to neonatal intensive care unit "NICU".
   c. Support respiration: suction of secretion and 100% oxygen.
   d. Support circulation :I.V.Fluid.

2. **Step Two:** Start to treat underlying cause.
   - Hypoglycemia: Glucose 10%.
   - Hypocalcemia: Calcium gluconate.
   - Hypomagnesemia: Mg^{+2}Sulphate.

4. **Step Three:**
Aim; specific anti convulsant agent, if seizure prolonged more than five minutes or re-occur.
   - Phenobarbitone.
   - If no response, add phenytoin.
   - If no response, use benzodiazepine.
   - If seizures resist to preceding drugs, give pyridoxine I.V.

3. **Step Four:**
After controlling the attacks:
   - Treat the cause e.g., antibiotic for meningitis.
   - Gradual withdrawal of anticonvulsant.
   - If maintenance therapy required, usually use phenobarbitone, clonazepam or sodium valproate and re-evaluate at 6-12 weeks intervals.

Neonatal Cyanosis

Bluish dis-coloration of skin and M.M, due to presence of more than 5 mg/dl reduced Hb in capillary blood, it may peripheral **"not affect tongue"** or central "affect the tongue as well".

Causes

- With increase work of breathing.
  - **Pulmonary:**
    - Severe RDS.
    - Cong. Pneumonia.
    - MAS.
    - Pneumothrax.
    - Diaphragmatic hernia.
    - Persistent pul.HPN in newborn.
  - **Cardiac:**
    - CCHD with increase pul.B.flow.
    - CHD with critical obstructions.
- With decrease work of breathing.
  - Central R.F.
- With normal work of breating.
  - CCHD with decrease pul. Blood flow.
  - **Hematology:**
    - Polycythemia.
    - Methemoglobinemia.

N.B.

Hyperoxia test: Differenciate between pul., and cardiac causes of cyanosis.
Performs ABG, then give 100% Oxygen and perform ABG again.
If PAO_2 becomes more than 150 mmhg after 100% oxygen, then it means
Pul.cause, if PAO_2 remainsbelow 10 mmhg despite 100% oxygen, means **Cardiac cause.** This patient should receive PGE_1 infusion to maintain ductusarteriousus patent.

Neonatal Septicemia

Clinical syndrome characterized by systemic illness with infection "multiplication of bacteria with their toxin in blood".

Q. Why Neonate more susceptible for infection?

1. Decrease HCL in stomach.
2. Skin is delicate and easy to invated.
3. Umbilical cord is the good gate to pass of organism and good media for multiplication.
4. Defective phagocytosis.
5. The absent IgM.
6. Defective cell mediated immunity.
7. Deficient compliment.

	Early Sepsis	Late Sepsis
Onset	In 1st week	After 1st week.
Risk factors	1.Premature. 2.PROM> 18 hours. 3.Chorioamnioitis. 4.Maternal UTI. 5.Maternal AP Fever >38 °C. 6.Maternalbacteruria. 7.Maternal malnutrition. 8.Difficultlabour.	1.Prematurity. 2.Hospitalization. 3.Umbilicalcatherization. 4.ETT intubation. 5.Mechanical ventilation. 6.Other disorders. * Meningomyocele. *TracheosphegealFistual. 7.Black male infant.
Organism	1.GroupB.B.Hemolytic streptococci. 2.E.Coli. 3.Listeriamonocytogens. 4.HIB. 5.Klebsiela. 6.TORCH.	1. Staph aurus. 2.Hitypeb. 3.Klebsiella. 4.Pseudomonous. 5.Enterovirus. 6.E.Coli
Main presentation	R.D.	Focal infection e.g., Meningitis.

C/P

a. **Early manifestation:** Non-specific.
   i. R.D and apnicattachs.

ii. Lethargy.
iii. Poor feeding and vomiting.
iv. Unstable temperature "mainly hypothermia".
v. Poor moro and sucking reflexes.

b. **Late manifestation:** Focal Infection.
i. **Respiratory:** pneumonia, RD.
ii. **Neurologic:** Meningitis with seizure, tense bulging AF, high pitched crying, hypotonia and hyporeflexia, irregular respiration.

iii. Cardiac:
A. Shock, pallor, cold skin and hypotension.
B. Oliguria, cyanosis.
C. Heart failure: Presented with tachypnea, tachycardia, tender hepatomegaly.

3. G.I.T: vomiting and diarrhea.
A. Direct hyperbilirubinemia.
B. Hepatosplenomegaly.
C. NEC.
D. Abd. Distension.

4. Hematologic:
A. Pancytopenia and bleeding tendency.
B. May be DIC.

Investigations

Septic screen.

Management

1. **Prophylaxis:** Maternal I.P ampicillin, if there is risk factors.
2. **Curative:**
   a. **Incubator care:** in NICU for:
   i. **Support respiration:** Oxygen & mech. ventilation.
   ii. **Support circulation:** I.V. fluid and packed R.B.C transfusion, Fresh plasma transfusion, Dopamine infusion.
   iii. **Support Nutrition:** Prolonged cases need T.P.N.
   b. **Specific treatment:** Immediate parental anti biotic initiated after taking appropriate cultures.
   - Ampicilin, gentamycin
   - For critically ill, give 34d generation cephalosporine.
   - All antibiotics should be given parentally for 2-3 weeks.

Infant of diabetic mother

Features

1. Commonly delivered preterm with increase in birth weight "large for GA", Why? Because maternal hyperglycemia → fetal hyperglycemia → increase fetal hepatic glucose uptake, glycogen synthesis and enhance lipogenesis and protein synthesis "hyperinsulinemia" → macrosomia "increase in growth of all organs except the brain".
2. Puffy plethoric face.

Complications

1. High fetal mortality especially in poorly controlled diabetes.
2. **Hypoglycemia** in 1st 48 hours, due to maternal hyperglycemia → fetal hyperglycemia → increase fetal insulin production, after birth → interruption of high maternal glucose to the neonate with hyper insulinemia is going on → hypoglycemia.
3. **R.D may be due to :**
   i. RDS.
   ii. TTN.
   iii. Hypoglycemia.
   iv. Polycythemia.
   v. Heart failure.
   vi. Cerebral trauma due to trauma or asphyxia.
4. **Hypocalcemia** due to : Transient hypoparathyrodism.
5. **Hyperbilirubinemia** due to : Polycythemia and reduced RBCs life span.
6. **Convulsion** may be due to: hypoglycemia, hypocalciumia, birth trauma.
7. **Cong anomalies** are 3 fold common especially CHD and NTD, sacroagenesis, Cleft lip and palate.
8. Hypertrophic Cardiomyopathy: uncommon.
9. **Macrosomia** may predispose to difficult labor and birth injury and asphyxia.
10. **R.V.T**: Rare.
11. Sudden I.D.S "IuFD".
12. Increase in incidence of C/S.
13. Small left colon syndrome complicated by delayed passage meconium.

Management

a.Proper control of maternal D.M.

✓ **For the baby :**

1. Immediate post-natal care and incubator care.
2. **Treatment of hypoglycemia:** initial treatment with glucose 10%, then maintenance glucose 10% continuous I.V. infusion.
3. Monitor blood glucose every 2 hours, if blood glucose is controlled, gradually withdraw I.V fluid and advance oral feeding.
4. Calcium every 6 hours.
5. Hydrocortison and glucagon can be added in poorly controlled cases.
6. Discharge if no hypoglycemia for 24-48 hours and no other complication

Hyperthermia

Hyperthermia elevation of body temperature to 38°C – 39°C in 2nd or 3rd day of life.

Etiology :

- ✓ High environmental temperature.
- ✓ Dehydration.
- ✓ Fever and neonatal infection.

C/P:

- ✓ Skin warm and flushed or pink initially then pale.
- ✓ Dehydration, I.C. Hge.
- ✓ Heat stroke, due to thermoregulatory center defect.

Hypothermia

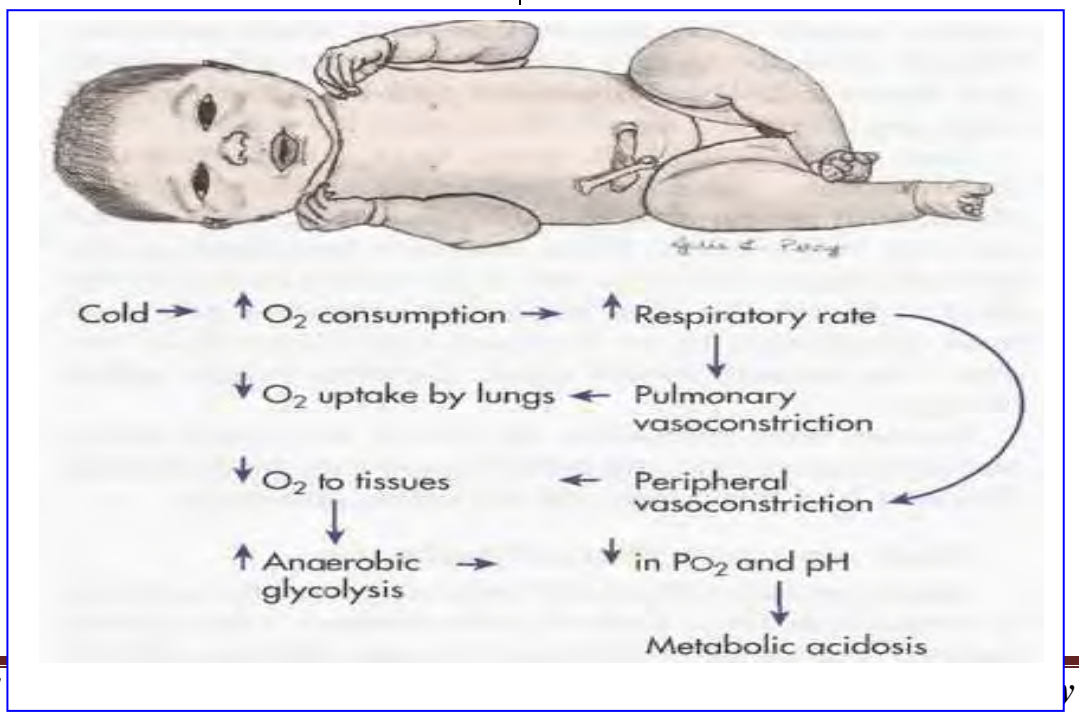
Body temperature less than 35.5°C.

Etiology:

- ✓ More common in premature.
- ✓ Cold environment.
- ✓ Inadequate drying clothing or warming.
- ✓ Sepsis.

C/P:

- ✓ Initial sign: weak suckling, lethargy, weakening, tachycardia, tachypnea, skin colour change.
- ✓ Later on: apathy, apnea and bradycardia, hypoglycemia, acidosis and RD.



Neonatal Hypoglycemia

Blood glucose less than 46 mg/dl in term or less than 40 mg/dl in pre term.

Causes

1. Increase insulin level as in I.D.M and insulinoma.
2. Premature babies and small for GA.
3. Increase requirement of glucose:
    a. Hypothermia.
    b. Sepsis.
    c. Polycythemia.
4. Decrease glucose production **as in** glycogen storage disease.
5. Panhypopitutrism.

C/P

Hypotonia, lethargic.
Poor suckling, Jitterness, Apnea, cyanosis, coma, convulsion.

Treatment

I.V fluid 10% dextrose or hydrocortisone, if not improved glucagon I.M.

Congenital Infection

Cong.Rubella: Presented with skin rash, **"purpuric rash"**, hepatosplenomegally and lymphadenopathy, jaundice, CHD (PDA, ps), R.D and SGA.

Cong.Rubella Syndrome:

- Cataract and glaucoma.
- Micro-ophthalmia& retinopathy.
- Deafness.
- CHD especially PDA.
- CP and microcephaly.

Toxoplasmosis: Caused by toxoplasma gondi.
70% of affected infant at birth are a symptomatic.

Manifestation:

- Chorioretinitis.
- Cerebral calcification.
- Hydrocephalus & microcephaly.

Cytomegallovirus: Presented with :

- SGA.
- Deep jaundice.
- Hepatosplenomegaly.
- Purpura.
- R.D.
- Microcephaly.
- Retinopathy.
- Peri-ventricular calcification.

Neonatal Herpes Simplex:

- Mostly type B, It is a serious disease and lead to high mortality rate.
- **Clinically:** Vesicular lesion of skin and eye and mouth generalized organ failure.

Tetanus Neonatorum:

Acute infection caused by clostridium tetani, which produce neurotropic exotoxins, the infection acquired by cutting the umbilical cord with unsterile instrument.

C/P:

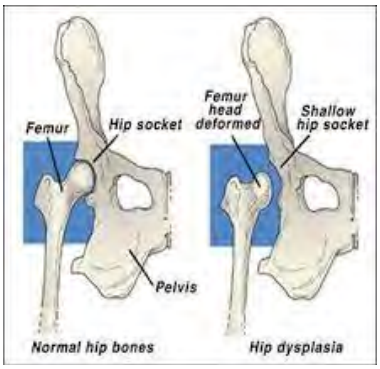
1. Refuse feeding by 2 days of age, because of lock jaw "difficult to open mouth and inability to suckling".
2. Stiffness of muscle lead to rigidity and opisthotonus.
3. Convulsion "tonic – clonic"
4. Baby may die **due to :**
   - Prolonged laryngeal spasm.
   - Laryngeal obstruction by secretion.
   - Pneumonia.

Management:

1. Prevent the noise to baby by isolation in separate room.
2. Start I.V fluid.
3. Give Oxygen for cyanosis.
4. May give tetanus Ig.
5. Antibiotic to prevent 2ndary bacterial infection.

Surgical Cases

Developmental Dysplasia of the hip (DDH)



It was called **"cong. dislocation of the hip".**

- ✓ 6 of 1000 live birth appear abnormal in initial examination, simply called clicky hip, many of these hips click for a few days only and they are almost always normal.

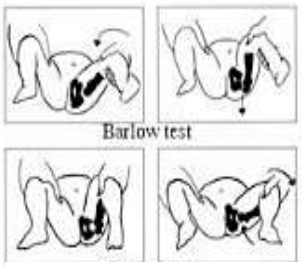
Risk factors:

i. Breech presentation "10 times".
ii. Female baby "4-6 times".
iii. Oligohydraminos.
iv. Big size baby, more than 4.5 kg.
v. 1st baby.
vi. Family history.

Diagnosis:

Every baby should be examined for hip dislocation at birth then 6 weeks then 3 months then 6 months and then 1 year.

a. **During early neonatal period.**
   i. **Barlow maneuver:** The hip dislocated posteriorly out of the acetabulum is positive, if the hip is manually dislocated **(hear click).**
   ii. **Ortholani maneuver:** The dislocated hip is relocated back in the acetabulum, is positive if the hip is lying in a dislocated position and is manually reducible **(feel a clunk).**
   iii. Ultrasound of hip.



b. **In older children:**
   - Decrease abduction of affected limb.
   - X-ray of both hips.
   - Higher greater trochanter.
   - Wide perineum.
   - A symmetric gluteal buttock creases.
   - Short leg.
   - Abnormal gait.

Management

Double or triple napkin in 1st 2 months. If not improved put baby in splint in abduction position and flexion for 6 months, if not improved baby needs surgical corrections.

In older children: Need surgical corrections.

N.B. *The final click, some time comes from the knee and may be felt to sublaxate smoothly without any clunk in 2-3 days of life. Therefore, require to assess the baby again.*

Cong. Club Foot: (Talipes equino Varus")

It is common cong. Abnormality of the foot, occur in 1:1000 live births, M:F is 2:1.
Bilaterally in 40% of cases and there is positive family history.

Management:

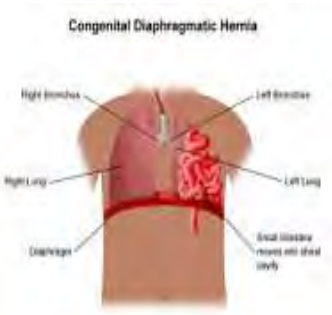
From 1st day of life, put baby in serial plaster, casting 6-12 weeks. If not improved, surgical corrections at 3-6 months.

Abdominal Wall Defects

1.Exomphalos (Omphalocele) Large defect in umbilicus with herniation of bowel and liver into a sac which is covered by fused amniotic membrane and peritoneum. The sac is translucent which becomes opaque by time. Associated with heart and kidney problems, associated also trisomy 13 or 18, Beckwith-Wiedemann.
Management: Incubator care and cover the exomphalos, insert NGT (to limit passage of air into the bowel) and surgical repair at birth.

2.Gastroschisis: Small defect usually to the right side of umbilicus through which bowel and some gonads herniation. Small and large bowel matted with each other due to ameniotic peritonitis before birth. The infant is in high risk of hypothermia.
Management: Urgent surgical correction.

Diaphragmatic Hernia



Occur in 1 per 3000 live births: Herniation of the intestine & other abd.contents through the Foramen of Bochdalek in the Diaphragm, into the thoracic cavity during early gestation leads to pul.hypoplasia.

Causes:

1. **Genetic factors:** More association with trisomy 13,18,21.
2. May be associated with drug toxicity as quinine and phenometrazine.

C/P:

b. **Prenatal:** Can be diagnosed in 45-95% of cases by Ultra sound, which shows:
- Absent intra-abd gastric air bubble.
- Mediastinal shift.
- Hydrops fetalis.

a. **Postnatal:**
- Severe R.D due to pul. parenchymal insufficiency.
- Cyanosis.
- Feeding intolerance.
- Tachycardia.
- **Scaphoid abdomen**.
- **By auscultation:** Decrease breath sound, bowel sound may be heard in the chest, heart sound displaced.
- Late presentation: I.O, Bowel ischemia, Necrosis following by volvulus.
- It is common defect in left side and posterolateral 85-90%.

Investigations:

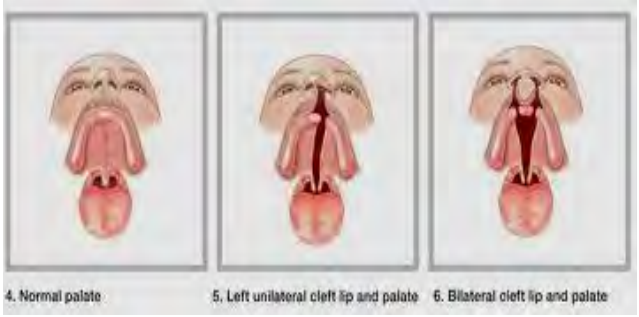
1. **Antenatal** decrease ∝ feto-protein.
2. **Postanatal.**
   - ABG, Hypoxia and hypercapnea, metabolic or resp. acidosis.
   - **Chest X-ray:** Reveals a bowel gas pattern in one hemithorax, with shift of mediastinal structures to the other side and collapse of the epsilateral lungs.
   - Echo.

Management:

Put baby on elective mechanical ventilation, during resuscitation avoid ambu bag and mask ventilation, the main treatment is surgical corrections.

Cleft Lip and Palate

One in each 1000 live birth. Cleft lip + cleft palate 1:750 birth, Cleft palate alone 1:2,500 birth.



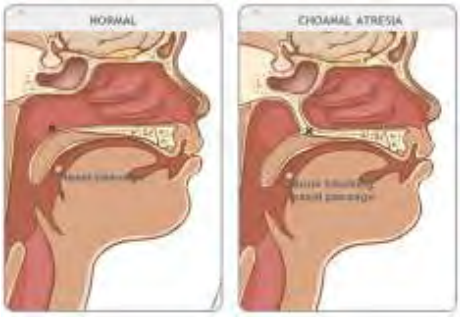
1. **Cleft lip:** More common in male, may be unilateral or bilateral. It results from failure of fusion of the frontonasal and maxillary process.
   Causes: Multifactorial.
   Treatment: Surgical repair within 1st week of life (2 plasty) & revised at 4-5 years of age.
   Complication if not treated:
   - Feeding problem.
   - **Phonation problem:** Speech defect due to inadequancies in function of palatal & pharyngeal muscles and emission of the air from the nose (hypernasal quality).
2. **Cleft Palate:** Result from failure of fusion of palatine process and nasal septum.
   Causes: Multifactorial.
   Treatment: Repair at several months (ideal 1 year of age).
   Complication if not treated:
   - Feeding problem.
   - Reccurent chocking attack.
   - Aspiration pneumonia.
   - Reccurent OM & hearing loss.

Diagnosis

Can be diagnosed on antenatal ultra sound scanning.

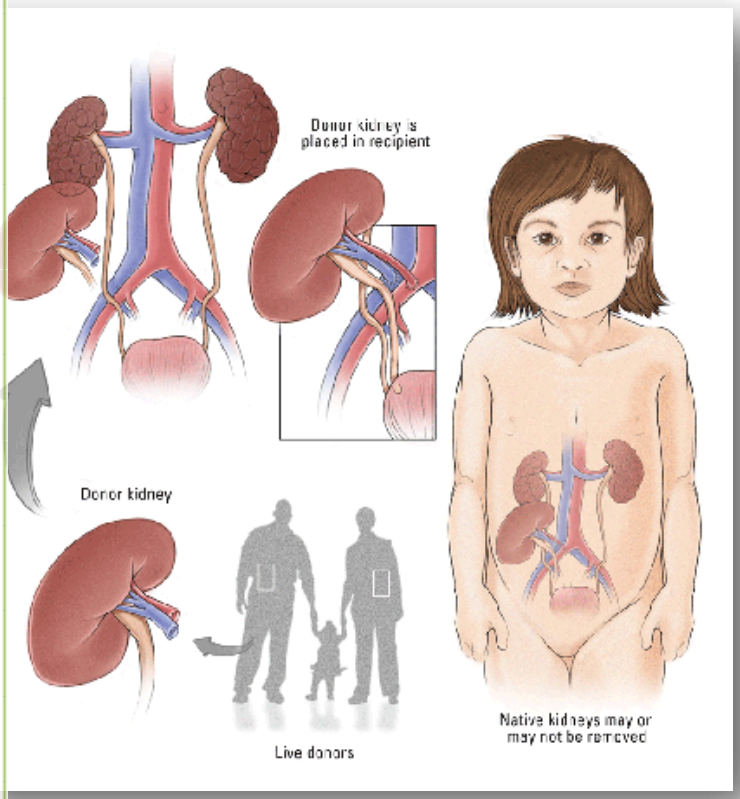
Choanal atresia

A rare bony obstruction between the nasal cavity and the nasopharynx. Main problem is bilateral lesions cause respiratory distress and cyanosis immediately after birth due to airways obstruction as newborn infants are obligatory nose breathers. The airway obstruction is **relieved** on crying or opening the mouth. **Diagnosed** clinically by difficulty to pass NGT during resuscitation.



Treatment: Initial – insert oral airway or tracheal tube, then surgical correction.

Nephrololgy By Dr.Asma Hadida



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]اكتب اسم الشركة[

Introduction to renal system

Normal urine analysis

Colour ; Yellow .
Appearance; Clear.
Specific gravity; 1010-1025 .
PH: Acidic 5-7
Protein; Nill .
Sugar "Glucose ": Nill .
Ketone : Nill .
Bilirubin : Nill .
Occult blood : Nill
Leukocyte esterase ; Nill .
Volume : 1-3 Ml /Kg /Hr .
Bacteria : No
Urobilinogen : 0.1-1mg/Dl .
WBC : 0-4 HPF .
RBC : 0-5 HPF .
Cast : 0-4 /LPF .

Abnormal urine analysis :

Colour :

Dark yellow : concentrated urine .
Brownish : high uro bilinogen **.**
Reddish : high blood .
Greenish :infection .
Drugs : rifambicine - orange.
Desferoxamine : red brown .
Methylene blue : green color .

Appearance: Turbid: crystals, infection.

Specific gravity :
Indicate the potency of kidney , high in case of dehydration .

High protein : renal problem , infection .

Glucose : appear when blood glucose more than 180 mg/dl

Ketone : increase in DM and starvation .

Bilirubin level : increase in liver disease and anemia .

Occult blood : in case of infection and renal problem .

Epithelial cells : if present in urine (normal

PH :

Alkaline:

- Infection by protus or pseudomonas.
- Vegetable dependent diet .
- Renal tubular acidosis .
- Respiratory alkalosis or metabolic alkalosis .

Acidic :

- Starvation .
- Respiratory acidosis and metabolic acidosis except RTA .
- Infected urine .
- Hypokalemic alkalosis .

Cast :

Hyaline cast :normally appear in urine . **increase in case of :**

- HF, febrile illness , post exercise .
- **Red cell cast** : indicate glomerular disease .
- **Granular cast :** in case of acute and chronic RF .
- **WBC cast :** in case of GN and UTI .
- **Epithelial cast** : in case of tubular damage .

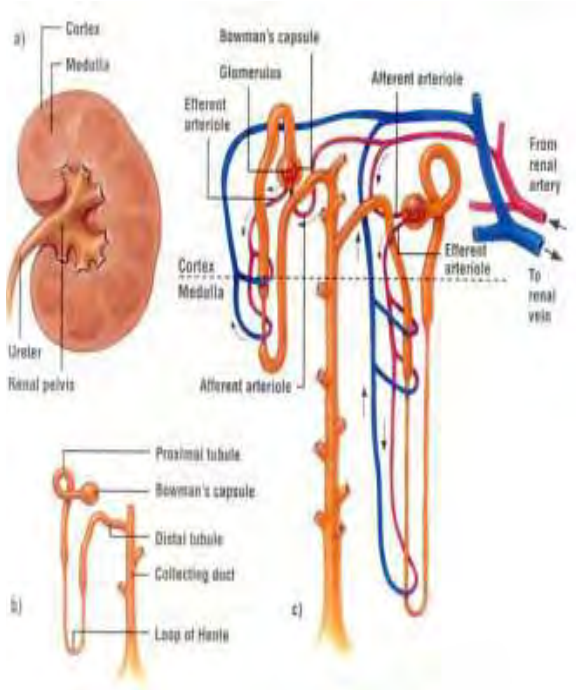
Crystals : phosphate / urate / oxalate / are common seen and usually no significant pathology ..**also may indicate** :

- Protus infection .
- Hyper urecemia .
- Primary oxalosis .

Common manifestation of renal diseases

During neonatal period :

- **Flank mass** : dysplasia , polycystic disease , hydronephrosis , and tumour .
- Hematuria : asphyxia , malformation , trauma , RVT.
- **Anuria and oliguria :**
  - **Anuria :** pt . not pass urine more than 24hr .
  - **Oliguria :** pass 0.5 ml /kg/hr .
  - * **Causes :** agenesis ,obstruction , asphyxia , vascular thrombosis .

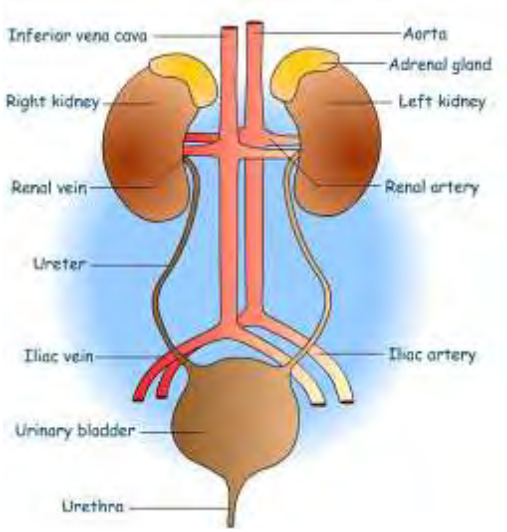


During childhood and adolescent period :

- **Cola-red colored urine :**
  - Hemolysis
  - Pigmenturia:(drug and urate) .
  - Hemturia :(HSP , GN , and hypercalcuria) .
- **Gross hemturia** :
  - GN , trauma , cystitis , tumour , and nephrolithiasis .
- **Edema :**
  - Nephritic syndrome .
  - Nephritis ,ARF , CRF .
  - Cardiac and liver disease .
- **Hypertension :**
  - AGN ,ARF and CRF ,Dysplasia ,COA, renal artery stenosis .
- **Polyuria** :DM, central and nephrogenic DI , hypokalemia and hypocalcaemia .
- **Oliguria:** dehydration and ATN ,AGN and HUS .
- **Urgency** : UTI and vaginitis , hypercalcuria .

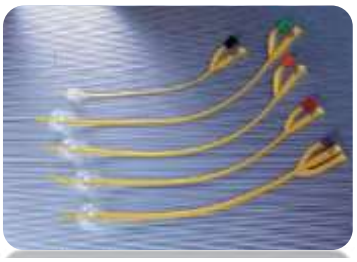
Assessment of renal function in children :

1. Plasma creatinine conc. .
2. GFR .
3. Creatinine clearance.



Radiological investigation :

1. **U/S**
2. **Functional scanning**: Radioisotope .
3. **Static nuclear medicine scanning** :use an isotope labelled substance e.g. **DMSA** ,good for detection of renal scar.
4. **Dynamic nuclear medicine** : give clue on blood flow, renal function and drainage, it is useful for detecting urinary obstruction ,VUR.
5. **Micturating cystourethrography (MCUG) ; for VUR .**
6. Intravenous urography **(IVU).**



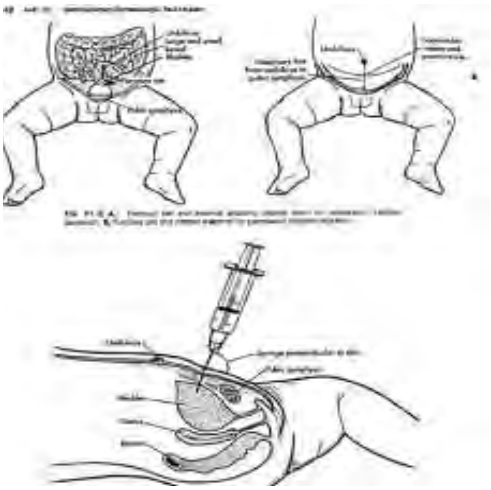
Collection of urine sample :

1.Clean catch : have sterile container ready , ensure baby has been feed , wait patiently , when baby start to void collect into the container , do not collect until a second or so after voiding has started, try gently tapping the supra pubic region to encourage micturation .

2. Urine bag method : wash genitalia and perineum with water "**not antiseptic "** and dry ,apply bag ,ensuring a seal all the way around the genitalia ,do not replace the nappy over the bag ,remove the bag as soon as the baby has voided .cut a hole in corner of the bag and allow urine to drain into a sterile container .

3. Urinary catheter .

4. Suprapubic aspiration : most accurate one b/c bladder sterile 100% and if detect only one bacteria significant .



Suprapubic aspiration procedure :

- ✓ Most accurate one b/c bladder sterile 100% and if detect only one bacteria significant , done if bladder full witch detected by percussion (***dullness)*** or radiologically by u/s . done before 2ys b/c bladder after 2ys become pelvic organ .
- **Site :** one finger above symphysis pubis on midline, direction of needle peripendicular and before that sterile the area by alcohol and betidine ***"from central to peripheral"***, under negative pressure just urine outcome do C/S.

Result of more than one organism indicate contaminated sample

• Incidence:-

↳ Urinary tract infection is common in infant and children.

- **In newborn**: Male : Female = 2:1 "up to 6 months".
- **In childhood**: Male : Female = 1:10.

Localization:

- Involvement of the renal paramchyma "**pyelonephritis**" (UUTI) or the bladder "**cystitis**" (LUTI).
- Most infections do not involve the renal parenchyma.

⇨ About 5% of school age girls develops a UTI and 80% of these patient experience a **recurrence.**

⇨ 3% of female and 1% of male will develop an attack of UTI before the age of 11 year.

✓ 50% → **recurrent UTI** , 50% → have structure abnormalities of urinary tract.

✓ UTI may damage the growing kidney that lead to renal scarring and hypertension, and if bilateral → **CRF.**

Predisposing factors:

1) Bacterial virulence: The fimbriae in some E. coli can attach to the mucosa of urinary tract.

2) Host factors:

A. Anatomical :

- Female has a relatively short urethera, urethral orifice can be easily contaminated by faces.
- **Primary "VUR"** : is present in 35% of children with UTI.
  - VUR increase the risk of pyelonephritis in most cases it is due to congenital abnormalities.
- **Mild VUR** : may occur transiently with cystitis.
  - patient with VUR have higher incidence of renal scarring.
- **Urinary tract obstruction**: as meatal stenosis, post urethral valves, diverticulum "urethra, bladder, or ureter" and ureteric stricture.
- Calculi.
- **Indwelling catheters:** especially with pseudomonas.

B. Functional :

- **Neurogenic bladder**: e.g. Spina bifida → in appropriate detusor contraction against closed sphincter, increase the pressure inside bladder → 2ry UVR and renal damage.

C. Immunologic :

- In immune deficiency.

D. Stasis : (Incomplete bladder emptying).

- ***causes***:
  1. Infrequent micturation → resulting in bladder enlargement
  2. Obstruction by load rectum.

Etiology:

1. **Ascending route (the most common):-**

- Fecal flora ascend through the urethera to the urinary bladder → cystitis.
- If pyelonephritis occurs, this means that organism have ascended the ureters by ***vesicuretic reflux "retrograde flow of urine from the bladder up the ureters".***
- The commonest organisms **are E. coli**, proteus, klebsiella, pseudomonas and streptococcus.

2. **Hematogenous route (common in neonatal period):-**

- Organism are blood borne to the kidney from GIT, respiratory tract, osteiomyelitis or skin infection.
- In addition to G -ve organism, G +ve organism as staph. aureus and streptococcus viridians may cause **the infection**.

Clinical pictures:

1. **Acute UTI:**

Large proportion of children with active urinary tract infection are asymptomatic..

➲ **In the neonatal period**: "non specific"

"insidious onset".

- Prolonged jaundice.
- Fever.
- Septicemia.
- Poor feeding.
- FTT.

➲ **In infancy**: " non specific"

"sudden onset".

- High fever.
- Vomiting.
- Irritability.
- Anorexia.
- Meningitis and even convulsion may occur.
- Frequency and foul smelling urine.
- Screening during micturation.
- Dripling (i.e. not good stream).

➲ **In older children:**

i. Lower UTI :

- Urgency.
- Frequency.
- Dysuria and burning micturation.
- 2ry nocturnal enuresis.
- Hematuria.
- Hypertension.

ii. Upper UTI :

- Fever.
- Lion pain and tenderness.
- Ill patient.
- Rigor.

Diagnosis of UTI :-

I. Investigation to prove UTI :

1. Collection of urine sample.
2. **Urine analysis**: The urine may be turbid due to the presence of excess pus cells.

∗ **Microscopic examination:** reveals:

a. **Pyuria :** > 5 WBC/HPF in the sediment of centrifugated urine sample or 10 or more WBC/nm^3 in non centrifugated sample.
b. Excess RBC may present.
c. **WBC cast** : suggest upper UTI.
d. Bacteria in fresh uncentrifugated sample "one organism /HPF = 10^5 colonies/ml.
e. Mild proteinuria.

3. **Rapid screening test for UTI**:

a. Dipstick nitrite test :

Urinary nitrate Nitrite,

(+ve test ➞ bacteremia).

b. Dipstick leukocyte esterase test :

(+ve test ➞ excess WBC in urine)

⇨ These tests have false +ve and false –ve results; So they must be confirmed by urine culture.

2. Chronic UTI :

- **General manifestation:**
  Fever, malaise, anorexia, anemia, growth failure, hypertension, and manifestation of chronic renal failure develop gradually.
- **Local manifestation:**
  Pain and tenderness in the renal angle, frequency of micturation, hematuria, enuresis, unilateral or bilateral kidney enlargement may be present.

II. Investigation to detect the etiology and predisposing factors:

1. **Microbiological study:**

a. Colony count : Determination of the number of colony forming units/ml of urine.

- If the count $>10^5$ colony forming units/ml → Significant bacteruria → UTI.
- If the count $<10^5$ → contamination of urine.
- If the count = 10,000-100,000 the case is doubtful and the sample should be repeated.

The colony count is **not needed** if the sample is taken **by suprapubic aspiration** B/C there is no chance of contamination from the urethra, so growth of any organism in specimen obtained by suprapubic aspiration = **UTI.**

- No antibiotic are used in treatment of UTI without C/S.

d. Blood culture:

- Should be obtained in neonate.

2. **Imaging of the urinary tract :**

a. Renal U/S :

- It become routine in every case of UTI especially if suspected upper UTI, it is helpful in Dx of congenital anomalies or obstruction e.g. by stone.

b. Voiding cysto urethrography "VCUG":

- To diagnose VUR, it is preferred to be done 1-2 months after complete treatment of the infection to avoid transient UVR that may accompany cystitis.

c. Intravenous pyelography :

- May be needed in some cases as ureteric stone.

d. Radio isotope scanning e.g. DMSA scan:

- Not routine.
- It is very accurate in differentiating upper from lower UTI, it is valuable also in detecting renal sarring.

e. Cystoscopy :

- May be needed in some cases.

III. Evaluation of renal functions:

By urea and creatinine.

Proper diagnosis of UTI needs the following:

1. Prove the presence of UTI.
2. Detect the causative organism and predisposing factors.
3. Differentiate B/W upper and lower UTI.

Indication of VCUG:

1. All male in first attack.
2. All female < 5 years in the first attack.
3. In female > 5 years if recurrent only or if U/S reveal abnormal finding.

Management :

Aim:

1. Detect underlying predisposing factors.
2. Detect if there is renal scar.
3. Any urinary obstruction which require surgical.

A) **< 2 yaers** :

- Addmition to hospital.
- Start IV Abs "3rd degeneration cephalosporin"
- During hospitalization: U/S.
- Discharge on prophylaxtic antibiotic.
- After 2 months of treatment → MCU and DMSA.

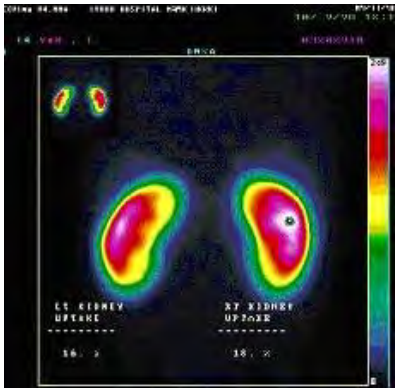
B) **> 2 yaers** :

- Addmition to hospital.
- U/S and DMSA mandatory.
- MCUG if :
  1. Detecting any abnormal by U/S or DMSA.
  2. Clinically pyelonephritis.
  3. Recurrent UTI.(w +ve FH of UVR)

∗ Differentiation B/W upper and lower UTI :-

The following are in favor of upper UTI :

- High fever and toxic look.
- Renal angle tenderness,
- Leukocytosis.
- Bacteremia.
- High ESR.
- Urinary WBC cast.
- Impaired up take or excretion of the radioisotope in DMSA scan.
- Impaired RFT.



⁂ **DMSA : Di mercapto succinic acid "Nuclear scan"**

1) To detect any kidney scar.
2) Function of kidney.

- It is static scan "isotope" filtered and returned to renal parenchyma.
- It divided function and detecting cortical scar.

Give 100ml dye → Normally uptake by both kidney
50% Rt , 50% Lt → Equal by both kidney.
⇨Any patient with abnormal function of kidney: uptake not equal in both kidney (Rt ≠ Lt).

⁂ **Micturating cysto uretrogram: (MCU)**

- Used for VUR.
- Normally : If bladder full with dye not return back B/C ureter form artificial angle but once dye enter ureter "due to VUR" indicate acute pyelonephritis.
- X-ray: See below "catheter".

So; MCU done for VUR and Post.urethral valve.

Any patient with acute pyelonephritis should be take prophylactic antibiotic e.g. Trimethoprime or Nitrofurantoin.

How to prevent UTI :-

1. Hygiene: clean perineum from front to back.
2. Avoid constipation.
3. Encourage fluid intake.
4. Avoid Nailon under wear.
5. Double micturating.
6. **Prophylactic treatment if:**
   - Investigation show abnormality.
   - No abnormality but frequent UTI in child less than 3 years.
7. Check urine after treatment of UTI.

N.B

⁂ **Proteus infection**: predispose to the formation of phosphate stone by splitting urea to ammonia and thus alkalinizing the urine.

⁂ **Pseudomonas**: may indicate the presence of some structural abnormality in the urinary tract affecting daring.

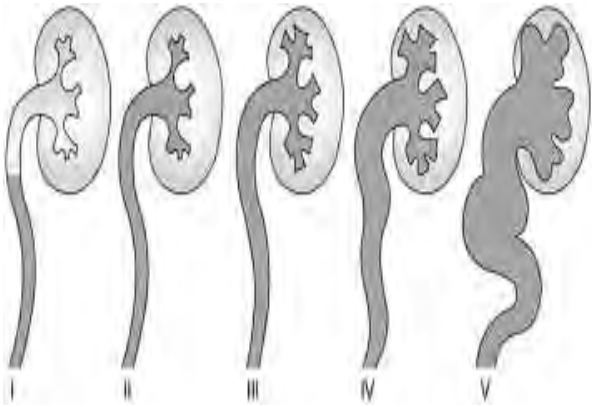
N.B

✓ **Circumcision**: decrease incidence of UTI in boys.

✓ Normal U/S does not exclude VUR.

Vesico-Uretric Reflux
V.U.R

- Backward reflux or flow of urine in to the ureter or kidney "may developmental anomaly of vesico uretral junction".
- Sever cases may be associated with renal dys
- It is familial : 30 - 50% in 1st degree relation.



∗ Causes:

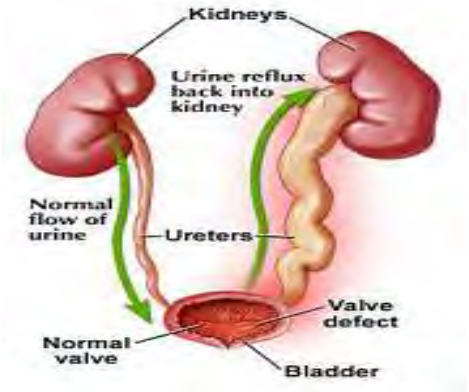
1) Congenital: 30% male - 50% female.
2) Acquired:
   - UTI.
   - Obstruction e.g. P.U.V 50%.
   - Neuro genic bladder.

⁂ Indication of surgery:

1. When prophylactic fall to control infection.
2. Where there is reflux nephropathy.
3. Compliance to medical treatment is poor.
4. There is large para uretric diverticulum.

➲ Grades of Vesico uretric Reflux : (By MCUG)

Grade I : Urine in ureter only "⅓ ureter"
Grade II : **I** + reflux to pelvis and calyces.
Grade III : **II** + mild dilatation of ureter and calyces.
Grade IV : **III** + moderate dilatation and blunting of calyces.
Grade V : **II** + sever dilatation with tortuously of ureter and intra renal reflux.



Treatment :

Grade I, II, III : Disappear spontaneously, Prophylactic Antibiotic continuous for 6 months.
Grade IV, V : Need surgery "re-implantatic of ureter".

☺ **Case history** :Ahmed a 2 months old infants, stopped feeding and had a high intermittent fever, he was referred to hospital where he had an infection screen.
- Urine examination showed > 100 WBC, > 10^5 E.coli/ml.
- He is treated with IV antibiotic.
- U/S : small Rt kidney with dilated renal pelvis and a dilated ureter.
- Discharged on prophylactic antibiotic.
- DMSA scan: 3 months later confirmed bilateral renal scarring.
- 3 years of age, the reflux had resolved and antibiotic prophylactic was stopped.
- His BP, renal growth and function continue to be monitored.

⁂ **Children with high risk to develop renal damage:-**

- Known AN renal anomaly.
- Infant.
- Boys.
- More than one UTI.
- Septicemia.
- Prolonged clinical course of fever >48hrs.
- FH of reflux.
- Unusual organism i.e. not E.coli.

Follow up of children with recurrent UTI, renal scarring or reflux :-

1. Urine culture.
2. Long term low dose Antibiotic prophylaxis.
3. Circumcision in boys.
4. Anti-reflux surgery.

Protein uria

- Normally child excrete 30-60 mg/day protein.
- If > 100 mg/day → Not normal.

Proteinuria

Physiological
- Fever.
- Exercise "active children"
- Orthostatic "Postural"
- Drug: Cephalosporine, Penicillin.

*(Excretion not exceed 4mg/hr/m^2)

Pathological
- Glomerular abnormalities.
- Minimal change NS.
- GN
- Anormal G.B.M "familial nephritides".
  - ↑ G. perfusion pressure.
  - Reduced renal mass.
  - Hypertension.
  - Tubular proteinuria.

Nephrotic syndrome

Definition:

Clinicolabratory syndrome characterized by :-

1. Massive proteinuria : More than 40 mg/m^2/hr. "**more than 2 grams/24hr**", according to wt > 50 mg/kg/day.
2. Hypoalbominemia : Less than 2.5 gram% "total plasma protein < 5.5 gram/dl"
3. Generalized edema.
4. Hyperlipidemia.

Etiology:

1**) Primary causes**:

- Idiopathic nephrotic syndrome 90% of cases:
  - Minimal lesion nephrotic syndrome.
  - Nephrotic syndrome with focal glomerulosclerosis.
  - Nephrotic syndrome with diffuse mesangial proliferation.

2) **Secondary causes**:

a. Infections: Shistosomiasis, malaria, syphilis.

b. Systemic disease: SLE, anaphylactoid purpura , D.M, sickle cell anemia, Malignancies "Hodgkins lymphoma", amyloidosis.

c. Drugs: Penicillamine.

d. Allergic conditions: Bee sting, serum sickness.

3) **Congenital :** Rare (AD).

Pathology:

The idiopathic "primary type" is the commonest in childhood .

- **It is divided pathologically into:**

1. **Minimal change NS** :
   No change is seen by light microscope, but by electron microscope reveal diffuse fusion of the foot processes of the glomerular epithelial cells.
2. **Mesangial proliferation :**
   As minimal change but shows increased cellularity and matrix of the mesangium.
3. **Focal aglomerulosclerosis :**
   As minimal change with sclerosis of some glomeruli.
4. **Membrano-proliferative GN :**
   Mesengial proliferation, epithelial proliferation with crescent formation and thickening of capillary wall.
5. **Membranous glomerulonephritis :**
   Thickned BM of the glomeruli due to deposition of IgG.

Pathophysiology:

1) Proteinuria :-

- Normally, healthy children may lose up to 100 mg/m^2/day protein in urine.
- If protein urine more than 100 mg/m^2/day, this is called **proteinuria** which is divided according to severity into:
  a. **Mild proteinuria** : if protein in urine of 24hr = < 0.5 g/m^2/day.
  b. **Moderate proteinuria :** 0.5 - 1 g/m^2/day.
  c. **Sever or massive** : > 1 g/m^2/day.

2) Hypo proteinemia :-

It is mainly due to urinary loss of albumin, there is also increased catabolism of protein during the acute phase.

3) Generalized edema :-

It is due to :

❶ Reduction in plasma colloid osmotic pressure secondary to hypoalbominemia, this lead to edema and hypovolemia.

❷ Retention of sodium and water secondary to increased secretion of aldosterone and ADH in response to hypovolemia.

4) Hyperlipidemia :-

Most lipid fraction in plasma are elevated in nephrotic syndrome, possibly as a result of increased synthesis and diminished catabolism of lipoproteins in response to hypoalbuminemia.

Proteinuria

Selective	**Non Selective**
Where protein of low molecular weight such as albumin are excreted more readily than are protein of high molecular weight.	Where protein of low and high molecular weight are lost in urine.

Clinical picture:

- The onset gradual from 1-10 year or 2-7 year , in minimal change type.
- The incidence in boys is twice that in girls.

1) <u>Generalized edema</u>:

- Is the usual presenting manifestation, it develops over several weeks and progress from periorbital puffiness to massive generalized edema often with ascitis and PE.
- The urine volume diminishes during the stage of edema formation.
- Respiratory difficulty may result from :
  a. Massive ascitis.
  b. PE or pul. edema.
  c. Chest infection as bronchopneumonia.
- Diarrhea may occur due to edema of the intestinal mucosa or the occurrence of slightly decreased but in 5-10% of cases it is elevated.

2) <u>Infection</u>:

- **Nephrotic patients are liable to infection b/c of :**
  a. Loss of Ig in urine.
  b. The edema fluid act as culture medium.
  c. Immunosuppressive effect of corticosteroid.
  d. Decrease perfusion of the spleen due to hypovolemia.
  e. Loss of complement in urine.

➲ The common infections are :
- Viral upper respiratory tract infection.
- Cellulitis of skin.

⏩ Spontaneous peritonitis particularly by pneumococcal.

Complication:

1) **Infection** : Spontaneous peritonitis or septicemia due to pneumococcal "capsulated bacteria", H. influenza or gram –ve organism e.g. E.coli.
2) **Hypercoagulabity of blood** : Leading to venous or arterial thrombosis.
   It is caused:
   - Hyperviscosity of the blood.
   - Elevated of fibrinogen and other clotting factors.
   - Loss of anti-thrombin III.
   - Decreased fibrinolysis.
   - Increased platelet aggregability "Thrombocytosis"
   - The hypercoagulability may lead to thromboembolic complication e.g. cerebral thrombosis or renal vein thrombosis.
3) Hypovolemic shock "↑HCT".
4) Renal failure due to decrease RBF.
5) S/E of treatment.
6) Hyperlipidemia may lead also to thrombosis and ischemic changes.
7) Loss of vitamin D binding globulin → high risk for Rickets.
8) Convulsion due to hypocalcemia → ionized part of Ca^{+} is bound to plasma protein.

Laboratory finding:

1. **Urine** :
   - Oliguria : during stage of edema formation.
   - Proteinuria : 24hr urine collection.
     "> 2 g/day" ">1 g/m^2/day".
2. **Blood :**
   - Hypoproteinemia (normal = 6.5-8 gm/dl)
   - Increase serum cholesterol (normal 100-200 mg/dl).
   - Total serum Ca^+ decrease.
   - Normal C3.

☑ **Clinical and laboratory criteria of minimal change lesion N/S :-**

❶ Age onset 1-10 year.
❷ No gross hematuria.
❸ No hypertension.
❹ RFT normal.
❺ Selective proteinuria.
❻ Normal serum complement.
❼ No evidence of a systemic disease.
❽ Good response to corticosteroids.

D.D of known cases of N/S presented with acute abdomen:

1. Infection " UTI or peritonitis"
2. Thrombus formation "ischemic pain"
3. S/E of steroid → gastritis or peptic ulcer.

Treatment:-

1. General:

a. Hospitalization :
- Measure wt, BP, urine output (to know if there is response to treatment or not).
- Proteinuria should be checked dialy.
- Patient should be examined for evidenced of infections.

b. Phsical activity :
- Should be as desired and tolerated by the child.

c. Diet :
- Salt is restricted (to avoid water retention and fluid over load).
- Normal protein intake .

d. Antibiotic : Prophylactic.

e. Diuretic :
- For mild edema: Hydrochlorothiazid + Sipronolactone .
- For marked edema: Furosemide "Lasix" + Metolazone
- For hypovolemia: If diuretic are not effective b/c of hypovolemia give salt-poor albumin to expand plasma volume IV over 2 hours then give diuretic.

f. Ca^+ supplement.

N.B:
Aggressive diuretic may cause Shock , Hypovolemia, Thrombosis .

2. Steroid therapy:

- steroid accelerate the occurrence of dieresis and disappearance of poteinuria.

* Precaution before starting steroid :
  1. Exclude impaired renal function.
  2. Exclude infection as T.B and UTI and exclude hypertension.

(A) **Treatment of 1st attack:**

- Prednisone "3 mg/kg/day in 3-4 divided doses" until urinary excretion of protein has returned to normal for 5 days.
- In 90% of MCNS this occur within 4 weeks:-

1. Patient who response to the initial daily prednisone therapy "5 consecutive day" with no proteinuria are switched to the alternate day regimen 2 mg/kg or prednisone are given every other day as a single dose with breakfast.
   ⋆ This alternate day regimen is for 4 weeks with tappreing over 3-6 months, the purpose of alternate day therapy is to maintain the remission using a relativity non-toxic dose of prednisone.
2. If the child continuous to have proteiuria after 1 month of daily prednisone, the case is considered steroid resistant and renal biopsy is indicate to determine the accurate pathology.

(B) **Treatment of relapse:**

- Same as 1st attack.
- A relapse is defined as recurrence of edema in addition to proteinuria, a small number of patient who respond to daily divided dose of prednisone relapse shortly after switching to or termination of alternate day therapy such patient are termed steroid dependent.

3. Cytotoxic therapy:

- It is indicate in :

❶ Steroid dependent patient (2 relapses occur or within 14 of stopped treatment) > 2 R/year.
❷ Steroid resistant patient .
❸ Frequency relapsing patient.

1. Cyclosporin (A): **Renal biopsy before start of treatment**.
2. Cyclophosphomide: Given in a dose of 3 mg/kg/day as a single dose, for a total duration 8 weeks
   • WBC should be monitored weekly (75% remission for one year & 50% remission for 5 years).
   • S/E: - Bone marrow suppression
     - Hemorrhagic cystitis
     - Alopecia.

N.B: WBC < 5000 treatment stopped.

➲ Urinary dipstick : Are zero or trace for 3 consective days.

➲ Clinical remission:

- ↓ wt.
- Child moving and active.
- No edema.
- ↓ proteinuria.

So, treatment by steroid
2 mg/kg/day for 4 wks
OR 60 mg/m^2 /day
⬇
2mg/kg/day
Alternate day for 4 wks
⬇
Then tapering for 1-3 months

⇨ Do urine dipstick for sugar, protein and ketone ; If normal → Remission stage.

So,

- ⅓ cases will remain in remission.
- ⅓ cases will relapse and become steroid dependence.
- ⅓ cases are steroid resistant.

Steroid resistant NS :-

1) Focal segment GS:
- Most common.
- Idiopathic.
- Familial.
- 30% develop end stage RF by age of 5 year.
- 20% relapse to cyclophosphamide and cyclosporine A.
- Recurrence post transplant is common.

2) Mesangio capillary "membrano proliferative" GN :
- In older children.
- Hematuria, low C3
- RF will develop over years.

3) Membranous Nephropathy:
- Associated with HBV.
- May proceed SLE.

4) Congenital N/S:
- AR.
- **Types:**
  1- Finish type AR.
  2- Heterogenous group DNA abnormality.
- Patient < 3 months.
- Poor prognosis (patient develop end stage RF rapidly).
- **Treatment** : Nephroectomy and transplantation.

⁂ Any edematous child → test for proteinuria to diagnose nephrotic syndrome.

Associated disorder with NS:
1. Atopy 34-60% → Increase IgE.
2. Malignancy e.g. H.lymphoma, Thymoma.
3. D.M.

x

D.D :-

1) Acute post. Stereptococcal GN.
2) **Other causes of generalized edema:**
   - Nutritional "Kwashiorkor"
   - Cardiac "HF"
   - Allergic "Angioneurotic edema"
   - Hepatic "Liver cirrhosis"
   - Protein losing entropathy.

Glomerulonephritis

- It is an inflammation of the kidney parenchyma.
- GN is common disease of childhood, characterized by edema, Hypertension, urinary changes "oliguria, hematuria, mild to moderate proteinuria, leucocyturia" together with circulatory congestion and azotemia.
- Acute GN usually result from the deposition of immune complexes in the glomerular structure.

Classification

Primary GN

1) **Immune complex GN**:
   - Post infectious GN may follow: Strepto. , Pneumoococci , HBV , Staph., Malaria , Mycoplasma , EBV , Mumps.
   - Membrano-proliferative GN.
   - IgA nephropathy "Berger disease" (most common 1ry cause)
2) anti-GBM antibody "Good pasture syndrome"
3) Uncertain cause e.g. Focal segmental GN.

Secondary GN

with systemic disease (commonest)

1) Immune mediated :
   - Lupus nephritis.
   - HSP.
2) Hereditary e.g. Alport syndrome.

Acute Post. Streptococcal GN

▪ Acute nephritic syndrome which follow pharyngeal or cutaneous infection with a nephritogenic stain of group B hemolytic streptococci.

Pathogenesis and pathology:

Streptococcal infection $\frac{Latent}{period}$ → Ab → Ag-Ab complex and complex C3

⬇

Deposited in glomerular BM

⬇

Acute inflammation

Proliferative of mesangeal and endothelial cells

⬇

Glomerular capillaries obstruction

⬇

Decrease GBF

↓ GFR → Oligouria; ↓ GFR → Fluid retention → Hypervolemia → Edema

++ Juxta GA → ++ Renin angiotensin system → Hypertension

Glumerular endothelial damage

⬇

Escape of RBC and protein in urine

⬇

Hematuria and Proteinuria

Clinical pictures:

(Peak age 5-12 year)

1. **Asymptomatic** : with microscopic hematuria that discovered on routine examination of the urine.
2. **Hematuria** : Painless, cola colored.
3. **Triad of** : edema, hematuria, oliguria.
4. Triad + one or more of the complication.
5. One or more of the complication mainly.

∗ Hematuria : Gross hematuria for 2 weeks then microscopic hematuria for about 6 weeks.

∗ Edema : Firm pitting edema, generally minimal or moderate in degree and noted as puffiness around the eyes and in the face associated with slight edema of the extremities.

∗ Hypertension : Present in 60-70%.

⇨ Before all of these symptoms, H/O pharyngitis or impetigo is present, or other systemic manifestation such as mild fever, anorexia, nausea, vomiting, headache and abdominal pain may be present.

Investigations:

A) For analysis :

- *Urine analysis*
  - Color : Cola colored.
  - Volume : Oliguria.
  - Specific gravity : High.
  - Proteinuria : Less than 1 gm/dl.
  - Microscopy : **Dysmorphic RBCs**.
  - Hyaline, granular and RBC cast.

B) For effect :

- ↑ K or dilutional, hypoNa^+.
- RFT is impaired.
- Anemia due to hemodilution.
- ↑ ESR.

Complication:

1) Circulatory overload leading to **HF** and pulmonary edema
   - **Manifested by** dyspnea, orthopnea, pul. crepitation, enlarged tender liver, and tachycardia.
2) **Hypertensive encephalopathy** :
   - Consequent loss of auto regulation of cerebral circulation leads to dilatation of vessels → brain edema.
   - It is manifested by irritability, headache, vomiting, blurring of vision, drowsiness, coma, and generalized convulsion may end in death.
3) **ARF** :
   - With sever oliguria or anuria, hyperkalemia, acidosis, impaired excretion of creatinine , urea, and phosphates.
   - Manifested by disturbed consciousness, vomiting, diarrhea, anemia and convulsion.

C) For etiology :

1. Low C3.
2. Evidence of recent streptococcal infection:
   - Throat or skin lesion swab and culture.
   - Anti-streptolysin O "ASO" titer may - ve after skin infection.
3. **Renal biopsy:**
   - ***Indicated for*** :
     - *i. Absent evidence of streptococcal infection or low C3.*
     - *ii. Nephrotic syndrome "Nephritic Nephrosis".*
     - *iii. ARF.*
     - *iv. Presistence of gross hematuria, proteinuria, low C3, or impaired RF beyond 2 months of onset "delayed recovery".*

Treatment :

1. **Bed rest :**
   - Till edema, hypertension and gross hematuria disappear.
   - All muscular activities are allowed when ESR is normal.
2. **Procaine penicillin :**
   - IM for 10 days.
   - Erythromycin for 10 days.
3. **Fluid intake and diet :**
   a. Fluid intake should be restricted :
      - If the urine output of 24 hour is less than 400cc/m2.
      - The fluid intake = 400cc/m^2 + urine output of the previous day.

   b. Salt "NaCl" and K "fruit juices" : should be restricted.
   c. Protein : Should be restricted as they increase K^+ and urea.
   - Diet consists mainly of CHO and small amount of protein with minimal salt.
4. **Treatment of hypertension :**
   a. If less than 140/90: No drug and follow up by measuring BP every 4 hours specially in the first 7-10 days.
   b. If more than 140/90 with no hypertensive encephalopathy:
      - Start with diuretic "as furosemide", if no response + vasodilator as prazosin, also may use captopril.

   c. If more than 140/90 with hypertensive encephalopathy:
      - Ca^+ channel blacker "Nifedipine".
      - Labetalol (IV).
5. **Treatment of ARF :** Conservative treatment with or without peritoneal dialysis.
6. **Treatment of HF :** Depend on underlying causes:
   - Hypertensive HF : Treatment of hypertension.
   - Hypervolemic HF : Diuretic ± Dialysis.
7. **Discharge** from hospital if :
   - No gross hematuria.
   - Normal RF, controlled or no HPN.

Course and prognosis :

- 85-90% of the case will make a complete recovery.
- Up to 5% die in acute stage from one of the complication and less than 5% develop end stage renal failure.
- Recurrence is extremely rare.

*** In patient who recover, the following occur**:

1. Edema disappear 1-2 weeks.
2. Gross hematuria and hypertension and ARF usually disappear in 2-3 weeks.
3. C3 normalize in 6-8 weeks of onset.
4. ESR return to normal in 3 month.
5. Microscopic hematuria may persist for 1-2 years.

D.D of low C3 :

1. Post streptococcal GN.
2. SLE.
3. Membranous proliferation GN.
4. Subacute bacterial endocarditis.
5. Shunt nephritis.

Case presentation:

- 3 years old boy presented with tea colored urine, periorbital edema, headache and vomiting proceed by H/O URTI 1-3 weeks. (may proceed by H/O pyoderma).
- Patient looks unwell, edematous.
- O/E : - BP 130/90mmHg.
  - Urine analysis : RBC 20 - 40 , RBC cast +ve, ↓urine output.

➡ Dx : Post strep. GN.

Hemolytic uremic syndrome

Include :

1. ARF
2. **MAHA** : microangiopathic hemolytic anemia.
3. Thrombocytopenia.

Pathogenesis:

Toxin "bacterial and others"

⇩

++ WBC

⇩

↑TNF and IL2

↑ O2 free radicles → ⇩ Endothelial damage

↑ Procoagulant → ⇩ Endothelial damage

Intravascular thrombosis

Damage circulatory obstruction → RBC and Plt → ↓ → Hematuria

Glomerular → • ARF •

MAHA Microangiopathic Hemolytic anemia and thrombocytopenia.

Caueses:

1. **Acute bloody diarrhea :**
   - Caused by shiga toxin producing E. coli.
   - In 80% of cases of HUS.
   - Transmitted from undercooked meat or unboiled milk.
2. **Other less common causes:**
   - Bacteria: Shigella, salmonella, campylobacter, pueumococci.
   - Viruses : Echo, coxahies, Influenza.
   - Drugs : Cyclosporin A.

Types:

1. Typical type of HUS : Epidemic, outbreak 95% good prognosis.
2. Atypical type if HUS : 5% "+ve FH", bad prognosis.

N.B :
Antibiotic should be avoided in HUS associated with shiga toxin producing E. coli as it kill bacteria → ↑↑ toxin release → deterioration.

Clinical pictures:
- Common age usually < 5 years.
- Bloody diarrhea or URTI is followed 5-10 days later by :
  1- Acute pallor and purpura.
  2- ARF with oliguria, edema, hypertension, and acidotic breathing.
  3- Hematuria.

∗Extra intestinal manifestation : due to:
- CNS : Seizure, infarction, coma.
- Intestinal : Perforation, intussusception.
- Liver : Hepatitis .
- Cardiac : Pericarditis.

D.D :
1. From other causes of intrinsic RF:-
   - Rapidly progressive GN.
   - Neglegted pre-renal failure.
   - Acute tubular necrosis.
2. From other causes of MAHA:-
   - Bilateral renal vein thrombosis.
   - Thrombotic thrombocytopenia purpura.

Investigation:
A) For ARF :
- High creatinine and urea.
- Increase K^+
- MR. (↓PH , ↓PaCO2 , ↓HCO3)

B) For MAHA:
- Low Hb.
- ↑ reticulocytosis.

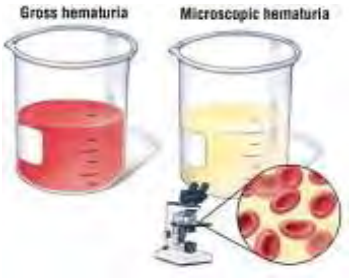
C) For bleeding:
- ↓ Plt
- Coagulation profile is normal.
- Coomb's test is -ve.

Treatment:
1. Prevention :
   a) Adequate cooking of meat.
   b) isolation of cases to avoid cross infection with E. coli.
2. Curative :
   a) Packed RBC if Hb < 6%.
   b) Treatment of ARF.
3. Long term follow up.

Prognosis:
With adequate treatment → 90% survive acute phase , 9% of them progress to end stage renal disease.

Hematuria

- Normal urine is yellow in color which due to the presence of the pigment urochrome.
- Hematuria is defined as the presence of **more than 5 red blood cells / high power field** in the sent of a 10 ml centrifuged fresh specimen.
- Gross or frank hematuria is visible to the naked eye.



Causes:

[1]. Glomerular diseases:

A) Acute post streptococcal GN: The most common cause of hematuria in a child.

B) **Other post infectious GN**: Caused by staph. , viral , measles, mumps , chiken pox, and influenza.

C) **Nephrotic of chronic infections:** As subacute bacterial endocarditis, syphilis, or malaria.

D) HUS "Hemolytic uremic syndrome".

E) **Nephrotic syndrome :** Especially types other than minimal change NS.

F) Syndromes of recurrent gross hematuria.

[2]. Interstitial nephritis:

- Which may be :
  - Drug induced. Cephalosporines, Sulphonamide.
- Infection :
  - Strept.
  - Brucellosis.
  - T.B.
  - Measles.
  - Toxoplasmosis.
  - EBV.
  - CMV.

[3]. UTI.

[4]. Congenital anomalies of the urinary tract:

a. Polycystic kidney.

b. Arterio-venous fistula.

[5]. Traumatic hematuria:

- Blunt trauma to the abdomen.
- Stones especially if spiky as oxalate stone.
- Meatal ulcer.

[6]. Tumours:

- Benign: Hemangioma.
- Polyp in urinary bladder.
- Malignant: Wilm's tumour.

[7]. Blood disease:

- Leukemia.
- Hemophilia.
- RVT.
- Thrombocytopenia.
- Sickle cell anemia.

N.B:

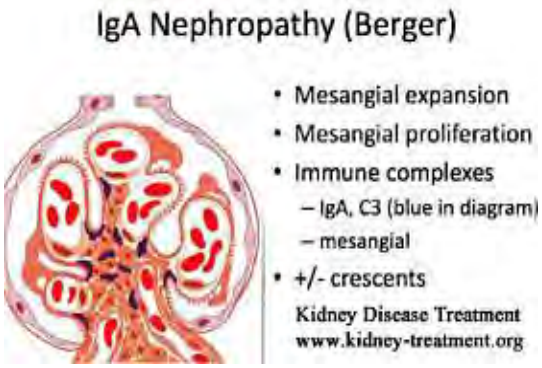
The younger patient is the more prone to hematuira.

⁂ **Syndromes of recurrent gross hematuria :**

Include :

➥1- **IgA nephropathy "Berger's disease":**

- Microscopic or recurrent painless hematuria short following URTI "1-2 days" and not associated with S and S of acute nephritic syndrome.
- Constant finding of renal biopsy is the presence of mesangial proliferation accompanied by mesangial deposits of IgA.
- The course is generally benign but may be recurrent following respiratory infection.
- RFT and serum complement are usually normal.
- 90% of children recover completely and 10% end by chronic renal failure.
- Most common cause of 1ry GN.
- **Associated with**:
  - Chronic liver diseases.
  - Psoriasis.
  - IBD.
  - HIV.
- Only 30% of patient with IgA nephropathy has progressive disease.
- Dx : Renal biopsy with IgA in the mesengeal of glomeruli.



Case presentation:

- The school nurse referred 7 years old boy b/c he failed his hearing test at school, the men in this patient family have H/O renal problem, and few of maternal uncle are deafness ; urine analysis is obtain show microscopic hematuria.

Dx: Alport syndrome.

2- Alport syndrome :

- Inherited as autosomal dominant or X-linked dominant.
- The condition is characterized by nephritic with progressive renal failure during adolescence particularly in male and bilateral sensory neural deafness. (After age of 6 years + eye manifestation "cataract or keratoconus").
- Electron microscope: Characteristic splitting of the BM of the glomeruli
- So; Clinical features:
  - Asymptomatic hematuria.
  - Progressiove impairment of renal function.
  - Deafness.
  - Ocular manifestation in small number.
- Treatment :
  - No response to treatment.
  - Renal dialysis.
  - Transplantation.

N.B: Alport syndrome is the most common hereditary nephritis.

Approach to diagnosis hematuria :

(I) Is this red urine hematuria ?

1. Hemoglobinuria: which may true or false.

- ∗True hemoglobiuria : is considered when hemoglobin is liberated in the circulation and pass through the kidney into the urine.
- ∗False hemoglobiuria : occur when RBCs already present in urine rupture due to long standing or by action of bacteria.
- ∗True hemoglobinuria : occurs in hemolytic anemia as favism or malaria when hemolysis is mainly intravascular and liberated hemoglobin is more than the amount of circulating haptoglobin "which is plasma protein which binds with any free Hb to prevent it its escape into urine".

▸ The urine appear portwine in color or reddish brown and sometimes black, tests for Hb are +ve but RBCs are not seen on microscopic film.

2. Hematuria with it's causes.

3. Dyes: Beet root "the condition is called **beeturia** in which urine and stool appear red"

4. Drugs: Rifampicin.

5. Urates: May cause slight pink or reddish staining of the diaper in newborn and young infant.

6. Bilirubinuria: In hepatitis, bilirubin regurges to the blood and then to urine which become tea like in color.

So; D.D of red urine without RBCs:

* Heme +ve

a. Hemoglobinuria in acute hemolysis.

b. Myoglobinuria due to rhabdomyolysis "in sk. ms. Injury" in viral mysitis, crush, prolonged fits, high serum creatinine kinase.

* Heme -ve

- **Food :** beet roots, black berries.
- **Drugs** : Rifampicine, Desferal, Nitrofurantoin.

(II) Diagnosis of the etiology of hematuria:

Clinical approach of hematuria :

A) History taking; ask about :

- Sore throat preceding the attack of GN.
- Passage of stones, lion pain, frequency and dysuria.
- Diarrhea and dehydration in RVT.
- Trauma.
- Intake of drugs.
- H/O living in endemic area for bilharziasis.

B) Clinical examination :

- Puffy face and edema are in favour nephritis.
- Fever denote infective cause.
- Hypertension occur in Gn, collagen vascular diseases, and poly cystic kidney.
- Purpura and ecchymosis are in favour of blood diseases.
- Abdominal mass as wilm's tumour.

(III) Laboratory approach of hematuria:

A) Localized hemauria :

	Glomerular	Extra glomerular
Color	Coa or tea	Bright red
Clots	Absent	May present
RBCs shape	Dysmorphic	Normal
RBC cast	present	Absent
Proteinuria	>100 mg/dl	< 100mg/dl

B) For all cases :

1. Analysis of fresh urine sample for cast, bacteria, crystals, and RBCs shape.
2. Abdominal U/S for : stone, tumors, or malformation.
3. RFT.
4. Blood picture with DLC.
5. Serum C3.
6. 24 hour urine Ca^{+}, uric acid, and oxalate for crystal urea and stone.

C) For glomerular hematuira :

1. Reduced C3.
2. For post strept. GN.
3. For lupus nephritis (ANA anti-double strain DNA).

D) For extra glomerular :

1. Urine culture for UTI.
2. Cystogram and renal scan for hydronephrosis.
3. Renal Doppler for RVT.

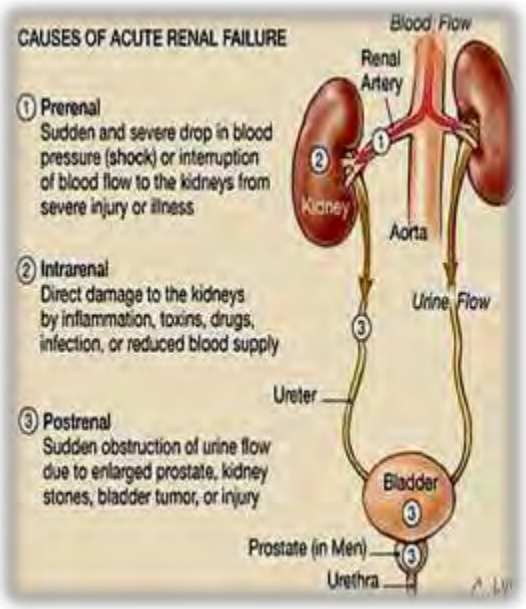
E) Renal biopsy

Acute Renal Failure

- It is a **clinical syndrome** characterized by sudden reduction in renal function to a level insufficiency to maintain homoestasis of blood fluid and electrolytes and demand.

Causes:

1. **Pre renal causes** : (60%)
   "Due to reduction of RBC"
   - Blood loss (Hge).
   - Plasma loss (burns).
   - Fluid loss (sever dehydration).
   - HF (impaired C.O.P).
   - Septic shock.
   - Sever hypoxia.
   - Hypoalbuminemia (NS).
2. **Renal causes**: (30%)
   "Due to renal parenchyma damage"
   - Acute GN.
   - Acute tubular necrosis.
   - Interstitial nephritis.
   - Renal vein thrombosis.
   - Untreated pre-renal failure.
   - HUS.



Clinical pictures:

I. Manifestation of the underlying etiology:-

- Pre renal: Dehydration, hypovolemia, hypotension, shock, and HF.
- Renal: Acute GN, HUS.
- Post renal: Congenital anomalies, stones, hematuria with clot.

II. Manifestation of ARF:-

- Decrease urine output "oliguria or anuria"
- Edema (due to salt and water retention).
- Pallor (due to anemia).
- Hypertension (due to ↑ salt and water retention and ↑ secretion of rennin) which may progress to hypertensive encephalopathy.
- Uremic encephalopathy, lethargy and drowsiness.
- GIT: Anorexia, nausea, vomiting and diarrhea.
- Acidotic breathing (deep and rapid).

III. Manifestation of complication:-

- CHD and pul. edema (due to volume overload).
- GIT hemorrhage (due to stress ulcers or gastritis)(due to MA).
- Arrhythmias (VF → Death).
- Behavioral changes, seizure, and coma.

Diagnosis

✓ **Dx of RF :**
- Urine volume : Oliguria and anuria.
- RFT : ↑BUN , ↑creatinine , ↑urea.
- ABG : MA.
- Electrolytes :
  - HyperK^{+} .
  - Dilutional hypoNa^{+} .

✓ **Dx of the cause :**
1. Post renal :
   - Palpable bladder and kidney.
   - Catheterization.
   - Abdominal U/S.
2. Pre renal :
   - Urine specific gravity > 1020.
   - Urine Na < 20mEq/L.
   - BUN/Creatinine > 2.

So;

- CBC :
  - Anemia → Dilutional - Hemolytic
  - Leucopenia → SLE nephritis.
  - ↓Plt → SLE , HUS , RVT
- RFT :
  - BUN ↑ due to decrease renal excretion (No. 20mg).
  - ↑ Serum creatinine due to decrease renal excretion.
  - ↑ Serum uric acid due to decrease renal excretion.
- Serum electrolytes :
  - ↓ Serum Na "dilutional"
  - ↑ Serum K due to excretion + 2ry to acidosis → ↑ release K from cells.
  - ↑ Serum phosphates due to ↓ excretion
  - ↓ Serum Ca → 2ry to hyperphosphates.
- ABG :
  - Metabolic acidosis due to ↓ ability of kidney to excrete acid + ↑ catabolism.
- Urine analysis :
  - RBCs , Protein.
  - Cast , tubular cells.
- BUN/Serum creatinine :
  - > 20 in pre renal.
  - < 20 in renal.
- ECG :For arrythemia.
- Chest x-ray : Pul. congestion , pul. edema.
- Studies for underlying causes :
  - Renal U/S.
  - Cystoscopy.

Treatment:

1) Hospitalization and monitoring:

- Clinical, body weight, urine output, fluid chart.
- ECG, blood urea, creatinine, electrolyte, PH, and bicarbonate, vital signs.

2) Treatment of underlying etiology:

- Remove obstruction by catheterization ± surgical for post renal failure.
- Correct pre renal failure :- (improve renal blood flow)
  - Fluid loss : Saline 20 ml/kg "over 1/2 -hr"
  - Blood loss : Fresh blood transfusion.
  - Plasma : Plasma transfusion.
  - Dopamine infusion 2-3 μg/kg/min can increase renal function.
- With good correction the patient will pass urine within 1-2 hrs.

3) Treatment of specific problem:

A) Fluid intake:

- Restrict fluids (daily intake 400ml/m^2) "insensible water loss" + urine output of previous day.

B) Nutrition:

- Calories 400 kcl/m^2/day (mostly CHO and fats).
- Proteins 0.5 g/kg/day as milk, meat, egg.TPN in prolonged cases.

C) Hyperkalemia : (ECG monitoring)

- Avoid K^+ containing foods and drinks as fruits.
- If K^+ level is more than 7 mEq/L :

1. **Shift K^+ intracellular**.
   - Regular insulin and dextrose to avoid hypoglycemia.
   - Salbutamol nebulizer "B_2 agonist".
   - Sodium bicarbonate 2ml/kg in concentration % IV over 10 minutes.
2. **Stabilize cardiac membrane** by Ca^+ glucanate:
   - 10% "1 ml/kg" slow IV "0.5 ml/kg IV over 5 minutes".
3. Remove potassium by dialysis.
4. **Ca^+ resonium enema** : (8hr)
   - It bind with K → intracellular Ca^+ free → support the heart.
   - It enhance K^+ loss in GIT.

D) Hyponatremia :

- Usually dilutional respond to fluid restrict.
- If not respond → dialysis.

E) Hypocalcemia and hyper phosphatemia :

- Tetany is rare as acidosis increase ionized Ca^+ .
- Reduce diatary phosphate and give phosphate binders (Ca^+ carbonate oral).

F) Treatment of complication:

- e.g Anemia : fresh packed RBC transfusion.

E) Dialysis (Peritoneal dialysis).

N.B:

* Blood urea 20-40 mg/dl.
* Serum creatinine 0.2-0.7 mg/dl.
* Blood urea nitrogen : blood urea/2
* K^+ 3.5-5.5 , If more than 6.5 need treatment.

Dialysis (Peritoneal dialysis):

1. Symptomatic uremia "vomiting , encephalopathy".
2. Symptomatic fluid overload especially cardiac failure or pericardial effusion.
3. Uncontrolled hypertension with symptomatic.
   *Electrolyte disturbances: Hyperk$^+$, MA.
4. Encephalopathy or seizure.
5. Prolonged oliguria.
6. Creatinine > 10 mg%.
7. Symptomatic hypocalcemia with sever hyperphosphatemia.

Chronic Renal Failure

- Irreversible damage of renal function over months or years → permanent reduction of GFR <75ml/1.73/m^2/min.
- In end stage renal disease, GFR < 10ml/1.73m^2/min.

$$GFR = \frac{\text{" t cm"} \times K}{\text{Plasma creatinine}} = \text{Schwartz formula}$$

K = > 1 year = 0.5
< 1 year = 0.45
Adolescents = 0.7

Classification according to GFR:

1. Mild CRF : GFR = 60-80 ml/min/1.73m^2.
2. Moderate CRF : GFR = 40-59 ml/min/1.73m^2.
3. Sever CRF : GFR = < 40 ml/min/1.73m^2.
4. End stage RF GFR = < 10 ml/min/1.73m^2.

Causes:

1) Congenital malformation, e.g. aplasia or hypoplasia, polycystic kidney.
2) Chronic GN.
3) Chronic pyelonephritis.
4) Chronic interstitial nephritis.
5) VUR.

GFR :-

- **At birth:** 25-60ml/min/1.73m^2.
- **1 year:** 50-160 ml/min/1.73m^2.
- **2-12 years:** 90-170 ml/min/1.73m^2.
- **> 12 years:** 90-150 ml/min/1.73m^2.

C/P :

1. FTT, poor growth, **short stature** due to: Resistance to growth hormone, anemia, anorexia "MA", ***renal osteodystrophy "Renal Rickt's".***
2. **Anemia**: due to ↓ erythropiotine, decrease intake of iron, B12, folic acid, and bone marrow depression by uremic toxin and defective iron utilization and frequent blood loss.
3. **Renal osteodystrophy**: Due to hyperphosphatemia and ↓ $1.25(HO)_2$ D_3 and 2ry hyperparathyroidism.
4. **Hypertension:** ↑ rennin and fluid retension.
5. Bleeding tendency: Due to platelet dysfunction.
6. **Infection:** Due to defective granulocyte function.
7. **Neurologic**: Fatigue, drowsiness, polyneuropathy due to uremic toxin.
8. **Pericarditis and cardiomyopathy**: Due to uremic toxin and hypertension.
9. **Hyperlipidemia**: ↓ lipoprotein lipase activity.

Diagnosis:

I) Diagnosis of renal failure "see before".

II) Is it CRF?

1. Renal U/S and DMSA scan show shrunken kidney.
2. Estimate GFR by creatinine clearance and DTPA scan.
3. **For complication**:
   a) CBC foe anemia.
   b) Echo.
   c) Ca, Pi, ALP, and parathyroid hormone.
   d) Bone x-ray.

N.B

Hemodialysis:

- S/E:-
  - ❶ Chronic blood loss.
  - ❷ Electrolyte disturbance.
  - ❸ Infection.
- Done in hospital 3 times/week in every 1-4 hrs duration, in infant and small children are better and suitable for peritoneal dialysis which more physiological and less hemodynamic changes.
- Peritoneal dialysis done in home daily usually at night by automatic machine.

Treatment:

I. Conservation:

1. **Diet:**
   - Restrict protein , ↑ calori intake.
     (↑fat - ↑COH).
   - Milk and milk product ↓ b/c high Pi.
   - ↓ salt and fluid if oliguria.
2. ***Treatment of hyperkalemia and metabolic acidosis***.
3. **Treatment of complication**: e.g.:
   - Hypertension.
   - Infection.
   - **Anemia:**
     → Fresh packed RBC transfusion.
     → Recombination erythropoitine.
   - **R.O.D**:
     - Phosphate binders "Ca+ carbonate".
     - Ca+2 0.5-2 gm/dl orally.
     - Vitamin D.
     - Partial parathyroidectomy.
   - **Growth failure**:
     - Correct acidosis.
     - Anemia.
     - R.O.D.
     - Treat anorexia, e.g. tube feeding.
     - Recomonent growth hormone.

II. Dialysis:

Indication:

1. As ARF.
2. Others:
   Pericarditis - Growth failure – Resistent anemia.

Types:

1. Hemodialysis.
2. Continous peritoneal dialysis.(The best in pediatrics).

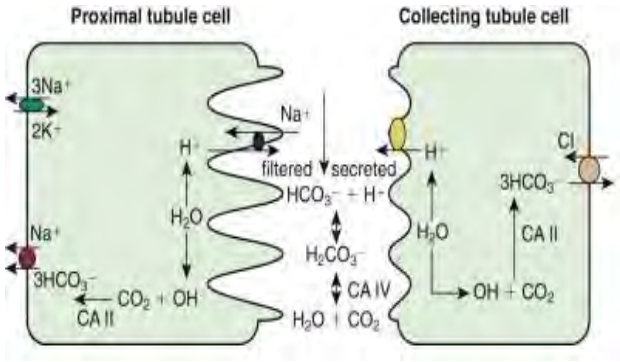
III. Renal transplantation: Definitive treatment.

Renal Tubular disorders

- Abnormalities of renal tubular function may occur at any point along the length of the nephron and affect any of the substances handled by it.

Generalized proximal tubular dysfunction "Fanconi syndrome"

- Excessive urinary loss of amino acids, glucose, phosphate, bicarbonates, sodium, calcium, potassium, and urate.



Causes:

1. Idiopathic.
2. Secondary to inborn error of metabolism:
   - Cystinosis "an AR disorder causing intracellular accumulation of cystine".
   - Glycogen storage disorders.
   - Lowe's syndrome "occulo-cerebro-renal dystropth".
   - Galactosemia.
   - Fructose intolerance.
   - Tyrosinemia.
   - Wilson's disease.
3. Acquired:
   - Heavy metals.
   - Drugs and toxin.
   - Vitamin D deficiency.

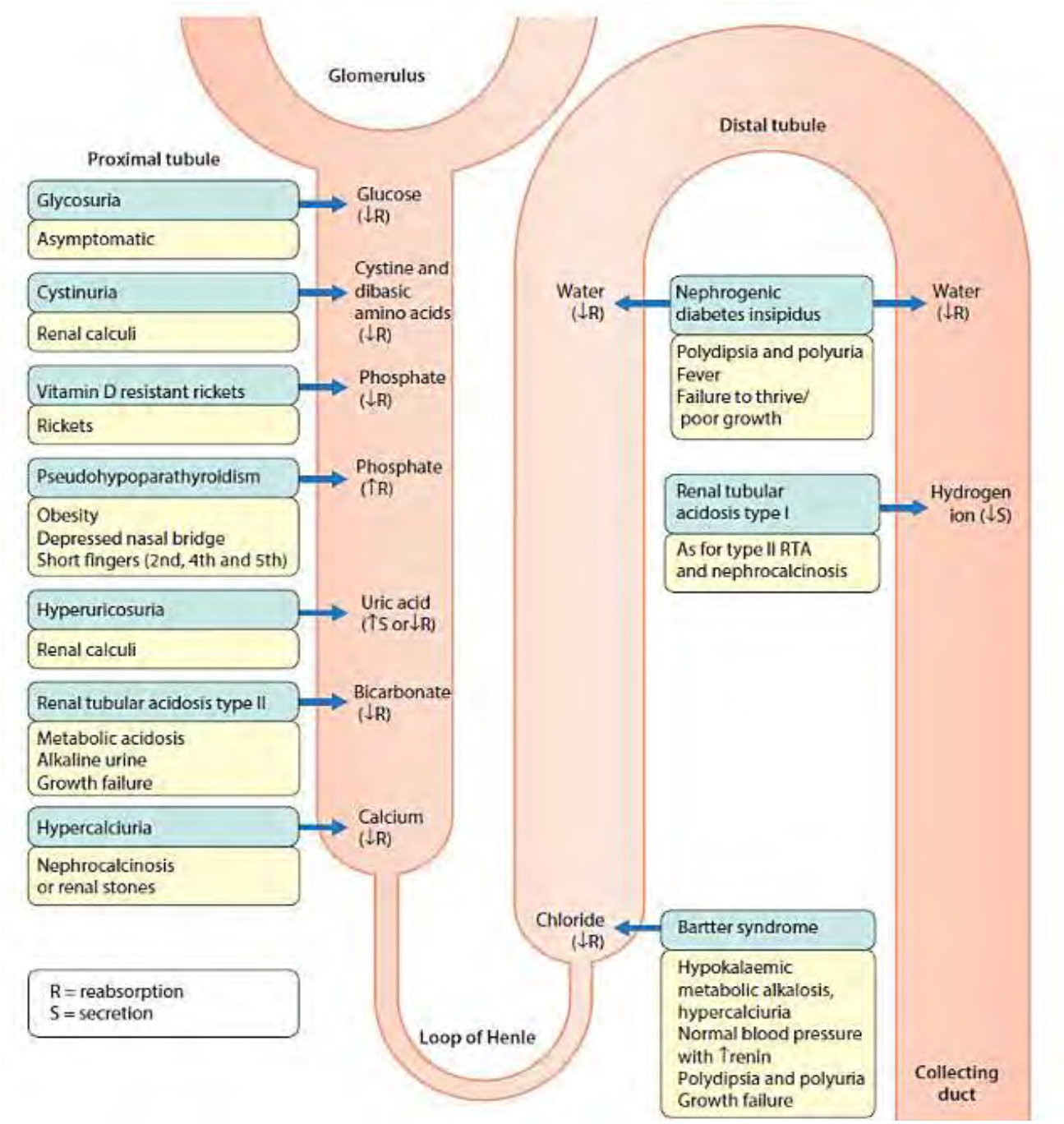
C/P :

- Polydipsia and polyuria.
- Salt depletion and dehydration.
- *Hyperchloraemic metabolic acidosis*.
- Rickets and osteoporosis.
- Failure to thrive/poor growth.

Bartter's syndrome:

Decrease CL reabsorption : hypokalemic metabolic alkalosis , hypercalcuria , normal

BP with increase rennin , presented with polyuria and polydipsia.



Renal tubular acidosis

Types

4 main types of RTA:-

- Type 1 : Impairment in distal acidification.
- Type 2 : Impairment in proximal tubule bicarbonate reclamation.
- Type 3 : Acombination of type 1 and 2.
- Type 4 : Occur as a result of lack of / or insensitivity to aldosterone.

*** All four types are associated with a hyperchloremic normal anion gap acidosis.**

Treatment :

- **Goal of RTA therapy**:
  - Improve growth.
  - Correct metabolic bone disease.
  - Prevent nephrolithiasis and nephrocalcinosis.
- **Principle therapies include the following**:

1. Alkali therapy "Na^+ citrate or Na^+ bicarbonate".
   - In all forms of RTA, with the goal of normal.
   - Plasma HCO_3 level.
   - Patient with distal RTA generally require "2-3 mEq/kg/day".
2. K^+ supplementation "Potassium citrate".

Clinical and laboratory manifestation :

- **Type 1 , 2 and 3** : Are associated with hypokalemia, **whereas type 4** is characterized by hyperkalemia in addition to hyperchloremic acidosis.
- **Hypercalciuria** is typical of *type 1 RTA* and in conjunction with hypocitraturia often leads to **nephrocalcinosis** and renal calculi.
- **Type 2 is** often part of a more global defect of proximal tubule function. The **fanconi syndrome** which in addition to bicarbonaturia is characterized by aminoaciduria, glycosuria, phosphaturia "hypophosphatemia", and Rickt's.
- **Type 4 RTA** is most commonly observed in pediatric patient with **obstruction uropathy**, tubular unresponsiveness to aldosterone "pseudoaldostrinism" that is often transient during infancy or ↓ aldosteron secretion "hypoaldostrinism"; all forms → growth failure, Polyuria, polydipsia, recurrent dehydration, and vomiting.

- **Determining the urinary anion gap helpful for the evaluation of metabolic acidosis**:
- It is indirect estimate of urinary ammonium excretion "and thus urinary acid excretion" and is calculated by the following formula after determining urinary electrolyte concentration:

  Anion gap = $(K+Na^+)$ - (Cl^- + HCO_3)= "10-12 mEQ/l"

Increase anion gap : MUDIPES

M: methanol.
U: uremia.
D: DKA.
I: idiopathic acidosis, iron & isoniazid.
P: paraldehyde, phenytoin .
E: ethanol , ethylene glycol .
S: salicylate.

Nocturnal enuresis

Children are usually dry by day by 2 years and dry by night by 3 years.

Types :

1) Primary NE 80% :-
   Never achieved night time dryness.
2) Secondary NE 20% :-
   Wet bed after a period 3-6 month of dryness.

Incidence:

- boys more than girls.
- 10% of 5 years, and 5% of 10 years old.
- Double incidence if +ve FH of NE.
- If both parent suffered → 75% chance that the off spring well be affected.

Causes :

Causes of 1ry NE:

1. Faculty training.
2. Delay micturation.
3. Low bladder capacity.
4. Genetic factor.

Causes of 2ry NE:

1. UTI.
2. DM.
3. DI.
4. CRF.
5. Constipation.
6. Neurogenic bladder.

Treatment :

(1) Baby < 5 years:

★ Complete urine examination and complete examination should be done before treatment.

- Reassurance.
- Family therapy.
- Practical help.

(2) Age 5-7 years:

- Chart therapy.
- Desmopresine.
- AD analogue.

Successful rate 60-70% by high relapsing rate when treatment stopped 30%.

(3) Baby > 7 years:

❶ Chart therapy.
❷ **Drugs**: Desmopresine.
❸ **Alarm**:
- 80% effective but may take 3 months.
- Should cont. utile totally dry for 6 weeks.
- 10% relapsing but well respond to further trial.

❹ T.A.D : Dilatation of urinary bladder.

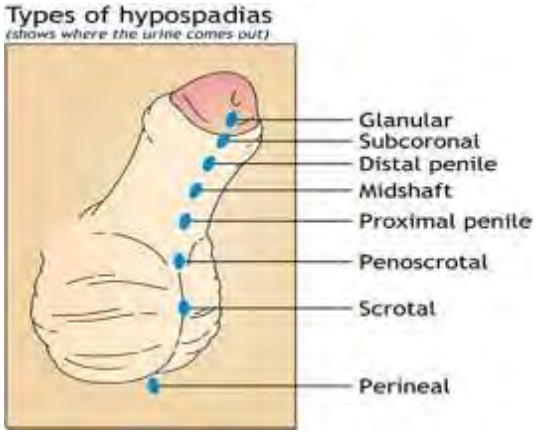
Congenital anomalies

Hypospidius

• It is congenital disorder of failure of urethral miatus to reach normal opening in penis.

Types :

1. Glander : Most common open on the dorsal of glanda penis.
2. Penile : Under saft of penis.
3. Corneal : At junction b/w glands and shaft of penis.
4. Peniloscrotal : At junction b/w penis and scrotum.
5. Perineal : On scrotum associated with UTI.



Treatment : ⇒ Urethral reconstruction or urethroplasty "1 year-18 month"

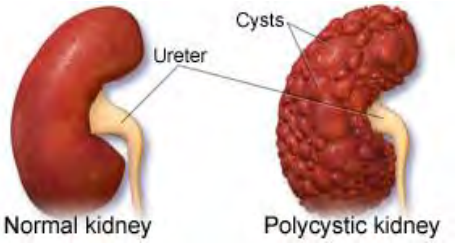
N.B

★Abnormalities associated with hypospediius:
1. Inguinal hernia.
2. Undescending testis.
3. Meatal stenosis.

➲ Circumcision is **contraindication.**

➲ Renal U/S should be done for any other congenital anomalies.

Polycystic Kidney



Types :

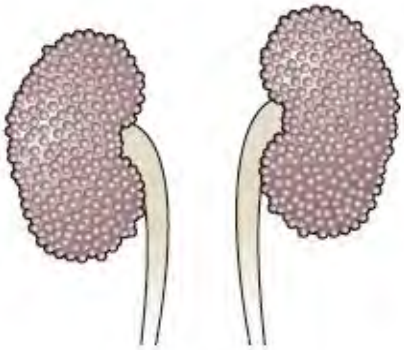
1) Infantile type: AR

a. At birth or early infancy: large palpable renal mass or abdominal distension.
- RD 2ry to pul. hypoplasia.
- AN U/S show large kidney and oligohydroaminos.

b. In advanced cases: S & S of ARF and HPN and P.HPN.
- Always ass. With con. Hepatic fibrosis

2) Adult type: AD

Usually present in adult life but can be rarely present in neonatal period.
- Associated with :
  - ❶ Cerebral aneurysim.
  - ❷ Subarachnoid hemorrhage.
  - ❸ Liver and spleen or pancreatic cyst.
- +ve FH and present in 4th - 5th decade.



Autosomal recessive polycystic kidney disease (ARPKD). There is diffuse bilateral enlargement of both kidneys.

Autosomal dominant polycystic kidney disease (ADPKD). There are separate cysts of varying size between normal renal parenchyma The kidneys are enlarged

Newborn examination by Dr.Asma Hadida

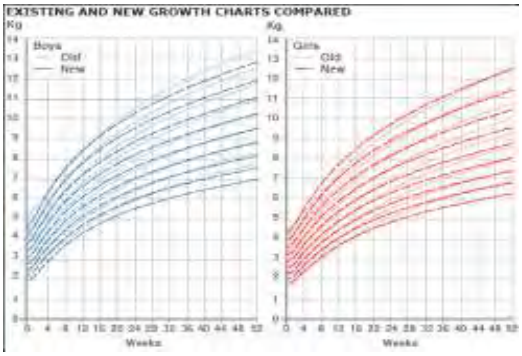
2016_2015

Newborn Examination

Before starting the examination, you must wash your hands and keep them warm and don't touch anything except the baby and expose the baby completely.

Steps:

- General look and posture.
- Anthropometric measurement.
- Vital signs.
- Skin examination.
- Head, face, neck.
- Upper limb.
- Chest & heart.
- Abd., and genitalia.
- Examination of neurobehaviour (primitive reflexes).
- Examination back & lower limb.
- Examination maturity sign.


EXISTING AND NEW GROWTH CHARTS COMPARED
Kg
Boys
Old
New
Weeks
Kg
Girls
Old
New
Weeks

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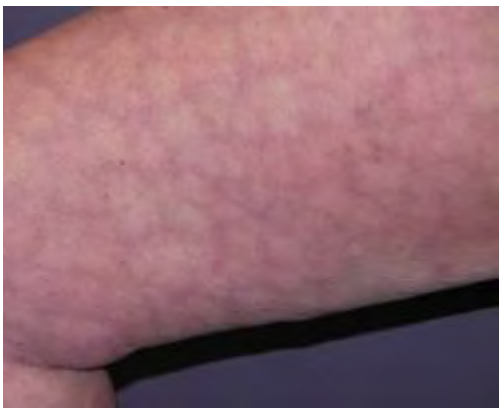
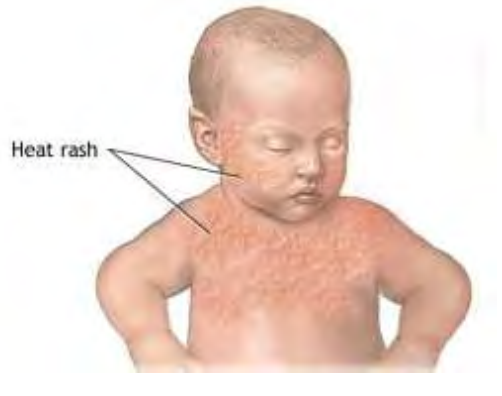
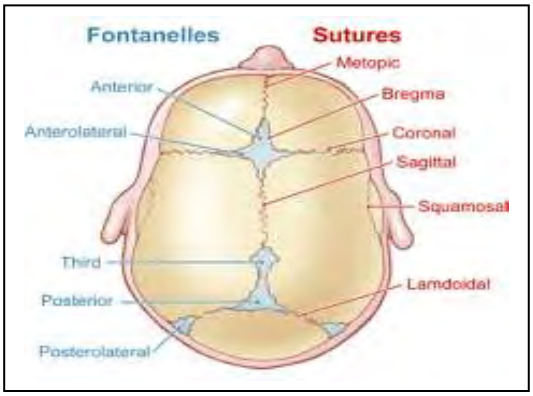
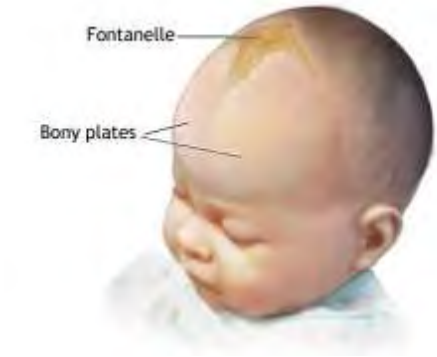
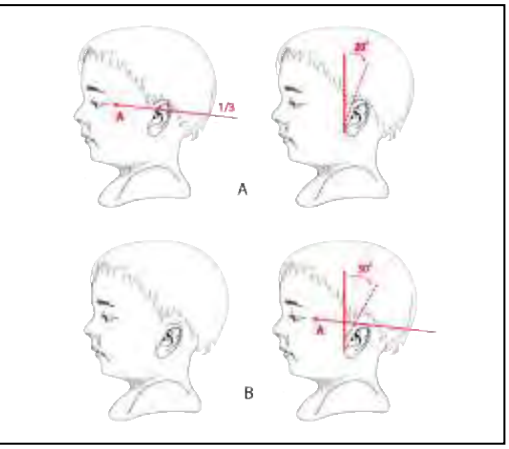


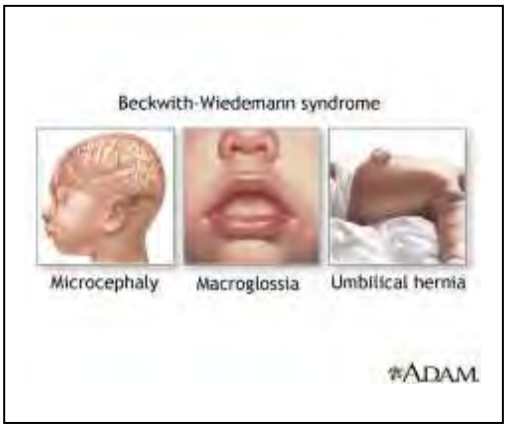

Heat rash

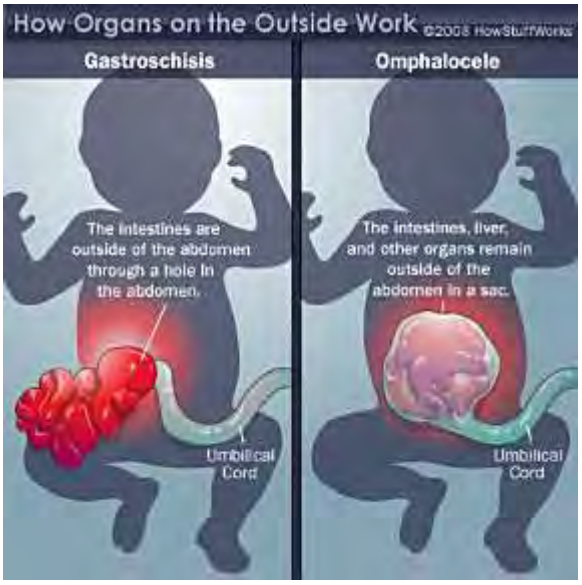

Fontanelles
Sutures
Metopic
Anterior
Bregma
Anterolateral
Coronal
Sagittal
Squamosal
Third
Lamdoidal
Posterior
Posterolateral


Fontanelle
Bony plates


A
B

the trochelmans


Beckwith-Wiedemann syndrome
Microcephaly
Macroglossia
Umbilical hernia
A.D.A.M.


How Organs on the Outside Work
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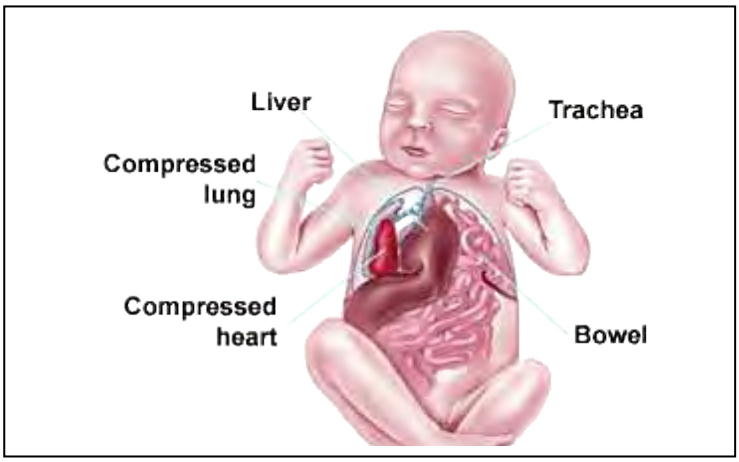
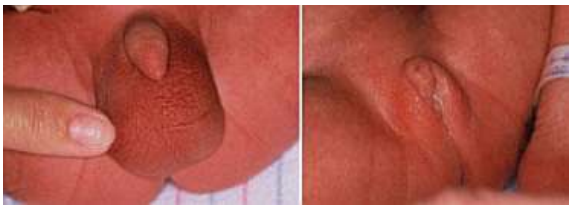
Gastroschisis
Omphalocele
The intestines are outside of the abdomen through a hole in the abdomen.
The intestines, liver, and other organs remain outside of the abdomen in a sac.
Umbilical Cord
Umbilical Cord

Lt


Liver
Trachea
Compressed lung
Compressed heart
Bowel

Umbilical Arteries
Umbilical Vein



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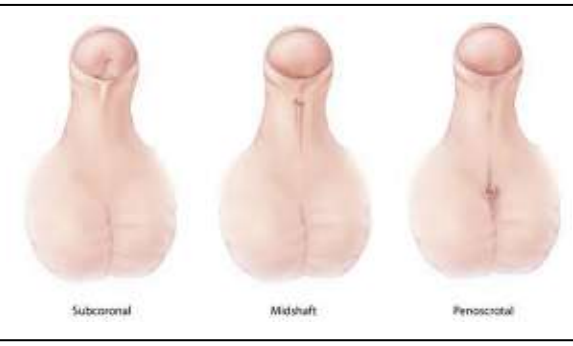
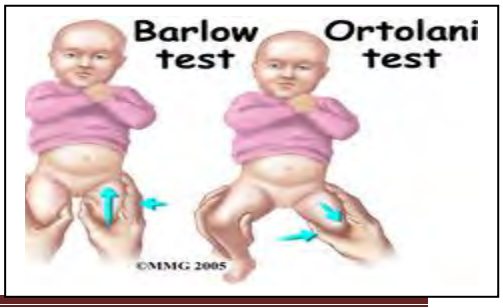

Subcoronal
Midshaft
Penoscrotal


Barlow test
Ortolani test
©MMG 2005

R
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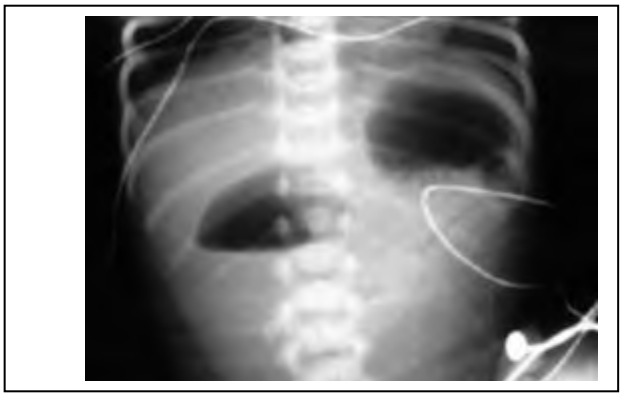
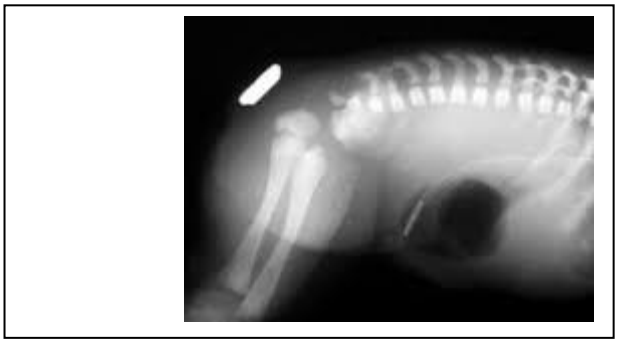
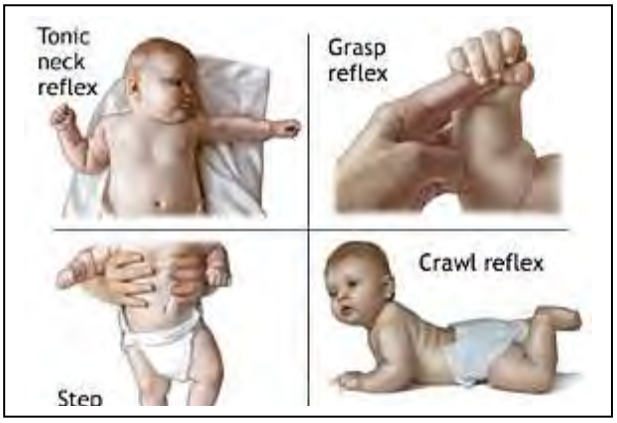
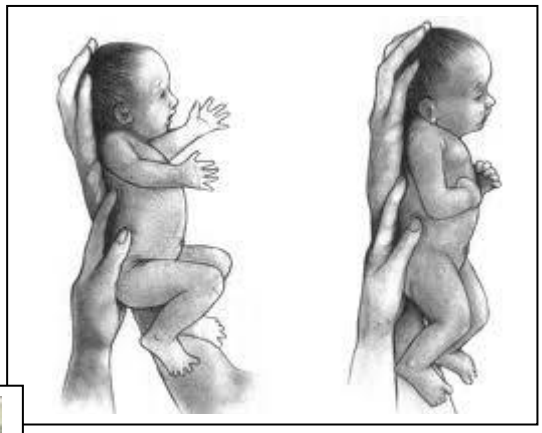
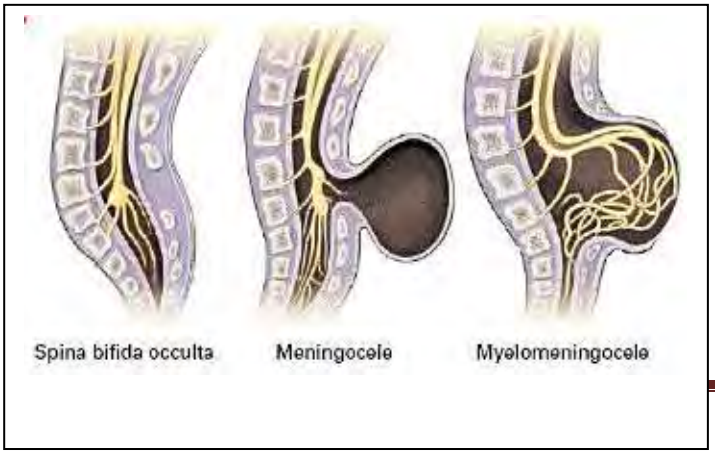
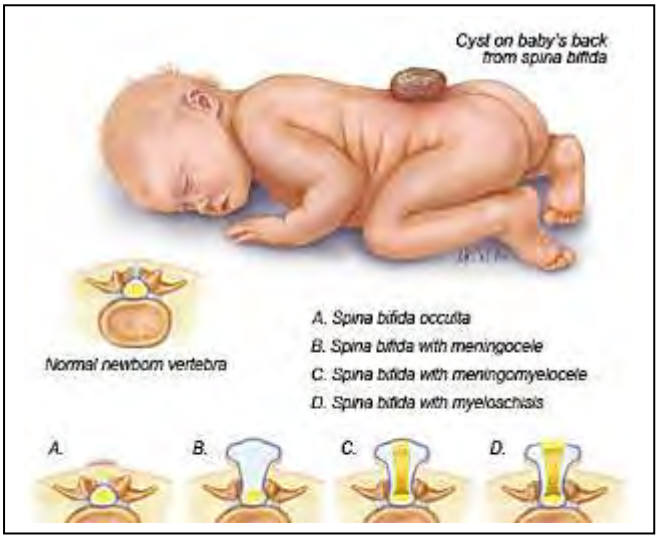




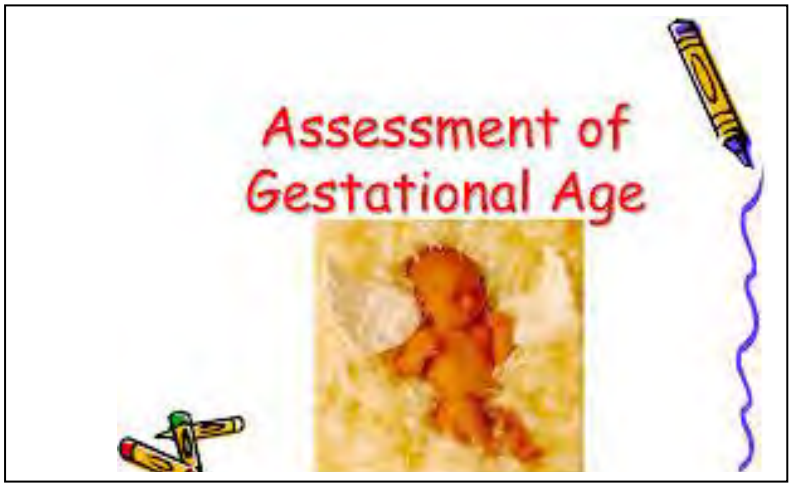

Tonic neck reflex
Grasp reflex
Crawl reflex
Step

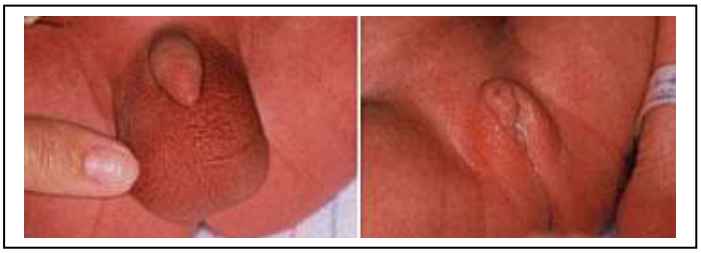


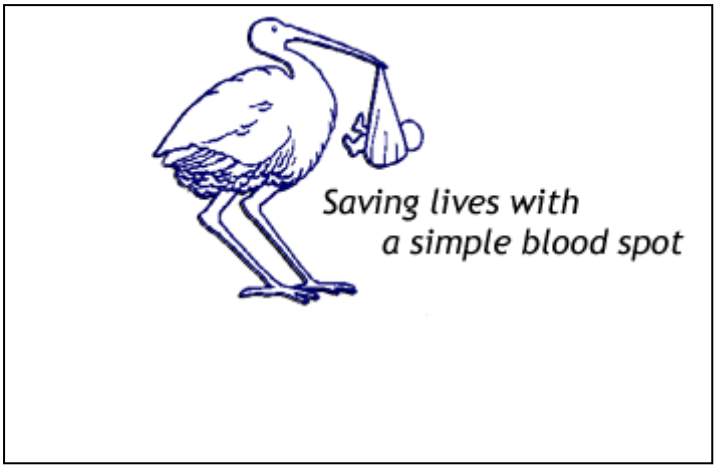

Spina bifida occulta
Meningocele
Myelomeningocele


Cyst on baby's back from spina bifida
Normal newborn vertebra
A. Spina bifida occulta
B. Spina bifida with meningocele
C. Spina bifida with meningomyelocele
D. Spina bifida with myeloschisis
A.
B.
C.
D.

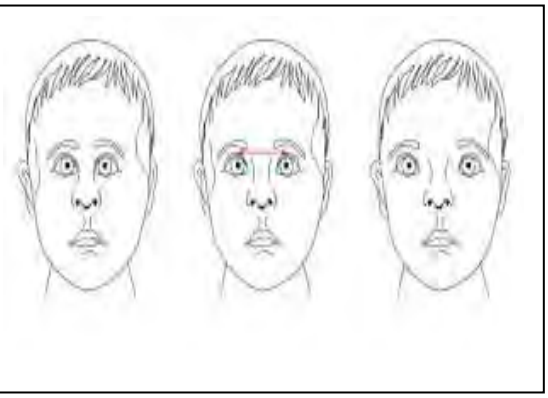
KLEAR-TRACE


Assessment of
Gestational Age




Saving lives with
a simple blood spot

Newborn Screening



Nutrition by .Asma Hadida

TOSHIBA

[اكتب اسم الشركة]

Nutrition, nutrition problems, and nutritional assessments

★Normal nutritional requirement for different ages:

- ***Adequate diet should supply***:
  - Adequate calories
  - Macronutrient "**CHO ,protein , lipid** ".
  - Micronutrient **"minerals, vitamins , and trace elements**

Age in years	Calories /kg/day
Below 6m	115
6m -1y	105
1-3y	100
4-6	90
7-9	80
10-12	70

Beyond one year calorie requirement decrease by 10cal /kg/day every 3years .

1. **Energy "calories " required for :**

- Basal metabolic 50% of total caloric intake .
- Physical activity 25%.
- Growth 12%.
- Fecal loss 8%.
- Specific dynamic action 5%.

Required ideal distribution :

- CHO:50% of total caloric intake

2. MACRONUTRIENTS

a) ***Protein : required for***:

- Growth and development .
- Formation of cells and cellular elements .
- Formation of enzyme and hormones
- **Vital function of blood**:
  - Hb .
  - Osmotic pressure .
  - Buffering system .
  - Coagulation system .
  - Immunoglobulins .
  - Carrying systems for iron , Ca, etc

Required amount "gram /kg /day":

- Infancy 2-2.5gm /kg /day .
- 2-6y "1.8-1.6".
- 7-12y 1.2.
- Adult 0.8.

Protein need are higher in infancy and childhood due to rapid rate of growth .

Types :

1. **High biological value protein**: contain all essential amino _acid "animal protein e.g milk , meat and egg ".
2. **Low biological value protein** : def. in some essential A.A" vegetable origin e.g cereals .

_it is recommended to use a mixture of both .

Related disorder:

KWK is the classic example of protein def.

b) *Lipid* :

Required for :

- Energy 1 gm -9Kcal .
- Oxidation -A.A.
- Essential F.A for growth and some metabolic processes.
- Fat soluble vitamins .
- Insulation and support of body structures .

Required amount:35% of the total caloric intake

CHO:

Required for: supplying the greatest part of caloric need of the body .

Required amount: 10gm /kg/day

3. MICRONUTRIENTS

a. ***Minerals and vit.***:

. Iron : 10mg /day .
. VitA: 2000 iu /day.
. VitD: 400iu /day.
.Vit. B1 : 1mg /day .
. B2 : 1mg/day .
. Vit. C: 50mg /day .

b. Fibers :

Increase fecal output and prevent constipation ,
Sources : fruits and vegetables .

Water :

Water content :

75-80% of body wt. which is higher than that of adult "55-60%".

Water requirement:

- Infancy ⇨ 120ml/kg/day.
- 2-6y ⇨100ml/ kg/day .
- 7-12y ⇨ 90-80ml/kg/day.
- 13-18y ⇨ 80-70ml/kg/day.
- Adult ⇨ 50-40ml/kg/day.

Nutritional requirement of normal infant:

1. Calori 100K/cal/kg /day .
2. Fluid 150ml/kg/day.
3. Vit. D 200-400iu/day.
4. Protein 2-2.4g/kg/day.
5. Ca: 2-4mmol /kg/day.
6. CL: 2-4mmol /kg/day .
7. K: 1-3mmol/kg/day.

⇨ Higher requirement of preterm baby

PROTEIN - ENERGY MALNUTRITION

By WHO, as a range of pathological condition arising from coincident lack, in varying proportion , of protein and energy , occurring most in infant and young children and commonly associated with infections and multiple vitamins and mineral def.

PEM comprises a spectrum of diseases:

1. Kwashiorkor.
2. Marasmus .
3. Marasmic kwashiorkor.

Classification of PEM:

1. **Jellife classification** :

% of expected wt/age	Level of PEM
90-81%	1st degree
80-71%	2nd degree
70-61%	3rd degree
<60%	4th degree

2. **Wellcome classification** : depending on wt as a percent of reference 50th percentile wt/age as a presence or absence of edema .

Syndrome	%of expected wt/age	edema
Underwt	60-80	-
Marasmus	<60	_
MK	<60	+
KWK	>60-80 or normal	++
Nutritional edema	Nomal>80	+++

3. Waterlow classification :

- **Wasted :** wt/lt < 80% of reference 50th percentile .
- **Stunted :** lt /age < 90% of reference 50th percentile .
- **Wasted and stunted** : wt /lt < 80% and lt /age < 90% of reference 50th percentile.

	Weight /age	Length / age	Weight/length
Wasted	Decrease	Normal	Decrease
Stunted	Decrease	Decrease	Normal
Wasted and stunted	Decrease	Decrease	Decrease

There is increasing evidence of the importance of growth and nutrition in relation to cognitive development.

Significant of weight /age and normal length /age in PEM:

- Decrease of wt/age and normal lt /age indicates recent nutritional deficiency
- Decrease of lt/age and normal wt/age indicate nutritional def. in the past .
- Decrease of both lt and wt/age indicate both recent and old nutritional def .

MARASMUS

Marasmus: is chronic state of total caloric deficiency. It is not disease by itself it is a syndrome.

Causes of marasmus:

1. **Nutritional**: sever under nutrition due to scanty breast milk or diluted formula , or may be due to prolonged breastfeeding without supplementation.
2. **Non nutritional organic marasmus**:
   - Recurrent or chronic infection e.g. repeated GE .
   - **Chronic systemic disease** : CHD, CRF.
   - **Metabolic disease** e.g. RTA
   - **Congenital** e.g. cleft lip or palate.

C/P: Characteristic features of marasmus:

1. **Loss of wt. "under wt. for age and sex ":**
   - 1st degree : 15-25% loss of wt.
   - 2nd degree : 25-35% loss of wt.
   - 3rd degree : more than 35% loss of wt.
2. **Loss of S/C fat :**
   - ***1st degree***: over the abd. Wall and trunk .
   - ***2nd degree***: over the limbs and buttock.
   - ***3rd degree***: over the face "***senile face***" it is buccal pad of fat its fat is unsaturated and less readily metabolized .
3. **Muscle wasting :** limbs look thin with a small atrophic muscle .

So; C/P of marasmus:

1. ***Manifestation of low caloric intake :***
   - ✓ Loss of S/C fat .
   - ✓ Hypothermia .
2. ***Manifestation of low mineral***:
   - ✓ Iron def. anemia .
3. ***Manifestation of low vitamins intake***:
   - ✓ Keratomalacia.
   - ✓ Rickets.
   - ✓ Scurvy.
   - ✓ Starvation , scanty dry greenish stool.

There no rickets as it is should occur in growing infants. if rickets is present, it is called **atrophic rickets**.

Complication of marasmus:

1. **Chest infection** : bronchitis, bronchopneumonia .
2. **GIT infection** : GE , and dehydration .
3. Cutaneous infection .

Investigations :

1. **Urinalysis** : ketonuria is usually present as the starvation and consumption of fat causes ketone bodies to be excreted in urine .
2. **Blood** : hypokalemia , anemia , hypoglycemia.
3. **Stool analysis** : low PH IN in case of lactase def.
4. Every system should be investigated.

Discussion:

Q. How to calculate the degree of wt. loss ?

By dividing = actual wt./expected wt.100

Q. Does under wt. mean marasmus?

No, marasmus is characterized by loss of wt., loss of S/C fat and muscle wasting , i.e. marasmus is wasted infant.

Under wt.:

1. **Nutritional dwarfism** :"small and stunted infant" no loss of S/C fat, no wasting, both wt. and lt. are equally affected .
2. **Nutritional marasmus** : "wasted infant" wt. is affected more than lt.

Relevant clinical finding in marasmus:

➲ **History :**

- **Dietetic error** : scanty breast feeding without supplementation or diluted formula .
- **Symptoms of chronic illness** as chronic cough , chronic diarrhea .
- **Social standard** : mostly in low socioeconomic classes.

➲ **Examination :**

- Under wt. with loss of S/C fat and wasting .
- Pallor and C/P of vit. def .
- Fever in infection .
- **Chest and cardiac exam**. for chronic disease .

KWASHIORKOR

Due to sever protein def. with almost adequate calori intake with other nutrient especially of vitamin B.

Characteristic features Of KWK

Constant :

1. **Edema :** Pitting "start in dorsum of hand and feet ,then spread to involve the eyelid, face, producing prominent cheeks **(moon face)**, limbs and genitalia, generalized in advanced cases, ascites and PE are unknown. The causes of edema are :
   - Hypoprotenemia with decreased osmotic pressure in capillaries .
   - **Salt and water retention due to** :
     - Increased activity of renin - angiotensin - aldosterone axis induced by hypovolemia .
     - Decreased inactivation mineralocorticoids by the fatty liver .
     - Increased activity of ADH.
     - Increased capillary permeability.
2. **Mental change** : .lethargy , apathy , disinterest , irritable, anorexia . the causes **due to** :
   - Serotonin def . due to def of its precursor tryptophan in diet .
   - B3 def simulating pellagra .
   - Disturbance in aromatic amino acid metabolism .

3. **Decreased muscle fat ratio** : Muscle atrophy with preserved S/C fat , demonstrated by palpation of biceps muscle and reduced MAC(less than 12cm
4. **Growth failure** : but it masked by edema and increase S/C fat, therefore , the growth retardation is marked in length more than weight.
5. Purpuric eruption "**bad prognosis**".

Variable :

1. **Hair changes**: Dyspigmentation result in streaky red or gray hair color "hypochromotrichia" , thin sparse easy detached ,flag sign : if there is period of remission and relapse (the darker parts for the former and the lighter parts for the later) .hair changes may be due to def. of sulfur containing AA or copper def.
2. **Skin change**: Pigmentary changes ,hyperkeratosis, atrophy, fissuring , scaling , desquamation more in lower limbs buttocks and napkin area, it is due to nicotinic acid and vitamin A def, suprarenal function disturbances and def of protein and essential FA .
3. **Hepatomegaly** : Due to fatty infiltration(due to def. of lipoprotein) , it is reversible with treatment .
4. **Vit. Def**: e.gkeratomalacia .
5. **Infection** due to secondary immune- def e.g persistent diarrhea.

Causes of KWK:

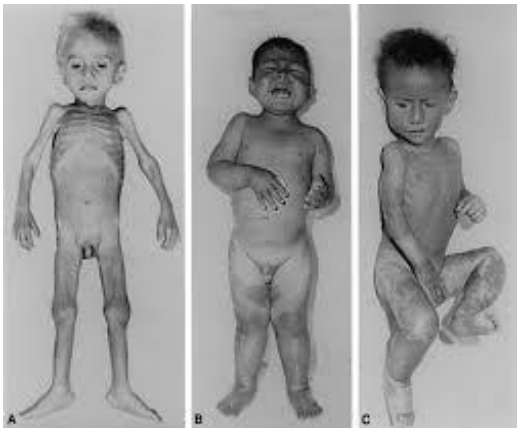
1. **Nutritional** : faulty weaning of low protein , high CHO diet .
2. **Non_nutritional** : infection" e.g pertussis , measles, T.B" and parasites "ascaris , ankylostoma , " sever GE with anorexia and prolonged restriction of milk .

Investigation:

1. Decrease in the conc. Of serum albumin is the most characteristic change , gamma gloublin is normal or high , low level of ceruloplasmin , transferrin and haptoglobulin.
2. Blood glucose value are low ,
3. Urinary excretion of hydroxyproline relative to creatinine may be decreased . urine urea is low.
4. Plasma values of ess. AA may be decreased
5. K, copper , mg, zinc, def.
6. Serum cholesterol and triglycerides levels are low , FA are generally elevated .
7. Serum amylase , esterase , lipase, ALK decrease .
8. Anemia may be normocytic "protein def " ,microcytic "iron def. "or macrocytic (folic acid and vit B12 def).
9. Bone growth is usually delayed , with reduction of total bone mass and osteoporosis .

Complication of KWK:

1. **Sever infection**: bronchopneumonia is a common cause of death , T.B, skin infection .
2. **Sever GIT infection**: GE and dehydration .
3. **Cut. Infection**: sever napkin dermatitis .
4. water and electrolyte imbalance " e.ghyponatremia and hypokalemia ".
5. Hypothermia which occur in early days .
6. Heart failure .



Discussion:

Q. What is the most important clinical finding in KWK?

Edema is an early constant sign .

Q. Which is more serious marasmus or KWK?

KWK "malignant malnutrition" is more serious for several reasons :

1. Associated water and electrolyte disturbances " hyponatremia , hypokalemia , hypomg +"

2. Associated anemia , vit. and minerals deficiency.
3. Associated immune_def witch increase susceptibility to serious infection .
4. Associated sever anorexia which may necessitate hospitalization and NGT feeding .

Q. What are the main causes of death in KWK?

- Severe dehydration .
- Electrolyte disturbance.
- Serious infection as septicemia and bronchopneumonia.
- Hypoglycemia.
- Hypothermia.
- Severe anemia.

Relevant clinical finding in KWK:

➲ **History :**

- Dietetic error .
- H/O GE with prolonged restriction of milk feeding e.g post .GE.
- Symptoms suggesting recent infection .
- Appetite usually poor.

➲ **Examination :**

- **Behavior :** apathic look with lack of interest .
- **Head :** puffy face , hair changes, pallor , angular stomatitis .
- Edema.
- **Skin :** napkin dermatitis .
- Abd, exa. Hepatomegaly .
- Chest exa. Bronchopneumonia .
- Sings of dehydration .

Q. What is marasmic KWK?

It is combined form of malnutrionwith manifestation of both marasmus and KWK "wasting and Edema It occur in 2situation :

1. **In marasmus infant**: if the dietetic error "under nutrition " is wrongly corrected with low protein , high CHO diet ,i.e protein malnutrition occur on top of under nutrition .
2. **In KWK infant**: if the sever anorexia interfere with food intake leading to under nutrition i.e under nutrition occurs on top of protein malnutrition .

☑ **Comparison b/w marasmus and KWK**:

	Marasmus	KWK
Dietetic error	Low calori	Low protein
Main essential finding	Under wt. , loss of S/C fat, muscle wasting	Edema , mental changes , growth failure ,
Behavior	Alert	Apathic , disinterest
Appetite	Hunger	Anorexia
Complication	Less common	More common
Biochemical changes	Minimal	Marked
Prognosis	Better	Worse

Q.How to correct the dietetic error in marasmus ?

- Corrected by increase total calori intake to 150cal /kg /day .
- Normal req. 100cal /kg/day .
- The amount is calculated by actual wt., then it increase gradually to reach the amount of the expected wt.

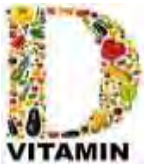
Q. How to correct the dietetic error of KWK?

Oral feeding start with small , frequent (every 2 hr) feeds (120 ml /kg/day , 66 calorie and 1 g protein /dl) of diluted milk . strength and volume are gradually increased and frequency decreased over the next 5 days. by days 6-8 , the child should receive 150 ml/kg/24hr in 6 feeds of high energy milk (114 calorie and 4 g protein / dl) . in the recovery period , high energy feeds made with milk , vegetable , oil , and sugar are needed , Protein def. is corrected by increase the protein intake to 4-6gm /kg / day “normal 2-3gm /kg /day"

High protein diet as yogurt , egg, ready-made cereals".

In sever anorexia “NGT” feeding .

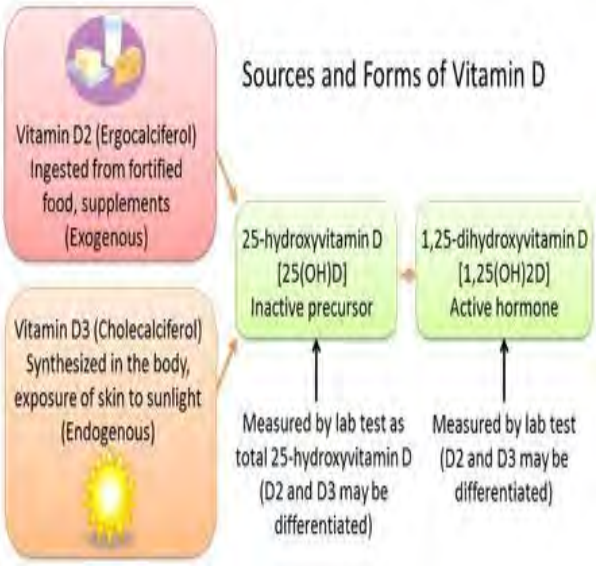
Vitamins



Two forms of vitamin D are practical importance :

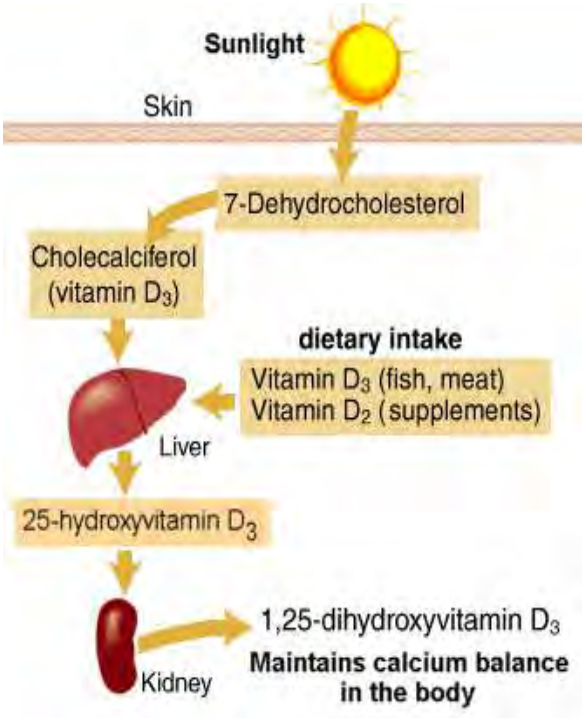
1.Vitamin D2or calciferol "plant origin" is a source of dietary and therapeutic vitamin D.

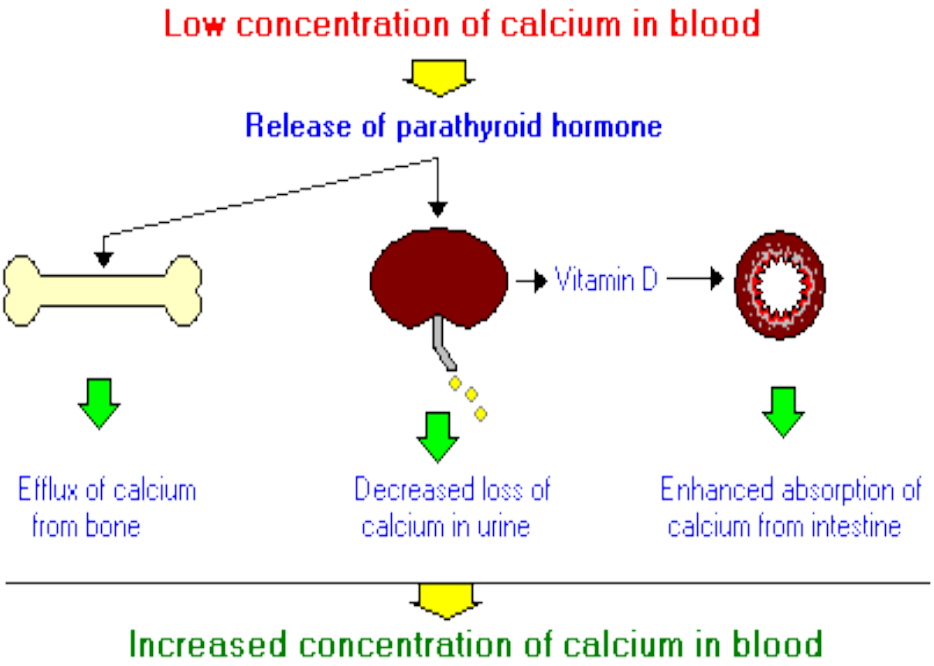
2.Vitamin D3 is naturally present in human skin in the provitamin stage **as 7-dehydrocholesterol** , which is activated photo chemically , by **UV rays** to **cholecalciferol** and transferred to the **liver** , to be hydroxylated into **25-OH-CC** and , subsequently , in the **renal** cortical cells to **1,25(OH)2-CC,** which functions as **a hormone** . receptors of the hormone are found in the kidney (renal tubular cells) , intestine , osteoblasts of bone and parathyroid gland .



Action of vitamin D and PTH, on Ca and phosphorus metabolism :

1. **Vitamin D** : promotes the absorption of Ca and therefore of phosphate from the gut , to control , with PTH , the mineralization of bone (PTH tend to de mineralize and vitamin D to mineralize) . it also promotes the renal tubular reabsorption of Ca and Pi (PTH inhibits renal tubular reabsorption of phosphate).
2. **Parathormone**(PTH) promotes intestinal Ca absorption (via its effect on vitamin D metabolism , i.e. renal hydroxylation either 1 or 24 hydroxylation) , in presence of hypocalcaemia .
3. **Calcitonin** is produced by the C- cell of the thyroid gland . it decrease the rate of bone demineralization and decrease renal tubular reabsorption of Ca and P(in presence of hypercalcemia) .





Rickets

Metabolic disorder of **growing bone** with defective mineralization **(of growing metaphysical cartilage and osteoid)** , in non growing bone **(of adult)** , vit D deficiency produce **osteomalacia** . when ossification does not improve with customary doses of vitamin D , the condition is called **vitamin D resistant rickets** .

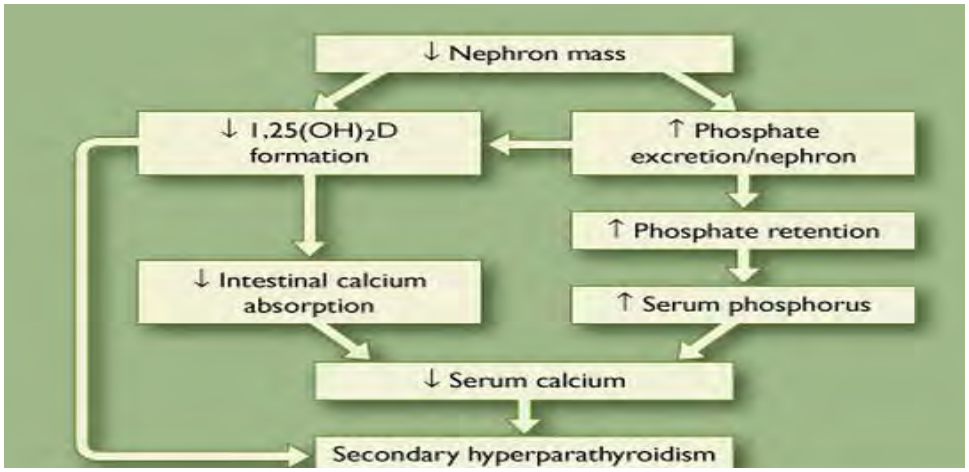
Classification of rickets :

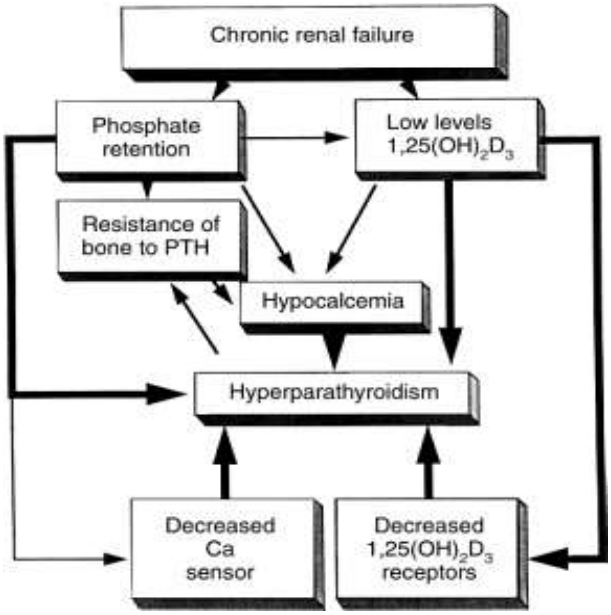
1. **Calcium Ca deficiency with secondary hyperparathyroidism** (*def . of vitamin D , low 25 OH –CC and no stimulation of higher 1.25 (OH)2 CC values* :
   a. Responsive to normal doses of vitamin D (**vitamin D deficiency rickets or nutritional rickets)**
   b. Responsive to high doses of vitamin D :
      1. **Hepatic disease and malabsorption of vitamin D** : As in :
         - Hepatitis .
         - Billiary atresia .
         - Anticonvulsant drugs .

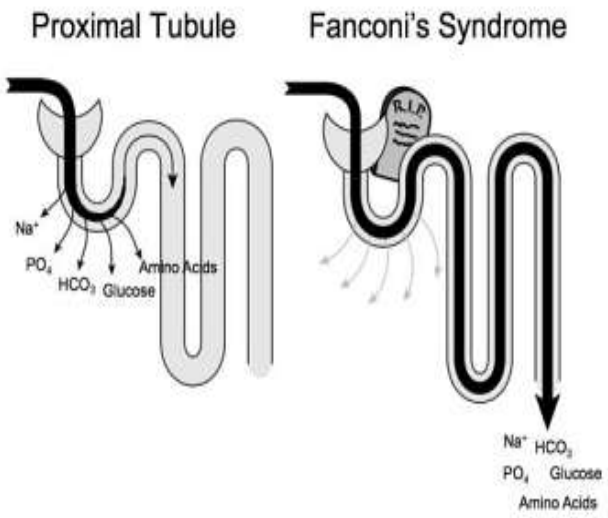
      These result in hepatic or celiac rickets .
      2. **Renal glomerular rickets (renal osteodystrophy due to defective hydroxylation of 25OH-CC in the kidney)** result from chronic glomerular insufficiency .
      3. **Vitamin D dependent rickets : type 1 AR** with defective formation of 1.25(OH)2 CC in the renal cortical cells .

2. **Primary phosphate deficiency (no secondary hyper-parathyrodism):**
   A. **Renal tubular rickets** :
      1. **Fanconi syndrome** : cystinosis or acquired fanconi , there are multiple renal tubular defects , with failure of tubular reabsorption of phosphate , amino acids , glucose , bicarbonate and water .
      2. **Renal tubular acidosis** : in which there is loss of calcium and phosphorus .
      3. **X-linked vitamin D resistant rickets** (familial hypophosphatemic rickets) : in which only renal tubular reabsoption of phosphorus is affected .
   B. **Oncogenic rickets** : in which there is phosphaturia (due to secretion of parathormone –like substance by the tumor cells) .

3. **End organ resistant** to 1,25 (OH)2 D : vitamin D depended rickets **, type 2** .
4. **Related conditions , resembling rickets** :
   a. essential bone matrix abnormality e.g hypophosophatasia .
   b. hyperparathyroidism , blood calcium high , S.P decreased .







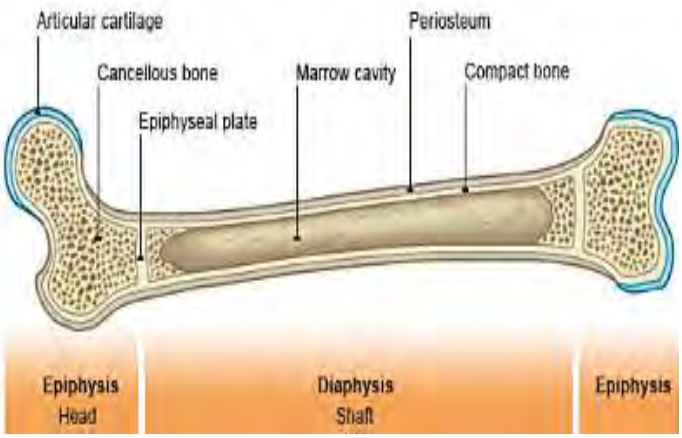
Rickets of vitamin D deficiency (nutritional rickets)

Predisposing factors :

1. **Age :** the usual age incidence is from 4 months to 2years .
2. **Atmospheric condition :**
   UV rays of sunlight are filtered off by smoke , dust and ordinary window glass . thus rickets is more prevalent in cities .
3. **Rapid growth** : rickets is a disease of growing bones . a marasmic child is least susceptible to develop rickets . but during successful treatment of marasmus , rachitic manifestation can develop . rapidly growing twin and premature are specially susceptible to rickets .
4. **Diet :**
   a. The Ca /P ratio in the diet of 2:1 is optimum for the absorption of both minerals . excessively low ratio as in cow's milk (ca/p: 3:4) impair Ca. absorption and makes the diet rachitogenic .
   b. Excess phytic acid : in the diet is rachitogenic as it hinders calcium absorption through the formation of insoluble calcium phytates .
   c. Citric acid (vitamin C) promotes calcium absorption .
5. **Race** : dark colored skin was supposed to predispose to rickets .
6. **In malabsorption syndrome** : vitamin D and calcium are not absorbed .

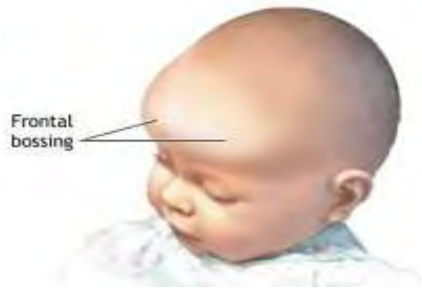
Rickets the following changes are observed :

1. The proliferating cartilage layer is enlarged with irregularity arranged cells and marked vascularity **(broadening metaphysic)**.
2. The layer of provisional calcification is devoid of calcium and is invaded by the proliferating cartilage . **thus the epiphyseal** line becomes irregular and frayed .
3. General rarefaction of bone shafts , which results in deformities and greenstick fractures.



Clinical picture :

1. **General manifestation :**
   a. **Early sign** : irritability , restlessness and profuse sweating over the head .
   b. **Generalized hypotonia and relaxation of ligaments** :
   These lead to delayed sitting and walking . weakness of abdominal muscle leads to pot-belly abdomen and tendency to constipation .Iron deficiency anemia is common



2. **Bony manifestation** : the skull is the most rapidly growing bone in the first year of life .thus , rickets starting in the 1st year mainly shows changes in the skull , while 2nd year onset rickets shows more changes in the long bones .

A. Head :

- Craniotabes or abnormal softening of skull bone is an early sign. It is detected by pressing firmly over the posterior parietal bone , a ping pong ball will be felt .
- Delayed closure of AF.
- Bossing of the skull due to osteiod deposition in the frontal and parietal region giving boxy skull (caput quadratum) .

B. Thorax :

❖ **Rosary beads** : is an early sign and is due to enlargement of costochondral junction . beading is 1st palpable , then becomes visible .

❖ **Harrisons sulcus** : a horizontal groove at the lower border of the chest along the attachment of the diaphragm . this results from dragging of the diaphragm during respiratory movement .

❖ A longitudinal groove develops posterior to rosaries . the sternum and its adjacent parts project forewords giving the so called pigon chest .

C. **Spine :** kyprhosis may be present due to hypotonia and relaxed ligamints .

D. **Pelvis** : contacted .

E. **Extremities :**

enlarged epiphysis are early especially at the wrist and ankles . a transverse groove is palpated at both maleoli **(marfans sign**) . walking on the softening shaft of long bones may cause deformity **(bow legs or knock knees).** upper limb deformities occur only in creeping infants .

Chemical pathology :

Due to the absence of vitamin D, less ca is absorbed from the intestine . lowering of serum ca result in stimulation of PTH . this leads to mobilization of ca and p from the bone . thus , serum ca level is maintained normal . PTH decrease tubular reabsoption of P so serum P is decreased .

Increased osteoblastic activity leads to rise in serum alkaline phosphatase which is mainly present in osteoblast. The enzyme may be normal in rachitic infants who are protein depleted.

In conclusion :

- Normal serum Ca (9-11mg /dl) .
- Decreased phosphorus to 1.5 -3.5mg /dl .
- Raised alkaline phosphatase .

X-ray changes :

1. Active rickets : the distal ends of bones appear widened , cupped or concave , frayed and , ill defined . the distance between the distal ends of radius and ulna and carpal bones is falsely increased since the large rachitic metaphysis , which is not calcified does not appear radio logically . the density of the shafts is decreased .
2. Healing rickets : a line of preparatory calcification appears which is separated from the distal end of the shaft by a zone of decreased calcification , the zone of osteoid tissue . as healing progresses and the osteiod tissue becomes calcified , the line becomes united with the shaft . healing rickets is observed 2-3weeks after giving vitamin D therapy .
3. **Healed rickets** : the only radiological evidence of past rickets of concavity and thickness of lower end of bones especially the radius . this is after 2-3m therapy.

Prevention of infantile rickets :

1. Health education to avoid over clothing and encourage sunlight exposure .
2. Normal infants should receive vitamin D 400-600 IU /day from the age of 4-6months. Premature are particularly in need of 800-1200IU/day from the second month of life .

Complication :

1. **Tetany (hypocalcemic tetany) :**
   Occur in sever advanced rickets , in the early stage of rickets before occurrence of secondary hyperparathyroidism , and with vitamin D shot therapy , when calcium level falls below 7mg/dl.
2. Bony deformities in the lower limb and pelvis.
3. Increased susceptibility to inter current infection.

Treatment of infantile rickets :

Vitamin D :

1. Oral : vitamin D in a dose of 1500-5000IU/day .
2. Shot therapy : 600,000 units given once orally or IM , a second and a third similar doses can be given each after 2 weeks period .

Other types of rickets :

Type	Mechanism	Clinical &laboratory features	Treatment
Hepatic rickets e.g chronic hepatitis , liver cirrhosis or biliary atresia	Defective 25-hydroxylation defective absorption of vitamin D with steatorrhea .	Similar to nutritional rickets .	Responsive to high doses of vitamin D.
Rickets due to prolonged use of anti-covulsant drugs.	Inhibit intestinal ca absorption induce liver enzymes to degredate vitamin D	Similar to nutritional rickets .	Vitamin D supplementation during anticonvulsent therapy
Celiac rickets e.g celiac disease .	Fat & vitamin D malabsorption	Similar to nutritional rickets but s, ca is lower and manifestation of celiac	Treatment of the cause , vitamin D in large doses or give 25OH-CC parental .

Renal glomerular rickets (renal osteodystrophy) e.g renal hypoplasia , chronic nephritis ,CRF.	1. Impaired glomerular function with retention of p lowering blood ca . 2. Impaired production of 1,25(OH)2CC. 3. Both 1&2 lead to secondary hyperparathyroidism .	Renal symptoms .	Treatment of renal condition vitamin D (1-OHCC), deit high in ca &low in p, with phosphate binders.
Vitamin D dependent rickets type 1	Defective formation of 1.25(OH)2CCby renal cortical cells .	AR c/p appear at 3-12m of age &similar to nutritional rickets + low S. 1,25 (OH)2.	High dose of vitamin D or 1,25OH-CC
Renal tubular rickets	Defective tubular reabsorption of p .	Manifest in 2nd year with bow leg & peridental abssesses. muscle tone normal , +ve FH.	Large doses of vitamin D 50,000IU
X-linked vitamin D resistant rickets (familial hypophosphatemic rickets).	Isolated defective reabsorption of P .	Similar to nutritional rickets	
Fanconi syndrome	Defective reabsorption of glucose , AA, k, water & bicarbonate in addition to P.	AGE 1-2Y with polyuria and potein uria ,kypokalemia and acidosis .	See renal system
Infantile renal tubular acidosis (lightwood syndrome).	Inadequacy of forming acid urine due to defective reabsorption of bicarbonate & P.	Hyperchloremic & metabolic acidosis	See renal system

Oncology by Dr.Asma Hadida

2015_2016

ONCOLOGY

- Leukaemia (commonest tumor in pediatrics age group).
- CNS tumours.
- Lymphoma.
- Neuroblastoma (commonest tumor in neonate).
- Wilm's tumour
- Soft tissue sarcoma.
- Bone tumours.
- Retinoblastoma.

Disease	Tumor associated.
Down syndrome	Acute leukemia.
Neurofibromatosis	Brain tumor. Pheochromocytoma. Leukemia.
Klinfelter syndrome	Germ cell tumor (Dysgermenoma) Breast tumor.
Tuberosclerosis	Rhabdomyoma (benign). Rhabdomyosarcoma.
WAGR syndrome	Wilm's tumor.
Backwith widman	Wilm's tumor.
Xeroderma pigmentosa	Basal cell carcinoma. Sequamous cell carcinoma.
Ataxia telagectasia	Leukemia and lymphoma.

Leukemia

• Group of malignant disease of hemobiotic cells in bone marrow giving rise to un control proliferation of the cell with subsequent bone marrow failure.

Classification :

1. Acute leukemia 97% :

Two types:

- Acute lymphoblastic leukemia 75% .
- Acute non lymphoblastic leukemia.
  (Mayeloblastic leukemia).

2. Chronic leukemia 3-5%.

Etiology :

Unknown but multifactorial.

1. Chromosomal defect: e.g. fanconi anemia, ataxia telangectasia.
2. Immune deficiency syndrome: e.g. Wiskott Aldrich syndrome.
3. Chemotherapy.
4. Down sundrome.
5. Immunized radiation or Benzen exposure.

Acute lymphoblastic leukemia (ALL)

Most common malignancy in childhood, peak age group 2nd - 4th year of life.

Morphological : 3 types

1. Small cell and small cytoplasm → Good prognosis.
2. Large cell and large cytoplasm and prominent neucli.
3. Large cell and vaculated cytoplasm → Bad prognosis.

Immunophenotype : 2 types according to blast cell membrane and cyto plasmic marker.

1. B cell >80% :

- Immature B cell :
  a. Early pre B cell called common acute lymphoblastic leukemia.
  b. Pre B cell.
- Mature B cell : worst prognosis.

2. T cell

C/P :

- Pateint presented with non specific viral illness, initially they may complain of anorexia, irritability, lethargic, weight loss, and fever.
- Manifestation of bone marrow failure:
  - Palor, thrombocytopenia purpural petichea, bruising.
- Manifestation of organ infiltration:
  - Generalized lymphadenopathy 25-50%.
  - Hepatosplenomegally 70%.
  - Medistinal mass "common with T cell type".
  - Manifestation of ↑ ICP.
  - Testis usually unilateral painless swelling.
  - Bone swelling and pain 30-40%.
  - Hematuria or renal failure.

Investigation :

- ➲ CBC: ↓ Hb, ↓ Plt count, WBC normal/low or high.
- ➲ Blood film:
  - Blast cell → leukemic cell which is never found in peripheral blood in normal.
  - In acute lymphoblastic leukemia, the blast cells are > 2% but absent blast cell does not exclude diagnosis.
- ➲ Bone marrow: Replaced by more than 25% blast cells (may reach up to 80-100%).
- ➲ To detect infiltration: Chest x-ray, CT scan, abd. U/S, lumbar puncture, LFT, RFT,

Treatment :

1. Induction of remission:

Aim: To eradicate malignant cell in bone marrow.

Duration of therapy: 4-6 weeks.

★ Mainly chemotherapy:

- Vincristine : weekly.
- Prednisolone : daily.
- L.Aspargenase : 9 doses.
- Intrathecal → Methotrixate, cytosine and arabinoside.

★ Criteria of remission:

- Physical improvement.
- Investigation : CBC normal, no blood cell in peripheral bone marrow blast cell < 5%.

2. CNS prophylactic :

Aim: Prevent later CNS relapse.

Duration of therapy: 4 weeks.

Drugs: - Intrathecal methotrixate.
- Craniospinal irradiation.

3. Maintaince phase:

Duration of therapy: 2-3 years.

Drugs:

- 6 mercaptopurine daily.
- Methotrixate weekly.
- Vincristine and steroid intermittent dose.

(If failed medical treatment ⇨ Bone marrow transplantation.)

Blast cell must be subject to :

- Immunophenotype assay:
  In most of cases have antigen on their surface of blast cell called common acute lymphoblastic leukemia.
- Cytogenic study.

Complication :

1. Relapsing : common relapse CNS and testis.
2. Tumour lysis syndrome :
   - Tumour cell very sensitive to chemotherapy → rapidly lysis leading to hyperurecemia, hyperkalemia, and hyperphosphatemia → ARF and hypocalcemia (manifest tetany).
3. Bone marrow suppression.
4. Peripheral neuritis "S/E of Vincristine".
5. Renal toxicity "S/E of Methotrixate".
6. Cardiomegally "S/E of Adriamycin".

Prevention of tumour lysis syndrome:

- Good hydration.
- Uric acid binding agent e.g. allopuranol.
- Treat hyperkalemia:
  - Ca glucanate.
  - Na bicarbonate.
  - Salbutamol.
  - Ca and resonium.
  - Insline and dextrose.
  - Dialysis.

	Good prognosis	Bad prognosis
Initial WBC count	< 50.000	> 50.000
Age	> 1 year , < 10 years	< 1 year , < 10 years
Sex	Female	Male
Race	White	Black
FAB classification	L1	L3
Immune type	Early pre B cell "common ALL"	Mature B cell type
Cytogenatic	Polyploidy	Hyploidy
CNS infiltration	-ve	+ve
Response to therapy	Rapid response	Slow response

Neuroblastoma

- Malignant tumor arise from neural crest tissue of sympathetic ganglion and adrenal medulla.
- Commonest tumor in neonate and infancy.
- Diagnosis :
  90% before 5 years.
  50% before 2 years.
- Common at time of diagnosis → already metastasis occurs. (Bone is common site).

Metastasis:
- Bone "common site".
- Bone marrow.
- Liver.
- Skin.

C/P :

- No specific manifestation.
- Irritable "due to high catecholamine"
- Localized tumor "abdominal mass" and usually cross midline "characteristic" which compared from wilm's tumor.
- Hypertension, diarrhea, sweating, palpitation, and tachycardia.
- Crying due to abdominal pain or pressure symptom.
- Bone pain due to metastasis.
- Proptosis , Horner syndrome.

Investigation :

➲ Abdominal U/S:
- Mass in adrenal medulla pushing kidney downward (usually kidney normal but sometimes hematuria).
- In > 50% detect calcification in tumour.

➲ CT scan abdomen.

➲ 24 hour urine collection for :
- Venyl manalic acid (VMA).
- Homovanyllic acid (HVA).
  If both were high → indicate tumor.

➲ Biopsy from tumor cell.

➲ Bone CT scan : to detect metastasis.

Stages :

Stage I : Tumor localized in organ of origin.

Stage II : Extended tumor beyond the structure with or without lymph node enlargement, I.E does not cross midline.

Stage III : Extended tumor and cross midline.

Stage IV : Metastasis.

Stage V : Localized 1ry tumor with discemination limited to liver, bone marrow in infant less than 1 year.
Good prognosis: relive spontaneous without treatment.

Treatment :

- Stage I and II :
  Only surgery "Good prognosis >80%".
- Stage III and IV :
  Surgery and chemotherapy with radiotherapy.

Prognosis :

- Stage I & II more than 90% cure with surgery alone.
- Stage III & IV 50% cure with surgery and chemotherapy with radiotherapy.
- Stage V more than 80% chance of long term survive with supportive treatment alone ; start treatment if patient symptomatic.

Wilm's tumour

Renal tumour occur in pediatrics age group.

- Wilms tumor originates from embryonal renal tissue and is the commonest renal tumor of childhood.
- Over80% of patients present before 5 years of age and it is very rarely seen after 10 years of age.

• Commonly associated with other congenital malformation:-
  - ✓ Anaridia.
  - ✓ Hemihyperatrophy.
  - ✓ Genitourinary malformation, e.g.
    - Cryptoarchidism.
    - Hypospadius.
    - Gonadal dysgenesis.
    - Horse show kidney.
  - ✓ Backwith widman syndrome.
  - ✓ WAGR syndrome.

Investigation :

- ➲ Abdominal U/S :
  - Tumor inside kidney , damage and destruction of all parenchyma.
  - < 10% calcification.
- ➲ CT scan.
- ➲ Biopsy.

Treatment :

- Surgery.
- Chemotherapy.
- Radiotherapy.

C/P :

Sometimes non specific manifestation.

1. Abdominal swelling and pain.
2. Hematuria.
3. Hypertension.
4. Genitourinary malformation.

Commonest site of metastasis is pulmonary , liver 15%.

CNS tumours

- Glioma : raised from supportive tissue "astrocyte".
- Ependyomoma : raised from ependymal cell that line in ventricle.
- Germ cell tumor.
- Rhabdoid tumor : unknown cell type.

Classification :

B) Histological classification :
Common are for each tumour include:

1. Glioma:
   - Cerebellum and optic nerve are mostly benign tumour.
   - Cerebrum and brain stem are mostly malignant.
2. Ependyomoma:
   - Forth ventricle, less common in spinal cord.
3. Rhabdoid tumour: Tumour in posterior fossa.
4. Craniopharangioma: Choroidal plexus.

A) Anatomical classification :

1. Supratentorial tumour:
   Organs: Cerebrum , basal ganglion, thalamus and hypothalamus.
   Tumours: Glyoma, epidymoma, germ cell tumour, craniopharangioma.

Supratentorial symptoms:
a. Manifestation of ↑ ICO.
b. Focal deficit manifestation e.g. memory loss, weakness, visual disturbance.

2. Infratentorial tumour:
   Organs: Cerebellum , brain stem.
   Tumours: Astrocytoma, medulloblastoma, Ependyoma.

Infratentorial symptoms:
a. Obstruction of CSF pathway.
b. Headache and vomiting.
c. Cranial nerve palsy "If brain stem involved".
d. Ataxia "If cerebellum involved"

N.B

- Approximaly 50% of brain tumor in children are infratentorial (2/3 in cerebellum).
- In majority of brain tumor ~~u in a ault~~

Craniopharyngioma

- It is rare tumor but important to excluded.
- It is one of the most important of supratentorial tumor.
- It's 2ry to expanded cyst reminant of Rathke pouch (dorsal protrusion of embryonic cyst).
- Presented with :
    1. Headache and visual field defect (due to pressure or injury to optic chaiasma).
    2. Short stature (due to pituitary hypothalamic dysfunction).
    3. Diabetes inspidus.
    4. Precocious puperty.
- Diagnosed by :
    - Skull x-ray : Calcification in over 99% of cases.
    - CT scan and MRI.

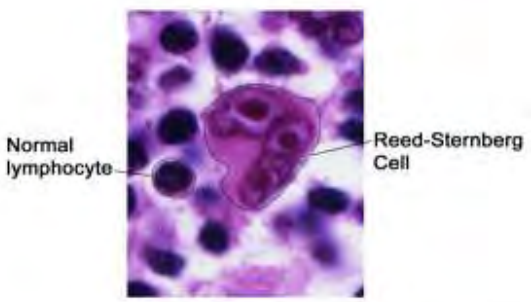
Lymphoma

Abnormal proliferation of lymphoid tissue outside bone marrow .
Male predominant in 1st decade 4:1 then changes > 12 year of age.

Types :

1. Hodgkin lymphoma.
2. Non Hodgkin lymphoma "Common".

Both are same presentation but different histologically ➡ Reed-Sternberg cell which is characteristic of Hodgkin lymphoma.



Etiology :

- Unknown.
- Certain associated with infection (EBV) in over 90% of Burkitt lymphoma.
- Human T cell leukemia virus type I and Retro virus associated with T cell lymphoma.
- Some cases of gastric lymphoma are known trigger by helicobacter pylori.

Hodgkin lymphoma

C/P :

1. Painless cervical lymphadenopathy (firm, not fixed to surrounding structure).
2. Constitutional symptoms : (⅓ cases)

B symptoms:
- Fever > 38°C.
- Weight loss of 10% over period of 6 month.
- Night sweating.

3. Asymptomatic mediastinal disease in 50% of cases (anterior mediastinal mass).
4. Pruritis is common.
5. Splenomegaly or hepatomegaly or both in advanced disease.

Investigation :

- CBC : ↓Hb , ↓ Plt count or usually normal.
- ESR , CRP : High.
- Excisional biopsy:

  Indication:
  - Lack of infection.
  - Nodules > 2cm in diameter.
  - Surpraclavicular adenopathy or abnormal chest x-ray .
  - Lymphadenopathy increse in size after 2 weeks.
  - Failed to resolve within 4-8 weeks.
- Chest x-ray.
- Abdominal CT scan or MRI.
- Bone marrow aspiration.

★ Diagnosis should be suspected in patient who has persistant unexplained lymphadenopathy with no evidence of inflammation or infection.

Stages : (Ann Arbor staging)

Stage I : Single lymph node region or single extralymphoid organ.

Stage II : 2 or more lymph node region at the same side of diaphragm.

Stage III: lymph node on both sides of diaphragm.

Stage IV: Discemminated :
- IV A: No B symptoms.
- IV B: with B symptoms.

Histological classification:

1) Lymphocyte predominant 10-20%.
2) Nodular sclerosing 40-60% "common type"
3) Mixed cellularity 20-40%.
   (Lymphocyte, plasma cells, Reed-Sternberg cells).
4) Lymphocyte deplation "bad prognostic type".

Treatment :

- Stage I and II : Chemotherapy mainly but can be treated with Radiotherapy alone.
- Stage III and IV : Mainly chemotherapy.

➲ Chemotherapy :

1. ABVD regime:
- **A**driamycin.
- **B**leomycin.
- **V**inblastin.
- **D**ecarbazine.

2. MOPP regime:
- **M**eclorethamine.
- **O**ncovein vincristin.
- **P**rednisolone.
- **P**rocarbazine.

Non Hodgkin lymphoma

- Heterogeneous group of disease characterized by neoplastic proliferation of immature lymphoid cells unlike all accumulation dry outside B.M.
- Most common type and common in pediatrics age group.
- Symptoms **mainly extra nodular** (compression symptoms).
- 10-15% nodular.
- Diagnosis: as Hodgkin lymphoma except in biopsy there is no Reed-Sternberg cell.
- Stages and treatment as Hodgkin lymphoma.

Classification :

1) Histological :
   - Lymphoblastic type (usually T cell).
   - Large cell type.
   - Small non cleared cell type.
2) Immunological :
   - T cell "common".
   - B cell.
   - Non T, non B.

C/P :

1. Abdominal lymphoma: 35%
   - Abdominal mass with abdominal pain and vomiting.
   - Ascitis , intussusceptions , hepatosplenomegaly, obstructive jaundice.
2. Anterior mediastinal mass: 25%
   - Mediastinal syndrome (cough, dyspnea).
   - Pleural effusion, pericardial effusion.
3. Others:
   - Lymphadenopathy in head and neck.
   - Bone marrow infiltration.

Risk factors

- Immunodeficiency.
- EBV play a role in pathogenesis of Burkitt lymphoma (derived from mature B cell).
- HIV (with immunoblastic lymphoma).
- Helicobacter pylori , predispose to the developed of NHL (gastric lymphoma).

Treatment :

1. Extensive surgical bedulking.
   - Only for small, easily resected tumour.
   - Systemic chemotherapy for all patient.

   CHOP : **C**yclophosphamide.
   Hydroxy adriamycin.
   Oncevin (Vincristine).
   Prednisolone.
2. Radiotherapy : for advanced N.H.L.
3. Cranial irradiation and intra thecal therapy in advanced (in addition to chemotherapy.

Diagnosis :

- Biopsy "Lymph node".
- CT scan.
- Abdominal U/S.
- LDH , ↑ ESR.
- Leukocytosis.

Prognosis :

G☺☺d Prognosis
90% cured

Acute myloid leukemia

- It is divided into 7 subtypes equal on morphology and immunology ($M_1 \rightarrow M_7$).
- More in older children with equally sex incidence.
- **C/P** : as Acute lymphoblastic leukemia "ALL".
- **Diagnosis** : Bone marrow show myeloblast.
- **Treatment** : is with a more intensive but shorter chemotherapy than use for ALL.

  $M_4 + M_5 \rightarrow$ CNS involvement.

  $M_6 + M_7 \rightarrow$ Worst prognosis.

Anterior mediastinal mass

5 Tumour

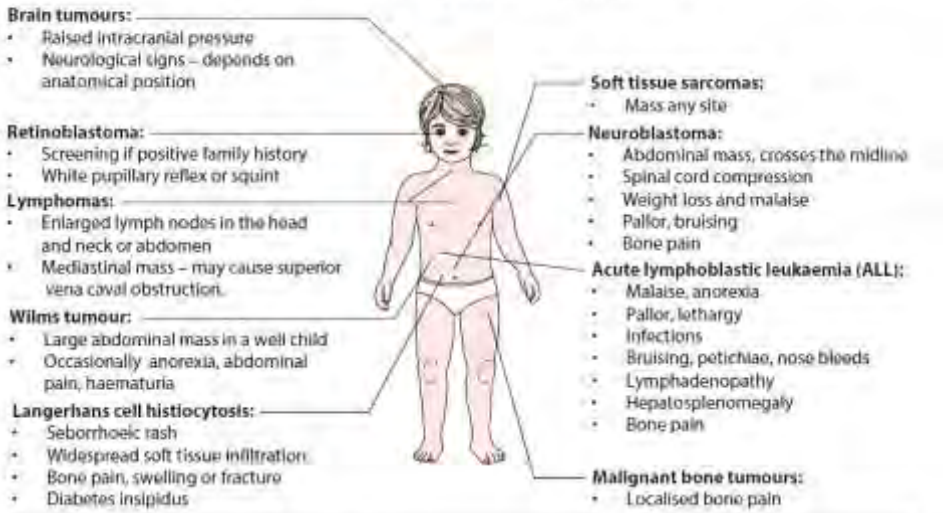
1. Teratoma (Germ cell tumour).
2. Thymoma.
3. Thyroid tumour.
4. T cell leukemia.
5. Lymphoma.

Most common tumour cause SVC syndrome is non Hodgkin lymphoma and less frequently Hodgkin lymphom, neuroblastoma and sarcoma.

Superior mediastinal syndrome:

- It's result from the presence of anterior mediastinal mass that compress the trachea and SVC.
- Patient have a cough, dyspnea particularly when supine.

Presentation of malignant disease in children



Pre-school (<5 years old)	School-aged	Adolescence
Acute lymphoblastic leukaemia (ALL) – peak incidence Non-Hodgkin lymphoma	Acute lymphoblastic leukaemia (ALL)	Acute lymphoblastic leukaemia (ALL) Hodgkin lymphoma
Neuroblastoma	Brain tumours	Malignant bone tumours
Wilm tumour		Soft tissue sarcomas
Retinoblastoma		

RESPIRATORY SYSTEM

CLINIC PART 1

BY DR.ASMA HADIDA

2014-2015

TO BE CONTINUE.....

Respiratory Examination

Respiratory examination:

1) **General examination** : Related to respiratory system.
2) **Local examination** : Chest examination.

What are respiratory sound heard by ear ?

Preparation of examination:

Stand on the right side of the bed.
Introduce yourself.
Permission taken.(inform relative or pt that examination will be done)
Good position for patient.(ideally child should be at 45c but small child can be examined in supine position)
Good exposure.(undress the child to top half (up to waist)

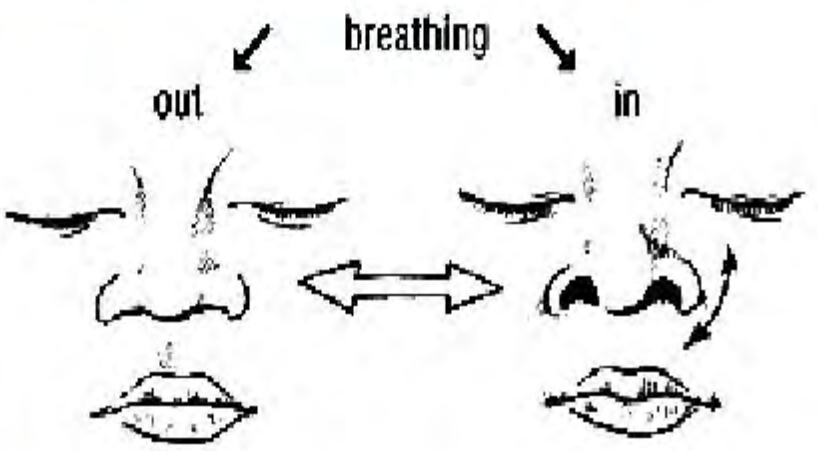
Before end the examination don't forget :

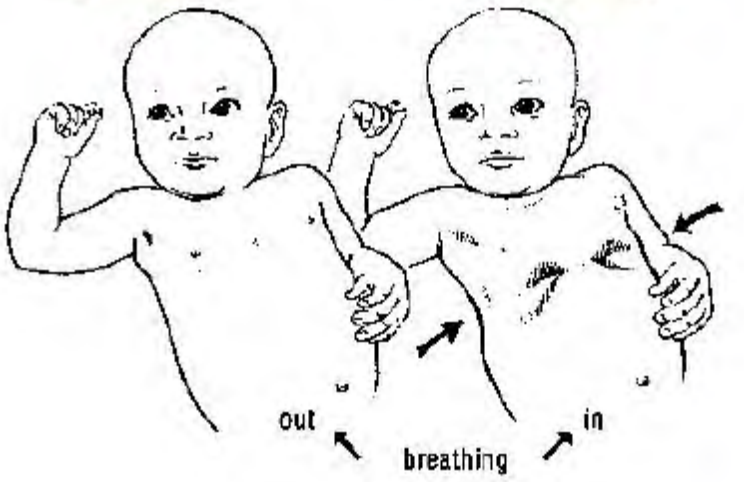
1. ENT examination.
2. General examination related to respiratory system.
   - **General condition:**
     Looks for :
     - Well or sick.
     - Distressed or no (Signs RD).
     - Dysmorphic features.
     - **Surrounding :** O_2 mask , pulse oximetery , IV fluid .
     - Any heard sound ⇨ e.g. : Wheeze , grunting , strider , and snoring
     - Built .
   - **Vital signs** : RR - Pulse - Temp - BP.
   - **Anthropometric measurement** : Length or Height - Weight - OFC.
   - **Abdominal examination** : False hepatomegly in hyper inflated chest.

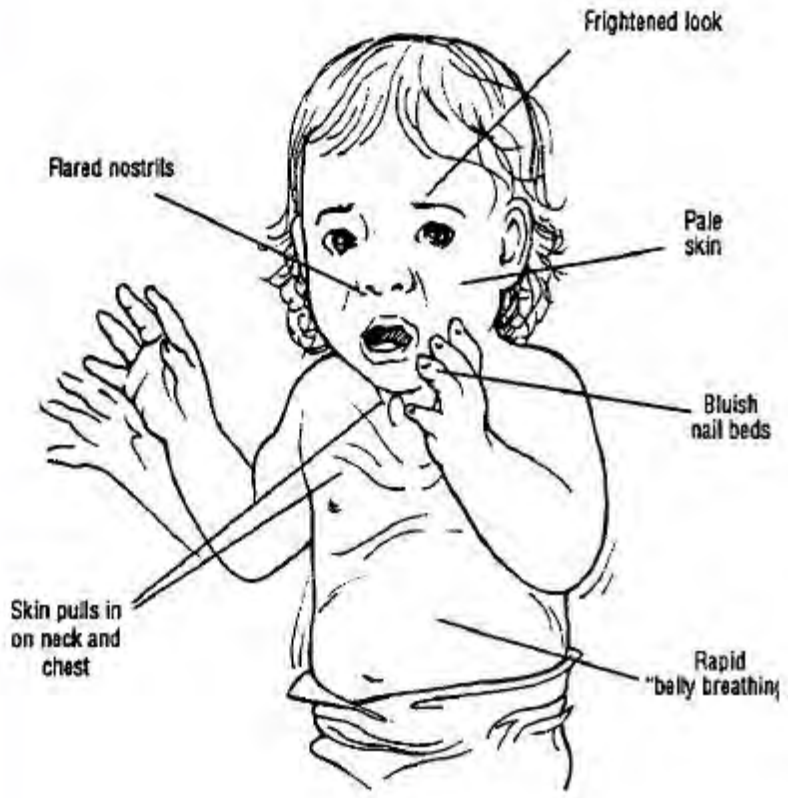
Why we have to look for dysmorphic features in respiratory examination ?

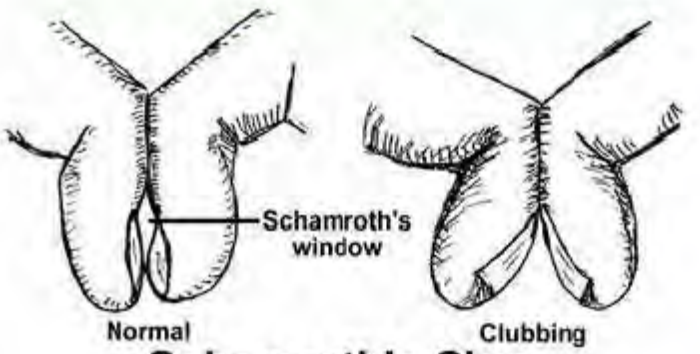
Inspection	What to be test ?	What is the procedures ?	What is the interpretation ?
	Respiratory rate	Count the chest movement for 1 minute	Respiratory rate: Depend on age: **Tachypnea:** Neonate - 2 mo → >60 cycle/min. 2 mo – 1 year → >50 cycle/min. 1 year – 2 year → >40 cycle/min. 2 year – 3 year → >30 cycle/min. 3 year – 4year → >20 cycle/min
	Inspiratory effort (RD)	Look for flaring of alanasi. Sub costal , intercostals	Insiratory obstruction Non compliant lungs Nasal obstruction
	Audible respiratory sounds .it gives clues to anatomy and pathology of the disease	Snuffles Strider : noise from trachea during inspiration Wheeze : musical sound during expiration Grunt : expiratory noise from مشقغیء	♦ insiratory obstruction ♦ upper air way obstruction ♦ expiratory obstruction ♦ breathing out against closed glottis – alveolar disease

	Growth parameter	↓ Height . weight HC	♦ if below normal – FTT (CF- chronic supprative lung disease)
	Nutritional status	↓ Loss of muscle bulk and sub cutaneous fat	♦ malnutrition
	Dysmorphic features	↓ Hypoplasia of the mandible , micrognatia , cleft palate ↓ Upward slant of palpebral fissure	♦ pierre robin syndrome ♦ down syndrome
	Intervention	↓ Oxygen mask ↓ Nasogastric tube	♦ hypoxia ♦ nability to swallow
	Digital clubbing	↓ Increase curvature in both dimensions loss of angle between nail and cuticle	♦ chronic supprative lung Disease ♦ cyanotic heart disease ♦ infective endocarditis
	Nial , tongue , conjunctiva	↓ Cyanosis	Alveolar disease Airway obstruction Decrease alveolar ventilation

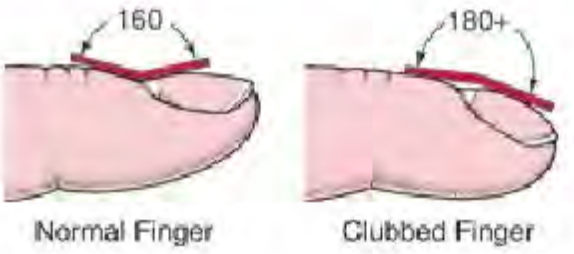


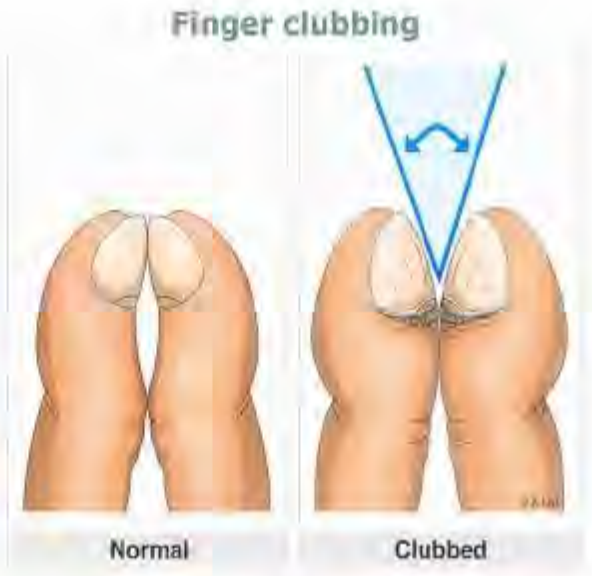






Schamroth's Sign





Looks for chest movement:
Breathing mainly abdominal up to 7 years then thoracoabdominal

- Recession or retraction:
  - Suprasternal recession.
  - Sub costal recession.
  - Intercostals recession.
  - Supraclavicular recession.

Chest examination

I) Inspection:

Skin : for any scar, dilated vein, cattery mark.
Symmetry and Shape of the chest:
(In newborn baby the chest is round in shape, then become epileptic in shape "Transverse > AP" and ↑ gradually in diameter with ↑ in age.

What is the signs of respiratory distress ?

What is the signs of chronic asthma ?

Abnormal shape:

i. **Barral chest(over expansion of chest):** ↑AP diameter ⇨ Chronic bronchial asthma, cystic fibrosis
ii. **Pigeon chest "Pectus carinatum"= prominent sternum :** The lower sternum protruded forward ⇨ Sever rickets, structural deformity in bronchectasis and chronic asthma .
iii. **Funnel chest "Pectus excavatum"= sternal depression** : The lower sternum is indented inward ⇨ Congenital anomaly, structural deformity in airway obstruction
iv. **Harrison salcus**: Horizontal groove at the costal insertion of diaphragm ⇨ Ricktes and chronic asthma.

Asymmetry:

i. **Bulge** : Plural effusion , Pneumothorax.
ii. **Retraction** : Collapse , Fibrosis.

Causes of limited chest movement:

Unilateral:

- Pneumonia.
- Effusion.
- Collapse.
- Fibrosis.
- Pneumothorax.
- F.B aspiration.

Bilateral:

- Sever bronchial asthma.
- Bronchopneumonia.
- Acute bronchiolitis.

- ***Syndromes associated with deformity of the chest*** :
- **Marfan syndrome** : pectus excavatum
- **Noonans syndrome** : pectus excavatum, shield like chest
- **Edward syndrome** : short sternum , small nipple
- **Pataus syndrome** : thin post rid , missing ribs

II) Palpation : (not suitable for infants only for older children)

Position of mediastanium:

- **Trachea** : Indicate position of upper mediastanium.

(Normally in the center or slightly shift to right).

- **Apex beat**: Indicate position of lower mediastanium.

Chest expansion:

From Front: Supra mammary.
Mammary line.
Costal cartilage.

From back: Above the scapula.
Below the scapula.

Tactile vocal fremitus (TVF):

↳ « after age of 7 years only »

Palpate transmitted sounds through chest wall by use the palm of hand symmetrically bilaterally on the back at 3 zones while child say (44).

Increased TVF in ⇨ consolidation.

Decreased TVF in ⇨ Pleural effusion.
⇨ Lung collapse.
⇨ Pneumothorax.
⇨ Asthma.

Causes of shifted Trachea and apex beat:

↘ Shifted to the same side ⇨ Collapse.
↘ Shifted to the opposite side ⇨ Pneumothorax or Pleural effusion.

III) <u>Percussion: inform the child before percussion that chest will tapped by fingers</u>

Anteriorlly:

Direct on the clavicle ⇨ for apical segment lesion.

On 2nd and 4th intercostals space mid clavicular line.

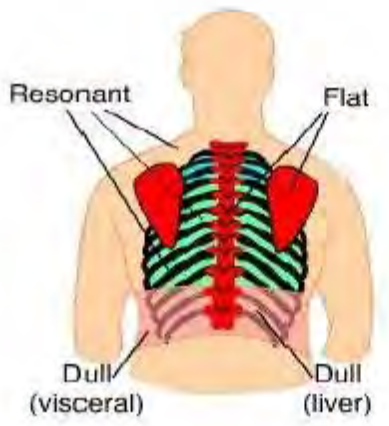
On the axilla.

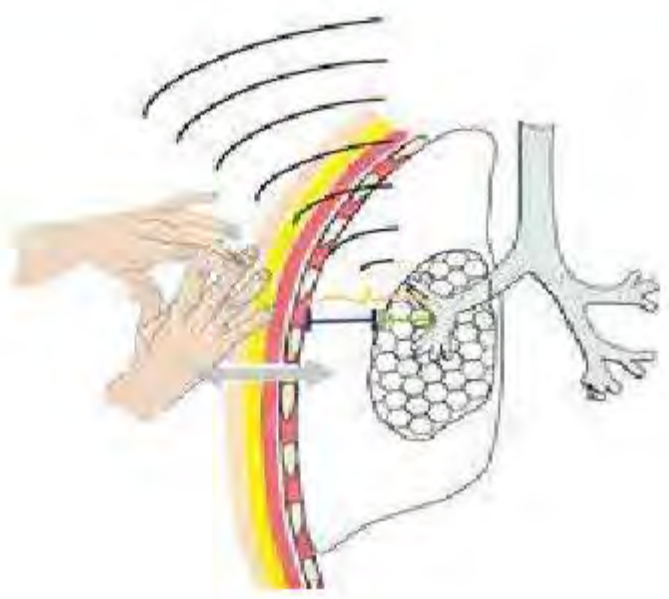
Posteriorlly:

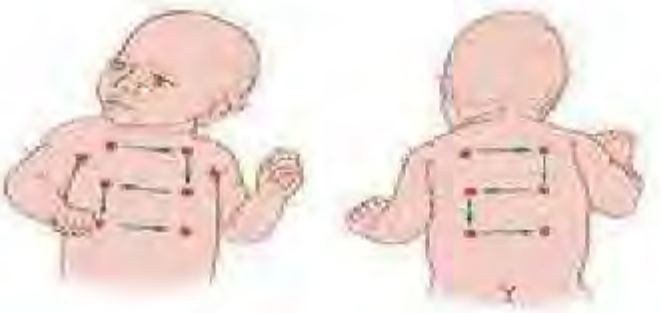
On the intercostals spaces bilaterally and symmetrically.

Result either:

- Resonance →Normal lungs.
- Hyper resonance → Hyperinflated chest as asthma, pneumothorax , emphysema
- Dullness → Consolidation or collapse.
- Stony dull → Pleural effusion.







What is the best sign of pt improving in respiratory diseases ?

IV) Auscultation:

At the same site of percussion you can auscultate and your comment should be about:

1. **Air entry:**
   Normal "good", or diminished "decreased", or absent "unilateral or bilateral".
2. **Type of breathing:**
   Vesicular breathing : (Normal breathing)
   Inspiration > Expiration without gap or pause
   Harsh vesicular breathing :
   Vesicular breathing with prolonged expiration (in bronchial asthma).
   Bronchial breathing:
   Inspiration = Expiration with gap (as in sever consolidation and upper level of pleural effusion.
3. **Added sound:**
   a) **Ronchi :**
   Indicate partial lower air way obstruction &narrowing.
   Heard mainly during expiration but if sever can be heard inspiratory and expiratory
   b) **Cripitation :**
   Fine : Heard at end of inspiration due to reopenition of distal alveoli.
   D.D: - Pulmonary edema
   - Early pneumonia.
   Coarse :Heard at inspiration due to air bubbling through fluid in large airway.
   D.D: - Bronchiectasis
   - Pneumonia.
   c) **Transmitted sound: conducted sound**
   Indicate nasal block.
   Doing by compare auscultation between chest and nose.
4. **Vocal resonance**

	Consolidation	Collapse	Pleural effusion	Pneumothorax
Chest expansion	Reduced in area of consolidation	Reduced in areas of collapse	Reduced	Reduced
Intercostals spaces	Normally spaced	Narrowed	Widened	Normal or widened
Percussion note	dull	dull	Stony dull	Hyperresonance
Breath sound	Bronchial	Absent	Diminished	Diminished
Added sound	Crepitation	None	Pleural rub	None
Vocal resonance	Increased	Decreased	Absent	Decreased
Anatomic location	Lobar upper or lower lobe	Segmental	Pleural distribution	Pleural distribution

Fine cripitation	Coarse crepitation
- Late inspiratory	- **Early inspiratory**
- Not relieved by coughing	- **Relieved by coughing**
- Due to sudden opening of small airway (loss of stability of peripheral airway , which collapse on expiration with hug inspiratory pressure)	- **Due to bubbling (passage) of air through secretion in major bronchi**
- Examples : ↓ Pulmonary edema ↓ HF ↓ Early pneumonia ↓ Interstitial lung disease	- **Examples :** ↓ **Bronchiectaisis** ↓ **Supprative lung disease** ↓ **Late pneumonia in stage of recovery**

By auscultation can you determine the severity of disease ?

Q. What is the causes of recurrent pneumonia ?
Q. What type of mask we use in pediatric ?
Q. what is the importance of nasal discharge ?
Q. What are the types of cough ?
Q. What are the complications of cough ?
Q. If you find nasal canula what does it indicate usually ?
Q. Why we use humidified oxygen ?
Q. How does the position help in the diagnosis ?
Q. How you can differentiate b/w acute bronchiolitis and pneumonia by chest X-ray and complete blood count ?
Q. How you can differentiate b/w palpable liver due to hepatomagaly or liver pushed due to hyperinflated chest ?
Q. Why you determine position of trachea and apex beat?

By otoscope we looking for :
- **Drum :** gray shiny = normal , red /bulging / fluid level →otitis media
- **Discharge indicate** →perforation

Definition of strider , types, causes?

Strider: It is audible, harsh, high pitched Inspiratory sound, due to upper airway obstruction "Larynx & Trachea".

Types: Acute and chronic.

Why usually admitted patient with strider?
To maintain the airway.

Causes

Larnygiotracheobronchitis "croup".
Epiglottis.
Laryngitis.
Foreign body aspiration.
Infectious: Measles , Diphtheria.
Angioneurotic edema.
Hypocalcaemia.
Retropharyngeal abscess.

Laryngiomalacia.
Laryngeal web.
Subglottic stenosis.
Vascular ring.
Vocal cord palsy.
Goiter , hypoglossal cyst.
Pier Robin syndrome :
- Glossoproptosis
- Micrognathia.
- Cleft palate.
- High arched palate.

Grunting:

It is expiratory sound due to forced expiration against partial closured of glottis.

Laryngomalacia:

- Presented with Inspiratory strider beginning t or shortly after birth due to exaggerated collapse of eiglottic structures.
- Diagnosed by history and examination .
- **Treatment**:
  - Self limiting usually **« reassurance of patient » .**
  - In severe cases : laryoepigllottoplasrty or tracheotomy.

	Croup	**Acute epiglottis**
Onset	Gradual	Sudden
Preceding coryza	Yes	No
Cough	Sever/barking	Slight/absent
Ability to drink	Yes	No
Salivation	No	Yes
Appearance	Well	Ill looking
Fever	< 38.5°C	>39°C
Stridor	Harsh	Soft
Voice	Hoarse	Muffled / unable to speak
Micro-organism	Para influenza	H.Influenza type B

Wheeze (sibilant rhonchi):

It is whistling musical sound mainly during expiration due to or produced by passage of airway through narrowed lower airway. (In sever case heard during inspiration and expiration).

D.D oF 1st attack wheeze chest :

- Bronchopneumonia.
- Acute Bronchiolitis.
- Bronchial asthma.
- F.B.A.
- GERD.
- CHF "Acute"
- T.B.

D.D oF Recurrent attack wheezy chest :

- Bronchial asthma .
- GERD.
- Cow milk allergy.
- Bronchiectasis .
- Cystic fibrosis.
- T.B.
- Chronic H.F.
- Immune deficiency.

Cough:

It is a sudden and often repetitively occurring reflex which helps to clear the large breathing passages from secretions, irritants, foreign particles and microbes.

Types:

1. Barky cough ⇨ croup
2. Spasmodic
3. Productive
4. Dry

D.D:

Infections : RTI
Reactive airway disease
Gastro esophageal reflux
Air pollution
Foreign body
Psychogenic cough
Other : Lung tissue disease.

D.D of Clubbing :

Respiratory cause:

Suppurative lung disease:

⇨ Bronchiactasis.
⇨ Empywmia.
⇨ Lung abscess.

Cystic fibrosis.
Interstitial lung fibrosis.
Lung cancer.

Cardiac cause:

Cyanotic CHD "TOF – TGA".
Infective endocarditis.

GIT cause:

Celiac disease.
IBD.
Liver cirrhosis.
Biliary atresia.

Familial idiopatic.

The important cases in respiratory system:

Bronchial asthma.
Acute bronchiolitis.
Pneumonia.
Cystic fibrosis and bronchiactesis.
Foreign body aspiration

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Assessment of the child with acute asthma

Determine the severity of the attack
- Mild
- Moderate
- Severe
- Life-threatening

Too breathless to talk or eat?

Increased work of breathing
- Tachypnoea – severe if >30 breaths/min

Chest recession:
- Moderate – some intercostal recession
- Severe – use of accessory neck muscles
- Life-threatening – poor respiratory effort

Auscultation:
- Wheeze
- Silent chest – poor air entry in life-threatening

Pulse:
- Severe – >120 beats/min

Level of consciousness – altered in life-threatening
Exhaustion

Tongue:
- Cyanosis in life threatening

Peak flow (% predicted):
- Moderate >50%
- Severe <50%
- Life-threatening <33%

0_2 saturation:
- Moderate >92%
- Severe or life-threatening <92%

Is there a trigger for the attack?:
- URTI or other viral illness
- Pneumonia
- Allergen, e.g. animal dander
- Exercise
- Cold air

Assessment of the child with chronic asthma

Clinical features to check

Growth and nutrition

Peak flow/spirometry

Chest for:
Hyperinflation
Harrison's sulcus
Wheeze

Are there other allergic disorders?
- Allergic rhinitis
- Eczema, etc.

If there is:
- Sputum
- Finger clubbing
- Growth failure

– If present, other causes should be sought

Monitor:
- Peak flow diary
- Severity and frequency of symptoms
- Exercise tolerance
- Interference with life, time off school
- Is sleep disturbed?
- Use of preventer and reliever medication – are they appropriate?
- Inhaler technique

Consider triggers:
- Allergic rhinitis needing treatment?
- Allergens - animal dander, etc.
- Stress

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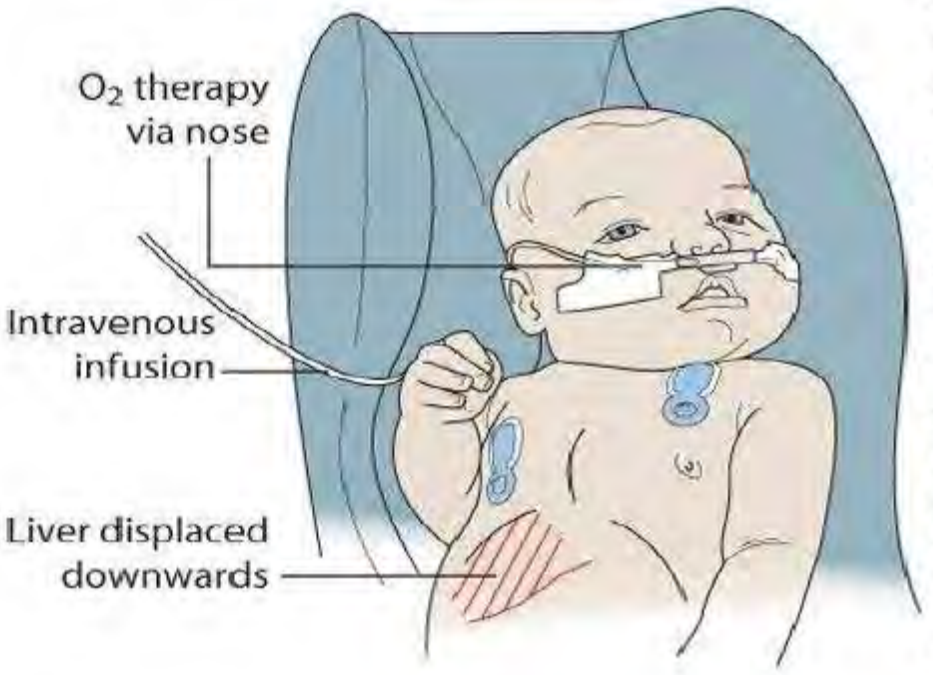
Bronchiolitis



Apnoea in infants <4 months

Sharp, dry cough

Cyanosis or pallor

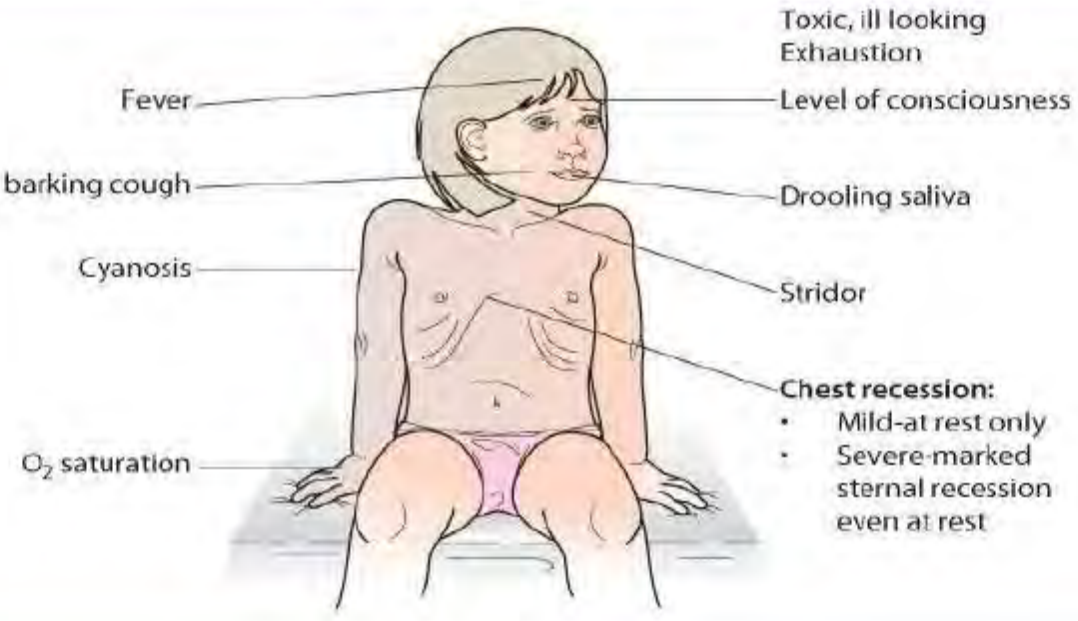
Hyperinflation of the chest:
- sternum prominent
- liver displaced downwards

Subcostal and intercostal recession

Auscultation:
- fine end-inspiratory crackles
- prolonged expiration

The child with stridor

Clinical features to assess



Clinical conditions

Croup
- Mostly viral
- 6 months – 6 years of age
- harsh, loud stridor
- coryza and mild fever, hoarse voice

Epiglottitis:
- Caused by *H. Influenzae* type b, rare since *Hib* immunisation
- Mostly aged 1 – 6 years
- Acute, life threatening illness
- High fever, ill, toxic-looking
- Painful throat, unable to swallow saliva which drools down the chin

Bacterial tracheitis:
- High fever, toxic
- Loud, harsh stridor

Inhaled foreign body
- Choking on peanut or toy in mouth
- Sudden onset of cough or respiratory distress

Laryngomalacia or congenital airway abnormality:
- Recurrent or continuous stridor since birth

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Assessment of the adolescent with cystic fibrosis

Siblings may be affected – autosomal recessive condition

Sweat - salty, may lead to dehydration in hot weather

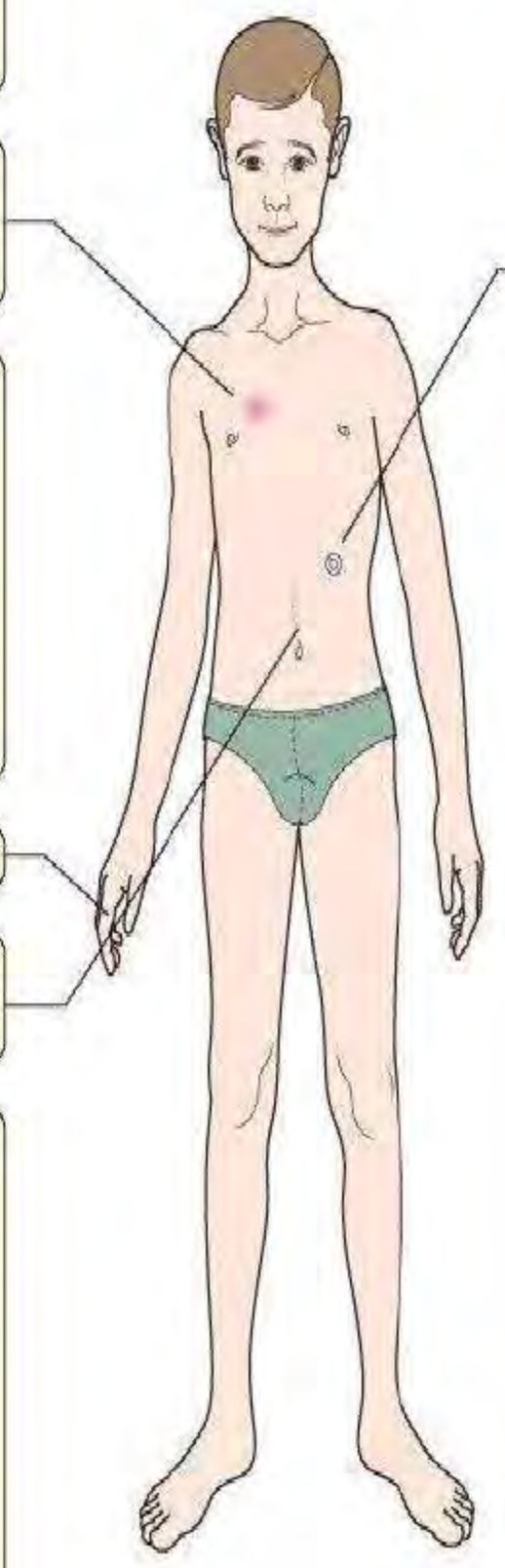
Central venous line, e.g. Portacath – for intravenous antibiotics to aggessively treat infection

Chest – determine if:
Hyperexpansion due to air trapping
Harrison's sulcus
Coarse inspiratory crepitations and/or expiratory wheeze
Chest infection

Clubbing of fingers

Scar from operation for meconium ileus as neonate (10-20%)

Monitor for potential complications:
Nasal polyps, sinusitis, rectal prolapse
Allergic bronchopulmonary aspergillosis (ABPA)
Diabetes mellitus (often insulin-dependent)
Cirrhosis and portal hypertension
Distal intestinal obstruction syndrome (DIOS, meconium ileus equivalent) Pneumothorax or recurrent haemoptysis
Concern about sterility in males



Sputum – evidence of infection – acute or chronic with *Pseudomonas* species or *Burkholderia cepacia*

Growth:
Aim for normal growth

Nutrition:
Gastrostomy for overnight feeding for extra calories
Pancreatic exocrine insufficiency
– taking sufficient pancreatic replacement therapy?
– Taking fat soluble vitamins?

Review of chest problems:
Spirometry – for deterioration
Regular breathing exercises?
Physiotherapy and exercise?
Bronchodilator therapy – is it optimal?
Chest infection - acute or chronic and its treatment?
Nebulised antipseudomonal antibiotics and DNase?
Avoidance of direct contact with other affected patients other than family members?

General overview:
School attendance and performance
Specific problems with managing their disease
Psychological needs

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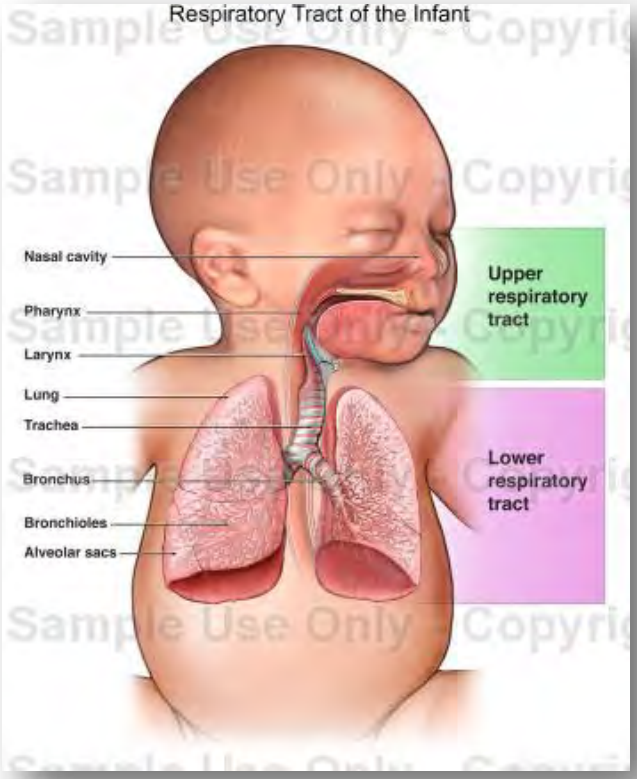


2015_2016

General Principles of Respiratory Disease inde Children.

A. Common pathologic features of respiratory disease:

1. Underlying pathologic process:

a. **Obstruction:** (i.e., **airwaynarrowing**) may be caused by intraluminal secretions, edema or inflammation of the airway wall, hypertrophy or contraction of the bronchial smooth muscle, or extrinsic compression.

b. **Restriction:(i.e., decreased lung volume)** may be caused by decreased lung compliance (stiff lungs), atelectasis or pneumothorax causing lung collapse, neuromuscular disease, or disorders of the chest wall.

2. Pathophysiology:

a. **Hypoxemia:(i.e., deficient oxygenation of blood)** most commonly is caused by ventilation-perfusion abnormalities but also may be due to intracardiac or intrapulmonary shunts, hypoventilation, or diffusion problems.

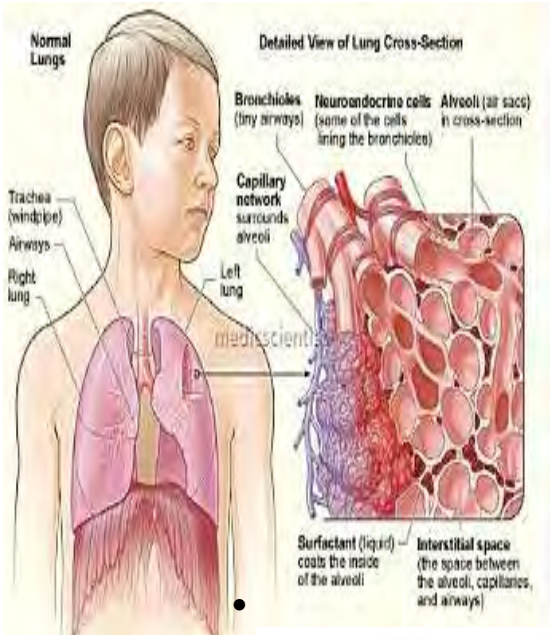
b. **Hypercapnia:(i.e., excess carbon dioxide in blood**) most commonly is caused by primary hypoventilation (due to upper airway obstruction, neuromuscular weakness, or central nervous system depression), but also may be seen with severe lower airway obstruction.

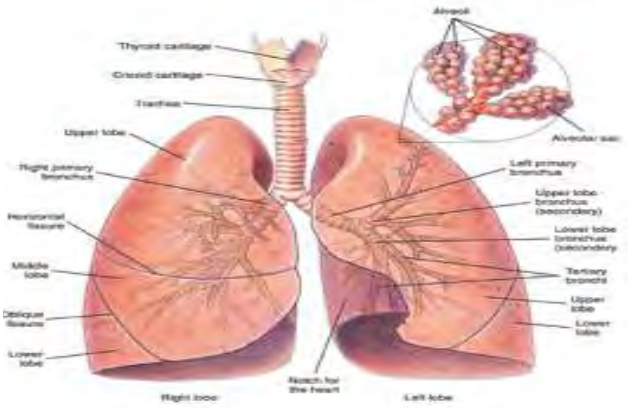
3. Pathogenic factors:

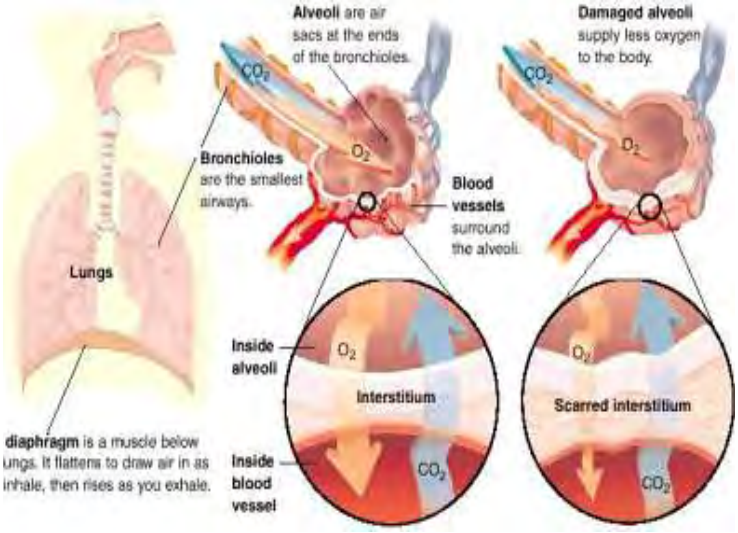
a. The small airways of the child result in high airway resistance and put the child at great risk to develop obstruction.

b. The young child lacks specific immunity (**organism-specific antibody**) and is relatively defenseless against invading microorganisms.

c. In children, most pulmonary disease has a single cause, whereas in adults, pulmonary disease is often multifactorial in etiology.







Anatomical differences :

- Prominent occiput can induce flexion of neck & airway obstruction
- The tongue is proportionately larger & obstructs the airway more easily
- Smaller airway diameter are more likely to be affected by obstruction /edema
- The larynx is more anterior , more caudad , funnel shaped and narrowest at the cricoid cartilage , making it more prone to sub glottis stenosis
- The epiglottis is larger and floppier & can lead to significant obstruction if infected

Phy -siological differences :

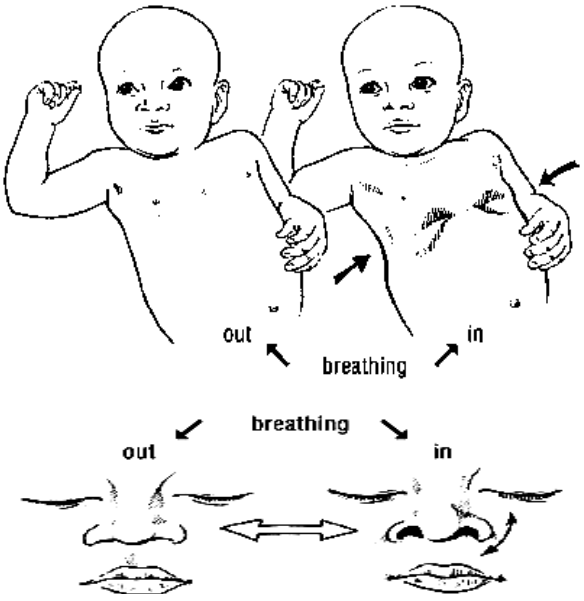
a. Infants are obligate nose breathers . blockage of nares (sputum – chonal atresia) can causes RD
b. Cartilage & chest wall muscles are less developed less compliant , so infants are unable to increase tidal volume as effectively as adult .

Toha_Tohty

B. Evaluation of Respiratory Disease:

History: It focuses on the following questions:

- Is the disorder acute, chronic, or recurrent?
- Is the disorder immediately or eventually life threatening?
- What are the symptoms?
- What factors affect the severity of symptoms?
- Is there a family history of pulmonary disease?
- Have any treatments been given?



Physical examination:

- **Respiratory rate**: Best indicator of respiratory function in young infant.
- **Effort of breathing**: It is a guide to respiratory dysfunction.

Grunting: Heard when vocal cords are adducted during expiration in an effort to maintain lung volume. It is expiratory sound due to forced expiration against partial closured of glottis.

  - **Chest retractions:(suprasternal, supraclavicular, intercostals, subcostal),** occur when airway resistance is increased.
  - **Nasal flaring:** It is a sign of increased airway resistance.
- **Breath sounds:**
  - **Crackles:** They are closed by the opening of small airways that closed on the previous breath **(inspiratory**).
  - **Wheezing**: It is produced by partial airway obstruction. It usually is expiratory in origin but can be inspiratory when the airway obstruction is fixed and rigid, as in airway edema.
  - **Stridor**: is a harsh inspiratory sound produced by obstruction of extrathoracic airways usually at the laryngeal level.
- **Anatomic changes:**
  - A change in tracheal position.
  - A change in thoracic configuration.
  - Clubbing of the fingers and toes.

Normal value of RR according to age:

- **Neonate : 40-60 c/min.**
- **Infant : 40-50 " "**
- **Toddler : 20-30 " "**
- **5 – 12 ys : 20-25 " "**
- **12 – 15 ": 15-20 " "**
- **> 15 years: 12-16 " "**

Laboratory studies:

- **Imaging procedures:**
  - Chest radiograph.
  - Ultrasonography.
  - Barium swallow.
  - Computed tomography (CT) scanning.

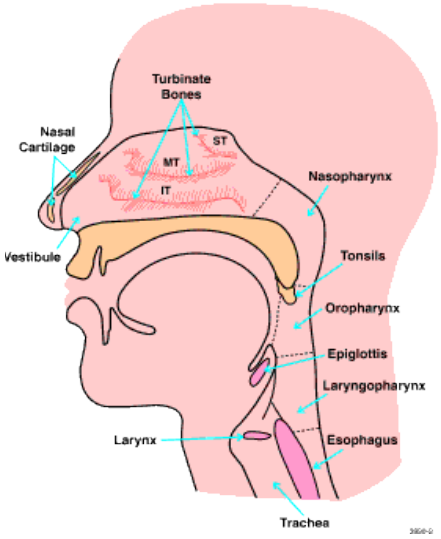
- **Pulmonary function tests:**
  - Spirometry and flow-volume curves.
  - ***Obstructive lung disease:*** Reductions in the forced expiratory volume in 1 second (FEV_1) .
  - ***Restrictive lung disease:*** (FVC) is reduced with a proportional reduction in FEV_1.
- **Analysis of gas exchange:**
  - ***Arterial oxygen tension (PO_2):*** It is a sensitive indicator of overall pulmonary function. The arterial **(PO_2)** and arterial carbon dioxide tension **(PCO_2)** provides information about the adequacy of alveolar gas exchange.
  - Capillary pH and **(PCO_2).**
  - **Pulse oximetry(amount of oxygen bound to HB)**
- **Tests for specific situations**:
  - **In children suspected of having asthma:** Pulmonary function tests before and after bronchodilator inhalation and allergy skin tests to specific environmental allergens are helpful **(skin prick test)**.
  - **In children suspected having cystic fibrosis:** Sweat testing for chloride levels is diagnosed.
- **Endoscopic procedures.**
  - Laryngoscopy.
  - Flexible bronchoscopy.
  - Rigid bronchoscopy.

Toha_Tohty

Upper Respiratory Tract

- Nasopharyngitis.
- Otitis Media.
- Tonsilits.
- Sinusitis.

Nasopharyngitis



- The commonest infections in children.
- Usually involves sinuses and nasophanyx.
- It is Common Cold"Acute infectious rhinitis ".

Etiology

Usually viral infection.
Viral : Rhino viruses, corona viruses and others.
Bacterial : H. Influenza, staph. Pneumococcus etc.

C/P

Sudden onset of

Fever:
Low grade fever.

Nasal symptoms:
- Sudden onset of sneezing.
- **Nasal discharge** : watery in the beginning then come mucoid, nasal obstruction, irritability may interfere with feeding.

Complications

1. Spread of infection:
- **To upper RT:** OM, sinusitis.
- **To L.R.T:** bronchitis, bronchiolitis &broncho-pneumonia

2. It may be:
- Prodroma of measles or pertussis.
- Precipitating factor for asthmatic attack in asthmatic patient.

Complications are suspected with:
- High fever "OM, sinusitis, or broncho-pneumonia".
- Significant cough "bronchitis".
- Persistent nasal discharge more than 10 days "sinusits"

Treatment: Antipyretic e.g., paracetamol.
Course: Fever subsides within 2 days – nasal discharge over a week.

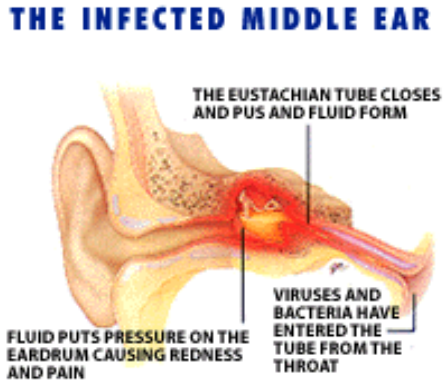
Otitis Media

"inflammation of the middle ear"
Common complications of Nashopharyngitis in infancy and childhood **"6 months-6year".**

Etiology

- Blockage of the Eustachian tube
- **Bacterial:**Streptococcus pneumonia and H.Influenze "the commonest 50% of cases".
- **Other organisms**: Staph, streptococcus, pseudomonas.
- **viral : rsv rinovirus**

C/P



Sudden onset of

Fever:
High, present few days after the onset of Nashopharyngitis.

Ear Manifestation:

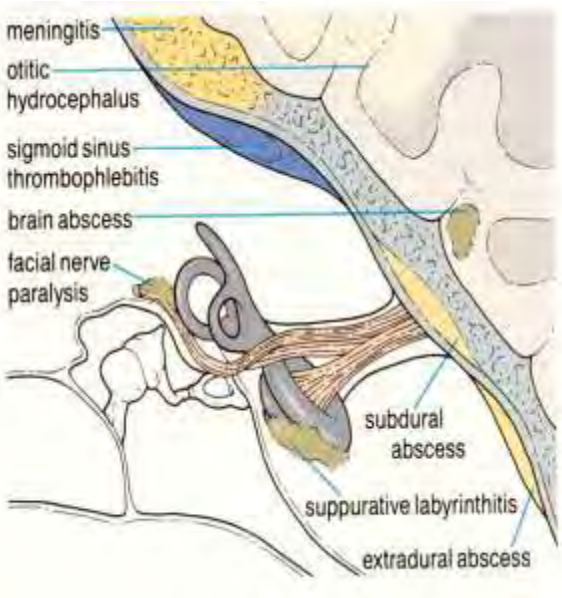
- **Earache:** Irritability and continuous crying.
- **Ear discharge"Otorrhea",** : may occur with drum perforation.
- **Ear Examination:** Congested bulging eardrum.,Signs of fullness, bulging, cloudiness and redness of the eardrum. **Viral otitis** may also result in blisters on the external side of the tympanic membrane, which is called ***bullous myringitis***

Complications

Local "ear":

- Drum perforation.
- Chronic secretory "OM" that may lead to hearing loss.
- Chronic suppurative OM.
- Acute labyrinthitis "present with ataxia"

Toha_Tohty



Spread of infection:

- ***To the bone:***
  - **"Acute or chronic"** with swollen, painful and erythematous mastoid area.
  - Petrositis.
- ***To the CNs:***
  - Meningitis, encephalitis.
  - Cerebral thromoboplebitis "may present with hemiplegic.
  - A.F.P.

Treatment: Immediate oral broad spectrum antibiotic/5 days to prevent complication e.g., ampicilin or amoxicilliln.

Secretory O.M. "glue ear, serous OM". :is simply a collection of fluid that occurs within the middle ear space due to the negative pressure produced by altered Eustachian tube function.

- *It is the most common cause of conductive hearing loss below the age of l0 years.*
- Sticky, serous material accumulates in the middle ear insidiously or after acute O.M.

Symptoms:

- Ear ache.
- Deafness, feeling of fullness or popping in the ear.
- It is especially common in children who have atopy freq. UR I Increase or cleft palate.

Treatment:

- Antibiotic.
- Anti-histamine.
- Decongestants.

Tonsillitis

Common infection above the age of 2 years.

C/P

	Strepto-coccal pharyngitis	Viral pharyngitis
History:		
Fever	High fever, may persist in severe cases up to 1st week.	Mild to moderate
Sore throat	Severe from the start with difficulty swallowing.	Moderate, occur 1-2 day after onset.
Others	• Vomiting &Abd.pain, cough ,headache bad breath (halitosis) .	○ Rhinitis, cough. ○ Conjunctivitis may help in differentiation between viral and bacterial.
Examination:		
Tonsilis	Diffuse redness of the tonsils and may follicular pus or membranous tonsillisare common.	○ Mild erythematic small pharyngeal ulcers may occur. ○ Follicular tonsillitis is rare.
Lympth node	Tender cervical lymphadenopathy	
Complications	▸ **Spread:**Peritonsillar abscess "quinsy" ▸ **Late:** after 2 weeks. ▸ **R.F:** 1-2 weeks. ▸ **G.N:** 2-3 weeks.	
Treatment:	○ Oral penicillin for at least 10days .	

Indication of adenotonsillectomy:

- Recurrent / persistent tonsillitis > 3 attacks a year.
- Obstructive sleep apnea.
- Recurrent otitis media.
- Recurrent aspiration pneumonia.

Indication for adenoidectomy:

- Persistent Nasal obstruction & mouth breathing.
- Chronic O.M.
- Hyponasal speech.
- Snoring/snorting.

N.B:

- Tonsils and adenoids are lymphoid tissue grown rapidly in the 1st 5 years of life, it usually are small in infant and reach their greatest relative size between 4-7 years, then progressive regress due to growth of skull..

Sinusitis

First well formed sinus is **maxillary sinus** that is formed by about 5-7 years which is usually infected by viruses ± secondary bacterial infection. It can be due to allergy, or autoimmune
The frontal sinus is well **formed by 10 year of age** And is rarely affected before that

Headache, fever, swelling and tenderness over the sinuses, + night cough.
Post nasal drip –cold last 10-14 days
Swelling around the eye

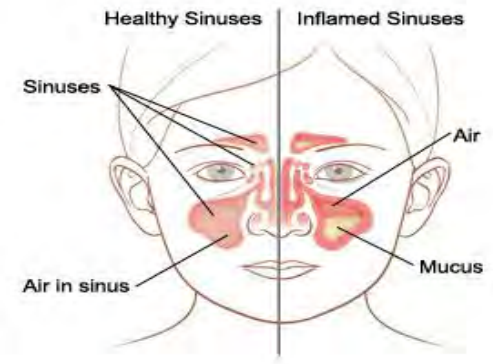
Diagnosis :

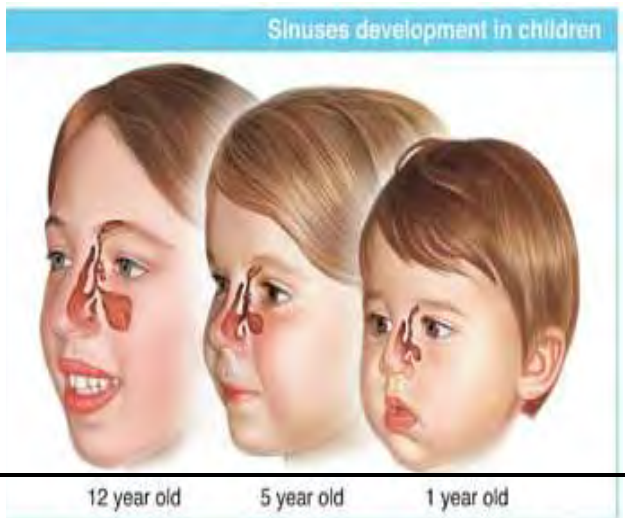
1. CT scan may help to determine how completely sinuses develop
2. X-ray may be used

Treatment :

Antibiotic for 14 days (amoxicillin – clarithromycin – azithromycin)

- Nasal decongestant





Strider

Harsh, continuous inspiratory sound due to partial obstruction in upper airways "**Larynx & Trachea".**

Causes

Acute

Infectious:

Viral: Laryngeotracheobronchitis "croup".
- Acute laryngitis.
- Spasmodic laryngitis.

Bacterial:
- Acute epiglotitis.
- Acute tracheitis "**stap.aureus"**
- Diphteritic laryngitis.

Non-Infectious:
- Laryngeal F.B.
- Laryngeospasm.
- Laryngeal edema.
- Laryngeal compression.
- Hypocalcemia .

Chronic

Congenital:
- Laryngeomalacia.
- Laryngeal web or cyst.
- Tracheomalacia.
- Vascular ring.
- Vocal cord palsy.
- Goiter , hypoglossal cyst.
- **Pier Robin syndrome**

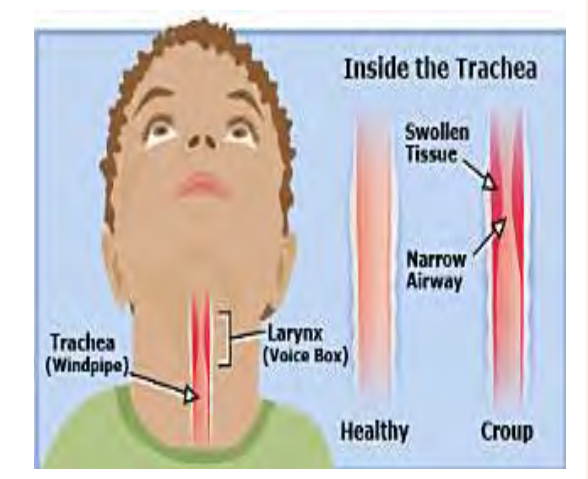
Acquired:
- Laryngeal Stenosis.
- Laryngeal tumor.
- Laryngeal paralysis.
- Tracheal stenosis.
- Hypocalcaemia.

Acute Infectious Strider

Laryngeo – trachea – bronchitis "Croup":

Age: 6 months – 6 years.

Cause: Viral esp. para-influenza,
others: RSV, influenza adenovines.



C/P:

- Preceded by URTI "Rhinitis".
- Moderate, severe.
- Croupy cough & hoarseness of voice.
- Low grade fever < 38.5^0C.
- Inspiratory & expiratory strider.
- RD may occur.

Investigation:

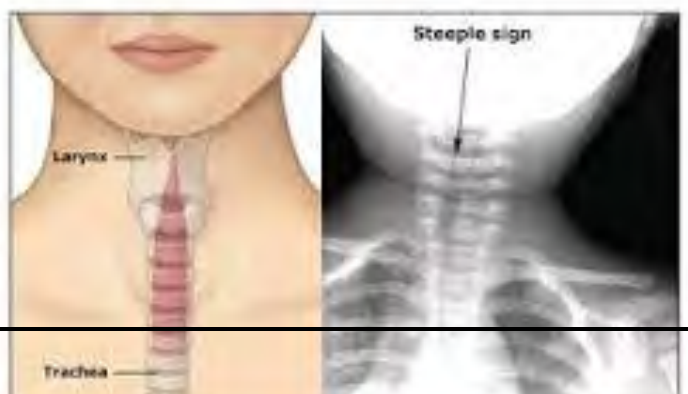
- Neck X-ray in AP view show sub-glottic narrowing "steeple sign".

Management:

- Calm the baby.
- Humidified O_2 & steam inhalation.
- **Adrenaline** Nebulizer Reduce need for intubation.
  - If patient receives adrenaline Neb. Patient must be kept for observation for 24 hours for rebound phenomena.
- **Corticosteroid** e.g., Dexamethsone
- In **severe cases:** Mechanical ventilation or rarely tracheotomy.

Complications:

- Bacterial tracheitis.
- Acute O.M.
- Lung infection "pneumonia".



N.B:

Grades of Strider:

- **Grade I** : Strider on exertion.
- **Grade II:** Strider at rest.
- **Grade III:** Strider with recession.
- **Grade IV**: Strider with recession, cyanosis, deteriotion of level of consciousness .

Spasmodic laryngitis:

Age: 1 – 3 years, not proceeding by coryza.

Cause: Viral but may be allergy or psychogenic.

C/P:

- Occurs at night.
- Mild to moderate.
- Croupy cough and hoarseness of voice.
- No fever.
- Inspiratory stridor.
- RD may occur.
- Recurrence is common.

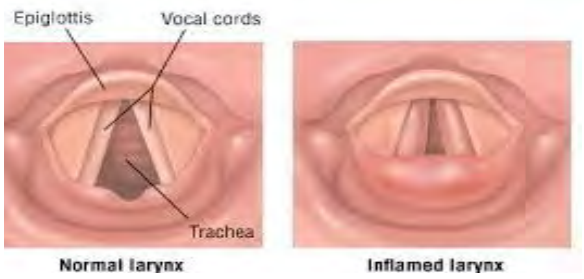
Acute Laryngitis:

Age: 1 – 3 years

Cause: Viral infection of the subglottic region.

C/P:

- Mild to moderate.
- Croupy cough with hoarseness of voice.
- Mild fever.
- Inspiratory stridor.
- Usually no RD.



Acute epiglottitis"Supra-glotitis":

Very important : Oral & MCQ.

Infection of the epiglottis by H.Influenza rarely by staph or pneumococci, or strepto-cocci.

C/P:

- **Peak age**: 2-7 years. No preceding cold symptoms.
- Toxic child with high fever > 39 ^{0}C.
- Drooling of saliva **"due to severe dysphagia"**.
- The child is severely exhausted: voice is muffled.
- Stridor is mild "soft".
- The child prefer upright posture, **"tripod"**.
- **Laryngeoscope** show large **"cheery red"** swollen epiglottis but his procedures and any stressful procedures may precipitate complete airway obstruction, also tongue depressor is C/I.

Management:

Medical emergency: Once suspected, patient must be admitted to **ICU.**

- **Artificial airway:**
  - ETT or "**less often tracheostomy"** is indicated regardless degree of RD. under **"GA".**
  - The tube kept in place for 2-3 days.
- O_2 inhalation.
- Culture of blood and if possible epiglottic surface should be done.
- **Antibiotic:**
  - Parental 3rd generation cephalosporin "**ceftriaxon or cefotaxime**".
  - Cont. Ab for 7-10 days.

N.B: *Lateral x-ray of the neck may show swollen epiglottis **"thumb sign".***

Fate:

- Severe condition with high mortality.
- Incidence decline due to HIB vaccination.
- **All contact must receive prophylaxis against HIB in the form of:**
  - 2 doses of rifampicin 12 hours apart.
  - Single I.M injection of ceftriaxone.
- **Complication:**
  - Meningits.
  - O.M.
  - Pneumonia.

	Croup	Acute epiglottis
Onset	Gradual	Sudden
Perceedingcoryza	Yes	No
Cough	Sever/barking	Slight/absent
Ability to drink	Yes	No
Salivation	No	Yes
Appearance	Well	Ill looking
Fever	< 38.5ºC	>39ºC
Stridor	Harsh	Soft
Voice	Hoarse	Muffled / unable to speak
Micro-organism	Para influenza	H.Influenza type B

Bacterial trachieitis:

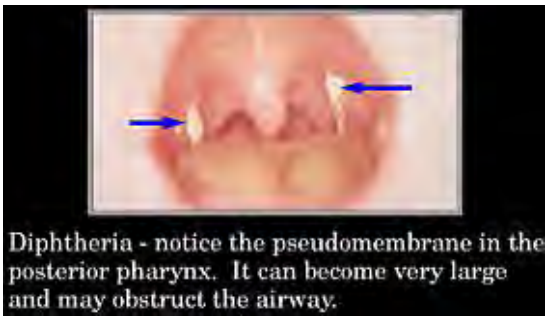
It is not distinguished as separate entity rather than a complication of viral croup and is characterized by toxic appearance of patient with obstructed air way and on, intubation copious amount of secretion and pus are seen.

Microorganism:

- Staph. Aureus "most common".
- Others e.g., morexellacatarrhalis.

Diphtherial Croup:

- Patient looks toxic with serous nasal discharge and grayish-white-yellow pseudomembrane on larynx and upper air way.



Diphtheria - notice the pseudomembrane in the posterior pharynx. It can become very large and may obstruct the airway.

Laryngomalacia: presence of soft laryngeal cartilage in epiglottis leads to collapse during inspiration

- Presented with Inspiratory strider beginning t or shortly after birth due to exaggerated collapse of eiglottic structures.
- Diagnosed by history and examination +flexible laryngeascopy
- It is most common cause of chronic stridor in infancy
- **Treatment**:
  - Self limiting usually « **reassurance of patient » .disappear at 18-24m**
  - **In severe cases** : laryoepigllottoplasrty or tracheotomy.

Laryngeal web :

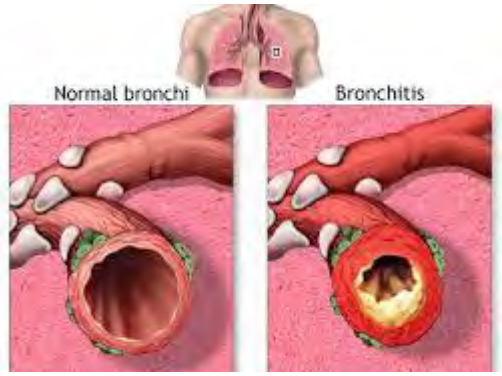
- It congenital presence of MM covered the CT between vocal cord
- Stridor beginning at birth (congenital)
- Diagnosis by rigid laryngoscopy
- **Management**
  1. Observation
  2. Lysed endoscopically using lasr

Lower respiratory tract

Infection

Common infections:
- Acute bronchitis.
- Acute bronchiolitis.
- Pneumonia.

Less Common infections:
- Lung suppuration.
- T.B.



Acute Bronchitis:
It is the most common cause of acute cough in children.

Etiology:
Viral "The commonest cause".
- Non specific, RSV, parainfluenza virus.
- Specific: Measles.

Bacterial:

Non specific.
- Pneumococci, streptococci, staph. Aures, H.Influenza.
- Malnutrition and immune deficiency is important predisposing factor.

C/P:

Initial stage:
- Nasopharyngitis "sneezing and nasal discharge" for 3-4 days before the onset.
- Fever is common, however it may be absent.

Diagnostic stage:

Early stage: The 1st few days.
- Cough and chest discomfort.
- Cough is dry, severe and metallic may be spasmodic with tracheitis.
- **Chest examination:** Free.

Productive stage " the next few days":
- Cough : productive and less severe "purulent sputum with bacterial causes.
- **Chest examination:**
  - Palpable Rhonchi.
  - Harsh vesicular breathing.
  - Scattered expiratory Rhonchi and may be coarse crepitations.

Convalescent stage "the last few days".
- **Cough:** decrease in frequency and severity.
- Chest signs disappear gradually.

Treatment:

Supportive Treatment:

- Expectorants and or mucolytics, drinking water is the best expectorant.
- Cough suppressants should be avoided.

Specific Treatment:

- Most causes are viral and recover spontaneously.
- Broad spectrum AB. E.g., ampicillin for 5 days in bacterial infection.

Bacterial Infections are suggested in presence of :

- High or persistent fever.
- Prolonged or purulent cough.
- Patient looking ill.

Acute Bronchiolitis

Acute inflammation of the bronchioles "Acute viral illness."

Incidence:

Infancy during the 1st two years peak incidence at **3-6 months of age**.
Season more at winter or spring.
Male > female.

Pathogenesis:

Mucosa and submucosa in small bronchioles are invaded by the virus leads to acute inflammation, bronchiolar obstruction by edema mucus and cellular debris, impaired pulmonary gas exchange may occur, early hypoxemia occur in sever disease, hypercapnea develops.

C/P

History:

- Prodroma of mild respiratory tract infection with nasal discharge and sneezing, then the infant gradually develops **respiratory distress, wheezing, cough and irritability and difficult feeding.** It is the most common cause of wheezing and RD in infancy.

Complications:

- Heart failure.
- Dehydration, due to tachypnea & anorexia.
- Lung collapse or pneumothorax, sudden deteriotion.
- Secondary bacterial infection "pneumonia".
- Recurrence..

Indication for hospitalization:

- Infant less than 6 months.
- Intolerance to oral feeding.
- Apnea.
- Severe RD with RR > 60 c/min.
- PaO_2< 65 mmHg.

Treatment:

- **Hospital management in infants with moderate to severe RD**:
  - O_2 therapy to correct hypoxemia "humidified".
  - IV fluid to prevent dehydration.
  - Nebulized salbutamol may be used with some benefit.
  - Corticosteroids are not beneficial.
  - Anti-viral are not necessary.
  - Assisted ventilation in cases with R.F.

Examination

Inspection:

- Manifestation of RD e.g., tachypnea and retraction.

Auscultation:

- Harsh vesicular breathing with prolonged expiration – expiratory Rhonchi.

D.D: Other causes of 1st attack wheezy chest.

Investigations:

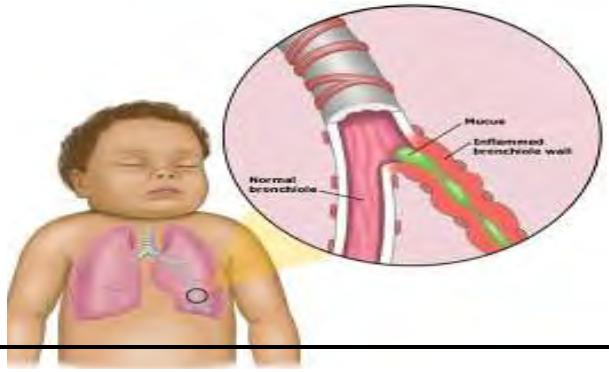
Chest X-ray:

- Hyper inflated chest "horizontal ribs+ flat diaphragm".
- Bilateral perihillar infiltrates.
- May be area of collapse.

- ESR & CRP and WBC count, usually normal.
- Detect RSV antigen in nasphyrngeal secretions by immunoflorescence.
- ABG to assess severity of the disease.

Risk factors for Bronchiolitis:

- Low birth weight.
- Low socio economic.
- Crowded living condition.
- Parental smoking.
- Chronic lung disease.
- CHD.
- Immuno deficiency cong. or acquired.



Home management in infant with minimal or mild RD:

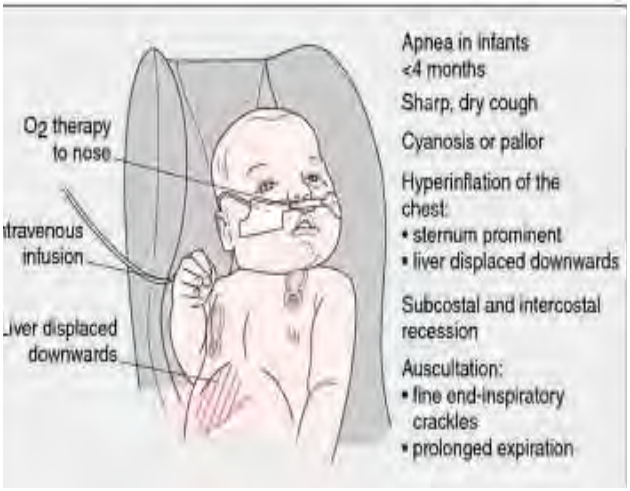
- Careful feeding to avoid aspiration.
- Close observation for the course of illness, increasing distress is an indication for hospitalization.

Prognosis:

- Recovery usually occur within 48 – 72 hours after the onset.
- Death in less than 1% from RF.

Prevention:

- RSV I.V Ig.
- Monoclonal AB I.M.



Pneumonia is an infection of the lower respiratory tract that involves the airways and parenchyma with consolidation of the alveolar spaces. The term **lower respiratory tract infection** is often used to encompass bronchitis, bronchiolitis or pneumonia or any combination of the three. **Pneumonitis** is a general term for lung inflammation that may or may not be associated with consolidation..

Causes & Pathological types:

Causes:

- *Infectious pneumonia:*
  - **Bacteria:**
    - G positive: Pneumococci "most common" in pre-school.
    - G negative: H.Influenza, klebsiella, pseudomonas.
    - T.B
  - **Mycoplasma:** 1 year atypical pneumonia.
  - **Viral:** RSV, adenovirus.
  - **Parasite:** Pneumocystis carini.
  - **Fungal:**Histoplasmosis.
- *Non-infectious pneumonia:*
  - **Aspiraton pneumonia:** of food, gastric acid… etc.
  - Allergic.

Pathological types & anatomical:

Lobar:

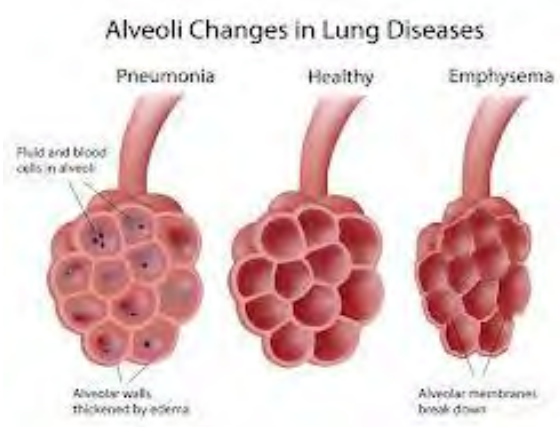
- Unilateral affection of one or more lobes.
- **Etiology:** bacterial may be due to mycoplasma or TB.
- **Chest X-ray**: lobar consolidation.

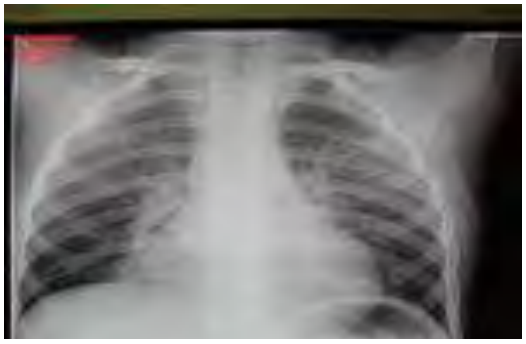
Broncho pneumonia:

- Bilateral affection of small areas of both lungs.
- **Etiology:** Bacterial or viral "scattered".
- **Chest X-ray:** Nodular or patchy infiltration.

Interstitial:

- Bilateral affection of interstitial lung tissues B/w alveoli.
- **Etiology:** mostly viral.
- **Chest X-ray:** Dense parahilar shadow with radiating streaks.





Other Classification

***Community acquired*:**

- **Bacterial:** Pneumococci, group B&A streptococci.
- **Viral:** RSV, parainfluenza, influenza and adenovirus.
- Mycoplasm& Chlamydia.
- Mycobacterial, T.B.
- Aspiration of milk or food.
- Allergic e.g., Loffler's syndrome.
- Rickettsial e.g., Coxiellaburneti.

Hospital acquired:

- **R.F:**
  - Micro aspiration of bacteria in seriously ill patient.
  - Contaminated equipment.
  - Impaired host defenses.
  - Coexisting cardiac, pulmonary, hepatic, renal in sufficiency.

***Causes of hospital acquired pneumonia*:**

- **G Negative bacterial:** The most common:
  - Enterobacter. Klebgsiella, E.coli. Protus.Hib. Pseudomonus, Legionella.
- **G positive:**
  - Strep. Pneumonia and staph. Aures.
- Opportunistic in immune suppressed individuals:
  - Fungal e.g., aspergellosis, Histoplasm, Candida.
  - Protozoal e.g., pneumocystis carni.

NB:

A typical pneumonia caused by: The most common:

- Mycoplasma pneumonia.
- Chlamyclia pneumonia.
- Chlamycliapsittaci.
- Legionella.
- Coxiella.

Toha_Tohty

C/P

Pneumonia should be considered in every case with RD, especially when there is Fever and Cough.

Manifestation of RD:

- **Grade I:** Tachypnea and working alanasi "increase depth of breathing".
- **Grade II**: The Intercostal and subcostal recession.
- **Grade III:** The expiratory grunting "increase pressure of respiration".
- **Grade IV:**The Cyanosis or alteredconsc. In severe cases.

Symptoms

General:

- Fever, malaise, poor general condition "worst in broncho pneumonia".
- May be abdominal pain referred from lower lobe pneumonia.

Chest:

- Cough "dry then productive"

Signs

1. Signs of R.D.
2. Chest Examination.

Inspection	Pneumonia	Bronchopneumonia
Movement	Decreased	Decreased.
Shape	Normal	Normal

- **Palpation /Tracheal Shift**: Central in both.
- **Tactile vocal Fremitus**: Increasednormal
- **PercussionNote** **Impaired** **Impaired**

Auscultation

Breath sound Diminished/bronchial Normal vesicular

Added sound Crepitations Bilateral wheeze/crepitations

Vocal resonance: Increased Bronchophony (May be normal).

Diagnosis

Radiological:

Chest X-ray:

- **Lobar pneumonia :** homogenous opacity in one or more lobes.
- **Broncho-pneumonia**: Scattered opacity in both lungs.
- **Interstitial pneumonia**: scattered bilateral perihilar pulmonary in filtrate, hyperinflation and atelectasis.

Complication:

- As abscess, effusion &pneumatocele may indicate S.Aureus, gram positive, or complicated pneumococcal pneumonia.
- ✓ ***Uncomplicated cases, repeat chest radiograph 6 weeks later to verify resolution.***

U/S: Guide thoracentesis of aloculated effusion.

Contrast CT Scan: For complicated cases.

Laboratory

- ABG with significant RD.
- **CBC & DLC**: Leucocytosis with predominant granulocyte, increase ESR and C-reactive protein positive suggest bacterial rather than viral pneumonia.
- Blood, pleural fluid or tracheobronchail secretions, aspirate culture and sensitivity.
- **Cold agglutinins** in 50% of mycoplasma pneumonia.
- Virus or virus antigen by immune Florescence.

Complications

(A) Respiratory:
- Pleural effusion.
- Empyema.
- Lung abscess.
- Pneumatocele.
- Septic thrombi or emboli in pulmonary veins.

(B) Systemic:
- Heart failure.
- Distant infections e.g., septrcemia, meningitis, pericarditis, osteomylitis.
- Metastatic abscess.

Differentiation between bacterial and viral pneumonia

Bacterial pneumonia:
- High fever.
- Leukocytosis, raised ESR and CRP

Viral pneumonia:
- Fever is usually not high.
- Leukocytosis count, ESR and CRP are normal or mildly elevated.

Viral Pneumonia: The commonest cause in pre-school children with peak at 2-3 years.
Causes: RSV, adenovirus, influenza and parainfluenza.
Clinically: Preceding URTI, Fever & RD, milder than bacterial pneumonia. May be widespread wheezes and crepitations.
Diagnosis:
Chest X-ray.
- Scattered bilateral infiltrate **"bronchopneumonia or interstitial pneumonia".**
- Hyper inflation.

CBC show normal or mildly elevated WBCs with predominant**lymphocytes.**
Detect **viral antigen** by immune florescence.

Bacterial Pneumonia:
Risk Factors:
- Respiratory viral infections "some are common in winter".
- **Splenectomy and hyposplenism**: Pneumococcal, H.Influenze pneumonia.

	Pneumococcal	Streptococcal	Staph.	H.Influenza	Klebsiell
Age	Commonest < 4 years.	3-5 years moderate	< 1 year	< 3 years gradual onset	Immuno def. severe.
C/P	○ Moderate ○ Usually lobar. ○ Broncho pneumonia in young infant.	○ High fever ○ Usually broncho pneumonia	○ High fever ○ Usually broncho pneumonia	○ Prolonged course. ○ Usually lobar	○ High fever. ○ Usually lobar
Treat ment	Penicillin	Penicillin	Cloxacillin or vancomycin	3rd generation cephalosporin	Amikacin

Bacterial infections depend on age:

Neonate :

- GroupB. B.Hemolyticstreptococcus., other gram negative bacilli e.g., E.Coli.

Infancy: Strept. Pneumonia, H. Influenza, also staph.Aures.

Older children: mycoplasma pneumonia, T.B.

Mycoplasma Pneumonia

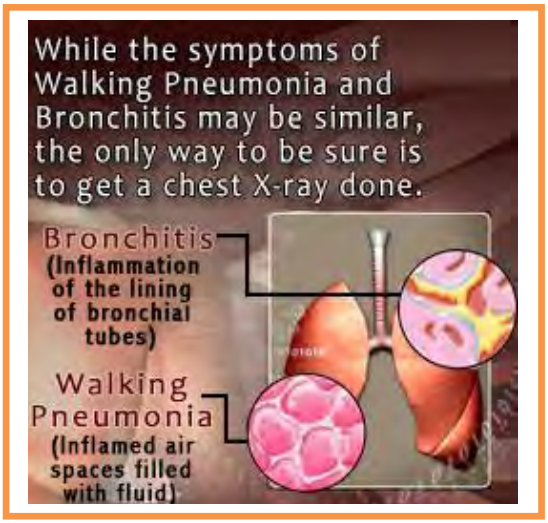
Primary atypical pneumonia.

- Common in school age 5-15 years.

- **Clinically:**
  - Severe non-productive cough without significant RD.
  - Low grade fever.
  - Pharyngitis is common.
  - Minimal physical sign **"walkingpneumonia"**. Radiological worse than clinically.
    - May be chest wheezes and inspiratory Rhonch.
- **Extra pulmonary manifestation**:
  - **Skin:**
    - Erythema multiform.
    - Steven Jhonson syndrome.
  - **CNS:**
    - Meningo encephalitis.
    - Aseptic meningitis.
    - Cerebellar ataxia.
    - Gullianbarre syndrome.
- **Joint**: Monoarticular arthritis.
- **Heart**: Myocarditis, pericarditis.
- **Blood:**
  - Hemolysis "autoimmune hemolytic anemia".
  - Thrombocytopenia.
- **GIT:** Protein losing enteropathy.

Walking Pneumonia

- **Diagnosis:** Mainly clinically.
  - **Chest X-ray show**: scattered bilateral perihilar pulmonary infiltrate.
  - **Rarely:** Lobar pneumonia ± effusions.
  - CBC usually normal.
  - Cold agglutinings may be detected.



Treatment:*Macrolid group*: Enthromycin, Azithromycin, Clithromycine.

Management

Supportive:

- **Bed rest**:
  - With severe RD, humidified O_2 inhalation and restricted I.V fluid.
  - Symptomatic treatment e.g., antipyretic.
  - Treatment of complication e.g., H.F.
  - Chest tube drainage for massive effusion orempyma.

- ***Specific Treatment***:
  - It is difficult to differentiate between bacterial and viral pneumonia " usually antibiotic are started based on child age, severity of illness, chest x-ray.
  - Antibiotic combination if the cause cannot be detected.
- **Mild Cases**: Amoxicillin or cefuroxime.

- **Hospitalized cases:**
  - Parental cefuroxime.
  - Vancomycin or clindamycin if staph.
  - Erytromycin or azithromycin or clarithromycin in mycoplasma pneumonia.
- **Viral pneumonia**: Antibiotic may be used as coexisting bacterial infections exists in 30% of cases.
- **Antiviral for immunodeficient**:
  - Ribavirin for RSV.
  - Amantidine for influenza A virus.

Recurrent Pneumonia

Causes:

1. **In the same lobe**: Local causes e.g. Foreign body Aspiration.
2. **In any site:**

- Immunodeficiencies cong. or acquired.
- Obstructing endobronchial lesion e.g., FBA, pulmonary sequestration.
- Decreased mucocilliary clearance e.g., cilliary dyskinesia cystic fibrosis.
- CHD with left to right shunt.

Slowly Resolving Pneumonia

Causes of slowly resolving pneumonia:

1. Inadequate therapy, in appropriate Ab choice, close or poor compliance.
2. Development of resistant organisms.
3. **Immunodeficiencies:** Cong. or acquired.
4. Obstructing endobronchial lesion e.g., FBA.
5. Decreased mucociliary clearance e.g., CF.
6. **Non-bacterial causes:** Viral, Fungi, parasites and mycobacteria.
7. **Non-infectious causes**: Hypersensitivity pneumonia, sarcoidosis and wegnergranulomatosis.

Workup:

1. **Identify the organism**: blood, sputum, pleural, BAL or lung biopsy can provide tissue for Gram stain and or C/S and thereby guide Ab. choice.
2. Chest CI scan.
3. Anti neutrophil cytoplasmic AB for Wegener granulomatosis.
4. Flexible Fiberoptic bronchoscopy.
5. Lung biopsy.

W.H.O Management of the child with cough or difficult breathing

SIGNS	CLASSIFY AS	IDENTIFY TREATMENT (Urgent pre-referral treatments are in bold print)
• Any general danger sign or • Chest indrawing or • Stridor in calm child	SEVERE PNEUMONIA OR VERY SEVERE DISEASE	► **Give first dose of Cotrimoxazole** ► **Refer URGENTLY to hospital**
• Fast breathing	PNEUMONIA	► **Give Cotrimoxazole for 5 days.** ► Soothe the throat and relieve the cough with a safe remedy. ► Advise mother when to return immediately. ► Follow-up in 2 days.
• No signs of pneumonia or very severe disease.	NO PNEUMONIA: COUGH OR COLD	► If coughing more than 21 days, refer for assessment. ► Soothe the throat and relieve the cough with a safe remedy ► Advise mother when to return immediately. ► Follow-up in 5 days if not improving.

A. Child 2 months – 5 years:

No Pneumonia "Cough or common cold"

- **Signs: No chest indrawing not tachypnic.**
- **Treatment:**
  - **Assess and treatment of problem e.g., ear or throat.**
  - **Advise mother do the following at home:**
    - **Clear nose if interfere with feeding.**
    - **Keep young infant warm.**
    - **Breast feed frequently "IgA".**
  - **Return quickly if :**

	▪ **Breathing become difficult.** ▪ **Chest indrawing and tachpnea.** ○ **Symptomatic treatment e.g., bronchodilator, antipyretic.**
Pneumonia	• **Signs:** No chest indrawing not tachypnic. • **Treatment**: ○ Advise mother to give home care. ○ Give antibiotic. ○ Symptomatic treatment. ○ Advise mother to return with her child in 2 days. • After 2 days: Assessment: ○ If improving, finish 5 days of Ab. ○ If the same, change Ab. ○ If worsen, refer urgently to hospital.
Severe pneumonia	• **Signs:** Severe chest indrawing, tachypnea.. • **Treatment**: ○ Refer urgently to Hospital. ○ Give 1st dose Ab. ○ Symptomatic treatment.
Very severe Disease	• **Signs:** ○ Convulsion. ○ Not able to drink. ○ Abnormal sleep or difficult to awake. ○ Strider at rest. ○ Relacting to feed, severe malnutrition. • Treatment: ○ Refer urgently to hospital. ○ Give 1st dose of Ab. ○ Symptomatic treatment.

Bronchial Asthma

- The most common respiratory disorder in children.
- Common in male > female.
- **It is chronic inflammatory disease of lung airways characterized by :**Hypersensitivity "hyper reactivity" of air way to varies stimuli leading to variable wide spread episodic air flow obstruction which is reversible either spontaneous or with treati

Associations:

- Family history of asthma may be present.
- Other allergic may be present as eczema, allergic Rhinitis or food allergy.
- Sinusitis and GERD.

Risk Factors:

- Genetic predisposition.
- Early weaning from breast milk before 4 months.
- Urban life.

Pathogenesis of asthma : "Atopy or type 1 hypersensitivity".

1. *Exposure to asthma triggers*: leads to **increase in IgE**→ activate mast cells, T-lymphycyte and easinophils.

Increased cytokines "IL4, IL5, IL3" in airway which result in:

- **Early phase:** Bronchoconstriction.
- **Late phase :** Edema, Increase mucus, increase Chronic .inf.Cells.
  - Airway narrowing especially in expiration.

2. *Persistent airway inflammation leads to*: airway remodeling:

- Collagen deposition beneath B.M.
- Hypertrophy of muscle and glands.

} Persistent airway narrowing

Asthma triggers:

- Respiratory viral infections.
- Exercise.
- Inhalation of :
  - Air pollutants e.g., dust, smoke.
  - Pollens, mite, animal dander.
  - Cold air.
  - Fumes & strong odors.
- **Drugs :** NSAID

C/P:

Asthma may have its onset at any age:

- 50% of patient usually start to suffer before 2 years.
- 80-90% have their attacks before 4-5 years.

Symptoms:

- Recurrent attack of dry cough, dyspnea and wheeze "worse at night".
- Attacks may be induced by certain triggers.
- In between attacks, the patient is either free or wheezing.

General signs: (during the attack):

- Irritability and restlessness.
- R.D.

Chest signs: "during the attack".

- **Inspection:** hyper inflated chest and prolonged expiration.
- **Palpation**: may be palpable wheezes.
- **Percussion:** Bilateral hyper resonance.
- **Auscultation:** Diminished vesicular breath sounds with prolonged expiration.
  - Bilateral expiratory Rhonchi.

Classification of severity of acute asthmatic attack "exacerbation"

	Mild	Moderate	Severe
Dyspnea	While walking	While walking	At rest
RD	Mild	Moderate	Severe
Pulse	< 100/min	100-120/min	>120/min
Pulsesparadoxus	< 10 mm Hg	10-20 mmHg	20-40 mmHg
Wheezes	End expiratory	Throughout exp	Exp.& inspiration
O_2 saturation	>95%	90-95%	< 90%
P_aCO_2	<45 mmHg.	Same	>45 mm Hg.
PEFR	>80%	50-80%	< 50%

Life Threatening Asthma:

- Wheeze + cyanosis + tachypnic& retraction
- O_2 Sat. < 92%.

Criteria:

- Restlessness "secondary to hypoxia".
- Drowsy and agitated.
- Exhaustion.
- Silent chest "Absent air entry".
- PEFR < 33%.
- Cyanosis.
- Poor resp. effort.
- Can't Talk

	Days with symptom	Nights with symptom	PEFR
Mild intermittent	< 2 / week	< 2 /months	>80%
Mild persistent	>2/week	>2/months	>80%
Moderate persistent	Daily	>1/week	60-80%
Severe persistent	Continual	Frequent	< 60%

PEFR, the greatest flow obtained on forced expiration after complete inspiration to total lung capacity. It is measured by spirometery.

Pulsesparadoxicus; is the different of systolic pressure during inspiration and expiration phase > 20 mmHg. Normally 10 - 20mHg.

Diagnosis:
It is clinical depends on history of symptoms of recurrent wheeze, cough and breathlessness "chest tightness".

- Nocturnal cough or wheeze could be the only symptom.
- History of trigger factors exposure, FH of atopy.

Severity of asthma:

Frequency of symptoms.
Sleeping disturbances.
Exercise tolerance.
Missing days at school.
Symptom free period.

- Are the symptoms reversible? Spontaneous or by treatment?

Examination: Usually normal between attacks.

- Signs of R.D, hyperinflation of chest.
- Chest deformity "**long standing asthma**" e.g., pectuscarnitum, Harrison sulcus.
- Associated conditions e.g., eczema, Rhinitis.
- **Auscultation:**
  - Decrease breath sounds on both sides.
  - Expiratory wheeze, prolonged expiration.
  - Try to exclude other causes of wheezing e.g., finger clubbing and sputum may indicate CF or CHD.

Investigations

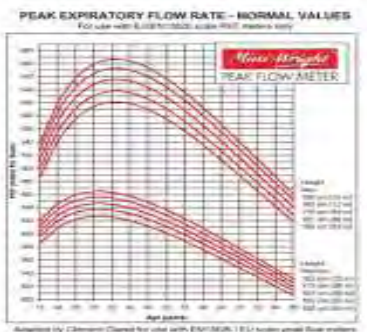
Usually not needed, because of diagnosis is mainly clinical.

- **Lung function test**: The most useful:
  - *Low forced expiratory volume in 1st second FEV_1".*
- Improved FEV, by ≥ 12% after inhaled B2 agonist.
- Worsening of FEV, by ≥ 15% after exercise challenge.

❖ **Peak expiratory flow rate "PEFR" lead to measure maximum air flow during expiration. "L/min".**
  - Measuring PEFR before & 15 min after inhalation of a bronchodilator "or exercise" can demonstrate reversible airways obstruction.

❖ **PEFR Variability "difference between morning and afternoon".**
  - Normally 5-10%.
  - In case of asthma ≥ 20%.

❖ PEFR is also expressed as percentage of predicted "useful in assessment of severity of asthma".



CBC: Esinophilia may be presented in blood or sputum of asthmatic patient.

ABG: Indication:

- Clinical detoriation.
- Suspected pneumothorax.
- Inability to maintain O_2 saturation more than 95%.

ABG in asthmatic patient:

- In early, hypocapnea, due to hyperinflation push out CO_2.
  - Later on, normal.
  - Hypercapnea, indicate severe asthma.

Chest X-ray: "during the attach" may show:

- Hyperinflation.
- Increased bronchovascular markings.
- May detect complications.

Chest X-ray is done if suspected:

- Pneumonia.
- Sudden deteriation patient.
- Suspected pneumothorax.
- Failed medical treatment and need ventilator.

Exercise tolerance test: for exercise induced asthma.

- Measure PEFR.
- Exercise for 5 min.
- Then repeated PEFR.
- In 2nd reading if drop more than 15%, > 1st reading indicate exercise induced asthma.

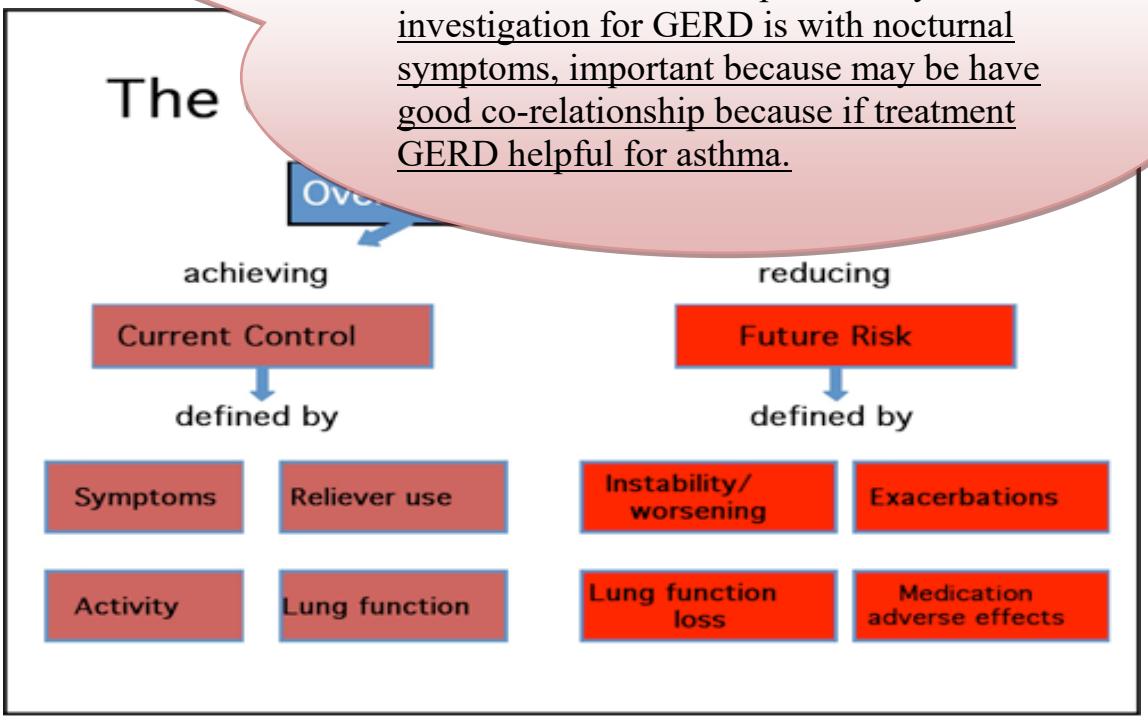
Skin prick test:

- Helpful to identify risk factors and irritant substance.
- Injected intradermally or inhaled to detect sensitivity.
- Helpful for Diagnosis, treatment and prevention.

The aim is to establish optimal symptom control with least possible medications.

N.B

with uncontrolled asthma particularly investigation for GERD is with nocturnal symptoms, important because may be have good co-relationship because if treatment GERD helpful for asthma.



ASTHMA DRUG THERAPY

RELIEVERS

1. Short-acting β_2-agonists

Asthavent® MDI / DP-Haler® / Revolizer® (Salbutamol)

Berotec® MDI (Fenoterol)

Venteze® MDI (Salbutamol)

Ventolin® MDI / Accuhaler® (Salbutamol)

2. Anticholinergics

Atrovent® MDI (Ipratropium Bromide)

Ipvent-40® MDI (Ipratropium Bromide)

Spiriva Handihaler® (Tiotropium)

CONTROLLERS

Long-acting β_2-agonists

Foratec DP-Haler® / Revolizer® (Formoterol)

Oxis Turbuhaler® (Formoterol)

Serevent® MDI / Accuhaler® (Salmeterol)

COMBINATIONS

DP-Haler® / Revolizer® (Budesonide + Formoterol)

Seretide® MDI / Accuhaler® (Fluticasone + Salmeterol)

Symbicord Turbuhaler® (Budesonide + Formoterol)

PREVENTERS

1. Inhaled Corticosteroids

Alvesco® MDI (Ciclesonide)

Beclate HFA® MDI (Beclomethasone)

Budeflam DP-Haler® / Revolizer® (Budesonide)

Budeflam HFA Gentle-Haler® (Budesonide)

Flixotide® MDI / Accuhaler® (Fluticasone)

Inflammide® MDI / Novolizer® (Budesonide)

Pulmicort Turbuhaler® (Budesonide)

QVAR® MDI (Beclomethasone)

2. Leukotriene receptor antagonist

Singulair® tablets (Montelukast)

Drugs used in Asthma:

Reliever:
- Short term symptomatic relief.
  - Short acting bronchodilators. B_2 agonist, salbutamol, "short acting", terbutaline.

Anti-cholinergic: ipratropium bromide

Controllers preventive:
- To decrease bronchial inflammation and hyper-reaction.
  - Inhaled steroids.
  - Long acting B_2 agonist.
  - $N_a$cromoglycate.
  - Leukotriene inhibitors.
  - Theophylline.
  - Oral steroid.

1.Bronchodilators

(A) *Short acting bronchodilators "4-6 hours"*
- Salbutamol.
- Terbutaline.

Mechanism: Selective B_2 agonist.
S/E: Hypokalemia, tremors, tachycardia.

(B) *Long acting broncho dilators "12 hours"*
- ❖ Salmeterol "For cases > 4 years"
- ❖ Formoterol "For cases > 6 years"

Mechanism: Selective B_2 agonist.
S/E: As short acting.

(C) *Theophyllin "long acting bronchodilator"*:
Mechanism: Increase CAMP in bronchial muscles.
S/E: Convulsions, Tachycardia, GIT upset.

(D) *Impratropium bromide*:
Mechanism: Parasympatholytic.
S/E: mild atropine like.

2.Anti-Inflammatory

1. Steroids:
Action: Potent anti-inflammatory, decrease secretion of mucus.
Routs:
- ❖ **Inhalation:**Fluticazone, budesonide, beclomethazone.
- ❖ **Oral** : prednisone.
- ❖ **I.V:** Methyl prednisolone.

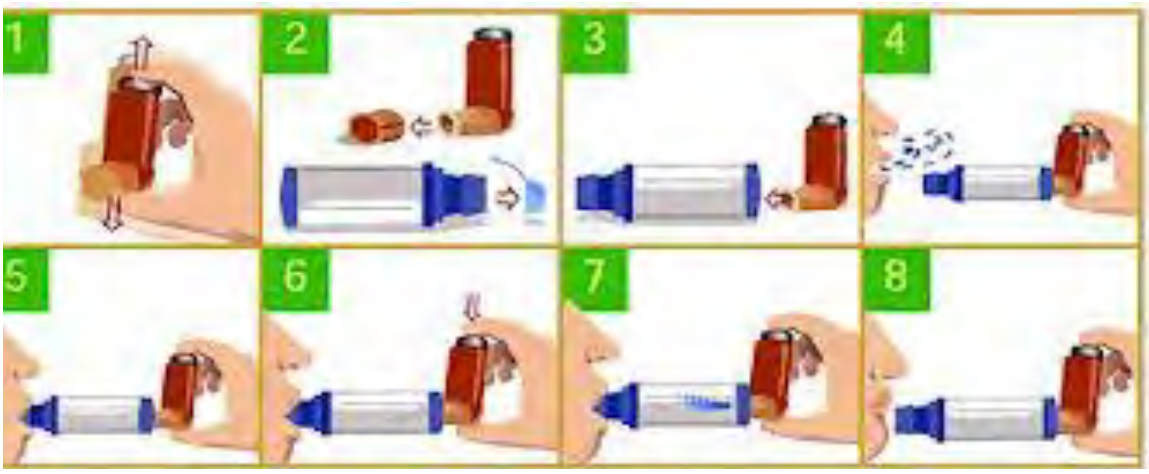
2. N_aCromoglycate:
Mechanism: Mast cell stabilizer.
Route: Inhalation.

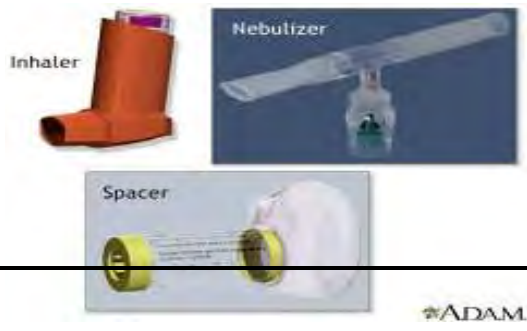
3. Monte Lukast: "Leukotriens"
Mechanism: Leukotriene receptor antagonist with bronchodilator effect.
Route: Oral "chewable tablet".

Devices for delivery of asthma

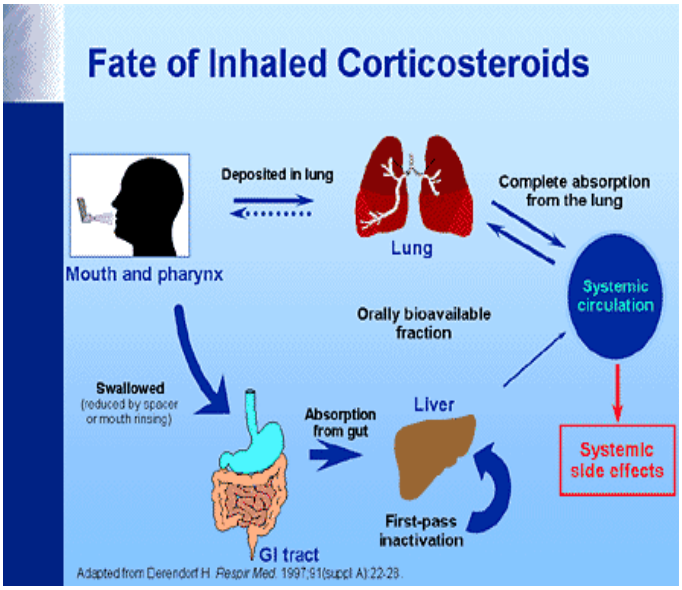
1. **Pressurized metered – dose inhaler "MDI"**: Need coordination, not very efficient. Not suitable for children < 5 years; MDI can be attached to a spacer device to increase amount of drug reaching to the airways and decrease impaction of drug on throat and require less coordination.
2. **In children < 2 years**: A face mask can be attached to the spacer.
3. Dry power inhaler.
4. **Nebuliser**: if child is unable to use any of these or if high dose are needed.

NB: ***If oral prednisolone is used, use the lowest possible dose "2 mg/kg perday"or give on alternate days to decrease the S/E.***





A.D.A.M.



Management of acute asthma mild to moderate:

- **If patient known case of asthma**;
  - Ventoline spacer and nebulizer at home.
  - 2 puffes every 2 min, repeated until improve "repeated around 5 times".
  - If not improved **hospital**.
  - In hospital detect severity.
- ***Mild: Detect*:**
  - R.R, signs of R.D, examine the chest.
  - Then give ventoline after half hour if not improved, repeat 3 times.
  - If not improved need admission.

In Hospital:

- Regular bronchodilator "ventoline" according to severity every 2 hours / 4 hours / 6 hours / 8 hours.
- O_2 therapy by O_2 mask.
- Regular anti-inflammatory "steroid" 6 hours 5mg/kg per dose.
- **If not improved**: another bronchodilator e.g., Ipratropium bromide nebuizer.
- **If not improved**: change to I.V.
- **If not improved**: + aminophylline I.V diluted with N/s over 20 min LD 5-6 mg/kg. Ifusion pump. MD = 0.7-1 mg/kg per 6 hours.
- **If not improved**: Direct to I.C.U close observation of vital signs and do ABG and X-ray **before** mechanical ventilation: to exclude **pneumothorax**.

Indication of admission in asthmatic patient:

1. Incomplete response with moderate retraction wheezing PEFR <60%, O_2 saturation ≤ 90%.
2. Pneumothorax.
3. Suspected pneumonia.
4. Social reason.
5. Baby Hypoxia.
6. Breath sound significantly reduced.

Status Asthmaticus:

- ❖ increasing severe asthma no responding to quick relievers

C/P : As severe acute asthma.

May signs suggesting RF:

- ❖ Drowsiness or confusion.
- ❖ Bradycardia.
- ❖ Absent pulses paradoxus "Resp. ms. Fatigue"
- ❖ Paradoxical thoraco-abd. Movement.
- ❖ Absent wheezes.

After treatment of Asthma: Advice to avoid exposure **to triggering agents**:

- Eliminate or reduce fumes, dust, smoke, animal dander.
- Avoid allergens as suggested by skin test.
- Treatment sinusitis, GERD, and rhinitis.
- Give annual influenza vaccine.

❖ **In exercise induced asthma give**:

- B_2 agonist inhalation.
- $N_a$cromoglycate inhalation before exercise.
- Montelukast oral.

Regular follow up:

- ❖ **Asthma checkups:**
- ❖ Every 2-4 weeks until good control.
- ❖ 2-4 times per year to maintain good control.
- ❖ Lung functions monitoring.

Treatment: Admit patient to I.C.U.

- ❖ **Monitor:** Vital data and chest signs. ABG.
- ❖ **Exclude complications**: by chest X-ray e.g., pneumothorax.
- ❖ **Supportive:**
  - Humidified O_2 inhalation.
  - I.V. fluid and correct dehydration.
  - Antibiotic for secondary infection.

Specific:

Bronchodilator:

Short acting B2 – agonist and ipratropium promide inhalation every 20 minutes for 3 doses then hourly or continuous with **cardiac monitoring**.Methyle prednisolone 1 mg/kg/6 hours I.V for 48 hours, other bronchodilators **may be used**:

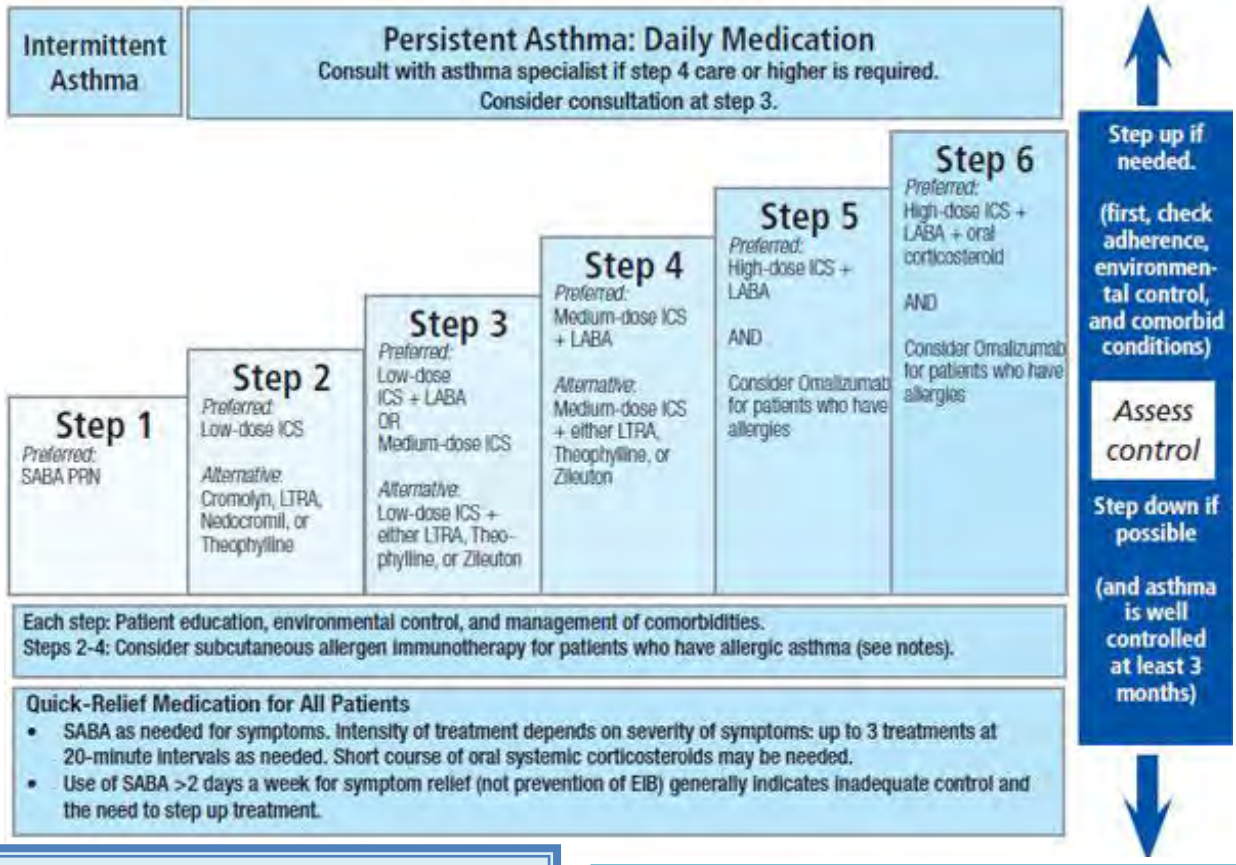
- Theophyline IV infusion 0.5 – 1 mg/kg/hr.
- Epinephrine S/c.

Mechanical Ventilator in R.F.

Prognosis:

- ❖ Gradual decline of the attacks with age occurs in 2/3 of patients.
- ❖ Atopic patient and steroid dependent carry a poor prognosis .

Stepwise approach to the Treatment of Chronic Asthma



Step (1) "mild intermittent"

- ❖ Symptoms < 2 times /week.
- ❖ Asymptomatic between attacks.
- ❖ Night time symptoms <2 timers / month.
- ❖ PEFR >80% predicted.
- ❖ PEFR variability <20%.

Treatment:

- ❖ No daily medication.
- ❖ Inhaled short acting B_2 agonist as needed "controller not needed".

Step (2) "Mild persistent"

- ❖ Symptoms > 2 times/week but <1 time per day.
- ❖ Night time symptoms > 2 times / month.
- ❖ PEFR >80% predicted.
- ❖ PEFR variability 20-30%.

Treatment:

- ❖ **Reliever:** short acting inhaled B_2 agonist as needed.
- ❖ **Controller:** low dose inhaled steroid or $N_a$cromocglycate or leukotriens.

Step (3) "moderate persistent"

- Daily symptoms with daily use of inhaled B_2 agonists.
- Night time symptoms > once / week.
- PEFR >60%, <80% predicted.
- PEFR variability >30%.

Treatment: Daily:

- **Reliever:** short acting B_2 agonist, as needed.
- **Controller:** high dose inhaled steroid or low dose inhaled steroid
- Long acting B_2-agonists.

Step (4) "severe persistent"

- Continuous symptoms.
- Limited physical activity.
- PEFR <60% predicted.
- PEFR variability >30%.

Treatment:

- Reliever same.
- **Controller:** high dose inhaled steroid + long acting B_2 agonist + oral prednisolone may be added.

Instruct and demonstrate to family & patient: How to use the inhaler, give advice about avoidance of environmental trigger factors.

- **1st review:** Patients medication technique, compliance, environmental control.
- If control of symptoms not maintained (**STEP UP)**
- After 3-6 months, if control maintained (**STEP Down)**

D.D OF Recurrent attack wheezy chest **:**

- **Bronchial asthma .**
- **GERD.**
- **Bronchiectasis .**
- **CF.**
- **T.B**
- **Chronic H.F**
- Cow milk allergy
- Immunedeficiency
- Foreign body aspiration

D.D of 1st attack wheeze chest :

- Bronchiolitis.
- Brochopneumonia.
- Bronchialasthma.
- F.B.A.
- GERD.
- CHF "Acute"
- T.B.

Suppurative Lung Disease

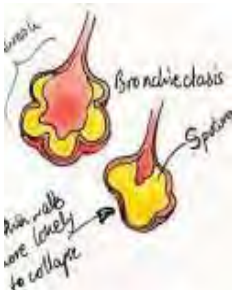
- Bronchiectasis.
- Lung abscess.
- Empyema with bronchopleural Fistula.

Bronchiectasis

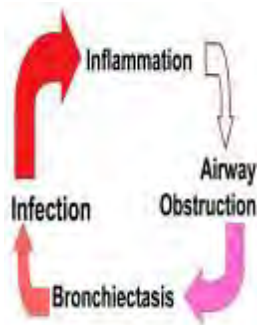
Permanent dilatation of medium sized bronchi with inflammatory destruction and accumulation of pus in bronchi "**obstruction and infections**

Etiology:

- **Congenital** : arrest in bronchial development.
- Immotile cilia syndrome (primary cilliraydyskinesia)
- **Acquired**: due to chronic infection and or obstruction.
- **Predisposing factors are**:
  - Foreign body.
  - Enlarged bronchopulmonary nodes due toT.B.
  - Cystic fibrosis.
  - Chronic asthma.
  - Immunodeficiency.
  - Severe pneumonia e.g. TB measles, whooping cough.



  - Chronic sinusitis and chronic ear infections.
  - Bronchiectasis.
  - Dextrocardia and situsinversus.



- ***Pathogenesis:***
  - Destruction of the bronchial epithelium interfere with cough reflex, retention of secretions, persistent

infection, destruction of musculoelastic tissue in bronchial wall.

- Infection of the peri-bronchial tissues, pneumonitis fibrosis that pulls on the bronchial wall, dilatation "**most frequent in the basal segments of the lobes"**

C/P

- **<u>General manifestation of toxemia</u>**: fever, anorexia, wt.loss.
- **<u>Respiratory manifestation:</u>**
  - Finger clubbing.
  - Cough with expectoration of excessive fetid sputum related to posture may be hemoptysis.
- **<u>Auscultation</u>**: Patchy area with bronchial breathing and crepitations.
- **<u>Investigations:</u>**
  - **<u>X-ray</u>** : Honey comb or soap bubble appearance.
  - Bronchography.
  - Bronchoscope.
  - **CT. most accurate and specific**.
  - Sputum C/S test.

<u>Treatment</u>

<u>Medical:</u>

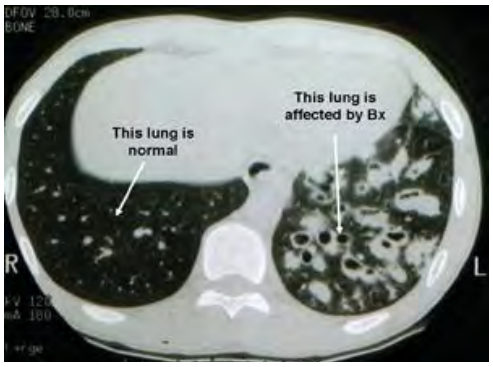
- Postural drainage.
- Parental Ab. According to C/S, bronchodilator and other symptomatic treatment for cough.
- Physiotherapy.

<u>Surgical</u>: Resection for localized bronchiectasis.

<u>Outcome</u>: Cong. causes carry bad prognosis.

<u>Complication</u>:

- Massive hemoptysis.
- Visceral abscess.
- Amyloidosis.
- Corpulmonale.



All pts with bronchiectasis should received yearly influenza vaccine and every 5 years pneumococcal vaccines

Lung Abscess

- Suppurative process resulting in destruction of lung parenchyma with formation of cavity containing pus.

Etiology

- Aspiration of infected material or F.B .
- ***Secondary lung abscess;***
  - Pneumonia e.g., staph. Aureus and klebsciella.
  - Metastatic lung abscess secondary to septic emboli e.g., from right sides infective endocarditis and septic thrombophlebitis.
  - Amebic lung abscess "rare".

C/P

- **General manifestation of toxemia** : F.M..A.H. weight loss.
- **Respiratory manifestation** : Cough with expectoration of fetid sputum, hemoptysis.

Investigation

- ***X-ray***: Cavity with fluid level surrounded by alveolar infiltration.
- Bronchoscopy.
- C.T.
- Suputum C/s.

Treatment:

1: **Medical** :Ab according to C/S.
Bronchoscopy if F.B is suspected.

2: **Surgical** : Resection of affected lobe in recurrent hemoptysis or repeated infection

Surgical drainage of lung abscess is never indicated.

Empyema "pleurisy"

- Accumulation of pus in the pleural spaces:

- ***Etiology***:
  - **Secondary diseases:**
  - pneumococcal, H.influenza.
  - Lung abscess or bronchiectasis.
  - **Pneumonia :** Staph., Mediastinal disease: Mediastinitis.
  - **Subdiaphragmatic disease**: Subphrenic abscess.
  - Contamination introduced from trauma or thoracic surgery.

- **History:** Acute onset usually with : high fever, pleuritic pain: chest pain increase with coughing or deep breathing and decrease by lying on the affected side.
- Manifestation of R.D.

Examination: Depend on the amount of effusion.
Inspection: Limitation of movement of the affected side.
Palpation: Mediastinal position.
- Small amount , no shift.
- Moderate amount, shift apex beat and trachea to the opposite side.

Percussion: Stony dullness.

Investigation:

X-ray : Obliteration of costopherenic angle by homogenous opacity raising to the axilla and pushing the mediastinum to the opposite side.

Lab :Thoracocenteses: The collected fluid is examined "C&S" to determine the causative organism.

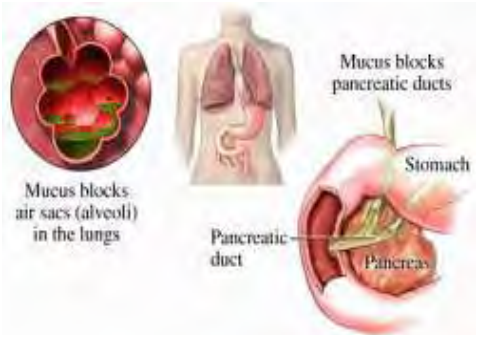
Treatment:

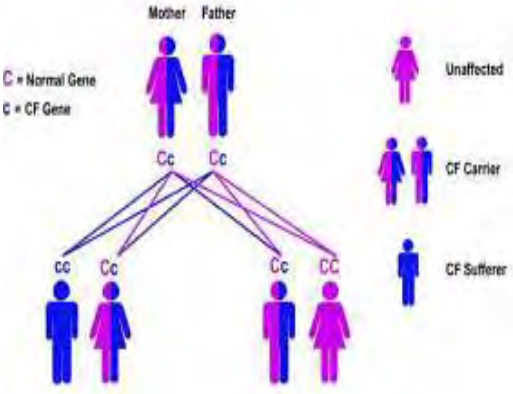
- Treatment of the cause e.g., pneumonia by proper Ab.
- Management of effusion, by drainage by underwater seal system

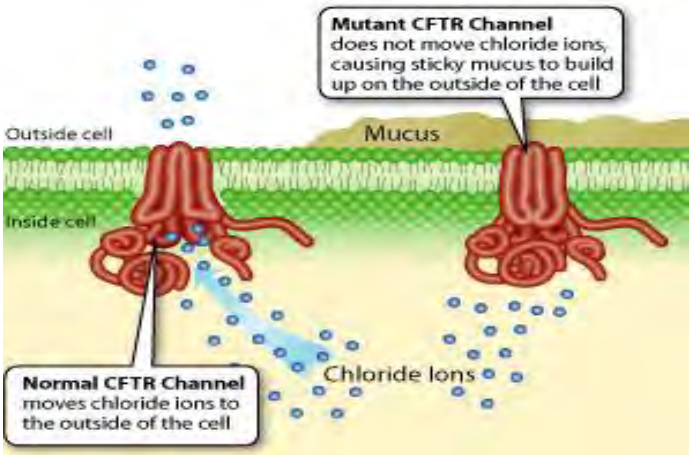
It is **AR** a gene on chromosome **7 codes** for the protein **"cystic fibrosis trans membrane regulator",** this is **absent** in C.F.

CFTR is a chloride channel located on the apical surface of epithelial cells. CFTR is **important for** the proper movement of salt and water across cell membranes and maintaining the appropriate composition of various secretions, especially in the **airway, liver, and pancreas**. The most common mutation is a deletion of three base pairs resulting in the absence of phenylalanine at the **508 position(ΔF508**). More than 1000 mutations of the *CFTR* gene have been identified.

The secretary and absorptive characteristics of epithelial cells are affected by abnormal CFTR, resulting in the clinical manifestations of CF. The lack of normal CFTR function alters chloride ion conductance in the sweat gland, resulting in **excessively high** sweat sodium and chloride levels







Pathogenesis: it is known that abnormal airway secretions make the airway more prone to colonization with bacteria. Defects in CFTR may also reduce the function of airway defenses and promote bacterial adhesion to the airway epithelium. This all leads to chronic airway infections and eventually to bronchial damage (**bronchiectasis**).

RESPIRATORY COMPLICATIONS	GASTROINTESTINAL COMPLICATIONS
Bronchiectasis, bronchitis, bronchiolitis, pneumonia	Meconium ileus (infants)
Atelectasis	Meconium peritonitis (infants)
Hemoptysis	Volvulus (infants)
Pneumothorax	Intestinal atresia (infants)
Nasal polyps	Distal intestinal obstruction syndrome (non-neonatal obstruction)
Sinusitis	Rectal prolapse
Airway reactivity	Intussusception
Corpulmonale (associated with chronic hypoxemia)	Fibrosingcolonopathy (strictures)
Respiratory failure	Appendicitis
Mucoid impaction of the bronchi	Pancreatitis
Allergic bronchopulmonaryaspergillosis	Hepatic cirrhosis (portal hypertension: esophageal varices, hypersplenism)
OTHER COMPLICATIONS	Neonatal obstructive jaundice
Infertility	Hepatic steatosis
Delayed puberty	Gastroesophageal reflux
	Cholelithiasis
	Growth failure (malabsorption)
	Fat soluble vitamin deficiency states (vitamins A, K, E, D)
	Insulin deficiency, symptomatic hyperglycemia, diabetes
	Malignancy (rare)

OTHER COMPLICATIONS
Infertility
Delayed puberty
Edema-hypoproteinemia (due to protein malabsorption)
Dehydration-heat exhaustion

Electrolyte disturbances (hyponatremia, hypokalemia, hypochloremia, and metabolic alkalosis)
Hypertrophic osteoarthropathy-arthritis
Digital clubbing

The respiratory epithelium of patients with CF exhibits marked impermeability to chloride and an excessive reabsorption of sodium. These alterations in the bioelectrical properties of the epithelium lead to a relative **dehydration of airway secretions**, which results in airway obstruction and impaired mucociliary transport. This, in turn, leads to **endobronchial colonization** with bacteria, especially ***Staphylococcus aureus*** **and** ***Pseudomonas aeruginosa*****.** ***Pseudomonas aeruginosa*** **predominates in older patients with advanced disease**. Chronic bronchial infection results in persistent or recurrent cough that is often productive of sputum, especially in older children. Chronic airway infection leads to airway obstruction and **bronchiectasis** and eventually to pulmonary insufficiency and premature death. The median age of survival (years) is currently in the **mid-30s**. **Digital clubbing** is common in patients with CF, even in those without significant lung disease. Chronic sinusitis and **nasal polyposis** are common. Any child with nasal polyps, especially if younger than 12 years of age, should be evaluated for CF.

Management:

- There is no specific treatment for CF
- **Aim of therapy:** to prevent progression of lung disease and maintain nutrition and growth.

Respiratory management|: Physiotherapy, "Chest percussion , postural drainage" to decrease accumulation of secretions.
Also physical exercise should be increased.

- Continuous oral antibiotic + I.V antibiotic for acute exacerbations.
- Nebulizer antibiotic could be used for pseudomonas infection.
- Bronchodilator and inhaled steroids are helpful.
-

Respiratory :

Allergic broncho-pulmonary aspergillosis (ABPA) is a hypersensitivity reaction to *Aspergillus* in the CF airways.

Minor hemoptysis is usually due to airway infection, **but major hemoptysis** is caused by bleeding from bronchial artery collateral vessels in damaged/chronically infected portions of the lung.

Pneumothoraces can occur in patients with advanced lung disease

Endocrine :

Ninety percent of patients with CF are born with **exocrine pancreatic insufficiency**. This leads **to malabsorption** of proteins, sugars (to a lesser extent), and especially fat. Fat malabsorption manifests clinically as **steatorrhea** (large foul-smelling stools), deficiencies of fat-soluble vitamins (**A, D, E, and K**), and failure to thrive. Protein malabsorption can present early in infancy as **hypoproteinemia and peripheral edema**.

G.I.T:

Approximately 10% of patients with CF are born with intestinal obstruction caused by inspissated meconium (**meconium ileus**).

In older patients, intestinal obstruction may result from thick inspissated mucus in the intestinal lumen (**distal intestinal obstruction syndrome**).

In adolescent or adult patients, progressive pancreatic damage can lead to enough islet cell destruction to cause **insulin deficiency.** This initially presents as glucose intolerance, but true diabetes that requires insulin therapy (**CF-related diabetes**) may develop.

Electrolyte disturbances :

The failure of the sweat ducts to conserve sodium and chloride may lead to **hyponatremia** and **hypochloremic metabolic alkalosis**, especially in infants.

Reproductive :

Inspissation of mucus in the reproductive tract leads to reproductive dysfunction in both males and females**.**

In males, congenital absence of the vas deferens and azoospermia are nearly universal.

In females, secondary amenorrhea is often present as a result of chronic illness and reduced body weight. Fertility is also diminished by abnormal secretions in the fallopian tubes and cervix, but women with CF **can conceive**

newborn screening for CF: which is based on elevated immune reactive trypsinogen (IRT) levels or DNA tests. Newborn screening identifies the majority of infants with CF, but there are both false positive and false negative results.
the diagnostic test of choice is the sweat test. The following criteria should be met to establish the diagnosis of CF:

1. **A positive result from a newborn screening test *and* at least one of the following**: Positive results on at least two sweat chloride tests (*positive if the value is > 60 mEq/L with adequate sweat collection, borderline if 40 to 60 mEq/L, and negative if < 40 mEq/L*) or
2. Two mutations known to cause CF identified by DNA analysis. or
3. A characteristic abnormality in ion transport across nasal epithelium demonstrated in vivo (**nasal potential difference testing**).

TREATMENT

The treatment of CF is **multifactorial**, but it is primarily directed toward the ***gastrointestinal*** and ***pulmonary complications***.

- **management of pulmonary complications** :is directed toward facilitating clearance of secretions from the airways and minimizing the effects of chronic bronchial infection:
- (**chest physiotherapy**) help remove mucus from the airways.
- **Aerosolized DNAse and 7% hypertonic saline:** both delivered by nebulizer, decrease the viscosity of mucus.
- **Antibiotic therapy** is important in controlling chronic infection. **(oral, aerosolized**, **and parenteral**) help to slow the progression of lung disease. Patients often **require 2- to 3-week courses** of high-dose intravenous antibiotics and aggressive chest physiotherapy to treat pulmonary exacerbations. Common infecting organisms are ***Pseudomonas aeruginosa* and *Staphylococcus aureus***.

☒ **Exocrine pancreatic insufficiency:**

- **Enteric-coated pancreatic enzyme capsules, which** contain lipase and proteases. Patients with CF are encouraged to follow high-calorie diets, often with the addition of nutritional supplements. *Lipase dosages exceeding 2500 U/kg/meal are contraindicated because they have been associated with* ***fibrosing colonopathy***.
- Fat-soluble **vitamins (A, E, D, and K)** are recommended.
- . Patients with CF-related diabetes are treated with insulin, primarily to improve nutrition and prevent dehydration, **as ketoacidosis is rare**.

The management of CF is complex and is best coordinated by personnel at accredited CF centers. As with other severe chronic diseases, management of patients with CF requires a multidisciplinary team. Physicians and other care providers, including nurses, respiratory therapists, nutritionists, and social workers must work with patients and their families to maintain an optimistic, comprehensive, and aggressive approach to treatment.

Foreign body aspiration

Aspiration of foreign bodies into the trachea and bronchi is **relatively common**.

Age : The majority of children who aspirate foreign bodies are under 4 years of age. Most deaths secondary to foreign body aspiration occur in this age group.

Site : Because the right mainstem bronchus takes off at a less acute angle than the left mainstem bronchus, foreign bodies tend to lodge in right-sided airways.

Some foreign bodies, **especially nuts**, can also lodge more proximally in the larynx or subglottic space totally occluding the airway.

Many foreign bodies are not **radiopaque**, which makes them difficult to detect **radiographically**.

The most common foreign bodies aspirated by young children are food **(especially nuts)** and small toys. Coins more often lodge in the esophagus than in the airways. Older children have been known to aspirate **rubber balloons**, which can be life-threatening

Clinical Manifestations

Many children who aspirate foreign bodies have clear histories **of choking, witnessed aspiration**, or physical or radiographic evidence of foreign body aspiration.

*a **small percentage*** of patients have a negative history because the aspiration went unrecognized. **Physical findings**:

include cough, localized wheezing, unilateral absence of breath sounds, stridor, and, rarely, bloody sputum.

- ***Complication :***

Most foreign bodies are small and quickly expelled, but some may remain in the lung for long periods of time and may come to medical attention because of **persistent cough, sputum production, or recurrent pneumonia**.

- Foreign body aspiration should be in the differential diagnosis of patients with ***persistent wheezing unresponsive to bronchodilator therapy, persistent atelectasis, recurrent or persistent pneumonia, or chronic cough without another explanatio***n.
- Foreign bodies may also lodge in the esophagus and compress the trachea, thus producing respiratory symptoms. **esophageal foreign bodies** should be included in the differential diagnosis of infants or young children with persistent **stridor or wheezing**, particularly if dysphagia is present.

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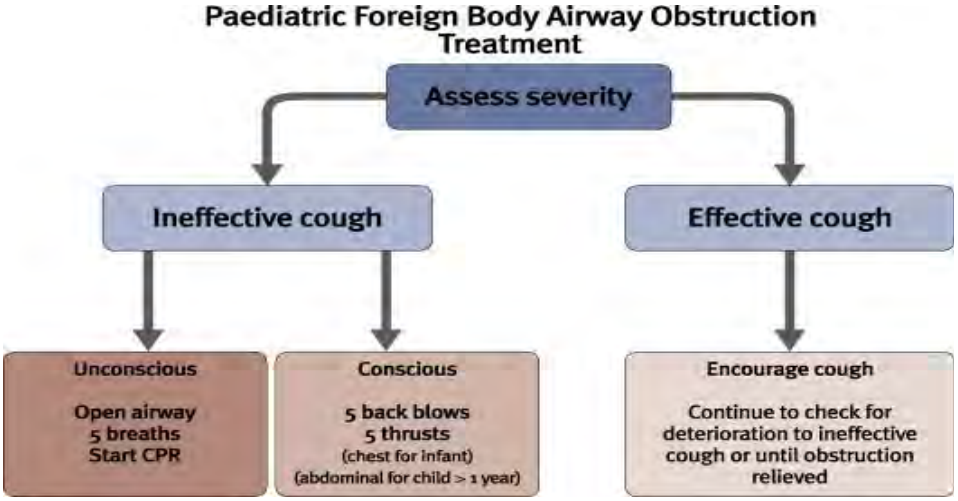
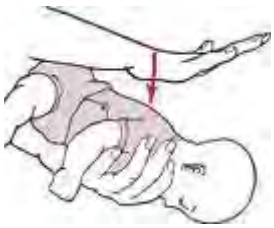
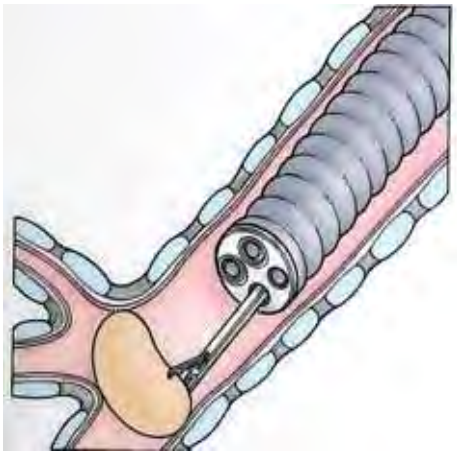
Diagnostic Studies

- ✓ **Radiographic** studies will reveal the presence of radiopaque objects and can also identify focal air trapping, especially on expiratory views.
- ✓ **Fluoroscopy** may be helpful, especially in children who cannot perform expiratory views.
- ✓ If there is strong evidence of foreign body aspiration, do **rigid bronchoscopy.**
- ✓ **Flexible bronchoscopy** can be used to locate an aspirated foreign body and may be useful when the presentation is not straightforward, but foreign body removal is best performed via **rigid bronchoscopy**







Place one fist just above the child's navel with the thumb side facing the abdomen

- ***Without RD***: Bronchoscopic extraction.
- ***With R.D***.
  - **If the child is breathing well**.
    - Let him attempt to clear the foreign body.
    - Encourage cough
  - **If cough becomes ineffective**:
    - Try to assist expulsion of the F.B.
    - Provide rescue breathing in between trials.
    - If trials fails and infant becomes unconscious attempt to visualize F.B and remove manually and repeat sequence as necessary.

First aid for the choking infant younger than 1 year:

- Hold infant prone with the head down.
- Give 5 interscapular back blow, using heal of hands.
- Turn the infant supine, with head dependent and perform 5 quick down ward chest thrusts in same location as external chest compression.

First Air for the chocking child older than 1 year:

A. **In consc. Patient**: Abd. Thrust in sitting or standing☹***(Heimlich maneuver)***

- Encircle the child chest with arms from behind.
- Place one fist against patients abdomen in midline just below tip of xiphoid.
- Grasp fist with other hand and exert 5 quick, upward thrusts.
- Continue until F.B is expelled or five thrusts are completed.

B. **In unconscious patients:Abd. Thrust in lying down:**

- Place the patient supine.
- Open patient airway using chin lift or jaw thrust.
- Place heel of one hand on child's abdomen just below costal margin.
- Place other hand on top of the 1st hand.

- Press both hands into abdomen with quick upward thrusts in midline.

Further interventions:

1. Laryngoscope removal with a Magill forceps.
2. If failed, push F.B more distally with intubation or bag and mask.
3. If failed, perform immediate cricothrotomy.

NB: CT scan for chest may detect material in airway , focal airway edema.

Vaccination By Dr Asma Hadida

2015_2016

Vaccine	Type	Route	Diluents
BCG	Live attenuated	ID	+
OPV	Live attenuated	Orally	
HBV	Recombinant from DNA	IM	
MMR	Live attenuated	SC	+
DPT	D&T (toxiod) P(killed)	IM	
IPV	Killed (inactive)	IM	
HIB	Conjugated polysaccharide	IM	+
Pneumococcal V	Conjugated polysaccharide	IM	
Meningococcal V	Conjugated polysaccharide	SC	
HPV	Recombinant from DNA	IM	
Rota virus V	Live attenuated	Orally	

Hexa vaccine : DPT , HIB , HBV, IPV , given 2,4,6 months IM

Recommended vaccination sites

- Vaccines should not be given intravenously
- The anterolateral thigh is the recommended site for **IM** vaccination in infants <12 months of age, due to its larger muscle size.
- The deltoid muscle is the recommended site for **IM** vaccination in children ≥12 months of age.
- **For SC injection:** Fatty tissue over the Anterolateral thigh in infants (under one year of age).
- Fatty tissue over triceps(upper arm) in children one year or older.

- Positioning

Classification of Immunity

- **Innate (natural) Immunity:** This is immunity which the individual inherits or possesses by virtue of his constitutional and genetic make-up. It is a rapid but non-specific response which includes inflammation, phagocytosis and activation of the complement system

- **Passive Immunity differs from active immunity in the following:**
  - Immunity is rapidly established.
  - Immunity produced is of short duration.
  - There is no education of the reticulo-endothelial system.

- **Accquired(induced)Immunity:** This acquired or induced immunity through the specific recognition of foreign antigens or antibodies. Could be:

Active immunity: Recognition of foreign antigens, which are two types :
  - Natural as result of infection e.g., measles disease.
  - Artificial when one is immunized with an antigen.

Passive immunity: When antibodies produced in one body (human or animal) are transferred to another to induce protection against diseasee.g.,Maternal antibodies to the foetus via placenta or breast milk (natural).
  - By administration of antisera, e.g., anti-rabies serum (artificial).
  - By administration of gamma globulin (artificial).

Vaccination

Vaccine:

Preparation of an antigen for preventive inculation with when administered stimulates specific antibody formation in the body.
Vaccination is the deliberate stimulation of the immune system to produce specific protection against a disease.

- Work by mimicking what happens during natural infection without causing illness.
- Use altered versions of viruses or bacteria to trigger an immune response.

- Are the most effective means of controlling infectious diseases.

Types of Vaccines:

Made by:

- Weakening the virus or bacteria
- Inactivating the virus or bacteria
- Using part of the virus or bacteria
- Inactivating a toxin (poison) made by the bacteria

1. **Live attenuated vaccines:**
   Live vaccines have all the major and minor antigenic components Severe reactions possible.. Usually produce immunity with one dose*.
   a. *Bacterial:* BCG, typhoid.
   b. *Viral:* Polio "Sabin", measles, mumps, rubella, yellow fever, influenza .. nasal spray.
2. **Inactivated "killed" vaccines:**
   - killed by heat or chemicals.
   - Cannot replicate in the host.
   - Generally not as effective as live vaccines.
   - Antibody titer may diminish with time Usually safe.
   - Usually require **multiple** doses to produce adequate antibody response

   a. *Bacterial:* Pertussis, typhoid, cholera, plague.
   b. *Viral:* Polio "salk", Rabies, influenza, hepatits A.
      - Needs booster dose.
3. **Toxoid :**
   Certain organisms produce exotoxins.
   •Toxins are detoxicated and used in the preparation of vaccine.
   •Toxoid vaccines are highly effective and safe immunizing agents.
   a. Tetanus, "toxin are detoxicated by Formaldahyde".
   b. Diphtheria.
4. **Sub-unit:**
   Antibody with less functional activity.
   •Immunogenicity improved by conjugation.
   a. Surface antigen "hepatits B"
   b. Capsular Polysaccharides "pneumococcus, meningoccoci , “

   Conjugated vaccine:
   ▫ A special type of subunit vaccine,
   ▫ Made by linking to a carrier protein to enhance the immune response. E.g. Meningococcal vaccine.

5. **Recombinant:**
   It is prepared by cloning genes for determined antigens in yeast cells where it is expressed.
   ▫then used for vaccine preparations.
   ▫genes for determined antigens are inserted into a vector, usually a virus, that is not dangerous.
   ▫HepB vaccine, HPV vaccine
6. **Combination:**
   a. MMR, DPT and penta vaccine.
   b. To simplify administration.
   c. Reduce cost & minimize the contact.
7. **Polyvalent:**
   More than one strain of the same species.
   e.g Poliomyletitis.

S/E:

1. **Local:** pain, redness, swelling, abscess.
2. **General:** Fever, vomiting, diarrhea, headache, malaise.
3. Hypersensitivity.
4. **Rare:** Meningitis, paralysis.

- **Contra indication of Vaccination:**
  - Very high fever.
  - Hospital admission "severe disease".

- **Contra indication to pertussis:**
  - Very high fever.
  - Severe respiratory distress and general weakness.
  - Severe crying of unknown cause.
  - Severe local reaction and swelling.
  - Convulsion within 72 hours of 1st dose.

- **Contra indication of live vaccines:**
  - Pregnant ladies. C/I, MMR.
  - BCG in severe skin disease e.g., eczema.
  - Measles vaccine in active TB "pulmonary" B/C activate gonus focus.
  - Immunosuppression e.g., corticosteroid postponed the vaccine for 3 months after stopped of the steroid.

False Contra Indication:

1. Simple respiratory infections.
2. Mild vomiting and diarrhea with temperature below 38.5^0C.
3. Antibiotic and local steroid therapy.
4. +ve H/O asthma in child and family.
5. H/O jaundice.
6. Premature child and low birth weight.
7. Malnutrition.
8. Minor surgery e.g., circumcision.
9. Stable neurological condition e.g. cerebral palsy.
10. Breast feeding.
11. HIV infection.
12. Chronic disease.

Polio vaccine OPV (Sabin):

- Live attenuated, oral route.
- uphold herd immunity by cross infection
- At birth, 9,18 ms, school entry and at 15 yrs.
- OPV is the only vaccine that still needs to be kept frozen at **-20C,**
- However, OPV may be stored at **+2 to +8C for up to 6 months**

Property	OPV	IPV
Mode of administration	By mouth	Injectable
Type	Live attenuated	Inactivated
Gastrointestinal tract immunity	Yes	No
Virus shed in feces	Yes	No
Requirements for transport and storage	Strict	Not strict

DPT, DT, Td or TT

- DPT **(NOT Hexa vaccine)** at 18 months.
- DT after 2 years
- Td contain small dose of diphtheria toxoid, recommended at school entry, 15 years.
- Do not give DPT to > 2 year olds.
- Do NOT give DTP or DT to child > 7 yrs.

Measles-Mumps-Rubella vaccine (MMR)

- Recommended 2 doses; at 12,18 ms. SC or IM route.
- **Possible side effects**:
  - Soreness and erythematic
  - Fever and rash
  - Mini-measles
  - Mini-mumps
- Egg allergy is NOT a contraindication

Meningitis A,C,Y,W135 vaccine

- Most invasive disease caused by A, C, Y, W-135 serotypes. Recommended at 9, 12 months and at school entry.
- Minimum interval between doses: **8 weeks**.

Rotavirus vaccine

- Live attenuated virus
- Recommended at 2, 4, 6 months
- Oral route
- Side effects may include diarrhea, fever
- Similar to OPV, the live virus is excreted in stool and can be a risk for immune suppressed people.

Pneumococcal vaccine

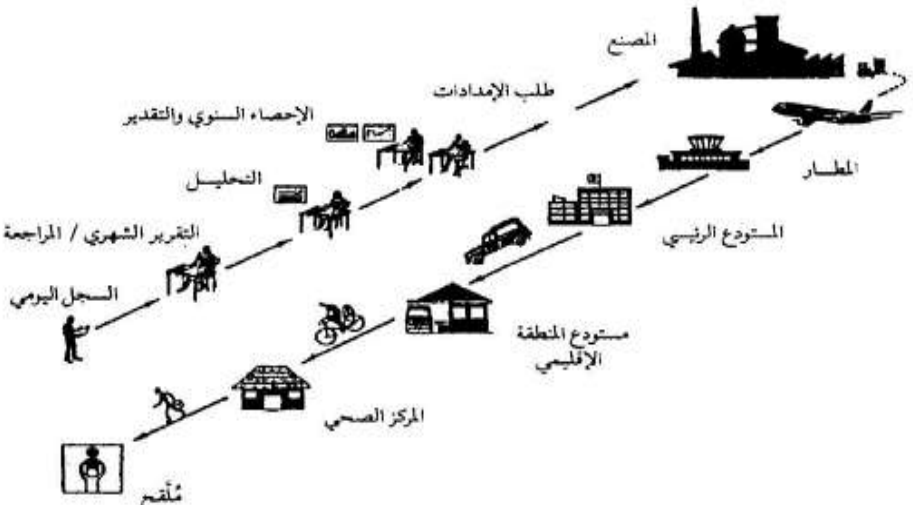
- 13-valent pneumococcal conjugate vaccine.
- Given at 2, 4, 12 months.
- **Human papiloma (HPV) vaccine**
- Recombinant DNA
- IM, 3-dose series recommended at 15 yrs old
- **Schedule**: 0, 1-2 months, 6 months after first dose

Delayed vaccinations

- Child not protected during delay.
- May require many additional visits.
- Catch up, do not restart.
- Complete the dose series.
- Minimum one month interval between doses.

Chain:

Strict system of transport and storage that assures vaccines are kept between 2 ^{0}C during delivery from the manufacture to the people to be vaccinated. " series of storage and transport links".



السن عند التطعيم Age at Immunization	vaccine الطعم												
	الدرن B.C.G	شلل الأطفال Polio		التهاب الكبد الوبائي Hepatitis B	الخناق Diphtheria	الكزاز Tetanus	السعال الديكي Pertussis	المستدمية نزلية ب Hib	العجيلة Rota	الالتهاب الرئوي Pneu	التهاب السحايا الرباعي المتقرن Meningitis A,C,Y,W135	الحصبة Measles	الحميراء Rubella	النكاف Mumps	اليوم الحميدي البشري للبنات H.P.V
		O . P . V	I . P . V												
بعد الولادة مباشرة															
شهرين															
4 أشهر															
6 أشهر															
9 شهر															
12 شهر															
18 شهر															
6 سنوات أو عند دخول المدرسة															
15 سنة أو الصف التاسع						طعم كبار T.d									
الجرعة الأولى															
الجرعة الثانية															
الجرعة الثالثة															

MCQ :

1. **<u>A young mother brought to you her baby 18m of age for vaccination , she told you that 2w back her baby admitted to hospital b/c of severe attack of bronchial asthma , on examination the baby active , a febrile with free chest what will be your opinion about vaccination of this baby</u> :**
    - **a.** Do not give any vaccine , bronchial asthma is contraindicated for all vaccine
    - **b.** Give OPV but not the others
    - **c.** Give only MMR but not the others
    - **d.** Give the all vaccines recommended in this age
    - **e.** Refer the pt to pediatrician for consultation
2. **<u>Live vaccines are used for active immunization against the following diseases except :</u>**
    - a. Measles
    - b. Yellow fever
    - c. Tuberculosis
    - d. **HBV**
    - e. Rubella

3. **<u>Which of the following vaccines is introduced recently to Libyan NPI</u>**:
    - a. MMR
    - b. DPT
    - c. BCG
    - d. OPV
    - e. **Rotavirus vaccine**

4. **<u>Regarding human papillomavirus vaccine , choose the wrong statement :</u>**
    - a. Newly introduced vaccine to Libyan NPI
    - b. **First dose given at birth**
    - c. In Libya it given to girls only
    - d. Recombinant DNA vaccine
    - e. IM rout of administration

5. **<u>Regarding MMR vaccine on statement is correct :</u>**

a. Is live attenuated v**acc**ine
b. **Given at 2,4&6m of age**
c. Is a combined **vaccine**
d. C/I in pregnancy and users of immunosuppressive drugs
e. There is no spread of the virus from vaccine to contact

6. **A mother of 2m aged infant came to the vaccination room for the 2nd shoot for her baby , when the nurse explain to her that her baby will receive 2 injection , she worry about pneumococcal vaccine b/c this is the first time one of her children receive it , now she asks what PCV is supposed to do ? what will be the answer ?**
   a. This vaccine will reduce the risk of pneumococcal bactremia
   b. This vaccine will reduce the risk of pneumococcal meningitis
   c. This vaccine will reduce the risk of all types of pneumonia
   d. This vaccines will reduce the risk of death from influenza
   e. **This vaccine will do A&B**
7. **Ali is 7 ms old , he missed 2 visit (at age of 4 & 6 ms) to vaccination center b/c travel to Egypt : asma (vaccine provider) she is confused how to deal with this baby ? your opinion will be :**
   a. Repeat the vaccination of at age 2m
   b. There is no vaccination at 7ms , ask him to come at age 9ms to complete vaccination
   c. Give missed dose of age 4m today then after one month give the other missed dose
   d. Repeat all vaccination from birth
   e. Give only OPV & HBV

8. **The following is not true regarding oral polio vaccine (OPV):**
   a. May protect contact of vaccinated people (increase herd immunity)
   b. Promotes antibody formation in the gut
   c. Rarely cause vaccine associated paralytic polio
   d. Is no longer used for routine vaccination in libya
   e. Is alive attenuated vaccine

ASSESSMENT OF DEHYDRATION

BY DR.ASMA HADIDA

2014-2015

Dehydration

Definition: Dehydration means loss of water and electrolytes from ECF, the ICF may be secondarily affected.

Etiology

1. **Diarrheal disease.**
2. **Decrease intake** e.g., starvation, coma.
3. **Vomiting** e.g., cong. pyloric stenosis, intestinal obstruction.
4. **Hyperventilation.**
5. **Excessive sweating.**
6. **Polyuria** e.g., DM, DI, CRF.

	Mild	Moderate	Severe
Infant	5%	10%	15%
Adolescent	3%	6%	9%
Infants & young children	Thirsty, alert, restless	Thirsty, restless or lethargic, irritable	Drowsy, limp, cold, sweaty, cyanotic extremities; may be comatose.
Older children	Thirsty, alert.	Thirsty, alert (usually)	Usually conscious (but at reduced level), cold, sweaty, cyanotic extremities; wrinkled skin on fingers and toes, muscle cramps.
Tachycardia	Absent	Present	Present
Palpable pulses	Present	Present (weak)	Decreased
Blood pressure	Normal	Orthostatic hypotension.	Hypotension
Cutaneous perfusion	Normal	Normal	Reduced & mottled
Skin turgor	Normal	Slight reduction	Reduced
Fontanel	Normal	Slightly depressed	Sunken
Mucous membrane	Moist	Dry	Very dry
Tears	Present	Present or absent	Absent
Respiration	Normal	Deep, may be rapid	Deep & rapid.
Urine output	Normal	Oliguria	Anuria & severe oliguria

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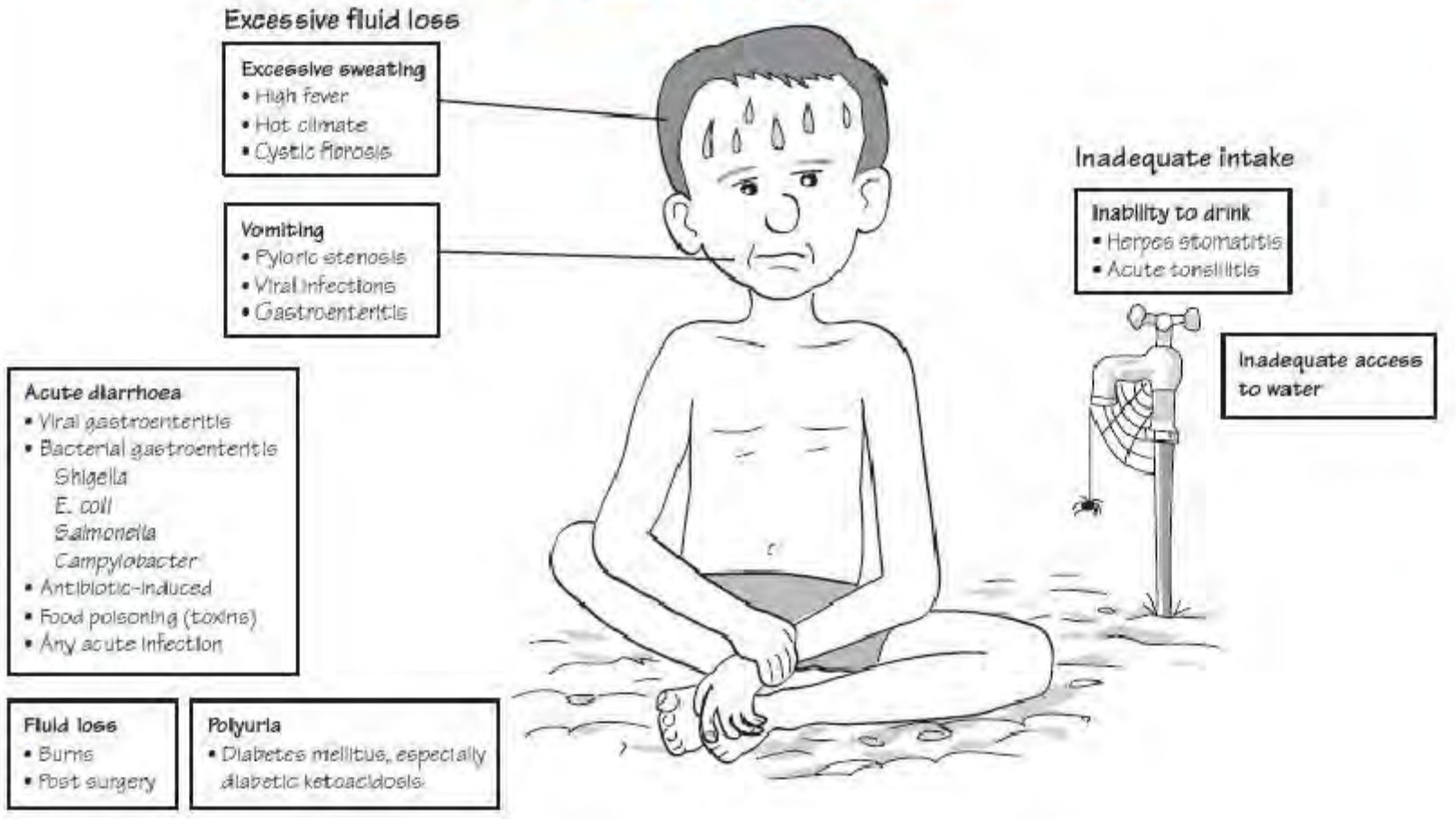
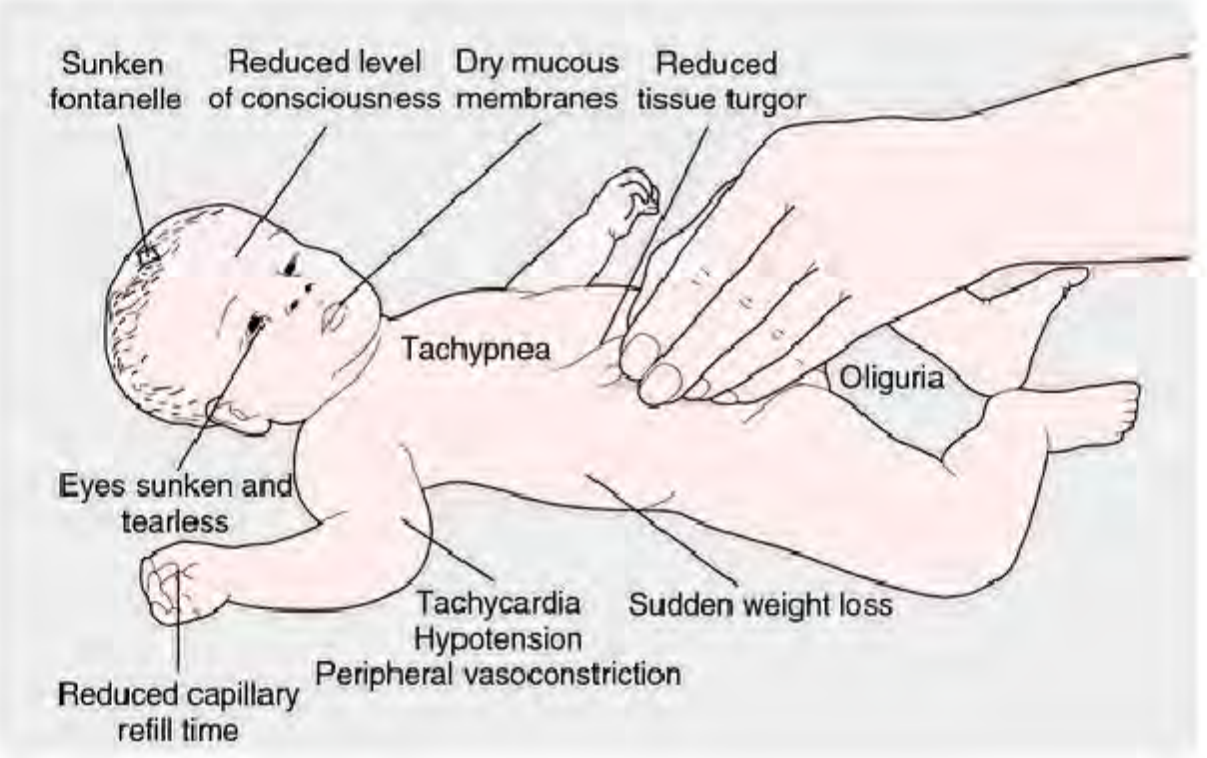
Causes of dehydration

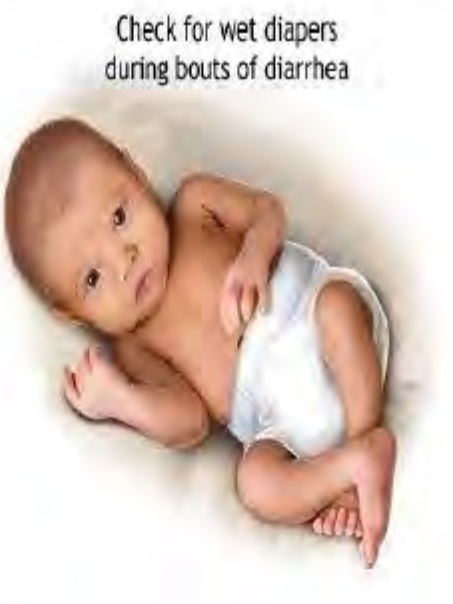




Key sign:

- General appearance
- Thirst
- Poor skin pinch
- One character (mild).
- If no tear + tachycardia "moderate"
- Poor pulse + cold extremities + severe ill (severe)

Hypotonic	Isotonic	Hypertonic
Serum Na <130 m eq/l.	130 – 150 meq/l	More than 150
Osmolality <275 m osml/l	275 – 295 m osml/l	More than 295
Excessive intake of water or hypotonic fluid during diarrhea	Balanced loss of N_a and Water	Excessive intake of hypertonic fluids during diarrhea
Net N_a^+ loss greater than water loss	No change of cellular hydration	Net water loss greater than Na loss
Over hydrated cells		Dehydrated cells
Marked collapse of ECF	Collapsed ECF	No collapse in ECF



How you can differentiate between hypotonic and hypertonic dehydration clinically ?

	Hypotonic	Isotonic	Hypertonic
Manifestation of I.C.F affection			
Tongue	Moist	Dry, thirsty	Very dry "woody" thirst
Brain	Lethergic	Irritable	Convulsion, irritable, hyper reflexia
Manifestation of ECF affection			
Shock	Rapid	Slowly	Absent
Skin turgor	Marked loss	Moderate loss	Normal or doughy
Fontanelle	Marked depressed	Moderately	Mild sunken

Steps of assessment of de hydration :

1. General condition :
   - Well or ill
   - Lethargic or irritable
   - Surrounding and canula (connected to IV fluid with infusion pump or not, type of IV fluid)
2. anterior fontanels (depressed or at level)
3. eyes (sunken or not) , if crying with tears or not
4. mucous membrane (dry or not)
5. tongue woody or not
6. extremities (cold or warm)
7. pulse
8. capillary refilling time (from sternum)
9. skin turger
10. urine out put
11. reflexes (do not do it except in hypertonic de hydration)
12. BP

Replacement Therapy

1. **Mild and moderate dehydration:** Treated with ORS.
2. **Severe dehydration:** Treated with IV rehydration.
   - IV fluid requirement:
     - Treat shock.
     - Deficit, maintenance, ongoing losses.

1. Mild to moderate dehydration:

- Type of fluid ORS.
- Amount 75 ml/kg "but if child wants more give more over 4 hours.
- How to give ORS?
  - One teaspoon full, 1-2 minutes for child below 2 years.
  - Freq. sips from cup for older child.
- **Mild dehydration:** 50 ml/kg over 4 hours then 10 ml/kg after each passed diarrhea or vomiting, 1st dose to correct dehydration, 2nd dose to avoid dehydration.
- **Moderate dehydration:** 100 ml/kg over 6 hours, then 10 ml/kg after each passed diarrhea or vomiting "you can manage at home but better at hospital".

C/I of ORS:

- Coma.
- Persistent vomiting.
- Abd. Distension.
- Paralytic ileus.
- Rapid loss of stool.
- Severe metabolic acidosis.
- CHO mal absorption.

If baby vomits, after giving ORS, wait 10 minutes and give at slower rate spoon every 2-3 minutes, if still vomit, may need IV fluid.

Liquid: e.g., juice, tea, are not suitable for use in ORS therapy because they may cause osmotic worsening of diarrhea and they contribute to hyponatremia, so water only is used.

Advantage of ORS:

- Cheap, easy to administer, used for all types of dehydration, can be given to any age, also any type of diarrhea.
- During diarrhea, do not stop breast feeding or any other type of feeding, because in GE 60% of food will be absorbed.
- Stop ORS when child doesn't want it any more.
- ORS should not be given after 24 hours of preparation.

Old formula of ORS	New formula of ORS (WHO)
Na = 90 mmol/l. Cl = 80 mmol/l. Bicarbonate = 30 mmol/l or citrate. K = 20 mmol/l. Glucose = 111 mmol/l *Dissolve in one liter of water, when added together, osmolarity 341 higher than body osmolarity, which may cause more diarrhea.*	Na = 70-75 mmol/l Cl = 65 mmol/l Bicarbonate or citrate = 20 mmol/l K = 10 mmol/l Glucose = 75 mmol/l. *Total osmolarity 245 less than body osmolarity, so less diarrhea.*

Note:
Glucose is need for N_a^+ absorption "also Cl and AA, and is used as energy with activation of N_a – K atpase pump lead to absorption of water and N_a."
If new formula not available, take old formula and dissolve in double amount of water.

Admission needed if:

- No home care.
- Continuous vomiting, ORS refusal or ORS not enough.
- Other illness (meningitis, pneumonia).
- Severe dehydration.
- Diagnosis not clear.

Calculation of IV fluid

1. Maintenance: Daily requirement of fluid

Less than 1 year	More than 1 year
1st day = 60-75 ml/kg/24 hours.	1st 10 kg = 100 ml/kg/24 hours.
2nd day = 90 ml/kg/24 hours.	2nd 10 kg = 50 ml/kg/24 hours.
3rd day = 90 ml/kg/24 hours.	Subsequent kg = 20 ml/kg/24 hours.
4th day = 120 ml/kg/24 hours.	
5th day = 150 ml/kg/24 hours.	
5th day to 1 year = 150 ml/kg/24 hours.	

2. Deficit: %age of body loss

	Mild	Moderate	Severe
Less than 1 year	5%	10%	15%
More than 1 year	3%	6%	9%

D = % x weight x 10

3. Ongoing loss = number of motions x weight x 10

Day 1 = M + D over 24 hours
Day 2 = M + Ongoing loss
} In hypo and isotonic dehydration

Management of shocked patient:

- **ABC :**
  - Call for help.
  - Cleaning air way.
  - Breathing by O_2.
  - Insert 2 large cannula for IV fluid.
  - Give 20 ml/kg IV fluid (normal saline over 20 minutes), it can be repeated 3 times, if not improved call ICU doctor.
- 1st day = M+D – treatment of shock.
- 2nd day = M + Ongoing loss.

Correction of hypertonic:

- The serum sodium concentration should not decrease more than 12 mEq/l every 24 hours. The deficits in severe hypernatremic dehydration may need to be corrected over 2 to 4 days.
- 1st day = 2/3 M + 1/2 D.
- 2nd day= 2/3 M + 1/2 D.
- **Type of IV Fluid:**
  - D $\frac{1}{4}$ saline.

Hypertonic dehydration:

Overly rapid treatment of hypernatremic dehydration may cause significant morbidity and mortality. ***Idiogenic osmoles*** are generated within the brain during the development of hypernatremia. These idogenic osmoles increase the ***osmolality*** within the cells of the brain, providing protection against ***brain cell shrinkage*** secondary to movement of water out of cells into the hypertonic extracellular fluid. These idiogenic osmoles dissipate slowly during correction of hypernatremia. With rapid lowering of the extracellular ***osmolality*** during correction of hypernatremia, there may be a new gradient created that causes water movement from the extracellular space into the cells of the brain, producing ***cerebral edema.*** Possible manifestations of the resultant ***cerebral edema*** include seizures, brain herniation, and death.

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Types of IV Fluids:

- Depend on type of dehydration:
  - **Hypotonic dehydration:** 1/2 D/s if N_a = 120-130, but if N_a less 120, normal saline.
  - **Hypertonic:** 1/2 D/s or 1/5 D/s.
  - **Isotonic:** 1/5 D/s.
- **Normal saline:** 0.9% N_aCl means every 100 cc of water contain 0.9 gr N_aCl. Na ions = 150 mmol/l, Cl ions = 150 mmol/l.
- **1/2 Dextrose saline:** Each 100 cc water contain 0.45 gr N_aCl and 5 gr dextrose, Na = 77 mmol/l, Cl = 77 mmol/l.
- **1/5 Dextrose saline:** Each 100 cc water contain 0.18 gr N_aCl and 5 gr dextrose, Na = 30 mmol/l, Cl = 30 mmol/l.

Complication of fluid therapy:

- Hypervolemia lead to edema and HF.
- Electrolytes disorders:
  - **Hypo K:** Aggravated by rapid correction of acidosis lead to intracellular shift of K.
  - **Hyper K:** Aggravated by acidosis and excessive K infusion in presence of renal impairment.

Complication of dehydration:

1. Shock.
2. ARF.
3. MA **due to:**
   a. Bicarbonate loss in motions.
   b. Renal impairment.
   c. Ketosis from starvation.
   d. Lactic acidosis from hypovolemic shock.
4. **Metabolic alkalosis:** If vomiting more severe than diarrhea.
5. Electrolytes disturbances.
6. Convulsion.
7. DIC.
8. Complication of fluid therapy.

Treatment of metabolic acidosis:

Give N_a bicarbonate N_aHCO_3 = base deficit x 0.3 x body weight (From ABG).

- Do not mix NaHCO3 with Ca-gluconate, because it leads to reaction.
- N_aHCO_3 = 1/2 of it given slowly IV and other 1/2 in drip.
- If rapid infusion lead to transformation of bicarb to carbonic acid and give water + CO_2. CO_2 leads to sudden vasodilation and rupture of blood vessels (interventricular Hge).

For Hypo K:

Give KCl 15% in saline bottle.

- Never give KCl by I.V route directly.
- Do not give K unless serum K level is known.
- Do not give K unless child passes urine.

Monitoring Therapy

- **Vital signs.**
  - Pulse
  - Blood pressure
- **Intake and output**
  - Fluid balance
  - Urine output and specific gravity
- **Physical examination**
  - Weight
  - Clinical signs of depletion or overload
- **Electrolytes.**

Infants are more susceptible to dehydration than adults, Why?

- Relative excess of total body water.
- Higher daily requirement of water in infant (150 ml/kg/24 hr).
- Higher freq. of diarrheal diseases in infants and children.
- Smaller metabolic reserve of water and electrolytes.
- Limited power of the kidney to concentrate urine.
- Greater surface area:weight ratio leading to greater insensible water loss.

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CNS by Dr.Asma hadida

2015_2016

Hydrocephalus

Definition : Dilatation of the ventricular system due to **either** :

- Obstruction of CFS flow within the ventricular system.
- Impairment of its absorption within the subarachnoid space . **"↑ pressure in ventricular system".**

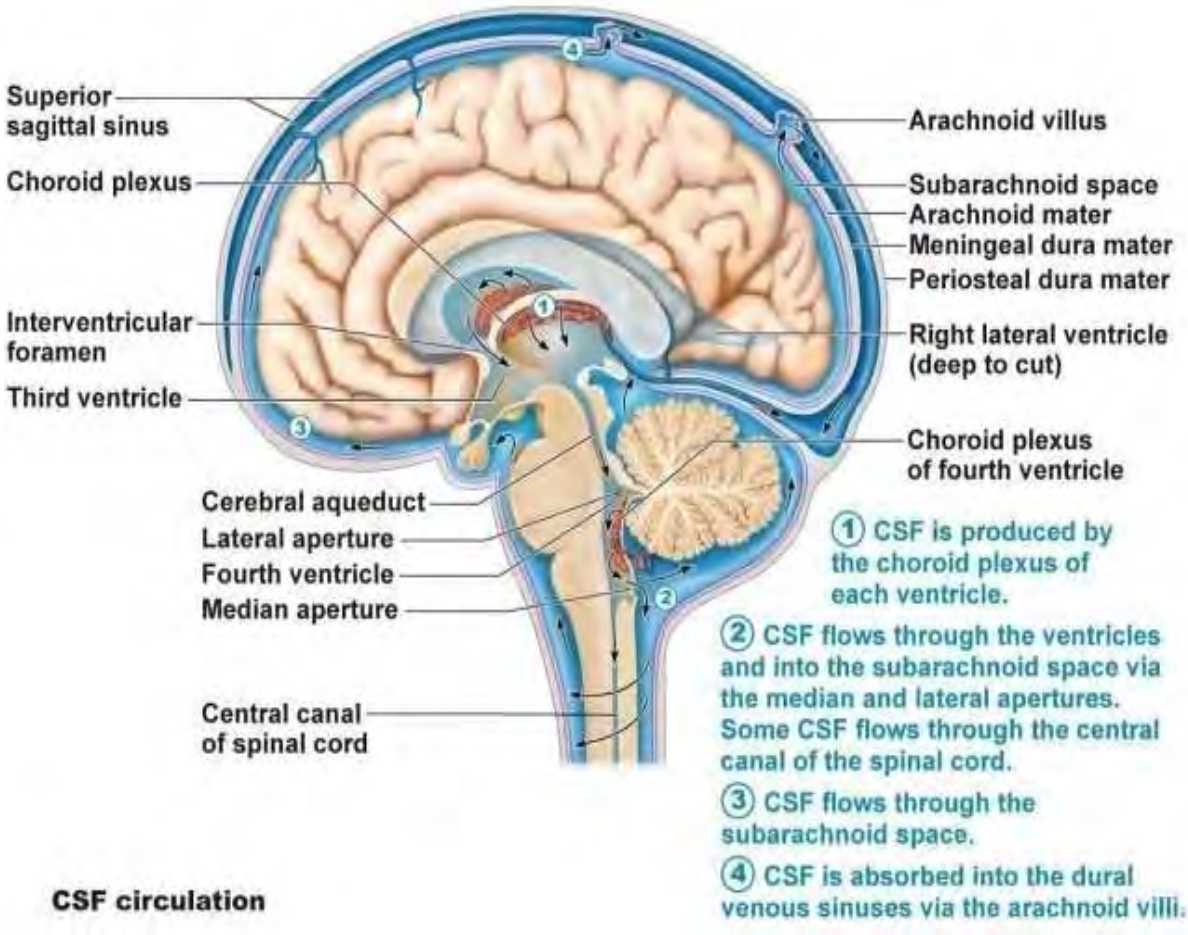
CFS amount in infant : 50 ml
CFS amount in adult : 150 ml

CSF circulation :

✓ ***Secretion*** : Choroidal plexus in the ventricles "Lateral ventricle mainly"

✓ ***Circulation*** :

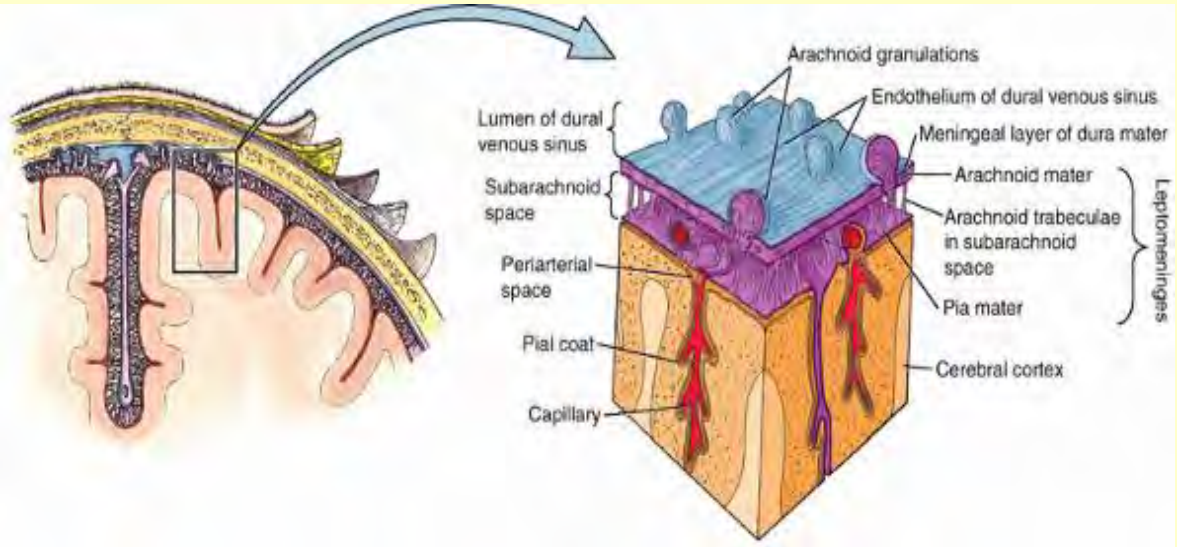
Lateral ventricle
↓ Foramina of monro
3rd ventricle
↓ Aqueduct of syvius
4th ventricle Subarachnoid space
↓ Foremen of luschka and magendie
Subarachnoid space



CSF circulation

Re absorption :

CSF is re absorption into venous circulation from sub arachnoids' spaces by arachnoids' villas.



Etiology and Types

❶ Obstructive hydrocephalus "Non communicating"

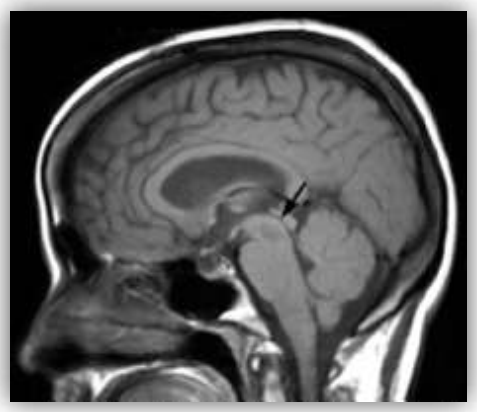
Obstruction of the CSF flow within the ventricular system

❷ Non Obstructive hydrocephalus "Communicating"

No obstruction in the CSF flow within the ventricular system but ventricular dilatation occurs due to impaired absorption of CSF within subarachnoid

1) Obstructive "non communicating" hydrocephalus :

1. **Congenital**: Aqueduct stenosis "commonest cause"
   · Congenital stenosis or due to toxoplasmosis.
2. **Traumatic**: Post-hemorrhagic " especially in premature"
3. **Inflammatory**: Post meningitis gliosis
   "T.B, pneumococcal, mumps"
4. **Neoplastic**: Post-fossa tumor.



★**Dandy walker malformation:**

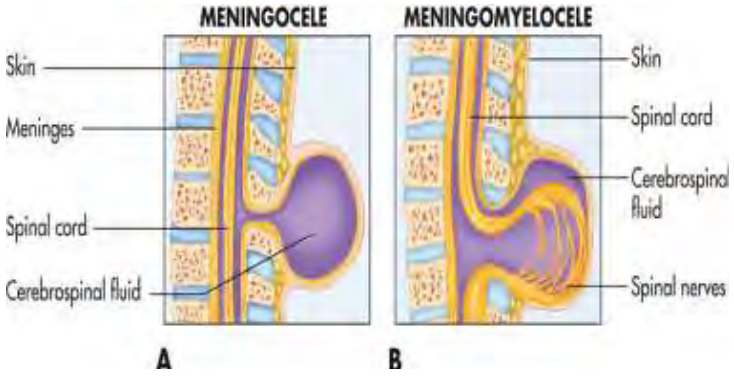
- Obstruction of foramina of luschka and magendie , cystic dilatation of 4th ventricle usually with agenesis of cerebellar vermix .

2) Non obstructive "communicating" hydrocephalus : (Impaired absorption)

1. **Congenital**: Arnold-Chiari malformation « Type 2 » .
2. **Traumatic**: Subarachnoid hemorrhage.
3. **Inflammatory**: Post. meningitic gliosis.
4. **Neoplastic**: Leukemia infiltration.
5. **Achondroplasia**: due to narrow jugular formen.

⁂ ***Arnold-Chiari malformatio***

Downward displacement of cerebellum and medulla; and pushed through the opening of the base of skull associated with meningocele or meningomyelocele.



N.B

Hydrocephalus may be **rarely** due to over production of CSF due to choroidal plexus papilloma

Clinical Features

A) Before closure of sutures and fontanels "Infancy"

1. General examination:

- Head examination :
  - **Circumference** : Progressive increase in all skull diameters.
  - **Scalp skin** : Thin and shiny.
  - **Scalp vein** : prominent.
  - **Ant. Fontanel** : Widely opened " Bulging of AF".
  - **Sutures :** Widely separated.
  - **Face** : Globular and prominent forehead.

- **Eye :**
  - **Sunset** appearance "downward displacement of the eye due to distortion of nerve supply.
- **Back** :
  - For any swelling **"meningocele or meningomyelocele.**

2. Neurological examination:

- Motor system:
  - **Tone :** Spasticity of limbs « compression on motor area may be hypotonia with meningomyelocele ».
  - **Power** : For evidence of paralysis.
  - **Reflexes** : Exaggerated.
  - **Cranial nerves** : Optic nerve atrophy in chronic cases.

* Convulsion and delayed motor and mental development may be present.

A) After closure of sutures "Older children"

1. General examination:

- Head enlargement is less evident because sutures do not separate.

2. Neurological examination:

- Signs of ↑ I.C.P are marked.

✓ Sings of ↑ I.C.P :

1. Sever bursting headach.
2. Blurring of vision.
3. Projectile vomiting.
4. Cushing response "**hypertention & bradycardia**".

Other finding :

- **Macc wan sign: percussion by one finger (resonant)**
  _ - prominent occipit in Dandy Walker and fore shortened occipit in Arnold Chiari.
- **Craniotapes** in all bones.
- Fundus examination for chorioretinitis in toxoplasmosis.
- Lower limb examination.
- **Cerebellar ataxia** Dandy walker malformation.
- May associated with **obstructive uropathy.**

Diagnosis :

Clinical pictures.
Cranial x-ray :
- **Before closure of sutures** : wide fontanels, wide separation of sutures, cranio-facial disproportion with large cranium.
- **After closure of sutures :** ↑ I.C.P " Beaten silver appearance , wide sella".

Transfacial U/S.
CT scan and MRI :
- Detect ventricular dilatation.
- Detect degree of cortical atrophy.
- Detect the cause.

Simultaneous lumbar and ventricular manometry.
- Normally both are equal.
- Ventricular pressure > spinal pressure.
  " in obstructive hydrocephaly"

➲ **Trans illumination** :
"hydrocephaly x-ray"
Film on one side of head → torch on other side of head → red light on x-ray film indicate hydrocephaly (**+ve test**).

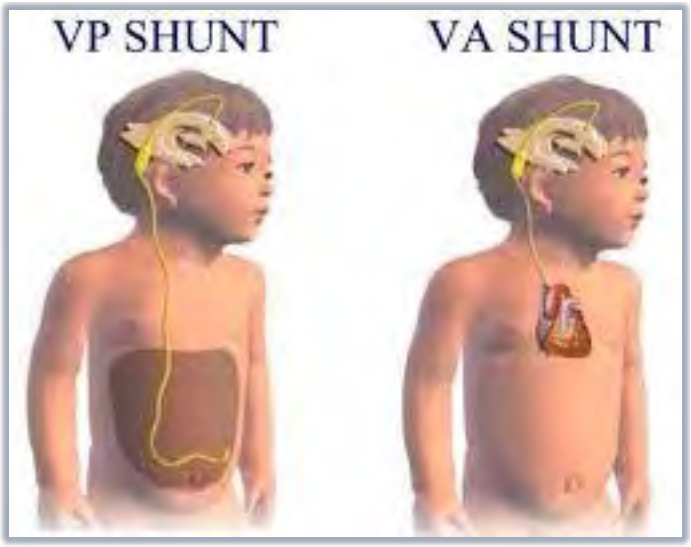
∗ **Communicating hydrocephally** : all ventricles are dilated.
∗ **Obstructive hydrocephally** :Dilatation only proximal to the site of obstruction.

Treatment

1. Medical : Not successful alone.
- Carbonic anhydrase inhibitors , acetazolamide " Diamox"
- Furesemide.

* May be effective in mild and slowly progressive hydrocephaly

S/E : - Transient effect
- Electrolyte disturbances.



2. Surgical :

- Removal of cysts or tumor that cause obstruction.
- Shunt operation " **main treatment**" :
- Types : ❶ Ventriculo-peritoneal : most common.
  ❷ Ventriculo-atrial : rare used.
  B/C cause arrhythmia
  ❸ Ventriculo-pleural.
- Complication :

① **Infection :** commonly with staph. Epidermis, may cause septicemia or meningitis.
② **Obstruction** , kinking or separation.
③ Shortening of the tube with time that necessitate to repeat the operation ."Dislodgment"
④ **Shunt nephritis "** Immune complex mediated"
⑤ Infective endocarditis
⑥ Perforation of liver or intestine.

Prognosis :

With early management prognosis is better as regard life expectancy , motor and mental development.

✱ **Three types of shunt according to age :**

- ✓ Low pressure.
- ✓ Medium pressure.
- ✓ High pressure: if used with low pressure hydrocephalous leads to intracranial bleeding due to rapid shifting of CSF.

Assessment of child with shunt :

Ask about *:*

- Headache
- Vomiting
- Visual problem
- Measure BP

⁂ **D.D from causes of macrocephally** (HC > 2 SD above mean)

A) Cranial causes :

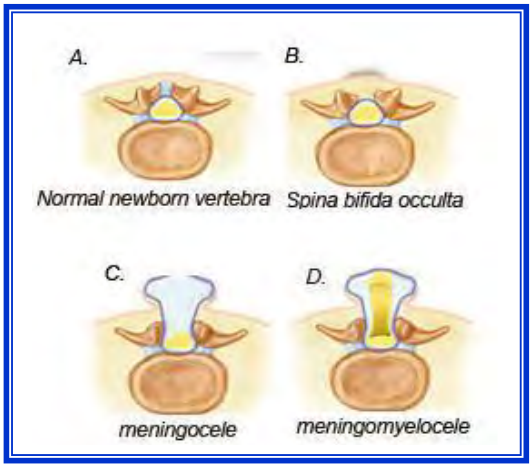
1. Constitutional.
2. Achondroplasia.
3. Familial.
4. Chronic hemolytic anemia.
5. Rickets' .
6. Osteogenesis imperfect.

B) Intracranial causes :

1. Hydrocephalus.
2. Tuberousclerosis.
3. Galactosemia.
4. Cerebral tumor.
5. Subdural hematoma.

NEURAL TUBAL DEFECTS

- Result from failure of normal fusion of the neural tube at the end of the **4th week of gestation**.
- The incidence has decrease due to improved maternal nutrition.
- If a mother has a fetus with NTD , folic acid supplements will decrease the risk of having a 2nd affected fetus.
- Pre conceptual folic acid supplements are recommended for all pregnancies.



Spina bifida occult :

- Failure of fusion of vertebral arch only.
- It is associated tuft of hair , lipoma, birth mark or a small dermal sinus.
- It could be also discovered by x-ray.
- There may be **diastematomyelia** → fibrous band or bone tethering and dividing the spinal cord → it may cause neurological problems of the bladder and L.L.

✱ C/F :

- Weakness
- Numbness of lower limb.
- Decrease sensation "ulceration"
- Repeated UTI.
- Reflux nephropathy.
- Renal failure.

✱ Diagnosis:

- **CT and MRI** : show details of the underlying lesion.
- **Neurosurgery:** is needed for disstematomelia.

Meningocele and meningomyelocele
(Spina bifida cystic)

- **Meningocele**: skin cover defect , it usually has good prognosis after surgical repair.
- **Meningomyelocele** : It is an open defect , not covered by skin , the spinal cord and surrounding nerves are exposed.

✱ C/F :

- Paralysis of legs (weakness, wasting, hypotonia, decrease or absent reflexes)
- Muscle imbalance may cause hip dislocation or talipes "**club feet**"
- Sensory loss → may lead to ulceration and bed sores.
- Bladder denervation "neuropathic bladder" →Retention of urine → Reflux nephropathy .
- Bowel denervation : chronic constipation.
- Scoliosis.
- **Hydrocephalus** : Arnold Chari malformation.

- Examination of the back is very important :
  What region is the swelling in? Size ? Leaking ? Stitches ?

* It is commonly located over the lumbar vertebrae, the more number of vertebrae it involves or the higher the swelling → much worse prognosis.

Management :

1. Lesion in back needs surgical closure especially in meningomyelocele , they will also need a ventricle peritoneal shunt for progressive hydrocephalus.
2. **Physiotherapy :** to prevent joint contractures and strengthen paralyzed muscles.
3. **Skin care**.
4. **Urinary catheterization :** is needed for bladder denervation.
   ✓ Urine sample should be checked regularly for UTI. ✓ Continuous antibiotic prophylaxis is needed , the child should be monitored for early evidence of hypertension and renal failure.
5. **Bowel denervtion** needed regular toileting and laxative.

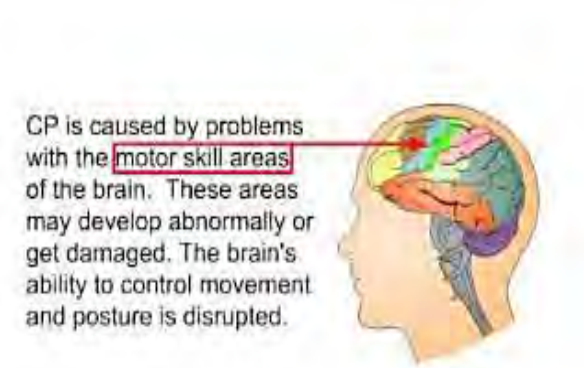
If surgery not done, child will die from :

- Meningitis.
- Progressive hydrocephalus
- Renal failure.

Little disease

Static encephalopathy :

- Non progressive disorder of posture and movement **" motor system"** , non curable involved immature brain of unknown etiology.
- Often associated with epilepsy and abnormalities of speech and vision.
- Can be compared from **neurodegenerative disease** which is progressive disease and genetic cause.
- The lesion **non-progressive** but clinical manifestation increase with cerebral maturation.



Etiology

1. Perinatal 80%
- CNS malformation.
- TORCH infection.
- Placental insufficiency.

2. Natal 10%
- Birth asphyxia.
- Birth trauma.
- I.C.Hge

3. Post natal 10%
- Kernicture
- Meningitis.
- I.C.Hge.
- Sever head trauma.
- Metabolic, e.g. : PKU – hypoglycemia

Clinical classification

According to site of damage to specific motor pathway

1) Spastic CP 70%

Damage to the upper motor neuron **"pyramidal pathway"**

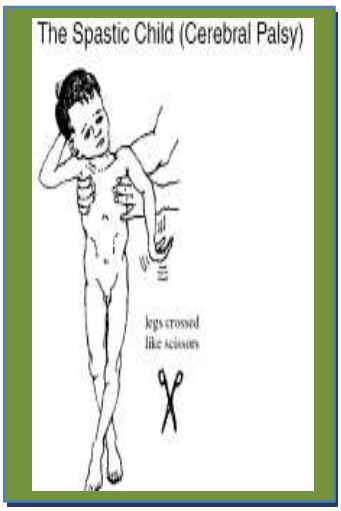
Symptoms :

- Feeding difficulty.
- Abnormal posturing.
- delayed developmental « **esp. motor milestone** »
- Abnormal gait.
- Early hand preference.

Sings :

- Limb tone is high **" Clasp - Knife spasticity "**
- Hyper-reflexia.
- Weakness of muscles.
- Extensor planter response **"+ve Babinski"**

* Before spasticity appears , there may be truncal hypotonia.



According to the power " degree of motor area damage"

Spastic CP is classified into **3 types**:

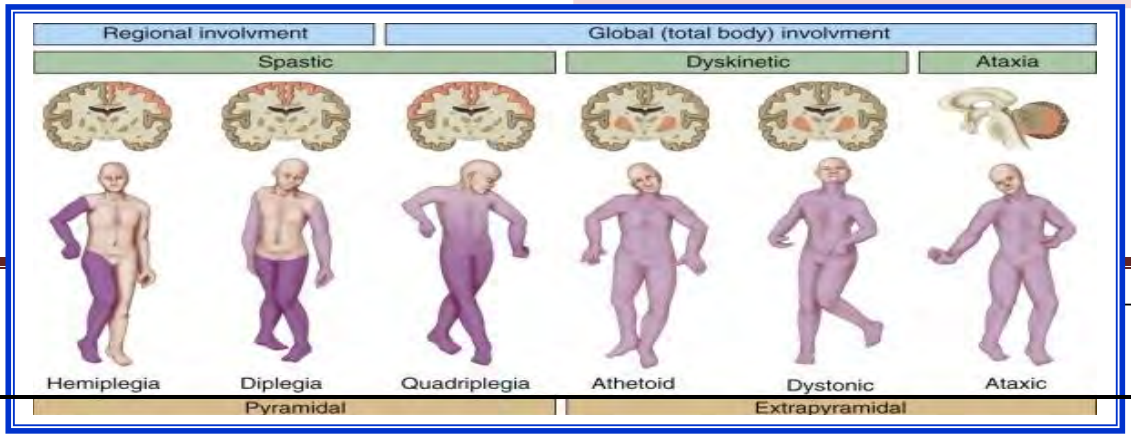
1. **Quardri plegic** : Paresis affects both sides of the body (UL & LL) , the arm are affected equal to or more than the leg. "The most sever associated with mental retardation ".
2. **Hemiplegic** : Paresis affects one side of the body and the arm is usually affected more than legs.
3. **Di plegic** : Paresis affects both lower limbs with arms less affected or not affected.

2) Dyskinetic "dystonic or athetoid" CP 10%

- Irregular involuntary movement of some or all muscles group.
- Athetosis is commonest abnormal movement it is purposeless snake like distal movement.
- Extra pyramidal **"dyskinetic"** common cause : kernictures and birth asphyxia.

Criteria :

- Hypotonic then replaced with time hypertonic and rigidity.
- Supra nuclear palsy → feeding and speech disorder.
- May be deafness.



Persistence of primitive reflexes indicate brain damage , e.g : CP.

Presentation of CP

1. Delayed mental milestones.
2. Delayed motor milestones.
3. Feeding difficulties.
4. Abnormal gait.

- Onset of CP usually detected after 1 year
- Most common complication of both CP and neurodegenerative disease → social problems.

Risk factors for CP :

- Small for gestational age.
- I.U bleeding.
- Infection.
- I.C.Hge.
- Congenital malformation.
- Neonatal hypoglycemia.
- Kernicture.
- Birth asphyxia.

3) Ataxic CP 10%

- Usually due to genetic cause.

Presented with :

- Truncal ataxia.
- Nystagmus.
- Staccato speech.
- Hypotonic.
- Intention tremor.
- In coordination of voluntary movement.

4) Atonic CP <10%

- Hypotonic → Floppy infant.
- Preserved deep tendon reflexes may be exaggerated.
- UMNL.
- No tremor.
- No in coordination.

5) Mixed CP <1%

Associations

1. Mental retardation 50% .
2. Epilepsy 30% .
3. Impaired learning.
4. Impaired vision.
   - ✓ Strabismus 30%.
   - ✓ Ampyliopia.
   - ✓ Nystugmus .
5. Supra nuclear bulbar palsy .
6. Speech disorder.
7. GERD.
8. Behavioral disturbances.
9. FTT.
10. Communication disorder.

Investigation

1. **CT and MRI** :
   Detect degree of brain atrophy.
   May detect the cause e.g. malformation.
   Rule out brain tumor and degenerative disease.
2. TORCH screen.
3. Metabolic screen.
4. **For association :** audiometry , fundus examination , EEG.

Management

"Rehabilitation"

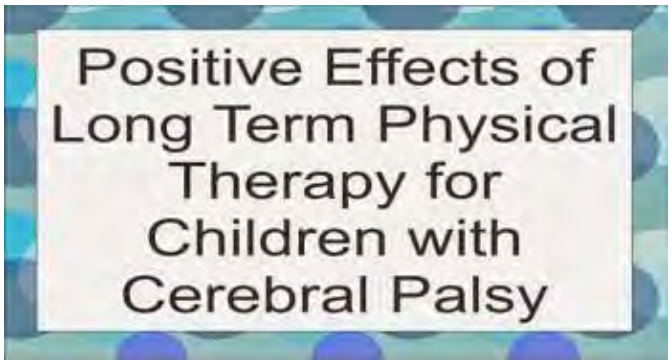
- Multidisciplinary (**need a team**)
- Pediatrician.
- **Physiotherapy** : To improve motor skills and prevent muscle contractures and abnormal posturing.
- Psychologist.
- Speech and language therapy.

⁂ The use of drugs in CP is very limited

- Baclofen « muscle relaxant » could be used to ↓ spasticity (C/I in epilepsy) .

✱ Degree of functional disability:

Class I : No limitation of activity.
Class II : Slight to moderate limitation.
Class III : Moderate to great limitation.
Class IV : No useful physical activity.



SEIZURES (CONVULSION)

Clinical event that results from an abnormal excessive neuronal activity and characterized by any of the following:

- Alternation of consciousness.
- Abnormal motor activity.
- Behavior abnormalities.
- Sensory disturbances or autonomic dysfunction.

*** It should be viewed as a symptom of underlying disease process.**

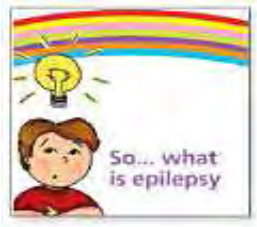
Etiology

1. **Neonatal convulsion** : see neonatology.
2. Febrile convulsion.
3. **Non recurrent "occasional" symptomatic convulsions.**

* one or more convulsive fit with acute neurological disease and stop with improvement of the disease.

CNS infection: meningitis , encephalitis , brain abscess , I.C. Hge , I.C.tumor.
Acute cerebral edema : hypertensive encephalopathy or head trauma.
CVA : thrombosis , embolism.
Toxic : tetanus , post immunization "DPT" , GE "**Shigellosis , Salmonellosis".**
Metabolic : hypoglycemia , hypocalcaemia , hypomagnesaemia.

4. **Reccurent convulsion.**
5. **Neurocutaneous disease** : neurofibromatosis, T,S , sturge weber syndrome .



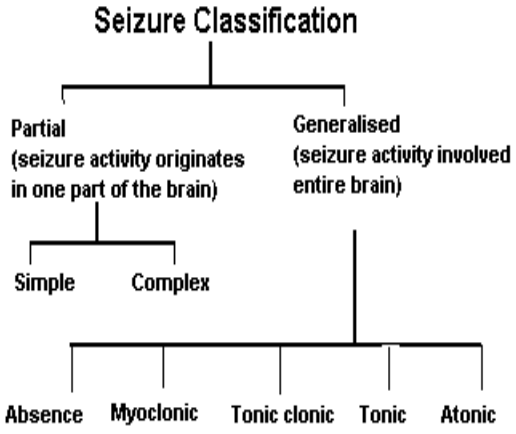
A) Partial "Focal"
1. Simple.
2. Complex.

B) Generalized
1. Tonic colonic.
2. Infantile spasm.
3. Petite mal **"absent"**
4. Myoclonic.

Etiology

1. Idiopathic "cryptogenic" 80%.
2. Genetic
3. **Symptomatic** :

A) Heredity cause :
- **Neuro metabolic** , e.g.: aminoacidopathy.
- **Neurodegenerative brain disorder**, e.g.: Lipidosis.
- **Neuro cutaneous disorder :**
  - ✓ Tuberous sclerosis.
  - ✓ Sturge Weber disease.
  - ✓ Neurofibromatosis.



B) Acquired causes:
1. **Prenatal** : I.U infection (CMV , Rubella , Toxoplasmosis).
2. **Prenatal** :
   - During pregnancy : placental insufficiency.
   - During labour : HIE , I.C.Hge .
3. **Postnatal** :
   - **CNS infection:** (post meningitis, post encephalitis).
   - Kernicterus.
   - Trauma.
   - **Metabolic disorder** : hypoglycemia, hypocalcemia.

Table 1. Clinical factors associated with idiopathic versus symptomatic epilepsy.

Idiopathic epilepsy	Symptomatic epilepsy
• Normal development	• Developmental delay
• Normal neurological examination	• History of brain injury
• Family history of epilepsy	• Abnormal neurological exam results
• No history of brain injury (e.g., head trauma, meningitis)	• Dysmorphic features
• Characteristic EEG abnormalities	• Other congenital malformations

Partial seizures

(Limited to one hemisphere)

1. Simple partial:

- **Pre ictal** : aura may be present.
- **Ictal phase** :
  - Twitching or jerking of one side of the body one arm or one leg "Partial"
  - Consciousness is usually retained.
  - May proceed to generalized tonic clonic
- **Post ictal** :
  - Weakness of the involved side **"Todd paralysis"**e.g.: Benign childhood epilepsy "Rolandic epilepsy".
    - Common in male , the group 5-10 years.
    - May be confused with tics.
    - Rarely with sensory symptoms.
    - Excellent prognosis for spikes.

↳ **Diagnosis :**

- **EEG** : Centro-temporal spikes.

↳ **Treatment** : Carbamazepine "Tegretol".

2. Complex partial: Epileptic discharge in the **temporal lobe**.

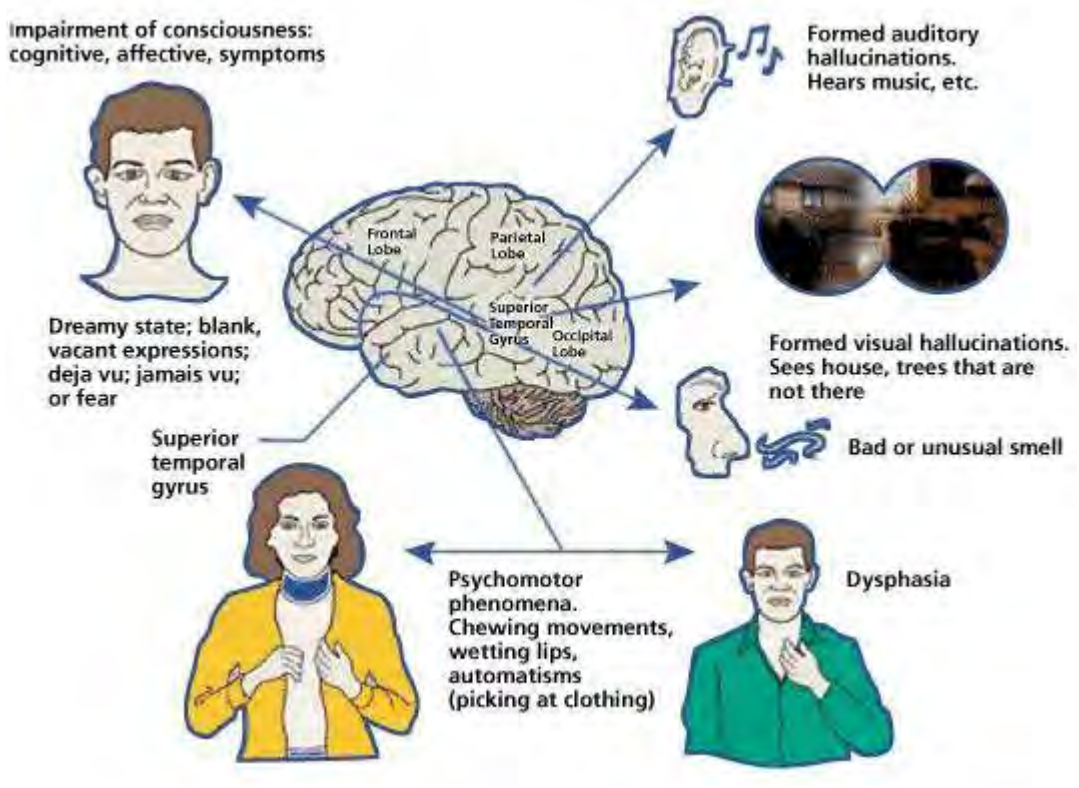
- **Pre ictal** : aura common.
- **Ictal phase** :
  - **Few minutes** ; Impaired consciousness usually without falling in most cases , the child unable to describe the period of altered consciousness, behavioral changes.
- **Post ictal** :
  - Amnesia for the attack.

↳ **Diagnosis** :

- **EEG** : Fronto-temporal lobe "sharp wave or spikes.

Partial seizure

fore the seizure. Jerking of the right half of the face. The patient is fully conscious.



Generalized seizures

1. **Absence "Petite mal"** :

- **Pre ictal** : No aura.
- **Ictal phase** :
  - There cessation of motor activity or speech with blank facial expression for 5-20 second then continuation of original activity as if nothing happened.
  - The attack can be triggered by **hyperventilation**.
- **Post ictal :** No post ictal manifestation

* This form of epilepsy is usually idiopathic or genetic and has **good prognosis.**

Diagnosis :

- **EEG** : 3 spike per second.

Treatment :

- Ethosuxamide is D.O.C
- Sodium valproate "Depakin".

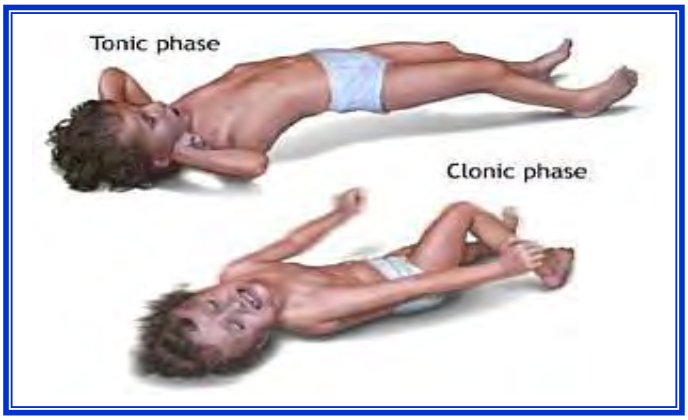
2. **Generalized tonic clonic "Grand mal"** :

- **Pre ictal** : Aura.
- **Ictal phase** :Sudden loss of consciousness and falling
  - ***Tonic phase***:
    - Limbs are extended, eyes are rolled up and the head is thrown backward or to one side.
    - Cyanosis of face, biting the tongue, urination or defecation may occur during this phase
  - ***Clonic phase***:
    - Starts within minutes with repetitive contractions and relaxation of all muscles.
- **Post ictal :** Deep sleep for hours.

* This form of epilepsy is usually idiopathic or genetic and has good prognosis.

Treatment :

- Sodium valproate "Depakin".
- Phenobarbitone



3. **Myoclonic :**

- Repetitive seizures consisting of sudden brief often symmetrical muscular contractions "flexion or extension" and falling with tendency to cause injuries to the face or to the back of his head.
- ***Consciousness intact***.

Diagnosis :

- **EEG** : 4-6 per second irregular poly spikes wave.

Treatment : Clonazepam , Ethosuxamide.

* It may be idiopathic or symptomatic e.g.: with degenerative brain disorder.

4**. Infantile spasm** :

(Serious type of myoclonic epilepsy)

- Infant 4-8 month.
- Brief symmetric contractions usually flexion of neck, trunk and extremities lasting 15-30 minute, several times per day.

"**Salam fit appearance**" common in male.

- 15% of cases have underlying neurological disorder.

Diagnosis :

- **EEG :** Diagnostic → hypsarrhythmia → high voltage with multifocal spike and wave discharge.

Treatment : • ACTH "D.O.C".

• Vigabartine : In infant <3m with tuberous sclerosis.

Prognosis :

Poor except if idiopathic cause.

Status epilepsy

Seizure lasting more than 30 minute or repetitive seizure without return to consciousness.

Causes :

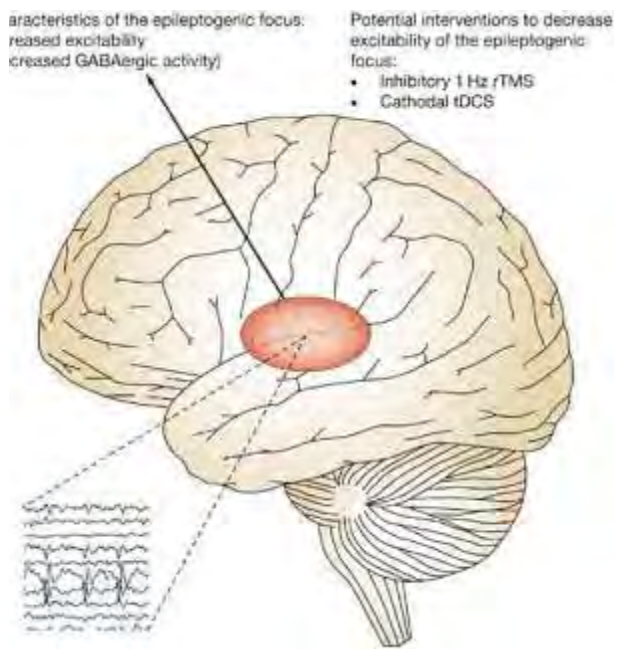
1. Sudden withdrawal of anti-convulsion therapy in epileptic pt.
2. prolonged febrile seizure "commonest"
3. CNS abnormality ,e.g: Encephalitis or tumor.
4. Metabolic disorder , e.g Hypoglycemia or Hypocalcemia.
5. Inborn error of metabolism.

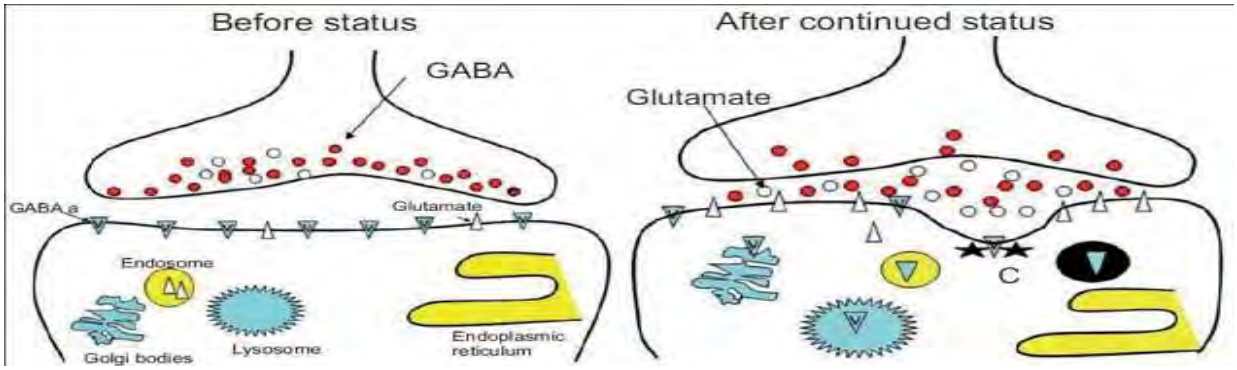
Complication:

1. **Neurological** : Cerebral edema, ischemia, Hge, and damage.
2. **Respiratory** : Apnea, air obstruction, pulmonary edema, aspiration pneumonia.
3. **CVS** : Shock, Heart failure, hypertension, and cardiac arrest.
4. **Metabolic** : Hyperpyrexia, metabolic acidosis, hypoglycemia, and hyponatremia.

Treatment :

- **ABC** :
  - Patent airway
  - Put in lateral position.
  - **O_2 and IV line**
    - * If no IV line → rectal diazepam.
    - * If there is IV line → take blood sample for glucose, Ca^{+} , acid-base balance, electrolytes.
- Then give IV diazepam or midazolam "these drugs cause respiratory depression (MV may be needed).
- **If no response** → call anesthetist : induction of general anesthesia and mechanical ventilation.





Management of epilepsy :

The following rules are important for long term drug therapy:

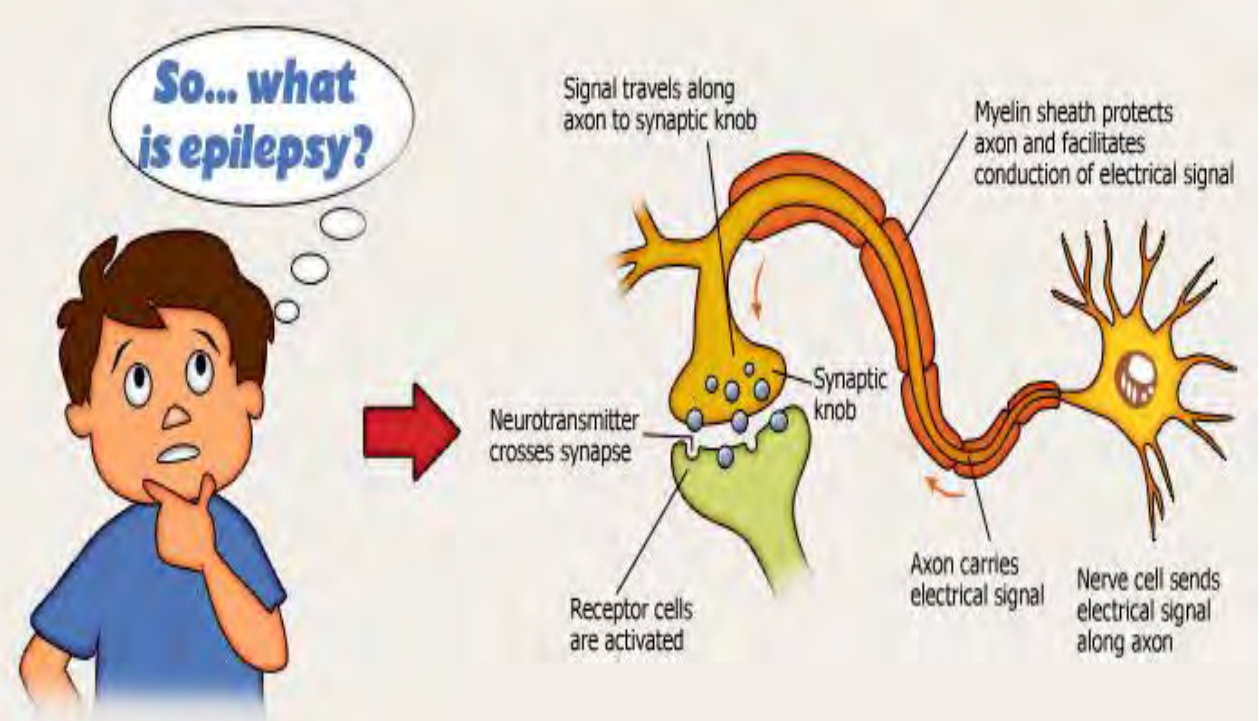
1. Initiation of therapy only after accurate diagnosis.
2. Choice of drugs according to clinical and EEG finding.
3. **Number of drugs** : start with one drug in small dose, then increase gradually until seizure control or max. dose is reached.
   * Failure of 1st drug is an indication to add the 2nd drug.
4. Duration and termination of therapy:
   - At least 2 years after the child being seizure free.
   - Termination should be gradually.
5. **Parent counseling:**
   - Avoid watch TV except in lighted room and far enough from the screen.
   - Computer games should be done under supervision.
   - Regular follow up.
   - Regular investigation LFT , RFT.
   - Jumping from one doctor to another not recommended
   - Avoid R.F ,e.g. : driving or high building.
   - Swimming under observation.

S/E of anti-convulsents drugs:

1) **Phenytoin :**
   - Coarsened facial features.
   - Hirsitism.
   - Gum hypertrophy.
   - Lymphadenopathy.
   - Cerebral atrophy.
   - Hepatic dysfunction.
   - Steven Jonson syndrome.
2) **Phenobarbitone** :
   - Skin rash.
   - Respiratory depression.
   - Steven Jonson syndrome.
   - Hepatic dysfunction.
   - Behavioral dysfunction.
3) **Na valoperate** :
   - Fulminant hepatic failure → monitor by LFT every 6 month.
   - Hyperamonia.
   - Pancreatitis.
   - Thrombocytopenia.
   - Weight gain.
4) **Ethosuxemide** :
   - Leukemia.
   - Pancytopenia.
   - Skin rash.

Prognosis:

- 70% of children with epilepsy are seizure free by age of 16 years.
- Remission is less likely in partial seizure.
- Infantile spasm also B/C usually symptomatic.



Febrile convulsions

- Convulsion due to rapid rise in temperature due to extra cranial infections.
- The most common seizure disorder during childhood, 3-5% of all children.

Etiology:

- **Genetic background :** in families with +ve history of convulsion, abnormal genes have be located on chromosomes 8, 19.
- Recurrent in 30-50% of cases especially those with family history.

Pathophysiology :

Fever increase the metabolic rate of cerebral neurons → lowering the seizure threshold.

Diagnostic criteria :

1. **Age:** 6 months - 6 years
   "Rare before or after this age".
2. **Temperature**: > 39°C , fits occur within 8-12 hr from the onset of fever.
3. **No evidence of CNS infection.**
   e.g meningitis, abscess.
4. **Evidence of extra cranial infection.**
   e.g tonsilitis, otitis media.

Clinical pictures

1. Typical simple FC :

- **Age :** 6 month - 5 years.
- **FH: +ve .**
- **Developmental history** : Normal.

Preictal :

Fever "the core temperature is 39°C or more.

Convulsion occur few hours after the fever.

Ictal :

- Generalized tonic clonic.
- **Duration** : Short < 15 minute.
- One fit only during the same illness.

Post ictal : Short. **"The commonest form"** .

2. Atypical complex FC :

- **Age** : < 6 month or more than 5 years.
- **FH: +ve** FH of epilepsy may present.
- Delayed developmental milestone.

Preictal :

- Fever "the core temperature >38.5°C.

Ictal :

- May be focal.
- **Duration :** > 15 minute.
- Recurrent during the same illness.
- Risk factor for developing epilepsy later on.
- "Uncommon".

Investigations:

- Usually not needed in children with simple febrile convulsion.
- **For complex**:
  - L.P.
  - CT - MRI.
  - **Others:**
    - Blood glucose.
    - Serum electrolytes.
    - Toxicology screen.
    - EEG.

Treatment :

- **ABC :**
  - Patent airway.
  - Diazepam rectal or IV 0.3-0.5mg/kg/dose.
  - Measure to lower the temperature .
    e.g.: Removing clothes
    Tapid sponging.
  - Giving antipyretic e.g. paracetamol.
  - Treat the cause , e.g. antibiotic in acute tonillitis.
  - Prophylactic therapy.

Indication:

Atypical "complex" FC.

High risk of recurrence:

Interval less than 3 month between the 1st and 2nd seizure or between 2nd and 3rd .Young age < 1 year , & +ve FH.

Drugs: Rectal diazepam , Na vaproate.

* Daily until the child is 3 years old or 2 years after being seizure free.

It is essential to know

- If the child has meningitis or another serious infection that need treatment.
- **If child is < 1 year** with febrile convulsion → exclude meningitis always do lumbar puncture.

Table. Factors that influence recurrence of seizures.

	Recurrent febrile seizure	Later afebrile seizure
Febrile seizure in first-degree relative	Yes	Undetermined
Afebrile seizure in first-degree relative	No	Undetermined
Developmental delay	No	Yes
Neurological abnormality	No	Yes
Age < 15 months at febrile seizure	Yes	Undetermined
Shorter duration of fever prior to seizure	Yes	Undetermined
Height of temperature	Inversely	Undetermined
Focal or prolonged febrile seizure	No	Yes
Multiple seizures in 24 hours	Possibly	No

Clinical evaluation and investigation in case with convulsions:

➲ **History :**

Personal history : Age

- **Neonate** :Certain causes are more common in neonate , e.g hypoxia.
- Infancy and childhood, e.g. febrile convulsion are age dependent.

➲ **Present history :**

- Did the child have s seizure before?
- Immediate preceding symptom, e.g fever .
- **Seizure description :**
  - Generalized or not .
  - Associated
  - Duration.
  - FH of epilepsy and FC.
  - PMH.
- Try to search for the etiology considering age of the patient.
- Manifestation of CNS infection "*fever , vomiting , loss of consciousness*"
- Criteria of febrile convulsion.
- H/O trauma.
- Intellectual deterioration → degenerative brain disorder.
- Treatment history (peior anti convulsant medication and the child response)

➲ **Examination :**

1. **General** :
   - **Vital sign** : BP , Temperature.
   - **Measurement** : OFC .
   - **Head and neck** : for signs of ↑ I.C.P and neck rigidity.
   - **Trunk and limbs** : Skin lesion in neurocutaneous syndrome.
2. **Systems** :
   - **CNS examination :** Motor , reflexes , sensation , and cranial nerves to diagnose brain tumor , Hge or infarction.
   - **Fundus examination :** Papilledema with ↑ I.C.P .
   - **Abdominal examination** : Organomegally may suggest metabolic or storage disease .

➲ **Investigation :**

- **Laboratory** :
  - CBC.
  - **Serum electrolytes :** Na , K , Ca , Glucose , Toxicology
  - L.P.
  - Metabolic screen.
  - TORCH screen.
- **Imaging :**
  - EEG.
  - **CT - MRI.**

MENINGITIS

Acute bacterial meningitis

Acute bacterial inflammation of the meninges covering CNS "pia and archnoid" , it is a serious disease with high incidence of acute complication and chronic morbidity.

Etiology: "Organisms"

1. During the 1st 2 months of life :

- **Gram +ve** : B hemolytic streptococci. , Listeria monocytogenes.
- **Gram –ve :** Enteric bacteria. H.I type B **"less commonly"**

2. From 2 months to 12 year :

- **Gram +ve** : Strept. Pneumonia Staph.
- **Gram –ve** : H.influenza type B. Neisseria meningitides.

N.B : Peak of H.influenza infection between 6-12 month incidence declined by vaccination.

Mode of transmission:

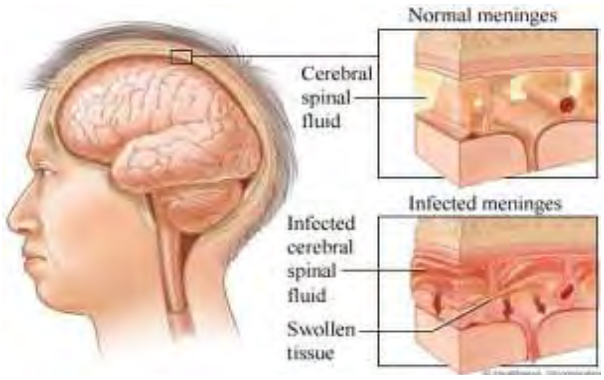
- Droplet infection : during epidemic.
- Blood spread : from adistant site.
- Direct spread : from O.M or mastoiditis orbital cellulitis.

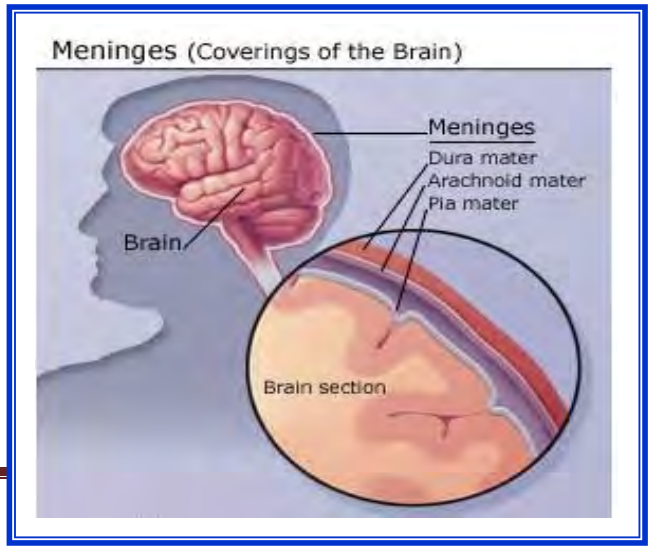
Pathgenesis :

- Bacterial colonization of nasopharynx is the usual initial site of infection , bacteria enter CSF through the choroid plexus and meninges.
- Bacteria then circulate into the extracerebral CSF and subarachnoid space and rapidly multiply B/C the CSF conc. Of complement and Ab. Are inadequate to overcome bacterial proliferation.

Pathology:

- The pia and arachnoid are congested and infiltrated with inflammatory cells → thin layer of pus forms and this may later form adhesions or may damage cranial nerves at the base of the brain.
- Vascular and parenchymal cerebral changes → vasospasm, vasculitis, thrombosis → cerebral infarction.
- Inflammation of the spinal nerve roots produces meningeal sings.





C/P:

1. Systemic manifestation :

- Fever , anorexia , malagia , and arthralgia . (Hypothermia in neonate)
- URTI symptoms.
- Purpura, DIC, and may be shock "meningococcemia"

2. Manifestation of ↑ I.C.P :

- **In infant:** - Bulging AF.
  - Separated sutures.
- **In older children:**
  - Projectile vomiting.
  - Sever headache.
  - Blurring of vision.
  - Abducent or oculomotor nerve palsy.
  - Hypertension and bradycardia. "Cushing response"

3. Manifestation of meningeal irritation :

(absent in infant less than 2 years)

- **Neck rigidity:** Passive flexion is difficult and painful.
- **Kernig sign** : Child in supine position, hip and knee joint are flexed 90° extension of the knee is limited.
- **Brudzinski sign :**
  - Neck sign: Passive flexion of neck → flexion of hip and knee.
  - Leg sign: Passive flexion of one leg → flexion of the other.

4. Manifestation of neurological deficits :

- Disturbed conscious level.
- Focal paralysis.
- Seizures.
- Coma.

Investigation:

- Complete septic screen.
  1. **CBC** : marked leukoytosis.
  2. **CSF examination**.
     - C & S is essential.
     - Exclude viral meningitis.

N.B :Mumps only viral infection cause low CSF sugar .

	Normal	Bacterial	T.B	Viral
Appearance	Clear	Turbid	Yellow	Clear
Tension	Normal	↑	↑	↑
Protein	10-40mg/% 5-20mg/dl	↑	↑	↑ or normal
Sugar	40-6-mg/dl 60% of bood glucose	↓	Very ↓	Normal
Chloride	720-750mg%	↓	↓	↓
Cells	Lymphocyte 0-5/mm^3	Polymorphis 10-100,000	Lymphocyte 250-500	Lymphocyte 15-2000

Complication:

1. Early : "during the course of illness".

1- CNS:

- Subdural effusion "enlarged head circumference and bulging AF".
- cerebral abscess.
- Convulsion "focal or generalized"

2- Vascular:

- DIC.
- Shock due to acute adrenal insufficiency
  "Waterhouse Fredrichen syndrome".

*Inappropriate secretion of ADH resulting in hyponatremia and water intoxication. "prevented by fluid restriction".

- 8^{th} cranial nerve damage → deafness.

2. Late : "occur after recovery and require follow up"

- CNS:
  - Hydrocephaly due to inflammatory obstruction of CSF pathway.
  - Subdural empyema with developmentof ↑ICP.
  - Epilepsy due to focal adhesion with underlying cortical irritation.
  - Mental retardation and learning deficits.

Treatment :

1) **Prevention :**
   - **Vaccination:**
     - Infant in the first year of life : HiB vaccine 3 doses against H. influenzae type B.
     - Children: Meningococal polysaccharide vaccine "A and C" at 3 year or school entry.
   - **Chemoprophylaxis:**
     - Rifampicin used to eradicate meningococci from the nasopharynx of carriers and minimize the risk of contact infection.
2) **Supportive :**
   - IV fluid if meningitis is complicated by shock (other wise it should be restricted to minimize cerebral edema)
   - Blood transfusion for cases with DIC.
   - Anti-convulsants : Diazepam and phenobarbitone.
   - **Measure to ↓ICP :**
     - Manitol 0.5-1mg/kg IV
     - Furosemide 1mg/kg IV
   - **Corticosteroids "Indication":**
     a- H.influenza meningitis to reduce inflammatory response caused by cell lysis.(use Dexamethazone 0.15mg/kg/dose/6hrs for 2 days).
     b- Septic shock to improve general condition.
     c- Adrenal failure.
3) **Specific treatment :**
   - **Antibiotic** : 3rd generation cephalosporin + ampicillin "to cover listeria"
   - If 3rd generation cephalosporin not available → Chloramphenicol + ampicillin.
   - If the organism is N.meningitidus → penicillin or vancomycin.

N.B: Monitor general condition , temperature , weight "they indicate improvement".

Note:

- If child improves then after a few days has fever think of nasocomial infection or subdural collection.
  (Fever , ↑OFC , focal neurological signs → need CT scan)
- If there SIADH → weight gain , hypoNa^+.

Pronosis :

Depend on :

1. Age: Younger age is the worse prognosis.
2. Course: Fulminant meningitis has worse prognosis.
3. Cause:
   - E.coli and staph. → ↑fatality.
   - H.influenza type B and pneumococci → moderate prognosis.
   - Meningococci → < 5% fatality and no residual disability.

Poor prognosis associated with :

1) Age young.
2) Long duration of illness without received Ab.
3) Seizures.
4) Coma at presentation.
5) Shock.
6) Leucopenia.
7) Immunocompramized.

⁂ **D.D :**
1. From other causes of meningitis.
2. Meningism : non infectious meningeal irritation due to extra cranial infection , e.g OM
   * CSF is normal.
3. Brain abscess.
4. Encephalitis.

Aseptic meningitis

↳ Meningitis with no micro-organism detected in CSF by gram stain or bacterial culture.

★ **Causes :**

1. Mostly viral : HSV , Enterovirus "Echo-coxachi" , Mumps , EBV , Adenovirus , HIV , and CMV.
2. Protozoa : Malaria , Toxoplama.
3. Non infectious :
   - CNS leukemia.
   - IC injection.
   - Post vaccination.
   - NSAID.
   - IVIG

★ **Treatment :**

Supportive and antiviral.

⇨**Risk factors for meningitis:**
① Young age "< 5 years".
② Male.
③ Overcrowding.
④ Splenectomy.
⑤ Sickle cell anemia.
⑥ AIDS.
⑦ Malignancy.
⑧ Meningomyelocele.
⑨ Penetrating trauma.
⑩ CSF shunt " by staph. Epidermis".

★ **Lumbar puncture** :
- Put baby in lateral position , fully flexion of the back.
- Sterilization of skin "from inside to out".
- Draw an imaginary line between the 2 iliac crest at middle of this line palpate for the intervertebral space (L3-L4 or L4-L5), then insert the L.P needle peripendicular.

⁞ **C/I :**
1. Sign of ↑I.C.P b/c there is a risk of coning of the cerebellum through the foramen magnum.
2. Coma or shock.
3. Local infection at site of L.P.
4. Coagulation disorder.
5. Skeletal deformity.

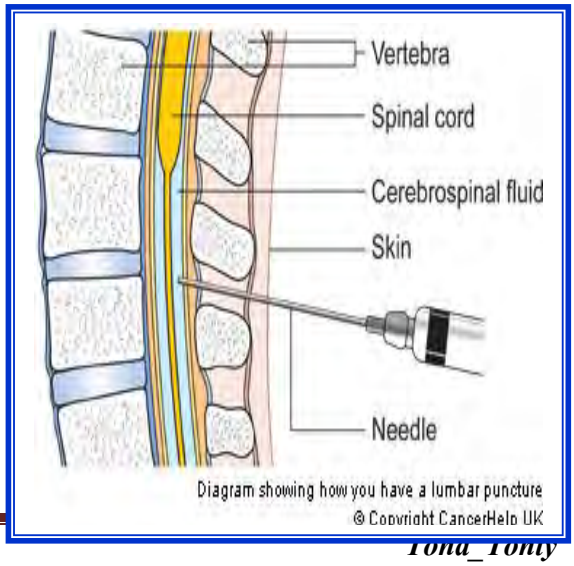
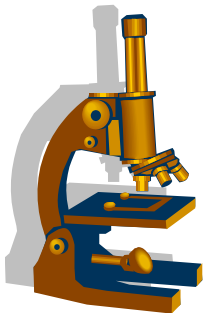


Diagram showing how you have a lumbar puncture
© Copyright CancerHelp UK



⁂ **Specimens of CSF should be collected :**

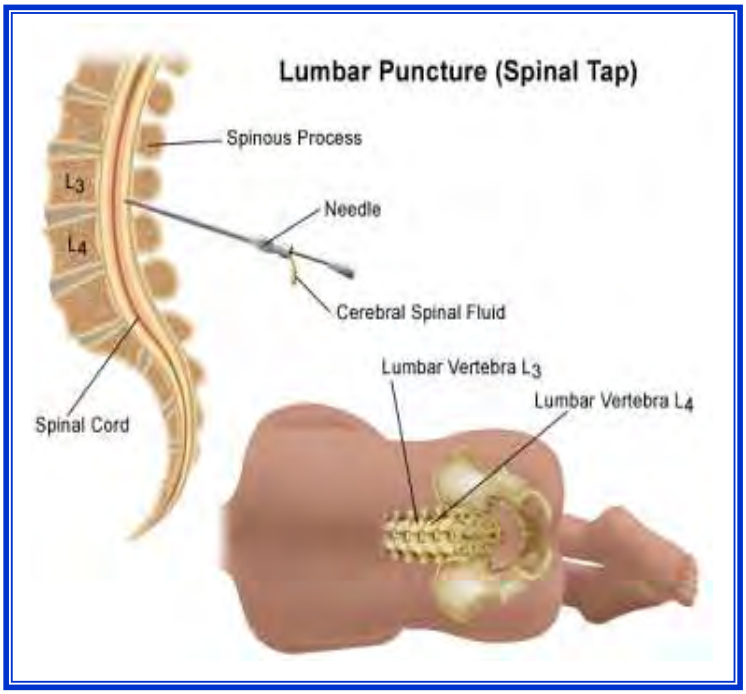
- Microscopy "GS to identify organism" and culture.
- Cells "polymorphs ,...etc" → fastest result to appear.
- Biochemistry : sugar and protein.

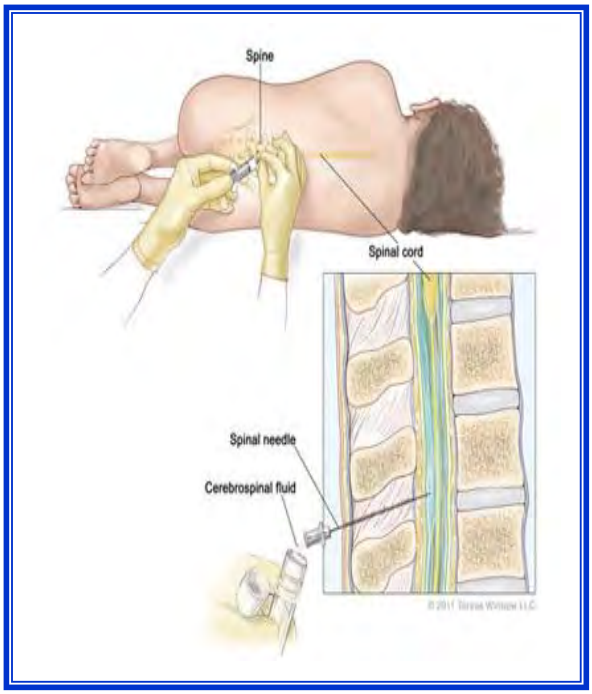
⁂ CSF should be transported in special transport media "otherwise bacteria will be lysed".

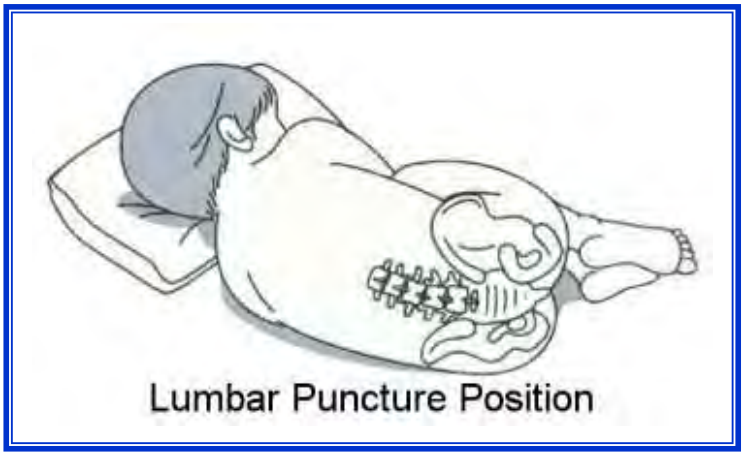
* Blood sugar should be taken "to compare with CSF sugar.

CSF sugar will ↓ → due to damage of B.B.B.

If cells ↑ , protein ↑ , sugar ↓ and no bacteria → partially treated meningitis or CSF not transported to lab.







ENCEPHALITIS

Inflammation of the brain commonly due to viral infection.

★ **Etiology:**

1. **HSV type I and type II :** (Are the commonest cause).
   - **Type I :** Asymptomatic infection or gingivostomatitis or ketatoconjunctivitis.
   - **Type II:** Vulvovaginitis. (the usual cause of encephalitis in neonate).
2. EBV.
3. CMV and HIV.
4. **Enterovirus polio-echo and coxsackie :**
   - Man is the reservoir of infection.
   - Transmitted by feco-oral route.
   - May cause epidemic in neonatal nurseries.
5. **Viruses transmitted from animal to humans** :
   - Arboviruses "arthropod-borne".
     - Animal and birds are R.O.I .
     - Transmitted by mosquitoes.
   - Rabies.
6. **Viruses associated with childhood illness :**
   - MMR , chicken pox → cause acute or post infectious encephalitis.
   - Post infectious illness is mediated by immune mechanism.
7. **Non viral cause of encephalitis :**
   - Mycoplasma pneumonia.
   - T.gondi.

★ **Clinical picture :**

1) **Manifestation of viremia** : fever , nausea , URTI .
2) ***Manifestation of meningeal irritation but less severe than meningitis***.
3) **Manifestation of ↑ICP** .
4) **Manifestation due to focal neurological deficits :**
   - Disturbed level on consciousness.
   - Irritability ,screaming , bizarre movement.
   - Confusion up to deep coma.
   - Paralysis or paresis and spasticity are common.
5) **Post encephalitic may develop later in the form of :**
   - **Motor** : Spastic cerebral palsy.
   - **Mental** : Speech defect , visual or auditory handicapped and behavioral abnormalities.
   - Epilepsy and MR.

★ **Investigation :**

- Serological test to detect antiviral antibodies.
- **Brain biopsy :** HS best confirmed by brain biopsy.

★ **Imaging :**

- **MRI** : In post infectious encephalitis " may show demyelination"
- **EEG** : Focal or generalized abnormalities. **(In HS there is temporal lobe abnormalities)**

★ **Treatment :**

✓ **Supportive :**

- Control of convulsion "diazepam and phenobarbitone"
- Measure to lower the ↑ in ICP"
  - Dehydrating measures to lower the ↑ed ICT.
  - Head elevation 30° in neutral position.
  - Reduction of fluid intake to 70% of daily requirement.
  - Dexamethazone and diuretics.
  - Mechanical ventilation in severe cases.

✓ **Specific therapy :**

- Only if HS → Acyclovair.

BRAIN ABSCESS

- Most common between 4-8 years.

★ **Etiology :**

Staph. aureus , streptococci , and others.

➲ **Origins** :

- Direct spread from sinusitis , orbital cellulitis , chronic O.M , mastoiditis.
- Direct spread from face , scalp , dental infection ,and penetrating head trauma.
- CCHD, meningitis, infected ventricloperitoneal shunt.

★ **Pathology:**

- Site : Anywhere frontal , parietal , and temporal lobes are the most commonly affected.
- Number : Single or multiple "multiple common with meningitis and CHD"

D.D of restriction of fluid intake:

1) Meningitis.
2) Encephalitis.
3) DKA.
4) CHD.
5) Birth asphyxia.
6) ARF.
7) Hypertonic dehydration.

★ **Clinical Pictures:**

1. Manifestation of toxemia.
2. Manifestation of ↑ICP.
3. Focal neurological signs : Hemiparesis, convulsion , and may be coma.

★ **Investigation :**

- CT and MRI are the most reliable.
  - Plain CT : Rounded hypodense lesion.
  - Contrast CT : The abscess capsule shows ring enhancement.

★ **Treatment**:

- Combination of Antibody and drainage.

MOTOR NEURON LESION

- **Upper motor neuron lesion (UMNL)**
  - Pyramidal
  - Extra pyramidal
  - Any lesion above anterior horn cell.
- **Lower motor neuron lesion (LMNL) Neuromuscular**
  - Lesion from anterior horn cells to the muscle.

⁂ Clinically ; How to differentiate b/w UMNL and LMNL :

1) By inspection :

	UMNL	LMNL
Posture	• Scissor position. • Fixed flexion contraction • Decorticate: flexion UL and extension LL. • Decrebrate : Extension of UL and LL. • Frog like position "rare".	Frog like position b/c hypotonia.
Muscle wasting	-ve except in long standing.	+ve
Fasciculation	-ve	+ve

2) By palpation

Muscle tone	Hypertonia.	Hypotonia
Power	Not differeniate b/w upper and lower Ask pt. to elevate his leg up: Grades: 0 → Complete paralysis. 1 → Only fasciculation. 2 → With gravity. 3 → Against gravity. 4 → With mild resistance. 5 → Complete resistance.	

Reflexes	• In upper limb examine biceps and triceps • In lower limb examine ankle and knee	
	Hyper reflexia " Brisky"	Hypo reflexia or abscent
Babenski sign	+ve	-ve
Clonus	Slightly flexion of knee joint and flexion of ankle joint then sudden extension. (If tremor +ve)	
	+ve	-ve

N.B:

Hypertonia is 2 types:

1. Pyramidal : Klasp knife spasticity.
2. Extra pyramidal : Rigidity which is two types:

a) Cog wheel → Rigidity and tremor.

b) Lead pipe → Rigidity only.

⇨Differentiation b/w pyramidal and extra pyramidal:

	Pyramidal	**Extra pyramidal**
Tone	Clasp knife spasticity	Lead pipe or Cog wheel rigidity
Reflexes	Hyper reflexia	Normal reflexes

Neuromuscular Disorders

1) Anterior horn cells :

1. Werdnig-Hoffmann disease: Genetic disease with sever hypotonia.
2. Poliomyelitis: Infectious disease present with acute paralysis.

2) Peripheral nerve :

1. Chronic hereditary polyneuropathy.
2. Infectious polyneuropathy: " Guillain-Barré syndrome": post infectious present with acute paralysis.

3) Neuromuscular junction :

1. Myasthenia gravis: Autoimmune disease present with motor weakness.
2. Botulism: present with acute paralysis.

4) Muscles "Myopathy" :

1. Muscular dystrophies: Genetic disease present with progressive motor weakness.
2. Dermatomyositis: Autoimmune presents with myalgia.

☑ **Clinical pictures of neuromuscular disorders :**

1. Floppy infant : Sever hypotonia at birth or early infancy.
2. Acute paralysis : Acute loss of previously acquired motor skills.
3. Progressive motor weakness : Gradual progressive loss of motor skills.
4. Delayed motor development : Delayed head support , sitting and standing.

1. Floppy infant

Sever persistent hypotonia presenting at birth or in early infancy.

★ **Etiology :**

1. Central "cerebral disorders" :

• Atonic cerebral palsy due to : **perinatal asphyxia ,birth trauma or CNS infection.**

• Chromosomal defect : e.g Down syndrome.

2. Peripheral "neuromascular disorders" :

• AHC : Werdnig-Hoffmann disease.

• Muscle disease : Congenital myopathy - Congenital muscular dystrophy**.**

★ **Clinical picture :**

1. **Manifestation of hypotonia :**
   - **Hypotonia of neck muscles "head lag"** : When baby is pulled up from his hands while in supine position the head lag backward.
   - **Hypotonia of trunk muscles** : Curved trunk on horizontal suspension → when baby is suspended in prone position over the palm of the examiner he droops around it like arag doll " C shaped ".
   - **Hypotonia of limbs** : Frog like position.
2. **Manifestations that differentiate central from peripheral :**
   - **In central group :** Reflexes are normal or exaggerated.
   - **In peripheral group :** Reflexes are absent or depressed.

Werdnig-Hoffmann disease

Genetic disorder in which AHC in the spinal cord are progressively lost begings in fetal life and continue in infancy and childhood.

★ **Etiology :**

- Autosomal recessive gene mutation on chromosome 5.
- Degeneration is due to failure to arrest apoptosis.

★ **Clinical picture :**

- Neonatal period: Hypotonia and absent grasp reflex.
- Later on: Bell shaped chest, diaphragmatic breathing.
- **Motor system:**
  - State : Muscle wasting, fasciculation "best seen in the tongue"
  - Power : Sever weakness with weak cough and weak cry.
  - Tone : Sever hypotonia → Forg like position and head lag.
  - Reflexes : Absent.
- Normal mentality and normal eye movement.
- Death from aspiration pneumonia and renal failure in late infancy.

★ **Investigation:**

- Electromyography (EMG) and muscle biopsy confirm the diagnosis → show features of denervation.

★ **Treatment :**

" No treatment is available ".

- Pharyngeal suction if there is weakness in pharyngeal muscles.
- Spinal brace to maintain sitting posture.
- Supportive treatment of pneumonia.

N.B:

⏩ **Spinal muscular atrophy :**

✔ Group of genetic disorder in which there is degeneration of AHC, it affect leg more than arms, proximal muscles more than distal.

✔ Classification:

- Mild.
- Moderate.
- Sever → Werdnig-Hoffmann disease.

2. Acute paralysis

Rapid loss within hours or days of the previously acquired motor skills.

Etiology :

1. **Spinal cord entral :**
   - Transverse myelitis.
   - Spinal cord trauma.
2. **Ant. Horn cells :**
   - Polio (asymmetric ascending paralysis).
3. **Peripheral nerves :**
   - Guillain-Barré syndrome : "commonest cause"
     ↳ Symmetric ascending paralysis.
   - **Post diphtheric paralysis :**
     ↳ Symmetric descending paralysis.
4. **Neuromuscular :**
   - **Botulism** : Symmetric descending paralysis.

Acute infection polyneuropathy
Guillain-Barré syndrome

Etiology : unknown.
- It Follow viral infection, e.g. measles, CMV, and EBV
- Post infectious demyelination.
- Also Follow: Cambylobacter pylori , mycoplasma pneumonic .

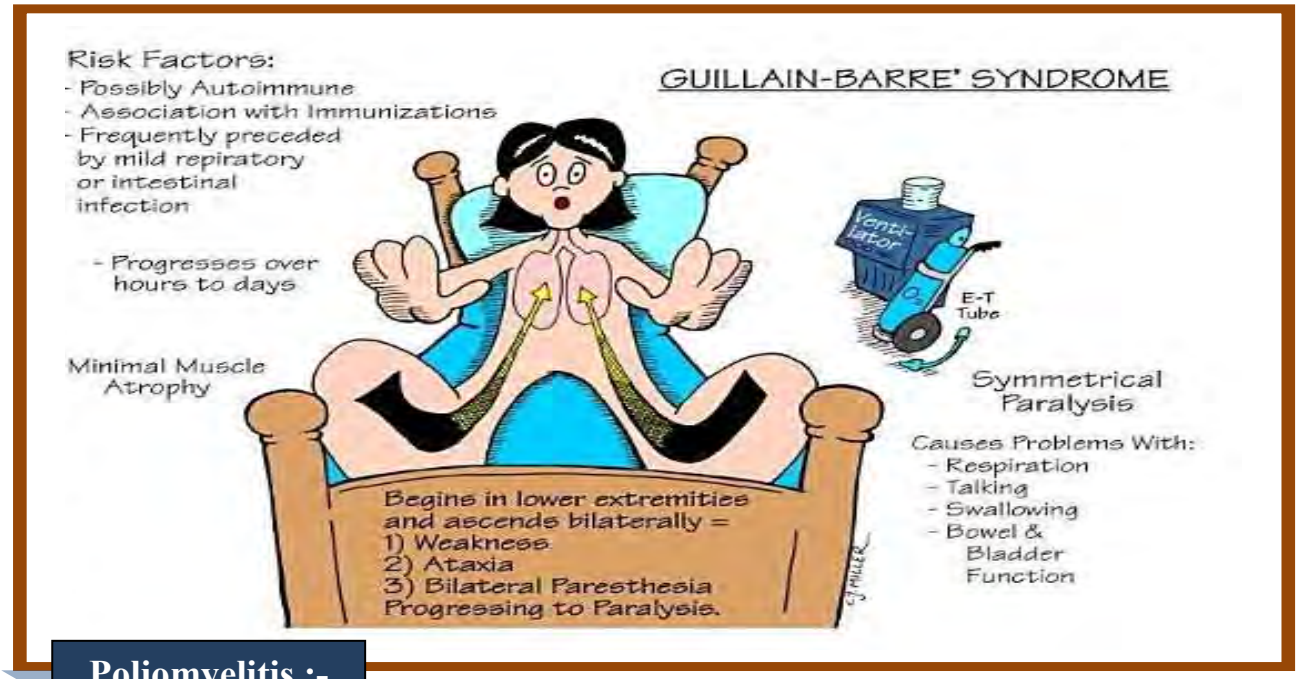
Clinical picture :

1) H/O proceeding in faction Few weeks before onset of paralysis.
2) **Neurological examination**: 20% has bladder dysfunction.
   - Motor : acute paralysis which is:
     - Ascending: begins in the lower limb and progressively ascend to involve the trunk and upper limb over days or weeks, respiratory muscles affection leads to paradoxical breathing → RF .
     - Symmetrical .
     - Reflexes: hyporeflexia or areflexia.
     - Sensation: parasthesia in some cases.
     - Cranial naves: bulbar and facial paralysis may occur.
3) Gradual complete recovery over few or several weeks in some pt. paralysis may persist For several months.

Investigation:
- CSF analysis: "2 weeks after onset of paralysis"
  ✓ Cytoalumious dissociation (↑protein but with normal cell count).
- EMG: ↓ed nerve conduction velocity.

Treatment:
- ICU and mechanical ventilation for cases with Resp. ms paralysis (life saving).
- IV gamma globulin in all pt. for 5successive days.
- **Plasma pharesis**.
- Physiotherapy should start from the 2nd week of illness

Prognosis:
- Most of cases 75% → spontaneous recovery within 2-3 weeks.
- 35% with some permanent neurological deficit.
- 5% death rate



Poliomyelitis :-

➥ B/C of vaccination it become rare disease.

Etiology :

- Acute viral infection of AHC (3 viral strain).
- Transmission : Feco-oral and droplet.
- Man is only reservoir.

Diagnosis of paralytic polio :-

1- Acute paralysis:-

⏩ **Type**:

- LMNL (hypotonia and hyporeflexia).
- Pure motor no sensory affection as the site of destruction is the ant. HC or cranial nerve nuclei.

⏩ **Distribution** :

- Unilateral, and if bilateral it is asymmetrical.
- Proximal more than distal. "start in L.L and may extend to affect abdominal muscles, trunk and neck" in some pts, bulbar and respiratory paralysis may occur.

2- Gradual incomplete recovery:-

- Usually follows the paralysis.

Treatment:

- Acute stage (1st 2 weeks) ➞ isolation and treatment of bulbar and Resp. affection.
- Restoration stage (2 week - 6 month) ➞ physiotherapy.
- Residual stage (After 6 month) ➞ orthopedic correction of contractures.

3. Progressive motor weakness

- Gradual loss of previously acquired motor skills.
- Progression may be rapid over weeks or months (tumour) **or** slow over months or years as in "degenerative brain disease and muscular dystrophy".

Etiology :

1- Cerebral causes:-
- Brain tumour and cyst.
- Degenerative brain diseases.

2- Spinal cord causes:-
- Compression paraplegia (spinal cord tumor or Pott's disease).
- Degenerative disease of the spinal cord.

3- Muscular cause:-
- Muscular dystrophies (Duchenne muscular dystrophy is the most common).

Clinical evaluation :

By detailed History and examination :

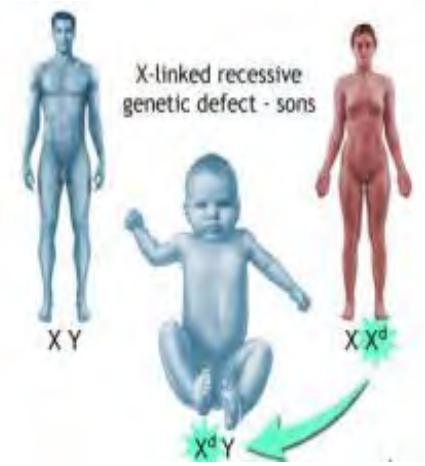
1- Differentiation b/w UMNL and LMNL:-
- **UMNL :** Brain and spinal cord → hypertonic & hyper reflexia.
- **LMNL** : AHC, PN, MS → hypotonia and hypo reflexia.

2- Diagnostic of the cause:-
- Cerebral causes :↑ICP and/or conclusion and mental affection are usually present.
- Spinal cord lesion : LL only are usually affected other manifestation as urinary incontinence of urine.
- In muscular dystrophy progression is very slow over years and in Duchenne type pseudohypertrophy of the calf muscles is present.

Investigation :
- CT and MRI → in brain or spinal cord lesion.
- Serum CPK, EMG, muscle biopsy in muscular disease.

Duchenne Muscular Dystrophy

- X- linked recessive male only affected "some new mutation "
  The abnormal gene "dystrophin gene" is present on X chromosome.



Clinical picture :

• Dystrophin deletion which is needed for stability of muscle cell wall

➲ **History:**

• Rarely given before the age of 3 years.

• Difficulty in getting up stairs and getting up from sitting position of gradual onset and slowly progressive course up to difficulty in walking.

• FH may be + ve from the mother side (the disease is transmitted from carrier mother to their sons).

N.B:

⁂ When dystrophin decrease → influx of Ca^+, irreversible destruction of muscle cells.

➲ **Examination:**

1) Neurological examination :-

• State: Pseudohypertrophy of calf muscles, early and constant sign followed by muscle of the forearm.

• Power and tone: Weakness and hypotonia of pelvic girdle.

• Waddling gait.

• **Gower sign**:

- Characteristic manner in getting up from bed "the child climbs his legs".

- Weakness and hypotonia of shoulder girdle muscles.

- Inability to comb the hair.

- Inability to raise the arm above the head.

• Mild mental affection.

2) Other systemic examination :-

• Heart: Cardiomyopathy → Cardiomegally.

• Death occur usually in 2nd decade "15-25 year" from respiratory failure or heart failure.

• Wheel chair by age of 11 year.

Investigation :

- Serum CPK ↑ "10-200" times higher than normal.
- **EMG :** Myopathic changes.
- **Muscle biopsy :** Diagnostic "absent dystrophin".
  - ↳ Necrotic tissue, fat cells, fibrous tissue.

Treatment:• Mainly supportive :
- ↳ Good nutritional support, physiotherapy.

Case presentation :

- 3 years old child (boy) is brough to physian with H/O frequent falling down and according to the young boy parent, he is having difficult climbing staires.
- O/E : Hypertrophy of calf ms is noted.
- CBC : Hb = 12g/dl WBC = $10x10^3$
  Na = 130mg/dl K = 4mg/dl
  Urea = 25mg/dl CPK = 1750
  (normal <100)Dx: Ducchanne muscle dystrophy.

D.D " Other muscular dystrophies":-

1) Baker dystrophy:
- Similar to Duchanne ms. Dystrophy.
- X-linked recessive disease.
- Low dystrophin.
- Clinically same like duchanne disease.

2) Limb girdle dystrophy:
- Autosomal recessive.
  "girls and boys affected"
- Normal muscle dystrophin.

3) Metabolic myopathies:
- e.g. Glycogen storage disease.

Myotonia dystrophica

↳ Myotonic muscle dystrophy.
- Autosomal dominant disorder.
- Distal atrophy.
- In neonatal period may presented with hypotonia.
- Onset early.
- Associated with:
  - Frog like position.
  - Difficult feeding.
  - FTT.

☆ **Definition of myotonia**: Disorder of muscle fiber that result in abnormality prolonged contraction with difficult to relaxing after any vigrous effort → Failure of muscle relaxation after forceful contraction.

Clinical pictures:
- Hypotonia.
- FTT.
- Delay eye opening after closure.
- In majority cases ; patient have learning difficulties associated with :
  - Cataract.
  - DM.
  - Testicular atrophy.
  - MR.

Treatment:
- Symptomatic treatment.
- Most of them die due to respiratory failure.

Diseases of neuromuscular junction

Myasthenia gravis :

- Autoimmune disorder due to binding of antibody against acetylcholine receptors at the post synaptic membrane.
- More common in female.
- Onset after 1st year.

Clinical pictures:

- Onset usually gradually with involvement of extra-ocular muscles "unilateral or asymmetric ophthalmoplegia or ptosis", strabismus.
- Involvement rapidly extends to proximal muscle bulbar muscle and respiratory muscles.
- Weakness appear after exercise or at the end of the day.
- UL more than LL.
- Difficulty in chewing.
- Loss of facial expression.

Diagnosis:

① Tensilon test "edrophonium test": Rapid relieve of symptom with IV cholinergic medication "edrophonium" (anti cholinesterase).

② EMG : Show diminished amplitude of muscle action potential.

③ Measurement Ab to ACh receptors.

Treatment:

Symptomatic treatment:

1. Oral long acting anti-cholinesterase.
   e.g. Neostigmine, Pyridostigmine.
2. Thymectomy . "May associated with thymoma".
3. Ig may play a role in treatment.

Heredity motor sensory neuropathy "Charcot-marie-tooth disease"

Clinical pictures:

- Symmetrical slowly progressive muscular wasting« more in distal muscle of lower limbs ».
- Autosomal dominant.
- Present in first 10 years.
  - Distal atrophy in legs, which leads to gait disturbances foot deformity "pes cavus".
  - Rarely sensory loss.
  - ↓ reflexes.

Investigation :-

- **Nerve biopsy:**

Affected nerve show onion bulb formation and remyelination → hypertrophy of the nerve due to demyelination and remyelination.

Neurocutaneous diseases

1) Neurofibromatosis "Von reckling Hausen disease"

- Autosomal dominant but must with new mutation.
- Every organ can be affected, features present from birth but complication may be delayed into adulthood.

⇨**NF 1**"on chromosome 17":-

*** To make the Dx two of the Following are needed:**

1. At least 5 café-au-lait sport > 5mm pre-pubertal or at least 6 café au lait sports > 15mm postpubertal.
2. Axillary or inguinal freckling .
3. > 2 iris lish nodules, pigmented hamartoma of the iris."seen on slit lamp only".
4. > 2 neurofibroma or one plexiform neuroFibroma.
5. Osseus lesions → Kyphoscoliosis, splenoid dysplasia.
6. Optic glioma.

Complication:

1. Low grade gliomas hamartomas.
2. Malignant neoplasm "astrocytoma, neurofibrosarcoma and others.
3. Cerebral vessel stenosis, aneurysm → TIA, hemiparesis Hag.
4. Complex partial or generalized seizures.
5. Cognitive defects → learning disability, attention deficit, speech abnormalities.
6. Pheochromocytoma.
7. Increase incidence of leukemia, rhabdomyosarcoma, Wilms tumour, neuroblastoma.
8. Macrocephaly.

Treatment:

- Genetic counseling.
- Early detection of treatable conditions.
- Annual ophthalmologic examination .
- Examine family members.

⇨**NF 2**"on chromosome 22":-

Clinical picture:

- Bilateral acoustic neuromas.
- Hearing loss .
- Facial weakness.
- Headache .
- Unsteady gait .
- Skin finding much less common.
- CNS tumors common.

*** To make Dx of N.F2 one of the Following are needed:-**

- Bilateral acoustic neuromas.
- Parent or silbling or child with NF2 + unilateral 8^{th} nerve mass or any 2 of the following:-
  - Neuro Fibroma.
  - Meningioma.
  - Glioma.
  - Schwannoma.

2) Tuberous sclerosis:-

- Autosomal dominant, half with new mutations.
- Wide Range of manifestation within same family.

Clinical picture:

- ***Infancy***: Infantile spasm and characteristic skin lesion:-
  - Ash- leaf macule → hypopigmented lesion on the trunk seen by woods lamp.
- ***Childhood***: Generalized seizures and skin lesions:
  - Sebaceous adenoma → small red modules on nose and cheeks "looks like acne".
  - Shagreen patch → rough, raised lesion with orange.
  - Peel consistency most in lumbosacral area.

Complications:-

May ass. with autism.

1- Retinal lesion "hamartoma" → phakomas (round, flat gray lesions in area of disc).
2- Cardiac Rhabdomyoma → can be detected in fetus with Echo , most spontaneously regress over 1^{st} 2 years.
3- Renal lesion → either hamartoma or polycystic kidney.
4- Pul.→ cystic or fibrous changes.
5- Subangual fibroma → beneath the nail.

Diagnosis:

- Head CT scan → periventricular calcification.
- Echo.

3) Sturge-Weber Syndrome:-

- Have Facial nevus "port wine stain".
- Seizures.
- Hemi paresis .
- I.C calcification and MR.
- Nevus is always present at birth and always involve at least the upper face and eyelid.
- Glaucoma in ipsilateral eye.

Presentation:-

⏩ Seizures in most cases (Focal tonic clonic, contra lateral to the nevus)) becomes refractory and slowly develops hemi paresis, MR.

Dx: • Skull x-ray shows occipital - parietal calcification.
• CT scan → show unilateral cortical atrophy and hydrocephalus.

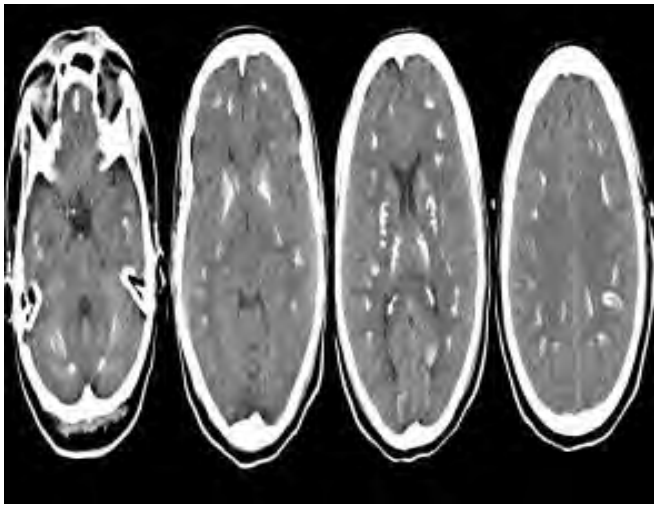
R : - Control seizures. – Hemispherectomy. - Regular IOP evaluation.

D.D of intra cranial calcification:

1. **Neurodermatosis** e.g. : Sturg Weber syndrome.
   Tuberous sclerosis.
2. Trauma "Hge"
3. **Inflammation:**
   - Congenital toxoplasmosis.
   - CMV.
   - Rubella.
   - Pyogenic abscess.
   - HS.
   - Echinococcus.
4. **Tumors:**
   - Craniopharyngioma
   - Teratoma.
   - Astrocytoma.
5. **Metabolic:**
   - Vit A. D excess
   - Hypo or hyper parathyroid.

D.D of Cafe' auliat spot :

1. NF commonest.
2. Ataxia telangectasis.
3. Multiple endocrine disorder.
4. Tuberous sclerosis.
5. Russell silver syndrome.
6. Noonan syndrome



Pseudo-tumor cerebri :

- Idiopathic intracranial hypertension in over 90% of cases.
- Clinically manifestation of ↑ ICP with normal CSF and normal CT scan and in the absent of mass lesion.

Clinical pictures:

1. Headache.
2. Fatigue.
3. Projectile vomiting.
4. Anorexia.
5. Neck stiffness.
6. Diplopic.

Normal neurological examination except for papilledema or 3rd / 6th cranial nerve palsy.

Normal CT scan, normal CSF profile with exception of ↑ pressure which measured by manometer.

Risk factors:

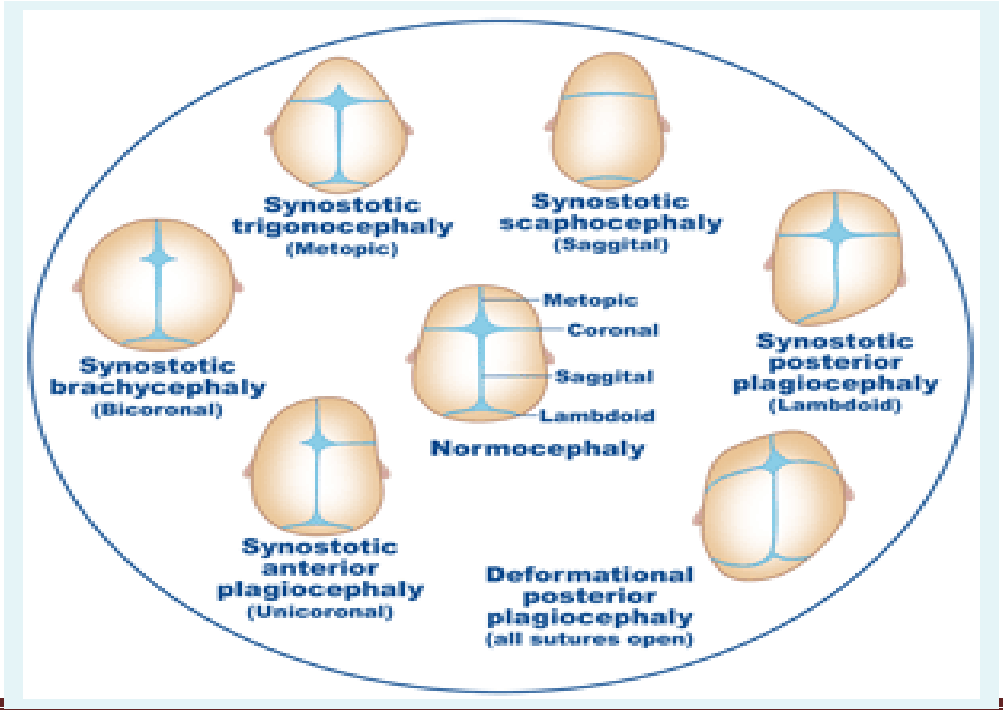
(1) **Drugs :**
- e.g.: - Nalodoxic acid.
- Nitrofurinton.
- Corticosteroid.
- Execs vita. A.

(2) **Endocrine disorder:**
- Hyperthyroidism.
- Cushing syndrome.
- Hypoparathyrodism.

(3) **Thrombosis:**
- Venous sinus thrombosis may be due to head trauma or OM.

Treatment:

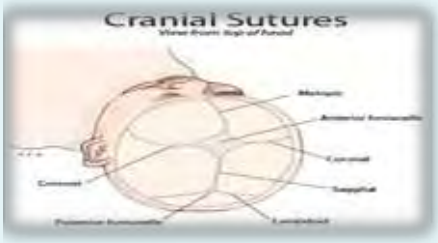
1. Removal of risk factor.
2. Medication e.g. lasix or acetazolamide.
3. Spinal tap.
4. In severe cases: shunt.



Microcephaly:

HC that measures more than 3SD below the mean for age and sex

- Head circumference below the 3rd percentile for age and sex.
- **Causes:**
  1. **True microcephaly**: due to small sized brain.
  2. **Craniosyntosis**: due to early fusion of sutures.



1. <u>True microcephaly:</u>

- **Criteria :**
  - Skull sutures and fontanels are normal.
  - No increase in ICP.
  - Skull x-ray show small vault.
  - CT scan show brain atrophy.
- **Etiology :**

A) **Genetic :**

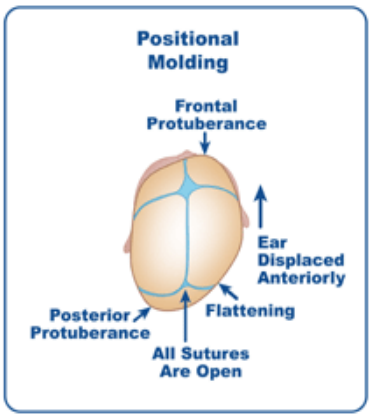
- Familial AR "**sever atrophy of frontal lobes "camel head"**
- **Chromosomal**: Trisomy 21 , 18 , 13.

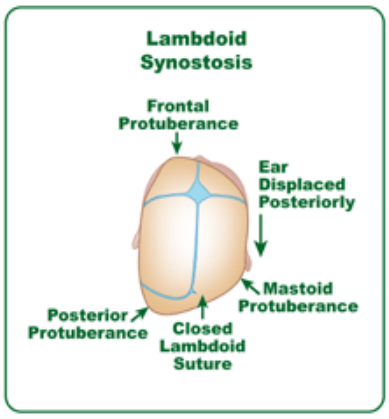
B) **Non genetic :**

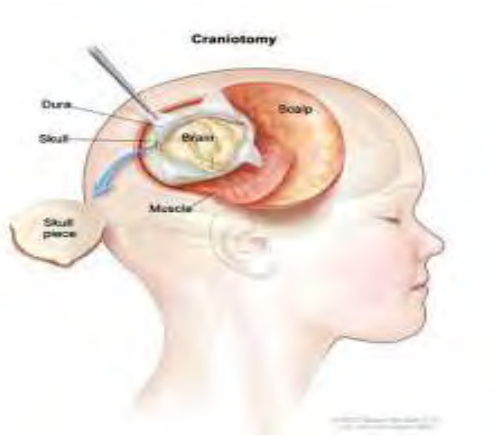
- **Prenatal:** TORCH infection, fetal irradiation, maternal diabetes or PKU, maternal drugs e.g. phenytion.
- **Natal**: HIE.(hypoxic ischemic encephalopathy)
- **Post natal**: Early meningitis and encephalitis.

2. <u>Craniosyntosis</u>:

- **Criteria :**
  - Palpable ridge is felt at the affected sutures.
  - If multiple sutures are affected :
    - Microcephaly → Brain atrophy.
    - ↑ICP → Hydrocephaly.
  - **Abnormal skull shape which may be :**
    - Scaphocephaly**.(sagittal)**→frontal bossing , prominent occipital
    - Brachycephaly**.(coronal)** →broad flattened forehead
    - Acrocephaly.
    - Trigonocephaly**.(metopic)**→pointed forehead and midline ridge
- **Treatment :** Surgical separation of skull sutures **"Craniotomy"** in cases with ↑ICP, hydrocephaly, cosmetic reason.







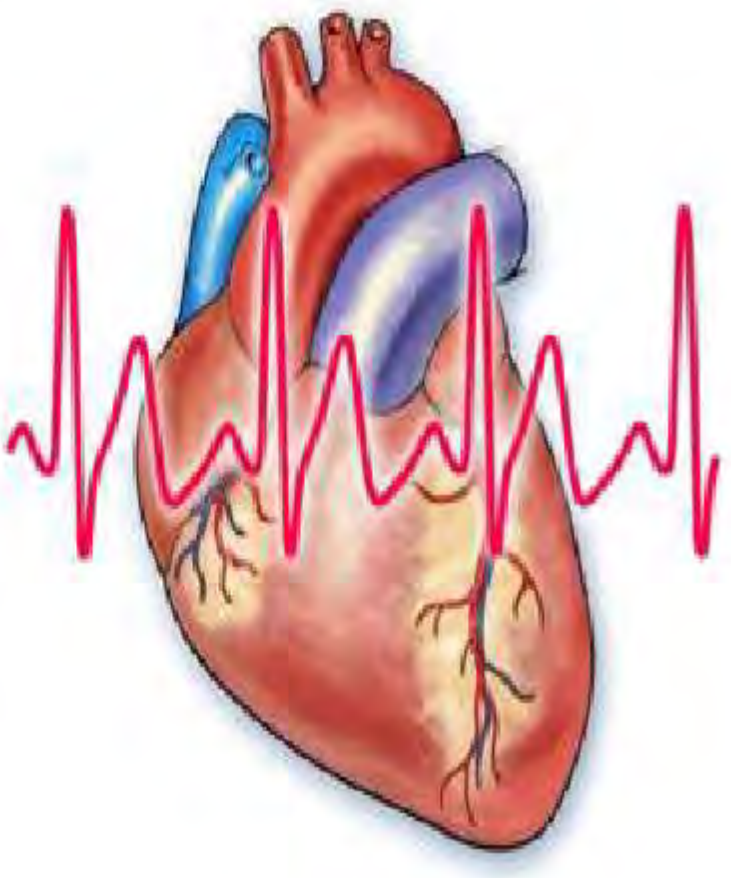
Sudden Infant Death Syndrome

Definition: Unexplained death by history or through post mortem examination in infant of one year or less.

* **Etiology**: Unknown.
  - Death commonly occur :
    - From mid night up to 9am.
    - More in winter season.
  - **peak incidence**: from 2nd month - 3rd month.

* **Risk factors**:
  1. Preterm.
  2. Lack of prenatal care.
  3. Maternal smoking during pregnancy.
  4. Low socioeconomic state.
  5. Prone side sleep position.
  6. Sibling of SIDS have 4-5 times increase risk.

CARDIOVASCULAR SYSTEM CLINIC PART 1



BY DR.ASMA HADIDA

2014-2015

TO BE CONTINUE.....

CVS Examination

- **CVS examination:**
  1) **General examination** : Related to CVS system.
  2) **Local examination** : Pericardial examination.

- **Preparation of examination:**
  Stand on the right side of bed.
  Introduce yourself.
  Permission taken.(inform relative or pt that examination will be done)
  Good position of the pt *(ideally child should be at 45c but small child can be examined in supine position)*
  Good exposure.(undress the child to top half up to waist)

- the teeth should be inspected for caries, which are a risk factor for infective endocarditis

Before end the examination don't forget :

- General examination related to cardiovascular system.

Inspection:

Looks for:

Level of conscious .
Well or unwell
Color (pink , cyanosis , pallor)
Position of child.(example : squatting)
Built : average or under wt .
Cyanosis "**lips, tongue** and extremities "
Any dysmophic features number of dysmorphic syndromes are closely associated with specific heart lesions, and their presence lends a strong clue.
Any sign of RD
Surrounding : O_2 mask , pulse oximetery , IV fluid .
Any heard sound (weak cry –audible wheeze)
Icterus : chronic CHF with cardiac cirrhosis

DD of weak crying ?

1

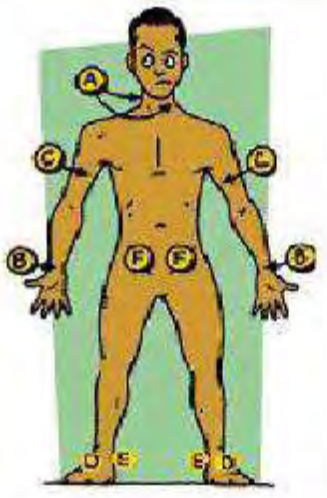
Palpation:

Vital signs:

- Pulse **"rate and look for radio femoral delay " and capillary refilling** **(very important)**

Vital sign

1. Pulse (Rate , rhythm , volume , special character , Radio-femoral delay)
2. Blood pressure .
3. Respiratory rate .
4. Temperature .



A. Carotid

B. Radial

C. Brachial

D. Dorsalis Pedis

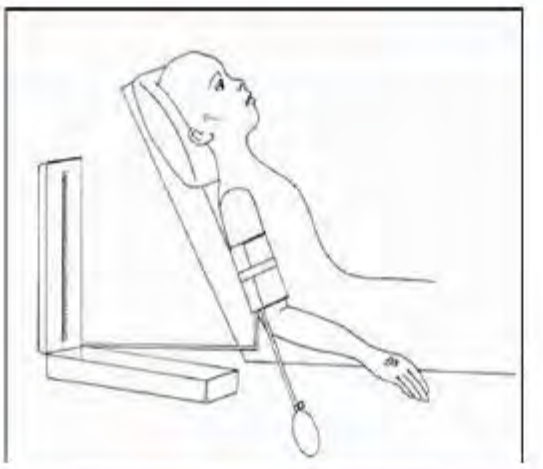
E. Posterior Tibial

F. Femoral

If you detect **radial pulse** , the second step examine **femoral pulse** to exclude coarcatation of aorta .

Blood pressure :

- The width of the cuff should be 2/3 the length of upper arm .
- Must use suitable cuff and child must be relaxed not crying
- Normal range of BP determined by body size gender , and age
- Average BP in **newborn** =70/40 mmhg
- Average BP in **infancy** = 80/50 mm HG
- Average BP in **school children**= 90/60 mm HG
- Put reading in BP chart
- Measure BP of 4 limbs



Methods of BP measurement :

- Sphengomanomerty
- Automated device (dinamap)

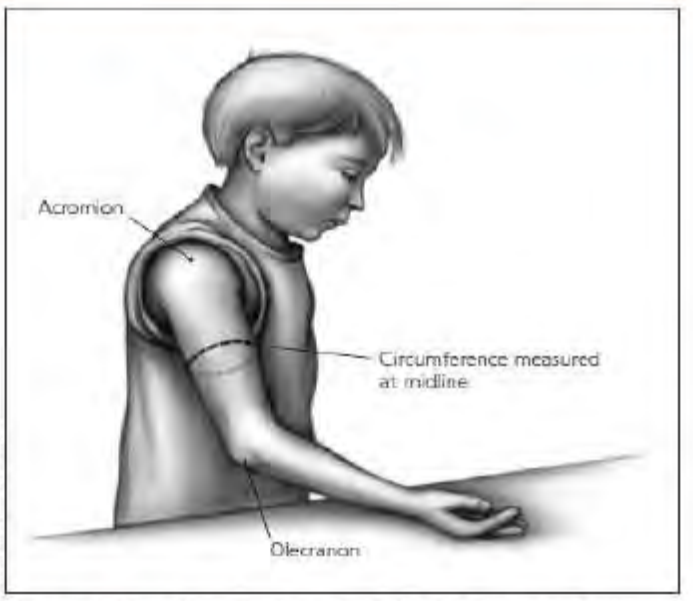
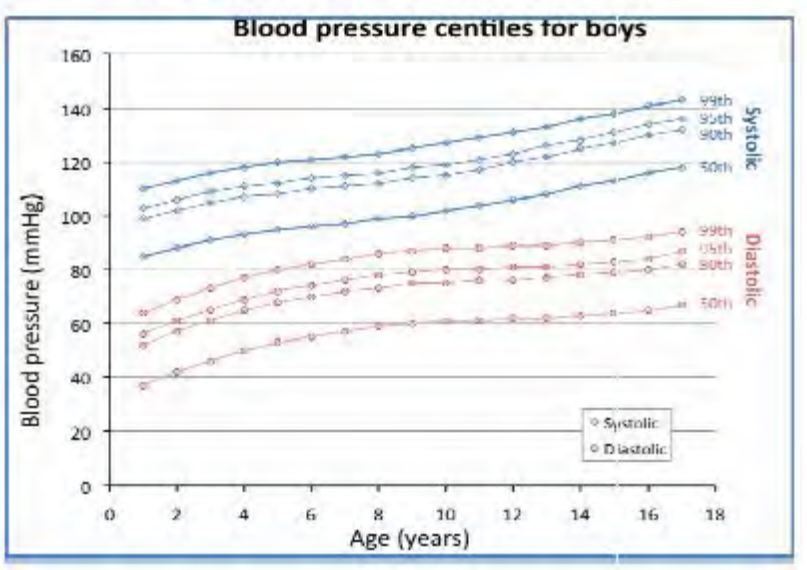


Figure 1. Arm circumference should be measured midway between the olecranon and acromial process.



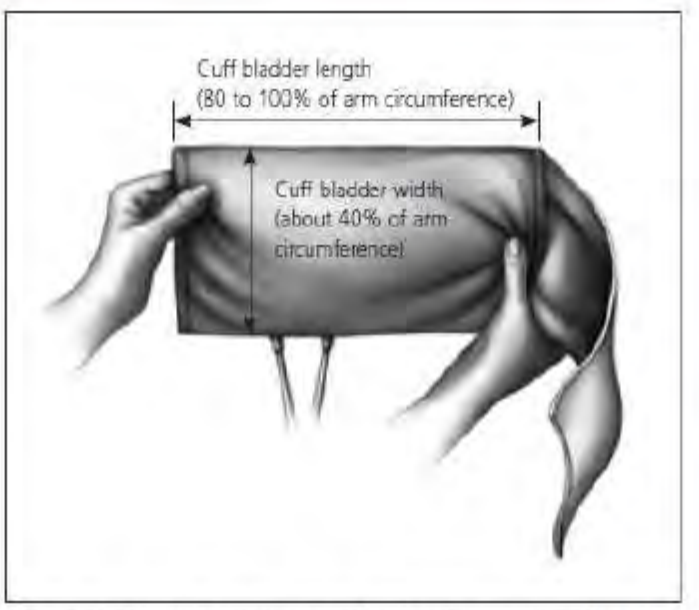


Figure 2. Blood pressure cuff showing size estimation based on arm circumference.

- ***Neck short in infant difficult to assess JVP***
- Signs of anemia and capillary refilling
- **Hand :**
- Clubbing "**In age >6m with prolonged hypoxia** "
- Palmer creases .
- **Nails** :
  Oslers nodules, and splinter hemorrhage **(in discussion only or according to case)**
- ***Lower limb:***
- **Edema** " sacral mainly in infants "
- Clubbing " **deferential clubbing** "
  ↳ "first seen in the thumbs and big toe"
- surrounding *(if pt is connected to pulse oxymeter comment on oxygen saturation and HR)*

- **Anthropometric measurement** ***(baby with congenital heart disease usually FTT)***
- Examine the abdomen for tender hepatomegaly
- Examine the abdomen for splenomegaly (**in infective endocarditis**)
- **Lung** : basal crepitation "congestive HF ".

Sings of heart failure :

- Tachypnea
- Tachycardia
- Tender hepatomegally
- Cardiomegaly on x-ray

To avoid missing the examiner's favourite - dextrocardia - a two handed approach may be used.

Pericardium examination:

I) Inspection:(After proper exposure)

Look for:

Any chest deformity

Any scar: "sternotomy scar, Lt inframamary scar".

Any visible pulsation:

- **Apex beat** ⇨ *(Thin child and Lt V.H.)*
- **Epigastrum** ⇨ *(Rt V.H, T.R, aortic aneurysm, thin children.)*
- **Suprasternal** ⇨*(AR and hyperdynamic circulation.)*

Any localized bulging.

Any superficial dilated vein.

Signs of RD.

Cautary marks.

1) **Apex beat position:**
   - Outer most and lower most palpable beat.
     - **Infant:** 3rd & 4th inter costal space.
     - **2-5 yr:** 4th inter costal space.
     - **> 5 yr:** 5th inter costal space.
   - ***When palpate apex beat should be look for:***
     - Apical beat position.
     - Localized or diffuse.
     - Characters.
   - If can't palpate it may be shifted ⇥ put the child in left lateral position and palpate, if still impalpable think for DD.
2) **Thrill:**
   - Palpable murmur (**palpate in all areas**).
   - Found in (MS - VSD – MR)
3) **Parasternal heave:**
   - **Indicate Rt ventricular hypertrophy.**
   - Using the palm of your hand applied firmly to the Lt parasternal areas.

Right thoracotomy scar

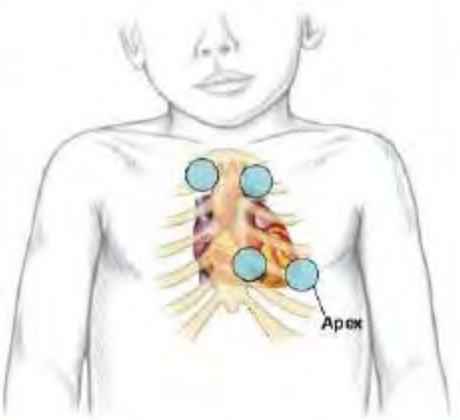
Left thoracotomy scar

Midline sternotomy scar

What is the causes of the scar in the back ?

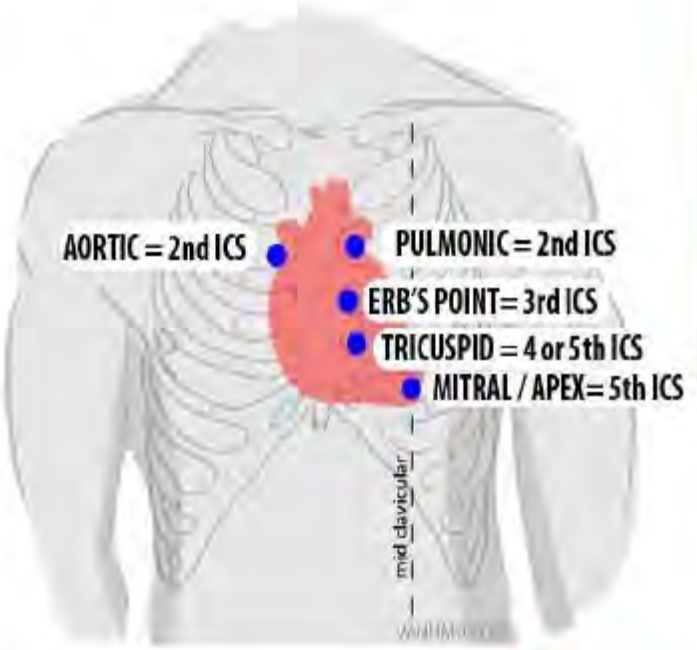
D.D of non palpable apex beat :

1. Obese.
2. Dextrocardia.
3. Pericardial and pleural effusion.
4. Hyper inflated chest
5. Shifting mediastinum to Rt side in pneumothorax.



D.D of shifting apex beat :

1. **Shifted laterally**: Rt ventricular hyperatrophy.
2. **Shifted laterally and downward**: Lt ventri. hyperatrophy
3. **Displaced to the Rt** :
   i. Dextrocardia.
   ii. Pleural effusion.



II) Auscultation:

In all valve area and auscultate for :

- $S_1 + S_2$.
- Murmur.
- Added sounds.

S_1 :

- Closure of M and T valve, occur with ventricular contraction.
- **Muffled S_1** ⇨ VSD and MR.
- **Loud S_1**⇨ MS

S_2 :

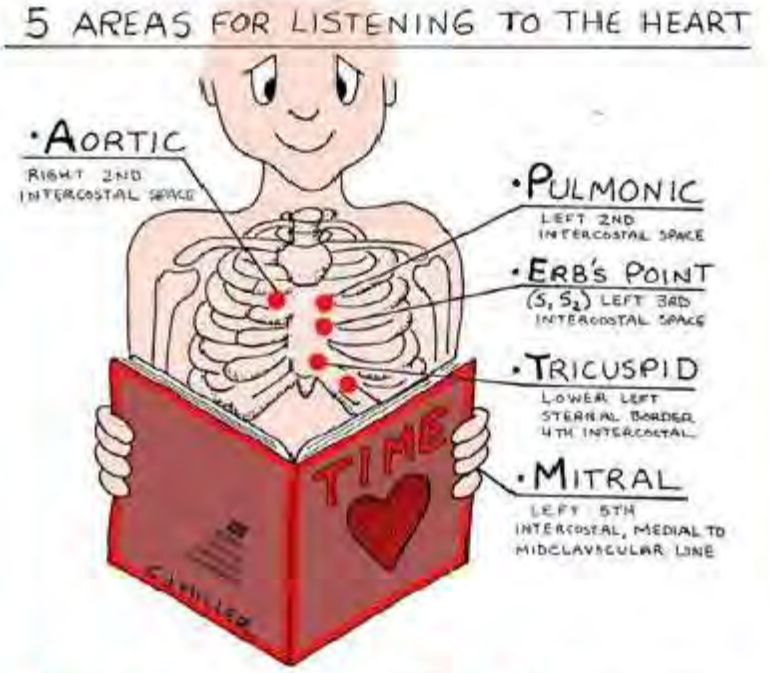
- Closure of A and P valve, occur with ventricular relaxation.
- **Muffled S_2** ⇨ AS and low COP.
- **Loud S_2**⇨ Systemic and pulmonary hypertension and PS.
- **Widely fixed split S_2** ⇨ ASD.

Murmur :

- Site.
- **Timing:-**
  - Systolic "ejection - pansystolic - late".
  - Diastolic "early - mild - late".
  - Continuous murmur.
- **Radiation:-**
  - PSM "MR" radiated to axilla.
  - PSM "TR" radiated to epigastrium.
  - ESM "AS" radiated to carotid artery.
  - PSM "VSD" radiated to all over pericardium.
- **Grade.**

Adding sounds :

- ***S_3 :*** Rapid ventricular filling in diastole, it's may normal in children, young adult, and hyperdynamic state, while abnormal as in heart failure (Gullop rhythm).
- ***S_4 :*** Always abnormal, occur just before S_1.
- Pericardial rub.



⁂ **Systolic murmur:**

Pan systolic: VSD - MR - TR.
Ejection systolic: ASD - AS - PS.

⁂ **Diastolic murmur:**

Early diastolic: AR - PR
Mid diastolic: AR - MS - TR.

⁂ **Continuous murmur:**

PDA "machinery murmur heard in infra clavicular area".

⁂ **Innocent murmur:**

- Systolic murmur with no radiation.
- Short and Soft.
- Mainly in Lt sternal area.
- May change with posture.
- Asymptomatic.

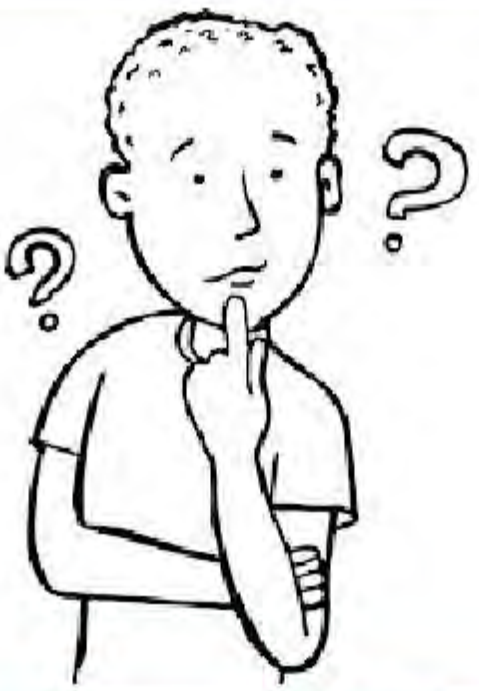
Rremember to:

Ascultate the back of chest :
Between 2 scapula "**CVA".**
Base of lung "**crepitation in HF**".

Systolic murmur "load" + no cyanosis ⇨ **VSD**.
Murmur + cyanosis + scar or no scar ⇨ **TOF**.
Continuous murmur "low pitched" ⇨ **PDA**.
Murmur from the back + no cyanosis ⇨ **COA**

Grades of murmur (intensity):

Grade I	Soft heard in quite room (S_1 &S_2 > murmur).
Grade II	Easily heard but not loud (S_1 &S_2 = murmur).
Grade III	Moderately loud but not ass. with trill (S_1 &S_2 < murmur).
Grade IV	Loud and accompanied with trill.
Grade V	Very load heard with only rim of diaphragm.
Grade VI	Heard with stethoscope 1cm away from chest.



Check capillary refill time :
- If more than 2 seconds consider shock .
- Indicate decrease in peripheral perfusion .

Neonate and infant from sternum.

Toddler from dorsum of hand.

Preschool and school from nail.

- *Most common congenital disease is VSD*
- *Most common cyanotic heart disease is TOF (Tetrology of fallate).*
- *Most cyanotic heart disease in neonate is TGA .*

Analysis of the type of Apex beat

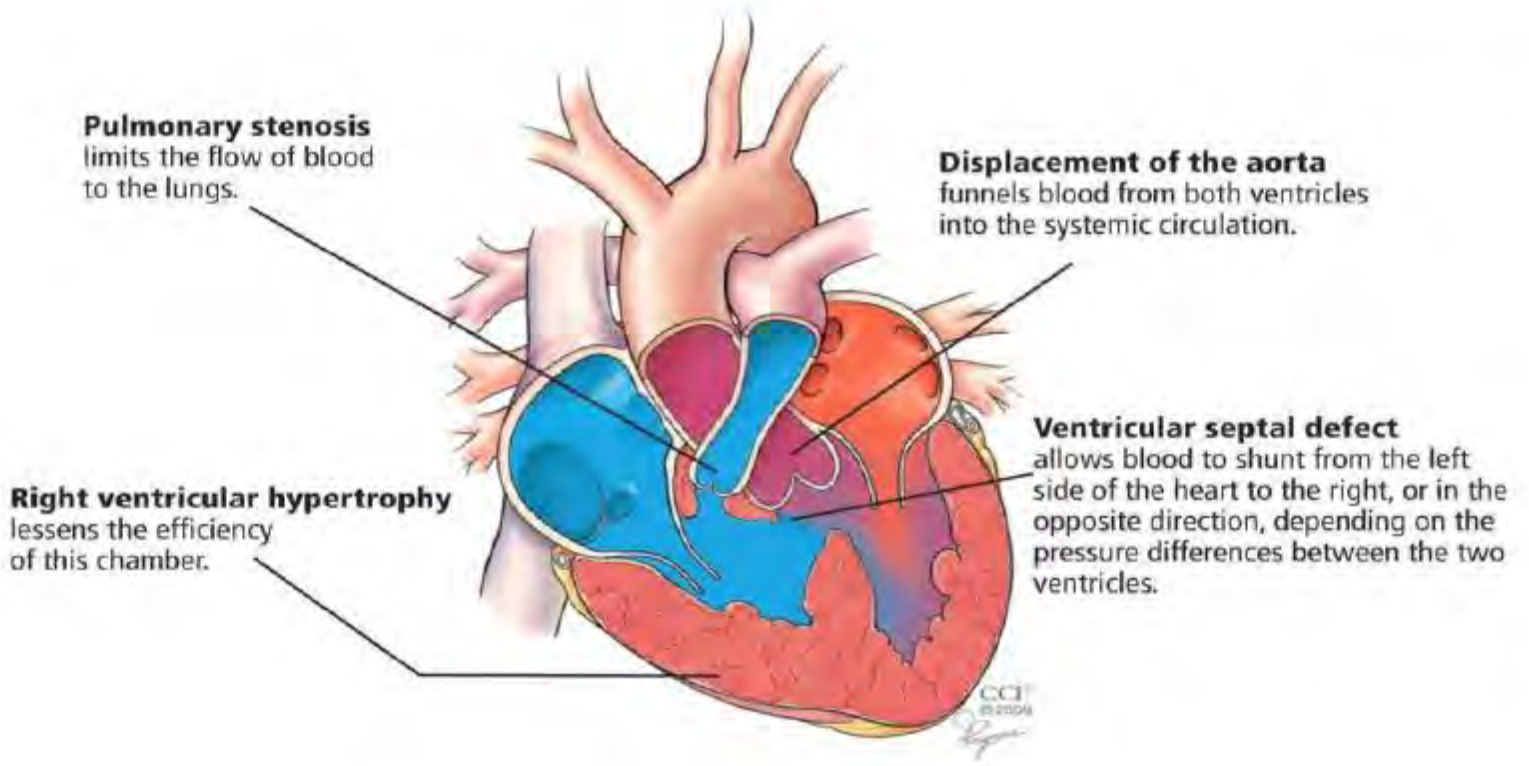
*Is it **forceful**?*

No → *Is S_1 **palpable**?*

- No → **Normal Apex**: Only if position is also normal. (Left 5th ICS, just medial to MCL) If displaced, may be due to non-cardiac causes.
- Yes → **Tapping Apex**: Generally in normal position. Seen in Mitral stenosis due to loud S_1

Yes → *Is it **sustained**?*

- No → **Hyperdynamic Apex**: Will be displaced too. Diffuse apex is characteristic. Ventricular dilatation (volume overload)
- Yes → **Heaving Apex**: Will be displaced too. Localized apex generally. Ventricular hypertrophy (pressure overload)



Children with Tetralogy of Fallot exhibit bluish skin during episodes of crying or feeding.

Low oxygen levels in the blood cause the lips, fingers, and toes to look blue (cyanotic)

Heart rate :

- 0-5y →120-+20 (100-140)
- 5-9→100-+20(80-120)
- 9-10→90-+20(60-120)
- >10→80-+20(60-100)

Rate :

Bradycardia <60 beat /min :

1. Complete heart block
2. Drugs : digoxin , B. blocker
3. Hypothyroidism

Tachycardia ->160beat /min infant

>140beat /min toddler :

1. fixed tachycardia in older children with RF
2. tachyarrythmias

Dr. Asma Hadida 2015